

1 NEOCLASSICAL CONTROL: CONTROL
2 SYSTEMS AND ESTIMATION DESIGN
3 USING THE YOULA PARAMETERIZATION
4 TECHNIQUE

NEOCLASSICAL CONTROL: CONTROL SYSTEMS AND ESTIMATION DESIGN USING THE YOULA PARAMETERIZATION TECHNIQUE

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To Professor John Brewer

6 CONTRIBUTORS

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PREFACE

In the past several decades, there have been many published control system design books in classical (PID), modern (state-space), and neo-classical and optimal robust control. Then, why is it necessary to publish yet another control system design book?

To clearly answer this question, I need to say a few words about my own background. Before becoming a Professor in System Dynamics and Control, I have been a control engineer with both energy and automotive industries for over thirty years. In these industries, feedback is still a novelty and majority of feedback control technique is around designing PID controllers. Control theory has drastically advanced and yet with these advancements, the gap between industrial application of controls and theoretical controls have been increased. Majority of the current control publications are very much math heavy and make the assumptions that all practitioners and students have the right background to fully appreciate these theoretical development. Most importantly, in all these developments, the beauty of control design at times is not clearly addressed and hence, large number of the students and the practitioners are not still have a good understanding of why feedback control is needed.

In this book, we try to close this gap by illustrating that feedback control could be elegantly designed in the frequency domain using Youla parametrization approach. It is

the system bandwidth which elegantly defines a trade-off between performance and robustness. The father of Quantitative Feedback Theory (QFT), Issac Horowitz, said that the bandwidth is like an open window during a hot summer night. If the window is kept widely open then one will let the flies in and if the window is kept shut, then one will limit the summer night breeze. This simple and elegant metaphor defines the trade-off or balance between performance (speed) and noise rejection capability in feedback control design.

Youla parameterization technique is not new, see references[]. However, in this book, we will provide deeper insights with many practical applications in utilizing this technique in both Single Input Single Output (SISO) and Multiple Input Multiple Output (MIMO) frequency domain feedback control design. In addition, an estimation technique using Youla parameterization and controller output observer [] both for SISO and MIMO plants, for the first time, is introduced. Although, the emphasis of these two volumes is on frequency domain, we will discuss optimal robust control approaches such as H_2/LQG and H_∞ using state-space approaches in chapter 5 of this book. We will provide comparative results through practical examples between the optimal robust control methods and the Youla parameterization approach.

We have to apologize for the level of mathematical rigor provided in this book. It goes without saying that control system design is a subject which requires mastery of basic mathematical concepts. We have tried hard to avoid theorem and proof format when introducing mathematical concepts and provided the level mathematics to give the readers the necessary background for understanding and appreciating the control theoretical concepts discussed in this book.

This book is divided in two volumes, Volume I covers SISO control and estimation design using Youla Parametrization approach, while Volume II covers MIMO control and estimation design using Youla Parametrization approach. Volume I of this book includes eight chapters,

In Chapter one, a review of the Laplace transform is given. In this chapter, we tried to avoid many details in introducing the Laplace transform and concentrated on providing the properties of the Laplace transform. It is these properties that will be utilized throughout this book for developing the main concepts.

In Chapter two, the response of linear time-invariant (LTI) dynamic systems is provided. A review of stability of LTI dynamic systems is given and frequency response including Bode and Nyquist diagrams are discussed.

In Chapter three, we provide a discussion of feedback principals, a review of well-posedness and internal stability is given, and the role of Youla parameter and interpolation conditions in assuring internal stability is provided. Complementary sensitivity transfer function T , and sensitivity transfer function S are derived and modeling uncertainty is discussed.

In Chapter four, we provide the fundamentals of feedback design using Youla parameterization. We discuss algebraic and analytical constraints in feedback design and we reveal how Youla parameter could be utilized in designing a feedback control by considering the constraints and trade-offs in SISO control design.

In Chapter five, a review of signal and system size (norms) is provided. Linear Fractional Transformation (LFT) platform and its use in control design is discussed including uncertainty modeling with the use of this platform. A brief review of state-space method including concepts such as controllability and observability are discussed.

In Chapter six, we discuss Mixed Sensitivity Method and H_∞ method for SISO robust control design. In this chapter, we provide insights to the frequency domain solution of H_∞ control and selection of the weighting filters in shaping complementary sensitivity, sensitivity, and Youla transfer functions.

In Chapter seven, we discuss how Youla parametrization could be utilized in deriving state and input estimation method. In this chapter, an introduction to Luenberger observer and Kalman filtering are provided including a discussion about the Controlled Output Observer approach []. A step by step transformation approach from Luenberger observer to Youla parameterization estimation using controlled Output Observer framework is developed.

In Chapter eight, we present three automotive control examples, the application of active differential in vehicle yaw control, active front steering in vehicle yaw control, Anti-lock Braking System (ABS) development. The estimation example consists of battery State Of Charge (SOC) estimation using Youla Controller Output Observer (YCOO) method. In the control examples, we reveal the differences between using SISO robust control design using the standard Matlab toolbox and Youla parameterization hand-computation technique. In the estimation example, we compare the result of YCOO to both linear and non-linear Kalman filter method.

Volume II of this book includes ten chapters,

In Chapter one, we provide an overview of different multivariable feedback control methods and the importance of Youla parametrization approach in designing multivariable

feedback control system. In the same chapter, we introduce multivariable transfer function matrix, Schur complement, and Rosenbrock's system matrix including its application for gaining insight into poles and zeros of a transfer function matrix.

In Chapter two, Matrix Fractional Description, unimodular matrices, and the concept of polynomial matrices' equivalency with several insightful examples are introduced. A discussion on Smith canonical form and Smith-Mcmillan form of a transfer function matrix including several examples are provided. The idea of coprimeness and several approaches for transforming a transfer function matrix to its equivalent stat-space form are discussed.

In Chapter three, we discuss the eigenvalue problem and matrix diagonalization using eigenvectors and eigenvalues. The diagonalization issues using eigenvalues and eigenvectors are discussed and the use of singular value decomposition as a general method for transfer function matrices diagonalization is reviewed.

In Chapter four, we discuss MIMO feedback principals. We introduce MIMO return ratio and return difference, complementary sensitivity and sensitivity transfer function matrices and Youla transfer function matrix. Concepts such as Nyquist stability, wellposedness and internal stability for MIMO feedback systems are discussed. Some of the properties of singular value decomposition are reviewed and their implications on the limit of MIMO feedback performance are revealed.

In Chapter five, we introduce the concept of MIMO feedback control design using Youla parametrization. We illustrate the simplicity of this technique by showing that for low order transfer function matrices, the MIMO control design could be easily performed by hand.

In Chapter six, a review of vector signal and MIMO system size (norms) is provided. Linear Fractional Transformation (LFT) platform is revisited and the uncertainty modeling for MIMO feedback systems with the use of this platform is discussed

In Chapter seven, we discuss optimal control methods. We derive the Hamilton-Jacobi equation using Bellman optimality principal. We review the derivation of Linear Quadratic Regulator (LQR) and Linear Quadratic Gaussian (LQG) from Hamilton-Jacobi equation. We discuss the equivalency between H_2 and LQG optimal control and provide the state-space solutions to H_2 and H_∞ optimal robust control.

In Chapter eight, we discuss two fundamental concepts in uncertainty modeling, μ synthesis and Integral Quadratic Constraints, IQC . We will discuss the advantages and drawbacks of these concepts. The importance of physical system modeling, bond graphs, in identifying uncertainty and designing robust controllers are provided.

In Chapter nine, we discuss how Youla parametrization could be utilized in deriving state and input estimation method for MIMO systems. A step by step design approach for MIMO Youla parameterization estimation using controlled Output Observer framework is developed.

In Chapter ten, we present three automotive control examples, the application of MIMO control design for integration of active differential and active roll control in vehicle yaw control, A new energy management control concept utilizing robust MIMO control design. MIMO robust control design for engine speed control development of a hybrid powertrain vehicle. The estimation example consists of a vehicle semi-active suspension system using MIMO Youla Controller Output Observer (YCOO) method. In the control examples, we reveal the differences between using MIMO robust control design using the standard Matlab toolbox and MIMO YCOO hand-computation technique. In the estimation example, we compare the result of MIMO YCOO to both linear and non-linear Kalman filter method.

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July, 2017

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My sincere thanks and appreciation to Professor John Brewer who introduced me to the concept of Youla parameterization in 90s. Unfortunately, Professor Brewer passed way on Oct. 17, 2003 at the age of 65. His enthusiasms and passion for teaching control system design to his students was exemplary. He shall always be remembered as one of the best teachers by his students.

F.A.

ACRONYMS

COO	Control Output Observer
YCOO	Youla Control Output Observer
MIMO	Multiple Input Multiple Output
SISO	Single Input Single Output

GLOSSARY

265	NormGibbs	Draw a sample from a posterior distribution of data with an unknown
266		mean and variance using Gibbs sampling.
267	pNull	Test a one sided hypothesis from a numerically specified posterior CDF
268		or from a sample from the posterior
269	sintegral	A numerical integration using Simpson's rule

270 SYMBOLS

271 A Amplitude

272 $\&$ Propositional logic symbol

273 a Filter Coefficient

274 $\mathcal{B}$ Number of Beats

INTRODUCTION

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“It seems that without electronic controls nothing works”

I.1 Why Feedback Control?

Control systems are increasingly becoming an integral part of every piece of equipment that we humans use. Without control systems our lives would drastically change. From aircrafts to household appliances, control systems play a vital role in the operation of these machines. In the automotive domain, nowadays, the majority of vehicle attributes are software and control-system based. It is therefore important for everyone, especially machine operators and technical managers, to have a basic understanding of control system operation and their impact. It is also becoming significantly important that all engineers, from every specialty, should be required to take at least one good control system design class throughout their academic education.

When it comes to control system design, our emphasis in this book is on feedback control system design, particularly model-based feedback control design. We will describe what we mean by model based in the next few paragraphs. But, first let's clarify the need for feedback controls. It is interesting, especially in industry, that there is still not a good understanding about why feedback control is necessary.

Let's take a look at Figure I.1(a), where a person without eyesight is trying to pick up a book, which is located on a table. From this person's point of view, there is an uncertainty about the exact position of this book. To reduce this uncertainty, the person would have to find the book by examining different locations on the table with their hands. Hence, it normally takes more effort or energy to reduce overcome the uncertainty and locate this book. Now, imagine a person with eyesight, as in Figure I.1(b). For this person, the exact location of the book is found by making a measurement using the person's eyesight and feeding back this information to the brain. Therefore, uncertainty about the position of the book is reduced by this feedback measurement. Interestingly, the effort and energy required to locate the book is also reduced when compared to the person without eyesight. We can say then that the fundamental reason for feedback in control system design is simply to reduce **uncertainty**. We will see that this uncertainty exist in many sources including the act of the measurement itself. Amazingly, the same uncertainty that gives rise to the use of feedback control itself, itself can result in its demise!

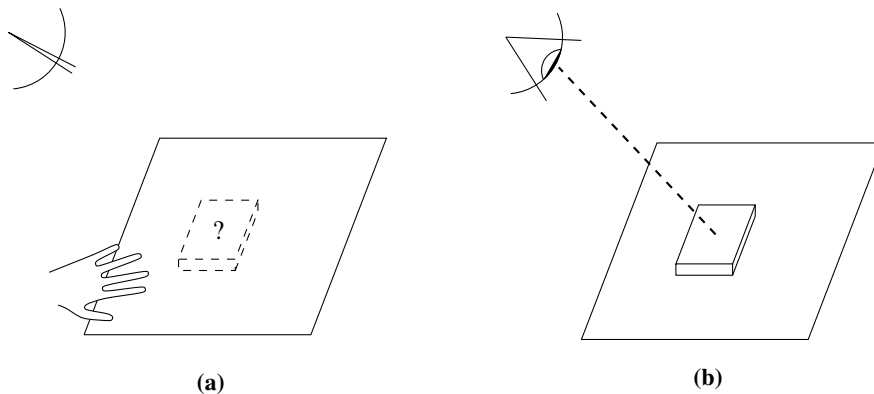


Figure I.1: There is more uncertainty about the book's location with **a** closed eyes, compared to **b** open eyes.

Another important takeaway from this discussion is that reducing uncertainty requires effort and energy and the source of this effort and energy is the actuator. We will relate

the actuator effort and energy to system robustness when discussing robust control development in the following chapters.

There are several different methods that could be utilized in designing feedback controllers []. Our intent in this book is to illustrate that the use of Youla parametrization technique is one of the most straightforward methods in designing model based feedback control system, especially for the SISO control system design. We will show that control system design becomes procedural with minimal calibration effort when implementing these controllers on the actual systems. Reducing controller calibration is significantly important to industry, as reducing calibration effort directly reduces development costs. We will argue that the elegance of Youla parametrization is its simplicity and in reducing controller calibration overhead.

Imagine designing a control system for a very complex system which could be mathematically described by nonlinear partial differential equations (PDEs). What are the choices to design a controller for this complex system?

- Direct control of nonlinear PDEs: As far as the authors of this book are aware, there are no methods which directly address the control design of nonlinear PDEs, especially the open loop unstable kinds. In addition, if there were control methodologies to address nonlinear PDEs directly, quantifying model errors and uncertainties would have been an extremely difficult task. The quote by George Box “Essentially, all models are wrong, but some are useful” should be kept in mind when complex systems are modeled using nonlinear PDEs.
- Indirect control by model reduction of the nonlinear PDEs to (LTI) PDEs: There are methods which address the control of LTI PDEs, such as boundary control of LTI PDEs using backstepping control method [], or robust control of infinite dimensional LTI using H_∞ control [].

It should be noted that at times, linearization of particular nonlinear systems results in loss of controllability. We will not address these systems in this book.

We will use model reduction techniques to go one step further than indirect control by reducing the original complex model to LTI ordinary differential equations (ODEs). Here stands the trade-off between complexity and uncertainty: As the complex model is reduced to a simpler model, naturally the complexity is reduced. However, uncertainty is increased due to loss of information. Therefore, the entire idea of robust control is to quantify these uncertainties in the control design phase.

There are generally two objectives in designing a feedback control system: performance and robustness. “Performance” indicates the ability of the control system to respond quickly and accurately under ideal conditions. “Robustness” indicates the ability of the control system to respond accurately in the face of disturbances, noise, or other sources of uncertainty. Generally (but not always), there is a trade-off between robustness and performance: Speeding up the response requires sacrificing robustness, while increasing robustness requires sacrificing the nominal performance. This concept — the trade-off between robustness and performance — is illustrated in Figure I.2.

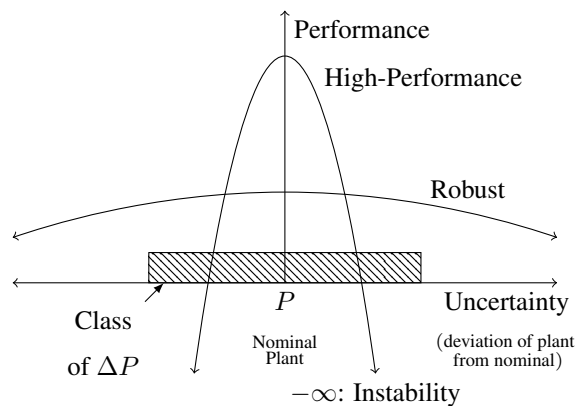


Figure I.2: Robustness and performance.

More specifically, the conflicting measures are:

- Stability (is the controller robustly stable?)
- Command tracking and disturbance rejection
- Sensor noise rejection
- Actuator effort and saturation

The history of control systems theory and the path to robust control is briefly outlined in Figure I.3.

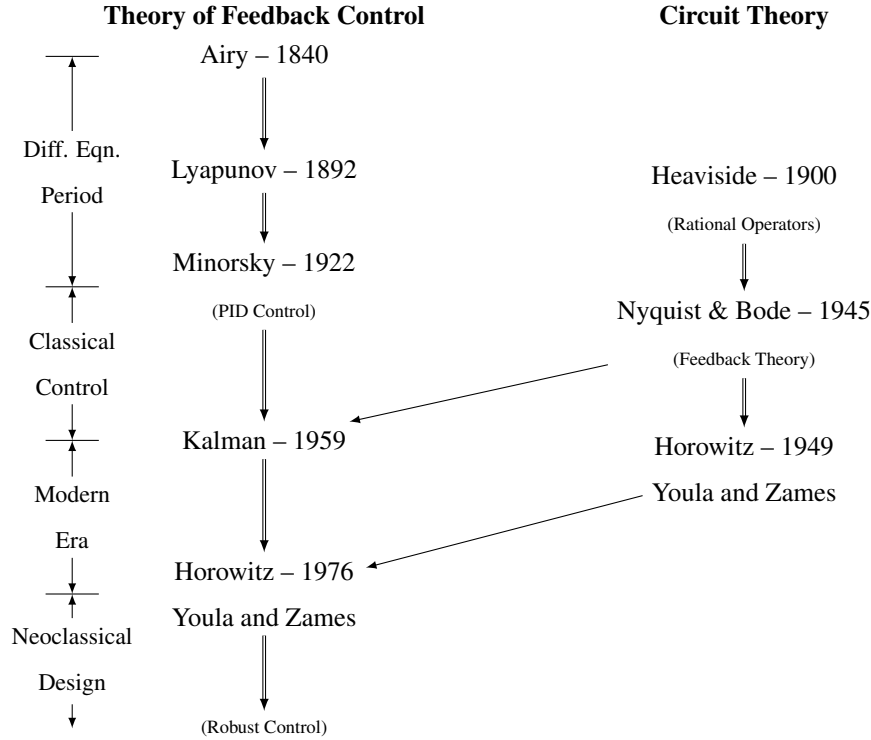


Figure I.3: Brief history of feedback control.

I.2 Why Youla Parameterization?

In the prior section, we briefly discussed the benefits of feedback control and various options for model based control development. In the next few paragraphs, we will elaborate more on the benefits of feedback control and the utility of Youla parametrization in feedback control design. This section presumes some knowledge of the Laplace transform, transfer functions, and block diagram algebra.

When dealing with model-based controls, the starting point is a mathematical description of the system. Consider the simple system shown in Figure I.4, where P is the plant model and Q is the control.,

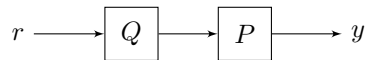


Figure I.4: System model.

In order to enforce perfect target following, where $y(s) = r(s)$ and s is the Laplace operator, we have to select the controller $Q(s) = P^{-1}(s)$. This very simple example illustrates that plant invertibility is very important in control system design. We will elaborate on this point when designing feedback controllers using Youla parametrization method. Let's assume for a moment that this plant is invertible. What happens when we implement this simple controller on an actual plant?

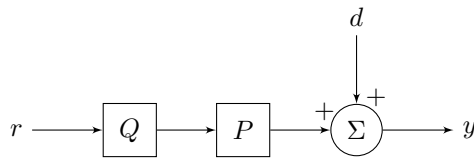


Figure I.5: System model with output disturbance.

In order to capture the reality a bit better, we added a disturbance signal at the output of the plant, as illustrated in Figure I.5. This disturbance signal could, for example, model a wind gust acting on a vehicle while driving. Based on the previous feedforward controller, we can write the following equations.

$$y(s) = d(s) + P(s)Q(s)r(s), \quad (\text{I.1})$$

but

$$Q(s) = P^{-1}(s) \quad (\text{I.2})$$

so therefore,

$$y(s) = d(s) + r(s). \quad (\text{I.3})$$

So, we are here stating the obvious point that if there is a disturbance or a source of uncertainty, the feedforward controller is not capable of coping with this. As a result, the target following in a feedforward loop deteriorates when compared to the situation when perfect knowledge of the plant was available. The output y no longer follows the commanded input r .

Let's now add a feedback to Figure I.5 as illustrated in Figure I.6 below.

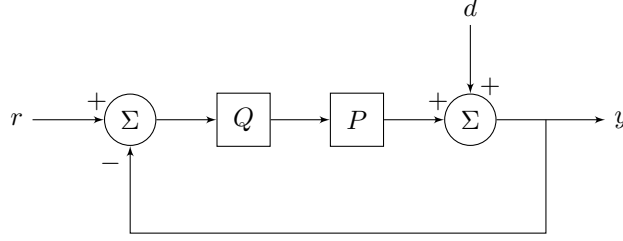


Figure I.6: System model.

The output $y(s)$ can be written as a function of the target signal, $r(s)$, and disturbance signal, $d(s)$, as follows.

$$y(s) = \frac{P(s)Q(s)}{1 + P(s)Q(s)}r(s) + \frac{1}{1 + P(s)Q(s)}d(s) \quad (\text{I.4})$$

Two important points should be evident from this equation. First, we cannot simply set $Q(s) = P^{-1}(s)$, as we did in the feedforward case, if we are designing for good target following and disturbance rejection. Secondly, if we set $P(s)Q(s) \gg 1$ then we will recover the target following of the feedforward controller by rejecting or minimizing the disturbance or uncertainty effect. There are other possible disturbances and uncertainties to consider, but the value of feedback controls in the presence of uncertainties has been made clear.

In the next few paragraphs, we provide a very basic introduction to Youla parameterization. The feedback control loop is revisited in Figure I.7. However, this time, we replaced the dynamic model of the plant $P(s)$ with a generalized symbol $G_p(s)$ and replaced the feedforward controller, $Q(s)$, with a generalized symbol for feedback controller $G_c(s)$.

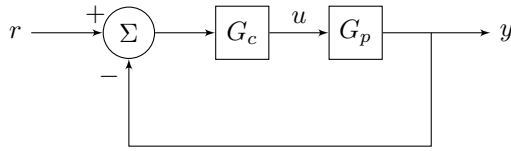


Figure I.7: Closed-loop feedback system.

The closed loop transfer function or the complementary sensitivity transfer function, $T(s)$, can be written as a function of $G_p(s)$ and $G_c(s)$ as follows,

$$T(s) = \frac{G_p(s)G_c(s)}{1 + G_p(s)G_c(s)} \quad (\text{I.5})$$

Equivalently, we define the Youla transfer function[] $Y(s)$ to be the transfer function from the input target, $r(s)$ to the actuator input, $u(s)$,

$$Y(s) = \frac{G_c(s)}{1 + G_p(s)G_c(s)} \quad (\text{I.6})$$

At the first glance, defining the Youla transfer function seems to be trivial and not useful. However, as we discuss below and in the following chapters, this simple mapping will have important consequences both in SISO and MIMO control and estimation design.

The feedback control loop presented in Figure I.8 can be easily manipulated to be drawn as a function of $Y(s)$ as follows,

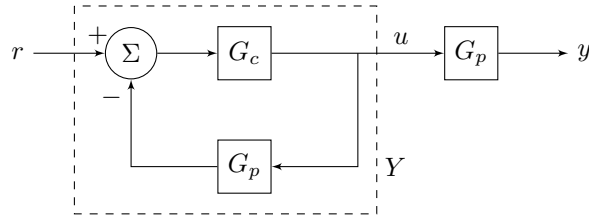


Figure I.8: Block diagram manipulation of system.

It can be seen, from this manipulation, that the complementary transfer function, $T(s)$ can be written as a function of $Y(s)$ as follows,

$$T(s) = Y(s)G_p(s) \quad (\text{I.7})$$

The transfer function $Y(s)$ is simply equal to the original feedforward controller, $Q(s)$. In Equation (I.7), the linearity in the free stable transfer function $Y(s)$ is what gives its practicality in hand computing or optimizing for a SISO or MIMO controller. Given a particular plant transfer function $G_p(s)$ one typically fixes a preferred closed loop transfer function $T(s)$ as described in the following chapters, and then computes a proper and stable $Y(s)$ using (I.7).

From Equation (I.6), the feedback controller, G_c , can be computed as follows,

$$G_c(s) = \frac{Y(s)}{1 - G_p(s)Y(s)} \quad (\text{I.8})$$

The details for addressing the feedback loop internal stability are addressed in the following chapters.

Before finishing this introduction, we illustrate the utility of using $Y(s)$ and the Youla parameterization method through a very simple classical example.

■ **EXAMPLE I.1**

Consider a plant with an unstable pole

$$G_p(s) = \frac{1}{s-1} \quad (\text{I.9})$$

and assume the controller

$$G_c(s) = K \quad (\text{I.10})$$

where K is a simple gain (proportional type) controller, as shown in Figure I.9,

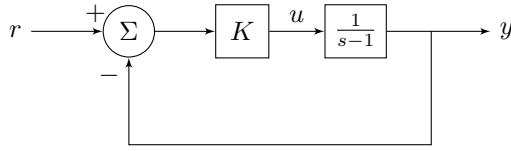


Figure I.9: Closed-loop feedback system.

The closed loop transfer function, $T(s)$, can be written as,

$$T(s) = \frac{K}{s-1+K} \quad (\text{I.11})$$

The simple gain can be computed to stabilize the feedback control loop, by selecting the controller K to be $1 < K < \infty$. The root locus is illustrated in Figure I.10,

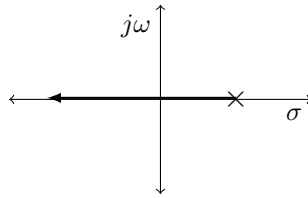


Figure I.10: Plant s -domain

Stabilizing this plant is a simple and straightforward task. However, students who are given this simple problem, traditionally, are not given the proper reasons as of why,

1. The gain, K , could not be chosen freely without additional consequences, e.g., actuator saturation.
2. A simple gain can stabilize the plant but can not accomplish an appropriate target following, a performance measure.

For this simple example, let's compute the $Y(s)$ transfer function, which is equal to the ratio of the actuator input to the target input, $\frac{u(s)}{r(s)}$.

$$Y(s) = \frac{K(s-1)}{s-1+K} \quad (\text{I.12})$$

It is a straightforward matter to obtain a Bode plot of this transfer function for various gains, K , as shown in Figure I.11 below,

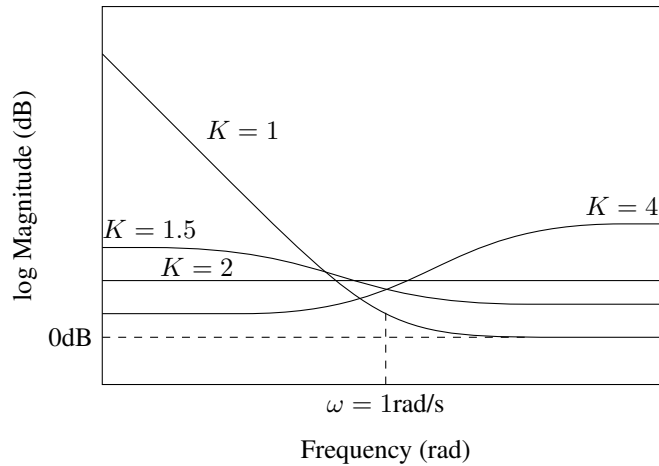


Figure I.11: Frequency response of $Y(s)$ for various gains K .

Now, it should be obvious, by looking at this plot, as we increase the gain, K , we are increasing the high-frequency demand on the actuator. More importantly, by increasing this gain, we are introducing noise to our system, as the $Y(s)$ transfer function is a high pass filter with a simple gain controller type, K .

Now, we use the Youla approach: we compute a stabilizing controller for this plant. According to this method, first, we need to select a closed loop transfer function, $T(s)$. For this example, we select $T(s)$ to be equal to

$$T(s) = \frac{K}{\tau s + 1}, \quad (\text{I.13})$$

where K is some gain and is not equal to the previously computed controller gain. Preferably, we would like to have $K = 1$, to ensure good target following as the closed loop transfer function becomes equal to 1 at steady state with this selection for K . But as we shall soon see, this won't be possible as internal stability will be compromised if we select $K = 1$, so we cannot simultaneously satisfy both target following and stability with this choice of controller, a simple gain.

Using Equation (I.7), we can compute the $Y(s)$ transfer function as $Y(s) = \frac{T(s)}{G_p(s)}$,

$$Y(s) = \frac{K(s-1)}{\tau s + 1} \quad (\text{I.14})$$

Now using Equation (I.8), we can compute the controller, G_c ,

$$G_c(s) = \frac{K(s-1)}{\tau s + (1-K)} \quad (\text{I.15})$$

We are free to select K and τ controller coefficients. However, as we know from classical control, the controller G_c must not cancel the unstable pole of the plant as in this case, the internal stability of the feedback control loop will be compromised. Therefore, the coefficients K and τ must be selected as such to cancel $s-1$ polynomial in the numerator of G_c . There is only one unique solution for the coefficients K and τ , to see this, let's rearrange Equation (I.15) and write it as,

$$G_c(s) = \frac{K}{\tau} \frac{s-1}{s + \frac{1-K}{\tau}} \quad (\text{I.16})$$

By looking at this Equation, it is evident that we must select $\frac{1-K}{\tau} = -1$, to cancel the numerator polynomial, with this selection, the controller G_c can be written as $G_c = \frac{K}{\tau}$ where $K = \tau + 1$, hence,

$$G_c(s) = 1 + \frac{1}{\tau} \quad (\text{I.17})$$

Again this controller is a pure gain controller, but this time, this controller is a function of τ , the first order time constant. So, if we make τ small or the closed loop time response fast, according to our selection of closed loop transfer function, $T(s)$, we have a higher gain controller. In addition, when we computed this controller using the proposed method, we saw that we are unable to simultaneously satisfy both target following and stability requirements using a simple gain controller. It should be evident that the Youla method exposed several important feedback control concepts along the way, which were not obvious using the classical approach. We will discuss these points and more about the utility of the Youla approach in the next chapters.

CONTROL DESIGN USING YOU LA PARAMETRIZATION SINGLE INPUT SINGLE OUTPUT (SISO)

CHAPTER 1

REVIEW OF THE LAPLACE TRANSFORM

Our emphasis in this book is on constant parameter linear time invariant dynamic systems. Laplace transform theory is most useful when applied to analysis of linear systems and this transformation allows the analyst to reduce differential equation problems to algebraic problems. The theory discussed below will be utilized in both the dynamic system analysis and control system design portions of the text. It is demonstrated in this chapter that linear, time-invariant dynamic equations can be transformed into rational functions. Many important input (test) signals can be transformed in this way as well.

Our emphasis in this book is on constant parameter linear time invariant dynamic systems. Laplace transform theory is most useful when applied to analysis of linear systems and this transformation allows the analyst to reduce differential equation problems to algebraic problems. The theory discussed below will be utilized in both the dynamic system analysis and control system design portions of the text.

It is demonstrated in this chapter that linear, time-invariant dynamic equations can be transformed into rational functions. Many important input (test) signals can be transformed in this way as well.

1.1 The Laplace Transform Concept

One of the most important problems in dynamic systems analysis is to predict the outputs of a physical system when the inputs are known. We can then ask the following question, Does a transformation scheme exist to help in the tasks of dynamic system analysis? The answer is yes. In fact, there are several transformations which could be used in addressing this problem. For example, the Fourier transformation can be used; however, the domain for the Fourier transformation is "limited". That is, too many important types of inputs do not belong to the domain of the Fourier transform. On the other hand, the Laplace transform has a very broad domain and the existence of the Laplace transform tables aid the execution of the steps in the indirect path shown in Figure 1.1.

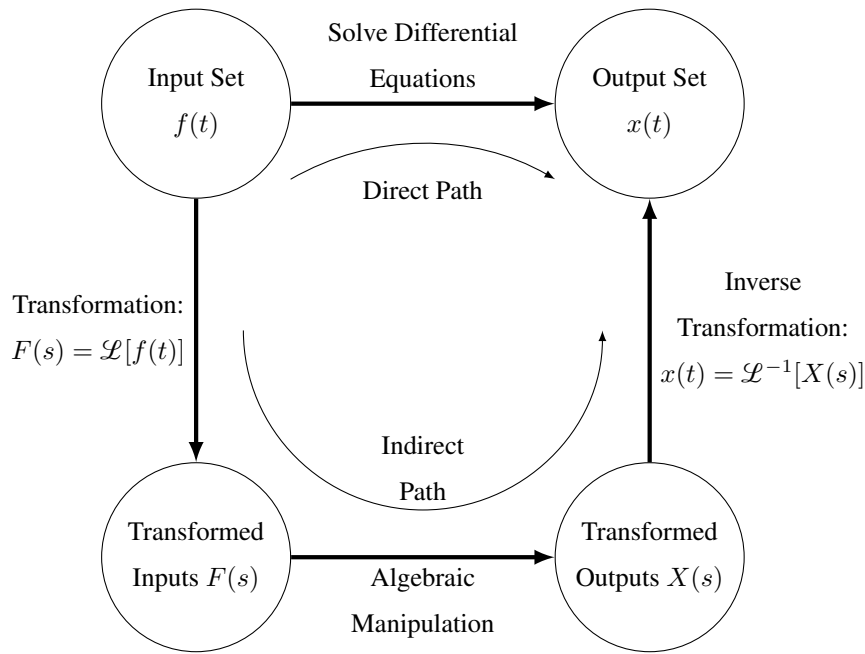


Figure 1.1: Laplace Transform Process

1.2 Singularity Functions

Before we can take up the definition of the Laplace transform, it is necessary to consider those functions of time which have a discontinuity at some instants. Especially important is some understanding of what we mean by an impulse function or the so called Dirac- δ function.

Our first definitions provide a terminology for values of a discontinuous function at either side of the instant of discontinuity.

Definition 1.1. The initial condition, denoted $y(0^-)$, is the limit from the left at $t = 0$:

$$y(0^-) = \lim_{t \rightarrow 0^-} y(t) \quad (1.1)$$

Definition 1.2. The initial value, denoted $y(0^+)$, is the limit from the right at $t = 0$:

$$y(0^+) = \lim_{t \rightarrow 0^+} y(t) \quad (1.2)$$

Notice that the restrictions are strict inequalities. In other words, the initial condition is the limiting value of the function just before a possible jump discontinuity at $t = 0$ and the initial value is the limiting value of the function just after a possible jump discontinuity at $t = 0$.

EXAMPLE 1.1

We consider the initial condition and initial value of a unit step function,

$$u(t) = 1(t) \quad (1.3)$$

$$u(0^-) = 0, \quad u(0^+) = 1, \quad \mathcal{L}[u(t)] = \frac{1}{s} \quad (1.4)$$

1.2.1 Definition of The Impulse Function

The impulse (or Dirac Delta) function $\delta(t)$ is defined such that

$$\delta(t) = 0 \text{ for } t \neq 0 \quad (1.5)$$

$$\int_{t_0}^{t_f} \delta(t - \tau) y(t) dt = \begin{cases} y(\tau), & t_0 < \tau \leq t_f \\ 0, & \text{otherwise} \end{cases} \text{ for any integrable } y(t) \quad (1.6)$$

An important question is why the limits of this integration are chosen such that $t_0 < \tau \leq t_f$. Suppose we write things more symmetrically: This means that the boundary on τ is $t_0 \leq \tau \leq t_f$. Then

$$\begin{aligned} y(\tau) &= \int_{t_0}^{t_f} \delta(t - \tau) y(t) dt \\ &= \int_{t_0}^{\tau} \delta(t - \tau) y(t) dt + \int_{\tau}^{t_f} \delta(t - \tau) y(t) dt \\ y(\tau) &= y(\tau) + y(\tau) = 2y(\tau) \end{aligned} \quad (1.7)$$

A contradiction! Similarly, suppose the boundary on τ is $t_0 < \tau < t_f$. Then

$$\begin{aligned}
 y(\tau) &= \int_{t_0}^{t_f} \delta(t - \tau)y(t)dt \\
 &= \int_{t_0}^{\tau} \delta(t - \tau)y(t)dt + \int_{\tau}^{t_f} \delta(t - \tau)y(t)dt \\
 y(\tau) &= 0 + 0 = 0
 \end{aligned} \tag{1.8}$$

So, we must have $t_0 < \tau \leq t_f$ or $t_0 \leq \tau < t_f$.

Another important point is even with the right definition of the boundary on τ , the definition of the impulse function given in Equation 1.6 is inconsistent with the definition of the Riemann integral, the definition used in introduction to calculus. Without covering the details of this inconsistency (as this is outside the scope of this text), this issue is briefly discussed in the following subsection.

1.2.2 The Impulse Function and the Riemann Integral

The impulse function is inconsistent with the definition of the Riemann integral (illustrated in Figures 1.2 and 1.3).

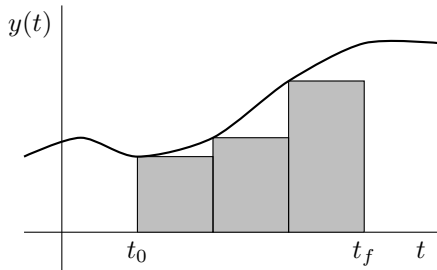


Figure 1.2: Riemann integral lower sum, S_L

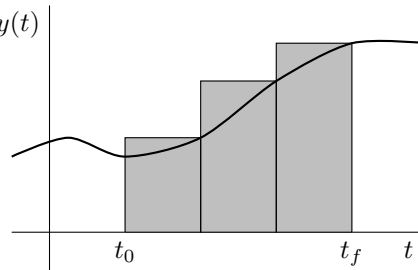


Figure 1.3: Riemann integral upper sum, S_U

The Riemann integral (with lower sum S_L and upper sum S_U) is the common limit as the partitioning is refined. For the impulse function with $t_0 < \tau \leq t_f$, however, the lower limit is defined as $S_L \equiv 0$ while the upper limit is nonzero. The Riemann integral simply does not work. There are several ways to resolve this mathematical difficulty, the simplest is to use the Stieltjes definition of the integral [?].

1.2.3 The General Definition of Singularity Functions

The k -th order derivative of singularity function $\delta^{(k)}(t)$ is defined such that:

$$\delta^{(k)}(t) \text{ for } t \neq 0 \quad (1.9)$$

$$\int_{t_0}^{t_f} \delta^{(k)}(t - \tau) y(t) dt = \begin{cases} (-1)^k \frac{d^k y(\tau)}{dt^k}, & t_0 < \tau \leq t_f \\ 0, & \text{otherwise} \end{cases} \quad (1.10)$$

Once again, it can be shown that this definition is inconsistent with Riemann's definition of integral.

“Graphs” of some singularity functions: The main reason for the persistent use of the singularity functions in the control-system literature, in spite of the mathematical problems, is the simple fact that they are useful theoretical devices. In addition, we are able to manufacture signals in the laboratory or on the computer which seem to be endowed with all of the theoretical properties of the singularity functions. A good way to think of all this is that the physical notion of the impulse is not what creates the difficulty; rather, it is the notion of what we mean by the integral that lies at the root of the theoretical problems.

The laboratory impulse is illustrated as dash lines overlaid on the function, $\rho(t)$ in Figure 1.4. (Notice that this pulse-like function ends at $t = \tau$.) An apparent mathematical justification for this claim derives from use of the mean-value theorem provided by the following equation,

$$\int_a^b y(t) dt = y(\zeta)(b - a), \quad a \leq \zeta \leq b \quad (1.11)$$

That is, imagine multiplying this pulse by $y(t)$ and integrating. It then seems to follow from the mean-value theorem,

$$\int_{t_0}^{t_f} \rho(t) y(t) dt = \int_{\tau-\epsilon}^{\tau} \rho(t) y(t) dt = \frac{1}{\epsilon} \cdot y(\zeta) \cdot \epsilon = y(\zeta), \quad \tau - \epsilon \leq \zeta \leq \tau. \quad (1.12)$$

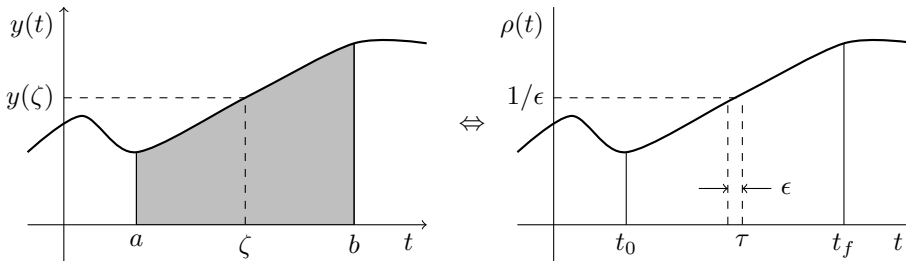


Figure 1.4: Mean value theorem development of singularity functions.

If we continue this line of reasoning, we can imagine laboratory versions of the higher order singularity functions. For example, a finite difference approximation for the first singularity function. To “construct” $\delta^{(1)}(\tau) \Rightarrow -\frac{dy}{dt}(\tau)$, refer to the finite difference approximation:

$$\frac{dy(\tau)}{dt} \approx \frac{y(\tau) - y(\tau - \epsilon)}{\epsilon} \quad (1.13)$$

The following example provides an illustration of an impulse function and a finite difference approximation for the first singularity function and their associated dynamic responses.

EXAMPLE 1.2

We compare the impulse and doublet response for a moving mass. Consider applying a force $F(t)$ to the mass shown in Figure 1.5.

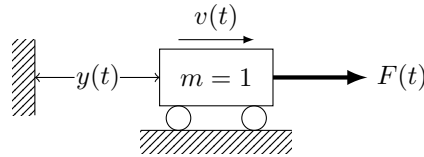


Figure 1.5: An mass with an applied force.

First, let $F(t)$ be an approximate impulse or doublet function, as shown in Figures 1.6 and 1.7. In these cases, ϵ is considered to be a very small (but finite) number.

The approximate impulse function is zero except for a length of time ϵ starting at $t = -\epsilon$, where the magnitude is $1/\epsilon$. The velocity response is a ramp between $t = -\epsilon$ and $t = 0$, ending at a magnitude of 1. In turn, the displacement response is a parabola between $t = -\epsilon$ and $t = 0$ and a ramp with slope equal to 1 for $t > 0$.

The approximate doublet is zero except for a length of time 2ϵ starting at $t = -2\epsilon$. For the first half of the duration, the approximate doublet has a magnitude of $1/\epsilon^2$. For the second half, it has a magnitude of $-1/\epsilon^2$. The velocity response is a positive ramp from $t = -2\epsilon$ to $t = -\epsilon$ and a negative ramp from $t = -\epsilon$ to $t = 0$, ending with a magnitude of $1/\epsilon$. Further integration gives the displacement response, a parabolic increase from $t = -2\epsilon$ to $t = -\epsilon$ and parabolic decrease from $t = -\epsilon$ to $t = 0$, such that the magnitude at $t = -\epsilon$ is $1/2$ and the magnitude at $t = 0$ is 1.

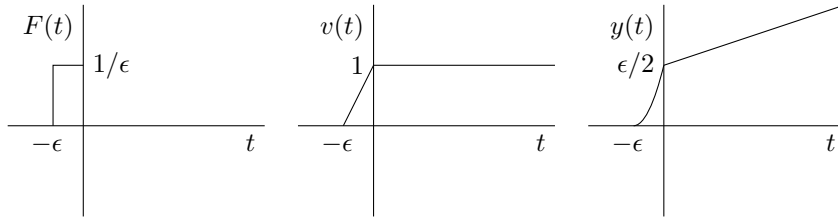


Figure 1.6: Impulse approximate response.

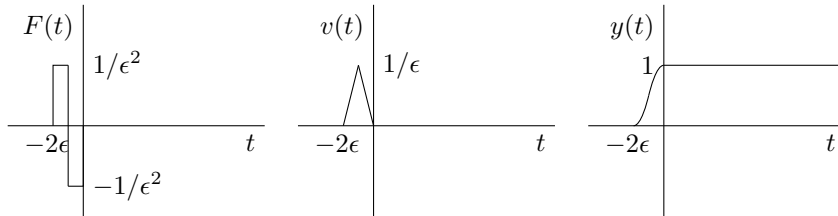
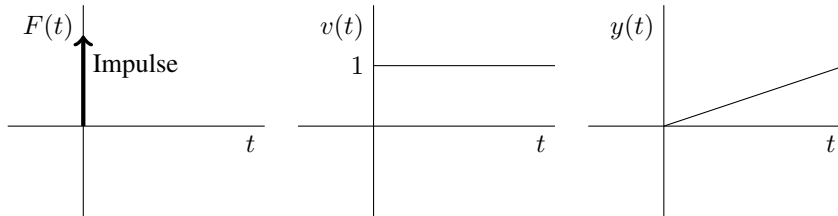
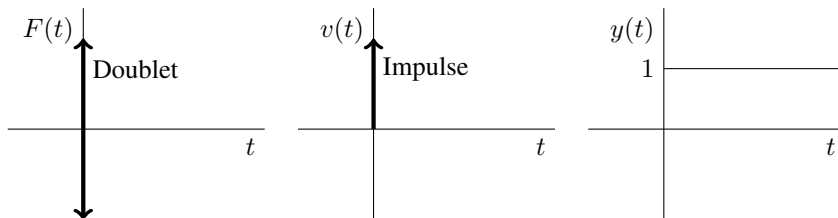


Figure 1.7: Doublet approximate response.

Next, consider the exact response to an impulse or doublet force input, as shown in Figures 1.8 and 1.9. As $\epsilon \rightarrow 0$, the approximate impulse and doublet functions approach true impulse and doublet functions, and integrate into the more familiar step and ramp functions.

Figure 1.8: Impulse exact response ($\epsilon \rightarrow 0$).Figure 1.9: Doublet exact response ($\epsilon \rightarrow 0$).

Similarly, we can construct a finite difference approximation for the second derivative and then infer a laboratory version of the third singularity function. In a similar way, we can approximate all the singularity functions.

1.3 The Laplace Transform

1.3.1 Definition of The Laplace Transform

The Laplace transform converts a function from the time domain with variable t , to the Laplace domain with variable s (the Laplace domain is sometimes referred to as the s -domain). The Laplace variable s is a complex number and can be written in terms of its real part σ and imaginary part ω .

$$s = \sigma + j\omega \quad (1.14)$$

The choice for s is arbitrary but proves to be very convenient in the application of dynamic systems theory [?]. The Laplace transform of $y(t)$ is denoted by $\hat{y}(s)$ or $\mathcal{L}[y(t)]$ and is defined as [?]

$$\mathcal{L}[y(t)] = \int_{0^-}^{\infty} e^{-st} y(t) dt \quad (1.15)$$

Notice the lower limit of 0^- , which reads “slightly before zero.” The Fourier transform is defined by

$$\mathcal{F}[y(t)] = \int_{-\infty}^{\infty} e^{-j\omega t} y(t) dt \quad (1.16)$$

The Laplace transformation domain [?] is given as follows: for a function to be Laplace transformable, $f(t)$ must be

1. Sectionally continuous; i.e., $f(t)$ may have only a finite number of discontinuities in any finite interval; and
2. Of exponential order; i.e., for some real σ_0 and t_1 , $e^{-\sigma_0 t} |f(t)|$. The region of the s plane for which $\text{Re}(s) > \sigma_0$ is called region of convergence.

These conditions are sufficient conditions for the existence of the integral in Equation 1.15.

EXAMPLE 1.3

We consider the Laplace transform of the unit step function.

$$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases} \quad (1.17)$$

The unit step function, $u(t)$, can be written as $1(t)$.

$$\begin{aligned}
 \hat{u}(s) &= \int_{0^-}^{\infty} e^{-st} 1(t) dt \\
 &= \int_0^{\infty} e^{-st} dt \\
 &= \left. \frac{-e^{-st}}{s} \right|_0^{\infty} \\
 &= +\frac{1}{s} \text{ (if } \operatorname{Re} s > 0, \text{ where } \operatorname{Re} s = \sigma)
 \end{aligned} \tag{1.18}$$

Plotting $\hat{u}(s)$ vs. s cannot be done in a very convenient manner, as was done with $u(t)$ vs. t , because s is complex. However, it is very useful to plot the critical points of $\hat{u}(s)$, i.e., points where $\hat{u}(s)$ is infinite (called poles) and is zero (called zeros), on the s plane. For the unit step function, $\hat{u}(s)$ has a single pole at origin as illustrated in pole-zero diagram of Figure 1.11.

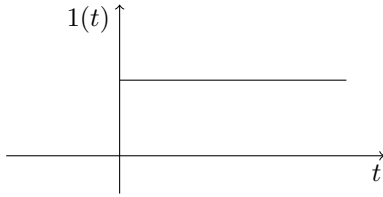


Figure 1.10: Unit step time domain

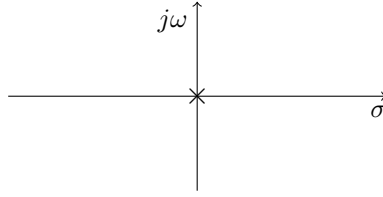


Figure 1.11: Unit step s -domain

1.3.2 Laplace Transform Properties

In this section, we discuss the properties of Laplace transform by series of propositions with their associated proofs. This should aid in quickly deriving Laplace transforms for set of functions.

Proposition 1.1. *The Laplace transform is linear, i.e.*

$$\mathcal{L}[c_1 y(t) + c_2 g(t)] = c_1 \mathcal{L}[y(t)] + c_2 \mathcal{L}[g(t)] \tag{1.19}$$

for any $c_1, c_2, y(t), g(t)$.

Proof: Linearity of the integration.

$$\int [c_1 f(t) + c_2 h(t)] dt = c_1 \int y(t) dt + c_2 \int h(t) dt,$$

Then, per the definition of the Laplace transform in Equation (1.15), the Laplace transform is linear. ■

Proposition 1.2. *The Laplace transform is non-multiplicative, i.e.*

$$\mathcal{L}[y(t)g(t)] \neq \mathcal{L}[y(t)]\mathcal{L}[g(t)], \quad (1.20)$$

due to the properties of integration.

Proof:

$$\int f(t)h(t)dt \neq \int f(t)dt \int h(t)dt$$

Therefore, per the definition of the Laplace transform in Equation (1.15), the Laplace transform cannot be multiplicative. ■

1.3.3 Shifting the Laplace Transform

The definition of the Laplace transform is manipulated to yield results which are useful for obtaining the transforms of an important set of functions and for manipulating and interpreting mathematical models of physical systems.

Proposition 1.3. *(Complex shift of the Laplace transform): If $\hat{y}(s) = \mathcal{L}[y(t)]$, then*

$$\mathcal{L}[e^{-\sigma t}y(t)] = \hat{y}(s + \sigma). \quad (1.21)$$

Proof:

$$\begin{aligned} \mathcal{L}[e^{-\sigma t}y(t)] &= \int_{0^-}^{\infty} e^{-st}e^{-\sigma t}y(t)dt \\ &= \int_{0^-}^{\infty} e^{-(s+\sigma)t}y(t)dt \\ &= \hat{y}(s + \sigma) \end{aligned}$$

Therefore, multiplying $y(t)$ by $e^{-\sigma t}$ shifts $\hat{y}(s)$ by σ . ■

This result shares a symmetry with the following proposition.

Proposition 1.4. *(Time shift of the Laplace transform): If $\hat{y}(s) = \mathcal{L}[y(t)]$ and $y(t) \equiv 0$ for $t < 0^-$, then*

$$\mathcal{L}[y(t - \tau)] = e^{-\tau s}\hat{y}(s). \quad (1.22)$$

Proof: Let $\zeta = t - \tau$, so

$$\begin{aligned}
 \mathcal{L}[y(t - \tau)] &= \int_{0^-}^{\infty} e^{-st} y(t - \tau) dt \\
 &= \int_{-\tau^-}^{\infty} e^{-s(\zeta + \tau)} y(\zeta) d\zeta \\
 &= e^{-s\tau} \int_{-\tau^-}^{\infty} e^{-s\zeta} y(\zeta) d\zeta \\
 &\quad -\tau^- = 0^- \text{ because } y(t) \equiv 0 \text{ for } t < 0^-, \text{ so} \\
 &= e^{-s\tau} \int_{0^-}^{\infty} e^{-s\zeta} y(\zeta) d\zeta \\
 &= e^{-s\tau} \hat{y}(s)
 \end{aligned}$$

627 Therefore, multiplying $\hat{y}(s)$ by $e^{-\tau s}$ shifts $y(t)$ by τ . ■

EXAMPLE 1.4

Consider the function

$$y(t) = e^{-\sigma t} 1(t) \quad (1.23)$$

$$\mathcal{L}[1(t)] = \frac{1}{s}, \text{ so } \mathcal{L}[e^{-\sigma t} 1(t)] = \frac{1}{s + \sigma} \quad (1.24)$$

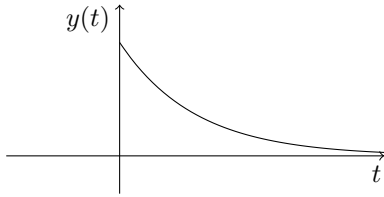


Figure 1.12: Time domain.

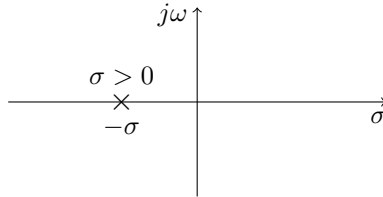


Figure 1.13: s-domain.

EXAMPLE 1.5

Consider the function

$$y(t) = \cos(\omega t) 1(t) \quad (1.25)$$

If z is complex, then by definition $e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots$. Suppose $z = j\theta$:

$$e^{j\theta} = 1 + j\theta - \frac{\theta^2}{2!} - \frac{j\theta^3}{3!} + \frac{\theta^4}{4!} + \frac{j\theta^5}{5!} + \dots \quad (1.26)$$

$$e^{j\theta} = \cos(\theta) + j \sin(\theta), \quad e^{-j\theta} = \cos(\theta) - j \sin(\theta) \quad (1.27)$$

$$\cos(\theta) = \frac{e^{j\theta} + e^{-j\theta}}{2} \quad (1.28)$$

Therefore:

$$\begin{aligned} \mathcal{L}[\cos(\omega t) 1(t)] &= \frac{1}{2} \mathcal{L}[e^{j\omega t} 1(t)] + \frac{1}{2} \mathcal{L}[e^{-j\omega t} 1(t)] \\ &= \frac{1}{2} \frac{1}{s - j\omega} + \frac{1}{2} \frac{1}{s + j\omega} \\ &= \frac{s}{s^2 + \omega^2} \end{aligned} \quad (1.29)$$

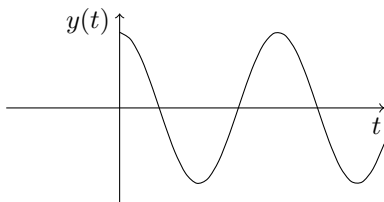


Figure 1.14: Time domain.

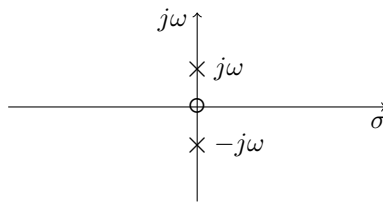


Figure 1.15: s-domain.

■ **EXAMPLE 1.6**

Consider the function

$$y(t) = e^{-\sigma t} \cos(\omega t) 1(t) \quad (1.30)$$

$$\mathcal{L}[e^{-\sigma t} \cos(\omega t) 1(t)] = \frac{s + \sigma}{(s + \sigma)^2 + \omega^2} \quad (1.31)$$

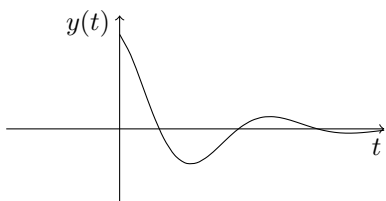


Figure 1.16: Time domain.

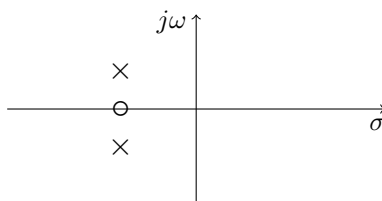


Figure 1.17: s -domain.

1.3.4 Laplace Transform Derivatives

The next two propositions are also symmetrical.

Proposition 1.5. (*Derivative of a transformed function*): If $\hat{y}(s) = \mathcal{L}[y(t)]$, then

$$\mathcal{L}[ty(t)] = -\frac{d}{ds} \hat{y}(s) \quad (1.32)$$

Proof:

$$\begin{aligned} -\frac{d}{ds} \hat{y}(s) &= -\frac{d}{ds} \int_{0^-}^{\infty} e^{-st} y(t) dt \\ &= \int_{0^-}^{\infty} e^{-st} (ty(t)) dt = \mathcal{L}[ty(t)] \end{aligned}$$

Therefore, when a function $y(t)$ is multiplied by t , the Laplace transform of the resulting output is the negative derivative in s of $\mathcal{L}[y(t)]$. ■

645 **EXAMPLE 1.7**

646 Consider the ramp function, $t(1(t))$.

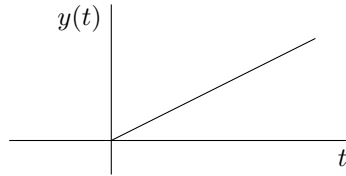


Figure 1.18: Ramp input.

$$\mathcal{L}[t(1(t))] = -\frac{d}{ds} \left[\frac{1}{s} \right] = \frac{1}{s^2} \quad (1.33)$$

647 Apply this proposition in an iterative fashion to establish

$$\mathcal{L}[t^n y(t)] = (-1)^n \frac{d^n}{ds^n} \hat{y}(s) \quad (1.34)$$

649 **Proposition 1.6.** (Transformation of a derivative): If $\hat{y}(s) = \mathcal{L}[y(t)]$, then the Laplace
650 transform of the derivative is related to the initial condition by

$$\mathcal{L} \left[\frac{dy}{dt} \right] = s\hat{y}(s) - y(0^-) \quad (1.35)$$

651 *Proof:* *Reminder: Integration by parts.*

$$\int u dv = uv - \int v du \quad (1.36)$$

Then,

$$\mathcal{L} \left[\frac{dy}{dt} \right] = \int_{0^-}^{\infty} e^{-st} \frac{dy}{dt} dt$$

Integration by parts:

$$u = e^{-st}, du = -se^{-st}, v = y(t), dv = \left(\frac{dy(t)}{dt} \right) dt.$$

$$= e^{-st} y(t) \Big|_{0^-}^{\infty} - \int_{0^-}^{\infty} -se^{-st} y(t) dt$$

knowing that

$$e^{-st}y(t)\Big|_{0^-}^{\infty} = 0 - y(0^-)$$

so

$$\mathcal{L}\left[\frac{dy}{dt}\right] = s\hat{y}(s) - y(0^-)$$

Therefore, when a function $y(t)$ is differentiated with respect to time, the Laplace transform of the resulting output is s times $\mathcal{L}[y(t)]$ minus the initial condition $y(0^-)$. ■

Applying this result in an iterative way establishes a corollary to this proposition.

Corollary 1.1.

$$\mathcal{L}\left[\frac{d^n y(t)}{dt^n}\right] = s^n \hat{y}(s) - \sum_{k=1}^n s^{n-k} \frac{d^{k-1}}{dt^{k-1}} y(0^-) \quad (1.37)$$

EXAMPLE 1.8

Consider a force applied to a mass, as shown in Figure 1.19.

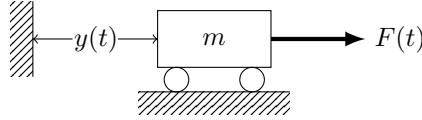


Figure 1.19: A mass with an applied force.

Newton's Law: $m \frac{d^2 y}{dt^2} = U$, where $U = F(t)$

$$\mathcal{L}\left[\frac{d^2 y}{dt^2}\right] = s^2 \hat{y}(s) - sy(0^-) - \frac{dy}{dt}(0^-) \quad (1.38)$$

$$m[s^2 \hat{y}(s) - sy(0^-) - \frac{dy}{dt}(0^-)] = \hat{u}(s) \quad (1.39)$$

$$\hat{y}(s) = \underbrace{\frac{sy(0^-) + \frac{dy}{dt}(0^-)}{s^2}}_{\text{I.C. Response}} + \underbrace{\frac{1}{ms^2} \hat{u}(s)}_{\text{Forced Response}} \quad (1.40)$$

When all initial conditions are zero, we are left with the transfer function, $G(s)$, from input $\hat{u}$ to output $\hat{y}$.

$$G(s) = \frac{\hat{y}}{\hat{u}}(s) = \frac{1}{ms^2} \quad (1.41)$$

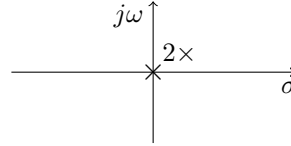
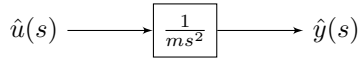


Figure 1.20: Forced mass block diagram. Figure 1.21: Poles of the forced mass.

1.3.5 Transforms of Singularity Functions

To this point, our example transforms have been rational functions. Here, we determine the transforms of the singularity functions which are rational functions (polynomials actually) for the special case of $\tau = 0$. As it turns out, it is this value of τ that is used most often in our discussions. A simple application of Equation 1.15 and the definition of an impulse function of the last section yields the transform

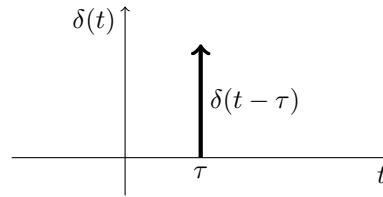


Figure 1.22: Impulse function.

$$\mathcal{L}[\delta(t - \tau)] = \int_{0^-}^{\infty} \delta(t - \tau) e^{-st} dt = e^{-s\tau} \quad (1.42)$$

Note: If $\tau = 0$ then $\mathcal{L}[\delta(t)] = 1$. By linearity,

$$\mathcal{L}^{-1}[R_0] = R_0 \delta(t). \quad (1.43)$$

More generally, application of Equation 1.15 and the previous definition of an impulse function of the last section results in

$$\mathcal{L}[\delta^{(k)}(t - \tau)] = s^k e^{-s\tau}, \quad \mathcal{L}[\delta^{(k)}(t)] = s^k \quad (1.44)$$

1.4 Inverse Laplace Transform

At this point, it is clear that the differential equations of linear, time invariant, systems can be converted to polynomials and rational functions where the Laplace transform is

effected; the Laplace transform can be used to solve differential equations by purely algebraic means. Nearly all that is needed to complete the solution of ordinary differential equations with constant coefficients by Laplace transformation is contained in the propositions and examples of previous sections. The final ingredient is a method for obtaining the inverse transform, $y(t) = \mathcal{L}^{-1}[\hat{y}(s)]$. One can certainly utilize the Cauchy integral formula, which will be introduced in the future chapters, for this purpose. Other, more mundane but useful methods are developed in this section.

1.4.1 Inverse Laplace Transformation by Heaviside Expansion:

In this section, we shall use the simpler procedure of partial-fraction or Heaviside expansion to obtain an inverse transformation. That is to say, we exploit the properties of rational functions as follows,

Suppose we have a transfer function, $G(s)$, and want $g(t)$. Also suppose that $G(s)$ is strictly proper, where number of poles, m , is greater than number of zeros, n , $m > n$. Write any strictly-proper transfer function as

$$G(s) = \frac{q_{n-1}s^{n-1} + q_{n-2}s^{n-2} + \dots + q_1s + q_0}{(s + p_1)(s + p_2)\dots(s + p_m)} \quad (1.45)$$

The problem of converting this form with a Heaviside expansion is divided into subcases according to whether or not the poles, p_k , are repeated.

Distinct Poles: Where the p_k are distinct, it is possible to make the Heaviside expansion as,

$$G(s) = \frac{R_1}{s + p_1} + \frac{R_2}{s + p_2} + \dots + \frac{R_n}{s + p_n} \quad (1.46)$$

The R_k are called residues. Multiply this by $(s + p_k)$ and note that result must apply to all values of s . In particular, we can set $s = -p_k$ to obtain the formula

$$R_k = (s + p_k)G(s) \Big|_{s=-p_k} \quad (1.47)$$

The residuals can then be independently inverted.

$$g(t) = [R_1e^{-p_1t} + R_2e^{-p_2t} + \dots + R_ne^{-p_nt}]1(t) \quad (1.48)$$

Distinct Poles with $G(s)$ being proper: When $G(s)$ is proper, the degree of the numerator is equal to the degree of the denominator. If $G(s)$ is proper, then it can be written as:

$$G(s) = R_0 + \frac{R_1}{s + p_1} + \frac{R_2}{s + p_2} + \dots + \frac{R_n}{s + p_n}, \text{ where } R_0 = G(\infty) \quad (1.49)$$

$$g(t) = R_0\delta(t) + [R_1e^{-p_1t} + R_2e^{-p_2t} + \dots + R_ne^{-p_nt}]1(t) \quad (1.50)$$

Remember that $g(t) = \mathcal{L}^{-1}[G(s)]$.

Repeated Poles: Suppose one of the poles of $G(s)$, call it $-p_1$, is repeated a times,

$$G(s) = \frac{q_{n-1}s^{n-1} + q_{n-2}s^{n-2} + \dots + q_1s + q_0}{(s + p_1)^a(s + p_2)\dots(s + p_{n-a+1})} \quad (1.51)$$

The Heaviside expansion is modified in the following way,

$$G(s) = \frac{R_{11}}{s + p_1} + \frac{R_{12}}{(s + p_1)^2} + \dots + \frac{R_{1a}}{(s + p_1)^a} + \frac{R_2}{s + p_2} + \dots + \frac{R_{n-a+1}}{s + p_{n-a+1}} \quad (1.52)$$

It is easy to verify that the residues associated with linear denominator terms are calculated as in Equation (1.47). For a single repeated pole ($a = 2$), it is easily verified that

$$R_{12} = \left. \frac{d}{ds} [(s + p_1)^a G(s)] \right|_{s=-p_1} \quad (1.53)$$

In addition, for values of $a - l \geq 2$, we find that

$$R_{1l} = \frac{1}{(a - l)!} \left. \frac{d^{a-l}}{ds^{a-l}} [(s + p_1)^a G(s)] \right|_{s=-p_1} \quad (1.54)$$

Then $g(t)$ can be written as,

$$g(t) = [R_{11}e^{-p_1t} + R_{12}te^{-p_1t} + \dots + R_{1a}t^{a-1}e^{-p_1t} + R_2e^{-p_2t} + \dots + R_{n-a+1}e^{-p_{n-a+1}t}]1(t) \quad (1.55)$$

The examples below illustrate the utility of Heaviside expansion in inverse Laplace transformation.

EXAMPLE 1.9

Consider the transfer function (distinct poles - proper)

$$G(s) = \frac{s^2 + 5s + 6}{s^2 + 3s + 2} \quad (1.56)$$

This is the case where $G(s)$ is proper. First, carry out any pole/zero cancellations.

$$G(s) = \frac{(s+2)(s+3)}{(s+1)(s+2)} = \frac{s+3}{s+1} \quad (1.57)$$

Therefore,

$$G(s) = \frac{s+3}{s+1} = R_0 + \frac{R_1}{s+1} = R_0 + \frac{R_1}{s+1} \quad (1.58)$$

According to Equation (1.49),

$$R_0 = G(\infty) = 1 \quad (1.59)$$

And according to Equation (1.47)

$$R_1 = (s + 1)G(s) \Big|_{s=-1} \quad (1.60)$$

yields $R_1=2$. Therefore, the partial fraction expansion of $G(s)$ is

$$G(s) = 1 + \frac{2}{s + 1} \quad (1.61)$$

and therefore,

$$g(t) = \delta(t) + [2e^{-t}]1(t) \quad (1.62)$$

EXAMPLE 1.10

Consider the transfer function (distinct poles, strictly proper)

$$G(s) = \frac{4s^2 + 13s + 12}{s(s^2 + 5s + 6)} \quad (1.63)$$

First, factor $G(s)$.

$$G(s) = \frac{4s^2 + 13s + 12}{s(s + 2)(s + 3)} \quad (1.64)$$

Therefore, the partial fraction expansion is of the form

$$G(s) = \frac{R_1}{s + p_1} + \frac{R_2}{s + p_2} + \frac{R_3}{s + p_3} = \frac{R_1}{s} + \frac{R_2}{s + 2} + \frac{R_3}{s + 3} \quad (1.65)$$

According to Equation (1.47)

$$R_k = (s + p_k)G(s) \Big|_{s=-p_k} \quad (1.66)$$

This yields

$$\begin{aligned} R_1 &= sG(s) \Big|_{s=0} = 2 \\ R_2 &= (s + 2)G(s) \Big|_{s=-2} = -1 \\ R_3 &= (s + 3)G(s) \Big|_{s=-3} = 3 \end{aligned} \quad (1.67)$$

Therefore, the partial fraction expansion of $G(s)$ is

$$G(s) = \frac{2}{s} - \frac{1}{s + 2} + \frac{3}{s + 3} \quad (1.68)$$

and

$$g(t) = [2 - e^{-2t} + 3e^{-3t}]1(t) \quad (1.69)$$

724 **EXAMPLE 1.11**

725 Consider the transfer function (repeated poles)

$$G(s) = \frac{4s^3 + 16s^2 + 23s + 13}{(s+1)^3(s+2)} \quad (1.70)$$

726 The partial fraction expansion is of the form

$$G(s) = \frac{R_{11}}{(s+1)} + \frac{R_{12}}{(s+1)^2} + \frac{R_{13}}{(s+1)^3} + \frac{R_2}{(s+2)} \quad (1.71)$$

727 According to Equations (1.53) and (1.54)

$$\begin{aligned} R_{11} &= \frac{1}{2!} \frac{d^2}{ds^2} [(s+1)^3 G(s)] \Big|_{s=-1} = 3 \\ R_{12} &= \frac{d}{ds} [(s+1)^3 G(s)] \Big|_{s=-1} = 1 \\ R_{13} &= [(s+1)^3 G(s)] \Big|_{s=-1} = 2 \\ R_2 &= [(s+2)G(s)] \Big|_{s=-2} = 1 \end{aligned} \quad (1.72)$$

728 Therefore, the partial fraction expansion of $G(s)$ is

$$G(s) = \frac{3}{(s+1)} + \frac{1}{(s+1)^2} + \frac{2}{(s+1)^3} + \frac{1}{(s+2)} \quad (1.73)$$

729 and

$$g(t) = [(3+t+2t^2)e^{-t} + e^{-2t}]1(t) \quad (1.74)$$

730 **1.5 The Transfer function and The State Space Representations (State Equations)**
731

732 The true value of the Laplace transform is that it provides an important, mathematical
733 characterization of a dynamic system, the transfer function. In this text, for control system
734 design with Youla parametrization, we discover that the transfer function is more valuable
735 than the underlying state space representation. Of course the state equations are the usual
736 product of dynamic system analysis and as discussed below, the transfer function is ob-
737 tained from the state equations. The state equations tell us much that is important about
738 the dynamics of any system and are used here to draw-out crucial mathematical properties
739 of the transfer function and the transfer function approach to feedback design.

1.5.1 The Transfer Function

Our picture of the dynamic system is that of Figure 1.23. We shall use “u” to denote input and “y” to denote output throughout the text. In continuous-time application, the transfer function is the ratio of the transformed output to the transformed input $G(s) = \frac{\hat{y}}{\hat{u}}$ and all initial conditions are zero.

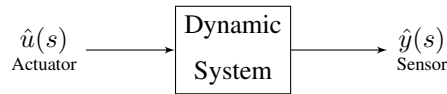


Figure 1.23: Block diagram of a dynamic system.

One can obtain this ratio anywhere the transforms exist. In some cases, the term “transfer function” is utilized to mean some input-output characteristic with direct physical measures rather than the “s-plane” transform. In these cases, the transfer function depends upon the particular input. In the case of the linear, time-invariant system (LTI), the transfer function is completely specified by the parameters of the system and is independent of the input.

This independence of the transfer function for the input, for the LTI systems, is easily established if one considers the differential-equation describing a dynamic system. Since such an equation transforms to a polynomial operator times the output equal to a polynomial times the input, the ratio of output to input is a rational function. Clearly, this rational function is independent of the particular input.

We have shown through several examples, one can always derive the equation of motion (the differential equation) and then write the transfer function by inspection. That is almost the way things go, but not quite.

1.5.2 The State Equations

Rather than a single, high-order equation, it is usually more convenient to write a system of simultaneous equations to describe the dynamics of a physical system. In mechanical engineering, for example, Hamilton’s principle leads to a system of first order equations. In general, the dynamics of a system can be written in matrix form ¹.

¹The matrix notation used here is conventional: column matrices (“vectors”) are represented by lower case and rectangular matrices are represented by upper case. The “T” superscript represents the matrix transpose so c^T represents a row matrix.

$$\begin{aligned}\dot{\underline{x}} &= A\underline{x} + \underline{b}u \\ \underline{y} &= \underline{c}^T \underline{x} + D\underline{u}\end{aligned}\tag{1.75}$$

The first equation is called the dynamic equation while the second is referred to as the message equation. The combination is called the *state variable model*.² A great deal of computer software is based upon the assumption that the state equations are an important part of the user's natural language.

The vector $\underline{x}$ represents a set of intermediate variables (momenta and generalized displacements in the context of Hamilton's principle) needed to complete the description of the relation of input to output. These intermediates are called *state variables*. If one takes the input-output point-of-view, as is often the case in control studies, the identities of the state variables are immaterial.

The state equations are represented in the block diagram of Figure 1.24.

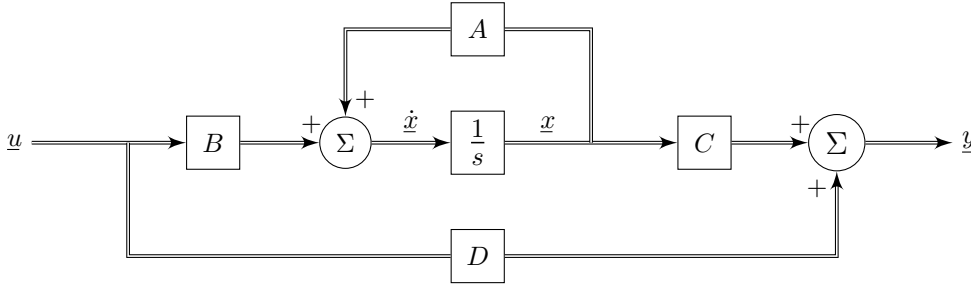


Figure 1.24: State-space block diagram.

The transfer function is related to state variable models after some preparation. For one thing, we must decide what we mean by the Laplace transform of a vector or a matrix. We let the Laplace transform of a vector be a vector of Laplace transforms.

$$\mathcal{L}[\underline{x}(t)] = \mathcal{L} \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix} \triangleq \begin{bmatrix} \mathcal{L}[x_1(t)] \\ \mathcal{L}[x_2(t)] \\ \vdots \\ \mathcal{L}[x_n(t)] \end{bmatrix}$$

²Kalman called the combination a state-variable realization. His explanation is that an “analog” computer circuit could be directly wired from these equations and such a circuit is a physical synthesis or “realization” of the equations. The analog computer is antiquated at this date so the modern student may have difficulty interpreting these ideas.

The k^{th} component of the Laplace transform of the vector is the Laplace transform of the k^{th} component. The Laplace transform of a matrix is defined in a similar fashion. The linearity property has implications for the Laplace transform of matrix products. Notice that by linearity, $\mathcal{L}[A\mathbf{x}(t)] = A\mathcal{L}[\mathbf{x}(t)]$. In fact, we can use the linearity property to obtain many matrix-transform relations similar to the scalar counterparts.

The state variable model of (1.75) can be transformed into the Laplace domain.

$$\begin{aligned} s\hat{\mathbf{x}} - \mathbf{x}(0^-) &= A\hat{\mathbf{x}} + \mathbf{b}\hat{u} \\ \hat{y} &= \mathbf{c}^T \hat{\mathbf{x}} + D\hat{u} \end{aligned} \quad (1.76)$$

These equations are manipulated to obtain

$$\hat{y} = [\mathbf{c}^T(sI - A)^{-1}] \hat{\mathbf{x}}(0^-) + [\mathbf{c}^T(sI - A)^{-1}\mathbf{b} + D] \hat{u} \quad (1.77)$$

where for any variable v , $\hat{v} = \mathcal{L}[v(t)]$.

It is here that we can justify the name “state”. In order to obtain the output, $\hat{y}(s)$, we must know the input, $\hat{u}(s)$ (that is, $u(t)$ for $t \geq 0$) and the initial conditions, $\mathbf{x}(0^-)$. These initial conditions contain all the information that is required about what had happened before the input was applied; they completely describe the current state of the system.

In order to obtain a connection with the transfer function, set $\mathbf{x}(0^-) = \mathbf{0}$ and knowing that $G(s) = \frac{\hat{y}}{\hat{u}}$, we obtain

$$G(s) = \mathbf{c}^T(sI - A)^{-1}\mathbf{b} + D \quad (1.78)$$

1.5.3 Transfer function properties

Consider a rational function $G(s) = \frac{\hat{y}}{\hat{u}}$,

Definition 1.3. $G(s)$ is proper if $\deg(\text{numerator}) = \deg(\text{denominator})$.

Definition 1.4. $G(s)$ is strictly proper if $\deg(\text{numerator}) < \deg(\text{denominator})$.

Definition 1.5. $G(s)$ is improper otherwise (i.e., if $\deg(\text{numerator}) > \deg(\text{denominator})$).

Recall the formula for matrix inversion.

$$M^{-1} = \frac{\text{adj } M}{\det M} \quad (1.79)$$

We can then write

$$(sI - A)^{-1}_{n \times n} = \frac{\text{adj } (sI - A)}{\det(sI - A)} = \frac{\text{poly. with } \deg \leq n-1}{\text{poly. with } \deg = n} \quad (1.80)$$

where n is dimension of matrix A . Then we can write $G(s)$ as

$$\begin{aligned}\frac{\hat{y}}{\hat{u}} = G(s) &= \underline{c}^T (sI - A)^{-1} \underline{b} + D \\ &= \frac{\underline{c}^T \text{adj}(sI - A) \underline{b}}{\det(sI - A)} + D \\ &= \frac{\underline{c}^T \text{adj}(sI - A) \underline{b} + D \det(sI - A)}{\det(sI - A)}\end{aligned}\quad (1.81)$$

An element of the adjugate matrix, $\text{adj}(sI - A)$, is formed by crossing out a single row and a single column and applying the determinant operation. This means that the elements of $\text{adj}(sI - A)$ are polynomials of degree no greater than $n - 1$. Using this observation, we can say the followings about $G(s)$:

- Strictly proper if $D = 0$.
- Proper if $D \neq 0$.
- Never improper.

The question arises whether any transfer function is improper: Is there another formulation of the dynamic equations such that such a thing is possible? One does see such formulations from time to time but one should be skeptical on physical grounds. The Paley-Wiener Theorem [17.2] establishes that an improper transfer function is physically impossible. In the last section of this chapter, we argue in a more heuristic way that the transfer function of a physical system is not improper.

1.5.4 Poles and Zeros of a Transfer Function

Given the state equations below, the transfer function $G(s)$ can be written as,

$$\begin{aligned}\dot{\underline{x}} &= A\underline{x} + \underline{b}u \\ \underline{y} &= \underline{c}^T \underline{x} + D\underline{u}\end{aligned}\quad (1.82)$$

$$G(s) = \frac{\underline{c}^T \text{adj}(sI - A) \underline{b} + D \det(sI - A)}{\det(sI - A)} = \frac{\eta(s)}{\Delta(s)}, \quad \deg \eta \leq \deg \Delta \quad (1.83)$$

Poles of $G(s)$ are roots of: $\det(sI - A) = 0$, and depend only upon the internal connections of the system; that is, the poles depend only upon the A-matrix. The zeros of $G(s)$ are the roots of: $\underline{c}^T \text{adj}(sI - A) \underline{b} + D \det(sI - A) = 0$. In contrast to poles, zeros also depend upon the way that the system connected to its environment (characterized by $\underline{c}^T$ and $\underline{b}$). The physical interpretation of these state matrices are provide in Table 1.1.

Table 1.1: Matrix Form Parameters

Matrix	Function
A	State interaction
B	Connection to actuator (“environment”)
C^T	Connection to sensor (“environment”)
D	Direct connection from actuator to sensor

Let $G(s)$ be the transfer function from $\hat{u}(s)$ to $\hat{y}(s)$ without any repeated poles,

$$G(s) = \frac{\eta(s)}{\Delta(s)} = \frac{(s + z_1)(s + z_2) \cdots (s + z_m)}{(s + p_1)(s + p_2) \cdots (s + p_n)}, \quad m \leq n \quad (1.84)$$

then, as shown in the previous sections, the output response $\hat{y}(s)$ is,

$$\hat{y}(s) = G(s)\hat{u}(s) = \dots + \frac{R_k}{s + p_k} + \dots \quad (1.85)$$

where,

$$\mathcal{L} \left[\frac{R_k}{s + p_k} \right] = R_k e^{-p_k t} \quad (1.86)$$

In this context, a pole and zero cancellation could be problematic in feedback design and it will be discussed in the following sections. The example below introduces the idea of a pole and zero cancellation.

EXAMPLE 1.12

Pole cancellation. Consider a system with dynamics

$$\hat{y}(s) = \frac{(s + 1)}{(s + 1)(s + 2)} \hat{u}(s) \quad (1.87)$$

Partial fraction expansion without carrying out pole-zero cancellation will result in a term with denominator $(s + 1)$ and numerator 0.

$$\hat{y}(s) = \frac{(s + 1)}{(s + 1)(s + 2)} \hat{u}(s) = \dots + \frac{0}{s + 1} + \dots \quad (1.88)$$

Inverse Laplace and re-transform:

$$\ddot{y} + 3\dot{y} + 2y = \dot{u} + u \quad (1.89)$$

$$s^2 \hat{y}(s) - \dot{y}(0^-) - sy(0^-) + 3s\hat{y}(s) - 3y(0^-) + 2\hat{y}(s) = s\hat{u} + 1\hat{u} \quad (1.90)$$

$$(s^2 + 3s + 2)\hat{y}(s) = (s + 1)\hat{u}(s) + \dot{y}(0^-) + sy(0^-) + 3y(0^-) \quad (1.91)$$

$$\hat{y}(s) = \frac{(s + 1)}{(s + 1)(s + 2)} \hat{u} + \frac{[y(0^-)]s + [3y(0^-) + \dot{y}(0^-)]}{(s + 1)(s + 2)} \quad (1.92)$$

$$\hat{y}(s) = \dots + \frac{R}{s + 1} + \dots, \text{ where } R \neq 0 \quad (1.93)$$

The pole is canceled from the forced response, but not the initial condition response.

This is called a “hidden mode.”

1.5.5 Physical Realizability:

It is important to obtain results which indicate the range of possibilities for real, physical transfer functions. the most satisfactory results are obtained by imposing a constraint of "cause and effect": the output of a causal physical system cannot anticipate the input. We claim that these causal physical systems are proper, and we ask the following question, Can we build a system for which $G(s)$ is improper? Is an improper transfer function realizable?

The simplest improper $G(s)$ is illustrated in Figure 1.25, this system is a simple differentiator.

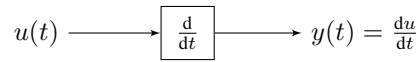


Figure 1.25: An improper system.

Let $u(0^-) = 0$, $\hat{y}(s) = s\hat{u}(s)$, $G(s) = \frac{\hat{y}(s)}{\hat{u}(s)}$. More generally, suppose $G(s) = \frac{\eta(s)}{\Delta(s)}$, where $\deg \eta \geq \deg \Delta + 1$. We can expand $G(s)$ such that

$$G(s) = R_0 + R_{-1}s + \dots + \underbrace{G'(s)}_{\text{Strictly Proper}}, \quad (1.94)$$

as illustrated in Figure 1.26.

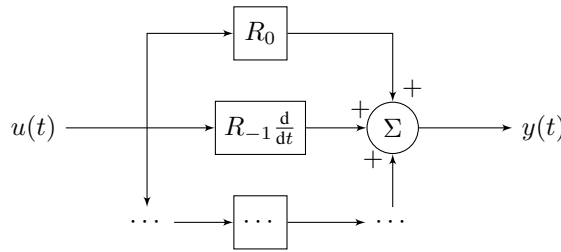


Figure 1.26: An expansion of $G(s)$ into proper and improper components.

But, what is the definition of a derivative? The derivative at time t is the common limit (if it exists) of the slope from the right and left.

$$\frac{dy(t)}{dt} = \lim_{\Delta t \rightarrow 0^-} \frac{y(t) - y(t - \Delta t)}{\Delta t} = \lim_{\Delta t \rightarrow 0^+} \frac{y(t + \Delta t) - y(t)}{\Delta t} \quad (1.95)$$

This is shown in Figure 1.27.

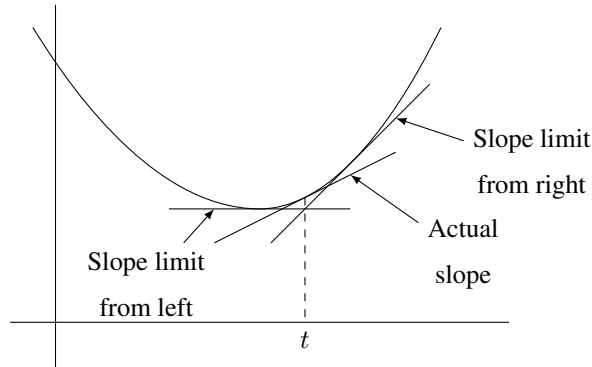


Figure 1.27: Derivative common limit.

We can determine the limit from the left, $\Delta t \rightarrow 0^-$ (from past values), but it is impossible to measure the limit from the right $\Delta t \rightarrow 0^+$ (and therefore the common limit), because no device cannot predict the future.

Instead, we can implement some delay, e.g. $\frac{s}{\tau s + 1}$, (making the $G(s)$ proper), in order to approximately measure the derivative. This delay results in using only the past values to approximate the computation of a differential operator.

It should be noted that some limit processes such as integration, do not require a device that can foresee the future, as discussed below.

One can ask, is integration physically realizable? Integration is the common limit (if it exists) as partitioning is refined 1.29. Information about future states is not required for integration, so integration is realizable.

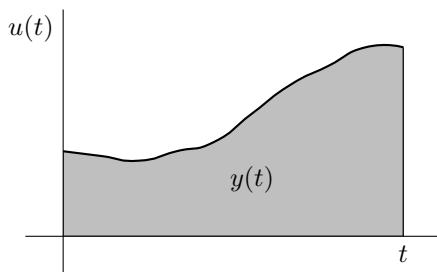
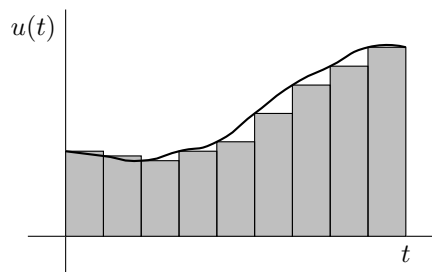
Figure 1.28: Integral of $u(t)$ 

Figure 1.29: The integral is the common limit of the upper and lower sum.

The integral operator is shown in Figure 1.30.

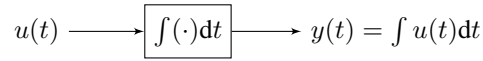


Figure 1.30: Integration block diagram.

The Laplace transform of this operator is given by

$$\frac{dy}{dt}(t) = u(t)$$

$$\text{Transform: } s\hat{y}(s) = \hat{u}(s)$$

$$G(s) = \frac{\hat{y}}{\hat{u}} = \frac{1}{s}$$

A consideration of frequency response allows one to anticipate the more rigorous result laid down by the Paley-Wiener Theorem [17.2], which is outside the scope of this text. Intuitively, it seems impossible for a physical system to have increasing gain with increasing frequency: If this sort of thing were possible, the power requirements would be enormous. This leads to speculate that *the degree of the numerator of a transfer function of a physically realizable dynamic system is never greater than the degree of the denominator.*

Notice that we satisfy this constraint wherever formula (1.78) is used: State equations (1.75) meet expectations for physical realizability. Something negative can be said about those formulations of the state equations from which improper transfer functions are calculated.

The same sort of reasoning leads some to argue that a physical system must be strictly proper (similar to our integral operator), where the D-matrix is zero, since it is hard to imagine that a real system can pass a very high frequency input without some attenuation. That is fair enough, but this conclusion is not supported by the more rigorous realizability theory based upon causality principles and complex analysis.

1.6 Problems

1.1 Find the Laplace transform of the two functions of time illustrated in Figure 1.31 (a) and (b). Hint: you can write an expression for these signals using step and delayed steps.

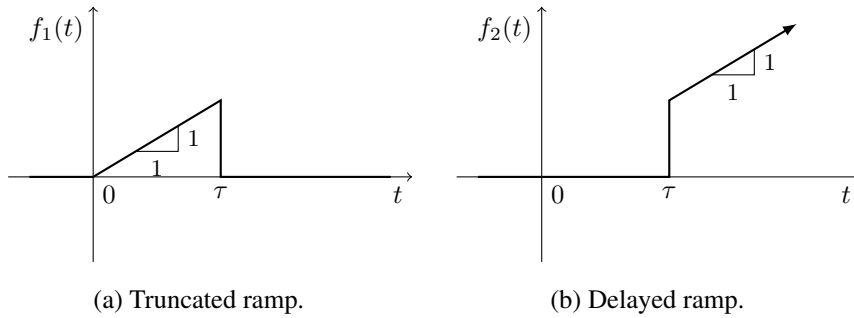


Figure 1.31

1.2 Obtain the Laplace transform of

$$y(t) = te^{-t}\cos(2t)$$

1.3 Find the inverse Laplace transforms of

(a) $\frac{7s-6}{s^2-s-6}$

(b) $\frac{2s^2+5}{s^2+3s+2}$

(c) $\frac{6(s+34)}{s(s^2+10s+34)}$

(d) $\frac{8s+10}{(s+1)(s+2)^3}$

1.4 Consider the following state-space equation,

$$\dot{x} = Ax + \underline{b}u$$

$$y = \underline{c}^T x$$

where

$$A = \begin{pmatrix} 0 & 1/J \\ 0 & -b/J \end{pmatrix} \quad \underline{b} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\underline{c}^T = (1 \quad 0) \quad D = 0$$

Find the transfer function $G(s)$ where $G(s) = \frac{\hat{y}(s)}{\hat{u}(s)}$.

1.5 For the system shown in Figure 1.32, the elements are designed so that

$$\frac{dc}{dt} = r(t) - c(t - \tau)$$

The unavoidable delay τ occurs because the measuring device must be placed downstream

of the roller. Use the time shifting property to find the transfer function $G(s) = \frac{C(s)}{R(s)}$.

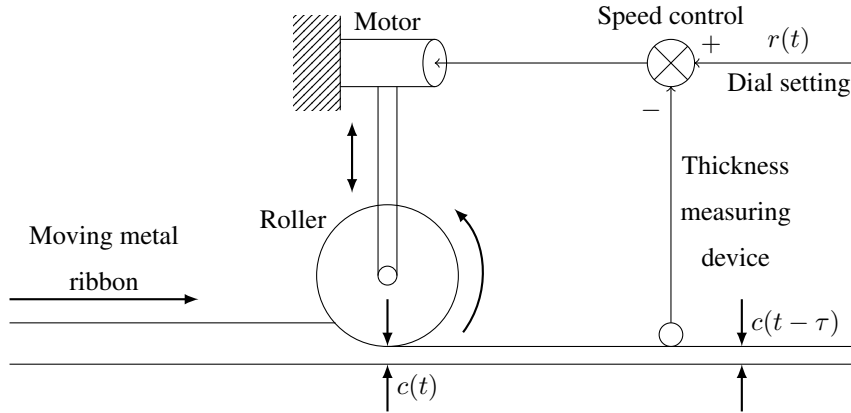


Figure 1.32

1.6 Apply the relevant propositions and proofs discussed in the class in a iterative fashion to establish the following corollaries:

(a)

$$\mathcal{L}[t^n y(t)] = (-1)^n \frac{d^n}{ds^n} \hat{y}(s)$$

(b)

$$\mathcal{L}\left[\frac{d^n}{dt^n} y(t)\right] = s^n \hat{y}(s) - \sum_{k=1}^n s^{n-k} \frac{d^{k-1}}{dt^{k-1}} y(0^-)$$

1.7 Consider the following state-space equation,

$$\dot{x} = Ax + \underline{b}u_1$$

$$y = \underline{c}^T x$$

where

$$A = \begin{bmatrix} 0 & \frac{1}{m} \\ 0 & -\frac{b}{m} \end{bmatrix}, \quad B = \begin{bmatrix} \frac{b}{m^2} \\ -\frac{b^2}{m} \end{bmatrix}, \quad \underline{c}^T = [1 \quad 0]$$

Find the transfer function $G(s)$, where

$$G(s) = \frac{\hat{y}}{\hat{u}_1}(s)$$

Explain how u_1 has to change so that the transfer function $G(s)$ will be equal to

$$G(s) = \frac{-\frac{1}{m}}{s(s + \frac{b}{m})}$$

1.8 Newton's second law for the submersible shown is

$$m \frac{d^2 y}{dt^2} = F$$

where F is regulated by pumping water in or out of tanks. The relationship between F and the pumping rate, $u(t)$, is

$$\frac{dF}{dt} = k_B u$$

(a) Find the transfer function $\frac{\hat{y}}{\hat{u}}(s)$. Indicate the pole-zero diagram.

(b) Actually, if the water is expelled or brought in from the bottom of the craft, then we get a little added thrust, so the second law becomes

$$m \frac{d^2 y}{dt^2} = F + k_T u$$

What is the transfer function $\frac{\hat{y}}{\hat{u}}(s)$ in this case? Indicate the pole-zero diagram.

(c) If, instead, the water is expelled or brought in from the top of the craft, then the second law becomes

$$m \frac{d^2 y}{dt^2} = F - k_T u$$

What is the transfer function $\frac{\hat{y}}{\hat{u}}(s)$ in this case? Indicate the pole-zero diagram.

CHAPTER 2

THE RESPONSE OF LINEAR, TIME-INVARIANT DYNAMIC SYSTEMS

This chapter is a review of the basics of dynamic-system response. In particular, we wish to relate the parameters of the transfer function or the state equations to the quantitative features of the time response of forced and unforced response. Special emphasis is placed upon the steady-state frequency response. The frequency-response relationships derived in this chapter are fundamental to the main issues of feedback design developed in Chapter 3.

Transient response is addressed as well. This is appropriate because improved dynamic, transient response can be thought of as an important, albeit a secondary, aspect of control system design. To stand to reason that one can improve dynamic response with feedback, so the designer might as well deal with these issues while tending to the more important matters, as will be discussed in the following chapters.

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2.1 The Time Response of Dynamic Systems

Begin with the elementary transfer-function relation where $\hat{y}(s) = G(s)\hat{u}(s)$, as indicated in Figure 2.1.

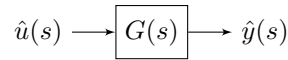


Figure 2.1: Dynamic system.

Consider $G(s)$ to be a dynamic system as indicated in Figure 2.2. Now consider an impulse input, where $\hat{u}(s) = \delta(s)$, so that $\hat{y}(s) = G(s)$. Then, $y(t) = g(t)$, where $g(t) = \mathcal{L}^{-1}[G(s)]$ is called the **impulse response of the dynamic system**.

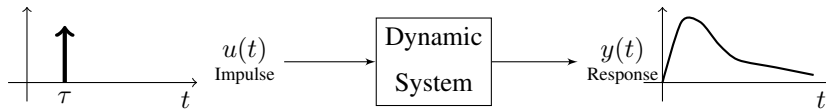


Figure 2.2: Response of a dynamic system.

Using an impulse response input, we will generalize the response of a dynamic system to any test input. However, before accomplishing this, we will go over three fundamental theorems.

2.1.1 Final Value Theorem

For a linear time invariant system, the Final Value Theorem can be used to find the value of $y(t)$ as $t \rightarrow \infty$.

Theorem 2.1 (Final Value Theorem) If $\mathcal{L}[y(t)] = \hat{y}(s)$ and poles of $s\hat{y}(s)$ lie strictly in the LHP, then the Final Value Theorem states

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} s\hat{y}(s) \quad (2.1)$$

Proof:

$$\begin{aligned} \mathcal{L}\left[\frac{dy}{dt}\right] &= s\hat{y}(s) - y(0^-) \\ s\hat{y}(s) &= y(0^-) + \mathcal{L}\left[\frac{dy}{dt}\right] \\ &= y(0^-) + \int_{0^-}^{\infty} e^{-st} \frac{dy}{dt}(t) dt \\ \lim_{s \rightarrow 0} s\hat{y}(s) &= y(0^-) + \int_{0^-}^{\infty} \frac{dy}{dt}(t) dt \\ &= y(0^-) + \lim_{t \rightarrow \infty} y(t) - y(0^-) \\ &= \lim_{t \rightarrow \infty} y(t) \end{aligned}$$

Therefore, the final value of $y(t)$ can be obtained from the Laplace transform of $y(t)$ using the Final Value Theorem. ■

2.1.2 Initial Value Theorem

For a linear time invariant system, the Initial Value Theorem can be used to find the initial value of $y(t)$ at the instant some input $u(t)$ is applied.

Theorem 2.2 (Initial Value Theorem) If $\mathcal{L}[y] = \hat{y}$, then the Initial Value Theorem states

$$\lim_{s \rightarrow \infty} s\hat{y}(s) = y(0^+) \quad (2.2)$$

Proof:

$$\begin{aligned} s\hat{y}(s) &= y(0^-) + \int_{0^-}^{\infty} e^{-st} \frac{dy}{dt}(t) dt \\ &= y(0^-) + \int_{0^-}^{0^+} e^{-st} \frac{dy}{dt}(t) dt + \int_{0^+}^{\infty} e^{-st} \frac{dy}{dt}(t) dt \\ \lim_{s \rightarrow \infty} s\hat{y}(s) &= y(0^-) + \int_{0^-}^{0^+} \underbrace{e^{-s \cdot 0}}_{=1} \frac{dy}{dt}(t) dt + \int_{0^+}^{\infty} \underbrace{e^{-st}}_{\rightarrow 0 \text{ as } s \rightarrow \infty} \frac{dy}{dt}(t) dt \\ &= y(0^-) + \int_{0^-}^{0^+} \frac{dy}{dt}(t) dt + 0 \\ &= y(0^-) + y(0^+) - y(0^-) \\ &= y(0^+) \end{aligned}$$

Therefore, the initial value of $y(t)$ can be obtained from the Laplace transform of $y(t)$ using the Initial Value Theorem. ■

■ **EXAMPLE 2.1**

Initial and final values for a step and ramp response.

$$G(s) = \frac{s+1}{s+2}, \quad u(t) = 1(t), \quad \hat{u}(s) = \frac{1}{s} \quad (2.3)$$

Given that initial condition $y(0^-) = 0$, so

$$\hat{y}(s) = \frac{s+1}{s(s+2)} \Rightarrow y(\infty) = \lim_{s \rightarrow 0} s\hat{y}(s) = \frac{1}{2}, \quad y(0^+) = \lim_{s \rightarrow \infty} s\hat{y}(s) = 1 \quad (2.4)$$

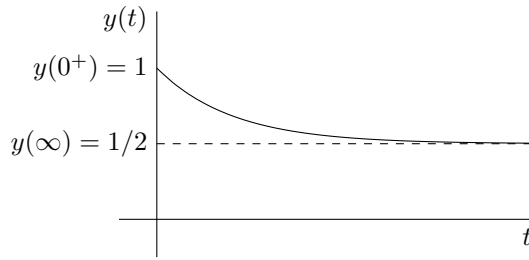


Figure 2.3: Response to step input.

Now suppose $u(t)$ is a ramp input.

$$u(t) = t1(t), \quad \hat{u}(s) = \frac{1}{s^2} \quad (2.5)$$

$$\hat{y}(s) = \frac{s+1}{s^2(s+2)} \Rightarrow y(\infty) = \infty, \quad y(0^+) = 0 \quad (2.6)$$

■ **EXAMPLE 2.2**

Initial and final values for a step response.

$$G(s) = \frac{s+1}{s^2+s+1}, \quad u(t) = 1(t), \quad \hat{u}(s) = \frac{1}{s} \quad (2.7)$$

$$\hat{y}(s) = \frac{s+1}{s(s^2+s+1)}, \text{ given that } y(0^-) = 0, \quad \dot{y}(0^-) = 0 \quad (2.8)$$

$$y(\infty) = \lim_{s \rightarrow 0} s\hat{y}(s) = 1 \quad (2.9)$$

$$y(0^+) = \lim_{s \rightarrow \infty} s\hat{y}(s) = 0 \quad (2.10)$$

$$\dot{y}(0^+) = \lim_{s \rightarrow \infty} s^2\hat{y}(s) = 1 \quad (2.11)$$

Note that there is a discontinuity in the slope: $\dot{y}(0^-) = 0$ and $\dot{y}(0^+) = 1$.

2.1.3 Convolution and the Laplace Transform

Consider a dynamic system $G(s)$ as indicated in Figure 2.1.

$$\hat{y}(s) = G(s)\hat{u}(s) \quad (2.12)$$

Theorem 2.3 (Convolution Theorem) If for $0^- \leq \zeta \leq t \leq \infty$, $\mathcal{L}[u(t)] = \hat{u}(s)$ and $\mathcal{L}[g(t)] = G(s)$, then

$$\mathcal{L} \left[\int_{0^-}^t g(t-\zeta)u(\zeta)d\zeta \right] = G(s)\hat{u}(s) = \mathcal{L} \left[\int_{0^-}^t g(\zeta)u(t-\zeta)d\zeta \right] \quad (2.13)$$

Convolution can be written as a short-hand notation using the “*” operator, i.e.

$$[g(t) * u(t)] = \left[\int_{0^-}^t g(t-\zeta)u(\zeta)d\zeta \right]. \quad (2.14)$$

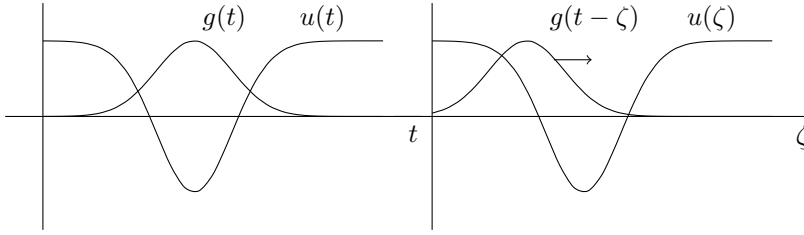


Figure 2.4: Convolution.

Proof: From the definition of the transform

$$\mathcal{L} [g(t) * u(t)] = \int_{0^-}^{\infty} e^{-st} \int_{0^-}^t g(t-\zeta)u(\zeta)d\zeta dt$$

Interchange the order of integration but preserve the ordering $0^- \leq \zeta \leq t \leq \infty$ so that

$$\mathcal{L} [g(t) * u(t)] = \int_{0^-}^{\infty} e^{-s\zeta} u(\zeta) \int_{\zeta^-}^{\infty} e^{-s(t-\zeta)} g(t-\zeta) dt d\zeta$$

Let $\eta = t - \zeta$ in the inner integration so $dt = d\eta$. Then,

$$\begin{aligned} \mathcal{L} [g(t) * u(t)] &= \int_{0^-}^{\infty} e^{-s\zeta} u(\zeta) \int_{0^-}^{\infty} e^{-s\eta} g(\eta) d\eta d\zeta \\ &= G(s) \int_{0^-}^{\infty} e^{-s\zeta} u(\zeta) d\zeta \end{aligned} \quad (2.15)$$

Therefore, $\mathcal{L}[g(t) * u(t)] = G(s)\hat{u}(s)$. ■

This means that for a dynamic system as shown in Figure 2.1,

$$y(t) = \int_{0^-}^t g(t-\zeta)u(\zeta)d\zeta \quad (2.16)$$

We are now in a position to write the general time response of a LTI dynamic system as,

From previous section, knowing that $\hat{y} = [\underline{c}^T(sI - A)^{-1}] \underline{x}(0^-) + [\underline{c}^T(sI - A)^{-1} \underline{b} + D] \hat{u}$

$$\begin{aligned} y(t) &= \underline{c}^T \Phi(t) \underline{x}(0^-) + g(t) * u(t) \\ &= \underline{c}^T \Phi(t) \underline{x}(0^-) + \int_{0^-}^t g(t - \zeta) u(\zeta) d\zeta \end{aligned} \quad (2.17)$$

then we can write

$$y(t) = \underline{c}^T \Phi(t) \underline{x}(0^-) + \underline{c}^T \left[\int_{0^-}^t \Phi(t - \zeta) u(\zeta) d\zeta \right] \underline{b} + Du(t) \quad (2.18)$$

where $\Phi(t) = \mathcal{L}^{-1}[\Phi(s)] = \mathcal{L}^{-1}[(sI - A)^{-1}]$ is known as the *State Transition Matrix* and $g(t) = \mathcal{L}^{-1}[G(s)]$.

There is a useful interpretation of $g(t)$ as the impulse response of the dynamic system. One can imagine a procedure where the impulse response is measured and the output to any other input predicted with the convolution (2.17).

The first term in (2.17) is called the *initial-condition response* and the second term is called the *forced response*. The initial-condition response will be a transient response for stable systems and the forced response makes some contribution to the transient response as well as to the steady-state response. The third term in (2.18) is called *direct transmission* or *static-gain* response and is most obvious where things are in terms of the state equations. We reason that there must be a similar response component in (2.17) but it is obscured in the convolution integral.

The stability concept requires some elaboration, which will be addressed in this section.

Proposition 2.1. *The step response of a dynamic system is the integral of the impulse response:*

$$y(t) = \int_0^t g(\zeta) d\zeta \quad (2.19)$$

Proof: From the simple fact that $\hat{y}(s) = G(s) \cdot 1/s$ and $G(s) = \mathcal{L}[g(t)]$. ■

2.1.4 Transmission Blocking Response

In this subsection, we provide an interpretation of a transfer-function zero. We propose the following definitions to distinguish between zeros and zeros with transmission blocking response properties.

Definition 2.1. A transfer function, $G(s)$, has a transmission zero at $s = z_k$ if there exists an input such that $u(t) = u_k e^{z_k t} 1(t)$ results in the output $y(t)$ not containing $e^{z_k t}$ terms.

Definition 2.2. A transfer function, $G(s)$, has a transmission zero at $s = z_k$ with the *transmission blocking response* property if there exists an input such that $u(t) = u_k e^{z_k t} 1(t)$ results in exactly zero output $y(t) \equiv 0 \quad \forall t$. These zeros are called transmission blocking zeros.

We don't make a distinction between zeros of a transfer function or transmission zeros. We will address this point in more details in the MIMO part of this text.

Now, suppose $u(t) = e^{-\mu t} 1(t)$. Is it possible to design $G(s)$ so that $y(t)$ has exactly zero response to $u(t)$, as shown in Figure 2.5? The answer to this question is *no* if all initial conditions are precisely equal to zero. The best we could do is to eliminate this particular input from the output response, but, we are unable to force the output $y(t)$ to be exactly equal to zero for all times.

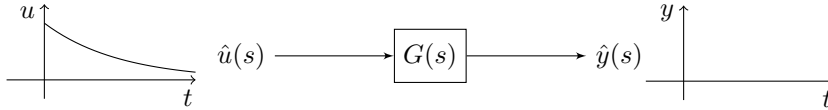


Figure 2.5: Transmission Blocking Response.

We then wonder, is possible to apply a nonzero input to a system with a nonzero gain and find that the output is exactly zero? The answer to this question is yes if both the *input and the initial conditions* are just right. The particular input we have in mind is the exponential $u(t) = e^{-\mu t} 1(t)$, so that $\hat{u}(s) = 1/(s + \mu)$ and the output is a superposition of forced and initial condition responses. Denote the transfer function $G(s) = \eta(s)/\Delta(s)$ so that

$$\hat{y}(s) = G(s)\hat{u}(s) + I.C. = \frac{\eta(s)}{\Delta(s)} \cdot \frac{1}{s + \mu} + \frac{q(s)}{\Delta(s)} \equiv 0 \quad (2.20)$$

where $q(s)$ is some polynomial due to initial conditions. In order to cancel the dynamics of $u(t)$, we require that $\eta(s)$ has $(s + \mu)$ as a factor, $(s + \mu)$ must be a zero of $G(s)$ (definition 1). However, appropriate dynamics from initial conditions are still required to fully block a response.

$$\frac{\eta(s)}{\Delta(s)} \cdot \frac{1}{s + \mu} + \frac{q(s)}{\Delta(s)} = 0 \quad \Rightarrow \quad q(s) = \frac{-\eta(s)}{s + \mu} \quad (2.21)$$

Knowing, the procedure for selecting the right combination of the input and initial conditions for transmission blocking response, we can then ask, are poles and zeros ever really canceled? We can factor out the $(s + \mu)$ term from $G(s)$ such that

$$G(s) = (s + \mu)G'(s) \quad (2.22)$$

Then, Equation (2.20) can be written as

$$\hat{y}(s) = G'(s) + I.C. \quad (2.23)$$

From Equation (2.23), we can state that even if we cancel the unwanted pole by a properly locating a zero in $G(s)$, this unwanted pole may still result in an output response if the initial conditions are not zero. The two examples below shed further light to the concepts of *transmission blocking response* and *pole-zero cancellation*.

EXAMPLE 2.3

Suppose we assume a step input, $u(t) = 1(t)$ to the following system,

$$G(s) = \frac{s}{s+1}, \quad u(t) = 1(t), \quad \hat{u}(s) = \frac{1}{s} \quad (2.24)$$

Then we ask, what should be the initial condition to generate an output, $y(t) \equiv 0 \quad \forall t$,

$$\frac{s}{s+1} \cdot \frac{1}{s} + \frac{q(s)}{s+1} = 0 \quad \Rightarrow \quad q(s) = -1 \quad (2.25)$$

For simulating this simple system, it is easier to write the output response $\hat{y}(s)$ in the following form,

$$\hat{y} = [\underline{c}^T(sI - A)^{-1}] \underline{x}(0^-) + [\underline{c}^T(sI - A)^{-1}\underline{b} + D] \hat{u} \quad (2.26)$$

As will be discussed later, the state matrices A , $\underline{b}$, $\underline{c}^T$, and D can be computed from $G(s)$. Knowing $G(s) = \frac{s}{s+1}$ then $A = -1$, $\underline{b} = 1$, $\underline{c}^T = -1$, and $D = 1$. Comparing Equations (2.26) and (2.25), it is not difficult to see that $q(s) = \underline{c}^T \cdot \underline{x}(0^-)$. Therefore knowing that $q(s) = -1$ and $\underline{c}^T = -1$, then $\underline{x}(0^-) = 1$. We can then simulate the following system,

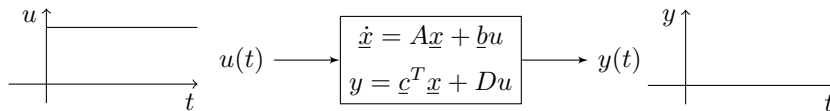


Figure 2.6: Transmission Blocking response.

we then observe that the the output $y(t) \equiv 0 \quad \forall t$.

EXAMPLE 2.4

Given the following system,

$$\hat{y} = \frac{(s+1)}{(s+1)(s+2)} \hat{u} = \dots + \frac{0}{s+1} + \dots \quad (2.27)$$

if we don't cancel the common pole-zero and utilize the Heaviside expansion to compute the residuals, as discussed in the previous section, we will find that the residual associated with the $s+1$ is equal to zero. we can equivalently write the system equations as

$$(s^2 + 3s + 2)\hat{y} = (s+1)\hat{u} \quad (2.28)$$

now, utilizing the inverse Laplace transform, we obtain

$$\ddot{y} + 3\dot{y} + 2y = \dot{u} + u \quad (2.29)$$

and re-transform,

$$s^2\hat{y} - \dot{y}(0^-) - sy(0^-) + 3s\hat{y} - 3y(0^-) + 2\hat{y} = s\hat{u} + \hat{u} \quad (2.30)$$

The output can be written in terms of the input and the initial conditions,

$$\hat{y} = \frac{(s+1)}{(s+1)(s+2)} \hat{u} + \frac{sy(0^-) + 3y(0^-) + \dot{y}(0^-)}{(s+1)(s+2)} \quad (2.31)$$

again, utilizing the Heaviside expansion,

$$\hat{y} = \dots + \frac{R}{s+1} + \dots, \quad \text{where } R \neq 0 \quad (2.32)$$

The pole is canceled by a common zero from the forced response, but not the initial condition response. We will study the impact of this type of pole cancellation from forced but not from the initial condition response in feedback control design in the following chapter.

2.1.5 Stability

In order to discuss the intuitive and important notion of stability, one must describe stability or, at least, instability in precise terms. There are some subtleties here, and these are related to *Inverse Laplace Transformation by Heaviside Expansion* discussed in the last chapter. It is clear that $y(t)$ will grow without bound where a pole exists in the right half-plane or where *multiple poles exist on the imaginary axis*. In the first case the *unbounded growth* is exponential while in the latter case the growth is some polynomial in t . Unbounded growth is certainly something we want to call *unstable*.

It is important to notice that the system poles directly define the components in the initial condition response (known also as unforced or homogeneous response) part of $y(t)$. This observation leads to the following working descriptions of stability.

Definition 2.3. If the initial condition response is unbounded we call the dynamic system *unstable*, otherwise we call the system *stable*. We call a system *asymptotically stable* if the initial condition response $\rightarrow 0$ as time $\rightarrow \infty$.

Clearly, a **stable** system¹ is one for which the poles of the transfer function are **(1) in the left half-plane**, and **(2) any pole on the imaginary axis is not a repeated pole**. Clearly, a system is **asymptotically stable** if and only if the poles of the transfer function are **strictly in the left half-plane**.

EXAMPLE 2.5

Consider the simple case of an “integrator”:

$$G(s) = \frac{1}{s} \quad (2.33)$$

The state equations are

$$\begin{aligned} A &= 0 & b &= 1 \\ c^T &= 1 & D &= 0 \end{aligned} \quad (2.34)$$

Since the impulse response is $g(t) = \mathcal{L}^{-1}[1/s] = 1(t)$, the output response $y(t)$ can be written as

$$y(t) = y(0^-) + \int_0^t u(\zeta) d\zeta \quad (2.35)$$

Where $y(0^-)$ is the initial value on the integrator. First of all, suppose that $u \equiv 0$. In this case, it is clear that $y(t)$ does not approach zero at $t \rightarrow \infty$, the system is stable but not asymptotically stable. This is a good point to state something about physical devices that we hope will serve as an integrator: Clearly, the true integral of the input,

$$\int_0^t u(\zeta) d\zeta$$

is achieved by our “integrator” only where

$$y(0^-) = 0$$

That is, in general case there is an unfortunate offset from the true integral!

¹note that based on the definition, there is no distinction between marginally stable and stable systems.

The stability discussions so far are around “initial condition response”. Still, there is another important way to view the stability condition by considering the effects of “inputs”.

Definition 2.4. Bounded-input, bounded-output (or BIBO) stability means that the output remains bounded for any input that is bounded.

If a transfer function is improper, the Heaviside expansion of the output will have terms in positive powers of “s” as we saw in the previous chapter. This will be true even when the input is bounded. However, s^k for $k \geq 0$ transforms to a singularity function, an unbounded output! Thus, the implication is that *even if an improper transfer function were realizable, it would be BIBO unstable*. Another implication of the definition is with regard to a transfer function with poles on the imaginary axis; for example, if there is a pole at the origin and the input is a step input (bounded) the Heaviside expansion will contain a term proportional to $1/s^2$, an output term proportional to $t^2/2$ (unbounded). That is, a system characterized by a transfer function with a pole at the origin is BIBO unstable. Similarly, if the transfer function has imaginary poles on the imaginary axis and the input is a sine or cosine function at the resonant frequency, The response is unbounded.

Thus, we have the next proposition:

Proposition 2.2. *A dynamic system is BIBO stable only if it is characterized by a proper transfer function. When a transfer function is proper, a necessary and sufficient condition for both BIBO stability and asymptotic stability is that all poles of the transfer function must lie strictly in the left half-plane.*

A necessary and sufficient condition for BIBO stability, of a proper transfer function, is the same as for asymptotic stability, all poles of the transfer function must lie strictly within the left half-plane.

We summarize our stability discussions by providing a table with stability definitions and several examples below.

Consider the transfer function $G(s)$.

$$G(s) = \frac{\eta(s)}{\Delta(s)} \quad (2.36)$$

In general, the output $\hat{y}(s)$ has a forced response and an initial condition response.

$$\hat{y}(s) = \underbrace{G(s)\hat{u}(s)}_{\text{Forced Response}} + \underbrace{\frac{q(s)}{\Delta(s)}}_{\text{I.C. Response}} = \frac{R_1}{s + p_1} + \frac{R_2}{s + p_2} + \dots \quad (2.37)$$

There are fundamentally three definitions of stability for LTI systems as given in Table (2.1). Stability in classical dynamics is based upon the initial condition response, **asymptotic stability**. For controls, however, it is more appropriate to look at **BIBO** stability based upon the forced response.

Table 2.1: Stability Definitions

Definition	Criterion
Asymp. Stable	Poles strictly in LHP
BIBO Stable	Poles strictly in LHP
Stable	Poles in LHP or on imag. axis (no repeated poles on imag. axis)

EXAMPLE 2.6

The definitive, mechanical-engineering example is

$$G(s) = 1/(ms^2) \quad (2.38)$$

which corresponds to

$$m \frac{d^2 y}{dt^2} = u(t) \quad (2.39)$$

Suppose that we apply the bounded input

$$u(t) = A \sin(\omega t)$$

It is a simple matter to demonstrate that the solution to the differential equation is

$$y(t) = -\frac{A}{m\omega^2} \sin(\omega t) + \left[\frac{A}{m\omega} + \frac{dy(0^-)}{dt} \right] t + y(0^-) \quad (2.40)$$

The second term is unbounded and nonzero if $A \neq 0$ where $dy(0^-)/dt = 0$ (BIBO unstable) and where $A = 0$ and $dy(0^-)/dt \neq 0$ (asymptotically unstable). Don't miss the point that even though the input force is bounded and has zero mean, the output displacement is unbounded and has a nonzero mean.

EXAMPLE 2.7

Consider the dynamics

$$\dot{y}(t) + \alpha y(t) = u(t) \quad (2.41)$$

with initial condition $y(0) = y_0$. Then,

$$s\hat{y}(s) - y(0) + \alpha\hat{y}(s) = \hat{u}(s) \quad (2.42)$$

$$\hat{y}(s) = \frac{1}{s + \alpha} (\hat{u}(s) + y_0) \quad (2.43)$$

Asymptotic stability is found by setting $\hat{u}(s) = 0$ and checking the initial condition response.

$$y(t) = \mathcal{L}^{-1} \left[\hat{y}(s) \Big|_{\hat{u}(s)=0} \right] = \mathcal{L}^{-1} \left[\frac{y_0}{s + \alpha} \right] = y_0 e^{-\alpha t} \quad (2.44)$$

For $\alpha > 0$, $y(t) \rightarrow 0$ as $t \rightarrow \infty$. Therefore, the system is asymptotically stable for $\alpha > 0$.

BIBO stability is found by setting the initial conditions to zero and considering the response to bounded inputs $\hat{u}(s)$. Letting

$$\hat{g}(s) = \frac{1}{s + \alpha} \quad (2.45)$$

and

$$g(t) = \mathcal{L}^{-1}[\hat{g}(s)] \quad (2.46)$$

Then,

$$y(t) = \mathcal{L}^{-1} \left[\hat{y}(s) \Big|_{y_0=0} \right] = \mathcal{L}^{-1} [\hat{g}(s) \hat{u}(s)] \quad (2.47)$$

$$= \int_{0^-}^{\infty} g(\tau) u(t - \tau) d\tau \quad (2.48)$$

$$\therefore |y(t)| \leq \int_{0^-}^{\infty} |g(\tau)| |u(t - \tau)| d\tau \quad (2.49)$$

If $u(t)$ is bounded, then

$$|u(t - \tau)| < K_1 < \infty \quad (2.50)$$

Then,

$$|y(t)| \leq K_1 \int_{0^-}^{\infty} |g(\tau)| d\tau \quad (2.51)$$

Hence for BIBO stability,

$$\int_{0^-}^{\infty} |g(\tau)| d\tau < \infty \quad (2.52)$$

In other words, if the impulse response is absolutely integrable then the system is BIBO stable. This is true if and only if the poles of $\hat{g}(s)$ are in the left-hand plane (that is, $\alpha > 0$).

1143 **EXAMPLE 2.8**

1144 Consider the dynamics

$$\ddot{y}(t) + y(t) = u(t) \quad (2.53)$$

1145 with initial condition $y(0) = y_0$ and $\dot{y}(0) = v_0$. Then,

$$s^2 \hat{y}(s) - \dot{y}(0) - sy(0) + \hat{y}(s) = \hat{u}(s) \quad (2.54)$$

$$\hat{y}(s) = \frac{1}{s^2 + 1} (\hat{u}(s) + v_0 + sy_0) \quad (2.55)$$

Asymptotic stability is found by setting $\hat{u}(s) = 0$ and checking the initial condition response.

$$y(t) = \mathcal{L}^{-1} \left[\hat{y}(s) \Big|_{\hat{u}(s)=0} \right] = \mathcal{L}^{-1} \left[\frac{1}{s^2 + 1} (v_0 + sy_0) \right] \quad (2.56)$$

$$= y_0 \cos(t) + v_0 \sin(t) \quad (2.57)$$

1147 For any nonzero initial conditions, $y(t)$ will not go to zero. It is therefore not asymp-
1148 totically stable.

1149 BIBO stability is found by setting the initial conditions to zero and considering
1150 the response to bounded inputs $\hat{u}(s)$. Following the same procedure as the previous
1151 Letting

$$\hat{g}(s) = \frac{1}{s^2 + 1} \quad (2.58)$$

1152 and

$$g(t) = \mathcal{L}^{-1}[\hat{g}(s)] \quad (2.59)$$

Then,

$$y(t) = \mathcal{L}^{-1} \left[\hat{y}(s) \Big|_{y_0=0} \right] = \mathcal{L}^{-1} [\hat{g}(s) \hat{u}(s)] \quad (2.60)$$

$$= \int_{0^-}^{\infty} g(\tau) u(t - \tau) d\tau \quad (2.61)$$

$$\therefore |y(t)| \leq \int_{0^-}^{\infty} |g(\tau)| |u(t - \tau)| d\tau \quad (2.62)$$

1153 If $u(t)$ is bounded, then

$$|u(t - \tau)| < K_1 < \infty \quad (2.63)$$

1154 Then,

$$|y(t)| \leq K_1 \int_{0^-}^{\infty} |g(\tau)| d\tau \quad (2.64)$$

Hence for BIBO stability,

$$\int_{0^-}^{\infty} |g(\tau)| d\tau < \infty \quad (2.65)$$

In other words, if the impulse response is absolutely integrable then the system is BIBO stable. In this case, we know that this is not true and Equation (2.65) is not finite. Therefore, the system is not BIBO stable.

EXAMPLE 2.9

BIBO stability (sine input).

$$G(s) = \frac{1}{s^2 + \omega^2} = \frac{1}{(s + j\omega)(s - j\omega)} \quad (2.66)$$

Apply a sinusoidal input:

$$\hat{u}(s) = \frac{\omega^2}{s^2 + \omega^2} \Rightarrow \hat{y}(s) = \frac{\omega^2}{s^2 + \omega^2} \cdot \frac{1}{s^2 + \omega^2} \quad (2.67)$$

Therefore, similar to the last example, a bounded input (sine wave) generates an unbounded response. Therefore, the system is BIBO unstable.

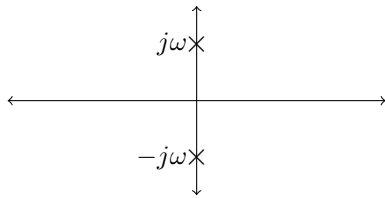


Figure 2.7: $G(s)$ roots.

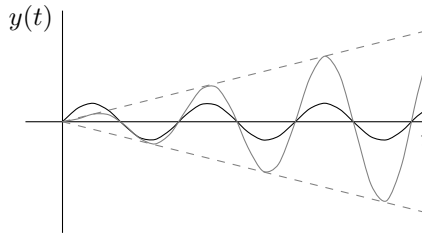


Figure 2.8: Unbounded response.

EXAMPLE 2.10

BIBO stability (step input).

$$G(s) = \frac{1}{s} \quad (2.68)$$

Suppose $u(t)$ is a step input.

$$\hat{u}(s) = \frac{1}{s}, \quad \Rightarrow \quad \hat{y}(s) = \frac{1}{s^2} \quad (2.69)$$

$$y(t) = \mathcal{L}^{-1}[\hat{y}(s)] = t1(t) \quad (2.70)$$

$y(t)$ has an unbounded (ramp) response to a bounded (step) input, therefore BIBO unstable.

1170 ■ **EXAMPLE 2.11**

1171 BIBO stability (improper transfer function).

$$G(s) = \frac{s^2}{s+1} \quad (2.71)$$

1172 $G(s)$ can be expanded into an improper residual and a proper transfer function.

$$G(s) = Rs + \underbrace{G'(s)}_{\text{proper}} = s + \frac{-s}{s+1} \quad (2.72)$$

1173 Suppose $\hat{u}(s) = \frac{1}{s}$ (step input).

$$\hat{y}(s) = 1 + \frac{1}{s} \cdot \frac{-s}{s+1} = 1 - \frac{1}{s+1} \quad (2.73)$$

$$y(t) = \delta(t) - e^{-t} \quad (2.74)$$

1175 A bounded input (step function) results in an unbounded output (impulse function).

1176 Therefore improper transfer functions are BIBO unstable.

1177 **2.1.6 Initial values and Reverse Action**

1178 Much can be learned from an examination of the record of the response of a dynamic
1179 system. In this subsection and the next, we examine the interpretation of observables at
1180 two extremes, $t = 0^+$ and $t \rightarrow \infty$. In order to be specific, we consider the response to step
1181 inputs

$$u(t) = 1(t) \quad (2.75)$$

1182 and we take n as the degree of the denominator of $G(s)$ and take m as the degree of the
1183 numerator. Taken together, this means,

$$\hat{y}(s) = G(s)/s \quad (2.76)$$

1184 Also, we write

$$G(s) = \frac{\eta_m s^m + \eta_{m-1} s^{m-1} + \dots + \eta_1 s + \eta_0}{\delta_n s^n + \delta_{n-1} s^{n-1} + \dots + \delta_1 s + \delta_0} \quad (2.77)$$

1185 **Theorem 2.4 (Derivative Discontinuity Theorem)** *Where all initial conditions are zero,*
1186 *a discontinuity occurs in the $(n-m)^{th}$ derivative of the step response of a dynamic system:*

$$\frac{d^{n-m}}{dt^{n-m}} y(0^+) = \frac{\eta_m}{\delta_n} \quad (2.78)$$

1187 *Proof:* Left as an exercise to the reader. Hint: Use the initial value theorem and (2.76). ■

The point is that where a discontinuity can be observed, the difference in degree can be determined empirically. In practice, only discontinuities in y and dy/dt can be perceived easily so that only small differences in n and m are inferred in this way. Refer to Example 2.2.

In particular, where the transfer function is proper, $n = m$,

$$y(0^+) = \frac{\eta_n}{\delta_n} \quad (2.79)$$

In this case, a discontinuity occurs in the output.

Some systems display reverse action, which means that the initial response is in the direction opposite to that of the steady-state response. If a system is stable and strictly proper, and if $\eta_0 \geq 0$ the system displays this unusual behavior only where an odd number of zeros of the transfer function are in the right half-plane. This assertion is an immediate consequence of (2.79) because it is this case where

$$\eta_m < 0 \quad (2.80)$$

Generally, we have reverse action only where the system is non-minimum phase. Example 2.12 shows the step response of a system with non-minimum phase zero.

2.1.7 Final Values and Static Gain

The theories of automatic control laid down in this text are strongly related to the way in which systems magnify signals. The quantification of this magnification is the **gain** of the system. We begin the study of the simplest form of gain in this subsection.

Knowing the Final Value Theorem and $G(s)$ in (2.77), we can write The final value of a response to a step-input of magnitude u_0

$$y(\infty) = \frac{\eta_0}{\delta_0} u_0 \quad (2.81)$$

or

$$y(\infty) = G(0)u_0 \quad (2.82)$$

Definition 2.5. A system is said to be of type l if δ_l is the denominator coefficient of lowest degree which is nonzero. The ratio

$$K = \frac{\eta_0}{\delta_l} \quad (2.83)$$

is called the *error constant*. If $l = 0$, K is called **the static gain**, if $l = 1$, K is called **the velocity error coefficient**, and if $l = 2$, K is called **the acceleration error coefficient**.

1213 This is consistent with (2.81) since for a type-0 system and the two Figures below, we can
1214 write

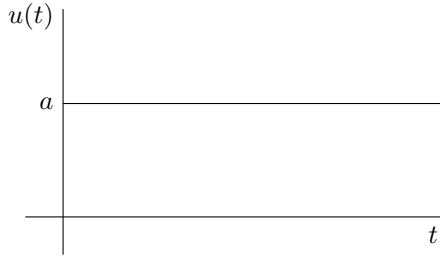


Figure 2.9: Step input.

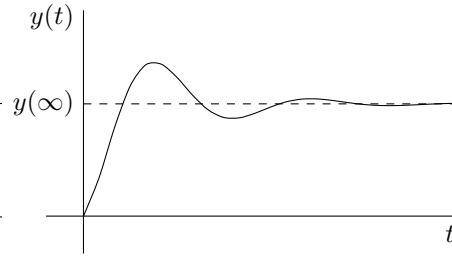


Figure 2.10: Step Response.

$$\hat{y}(s) = G(s)\hat{u}(s), \quad \hat{u}(s) = \frac{a}{s} \quad (2.84)$$

$$y(\infty) = \lim_{s \rightarrow 0} s\hat{y}(s) = \lim_{s \rightarrow 0} sG(s)\frac{a}{s} = a \cdot G(0) \quad (2.85)$$

$$K = \frac{y(\infty)}{a} = G(0) \quad (2.86)$$

EXAMPLE 2.12

Static gain for minimum and non-minimum phase transfer functions.

$$\text{Minimum Phase: } G(s) = \frac{s+1}{s^2+s+1} \Rightarrow G(0) = 1$$

$$\text{Non-minimum Phase: } G(s) = \frac{1-s}{s^2+s+1} \Rightarrow G(0) = 1$$

$$\text{Non-minimum Phase: } G(s) = \frac{s-1}{s^2+s+1} \Rightarrow G(0) = -1$$

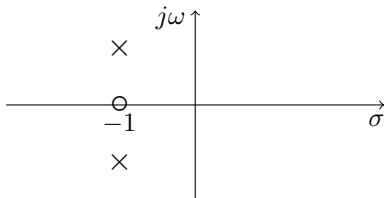


Figure 2.11: Minimum phase.

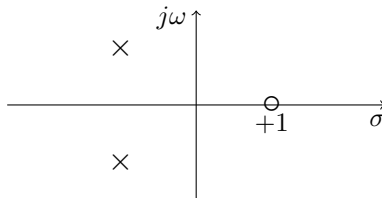


Figure 2.12: Non-minimum phase.

Consider a unit step ($a = 1$).

$G(s)$	$\frac{s+1}{s^2+s+1}$	$\frac{1-s}{s^2+s+1}$	$\frac{s-1}{s^2+s+1}$
$y(\infty)$	+1	+1	-1
$y(0^+)$	0	0	0
$\dot{y}(0^+)$	1	-1	1

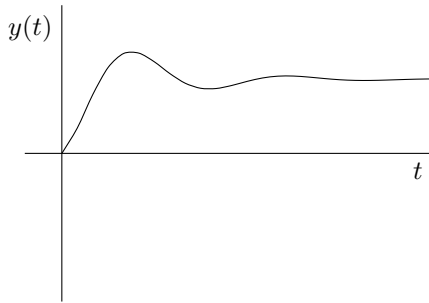


Figure 2.13: Minimum phase response.

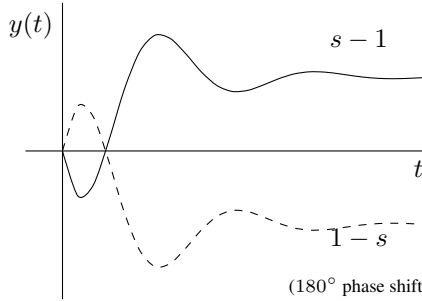


Figure 2.14: Nonminimum phase response.

Non-minimum phase transfer functions experience reverse action: the response initially moves *away* from the steady-state value. Negative static gain indicates a negative response.

2.1.8 Time Response Metrics

The performance of a dynamic system is often measured with time-domain response metrics. The response of first- and second-order systems are described in this section. At first, it will seem that too much effort is expended in this way. However, where we lay down the details of feedback design by “loop shaping”, it is clear that the developments in this section are indispensable.

1st-Order System (Single-Pole Response) Consider the transfer function $G(s)$, where K is the static gain and τ is the transfer function time constant.

$$G(s) = \frac{K}{\tau s + 1} \Rightarrow \text{Static Gain } G(0) = K \quad (2.87)$$

The pole-zero diagram for $G(s)$ is shown in Figure 2.15

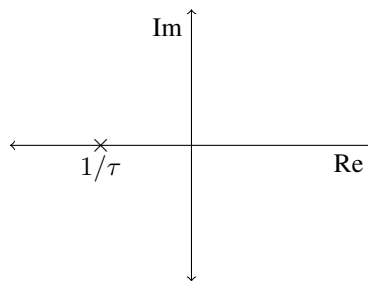


Figure 2.15: Pole-zero diagram of a 1st order system.

The system's step response can be found as follows.

$$\hat{y}(s) = G(s) \frac{1}{s} = \frac{K}{s(\tau s + 1)} \quad (2.88)$$

It is easy to verify that if the initial condition is zero, the time response to a unit step input is

$$y(t) = K(1 - e^{-t/\tau}) \quad (2.89)$$

This classic response is illustrated in Figure 2.16. Notice the graphical interpretations of the time constant, τ . The initial value of the slope of this response is,

$$\frac{dy}{dt}(t) = \frac{K}{\tau} e^{-t/\tau} \Rightarrow \frac{dy}{dt}(0) = \frac{K}{\tau} \quad (2.90)$$

$$y(\tau) = K(1 - e^{-1}) \approx 0.632 \cdot K \quad (2.91)$$

So, the step response will have an initial slope K/τ , a final value of K , and will rise to 63.2% of the final value at $t = \tau$.

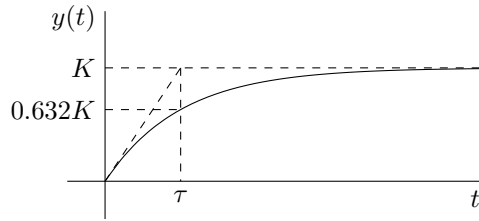


Figure 2.16: Single pole step response.

2nd-Order System (Quadratic Factor) The step response of the system under study is completely described by a static gain and a pair of complex-conjugate poles. We make this study because this transfer function is, surprisingly, a useful approximation of many closed-loop systems.

Consider the second-order transfer function $G(s)$, where K is the static gain, ω_n is the undamped natural frequency, σ is the damping factor, ζ is the damping ratio, and ω is the frequency, and where ω_n , σ , ζ , and ω are related according to

$$\omega = \omega_n \sqrt{1 - \zeta^2}. \quad (2.92)$$

and

$$\sigma = \zeta \omega_n. \quad (2.93)$$

Then,

$$G(s) = \frac{K\omega_n^2}{(s + \sigma)^2 + \omega^2} = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \Rightarrow \text{Static Gain } G(0) = K \quad (2.94)$$

In this case $\zeta < 1$, so that the poles are complex conjugates. The location of these poles in the s -domain is shown in Figure 2.17. Otherwise, the $G(s)$ could be split into two first-order poles and the solution would follow the process for first-order systems. The unit step response can be found as follows.

$$\hat{y}(s) = G(s) \frac{1}{s} = \frac{K\omega_n^2}{s((s + \zeta\omega_n)^2 + \omega^2)} \quad (2.95)$$

$$= \frac{R_1}{s} + \frac{R_2s + R_3}{(s + \zeta\omega_n)^2 + \omega^2} \quad (2.96)$$

(Partial fraction expansion steps not shown here.)

$$y(t) = K \left(1 - e^{-\zeta\omega_n t} \left(\cos(\omega t) + \frac{\zeta}{\sqrt{1 - \zeta^2}} \sin(\omega t) \right) \right) \quad (2.97)$$

1252 From Figure 2.17, we can see that there is a geometric relationship between the natural
1253 frequency ω_n and the damping factor σ .

$$\cos \theta = \frac{\zeta\omega_n}{\omega_n} \Rightarrow \zeta = \cos \theta \quad (2.98)$$

1254 Similarly,

$$\sin \theta = \frac{\sqrt{\omega_n^2 - \omega^2}}{\omega_n} = \frac{\sqrt{\omega_n^2 - \omega_n^2(1 - \zeta^2)}}{\omega_n} \Rightarrow \sqrt{1 - \zeta^2} = \sin \theta \quad (2.99)$$

1255 Then, Equation (2.97) can be rewritten as

$$y(t) = K \left(1 - \frac{e^{-\zeta\omega_n t}}{\sin(\theta)} (\cos(\omega t) \sin \theta + \sin(\omega t) \cos \theta) \right), \quad (2.100)$$

1256 which can be further simplified to

$$y(t) = K \left(1 - \frac{e^{-\zeta\omega_n t}}{\sin(\theta)} \sin(\omega t + \theta) \right). \quad (2.101)$$

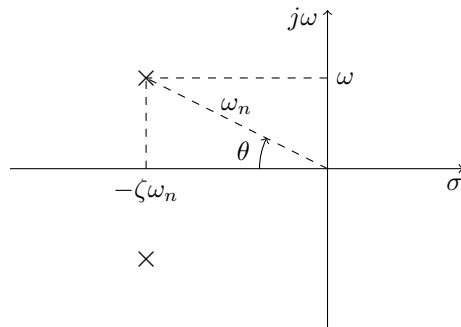


Figure 2.17: 2nd-order system s -domain.

We characterize this response in Figure 2.18. By that, we mean that figures of merit of the speed of response and the damping are defined in this figure. Both “rise time” and “peak time” are measures of the speed at which response first approaches the steady-state value. The rise time, t_R , is the time at which $y(t) = K$ (that is, the time for $y(t)$ to first reach the steady-state value), occurring at

$$\omega t_R + \theta = \pi$$

so that

$$t_R = \frac{\pi - \theta}{\omega_n \sqrt{1 - \zeta^2}} \quad (2.102)$$

A formula for the peak time, t_p , is obtained by setting $dy/dt = 0$:

$$t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} \quad (2.103)$$

Where the quadratic factor is used to approximate a closed-loop system, we are interested in the degree to which the response overshoots the mark. In many cases, we hope to control this value with our choices of design parameters. The overshoot occurs at $t = t_p$, so it is easy to find that the percent overshoot to be

$$100 \frac{m_1}{K} = 100 \exp \left[\frac{-\zeta \pi}{\sqrt{1 - \zeta^2}} \right] \quad (2.104)$$

The last three relations are plotted in Figure 2.18. Notice a curious fact, *the percent overshoot depends only upon the damping ratio* (alternatively, upon the pole angle).

A measure of the damping is the ratio of successive maxima, a simple analysis leads to

$$\frac{m_1}{m_2} = \exp \left[\frac{-2\zeta \pi}{\sqrt{1 - \zeta^2}} \right] = \frac{m_{k+1}}{m_k} \quad (2.105)$$

The similarity of the right sides of Equations (2.104) and (2.105) is obvious, in fact,

$$\frac{m_1}{K} = \sqrt{\frac{m_{k+1}}{m_k}}$$

Once again, a performance measure depends only upon the damping ratio.

Finally, another measure of the damping is the rapidity with which the exponential envelopes of the response decay. That decay occurs with the time constant

$$\begin{aligned} \tau &= 1/\sigma \\ &= 1/(\zeta \omega_n) \end{aligned} \quad (2.106)$$

One final step-response metric for 2nd-order systems is:

- **Settling Time:** The time at which the deviation of the response from the final value is bounded by $\pm 2\%$. Alternate definitions of the settling time may restrict or relax the error bound, e.g. a $\pm 5\%$ bound could be used in place of a $\pm 2\%$ bound.

These metrics are illustrated in Figure 2.18.

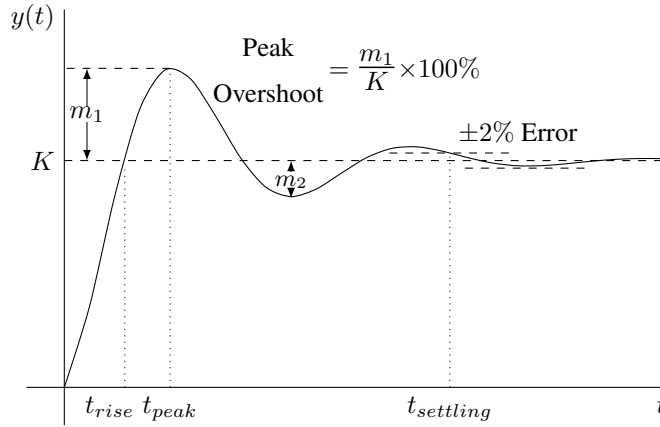


Figure 2.18: Quadratic pole step response.

2.1.9 The Effect of Zeros on Transient Response

The effect of an individual, transfer function pole on the transient response of a system is made clear in the inverse Laplace transformation by Heaviside expansion in the previous section: Each pole contributes an exponential decay term. The effect of a zero is more subtle. Obviously, the direct effect is on the residues, R_k . However, the indirect effect on the transient response can be profound. We study one example for the effect of a zero in this section.

Incidentally, the entire pattern of pole-zero location affects the value of an individual residue. For this reason a particular pattern will provide a response unanticipated if individual poles are examined in isolation.

Consider the system described by the transfer function

$$G(s) = \frac{K\omega_n}{\rho\zeta} \frac{s + \rho\zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (2.107)$$

Essentially, this is a quadratic factor with a zero.

Consider the step response of this system. In particular, consider the calculation of residues. The numerator contributes a multiplicative numerator term of the form $(-p_1 + \rho\zeta\omega_n)$ to a residue of the transient response. Here, " p_1 " is a complex pole of form shown

1295 in Figure 2.17. This same factor can be written in the polar form $(Pe^{j\beta})$ where P and β
 1296 are given the geometric interpretation in Figure 2.19 below.

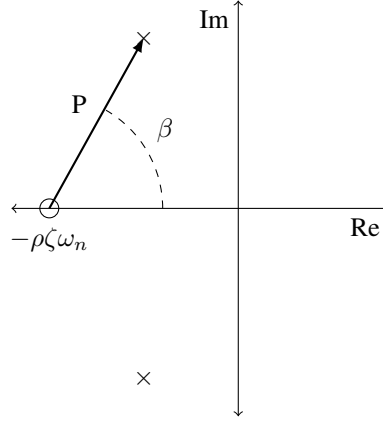


Figure 2.19: Polar form of the factor.

1296
 1297 Where one goes through the details, it is found that the zero in (2.107) modifies equation
 1298 (2.101) to become

$$\frac{y(t)}{K} = 1 - \frac{e^{-\sigma t}}{\sin(\theta + \beta)} \sin(\omega_d t + \theta + \beta) \quad (2.108)$$

1299 where

$$\beta = \tan^{-1} \left[\frac{\sqrt{1 - \zeta^2}}{|\rho - 1|\zeta} \right] \quad (2.109)$$

1300 This step response is plotted in Figure 2.20 below for various values of ρ but fixed value of
 1301 ζ . This corresponds to holding the poles fixed and moving the location of the zero.

1302 Clearly, the response can be sped-up by moving the location of the zero. That is, *the*
 1303 *rise and peak times* can be favorably influenced in this way. However, some things cannot
 1304 be influenced. For instance the frequency, time constant (2.106), or the damping-per-cycle
 1305 (2.105) are independent of the zero location. It is obvious from this Figure that the price
 1306 one pays for smaller peak or rise time is an increased overshoot. These facts are summa-
 1307 rized in the formulas

$$t_R = \frac{\pi - \theta - \beta}{\omega_n \sqrt{1 - \zeta^2}} \quad (2.110)$$

$$t_p = \frac{\pi - \beta}{\omega_n \sqrt{1 - \zeta^2}} \quad (2.111)$$

$$100 \frac{m_1}{K} = 100 \exp \left[\frac{-\zeta(\pi - \beta)}{\sqrt{1 - \zeta^2}} \right] \quad (2.112)$$

1310 Time constant and damping-per-cycle are not the only things which are independent of the
 1311 zero location. It is clear from Figure 2.20 that values of the response at a discrete set of

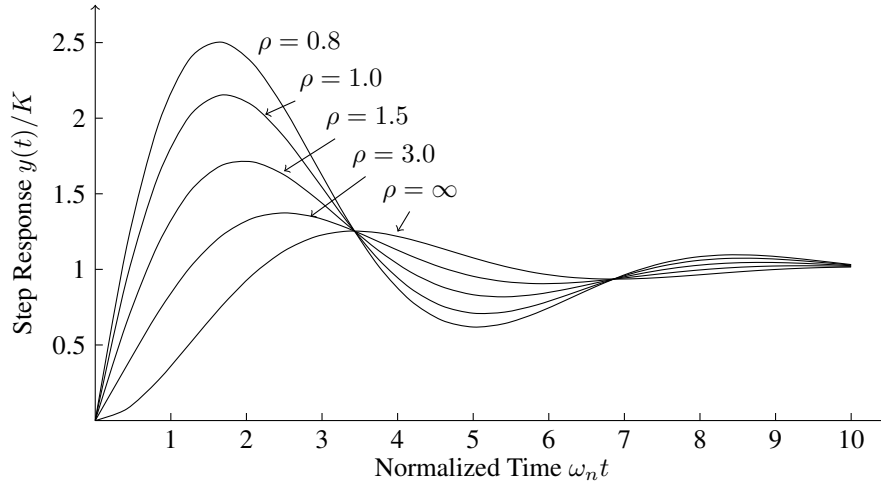


Figure 2.20: Effect of a zero on the 2nd-order response ($\zeta = 0.4$).

times are invariant to the zero location! This fact is established in the next theorem. In this theorem, the limiting case of $\rho \rightarrow \infty$ corresponds to (2.101).

Theorem 2.5 For the system characterized by a quadratic factor and a single zero, the values of the unit-step response at the times

$$t_k = \frac{k\pi}{\omega_n \sqrt{1 - \zeta^2}}, \quad k = 0, 1, 2, 3, \dots \quad (2.113)$$

are invariant, are independent of the zero location as characterized by ρ . In the limiting case of $\rho \rightarrow \infty$, these invariants are the relative maxima and minima of the response.

Proof: Substitute (2.113) into (2.108) to obtain the values

$$y(t_k) = K[1 + (-1)^{k+1} \exp(-\sigma t_k)] \quad (2.114)$$

which are independent of β . ■

Comment: Of course, the response invariants are not peak times for finite ρ .

2.1.10 The Butterworth Pattern

We continue our study of the effect of pattern on response with a study of a famous pole configuration, the *Butterworth pattern*. It is surprising how often this combination recurs in the control theory literature.

Definition 2.6. The *Butterworth* transfer-function is a transfer function with unity gain and with poles in the *Butterworth pattern*, which is a set of poles situated in the left

half-plane on a circle of radius ω_n and spaced at angles

$$\pm l \frac{\pi}{n}, \quad l = 0, 1, \dots, \frac{n-1}{2}$$

if n is odd and

$$\pm \frac{(l+1/2)\pi}{n}, \quad l = 0, 1, \dots, \frac{n}{2} - 1$$

if n is even. These angles are measured from the negative real axis.

Because of the importance of this pattern in control system design, we list the transfer functions for several values of n .

Table 2.2: The denominator of the k^{th} -order Butterworth transfer function, $k \leq 6$. (The numerator of the k^{th} -order Butterworth transfer function is ω_n^k .)

k	denominator
k=1	$s + \omega_n$
k=2	$s^2 + \sqrt{2}\omega_n s + \omega_n^2$
k=3	$s^3 + 2\omega_n s^2 + 2\omega_n^2 s + \omega_n^3$
k=4	$s^4 + 2.6\omega_n s^3 + 3.4\omega_n^2 s^2 + 2.6\omega_n^3 s + \omega_n^4$
k=5	$s^5 + 3.24\omega_n s^4 + 5.24\omega_n^2 s^3 + 5.24\omega_n^3 s^2 + 3.24\omega_n^4 s + \omega_n^5$
k=6	$s^6 + 3.86\omega_n s^5 + 7.46\omega_n^2 s^4 + 9.13\omega_n^3 s^3 + 7.46\omega_n^4 s^2 + 3.86\omega_n^5 s + \omega_n^6$

The step responses associated with *Butterworth* transfer function are illustrated in Figure 2.21 below for the first few values of n . These responses are characterized by fair speed-of-response with good damping-per-cycle and little overshoot.

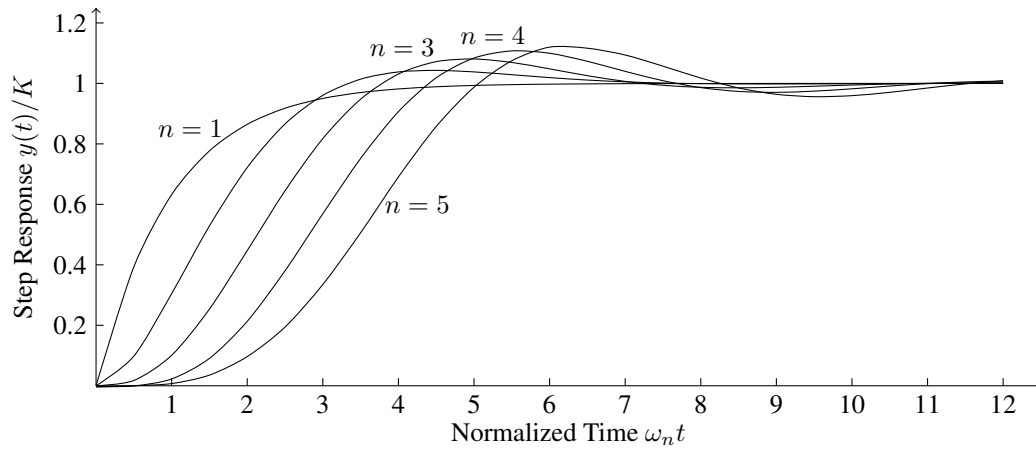


Figure 2.21: Step response of filters in the Butterworth pattern.

2.2 Frequency Response of Dynamic Systems

For the purposes of feedback control, it is crucial to recall the theory and the representations of frequency response. For one thing, this theory provides the generalization of gain, a generalization crucial to feedback design.

2.2.1 Steady-State Frequency Response of LTI systems

Suppose a stable dynamic system is subjected to a sinusoidal input.

$$u(t) = A \cos(\omega t) 1(t) \quad (2.115)$$

The system will experience both a transient and steady-state response, as shown in Figure 2.22.

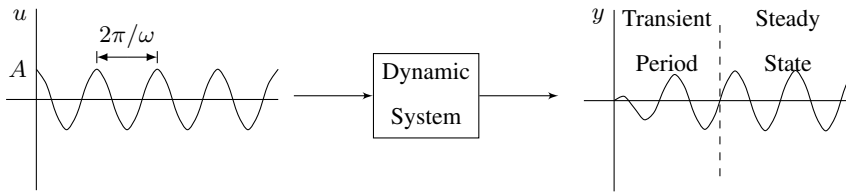


Figure 2.22: Frequency Response

We can determine the transient and steady-state responses analytically by writing the Laplace transform of the input as

$$\hat{u}(s) = \frac{As}{s^2 + \omega^2} = \frac{As}{(s + j\omega)(s - j\omega)} \quad (2.116)$$

It follows that the output is

$$\hat{y}(s) = \frac{As}{(s + j\omega)(s - j\omega)} G(s) \quad (2.117)$$

The response is a sum of a transient term and a steady-state response. We study the steady-state response, that portion of the total response which is due to the poles of the input, $\hat{u}(s)$.

The Heaviside expansion of this steady state response is

$$\hat{y}_{ss} = \frac{A}{2} \frac{G(-j\omega)}{s + j\omega} + \frac{A}{2} \frac{G(j\omega)}{s - j\omega} \quad (2.118)$$

Next suppose that $G(s)$ is a simple zero such as

$$G(s) = s + \alpha \Rightarrow G(j\omega) = \alpha + j\omega, \quad G(-j\omega) = \alpha - j\omega \quad (2.119)$$

Also, if $G(s)$ is a simple pole, we can write

$$G(s) = \frac{1}{s + \alpha} \Rightarrow G(j\omega) = \frac{1}{\alpha + j\omega}, \quad G(-j\omega) = \frac{1}{\alpha - j\omega} \quad (2.120)$$

In general, $G(-j\omega)$ is equal to the complex conjugate of $G(j\omega)$:

$$G(-j\omega) = \bar{G}(j\omega) \quad (2.121)$$

This result is perfectly general for any rational function since the conjugate of the product is the product of the conjugates. It is left as an exercise to prove that the result also applies where a dead-time factor is included in $G(s)$. So, in steady state,

$$\hat{y}_{ss} = \frac{A}{2} \left[\frac{\bar{G}(j\omega)}{s + j\omega} + \frac{G(j\omega)}{s - j\omega} \right] \quad (2.122)$$

We can consider $G(j\omega)$ to change the magnitude of the input by a factor of M and to shift the phase of the input by a phase angle ϕ . $G(j\omega)$ and its conjugate can then be written in polar form:

$$G(j\omega) = M e^{j\phi} \quad (2.123)$$

$$\bar{G}(j\omega) = M e^{-j\phi} \quad (2.124)$$

Note that $M = M(\omega)$ and $\phi = \phi(\omega)$,

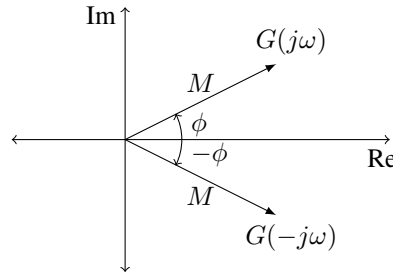


Figure 2.23: Frequency response magnitude and phase.

It follows that the inverse transform of the steady-state response is

$$y_{ss}(t) = MA \left(\frac{e^{-j\omega t} \cdot e^{-j\phi} + e^{j\omega t} \cdot e^{j\phi}}{2} \right) \quad (2.125)$$

That is,

$$y_{ss}(t) = M(\omega)A \cos(\omega t + \phi(\omega)) \quad (2.126)$$

For a frequency input with amplitude A , the steady state frequency response has an amplitude of MA (where $M(\omega)$ is the gain in the amplitude) and a phase shift $\phi(\omega)$. This is illustrated in Figure 2.24.

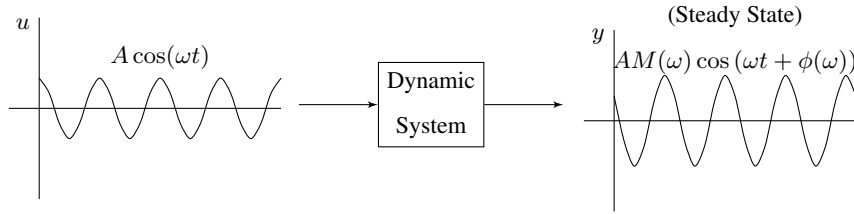


Figure 2.24: Frequency Response

Definition 2.7. The modulus of the transfer function evaluated on the imaginary axis, $M(\omega) = |G(j\omega)|$, is the gain and the argument of $G(j\omega)$ is the phase of the steady-state system response. $G(j\omega)$ for $0 \leq \omega \leq \infty$ is called the frequency response of the system.

What we call gain is called “amplitude ratio” in some texts. Our usage is consistent with our definition of “gain” in the previous section because in the limit $\omega \rightarrow 0$, the input to Equation (2.115) is the unit step.

There is no logical way to fix the range of the phase angle although there are conventions that are associated with certain specific types of transfer functions. In fact, there are an infinite number of equally-valid copies of the phase-angle variation each starting at a location 2π radians away from the next. We say the copies are *modulo* 2π . It is very useful to remember this fact as one tries to decipher simulation outputs and to interpret stability criteria. There is no reason to assign any more significance to one copy or another but one thing is not allowed: One cannot skip from one copy to another, one must be consistent. Unfortunately, where discontinuities occur in phase (at complex poles or zeros on the imaginary axis) computer programs can switch from one copy to another and that is an error. Simply put *be very careful of the use of software to obtain the phase angle where there are pole or zero pairs on the imaginary axis!* The frequency response of a system is so fundamental to feedback design that all computer aids to control design provide for the calculation and the plotting of $G(j\omega)$. There are a variety of useful graphical representations.

2.2.2 Frequency Response Representation:

We consider three ways of graphically representing the frequency response:

1. The graphs of $20 \log_{10}(M)$ and ϕ vs. $\log_{10} \omega$ is called the **Bode diagram**.

2. The graph of $20 \log_{10}(M)$ vs $\phi(\omega)$, with ω as parameter, is called the **magnitude-phase plot**. A magnitude-phase plot with two sets of certain contour plots is called a **Nichols Chart**.

3. The graph of $\text{Imag}[G(j\omega)]$ vs $\text{Real}[G(j\omega)]$ over the positive frequency range is called the **polar plot**. The same type of plot but versus all “frequencies” (including negative values of ω) is called a **Nyquist plot**.

Denote

$$\text{Lm } G(j\omega) \triangleq 20 \log_{10}[M(\omega)] \quad (2.127)$$

and read this as the *log-magnitude* of the gain. The units of log-magnitude are called *decibels* or, more simply, dB.

The discussion in this section concludes with the detailed study of frequency response of certain canonical transfer functions.

2.2.3 Frequency Response: The Real Pole

Begin with the transfer function

$$G(s) = \frac{1}{\tau s + 1} \quad (2.128)$$

So that the frequency response is

$$G(j\omega) = \frac{1}{1 + j\tau\omega} \quad (2.129)$$

It is a simple matter to demonstrate that the gain

$$M(\omega) = \frac{1}{\sqrt{1 + \tau^2\omega^2}} \quad (2.130)$$

while the phase is

$$\begin{aligned} \phi(\omega) &= \tan^{-1}(\Im G(j\omega)/\Re G(j\omega)) \\ &= -\tan^{-1}\left(\frac{\tau\omega}{+1}\right) \\ &= -\tan^{-1}(\tau\omega) \end{aligned} \quad (2.131)$$

These relations are useful for estimating the range of gain and phase as ω is varied from 0 to ∞ . For the present case, it is clear that, over this frequency range, ϕ varies from 0 to -90° while the gain varies from 1.0 (0 dB) to 0. Detailed results are obtained from standard computer software while a qualitative understanding is gained from the study of relations (2.130, 2.131).

It is useful to analyze the asymptotic properties of these equations by looking at the gain response on a logarithmic scale. The log of Equation (2.130) is

$$\begin{aligned}\text{Lm } M(\omega) &= 20 \log_{10} M(\omega) = 20 \log_{10}(1) - 20 \log_{10} \left(\sqrt{1 + \tau^2 \omega^2} \right) \\ &= -10 \log_{10}(1 + \tau^2 \omega^2)\end{aligned}\quad (2.132)$$

1410 so that, asymptotically at low frequencies, at $\omega \ll 1/\tau$,

$$\text{Lm } M \approx 0$$

1411 Similarly, where $\omega \gg 1/\tau$,

$$\text{Lm } M \approx -\text{Lm } (\tau) - \text{Lm } (\omega)$$

1412 The low-frequency asymptote is a straight line on the Bode diagram with a slope of 0
1413 dB/decade while the high frequency asymptote has a slope of -20 dB/decade. Denote the
1414 break frequency

$$\omega_b = 1/\tau \quad (2.133)$$

1415 so-called because the straight-line asymptotes of the log-magnitude curve appear to “break”
1416 at this frequency: The asymptotes intersect at this break frequency. Notice that the phase
1417 at the break frequency,

$$\phi(\omega_b) = -45^\circ \quad (2.134)$$

1418 Finally, we notice that at the break, the exact value of the gain is $M = 1/\sqrt{2}$ or

$$\text{Lm } M = -3 \text{ dB} \quad (2.135)$$

1419 This -3 dB correction is the maximum deviation of the gain from the straight-line asymptotes.
1420

1421 The Bode diagrams of a single pole are shown in Figures 2.25 and 2.26. An unstable
1422 pole will produce the same decrease in slope as a stable pole. However, it will cause a 90°
1423 *increase* in phase, rather than decrease.

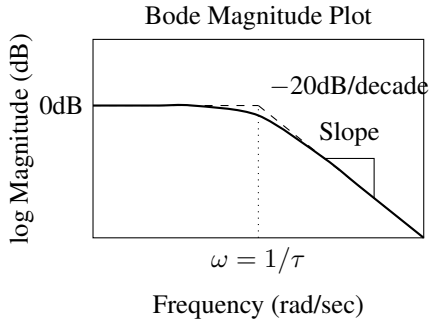


Figure 2.25: Bode magnitude plot, single pole.

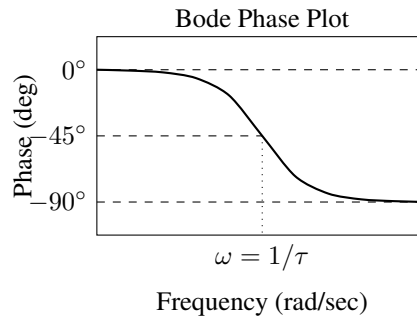


Figure 2.26: Bode phase plot, single pole.

2.2.4 Frequency Response: The Real Zero:

The transfer function

$$G(s) = \tau s + 1 \quad (2.136)$$

receives attention at this point, the case of a single real zero. Clearly, the frequency response of the zero is closely related to that of the pole. An important difference is that the zero is a numerator term so an opposite sign is applied to the argument: such that, the phase ranges from 0° to $+90^\circ$, a *phase advance*.

There is no violation in some principle of causality in this *phase advance* or *phase lead*, as it is called. Remember this is a steady-state frequency response and the transient build-up to the steady-state is such that there is no anticipatory action on the part of the system. Also, remember that for any version of a phase curve, there are equally valid copies at multiples of $\pm 2\pi$, including copies where are interpreted as *phase lag*. It may be that we are misled by terms like “lag” or “lead” because these phase curves are steady-state phase relations and so it is only *phase difference* that can be interpreted in any meaningful way.

Assume, at first, that the zero is in the left half-plane such that $s = -1/\tau$. Then for a frequency input $s = j\omega$, we have

$$G(j\omega) = 1 + j\tau\omega \quad (2.137)$$

Using the analogy with the pole to quickly outline the main features of the zero frequency response,

$$M(\omega) = \sqrt{1 + \tau^2\omega^2} \quad (2.138)$$

Thus,²

$$\phi(\omega) = \tan^{-1} \left(\frac{\tau\omega}{1} \right) \quad (2.139)$$

So one valid copy of the phase curve is in the first quadrant:

$$\phi(\omega) = (0 \leq \phi \leq 90^\circ) \quad (2.140)$$

1439 Additionally, it is conventional to look at the magnitude response on a logarithmic scale.

1440 The log of Equation (2.138) is

$$\text{Lm } M(\omega) = 20 \log_{10} \left(\sqrt{1 + \tau^2 \omega^2} \right) \quad (2.141)$$

1441 The magnitude formula is graphed in Figure 2.27, and the phase formula is graphed in

1442 Figure 2.28. In Figure 2.28, a left half-plane zero is referred to as *minimum phase* while a

1443 zero in the right-half plane is called *nonminimum phase*.

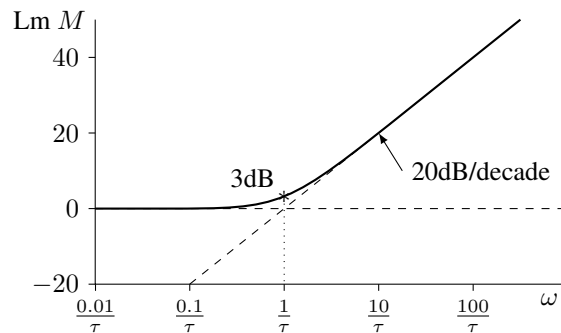


Figure 2.27: Magnitude response for a zero.

²A good habit is to write the arctan function as a function with two arguments, a numerator and a denominator.

To not do so can lead to error in the composition of a valid copy of the phase curve.

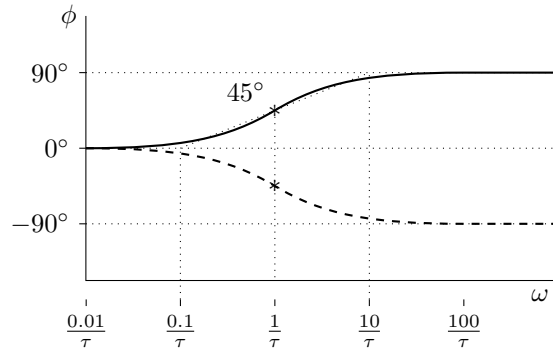


Figure 2.28: Minimum (solid) and non-minimum (dashed) phase response for a zero.

The *non-minimum-phase zero*, the zero in the right half-plane has a crucially different frequency response. This is also illustrated in Figure 2.28, while Figure 2.29 shows the zero locations in the s -plane. The difference is in the phase curves: First of all, notice that the *positive gain* version of the right half-plane zero transfer function is

$$G(s) = -\tau s + 1 \quad (2.142)$$

Thus, in the case of the non-minimum phase zero

$$\phi(\omega) = \tan^{-1} \left(\frac{-\tau\omega}{+1} \right) \quad (2.143)$$

So that the valid copy of the phase curve is in the *fourth quadrant*:

$$\phi(\omega) = (-90^\circ \leq \phi \leq 0) \quad (2.144)$$

Thus, the phase for the minimum phase zero is less *negative* than that for the non-minimum phase zero. It is for this reason that the terms minimum and non-minimum phase are used. (Clearly, all this a bit of a *misnomer* since a more precise terminology would be “minimum phase-lag zero”.)

The gain curves are the same in the two cases, the slope of the log-magnitude curve is 0 dB per decade up to the break frequency, $\omega_b = 1/\tau$, after which the slope increases to +20 dB per decade. The gain correction at this “corner” frequency is +3 dB and the phase is $+45^\circ$ for minimum phase and -45° for non-minimum phase.

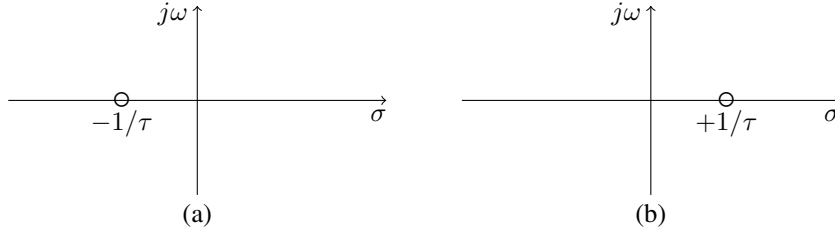


Figure 2.29: (a) Minimum and (b) non-minimum phase zeros.

Incidentally, where we write a right half-plane zero as

$$G(j\omega) = -1 + j\tau\omega$$

we have

$$\phi(\omega) = \tan^{-1} \left(\frac{+\tau\omega}{-1} \right) \quad (2.145)$$

So one valid copy of the phase curve is in the third quadrant:

$$\phi(\omega) = (90^\circ \leq \phi \leq 180^\circ) \quad (2.146)$$

However, this is the negative-gain version of the transfer function.

2.2.5 Frequency Reponse: The Quadratic Factor:

The quadratic factor of (2.147) is a polynomial factor of many physical transfer-functions.

The detailed study of the quadratic factor is so important because of its usefulness as an approximation for many closed-loop systems. The frequency response serves as both the explanation and the formulatin of this approximation.

Consider a transfer function that is a second-order complex pair of poles,

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (2.147)$$

now substitute $s = j\omega$,

$$G(j\omega) = \frac{\omega_n^2}{\omega_n^2 - \omega^2 + 2\zeta\omega_n j\omega} \quad (2.148)$$

where ω_n is the natural frequency and ζ is the damping ratio as before. Then,

$$M(\omega) = \frac{\omega_n^2}{\sqrt{(\omega_n^2 - \omega^2)^2 + 4\zeta^2\omega_n^2\omega^2}} \quad (2.149)$$

$$(2.150)$$

and the phase

$$\phi(\omega) = \tan^{-1} \left(\frac{2\zeta\omega_n\omega}{\omega_n^2 - \omega^2} \right) \quad (2.151)$$

The frequency response of the quadratic factor is illustrated in Figure 2.30 for several different values of pole angle, $\theta = \cos^{-1}\zeta$. Notice the *resonance*, the local maximum of the gain curve. We investigate this effect by *maximizing* M which is the same as minimizing the term under the radical of the denominator of (2.149). Set the derivative of this term, with respect to ω , to zero in order to obtain a formula for the *resonant frequency*, ω_r :

$$-4\omega_r(\omega_n^2 - \omega_r^2) + 8\zeta^2\omega_n^2\omega_r = 0$$

1465 The non-zero solution of this equation is

$$\omega_r = \omega_n \sqrt{1 - 2\zeta^2} \quad (2.152)$$

This frequency is real (resonance actually occurs) only if

$$\zeta \leq \frac{1}{\sqrt{2}} \quad \text{or} \quad \theta \geq 45^\circ \quad (2.153)$$

1466 There is an indication of this fact in Figure 2.30. Notice that resonant frequency is neither
1467 the damped natural frequency, ω_d , nor the undamped natural frequency, ω_n . Where the
1468 resonant frequency is substituted back into Equation (2.149), the result is the value of the
1469 local gain-maximum, the resonant peak

$$M_p = M(\omega_r) = \frac{1}{2\zeta\sqrt{1 - \zeta^2}} \quad (2.154)$$

1470 This relation is plotted in Figure 2.30 and is fundamental to the approximation of closed-
1471 loop systems with a quadratic factor. This idea is developed in the next chapter. In contrast,
1472 we have

$$M_{\omega_n} = \frac{1}{2\zeta} \quad (2.155)$$

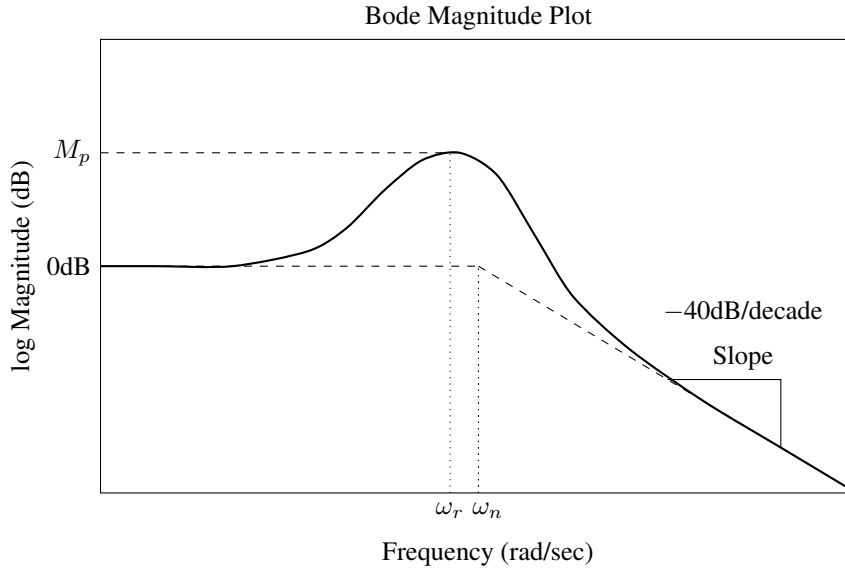


Figure 2.30: Bode magnitude plot, second-order pole.

It is left as an exercise to show that the break frequency is

$$\omega_b = \omega_n$$

The phase is -90° at this frequency and ranges between the asymptotic values of 0° and -180° . The slope of the low-frequency gain-asymptote is 0 dB/decade while that of the high frequency asymptote is -40 dB/decade. When these asymptotes are taken together with (2.155), we have one quick procedure for an empirical determination of the damping ratio from the frequency response curve: *Measure the magnification at a phase angle of -90° and take the damping ratio to be one-half the reciprocal of this magnification.*

Definition 2.8. The cross-over frequency, ω_c is that frequency where the modulus is 1 of the return ratio; that is

$$\text{Lm } G(j\omega_c) = 0 \quad \text{dB} \quad (2.156)$$

It is left as exercise to show that, where the static gain is equal to 1.0, the cross-over frequency for the quadratic factor is given by

$$\omega_c = \sqrt{2}\omega_r \quad (2.157)$$

In general, the cross-over point is varied by simply varying the static gain.

2.2.6 Frequency Response: Pure Time Delay:

Consider a dynamic system which represents a pure time delay (also known as dead-time), as shown in Figure 2.31.



Figure 2.31: Block diagram of time delay.

The impulse response of this system can be written as,

$$g(t) = \delta(t - T_d) \quad (2.158)$$

so that

$$y(t) = g(t) * u(t) \quad (2.159)$$

then using Laplace transform,

$$\hat{y}(s) = \mathcal{L}[y(t)] = \underbrace{e^{-sT_d}}_{G(s)} \hat{u}(s) \quad (2.160)$$

Substituting $j\omega$ for s ,

$$G(j\omega) = e^{-j\omega T_d} = \cos(\omega T_d) - j \sin(\omega T_d) \quad (2.161)$$

so that the gain and phase are

$$M(\omega) = 1 \quad (2.162)$$

$$\phi = \tan^{-1} \left(\frac{-\sin(\omega T_d)}{\cos(\omega T_d)} \right) = -\tan^{-1}(\tan(\omega T_d)) = -T_d \omega \quad (2.163)$$

Pure time delay is an all-pass filter: the transfer function has a gain of 1 (or 0dB) for all frequencies. The phase angle decreases linearly as a function of frequency (appearing exponential on a semilog plot). The main feature here is the continually decreasing phase (the increasing phase lag).

We demonstrate in the next chapter that excessive phase lag is the cause of instability in feedback control systems. It is for that reason that nonminimum phase zeros and dead-time are primary sources of challenge for the control engineer. Think of things this way: Poles contribute phase lag but at a decreasing amplitude ratio (tolerable), dead-time contributes phase lag at a constant amplitude ratio (bad), while non-minimum phase zeros contribute phase lag at an increasing amplitude ratio (the worst). Minimum phase zeros contribute phase advance (good) at an increasing amplitude ratio.

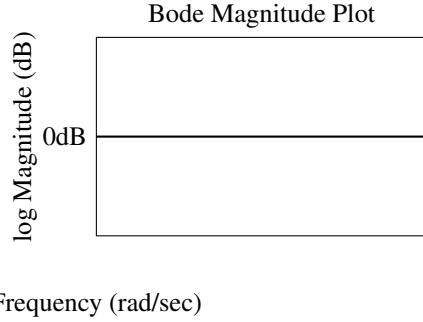


Figure 2.32: Bode magnitude plot, time delay.

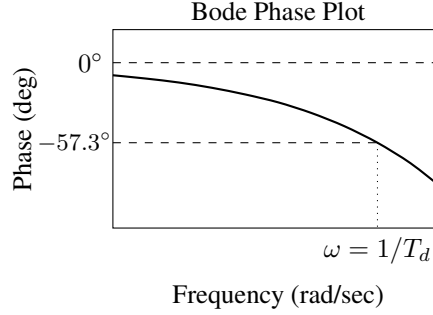


Figure 2.33: Bode phase plot, time delay.

Considering (2.161), dead-time is a nonlinear and irrational function. However, for control system design of LTI dynamic system, the dead-time is required to be linearized and rationalized. The best approximation for this rationalization was developed by Henri Padé in 1890, using the Taylor series expansion. Padé's method approximates functions as a ratio of two polynomial functions, the coefficients of which depend on the approximated function and the order of the approximation,

$$f(x) \approx R_{m,n}(x) = \frac{p_0 + p_1x + p_2x^2 + \dots + p_mx^m}{q_0 + q_1x + q_2x^2 + \dots + q_nx^n} \quad (2.164)$$

The full derivation of the Padé approximation is out of the scope of this text. In the case of pure time delay, we must consider the approximation of the function e^x . The approximation uses $n = m$, as

$$e^x = \frac{e^{x/2}}{e^{-x/2}}$$

so that the degree of the numerator and denominator polynomials are the same. The first-, second-, third-, fourth- and fifth-order approximations of e^x are shown below.

$$e^x \approx R_{1,1}(x) = \frac{1 + \frac{1}{2}x}{1 - \frac{1}{2}x} \quad (2.165)$$

$$e^x \approx R_{2,2}(x) = \frac{1 + \frac{1}{2}x + \frac{1}{12}x^2}{1 - \frac{1}{2}x + \frac{1}{12}x^2} \quad (2.166)$$

$$e^x \approx R_{3,3}(x) = \frac{1 + \frac{1}{2}x + \frac{1}{10}x^2 + \frac{1}{120}x^3}{1 - \frac{1}{2}x + \frac{1}{10}x^2 - \frac{1}{120}x^3} \quad (2.167)$$

$$e^x \approx R_{4,4}(x) = \frac{1 + \frac{1}{2}x + \frac{3}{28}x^2 + \frac{1}{84}x^3 + \frac{1}{1680}x^4}{1 - \frac{1}{2}x + \frac{3}{28}x^2 - \frac{1}{84}x^3 + \frac{1}{1680}x^4} \quad (2.168)$$

$$e^x \approx R_{5,5}(x) = \frac{1 + \frac{1}{2}x + \frac{1}{9}x^2 + \frac{1}{72}x^3 + \frac{1}{1008}x^4 + \frac{1}{30240}x^5}{1 - \frac{1}{2}x + \frac{1}{9}x^2 - \frac{1}{72}x^3 + \frac{1}{1008}x^4 - \frac{1}{30240}x^5} \quad (2.169)$$

It is important to note that additional coefficients cannot simply be “tacked on” to obtain a higher order system. Rather, the values of the coefficients change with the number of coefficients used.

In the case of pure time delay we have $x = -T_d s$. By making this substitution to an approximation of e^x to obtain $e^{-T_d s}$, time delay is approximated as a linear transfer function. For instance, a first-order approximation of delay is

$$e^{-T_d s} \approx \frac{1 - \frac{T_d}{2}s}{1 + \frac{T_d}{2}s} \quad (2.170)$$

while a second-order approximation is

$$e^{-T_d s} \approx \frac{1 - \frac{T_d}{2}s + \frac{T_d^2}{12}s^2}{1 + \frac{T_d}{2}s + \frac{T_d^2}{12}s^2} \quad (2.171)$$

Higher-order approximations increase the accuracy but add more complexity to the control system design.

Note that these approximations are all-pass filters: They are gain neutral (have no affect on the gain of the feedback loop), regardless of frequency. However, delay introduces an increasing phase lag, as stated earlier, that can drastically degrade the stability margins or even cause instability in a poorly designed feedback system (linear approximations introduce -180° phase shift per order of approximation), so it is very important to account for pure time delay in control system design.

In general, time delay is treated as lag between the controller and the actuator, as shown in Figure 2.34.

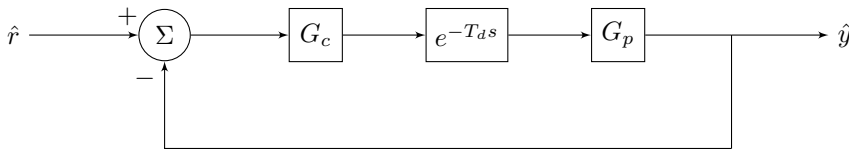


Figure 2.34: Time day in a block diagram.

EXAMPLE 2.13

Consider a plant that experiences a pure time delay of $T_d = 0.1$ seconds and has a single pole with time constant $\tau = 0.1$.

$$G_p = e^{-0.1s} \frac{1}{0.1s + 1} \quad (2.172)$$

Then, the first- and second-order approximations of G_p are as follows.

$$G_{p,1} = \frac{20 - s}{s + 20} \cdot \frac{1}{0.1s + 1} \quad (2.173)$$

$$G_{p,2} = \frac{s^2 - 60s + 1200}{s^2 + 60s + 1200} \cdot \frac{1}{0.1s + 1} \quad (2.174)$$

The true system and both approximate systems were simulated, and the results are shown in Figure 2.35.

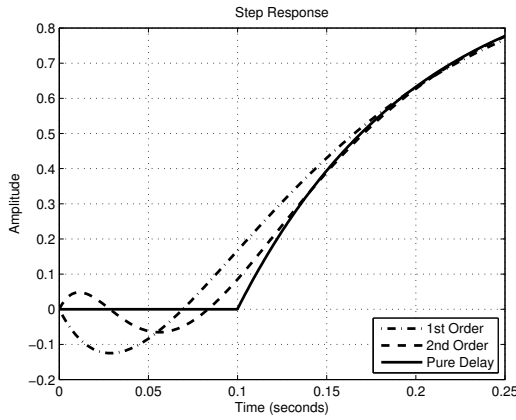


Figure 2.35: Various delay approximations.

2.2.7 Frequency Response: Static Gain:

We complete the catalog of frequency response components with the nearly trivial

$$G(s) = K \quad (2.175)$$

a pure gain. It is obvious that

$$\text{Lm } G(j\omega) = \text{Lm } K, \quad \phi \equiv 0^\circ. \quad (2.176)$$

2.2.8 Frequency Response: The Composite Transfer-Function:

The general transfer function can be written as

$$G(s) = G_1(s)G_2(s)G_3(s)\dots G_n(s) \quad (2.177)$$

where each $G_k(s)$ is one of the items cataloged above. Then, the frequency response

$$G(j\omega) = G_1(j\omega)G_2(j\omega)G_3(j\omega)\dots G_n(j\omega) \quad (2.178)$$

is obtained by superimposing the results obtained to this point. It is no accident that the Bode and magnitude-phase diagrams are structured in such a way that these superpositions can be fully exploited.

Rewrite the basic relation in polar form

$$\begin{aligned} G(j\omega) &= M_1 e^{(j\phi_1)} M_2 e^{(j\phi_2)} M_3 e^{(j\phi_3)} \dots M_n e^{(j\phi_n)} \\ &= M_1 M_2 M_3 \dots M_n e^{[j(\phi_1 + \phi_2 + \phi_3 \dots + \phi_n)]} \end{aligned} \quad (2.179)$$

Therefore, because $\log_{10}(\Pi M_k) = \sum \log_{10} M_k$,

$$\text{Lm } G(j\omega) = \text{Lm } G_1(j\omega) + \text{Lm } G_2(j\omega) + \text{Lm } G_3(j\omega) \dots + \text{Lm } G_n(j\omega) \quad (2.180)$$

and

$$\phi(\omega) = \phi_1 + \phi_2 + \phi_3 + \dots + \phi_n \quad (2.181)$$

Clearly, we have a superposition of both the gain and the phase curves on both Bode and Nichols diagrams. Thus, the catalog of curves delineated in the previous sections allows one to sketch the frequency response of a composite transfer function. These sketches are useful as checks to software and, more importantly, as intuitive predictors of system performance and of feedback requirements.

2.2.9 Frequency Response: Asymptote Formulas:

It is often sufficient, even useful, to use formulas for the frequency-response asymptote rather than a formula for the Bode magnitude itself. It is an easy matter to obtain the asymptote formula, simply apply the following rule to each factor of the form $(\tau s + 1)$ in both the numerator and denominator of the transfer function. Where $1/\tau < \omega$, replace the factor $(\tau s + 1)$ by $\tau\omega$ and where $1/\tau > \omega$, replace the factor by 1.

EXAMPLE 2.14

Consider the following transfer function,

$$G(s) = \frac{(\frac{1}{3}s + 1)}{(s + 1)(\frac{1}{10}s + 1)} \quad (2.182)$$

the asymptote formulas are indicated below:

Asymptote Formula	Frequency Range (rad/s)
1	$\omega < 1$
$1/\omega$	$1 < \omega < 3$
$1/3$	$3 < \omega < 10$
$10/(3\omega)$	$10 < \omega$

A check is that the asymptotes are continuous at the corner frequencies.

2.2.10 Physical Realizability:

It is important to obtain results which indicate the range of possibilities for real, physical transfer functions. The most satisfactory results are obtained by imposing a constraint of “cause and effect”: the output of a physical system cannot anticipate the input. In the previous chapter, we use the very definition of the limit to demonstrate that instantaneous limit processes, such as differentiation, are physically unrealizable in this sense. (Some limit processes such as integration, do not require a device that can foresee the future.)

A consideration of frequency response allows one to anticipate the more rigorous result of the *Paley-Wiener Theorem*, which was briefly discussed previously. Intuitively, it seems impossible for a physical system to have increasing gain with increasing frequency: If this sort of thing were possible, the power requirements would be enormous. This leads us to speculate that:

The degree of the numerator of a transfer function of a physically realizable dynamic system is never greater than the degree of the denominator.

Notice that we satisfy this constraint wherever formula (1.78) is used: State equations (1.75) meet expectations for physical realizability. Something negative can be said about those formulations of the state equations from which improper transfer functions are calculated.

The same sort of reasoning leads some to argue that a physical system must be strictly proper (the D-matrix is zero) since it is hard to imagine that a real system can pass a very high frequency without some attenuation. That is fair enough, but this conclusion is not supported by the more rigorous realizability theory based upon causality principles and complex analysis.

2.2.11 Non-Minimum Phase, All-Pass, and Blaschke Factors:

We present a factorization of a rational function which is useful to us in the discussion of feedback control. In particular, this factorization is used to generalize methods developed for *minimum-phase*, controlled objects to the *nonminimum-phase* case.

Definition 2.9. A system is called *all-pass* if the Bode-magnitude of the transfer function is identically a constant over all frequencies.

■ **EXAMPLE 2.15**

It follows from (2.162) that dead-time is an all-pass element.

■ **EXAMPLE 2.16**

The nonminimum-phase transfer function

$$G(s) = \frac{p - s}{s + p} \quad (2.183)$$

is all-pass as is the unstable transfer function

$$G(s) = \frac{s + p}{p - s} \quad (2.184)$$

Definition 2.10. A rational function either of form (2.183) or of form (2.184) is called a *Blaschke Factor*.

Any transfer function can be factored into minimum-phase factor and an all-pass factor, where the all-pass factor is a dead-time transfer function of a set of Blaschke factors. This obvious result is demonstrated with a simple example:

$$G(s) = \frac{1 - s}{s + 2} = \frac{(s + 1)(1 - s)}{(s + 2)(s + 1)}$$

We can write this symbolically as

$$G(s) = G_{mp}(s)G_{ap}(s) \quad (2.185)$$

The minimum-phase part, $G_{mp}(s)$, is formed by reflecting all poles and zeros of $G(s)$ in the right half-plane about the imaginary axis into the left half-plane. The Bode-magnitude of $G(s)$ is then the Bode-magnitude of G_{mp} because G_{ap} is all-pass. In this way, we can imagine that the non-minimum phase factors contribute only to the phase curve.

Where the right half-plane zero is complex, we prefer to write the Blaschke factor as

$$G(s) = \frac{(z - s)}{(s + \bar{z})} \quad (2.186)$$

and similarly for a right half-plane pole. That is to say, in such cases, the reflection is imagined to be about both the real and the imaginary axes.

2.3 Frequency Response Plotting

The control engineer must be able to estimate the response of the dynamic system from the appearance of the pole-zero diagram of the transfer function. The basic ideas for that sort of estimate are set down in this chapter. These estimates become especially valuable in the following chapters, where it is demonstrated that even high-order, very complicated feedback systems can be made to respond much like the simpler systems described in such detail in this chapter. In fact, it is this eventuality that motivates our detailed investigations here.

The trick to make a complicated feedback system respond much like a simpler system is to properly “shape” the magnitude and phase curves of the feedback system frequency response. As it turns out, this type of consideration is secondary to other more important aspect of control system design but the primary objectives are obtained from frequency-response shaping as well. That is, the steady-state frequency response developed in this chapter is among the most valuable analytic tools of the control engineer.

At this point, we must develop and justify the claims of this summary and, to do that, we turn to Chapter 3, the most interesting and the most important chapter of this volume.

2.3.1 Matlab codes for plotting system frequency response

Bode Plot: These plots can be drawn in Matlab using the `bode` command

```
>> bode(num, den) % (vectors of numerator, denominator coefficients)
OR
>> bode(A,B,C,D,1) % (state space representation)
OR
>> bode(G) % (transfer function with s=tf('s') operator)
OR
>> [MAG, PHASE, w] = bode(num, den);
>> LM=20*log10(MAG);
>> semilogx(w, LM)
>> semilogx(w, PHASE)
```

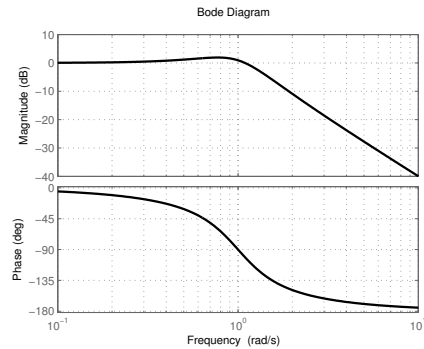



Figure 2.36: Bode Plot.

1646 **Polar Plot/Nyquist Diagram:** The polar plot has the gain of the system as the radial
 1647 coordinate and the phase as the angular coordinate. A Nyquist diagram plots the real and
 1648 imaginary parts over the range of positive and negative frequencies. These plots are useful
 1649 for stability analysis.

```
1650 >> nyquist(num,den) % Or other input types
1651 OR
1652 >> [RE,IM] = nyquist(num,den);
1653 >> plot(RE,IM)
1654
1655 >> [MAG,PHASE,w] = bode(num,den)
1656 >> PHASE=deg2rad(PHASE);
1657 >> polar(PHASE,MAG)
```

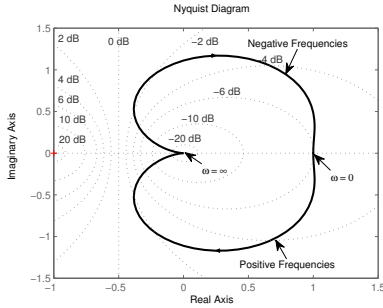


Figure 2.37: Nyquist Plot. The dotted lines show the closed-loop frequency response for the open-loop locus that is drawn.

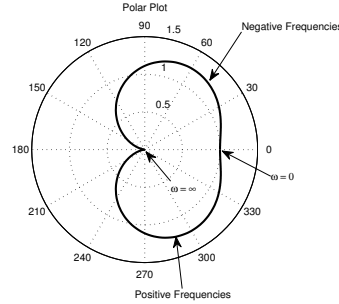


Figure 2.38: Polar Plot.

2.4 Problems

2.1 Using Heaviside (partial fraction) expansion to find $y(t)$, where

$$\hat{y}(s) = \frac{s^2 + 1}{s^2 - 1}$$

2.2 The hydraulic system of Figure 1.0 will displace pole-zero cancellation whenever the tank areas are equal and the hydraulic resistances are matched:

$$A_k = A, R_{i,k} = R$$

In this case, the standard model is

$$a = \begin{bmatrix} -\frac{2}{\tau} - \frac{1}{\tau_0} & \frac{1}{\tau} & \frac{1}{\tau} \\ \frac{1}{\tau} & -\frac{2}{\tau} & \frac{1}{\tau} \\ \frac{1}{\tau} & \frac{1}{\tau} & -\frac{2}{\tau} \end{bmatrix}, \quad b = \begin{bmatrix} \frac{1}{A} & 0 \\ 0 & \frac{1}{A} \\ 0 & 0 \end{bmatrix}$$

$$a = [0 \quad 1 \quad 0], \quad b = [0 \quad 0]$$

Here, the tank heads are taken to be the state variables and

$$\tau = RA, \quad \tau_0 = R_0 A$$

Determine the response $y(t) = x_2(t)$ to a step input $u_1(t) = 1(t)$. Take $\tau = 2.0 \text{ sec}$, $\tau_0 = 10.0 \text{ sec}$, and $A = 10.0 \text{ ft}^2$. Plot your results. Repeat for $u_2(t) = 1(t)$ (notice that you must rewrite the b -vector here). Plot both step responses on the same graph.

2.3 Find a state-variable description appropriate to the sixth-order Butterworth filter (type `>> help TF2SS, help BUTTER` in Matlab). Take $\omega_n = 5/6 \text{ rad/sec}$. Plot the impulse response of the Butterworth filter. Remember to put your system in the continuous time s -domain.

2.4 Suppose the impulse response of a certain system is te^{-t} . What is the step response? Do not use the Laplace transform here! What is the transfer function? Use the convolution integral to obtain the output if the input is e^{-t} . Sketch a graph of both factors in the convolution integrand as a function of t . Check this calculation by using the Laplace transform.

2.5 Consider the differential equation

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 4y = \frac{d^2u}{dt^2} + 4u$$

where the initial conditions are

$$y(0^-) = 0, \quad \frac{dy}{dt}(0^-) = 0$$

and the input is

$$u(t) = \cos(2t)h(t)$$

where $h(t)$ is the unit step function.

(a) Obtain $\hat{y}(s)$ and use the initial and final value theorems to estimate initial and final values and the initial slope of $y(t)$.

(b) Obtain $y(t)$ and verify the results of part (a).

2.6 The goal in this exercise is to give you a chance to become more familiar with Matlab. To start with, consider a dynamic system described by the state-variable equations

$$\frac{dx}{dt} = \begin{bmatrix} -2 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} u + \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} d$$

$$y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} x$$

Enter these matrices in Matlab and call them A , B , C , and D . Notice that you need two B-matrices, one for $u(t)$ and one for $d(t)$. Call them B_u and B_d . Use Matlab to determine the transfer functions $\hat{y}/\hat{u}$ and $\hat{y}/\hat{d}$. Ask for the help-screen on SS2TF. Determine the poles, zeros, and gains of these transfer functions. A natural choice is to apply the `ROOTS` command to the numerator polynomial to determine the zeros, but this is often inaccurate: we recommend you use `TZERO` or, even better, use `SS2ZP`.

CHAPTER 3

FEEDBACK PRINCIPALS

We develop a mathematical theory of feedback design in this chapter. The plan is that the principles are laid down in this chapter, based on which specific design methodologies are developed in the following chapters. The theory is made for continuous-time applications since these procedures carry over, with little modification, to the discrete-time control. Of course, the theory of discrete-time control requires additional ideas and techniques.

We start with a dynamic system that we call the “controlled object” or the “object”. (Another common name is the “plant”, apparently, a contribution of the chemical-process control industry.) This dynamic system has an input, $u(t)$, and an output, $y(t)$. The input is always assumed to be limited in some way, say a power limitation or some other type of limitation on its magnitude. The design problem is to place a feedback loop around the controlled object and improve its performance without exceeding the limitations of the input signal, $u(t)$.¹ *The limitation on the input to the object is the essence of the problem* and this should be kept in mind at all times when designing feedback control systems.

¹Nowadays, the feedback loop is becoming an inherent part of a dynamic systems, and these dynamic systems simply do not operate without some level an automation such as mechatronics, cyberphysical, and so on.

Think of it this way: Without a limitation of the input, a feedback loop can be effected with a little tinkering and there would be little need for someone trained in engineering science. We call this a *feedback control systems*.

We begin with a review of the reasons for using feedback. A special effort is made to not over-emphasize secondary considerations in the design of a control system, say the design for improved transient response. Instead, we emphasize the main points, the design of feedback (1) to attenuate signals input to the feedback loop, and (2) to provide the capability to perform well in spite of inevitable imperfections in our models of dynamic systems.

By “attenuation of signals” we mean the attenuation of every input signal, at least at some nodes in the feedback loop. It is demonstrated that the attenuation of some signals may be achieved with the use of very low gain while other signals are attenuated with the use of very *high gain*. Obviously, there are trade-offs in this regard. It is demonstrated that this single notion can be used to achieve all of the primary goals of feedback design. Further, it is shown that high gain is also desirable for “performance robustness”; that is, the continued good performance of the closed loop in the face of inadequacy or uncertainty in the mathematical models.

The limitation on $u(t)$, is paramount but there are other fundamental constraints on feedback design. There are stability constraints and the limitation on *bandwidth* (that is, a limitation on the speed of response of a closed loop relative to that of the open loop). Other feedback design trade-offs are delineated:

1. some signals can be attenuated only at the expense of not attenuating others,
2. assured stability is achieved at the expense of performance robustness,
3. feedback is achieved with the loss of gain and with the introduction of another unwanted signal, the sensor noise.

The primary mathematical objects in our developments are the poles and zeros of the closed-loop transfer functions and, more significantly, the “frequency response” functions, the frequency-dependent magnitude and phase of the transfer functions. The stage is set for succeeding chapters by introducing the notion of *parametrization*. In order to improve the performance of a feedback loop, the designer interjects another dynamic system into the loop, the dynamic component called a “compensator” or “controller”. In this chapter, it is demonstrated that all feasible compensators belong to a single family which is characterized by a single representative called a “parameter”. This idea has an antiquity but

it is only in the last quarter of the twentieth century that its true value was appreciated. Parametrization is the focus or point of departure of all the remaining chapters in our text. For example, it is demonstrated, in the following chapter, that, essentially, all SISO design problems are solved with the use of this parametrization. The only case where we must abandon this notion is that case where the compensator is required to be of a special form, say a PID controller.

3.1 The Value of Feedback Control

Why use feedback in the first place? A general feedback control system is shown in Figure 3.1.

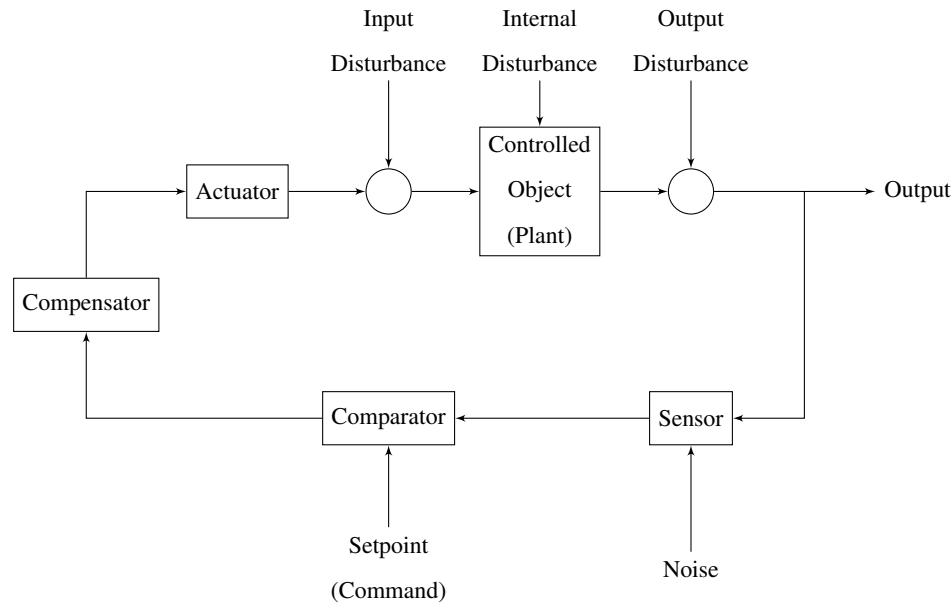


Figure 3.1: Feedback controller block diagram.

It is always possible to use an open loop system to accomplish many of the same goals often touted for feedback control. If one has achieved a feedback design merely to achieve stability or to speed up the response (such as for an optimized performance index, etc...) someone else could perform a computer simulation and record $u(t)$, the input to the controlled object. An open-loop synthesis can then be achieved in the manner indicated in the Figure 3.2 below.

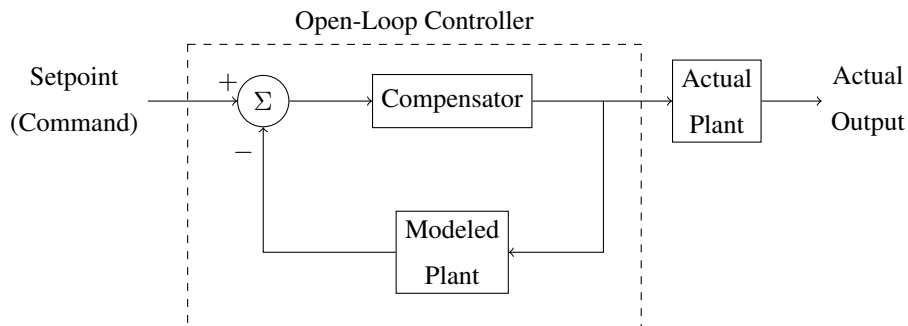


Figure 3.2: Open-loop synthesis.

Incidentally, Figure 3.2 is drawn for a linear, time-invariant system but the idea applies equally well to any type of control, say a nonlinear, time-varying, digital control.

Note, in passing, that the equivalent open loop has some attractive features, the sometimes-costly sensor is not required and the sensor-noise problem is completely avoided. In fact, open-loop control is an important way to realize an “automation” in certain instances. It is interesting to note that any method that is used to design a feedback control for a linear, time-invariant controlled object can also be employed to design a open-loop scheme (refer to the next section in this chapter).

We shall take as axiomatic that the primary goal of feedback design is to achieve those things that the equivalent open loop cannot do for us. It must be clear that the open-loop controller is blinded to disturbances and will operate properly only if the model used in the simulation is very close to the actual dynamics of the controlled object we eventually synthesize.

The primary goal of feedback design is to attenuate the effects of unwanted inputs and to provide, nearly, the desired response in spite of inadequacies in the model or in spite of changes in physical parameters. Horowitz² describes feedback design as the management of uncertainty.

“Feedback”, then, is

²Isaac Horowitz (December 15, 1920 - 2005) was a notable scientist with significant contributions to automatic control theory. He developed and championed the Quantitative Feedback Theory which for the first time introduced a formal combination of the genuine frequency methodology founded by Hendrik Bode with plant uncertainty considerations.

the information required to attenuate the unwanted inputs and to adjust for the mismatch of our model with reality.

There are other very good reasons for using feedback but we shall take damping, stability, or optimality for example, to be secondary goals because such things can be achieved equally well, theoretically, with the recording the input and playing it back into the open loop, as shown in Figure 3.2.

Incidentally, the open loop is a useful way to achieve design goals in certain instances. For example, the equivalent open loop is the way some manufacturing processes (some “automated” machining) are controlled. The open loop works here because the disturbance signals (workpiece applied to the tool) are predictable in all their particulars and the model of the machine tool is accurate enough for most purposes.

An open-loop control (*feedforward control*) can be combined with a feedback control in an effective over-all design; that is, a feedback loop is used to attenuate unwanted inputs and to provide a feedback loop which does vary with parameter change in the object and then an open-loop compensator can be used to improve transient performance. A prescribed speed-of-response, say, is often achieved with an open-loop compensator after all other objectives are met with a feedback loop. We shall see this sort of thing in the final chapter of this volume.

3.1.1 The Advantages of the Closed Loop:

To be more specific, we propose the following list as the main considerations in the use of feedback in control systems.

1. To eliminate the necessity of human intervention. This goal can be due to the incentive to reduce costs through the replacement of labor, or due to a need to control the object in an exotic, remote, or dangerous environment, or to achieve a reliability and accuracy uncommon to human operation. Finally, the need may be a simple desire to relieve humans of burdensome or tedious tasks.
2. To reduce the responses of the controlled object to unwanted external disturbances. One type of disturbance, the “sensor noise”, is actually introduced by the feedback designer (the first requirement for achieving feedback control is to provide a measurement device).
3. To desensitize the performance of the system to variations in the dynamics of the controlled object. These variations are unforeseen and unplanned for and are due to the inadequacies of the mathematical model or due to the actual variation of physical

parameters (such as friction constants) during the lifetime of the system. A property of a feedback system that is insensitive to parameter variation is said to be *robust*. That is, we speak of *robust stability* or *robust performance*. An interesting aspect to this is that the robustness of one property of a feedback system, say stability, can only be improved at the expense of another, say robust performance.

4. To improve the dynamic response of the controlled object. Particular improvements are the damping and the speed with which the steady-state is obtained. It is theoretically possible to speed up the response of a feedback control system but practical limitation, what we call the *bandwidth constraint*, is severe and one is more confident with the improvements in speed achieved with a redesign of the controlled object itself (an increase in power or some other “size” or capability).

This is not to say that stability and damping are to be ignored in feedback design: To be clear on this point, consideration (4) may be crucial for the controlled process itself but, theoretically at least, it can be achieved with out the use of feedback and so it is a secondary consideration in the feedback design. Our purpose in presenting this list is to put these things into proper order of priority.

3.2 Closed Loop Transfer Functions

The simplest and most common feedback arrangement is illustrated in Figure 3.3. We refer to this configuration as the single-loop feedback. There are many other useful configurations; a few others are mentioned here and those are introduced throughout this section.

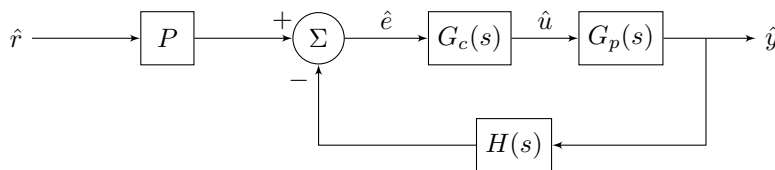


Figure 3.3: Single-loop feedback with prefilter P , controller G_c , plant G_p , and sensor H .

Take particular notice of the terminology and notation defined in this Figure. In particular, notice that the controlled object is that system with input $u(t)$ and output $y(t)$. We intend with this to conform to the state-variable notation which is previously introduced. Normally, the controlled object is a “given” in a design and is thought of as the equipment that is fixed by other considerations such as power-generation and power-transmission. The usual procedure is for a set of design criteria to fix the controlled object using the

rules of thermodynamics, or mechanics, and then call in the control engineer to manage the uncertainty and make it work. At least, this is the traditional approach.

The compensator and, perhaps, the prefilter are added to the control system in order to achieve the feedback design. The task of the feedback designer is to obtain the transfer functions of the compensators, $G_c(s)$, and, possibly, $P(s)$ or even $H(s)$. The transfer function of the controlled object, $G_p(s)$, may contain the dynamics of actuators or sensors. At other places, it may be convenient to allocate these effects to other blocks, say the sensor dynamics to the block denoted $H(s)$.

In Figure 3.4, the “disturbance”, $\hat{d}_y$, is shown to be injected at the output of the controlled object. This is an appropriate depiction where, say, the disturbance is a “load” such as the variation of tire forces due to different road conditions or the application of a part in a machining operation. We may also indicate a disturbance $\hat{d}_u$. This is appropriate where, say, the disturbance is at the actuator. (Note that we have $P \equiv 1$ in Figure 3.4.)

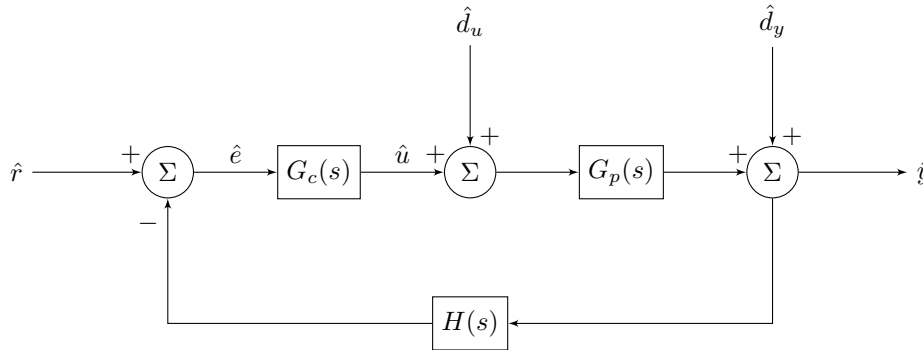


Figure 3.4: Single-loop feedback with disturbances.

Another kind of disturbance is the noise in the sensor, $\hat{n}$. A unique feature of this input is that it occurs after the output and should be thought of as the difference between the actual and measured outputs. Also, notice that “noise” is present only after we insert the sensor for purposes of feedback control:

The addition of sensor noise is a price we pay for feedback control.

The feedback system in which $H(s) \equiv 1$ is called *unity feedback* as shown in Figure 3.5. The reader is admonished to carefully note which developments in this chapter apply to unity feedback and which do not.

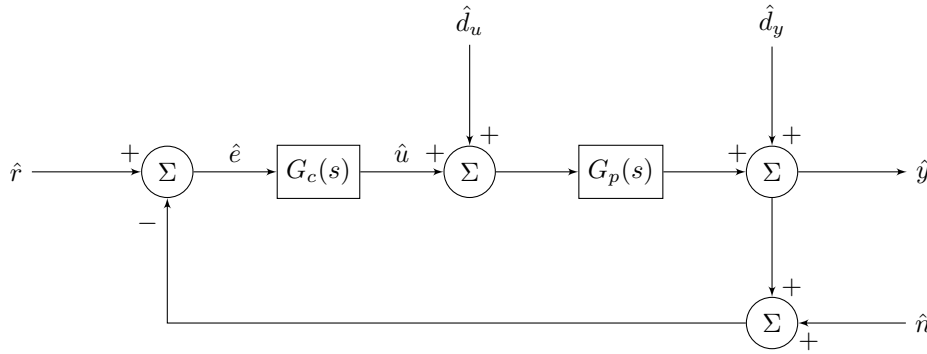


Figure 3.5: Single-loop feedback with unity feedback and with disturbances and noise.

Other common terminologies are: The controlled object is commonly called the “plant”, especially by those in process-control engineering. The command signal, $r(t)$, is sometimes called the “reference target” or, in more limited usage, the “set point”. We often refer to the manipulated variable, $u(t)$, as the “controlled input”.

Definition 3.1. The number of compensators and prefilters is called the degree of freedom of the feedback design.

3.2.1 The Return Ratio

The first order of business is to relate closed-loop transfer functions to the open-loop transfer functions. Our device for that purpose is the principle of superposition and the concept of the return ratio.

Definition 3.2. Any individual point in a feedback loop is called a *node*. The return ratio, denoted $L(s)$, is associated with a node and is obtained by imagining a cut made in the loop at the node, the injection of a signal, denote it $\hat{i}(s)$, and the measurement of the returned signal. We define the return ratio to be the negative ratio (the transfer function) of the returned signal to the injected signal. Thus, the returned signal $\hat{v}(s)$ may be written as

$$\hat{v}(s) = -L(s)\hat{i}(s) \quad (3.1)$$

The return ratio is also known as the open-loop transfer function because it is the transfer function from input (injected signal) to output (negative returned signal) where the loop is

cut or “opened up”. Other names are common such as loop transmission or loop transfer function.

We illustrate an imagined experiment in Figure 3.6 for a feedback system. The imagined cut occurs at the manipulated variable. It is obvious that the returned signal

$$\hat{v}(s) = G_c \cdot (-1) \cdot H(s) \cdot G_p(s) \cdot \hat{i}(s)$$

so that the return ratio is

$$L(s) = -G_p(s)H(s)G_c(s) \quad (3.2)$$

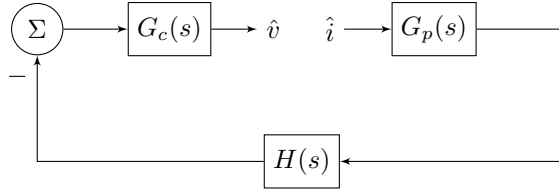


Figure 3.6: Block diagram for determining return ratio.

A moment of reflection is all that is needed to establish that all nodes in a single-loop feedback system have the same return ratio, which is the product of all individual transfer functions in the loop.

3.2.2 Closed-Loop Transfer Functions and the Return Difference

We begin with the **terminology**: By an input to a control system, we mean any autonomous signal injected into the loop. Commands, disturbances, and sensor noise are all inputs.

We wish to characterize the response of any node to any input by the transfer function between any such pair. A transfer function of this kind is given the obvious name, *closed-loop transfer function*. Use $\hat{O}(s)$ to denote some node variable and $\hat{I}(s)$ to denote some input signal. We seek the formula for the closed-loop transfer function

$$\frac{\hat{O}(s)}{\hat{I}(s)}$$

We could imagine that $\hat{I}$ be any input (e.g. $\hat{r}$, $\hat{d}$, or $\hat{n}$) and $\hat{O}(s)$ be any potential output (e.g. $\hat{e}$, $\hat{u}$, or $\hat{y}$).

The guiding principal here is superposition: The node variable is the sum of the signal coming directly from input and the returned signal. By definition the returned signal is

1891 $-L(s)$ times the node variable so that

$$\hat{O}(s) = G_I(s)\hat{I}(s) - L(s)\hat{O}(s) \quad (3.3)$$

1892 where $G_I(s)$ is the product of transfer functions in the *direct path* from the input $\hat{I}$ to the
1893 node (output) $\hat{O}$, the path in the direction of signal flow. Hence

$$\frac{\hat{O}(s)}{\hat{I}(s)} = \frac{G_I(s)}{1 + L(s)} \quad (3.4)$$

1894 This is a remarkably simple and general relationship between open-loop and closed-
1895 loop transfer functions. The term $1 + L$ occurs in the denominator of all closed loop
1896 transfer functions and so we give it a name, *the return difference*. The return difference
1897 obtains its name from the fact that it is the transfer function $\frac{\hat{i}-\hat{v}}{\hat{i}}$ in the imagined experiment
1898 where the loop is cut, Figure 3.6. That is, $1 + L$ is the transfer function from the injected
1899 signal to the difference between the injected and returned signal.

1900 3.2.3 Sensitivity, Complimentary Sensitivity and the Youla Parameter

1901 Classical feedback-design procedures focused on the return ratio, the open-loop transfer
1902 function, in order to achieve closed-loop objectives. Though seemingly indirect, these
1903 procedures are quite effective and receive some attention in the next chapter. However, the
1904 more recent (neo-classical) procedures focus directly on the closed-loop transfer functions
1905 themselves. These procedures are drawn out throughout this text, but this is a good place
1906 to introduce some definitions and propositions fundamental to both design methods.

1907 In addition, we encounter a closed-loop transfer function with a simple appearance but
1908 profound significance, the “Youla parameter”.

1909 Without loss of generality, we assume unity feedback, i.e

$$H(s) \equiv 1. \quad (3.5)$$

1910 Then, the return ratio can be simplified to

$$L(s) = G_c(s)G_p(s). \quad (3.6)$$

1911 **Definition 3.3.** : The sensitivity transfer function is the reciprocal of the return difference:

$$S(s) \triangleq \frac{1}{1 + L(s)} \quad (3.7)$$

1912 **Definition 3.4.** The complimentary sensitivity function is:

$$T(s) \triangleq \frac{L(s)}{1 + L(s)} \quad (3.8)$$

1913 **Definition 3.5.** The *Youla* parameter

$$\begin{aligned}
 Y(s)/G_p(s) &\triangleq \frac{G_c}{1+L(s)} \\
 &= \frac{1}{1+L(s)} \times \frac{L(s)}{G_p(s)} \\
 &= G_c(s)S(s) \\
 &= T(s)/G_p(s)
 \end{aligned} \tag{3.9}$$

1914 All of these names are justified in the in the ensuing parts of this chapter. In our develop-
 1915 ments, $Y(s)$ will play an especially fundamental role.

1916 Notice that Figure 3.5 can be used to show that

$$\frac{\hat{y}(s)}{\hat{n}(s)} = -T(s) \tag{3.10}$$

1917 and

$$\frac{\hat{e}(s)}{\hat{r}(s)} = S(s) \tag{3.11}$$

Many other closed-loop relationships are summarized in Table 3.1.

Table 3.1: Important Rational Functions for Characterizing Closed-Loop Performance

Transfer Function	From...	To...
$S = 1/(1+L)$	r d_y	e y
$T = L/(1+L)$	r $-n$	y y
$Y = T/G_p = G_c S$	r $-n$ $-d_y$	u u u

1918

1919 Notice that the complimentary sensitivity can be written as

$$T(s) = \frac{1}{1+1/L(s)} \tag{3.12}$$

1920 The denominator, $1+L^{-1}$ is called the inverse return-difference. This may be an unfortu-
 1921 nate convention because it would seem, intuitively, that “inverse return-difference” should
 1922 refer to $(1+L)^{-1}$.

1923

1924 The sensitivity function earns its name in the section on the Bode robustness measures
 1925 and the complimentary sensitivity earns its name there as well. The complimentary aspect
 1926 of these transfer functions is already hinted at in the next simple calculation. Consider the
 1927 obvious equation

$$\frac{1+L(s)}{1+L(s)} = 1$$

which means

$$\frac{1}{1 + L(s)} + \frac{L(s)}{1 + L(s)} = 1$$

We have established a basic constraint:

Proposition 3.1. *In a one-degree-of-freedom feedback system*

$$S(s) + T(s) = 1 \quad (3.13)$$

For example, it is demonstrated in the coming section that the designer will attempt to arrange $T(j\omega) \approx 0$ in order to achieve good sensor-noise attenuation at the output. Where that is achieved, a necessary side-effect is that $S(j\omega) \approx 1$ and as is demonstrated further on, that means the closed loop will not attenuate the effects of disturbance at the output in this frequency range. Equation (3.13) establishes an unavoidable trade-off in feedback design.

The general shape of S , T , and Y is drawn in Figure 3.7.

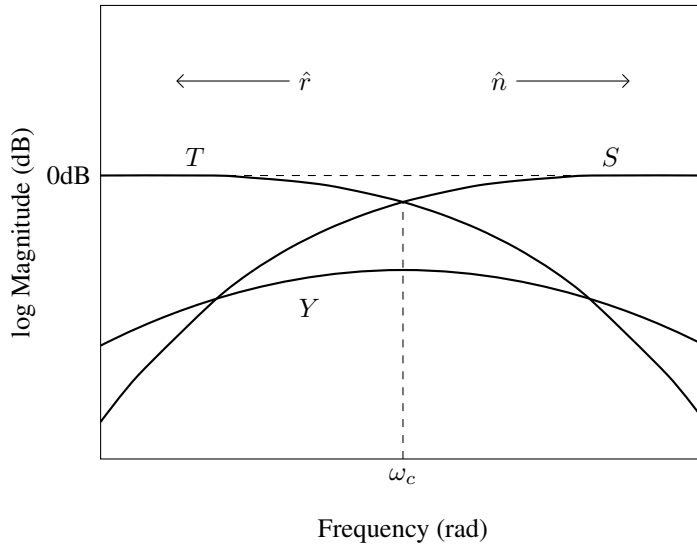
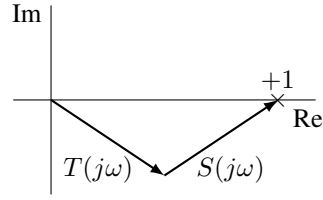


Figure 3.7: Bode magnitude plot for S , T , and Y . Controller commands $\hat{r}$ operate in the low-frequency range, while noise $\hat{n}$ tends to be high-frequency. S and T intersect at the crossover frequency, ω_c .

Equation (3.13) establishes that there is a constraint on $S(s)$ and $T(s)$. We must be careful, however, as $S(s)$ and $T(s)$ are complex. As shown in Figure 3.8, having $S+T = 1$ does **not** mean that $\|T\| + \|S\| = 1$.

Figure 3.8: S and T relationship.

1941 Additionally, we can see from Equations (3.6, 3.8, 3.9) that

$$T = \frac{L}{1+L} = \frac{G_c G_p}{1+L} = \frac{G_c}{1+L} \cdot G_p \Rightarrow T = Y G_p \quad (3.14)$$

1942 This is another relationship that will prove useful in designing and analyzing control sys-
1943 tems.

1944 ■ EXAMPLE 3.1

1945 Consider a mass with an applied displacement, $u(t)$ and a disturbance force $d(t)$, as
1946 shown in Figure 3.9. Suppose that G_c , P , and $H \equiv 1$. We will consider the stability
1947 of some feedback transfer functions.

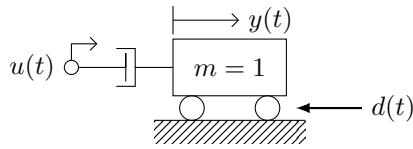


Figure 3.9: A mass with a disturbance and damper force.

1948 We will first consider the following plant G_p and disturbance G_d dynamics to be as
1949 follows.

$$G_p = \frac{1}{s+1} = \frac{s}{s(s+1)} \quad (3.15)$$

$$G_d = \frac{1}{s(s+1)} \quad (3.16)$$

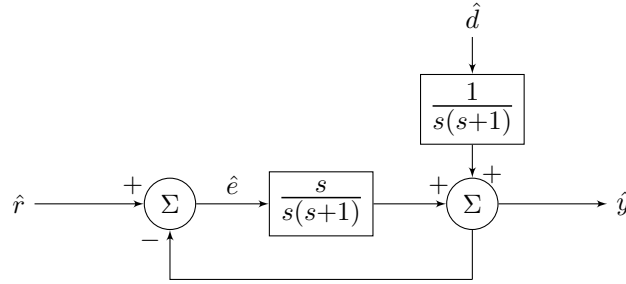


Figure 3.10: Block Diagram.

Then,

$$\frac{\hat{y}}{\hat{r}} = \frac{\frac{1}{s+1}}{1 + \frac{1}{s+1}} = \frac{1}{s+2} \quad (3.17)$$

$$\frac{\hat{y}}{\hat{d}} = \frac{\frac{1}{s(s+1)}}{1 + \frac{s}{s(s+1)}} = \frac{1}{s(s+2)} \quad (3.18)$$

The dynamics from $\hat{r}$ to $\hat{y}$ are stable. However, because of the integrator in the dynamics from $\hat{d}$ to $\hat{y}$ (that is, the s^{-1} or the pole at $s = 0$), the response to a disturbance input is BIBO unstable.

Now, we will consider new plant dynamics.

$$G_p = \frac{1}{s} = \frac{s+1}{s(s+1)} \quad (3.19)$$

$$G_d = \frac{1}{s(s+1)} \quad (3.20)$$

Then,

$$\frac{\hat{y}}{\hat{r}} = \frac{\frac{1}{s}}{1 + \frac{1}{s}} = \frac{1}{s+1} \quad (3.21)$$

$$\frac{\hat{y}}{\hat{d}} = \frac{\frac{1}{s(s+1)}}{1 + \frac{1}{s}} = \frac{1}{(s+1)(s+1)} \quad (3.22)$$

This time, both the response to a commanded input and a disturbance are stable.

What this example shows is that we can use feedback only if the controlled object has cancellations in the left-hand plane.

Internal disturbances can excite components of $G(s)$ that we had assumed were canceled.

Suppose we have a plant defined by state matrices A , B , and C (with $D = 0$ so that the system is strictly proper) and that there is a disturbance $d(t)$ with input matrix B_d , so

$$\dot{\underline{x}} = A\underline{x} + Bu + B_d d \quad (3.23)$$

1966

$$y = C^T \underline{x} \quad (3.24)$$

1967 The output y can be written in terms of u and d , by converting the state-space representation
 1968 into transfer function form.

$$\hat{y}(s) = G_p(s)\hat{u}(s) + G_d(s)\hat{d}(s) \quad (3.25)$$

1969 where

$$G_p(s) = C^T(sI - A)^{-1}B = \frac{\eta(s)}{\delta(s)} \quad (3.26)$$

1970 and

$$G_d(s) = C^T(sI - A)^{-1}B_d = \frac{\eta_d(s)}{\delta(s)} \quad (3.27)$$

1971 So, a control system that cancels unwanted dynamics from $G_p(s)$ may have no impact
 1972 on unwanted dynamics in $G_d(s)$. Various feedback transfer functions can provide insight
 1973 into how these dynamics are amplified or reduced by feedback systems. These transfer
 1974 functions will be derived in this section.

1975 First, consider the block diagram in Figure 3.11, which replaces the generic blocks of
 1976 Figure 3.1 with conventional transfer function and summing junction blocks. Here, G_c
 1977 represents the controller transfer function, G_p represents the plant, G_{du} and G_{dy} represent
 1978 input and output disturbance dynamics, respectively, and P represents any input signal
 1979 preconditioning.

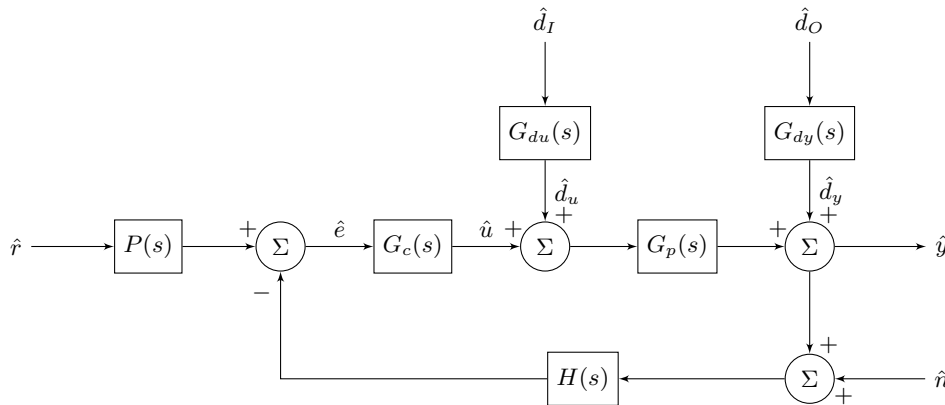


Figure 3.11: SISO block diagram.

3.3 Well-Posedness and Internal Stability

3.3.1 Well-Posedness

Consider the particular feedback system illustrated in Figure 3.12. It is easy to demonstrate that

$$\frac{\hat{y}}{\hat{r}} = \frac{L(s)}{1 + L(s)}$$

where

$$L(s) = \left[1 + \frac{1}{s} \right] \frac{(1-s)}{(s+1)} = \frac{(s+1)(1-s)}{s(s+1)} = \frac{1-s}{s}$$

so

$$\frac{\hat{y}}{\hat{r}} = \frac{(1-s)/s}{1 + (1-s)/s} = 1 - s \quad (3.28)$$

an improper, closed-loop transfer function! However, we have demonstrated that improper transfer functions are physically unrealizable. (If nothing else, a dynamic system with an improper transfer function is BIBO unstable.)

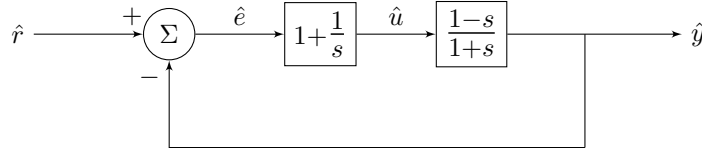


Figure 3.12: Block diagram of system.

Notice that G_c , G_p , and L , in this example, are proper while the closed loop transfer function, T , is not! We are motivated to establish the following definition:

Definition 3.6. A feedback system is *well-posed* where all the closed-loop transfer functions are proper or strictly proper. Otherwise the feedback is *ill-posed*.

The example presented above demonstrates that a feedback can indeed be ill-posed and this can happen even where the individual transfer functions, around the loop, are not improper!

Before giving a mathematical condition for well-posedness, consider a description in words as to what well-posedness means and why it would be useful. Referring to Figure(3.12), at first we must determine under what conditions this interconnection of components makes sense. That is, we need to know when these equations, which result from the feedback loop transfer function, have a solution for an arbitrary input, $\hat{r}$. Then we could say the system

of Figure (3.12) is *well-posed* if a unique solution exists for $\hat{y}$ for all input $\hat{r}$; and further, such a unique solution must exist if this realization is perturbed in some neighborhood of the polynomial coefficients of the closed-loop transfer function.

We can demonstrate a mathematical condition for a well-posed feedback. Consider the Heaviside expansion of the return ratio:

$$L(s) = L(\infty) + L'(s) \quad (3.29)$$

where $L'(s)$ is strictly proper and has the same denominator as $L(s)$. It follows from the development in the previous section that the closed-loop transfer function

$$\frac{\hat{e}(s)}{\hat{r}(s)} = \frac{1}{1 + L(s)} = S(s) \quad (3.30)$$

It is clear that if the denominator, $1 + L(s)$, is strictly proper (if $1 + L(\infty) = 0$) then $1/(1 + L)$ is improper. Thus, $\frac{\hat{e}(s)}{\hat{r}(s)}$ could be improper even where $L(s)$ is not improper. On the other hand, where $1 + L$ is proper, $S(s)$ must be as well. We have proven the following proposition:

Proposition 3.2. *A feedback system is well-posed if and only if $1 + L(\infty) \neq 0$.*

That is, we require that the return difference not have a zero at infinity. Additionally,

Corollary 3.1. *A feedback system is well-posed if $G_c(s)$ or $G_p(s)$ is strictly proper.*

Proof: For example, where there is a single loop,

$$L(\infty) = G_c(\infty)G_p(\infty)$$

where $G_c(s)$ (or $G_p(s)$) is strictly proper, then $G_c(\infty)$ (or $G_p(\infty)$) is zero, so

$$L(\infty) = 0 \neq -1.$$

■

Alternative interpretations of an “ill-posed” feedback are available. For example, if we use feedback about a realizable device, we can then ask, is the feedback assured to be realizable or well-posed? The surprising answer is **no**! The possibility for this counterintuitive result will be demonstrated with the next simple example.

EXAMPLE 3.2

Consider a closed-loop negative feedback system with the simple plant dynamics $G_p = -1$, and let the controller $G_c \equiv 1$, as shown in Figure 3.13.

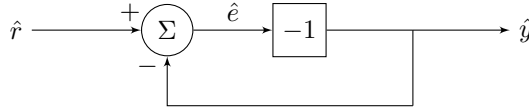


Figure 3.13: Ill-posed feedback.

Then,

$$\hat{e} = \hat{r} - \hat{y} = \hat{r} + \hat{e} \quad (3.31)$$

Or alternatively,

$$\hat{y} = -\hat{e} = \hat{y} - \hat{r} \quad (3.32)$$

Suppose $r = 1$, then what is the error and what is the output? The feedback loop in fact negates the input command, so that the error and output function independently of the input.

We could resolve this with a memory or a first order filter element between the error and the output.

$$G_c = \frac{1}{\tau s + 1} \Rightarrow \frac{\hat{y}}{\hat{e}} = \frac{-1}{\tau s + 1} \quad (3.33)$$

$$\hat{e} = \hat{r} - \hat{y} = \hat{r} + \frac{\hat{e}}{\tau s + 1} \Rightarrow \hat{y} = \frac{\tau s + 1}{\tau s} \hat{r} \quad (3.34)$$

It is important to note that “Well-posedness is a necessary but not sufficient condition for internal stability.”

3.3.2 The Internal Stability of Feedback Control

We outline one of our two primary mathematical characterizations of feedback systems in this section: The focus is on the poles and zeros of the closed-loop system. The second characterization, which is the focus of much of the remainder of the chapter, is the frequency response, the Bode functions of gain and phase, of the closed-loop transfer functions. Our feedback design theories are based upon a combination of these two characterizations.

The Closed-Loop Characteristic Equation and Closed-Loop Poles: It is clear from the previous definition of a pole that the following proposition is true.

Proposition 3.3. *The complex number, p_c , is a closed-loop pole if and only if*

$$1 + L(p_c) = 0 \quad (3.35)$$

That is, p_c must be a zero of the return difference.

Relation (3.35) is called the rational form of the closed-loop characteristic equation.

$L(s)$ is a rational function,

$$L(s) = \frac{\eta(s)}{\delta(s)} \quad (3.36)$$

where $\eta(s)$ and $\delta(s)$ are polynomials. The return difference $1 + L(s)$ can be written as a function of these polynomials,

$$1 + L(s) = [\eta(s) + \delta(s)]/\delta(s)$$

Notice that this rational function has the closed-loop characteristic polynomial as numerator and the open-loop characteristic polynomial as denominator. Hence, based on the previous proposition, where $1 + L(p_c) = 0$, we can write,

$$\eta(p_c) + \delta(p_c) = 0 \quad (3.37)$$

We call this the *polynomial form of the characteristic equation*.

For one thing, the need for stability requires that any root of this equation must lie in the left half-plane. In the past, research engineers have proposed feedback-design procedures based upon the fixing of the roots of this equation. There are the “root-locus” methods and the pole-placement, state-variable methods. These procedures are of dubious value partly because the focus is on a secondary performance issue, transient response, and partly because the locations of individual closed-loop poles, contrary to the belief of some, is not very a good indicator of transient response. That is to say, even on their own terms, pole-placement procedures are of uncertain value. We will discuss more on this in the future.

None of this is to say that we shall disregard the pole-oriented methods. However, their development is made only because these methods can be employed for interesting tasks for which they were not originally intended. In any case, we extract some important information from the location of the poles and zeros of the return ratio and from the location of the closed-loop poles and zeros.

Closed-Loop Zeros: We demonstrated in the previous chapter and will continue to demonstrate that zeros can have as profound but subtle effect on transient response. For that and for other reasons as well, we study the closed-loop zeros.

We begin with the Youla parameter closed-loop transfer function, presented in the previous section, and substitute the ratio of numerator polynomials to denominator polynomials for the various transfer functions:

$$G_k(s) = \eta_k(s)/\delta_k(s) \quad (3.38)$$

The result is

$$Y(s) = \frac{\eta_c(s)\delta_p(s)}{\delta_c(s)\delta_p(s) + \eta_c(s)\eta_p(s)} \quad (3.39)$$

The closed-loop zeros are easily discerned from this equation: The numerator of the closed-loop transfer function is the numerator of the compensator times the denominator of the plant. This proves the following:

Proposition 3.4. *The zeros of a closed-loop transfer function, are the zeros in the forward (or “direct”) path and the poles in the feedback path and differ from node to node.*

That is, since the “paths” are defined differently for the different nodes, the zeros are not invariant from node to node in the control system. This is in marked contrast to the nodal invariance of the closed-loop poles.

Pole-Zero Cancellation and The Internal Stability of Feedback Control: At times, we expect pole-zero cancellations in the transfer functions of some controlled objects, G_p . Pole-zero cancellations can occur in other way; for example, there is considerable incentive to use the poles or zeros of a compensator to cancel some of the zeros or poles of the controlled object so that a return ratio may actually be designed to be reducible.

Sometimes we are indifferent to pole-zero cancellations or even find them to be useful things; but, in other instances, they are unfortunate. The feedback theorist must learn the difference. This section is a description of the profound effect that reducibility of an open-loop transfer function or return ratio can have on the *closed-loop response*.³ (For one thing, it is demonstrated that a feedback system that is actually unstable can pass numerical tests for stability.)

The discussion also focuses on the clever use of pole-zero cancellation and on the not-so-clever use and that occurs near the end of the subsection. This discussion leads to

³The reducibility of transfer functions is intimately related to the “controllability” and “observability” properties of mathematical models described later in this text.

constraints and guidelines on the purposeful use of pole-zero cancellation in feedback design.

EXAMPLE 3.3

Consider the reducible transfer function

$$\frac{\hat{y}(s)}{\hat{u}(s)} = \frac{s+1}{(s+1)(s+2)} = \frac{s+1}{s^2+3s+2} \quad (3.40)$$

which corresponds to the differential equation

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = \frac{du}{dt} + u \quad (3.41)$$

Re-transform this equation but admit the possibility of nonzero initial conditions. The result is

$$\hat{y}(s) = \frac{s+1}{(s+1)(s+2)}\hat{u}(s) + \frac{\dot{y}(0^-) + 3y(0^-) + sy(0^-)}{(s+1)(s+2)} \quad (3.42)$$

a superposition of the force response with the IC response. The canceled pole at $s = -1$ does not play role in the forced response but it does, in general, in the IC response. The point is that a pole-zero cancellation eliminates the effect of the pole in the forced response only.

One might complain of this example that, after all, ICs are the result of past inputs and, for that reason, the particular ICs that would excite the canceled pole cannot be imposed. True enough, but that observation turns out to be more technical than real. The argument only means that the ICs cannot be excited by past values of the particular input, $u(t)$. Recall that the zeros depend upon external connections as well as the internal interconnections. This means that just because there is a pole-zero cancellation in one transfer function does not mean that the transfer function from some disturbance to $\hat{y}$ will contain that same cancellation. Now, in any real system, disturbance-like inputs are always acting and one must plan for the worst: *One must plan for the possibility that some unknown or unpredictable disturbance would affect the initial conditions so as to excite the pole we thought had been canceled.*

We turn here to another type of pole-zero cancellation, that cancellation which a designer purposely introduces in the product $L(s) = G_c(s)G_p(s)$ through the selection of a pole or zero of G_c to lie at same location as a corresponding term in G_p .

2125 **EXAMPLE 3.4**

2126 Consider an unstable system in which a controller cancels the unstable pole, such as
 2127 the one shown in Figure 3.14.

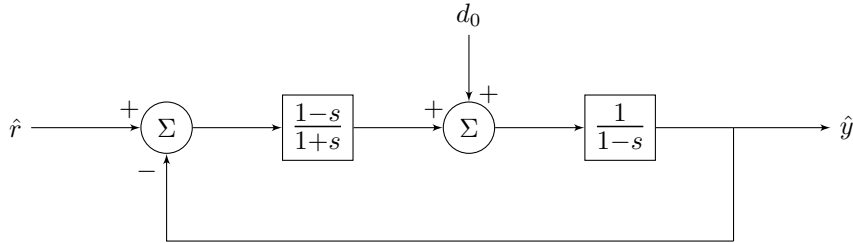


Figure 3.14: Feedback system with an unstable pole/zero cancellation.

2128 The closed-loop transfer function of $\hat{r}$ to $\hat{y}$ is

$$\frac{\hat{y}}{\hat{r}} = \frac{\frac{(1-s)}{(1-s)(1+s)}}{1 + \frac{(1-s)}{(1-s)(1+s)}} = \frac{\frac{1}{1+s}}{1 + \frac{1}{1+s}} = \frac{1}{s+2}, \quad (3.43)$$

2129 which is a stable response due to the cancellation of the unstable pole by the con-
 2130 troller's zero. However, now consider the impact of a disturbance at the plant input.

$$\frac{\hat{y}}{\hat{d}_0} = \frac{\frac{1}{1-s}}{1 + \frac{1}{s+1}} = \frac{s+1}{(s+2)(1-s)} \quad (3.44)$$

2131 The unstable pole is no longer canceled, and the result is an unstable response. Al-
 2132 though the closed loop system was nominally stable, a disturbance between the con-
 2133 troller and plant still excited the plant's instabilities.

2134 Notice that a pole-zero cancellation in the return ratio (the "open-loop" transfer function)
 2135 does not necessarily indicate the same cancellation in all closed-loop transfer functions.
 2136 Finally, notice that

2137 if a pole of one system component cancels the zero of the component which succeeds it
 2138 in the feedback loop (say a compensator zero is used to cancel a pole of the controlled
 2139 object) and if an input signal injected between these components, the signal will excite
 2140 the apparently canceled pole at some of the closed-loop nodes. If a pole of one system
 2141 component cancels the zero of the component which precedes it in the feedback loop (say
 2142 a compensator pole is used to cancel a zero of the controlled object) any signal will excite
 2143 the apparently canceled pole at the closed-loop node between the two components.

2144 **EXAMPLE 3.5**

2145 Suppose that, in Figure 3.14,

$$G_c = \frac{s + \alpha}{s + \beta} \quad \text{and} \quad G_p = \frac{s + \gamma}{s + \alpha} \quad (3.45)$$

2146 a zero is placed in the compensator to cancel a pole of the controlled object. It is easy
2147 to establish

$$\frac{\hat{y}}{\hat{r}} = \frac{s + \gamma}{2s + \beta + \gamma} \quad \text{and} \quad \frac{\hat{y}}{\hat{d}} = \frac{(s + \gamma)(s + \beta)}{(s + \alpha)(2s + \beta + \gamma)} \quad (3.46)$$

2148 So, the cancellation does not occur in the transfer function from the disturbance to the
2149 the output even though the cancellation is in effect in the transfer function from the
2150 command to the output!

2151 **EXAMPLE 3.6**

2152 Suppose that, in Figure 3.14,

$$G_c = \frac{s + \delta}{s + \gamma} \quad \text{and} \quad G_p = \frac{s + \gamma}{s + \alpha} \quad (3.47)$$

2153 a zero is placed in the compensator to cancel a pole of the controlled object. It is easy
2154 to establish

$$\frac{\hat{y}}{\hat{r}} = \frac{s + \delta}{2s + \alpha + \gamma} \quad \text{and} \quad Y = \frac{\hat{u}}{\hat{r}} = \frac{(s + \delta)(s + \alpha)}{(s + \gamma)(2s + \alpha + \gamma)} \quad (3.48)$$

2155 So, the cancellation does not occur in the transfer function from the reference to the
2156 controlled input even though the cancellation is in effect in the transfer function from
2157 the command to the output!

2158 Also, note that a computer-aided stability method that is based upon the assumption that
2159 the closed-loop poles are the zeros of $1 + L(s) = 0$ can give a false indication of stability
2160 if $L(s)$ contains a cancellation in the right half-plane. This includes the root locus and
2161 Nyquist criterion. This problem is avoided only if common factors in the denominator and
2162 numerator of L is not canceled.

2163 One thing is clear: Never effect a pole-zero cancellation in the right half-plane. The
2164 design may be adequate with a cancellation in the left half-plane but only simulation and
2165 testing can indicate that possibility.

2166 The concept of internal stability refers to the idea that for a system to be stable, all
2167 internal transfer functions must themselves be stable, not just the measurable outputs of

the system. For example: consider a room inside of a house that has caught fire. There may be no trace of the fire visible from outside the house, but it is burning nonetheless. Just as the fire will eventually spread to the rest of the house, so too will an internal instability eventually destabilize the system as a whole. To emphasize this point we give the following definition:

Definition 3.7. A well-posed closed-loop system is *internally stable* if the transfer function from every input to every node has poles strictly in the left half-plane and if there are no right-half-plane pole-zero cancellations in the return ratio.

We constrain ourselves to design only internally stable feedback systems. That is easy enough in SISO, as we constrain ourselves to pole-zero cancellations in the product $L = G_c G_p$ strictly in the left half-plane only. The task is shown to be more interesting in case of neo-classical design where one selects the Youla parameter instead of G_c . We turn to those ideas in the next section.

However, it is clear that there is one situation where no amount of feedback (or anything else) can be used to alleviate an internal instability: If a controlled object has pole-zero cancellation on the imaginary axis or in the right half plane, any control scheme is internally unstable.

we require that the controlled object not have a pole-zero cancellation on the imaginary axis or in the right half-plane in order to even contemplate the possibility of a viable feedback control.

In the older, state-variable methods, this requirement was known as the requirement of detectability and stabilizability.

Definition 3.8. A closed-loop system is internally stable if it is well-posed and if for all initial conditions $\underline{x}(0)$ of the plant $G_p(s)$ and $\underline{x}_c(0)$ of the controller G_c , the limits

$$\lim_{t \rightarrow \infty} \underline{x}(t) = \underline{0} \quad (3.49)$$

and

$$\lim_{t \rightarrow \infty} \underline{x}_c(t) = \underline{0} \quad (3.50)$$

hold when the inputs are zero.

Consider the simplified block diagram shown in Figure 3.5, in which the sensor dynamics $H(s)$, preconditioning $P(s)$, and disturbance dynamics $G_d(s)$ and $G_u(s)$ are all

neglected. What are the feedback transfer functions that must be stabilized for internal stability? First, notice that we have four inputs to the system: the commanded input $\hat{r}$, plant input and output disturbances $\hat{d}_u$ and $\hat{d}_y$, and noise $\hat{n}$. Although $\hat{y}$ is the only measured output, conventionally the error signal $\hat{e}$ and the controller output $\hat{u}$ are also treated as outputs. With 4 inputs and 3 outputs, a total of 12 transfer functions can be formed.

These transfer functions can be obtained from Figure 3.5 using block diagram algebra. Of course, some of these have already been derived in Section 3.1, and we know from those earlier derivations to expect that in some cases, different input/output pairings will produce the same transfer function. With that in mind, the 12 transfer functions for internal stability are given in Table 3.2.

Table 3.2: Transfer Functions for Internal Stability

		Inputs			
		$\hat{r}$	$\hat{n}$	$\hat{d}_y$	$\hat{d}_u$
Outputs	$\hat{e}$	$\frac{\hat{e}}{\hat{r}} = S$	$\frac{\hat{e}}{\hat{n}} = -S$	$\frac{\hat{e}}{\hat{d}_y} = -S$	$\frac{\hat{e}}{\hat{d}_u} = -G_p S$
	$\hat{y}$	$\frac{\hat{y}}{\hat{r}} = T$	$\frac{\hat{y}}{\hat{n}} = -T$	$\frac{\hat{y}}{\hat{d}_y} = S$	$\frac{\hat{y}}{\hat{d}_u} = G_p S$
	$\hat{u}$	$\frac{\hat{u}}{\hat{r}} = Y$	$\frac{\hat{u}}{\hat{n}} = -Y$	$\frac{\hat{u}}{\hat{d}_y} = -Y$	$\frac{\hat{u}}{\hat{d}_u} = -T$

Despite considering 12 different input/output pairs, we can characterize the system behavior with only 4 transfer functions: S , T , Y , and $G_p S$. In fact, additional input/output pairings could be derived from the system in Figure 3.5 — they would still simplify to one of these four transfer functions.

If we wish to consider the impact of more than one input on an output variable, we can rewrite the contents of Table 3.2 in matrix form.

$$\begin{bmatrix} \hat{e} \\ \hat{y} \\ \hat{u} \end{bmatrix} = \begin{bmatrix} S & -S & -S & -G_p S \\ T & -T & S & G_p S \\ Y & -Y & -Y & -T \end{bmatrix} \begin{bmatrix} \hat{r} \\ \hat{n} \\ \hat{d}_y \\ \hat{d}_u \end{bmatrix} \quad (3.51)$$

If we neglect the inputs $\hat{r}$ and $\hat{d}_y$ and the output $\hat{y}$, all 4 transfer functions can be fit into a 2×2 matrix expression.

$$\begin{bmatrix} \hat{e} \\ \hat{u} \end{bmatrix} = - \begin{bmatrix} G_p S & S \\ T & Y \end{bmatrix} \begin{bmatrix} \hat{d}_u \\ \hat{n} \end{bmatrix} \quad (3.52)$$

3.4 The Youla Parametrization of all Internally-Stabilizing Compensators

Given a plant, $G_p(s)$, one typically fixes a preferred closed-loop transfer function, $T(s)$, and then computes a proper and stable Youla parameter, $Y(s)$, knowing that $T(s) = Y(s)G_p(s)$. By shaping the closed-loop transfer function from the injection of the reference signal to the controlled input

$$\frac{\hat{u}}{\hat{r}} = Y(s) = \frac{G_c(s)}{1 + L(s)} \quad (3.53)$$

where $L(s) = G_c(s)G_p(s)$, we can then solve the above equation for $G_c(s)$:

$$G_c(s) = \frac{Y(s)}{1 - Y(s)G_p(s)} \quad (3.54)$$

Since $G_p(s)$ is fixed, $G_c(s)$ is easily calculated once the value of the rational function $Y(s)$ is given. For this reason we say Y **parameterizes** G_c and that (3.54) is the **Youla parametrization of G_c** .

This would not be a useful observation if it were not for a few remarkable facts that are developed in this subsection. Firstly, we make some incidental observations about the close association of the Youla parameter with the internal-model⁴ feedback scheme illustrated in Figure 3.15 below.

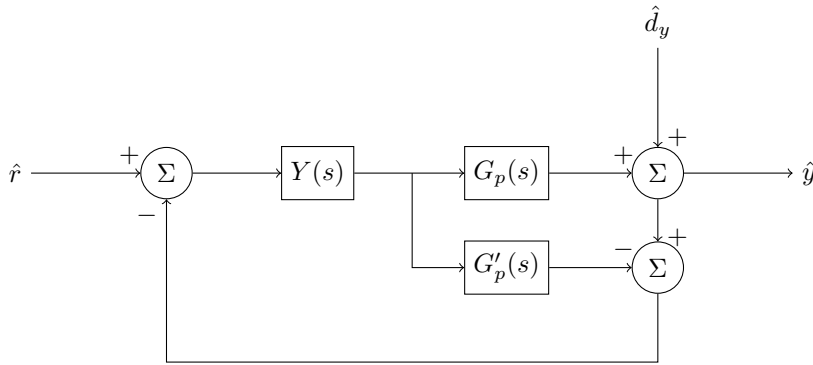


Figure 3.15: The internal-model control scheme.

Here, $G'_p(s)$ represents the modeled plant while G_p is the true plant. Then, if the plant is modeled well and $G'_p \approx G_p$ a signal equal to $G'_p(s)\hat{u}(s)$ is both added to and subtracted

⁴Notice that a “model” (a simulation) of the controlled object is included in the synthesis of the feedback. It is for this reason that the configuration is called *internal-model control*.

from essentially the same point in a single-loop feedback system. This means that the result is equivalent to single-loop feedback shown in Figure 3.16.

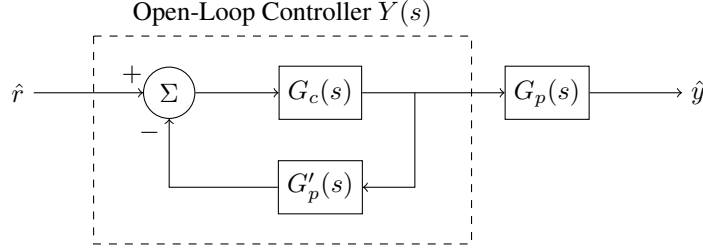


Figure 3.16: Open-loop synthesis.

However, we make the simple rearrangement indicated in Figure 3.17 and make it apparent that the single-loop feedback system is the same as the internal-model control if “ Y ” must be selected as in relation (3.54).

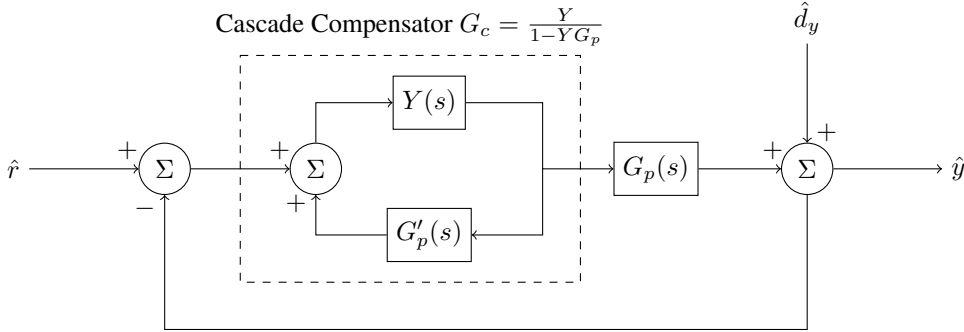


Figure 3.17: Dynamically equivalent single-loop feedback system.

We have established this proposition:

Proposition 3.5. *Single-loop and internal-model feedback control, with a model of the controlled object, are equivalent and the connection between the two is the equation pair (3.53) and (3.54).* ■

Next, as stated above, we can establish a simple closed-loop relation

$$\frac{\hat{y}}{\hat{r}} = T(s) = G_p(s)Y(s) \quad (3.55)$$

and conclude that the closed loop is stable only if the product $Y G_p$ is stable. We want to extend this simple observation to the case of *internal stability* and to the case of an unstable $G_p(s)$.

Remember that in classical control, we select the controller G_c to shape the return ratio L , and infer the impact on the closed loop transfer function T . The advantage of classical control is that we can avoid internal instability by forbidding pole-zero cancellations on the $j\omega$ axis or in the right-hand plane. How might we be able to design internally stabilizing controllers using neoclassical control?

In the previous sections, we pointed out that all closed-loop transfer functions can be reduced to only four transfer functions by the following map,

$$\begin{bmatrix} \hat{e} \\ \hat{u} \end{bmatrix} = - \begin{bmatrix} G_p S & S \\ T & Y \end{bmatrix} \begin{bmatrix} \hat{d}_u \\ \hat{n} \end{bmatrix} \quad (3.56)$$

We can rewrite these transfer function in terms of only $Y(s)$ and $G_p(s)$: Since $G_p Y = T$, we have that

$$G_p Y = 1 - S$$

or

$$S = 1 - G_p Y \quad (3.57)$$

This means that our array of transfer functions or the tableau for internal stability can be rewritten as

$$\begin{aligned} \frac{\hat{u}(s)}{\hat{n}(s)} &= -Y(s) & \frac{\hat{u}(s)}{\hat{d}_u(s)} &= -G_p(s)Y(s) \\ \frac{\hat{e}(s)}{\hat{n}(s)} &= -[1 - Y(s)G_p(s)] & \frac{\hat{e}(s)}{\hat{d}_u(s)} &= -G_p(s)[1 - Y(s)G_p(s)] \end{aligned} \quad (3.58)$$

We accomplished our task: The array of closed-loop transfer functions has been rewritten entirely in terms of $Y(s)$ and $G_p(s)$.

The closed loop is internally stable if and only if all transfer functions in this tableau are stable. We conclude that a closed loop is internally stable if and only if the following goals are obtained:

1. The Youla parameter, $Y(s)$, itself is BIBO stable.
2. $G_p(s)Y(s)$ is BIBO stable; thus, any unstable pole of the controlled object, $G_p(s)$, must be a zero of the Youla parameter, $Y(s)$.
3. $G_p(s)[1 - G_p(s)Y(s)]$ is BIBO stable; thus, any unstable pole of the controlled object, $G_p(s)$, must be a zero of

$$1 - G_p(s)Y(s) \quad (3.59)$$

It is emphasized in the following that an important feedback design procedure is the direct selection of $Y(s)$ with the requirements (3.59) in mind. It is easy to obtain condition (1), simply by selecting a stable $Y(s)$. Remember that the closed-loop transfer function is

$$T = G_p Y$$

so it is easy to obtain condition (2): Simply fix $Y(s)$ with a zero at each unstable pole of G_p . In fact, an unstable pole of G_p must be selected as a zero of $Y(s)$. On the other hand, since $Y(s)$ must be stable, an unstable pole cannot be selected for $Y(s)$ in order to cancel a non-minimum phase zero in G_p : It is forbidden to cancel a non-minimum phase zero in G_p with the selection of a pole of $Y(s)$. One implication is that the non-minimum phase behavior must be passed on to the closed-loop transfer function, $T(s)$. The tricky part is to specify Y such that condition (3) is also met! This is especially tricky where the unstable pole has a multiplicity greater than one. (That occurs most often as a multiple pole at the origin.) We say that the set of all $Y(s)$, that satisfy these conditions, is the *Youla parametrization* of all internally stabilizing compensators.

Again, condition (2) means that there is a unstable pole-zero cancellation in $G_p Y$ if G_p has an unstable pole. There might be some confusion generated here because we seem to prohibit right-half plane cancellations in the previous subsection. Careful reading leads to the following precise admonitions: A right half-plane cancellation is forbidden in $L = G_c G_p$ where the classical procedure is to select G_c . However, it is clear from the above development that such a cancellation is required in $T = G_p Y$ where the neo-classical procedure is to select $Y(s)$. Much more on this is provided in the next Chapter.

Once such a parameter is found, then there exists a compensator, $G_c(s)$ that provides an internally-stable closed loop and the form of this compensator can be calculated from (3.54). (Incidentally, after G_c is calculated, a check on internal stability calculations is available: merely check that there are no unstable-pole cancellations in the product $G_c G_p$.)

We assert that condition (2) is, in fact, an implication of (3). We prove that assertion by contradiction: Suppose that condition (3) holds while (2) does not. This means that $G_p Y$ is unbounded at a unstable pole of G_p , but if that were true, then $G_p(1 - G_p Y)$ would be unbounded there as well, a contradiction. All of this is summarized in the following proposition:

Proposition 3.6. *An internally-stabilizing compensator is parameterized as*

$$G_c(s) = \frac{Y(s)}{1 - G_p(s)Y(s)} \quad (3.60)$$

if and only if $Y(s)$ is any BIBO stable, proper rational-function such that $G_p(1 - G_p Y)$ is BIBO stable; That is, such that Y and $1 - G_p Y$ have a zero at each unstable pole of G_p . ■

Thus, we have:

Corollary 3.2. *Necessary conditions for $Y(s)$ to be internally stabilizing are*

$$Y(p_r) = 0 \quad (3.61)$$

and

$$G_p(p_r)Y(p_r) = 1 \quad (3.62)$$

where “ p_r ” is any pole of G_p in the right half-plane or on the imaginary axis. These conditions are sufficient only if “ p_r ” is not repeated. ■

(3.61, 3.62) may not be sufficient because there is no guarantee that the multiplicity of the zero in $1 - G_p Y$ will match the multiplicity of the unstable pole in G_p . Notice that, in this case, $G_c(s)$ may be stable or unstable. That is to say, a stable $T(s)$ is achieved by placing unstable poles in the compensator. Such a thing is always possible, *but note that some object to the use of an unstable compensator because if the feedback fails (a sensor or an actuator fails) then the opened loop system is driven out of bound by the compensator!*

Since $T = G_p Y$, relation (3.62) means that internal stability, an unavoidable requirement, is achieved only if

$$T(p_r) = 1 \quad (3.63)$$

Incidentally, condition (3.62) is not inconsistent with (3.61) because of the pole-zero cancellation of the p_r -term on the left side of (3.62). Some may also argue that condition (3.62) is undefined if we directly substitute for $(Y(p_r) = 0)$ and $(G(p_r) = \infty)$. To avoid any confusion, it better to write this condition as:

$$G_p Y(p_r) = 1$$

In either case, the Proposition (6) is a very remarkable result! Think of this: One is not assured of a stable feedback loop by placing some restriction on, say, the compensator $G_c(s)$. For example, the feedback about a stable compensator in cascade with a stable controlled object can be unstable. This remarkable neo-classical proposition is one of the motivations for focusing on the Youla parameter, $Y(s)$, during design. There are a number of other motivations as will become clear as we continue.

We have a simplification in a special case:

Corollary 3.3. *If the controlled object is stable, the closed loop is internally stable if and only if $Y(s)$ is stable (hence, not improper).* ■

This means that the “Youla parametrization of all stabilizing compensators” is the set of all stable, proper rational functions! That is a way to say that any G_c given by (3.60) will produce an internally stable closed loop provided that the Youla parameter is stable.

EXAMPLE 3.7

Suppose that the object is stable except for a single pole at the origin. In that case, “ Y ” is internally stabilizing if and only if “ Y ” has a zero at the origin and

$$G_p(0)Y(0) = T(0) = 1 \quad (3.64)$$

if and only if the closed-loop static gain is exactly 1.0. For example consider the velocity control of a mass for which

$$G_p(s) = 1/(ms) \quad (3.65)$$

Then, an internally stabilizing Y is

$$Y(s) = ms/(\tau s + 1) \quad (3.66)$$

for arbitrary values of “ τ ”. It follows from (3.60) that the compensator is

$$G_c(s) = \frac{ms/(\tau s + 1)}{1 - 1/(\tau s + 1)} = \frac{m\cancel{s}}{\tau\cancel{s}} = \frac{m}{\tau} \quad (3.67)$$

an arbitrary positive gain!

3.5 Interpolation Conditions

The internal stability condition of the last section, ($T(p_r = 1)$) is very interesting because it is a constraint that must be satisfied by the closed-loop transfer function, a constraint that can never be avoided.

We mentioned that in classical control, the controller G_c is designed to shape the return ratio $L = G_c G_p$ so that the performance of the closed loop transfer function $T(s)$ meet the given requirements. In designing G_c , pole-zero cancellation of unstable (right-hand plane) pole of G_p with an unstable zero(non-minimum phase zero) in G_c is forbidden in

order to preserve internal stability. However, we know from the previous section that internally stable feedback is parameterized by the Youla parameter Y in neo-classical control.

In addition, we provided necessary conditions for guaranteeing internal stability and we stated that these conditions become sufficient only when the unstable pole of G_p is not repeated. We pointed out that the tricky part is to specify Y such that the condition $G_p[1 - G_p Y]$ is BIBO stable, especially where the unstable pole of G_p has a multiplicity greater than one. We will further discuss this below to provide some insights and develop a set of conditions which are relatively easier to apply and that when satisfied guarantee the internal stability for all circumstances including repeated unstable poles of G_p .

Consider the case where G_p has a single unstable pole. We will let that unstable pole be at $s = p_r$, and define $\tilde{G}_p$ to be the remaining stable part of the transfer function, i.e.,

$$G_p = \frac{1}{s - p_r} \tilde{G}_p \quad (3.68)$$

Then to have a stable $T = Y G_p$, we must have

$$Y = (s - p_r) \tilde{Y}. \quad (3.69)$$

In other words, Y must have a non-minimum phase zero at $s = p_r$ to cancel the unstable pole of G_p , while the minimum phase and stable part of Y is denoted $\tilde{Y}$. Given this factorization for Y , we then have

$$G_p S = G_p (1 - Y G_p) = \frac{1}{s - p_r} \tilde{G}_p (1 - \tilde{Y} \tilde{G}_p) \quad (3.70)$$

For internal stability, $G_p S$ must be stable. Therefore not only must Y cancel the unstable pole in G_p , but so must S . In other words, after including a non-minimum phase zero in Y to cancel the unstable pole of G_p , the unstable pole of G_p must also be a zero of $(1 - Y G_p)$ as stated in the last section. This is especially tricky where the unstable pole has a multiplicity greater than one. To find a set of conditions which when satisfied guarantee internal stability under all conditions, we go back to our very interesting condition of $T(p_r) = 1$.

Suppose the open-loop transfer function L has unstable poles at $s = p_r$ with multiplicity a_p and has unstable zeros at $s = z_r$ with multiplicity a_z . If we let $\tilde{L}$ denote the stable, minimum phase factor of L then we have

$$L(s) = \frac{(s - z_r)^{a_z}}{(s - p_r)^{a_p}} \tilde{L}(s) \quad (3.71)$$

2367 Note that at the unstable pole p_r , the denominator of L goes to zero, and so L goes to
 2368 infinity:

$$L(p) = \frac{(p - z_r)^{a_z}}{0^{a_p}} \tilde{L}(p) = \infty \quad (3.72)$$

2369 Additionally, note that at the unstable zero z_r , the numerator of L goes to zero, and so L
 2370 goes to zero:

$$L(z) = \frac{(0)^{a_z}}{(z - p_r)^{a_p}} \tilde{L}(z) = 0 \quad (3.73)$$

2371 The same results apply to minimum phase or stable zero and stable pole of L , such that
 2372 $L(z) = 0$ at the stable zero or minimum phase zero and , $L(p) = \infty$ at the stable pole.
 2373 Therefore, we can write (the proof is a simple exercise),

2374 **Proposition 3.7.** *If “ z ” denotes a zero of the return ratio, L , of multiplicity a_z , and “ p ”*
 2375 *denotes a pole of the return ratio of multiplicity a_p , then*

$$T(p) = 1 \quad S(p) = 0 \quad (3.74)$$

and

$$\frac{d^k}{ds^k} T(p) = 0 \quad \frac{d^k}{ds^k} S(p) = 0 \quad 1 \leq k \leq a_p - 1 \quad (3.75)$$

while

$$T(z) = 0 \quad S(z) = 1 \quad (3.76)$$

and

$$\frac{d^k}{ds^k} T(z) = 0 \quad \frac{d^k}{ds^k} S(z) = 0 \quad 1 \leq k \leq a_z - 1 \quad (3.77)$$

2376 ■

2377 These very important relations are known as *interpolation constraints*. Condition (3.76)
 2378 subsumes our previous condition $T(p_r) = 1$. Thus, the closed-loop transfer functions,
 2379 $S(s)$ and $T(s)$, are constrained to take on certain values in the complex plane and these
 2380 constraints influence the values on the imaginary axis, $S(j\omega)$ and $T(j\omega)$. This proposition
 2381 imposes irrevocable limits to feedback design, and interestingly, it is these irrevocable
 2382 limits which guarantee internal stability of the feedback. Hence, *the feedback is internally*
 2383 *stable if and only if these constraints are met.*

2384 **EXAMPLE 3.8**

2385 In this example, we examine the “Interpolation Constraints” and the consequences of
 2386 violating these constraints in feedback design.

2387 Consider a force $F(t)$ applied to a mass to control velocity.

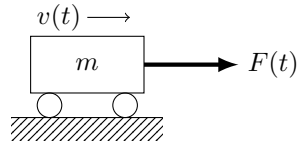


Figure 3.18: Schematic

$$\frac{\hat{v}}{\hat{u}} = \frac{1}{ms} \Rightarrow \text{BIBO Unstable} \quad \text{where} \quad \hat{u} = F(t) \quad (3.78)$$

2388 We will examine a Youla parameter that cancels the dynamics of the mass but does
 2389 *not* meet the interpolation condition.

$$Y = \frac{ks}{\tau s + 1} \quad (3.79)$$

2390 Then,

$$T = \frac{k}{m(\tau s + 1)} \quad (3.80)$$

Suppose that $k = \frac{m}{2}$. Then at $s = 0$, $T = \frac{1}{2}$, so the interpolation condition is *not* met. We will find the corresponding controller.

$$S = 1 - T = \frac{\tau s + \frac{1}{2}}{\tau s + 1} \quad \text{where} \quad S = (1 - G_p Y) \quad (3.81)$$

$$G_c = \frac{Y}{S} = \frac{ks}{\tau s + 1} \cdot \frac{\tau s + 1}{\tau s + \frac{1}{2}} \quad (3.82)$$

$$\Rightarrow G_c = \frac{ms}{2\tau s + 1} \quad (3.83)$$

2391 This means that the return ratio will have a cancellation of an unstable pole, so the
 2392 system will not be internally stable.

$$L = \frac{m\cancel{s}}{2\tau s + 1} \cdot \frac{1}{m\cancel{s}} \quad (3.84)$$

2393 Specifically, the transfer function $G_p(1 - YG_p)$ will be BIBO unstable.

2394 **EXAMPLE 3.9**

2395 Consider a force $F(t)$ applied to a mass again, but this time, to control position rather
2396 than velocity.

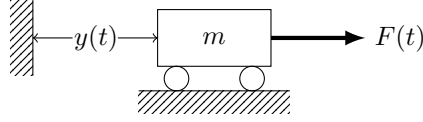


Figure 3.19: Schematic

$$\frac{\hat{y}}{\hat{u}} = \frac{1}{ms^2} \quad (3.85)$$

2397 We choose a Youla parameter to cancel the dynamics of the mass.

$$Y = \frac{mks^2}{(\tau s + 1)^2} \quad (3.86)$$

$$T = \frac{k}{(\tau s + 1)^2} \quad (3.87)$$

2399 Our interpolation conditions are $T(0) = 1$ and $\frac{dT(0)}{ds} = 0$.

$$T(0) = \frac{k}{(\tau \cdot 0 + 1)^2} = 1 \quad (3.88)$$

2400 Therefore $k = 1$. So,

$$\left. \frac{dT}{ds} \right|_{s=0} = \frac{-2\tau}{(\tau \cdot 0 + 1)^3} = 0 \quad (3.89)$$

2401 Therefore $\tau = 0$. This means we now have

$$Y = ms^2. \quad (3.90)$$

2402 Y is now improper, which is unacceptable.

2403 Next, we consider:

$$Y(s) = \frac{ms^2(\tau_1 s + 1)}{(\tau s + 1)^3} \quad (3.91)$$

2404 We added an additional pole and zero to Y with an extra tuning parameter τ_1 . Thus,

$$T(s) = \frac{(\tau_1 s + 1)}{(\tau s + 1)^3} \quad (3.92)$$

The first interpolation condition, $T(0) = 1$, is satisfied and (3.75) is

$$\left. \frac{dT(0)}{ds} \right|_{s=0} = \left[\frac{\tau_1}{(\tau s + 1)^3} - \frac{3\tau(\tau_1 s + 1)}{(\tau s + 1)^4} \right] \Big|_{s=0} \quad (3.93)$$

$$= \tau_1 - 3\tau = 0 \quad (3.94)$$

so that an internally stabilizing value of the parameter is

$$Y(s) = \frac{ms^2(3\tau s + 1)}{(\tau s + 1)^3} \quad (3.95)$$

An internally-stabilizing compensator is calculated similar to the previous example

$$G_c(s) = \frac{ms^2(3\tau s + 1)/(\tau s + 1)^3}{1 - (3\tau s + 1)/(\tau s + 1)^3} \quad (3.96)$$

$$= \frac{ms^2(3\tau s + 1)}{\tau^2 s^2(\tau s + 3)} \quad (3.97)$$

$$= \frac{m}{3\tau^2} \frac{(3\tau s + 1)}{(\tau/3)s + 1} \quad (3.98)$$

This could be a lead-lag controller depending of the selection of (τ) . As a check, we offer the fact that $G_c G_p$ does not contain the wrong kind of cancellation. Also, $T(s) = (3\tau s + 1)/(\tau s + 1)^3$ so that the close loop is stable; as a matter of fact, we have placed all three closed-loop poles at $s = -1/\tau$. Where we explore this further, we find some unsatisfactory aspects to this solution, but, at least, we have found an internally-stable closed-loop. We will discuss more satisfactory solutions of this problem in the next chapter.

3.6 Steady State Error

Before shifting our attention to the application of frequency methods in feedback design, it is important to discuss the time response steady state error when dealing with the closed loop transfer functions $S(s)$ and $T(s)$. Consider the block diagram in Figure 3.20.

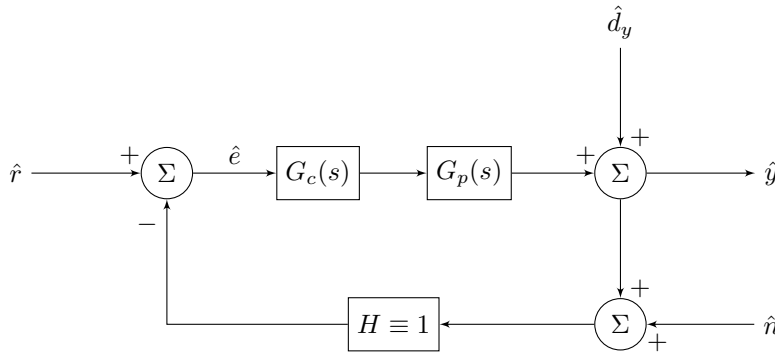


Figure 3.20: Single-loop system with unity feedback.

Steady state error is found by using the final value theorem.

$$\text{Step Input: } r(t) = 1(t) \Rightarrow \hat{r}(s) = \frac{1}{s}$$

$$\text{Ramp Input: } r(t) = t(t) \Rightarrow \hat{r}(s) = \frac{1}{s^2}$$

Then we can write,

$$\hat{e}(s) = S(s)\hat{r}(s)$$

So, for a step input,

$$e(\infty) = \lim_{s \rightarrow 0} s \cdot S(s) \frac{1}{s} = S(0)$$

Hence,

$$e(\infty) = 0 \quad \text{if} \quad S(0) = 0 \Rightarrow T(0) = 1$$

This means, $S(s)$ has at least a single zero at $s = 0$. For a ramp input,

$$e(\infty) = \lim_{s \rightarrow 0} s \cdot S(s) \frac{1}{s^2} = \frac{S(s)}{s} \Big|_{s=0}$$

This means that $e(\infty)$ is finite if $S(s)$ has a zero at $s = 0$ and is zero if $S(s)$ has at least two zeros at $s = 0$. If so, then

$$\frac{dS(s)}{ds} \Big|_{s=0} = 0$$

Therefore,

$$T(s) = 1 - S(s) \Rightarrow \frac{dT(s)}{ds} \Big|_{s=0} = 0$$

3.7 Feedback Design, and Frequency Methods: Input Attenuation and Robustness

In this section, we shift the focus from the poles and zeros of the closed-loop transfer functions to the frequency response. In some design procedures, the pole-zero concepts, such as well-posedness and internal stability, are kept in mind, as constraints, but the main details of the design of the compensators are obtained from the consideration of the frequency response. Those methods are laid down in the next chapter and we prepare for them here. There are two levels of mathematical rigor used in these developments. The more satisfying level is laid down in the later chapters. The less compelling interpretation is developed in the next subsection and is adequate for the design methods described in the remainder of this chapter and the next.

3.7.1 The Frequency Paradigm

One approach to feedback design is the “shaping”, or proper alternation, of the frequency response to provide desirable closed-loop performance. A *heuristic* justification for this is that input signals can be thought of, at times, as a superposition of sinusoidal terms of various frequencies, the Fourier series or transform. So that, if one shapes the entire frequency response, the response to all types of inputs are influenced. The point is that the frequency response represents a gain, an amplification or an attenuation, of the various Fourier components of the arbitrary input. This interpretation is an implicit justification for the formulas and techniques designed in the sequel.

The open-loop frequency-response gains are easily related to the feedback gains. These feedback gains allow one to infer the ability of the closed loop to attenuate unwanted signals entering at various points in the system.

This paradigm has led to an obviously successful design methodology and one can recognize a limited mathematical justification in some cases. One can also point to important cases where the paradigm breaks down. Some important inputs simply don’t have a Fourier transform, the ramp input for example. Even the step input has a Fourier transform in only a limited, least-squares sense.⁵ Also, the design methodology is applied equally well to open-loop transfer functions with poles on the imaginary axis or in the right half-plane, and these systems simply don’t have a steady-state frequency response.

3.7.2 Input Attenuation and Command Following

How are the design requirements, which are delineated in the previous sections, qualified where frequency methods are applied? That is the question addressed in this subsection.

A recurrent theme is that the answer is framed as the attenuation of some input signal or other at certain nodes of the feedback systems. That is, *we wish to design the feedback system so that all signals are to be attenuated at least at some nodes*. By “attenuation”, we mean that the gain (the Bode-magnitude) of the closed loop transfer function is as low as possible at all appropriate frequencies, that frequency range we expect to be present in the input signal. In fact, it is desirable to attenuate most inputs, such as disturbance and noise, at all nodes of the feedback system.

Obviously, we don’t wish to attenuate the command signal at the output node because we want $y(t) \approx r(t)$: We call this *good command-following* or *good tracking*. But even

⁵Refer to the “Gibbs phenomenon” in any text on the Fourier transform.

in this case, the objective can be stated as an attenuation, an attenuation of the *command signal at the error node*. Also, an attenuation of the command at the controlled input node, $u(t)$, is highly desirable in order to reduce the vulnerability of the feedback to a saturation of the controlled input.⁶

There is a difference in interpretation between attenuating unwanted input signals and the attenuation of the command at the error nodes. We wish to attenuate noise in order to prevent degradation of the output or to prevent saturation of the actuators. The attenuation of command at the error node means that the output is rapidly forced to the commanded value. This difference in interpretation is of not mathematical consequence.

Begin with a very general form of the generic closed-loop transfer function,

$$\frac{\hat{v}}{\hat{i}} = \frac{G_d}{1 + L} \quad (3.99)$$

Here, “ G_d ” is the product of transfer functions in the direct path, and “ L ” is the return ratio, $\hat{v}$ is an arbitrary node and $\hat{i}$ is an arbitrary input. From this follows:

Proposition 3.8. *An input signal is attenuated at a node where the closed-loop transfer function has a low gain and is obtained where*

1. *the gain of the product of direct path transfer function, G_d , is very small, or*
2. *the gain of the return ratio, L , is very high.* ■

The following facts should be kept in mind:

1. The gain is frequency dependent.
2. The product of transfer functions in the direct path, “ G_d ” is dependent upon the choice of the node.
3. The return ratio for unity feedback is the product of the compensator and the plant transfer functions, $G_c G_p$.

There exists some input-node pairs that $G_d = L$, such as the output to the sensor noise input. This means the noise is attenuated at the output only where the gain of the return ratio, “ L ”, is made low.

At this point, we refocus on the another formulation of feedback control, the S - T - Y formulation. In this way we come to a similar conclusions but with slightly different

⁶even though, this is a linear control design, this reduction in vulnerability equally applies to the nonlinear system.

interpretations. We want to attenuate all signals at the controlled input, $u(t)$, but that idea is so important in its ramifications and interpretations that the entire next section is devoted to the development of those principals. Here we are content to restrict our attention to the error and output nodes.

Consider the case of *unity feedback*, $H(j\omega) \equiv 1$. Also assume that a disturbance is injected as in Figure 3.20. Under these assumptions, the return ratio is

$$L(j\omega) = G_c(j\omega)G_p(j\omega) \quad (3.100)$$

and simple applications of (3.99) gives

$$\frac{\hat{y}(s)}{\hat{n}(s)} = \frac{-L(s)}{1 + L(s)} = -T(s) \quad (3.101)$$

and

$$\frac{\hat{y}(s)}{\hat{d}(s)} = \frac{1}{1 + L(s)} = S(s) \quad (3.102)$$

To study command following, we form

$$\frac{\hat{e}(s)}{\hat{r}(s)} = \frac{1}{1 + L(s)} = S(s) \quad (3.103)$$

Proposition 3.9. *The feedback loop will attenuate disturbances at the output and follow command signals at those frequencies where $|S(j\omega)|$ is low and it will attenuate noise at the output where $|T(j\omega)|$ is low.* ■

Knowing that

$$S(j\omega) = 1 - T(j\omega) \quad (3.104)$$

in a single-degree-of-freedom, single feedback-loop, the proposition indicates a fundamental trade-off in feedback design.

These equations could just as well be interpreted in terms of the return ratio. For instance, noise is suppressed at the output node at those frequencies where $|L(j\omega)| \ll 1$, but the error is quite large (poor closed-loop, command following), and the feedback does not attenuate disturbance at the output at those same frequencies. Similarly, where $|L(j\omega)| \gg 1$, we have good command following and good disturbance attenuation but poor noise attenuation.

Again, a trade-off between good noise attenuation and good disturbance attenuation (with command following) is implied in this statement.

Definition 3.9. The cross-over frequency is denoted ω_c and is that frequency at which $|L(j\omega)| = 1$ (0 dB).

We shall discover many significant insights into feedback design about ω_c . It suffices here to point out that this frequency usually separates the range of frequencies where we have good command following and disturbance rejection from the range where noise is suppressed at the output node. Of course, in that range of frequencies about the cross over, we are receiving *none of the benefits of feedback!*

Proposition 3.10. *The cross-over frequency is also characterized as that frequency where*

$$|S(j\omega_c)| = |T(j\omega_c)| \quad (3.105)$$

Proof: Immediate consequence of (3.101, 3.102) and Definition 3.9. ■

3.7.3 Bode Measures of Performance Robustness

In the previous sections, we listed as one of the primary goals of feedback design the assurance (or invariance) of performance in spite of modeling difficulties or parameter variations. This invariance in performance is called *Performance robustness*.⁷

We develop ways to quantify robustness in many places in this text. In the spirit of the axioms developed in this chapter, we develop measures of robustness from the work of Bode⁸. Consider the closed-loop system shown in Figure 3.21.

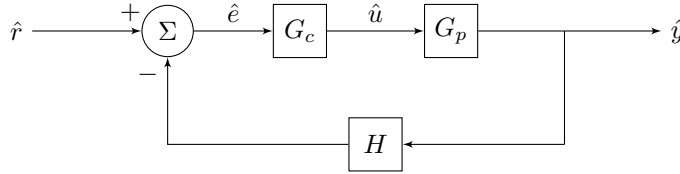


Figure 3.21: Block Diagram.

Two sources of variation in the closed-loop response are variations in the plant dynamics, G_p , and variations in the sensor dynamics, H . We then have a closed-loop response G_{cl} defined by

$$G_{cl} = \frac{\hat{y}(s)}{\hat{r}(s)} = \frac{G_c(s)G_p(s)}{1 + G_c(s)G_p(s)H(s)} \quad (3.106)$$

⁷In fact, this is a relatively new name, a name not used by Bode, for example, or by Perkins and Cruz who generalized this measure to the multivariable case. The older term is “feedback sensitivity”.

⁸Horowitz complains that Bode himself never used his measure in this way, and that this application of the Bode measure is a more recent invention. Horowitz is a persuasive critic of these popular robustness measures.

Definition 3.10. The Bode differential performance-robustness measure (more simply, the Bode measure) is the normalized partial derivative

$$S_{G_p}^{G_{cl}} = \frac{\% \text{ Change in } G_{cl}}{\% \text{ Change in } G_p} = \frac{\frac{\partial G_{cl}}{G_{cl}}}{\frac{\partial G_p}{G_p}} = \frac{G_p}{G_{cl}} \cdot \frac{\partial G_{cl}}{\partial G_p} \quad (3.107)$$

$$(3.108)$$

The Bode measure is read as the sensitivity of G_{cl} with respect to G_p . As given in the equation, this is interpreted as the ratio of a percent change in the closed loop with respect to a percent change that occurs in the controlled object, the change in the system without feedback,. The units of this measure are commonly taken to be “percent-per-percent” but it is demonstrated in the next proposition that we can take the units to be dB per dB. Implicit in this definition is the assumption that the variation occurs in the controlled object: Normally, the compensators can be constructed with precisely-known parameter values which do not vary in time. Variation in the measurement, $H(s)$ is a possibility and that possibility is accounted for later in this section.

We think of the Bode measure as a gauge of performance robustness as opposed to the stability robustness developed in the next chapter. This is because the Bode measure quantifies the variation in the entire closed-loop frequency response, which determines the response of the system to an arbitrary reference input by our frequency paradigm. The great limitation of this measure is that it only quantifies the effect of differential changes in the controlled object. (There are other derivations of Bode measure in which it seems that large variations are allowed; however, where carefully interpreted, it turns out that these derivations also require small variations.) The method of quantitative feedback theory, not covered in this text, speaks more directly to the situation where large parameter-variations are accounted for in design.

Proposition 3.11. *The Bode robustness measure can be rewritten as*

$$S_{G_p}^{G_{cl}} = \frac{\partial Lm G_{cl}(j\omega)}{\partial Lm G_p(j\omega)} \quad (3.109)$$

the partial derivative of the log-magnitude of the closed loop transfer function with respect to the log-magnitude of the open loop transfer function.

Proof:

$$\begin{aligned} S_{G_p}^{G_{cl}} &= \frac{\partial G_{cl}(j\omega)/G_{cl}(j\omega)}{\partial G_p(j\omega)/G_p(j\omega)} \\ &= \frac{\partial \ln G_{cl}(j\omega)}{\partial \ln G_p(j\omega)} \end{aligned} \quad (3.110)$$

Now multiply numerator and denominator by $20/\ln(10)$ to convert each to the log-magnitude function. ■

How does the Bode measure relate to our closed-loop transfer functions? We start with the definition to the closed-loop transfer function

$$G_{cl} = \frac{G_c G_p}{1 + G_c G_p H}$$

so that the Bode measure, denoted $S_{G_p}^{G_{cl}}$, is given by

$$\begin{aligned} S_{G_p}^{G_{cl}} &= \frac{G_p}{G_{cl}} \cdot \frac{\partial G_{cl}}{\partial G_p} \\ &= \frac{G_p(1 + G_c G_p H)}{G_c G_p} \cdot \frac{\partial}{\partial G_p} \left[\frac{G_c G_p}{1 + G_c G_p H} \right] \\ &= \frac{(1 + G_c G_p H)}{G_c} \cdot \left(\frac{(1 + G_c G_p H)G_c - G_c^2 G_p H}{(1 + G_c G_p H)^2} \right) \end{aligned}$$

Finally,

$$S_{G_p}^{G_{cl}} = \frac{1}{1 + L} = S \quad (3.111)$$

Notice the correspondence of this Bode robustness measure to the closed-loop transfer function presented in the previous sections. In fact, this correspondence is the reason that the closed-loop transfer function is called the “sensitivity”.

From the above, we have established:

Proposition 3.12. *The Bode robustness measure depends only upon the return difference, $1 + L(j\omega)$. Performance robustness is achieved at those frequencies where the gain of the return ratio is large, at those frequencies where error is suppressed.* ■

We have established that noise is attenuated at the output only where the command is not attenuated at the error node. It follows that noise attenuation must also be traded for performance robustness in the design process.

Next we consider parameter variation in the feedback path, in the measurement dynamics characterized by $H(s)$. Here the appropriate sensitivity is

$$\begin{aligned}
 S_H^{G_{cl}} &= \frac{H}{G_{cl}} \cdot \frac{\partial G_{cl}}{\partial H} \\
 &= \frac{H(1 + G_c G_p H)}{G_c G_p} \cdot \frac{\partial}{\partial H} \left[\frac{G_c G_p}{1 + G_c G_p H} \right] \\
 &= \frac{H(1 + G_c G_p H)}{G_c G_p} \cdot \left(\frac{-(G_c G_p)^2}{(1 + G_c G_p H)^2} \right) \\
 &= \frac{-G_c G_p H}{1 + G_c G_p H}
 \end{aligned} \tag{3.112}$$

Finally, the sensitivity to variations in the sensor is

$$S_H^{G_{cl}} = \frac{-L}{1 + L} = -T \tag{3.113}$$

the negative complimentary sensitivity! Once again the way in which $T(s)$ “compliments” $S(s)$ is made clear.

Proposition 3.13. *Performance robustness with respect to variation of parameters in the sensor cannot be achieved at the same frequencies as is the performance robustness with respect to parameter variation in the controlled object.* ■

Normally, $\text{Lm } T(j\omega)$ is near 0 dB at low frequencies and drops off at higher frequencies. This means that the sensor sensitivity is high at low frequencies (-1 dB/dB) and near the static gain condition and then improves at higher frequencies.

3.7.4 Graphical Interpretation of Return, Sensitivity, and Complimentary Sensitivity:

The return difference, $1 + L$, appears in the denominator of every closed-loop transfer function and in both Bode robustness measures, so it deserves a special notation and interpretation. We develop a graphical interpretation here. As side benefits, we develop graphical interpretations of the sensitivity and complimentary-sensitivity rational functions.

The interpretation is demonstrated in Figure 3.22, which is a picture of the L -plane. The complex number from the origin to any point in the plane is $L(j\omega)$ and the complex number from the -1 point to the polar plot of $L(j\omega)$ is $1 + L(j\omega)$, the return difference. That is because where this complex number is added to -1 , the result must be L .

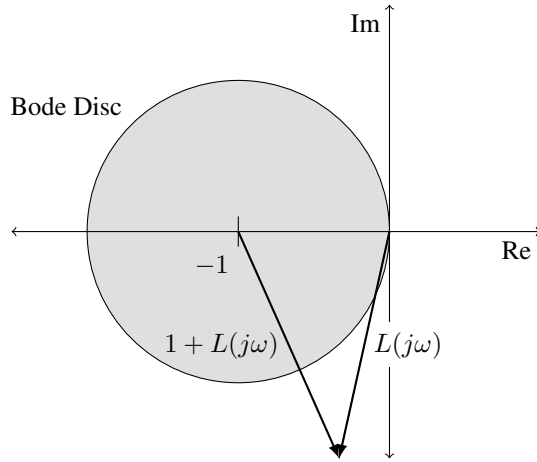


Figure 3.22: Graphical interpretation of the return ratio and return difference.

This simple interpretation provides insight into control-system design. Those frequencies where the polar plot of $L(j\omega)$ approaches the -1 point are those frequencies where the command signal is not attenuated at the error node and those frequencies for which performance robustness is not achieved. A more precise statement is the sensitivity, $S(j\omega)$, is greater than 1 dB per dB at those frequencies where the polar plot enters the unit circle centered at the -1 point in the L -plane.

Definition 3.11. Call the area inside the unit circle, centered at the -1 point in the complex L -plane, the Bode disc.

Incidentally, a MATLAB script for generating the Bode disc is

```
>> xdisc = 2*[logspace(-4,-1),logspace(-1,0)];
>> ydisc = sqrt(2*xdisc-xdisc.*xdisc);
>> mdisc = 10*log(xdisc.*xdisc+ydisc.*ydisc)/log(10)
>> for k = 1:100;
>> pdisc(k) = 180*(-pi+atan(ydisc(k)/xdisc(k)))/pi;
>> end;
```

The arrays “mdisc” and “pdisc” are the coordinates of the Bode disc on the $Lm G$ chart.

These arrays are used in a statement of the form

```
>> plot(phase,mag,pdisc,mdisc)
```


2615 where arrays “phase” and “mag” are the Lm G chart coordinates of some transfer function.
 2616 This call only display the disc from -90° to -180° . To display the disc from -90° to
 2617 -270° , generate

```
2618 >> pdiscp = -360*ones(100,1)-pdisc;
```

2619 and use the plot-command

```
2620 >> plot(phase,mag,pdisc,mdisc,pdiscp,mdiscp)
```

2621 The reader may decide to retain “xdisc” and “ydisc” rather than to clear them. These arrays
 2622 can be used to generate the Bode disc on the polar plot or Nyquist diagram with the call

```
2623 >> plot(-xdisc,ydisc,-xdisc,-ydisc)
```

2624 Next, we carefully interpret (3.104)

$$S(j\omega) + T(j\omega) = 1$$

2625 This relation places a constraint on any feedback design. For one thing, it means $|S(j\omega)|$
 2626 and $|T(j\omega)|$ cannot both be $\ll 1$ (0 dB) at the same frequency, although there are situations
 2627 where we would have preferred that this not be true. There are some tricky aspects to
 2628 the interpretation of this fundamental constraint because it is a relation between complex
 2629 numbers. For example, contrary to one’s first guess, (3.104) does not mean that $|S(j\omega)|$
 2630 and $|T(j\omega)|$ cannot both be $\equiv 1$ (0 dB) at the same frequency. Consider the vector diagram
 2631 in Figure 3.23, which is consistent with (3.104):

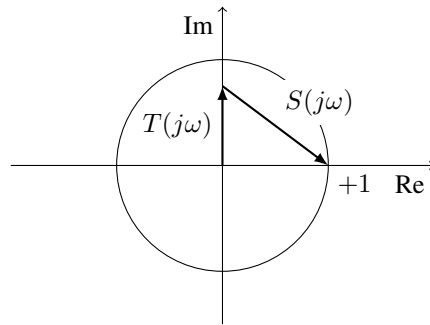


Figure 3.23: Potential S and T yielding $|S| > 1$.

2632 Clearly, $|S(j\omega)| = \sqrt{1 + |T(j\omega)|^2} > 1$. This is true even where $|T(j\omega)| > 1$: That
 2633 is, we can have the highly undesirable situation where both the sensitivity and complimen-
 2634 tary sensitivity magnitudes are both greater than 1 (0 dB). In fact, we find that this sorry

situation will always arise in the case of non-minimum phase objects if we are not careful to limit the closed-loop bandwidth. An extreme case is illustrated by the vector diagram in Figure 3.24, where the argument of the complimentary sensitivity is -180° :

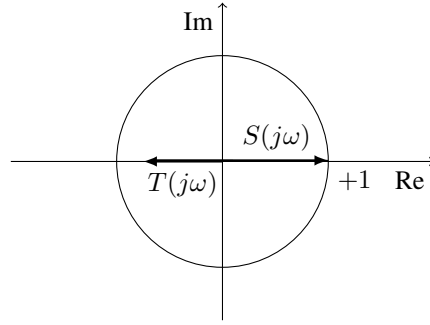


Figure 3.24: Potential S and T where $\angle T = -180^\circ$.

Obviously,

$$|S(j\omega)| = 1 + 2|T(j\omega)|$$

in this case. It is these diagrams and our desire to produce small values of $|S(j\omega)|$ and $|T(j\omega)|$ that indicate the importance of the phase of $|S(j\omega)|$ and $|T(j\omega)|$.

3.7.5 Weighting Factors and Performance Robustness:

One of the better-explored, neo-classical design methods is to establish upper bounds for the sensitivity and complimentary sensitivity. In this procedure, the designer writes

$$|T(j\omega)| \leq |W_{PT}(j\omega)| \quad (3.114)$$

and

$$|S(j\omega)| \leq |W_{PS}(j\omega)| \quad (3.115)$$

The “P” subscript means these constraints are for “performance” criteria. Here, the right sides, called weighting factors, are eventually based upon interpretations of the design criteria; for example, $|W_{PT}|$ is chosen to drop off sharply at high frequency and $|W_{PS}|$ is selected to be very low at low frequency. In addition, these weighting factors are chosen to constrain the peak values attained by $T(j\omega)$ or $S(j\omega)$ at any frequency (the so-called ∞ -norms).

The ability to make a proper choice of a weighting factor comes with practice. While a weighting factor could be selected to take any arbitrary mathematical form, there are

good reasons for taking them to be the Bode magnitudes of rational functions. In addition, it is useful to choose these rational functions to be proper with poles strictly in the left half-plane. That is to say, we select the weighting factors to be the Bode magnitudes of realizable, stable transfer functions. A common choice is the magnitude of the quadratic factor, which we explored in the previous Chapter. The reason for this becomes clear in later chapters.

It will also become clear that it is useful to redefine these constraints as

$$|W_{PT}(j\omega)T(j\omega)| \leq 1 \quad (3.116)$$

and

$$|W_{PS}(j\omega)S(j\omega)| \leq 1 \quad (3.117)$$

At this point, we choose to only make some interpretations. First of all, we translate the bound $|S|$ to a bound on $|L|$. Clearly, (3.115) can be re-written as

$$\frac{|W_{PS}(j\omega)|}{|1 + L(j\omega)|} \leq 1$$

or

$$|1 + L(j\omega)| \geq |W_{PS}| \quad (3.118)$$

The graphical interpretation of this is: At each frequency ω , the polar plot of the return ratio, $L(j\omega)$, must lie outside the circular disk centered at -1 with radius $|W_{PS}|$. This is remindful of the graphical interpretation of the Nyquist stability criterion (explained in the next Chapter).

Suppose that we extend this procedure by bounding the Youla parameter as well (after all, we reduce the vulnerability of the control system in that way):

$$\frac{|W_{PY}(j\omega)|}{|Y(j\omega)|} \leq 1 \quad (3.119)$$

However, $Y = T/G_p$, so this condition reduces to (3.116) with

$$W_{PT} = W_{PY}/G_p \quad (3.120)$$

The approach here is to take any original weighting on $|T|$, call it W'_{PT} , and compare it with W_{PY}/G_p and then take the new bound on $|T|$ to be the minimum of $|W'_{PT}|$ and $|W_{PY}/G_p|$ at each frequency.

2675 3.8 The Saturation Constraints

2676 It might appear that some design considerations are outside the realm of the feedback
 2677 designer. For example, it seems, to this point in our development, that power considerations
 2678 or the sizing of actuators, such as pumps or motors, and plants are left to others. The
 2679 impression is that these results are merely relayed to the feedback designer, after the fact,
 2680 in the form of the controlled-object transfer function, $G_p(s)$. This is far from the truth; in
 2681 fact, basic physical constraints are a fundamental part of the feedback design. If that were
 2682 not so, the academic designs produced on our sketch pads and desktops would likely be of
 2683 no value. This important matter is the subject of this section.

2684 These other considerations are usually expressed in the feedback design as the consid-
 2685 eration of a possible saturation of the controlled input, $u(t)$, in response to an input signal,
 2686 n , d , or r . (If the saturation were due to the sensor noise, the result would be especially
 2687 embarrassing because the sensor is added only for purposes of feedback.) In this section,
 2688 we learn that this consideration tends to limit the bandwidth (hence the speed of response)
 2689 of the closed-loop system.

2690 3.8.1 Bandwidth and Response Time:

2691 A somewhat slippery term in the control literature is that of bandwidth. “Slippery” because
 2692 the definition is not given in a consistent manner (if it is given at all) and the meaning and
 2693 use of the term is somewhat vague. Still, the term is useful and an attempt should be made
 2694 to describe it.

2695 The idea is to convey an estimate of the length of a frequency range which is important
 2696 on the Bode-magnitude curve of some dynamic system. This length is, roughly speaking,
 2697 proportional to the speed with which the system responds to, say, the unit step input.

2698 Of course, it is possible to give a precise mathematical meaning to bandwidth and we re-
 2699 produce and develop one such definition in Appendix B. There, ω_c denotes the bandwidth.
 2700 At that same place, we denote the effective duration of a transient T_r and demonstrate that

$$\omega_c \cdot T_r > 1/4 \quad (3.121)$$

2701 Clearly, the duration of a transient is reduced by increasing the bandwidth.

2702 For present purposes, we choose to give a description instead of a precise definition:
 2703 The bandwidth is that range of positive frequency bounded by those frequencies where
 2704 the Bode-magnitude falls to 3 dB below the maximum magnitude. Sometimes, 6 dB or
 2705 12 dB are used instead and so we will refer to the -3 dB- or to the -6 dB-bandwidth.

Additionally, only positive frequencies are used in this convention so that, as is often the case where the Bode diagram is monotonically decreasing, the bandwidth is taken to lie from the zero frequency to the -3 dB (or the -6 dB) frequency. We denote the bandwidth as “ ω_B ” (to distinguish it from a break frequency ω_b) and notice that -6 dB corresponds to a gain of $1/2$ and -3 dB corresponds to a gain of $1/\sqrt{2}$.

The relationship of bandwidth to speed of response is established in this subsection for two special cases. Begin with the first-order system whose Bode modulus is described by Equation (2.138). Clearly, the -3 dB bandwidth (the frequency where the magnitude is $1/\sqrt{2}$) is

$$\omega_B = 1/\tau \quad (3.122)$$

Higher bandwidth is related to smaller τ and faster system response to, say, step inputs.

Next, consider the quadratic factor whose Bode modulus is (2.149). A formula for bandwidth can be derived but it is overly complex for most purposes, so we consider the high-frequency asymptote instead. This asymptote is obtained by setting the condition $\omega \gg \omega_n$ to obtain

$$\left[\frac{\omega_n}{\omega} \right]^2$$

This approximation drops to the -6 dB value at

$$\omega_B = \sqrt{2} \cdot \omega_n \quad (3.123)$$

It follows (2.102, 2.103) that, at a fixed value of damping ratio, the rise and settling times diminish as bandwidth is increased.

3.8.2 The Youla Parameter and Saturation:

Consider the problem of attenuating disturbance, noise, and command at the node we call the “power-stage” or “controlled-input”, $u(t)$. Consider Figure 3.5 and obtain the closed-loop transfer function from the sensor noise to the controlled input,

$$\begin{aligned} \frac{\hat{u}(s)}{\hat{n}(s)} &= \frac{-G_c(s)}{1 + L(s)} \\ &= \frac{-1}{1 + L(s)} \left[\frac{L(s)}{G_p(s)} \right] \\ &= -Y(s) \end{aligned} \quad (3.124)$$

the “Youla parameter”. Similarly, the transfer function from the output disturbance signal to the controlled input is

$$\frac{\hat{u}(s)}{\hat{d}_y(s)} = -Y(s) \quad (3.125)$$

Finally,

$$\begin{aligned} \frac{\hat{u}(s)}{\hat{r}(s)} &= \frac{1}{1 + L(s)} \left[\frac{L(s)}{G_p(s)} \right] \\ &= Y(s) \end{aligned} \quad (3.126)$$

These relations can be rewritten in terms of another closed-loop transfer function, the “complimentary sensitivity”

$$Y(s) = \frac{T(s)}{G_p(s)} \quad (3.127)$$

The designer needs to know as much as possible about the quantitative properties of the noise signal in order to evaluate the effect of noise on the controlled input. In lieu of that, Equations (3.123, 3.124) can be used to obtain some useful observations. We first make some inferences about the closed-loop transfer functions, $Y(s)$, and then translate these observations into conditions and constraints on the return ratio, $L(s)$.

We take as axiomatic that $|L(j\omega)|$ will be chosen to be much greater than one in some range of frequency and much less than one in another. The exception is the neighborhood of the cross-over frequency where this Bode magnitude is ≈ 1 . This approach is certainly justified by the developments of prior sections.

Where $|L(j\omega)| \gg 1$ (at low frequencies), we have that

$$\begin{aligned} Y(j\omega) &\approx L/(LG_p) \\ &= 1/G_p(j\omega) \end{aligned}$$

Thus,

$$\text{Lm } Y \approx -\text{Lm } G_p \quad (3.128)$$

at low frequency (usually at frequencies below cross-over).

Where $|L(j\omega)| \ll 1$ (usually at high frequencies), we have

$$Y(j\omega) \approx L(j\omega)/G_p(j\omega)$$

So,

$$\text{Lm } Y \approx \text{Lm } L - \text{Lm } G_p = \text{Lm } G_c \quad (3.129)$$

Interpret this result to mean that $-\text{Lm } G_p$ is the low frequency asymptote of $\text{Lm } Y$ and $\text{Lm } L - \text{Lm } G_p$ is the high frequency asymptote. These asymptotes intersect at that frequency where $-\text{Lm } G_p = \text{Lm } L - \text{Lm } G_p$. That is, the frequency where $\text{Lm } L = 0\text{dB}$, which is the cross-over frequency ω_c of the return ratio.

A typical situation is illustrated in Figures 3.25.

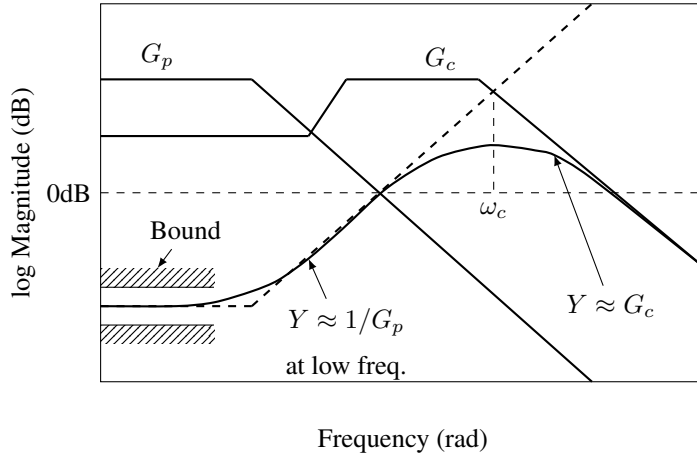


Figure 3.25: Youla design with high-gain plant.

We have the obvious implications where the cross-over frequency of the return ratio is higher than that of the controlled object:

Proposition 3.14. *Below the cross-over frequency of the controlled object, the tendency of the controlled input, $u(t)$, to saturate is dictated by the transfer function of the controlled object and is not affected by the feedback.* ■

and

Proposition 3.15. *Above the cross-over frequency of the return ratio, the tendency of the controlled input to saturate is dictated by Bode modulus of the return ratio less the Bode modulus of the controlled object (Bode modulus of the compensator) and, so, is affected by the feedback.* ■

and

Proposition 3.16. *Between the cross-over frequency of the controlled object and the return ratio, the tendency of the controlled input to saturate makes the transition from being independent of feedback to being dependent.* ■

The conclusions are modified where the cross-over frequency of the return ratio is set lower than cross-over of the controlled object (the closed loop responds slower than the open loop).

We expect the reference and the disturbance to be low frequency and we conclude that there is nothing the feedback designer can do about saturation to these inputs! This is a task for the designer of the controlled object: that person may be asked to increase the open-loop gain. It is intuitively satisfying that a saturation of this nature is due to the fact that the controlled object is not “powerful” enough.

This begs the question: is there a benefit to simultaneous design of the controlled object and the compensator? After all, a controlled object is a mechatronics system, where mechatronics systems are synergistic systems that include mechanical, electronics, computer, telecommunications and control systems. We don’t pursue answering this question in this book, but we should say that there are recent research studies 17.2 that showed the simultaneous design of a controlled object and a compensator could decrease the demand on requiring a more powerful controlled object.

On the other hand, the sensor noise is usually high frequency and something can be done about saturation in this frequency range. That “something” is to design the feedback loop so that $L(j\omega) \approx G_p(j\omega)$ after cross over. This has direct consequences when designing feedback loop using neo-classical technique with Youla parameter as will be discussed in the next chapter.

If the cross over frequency of the return ratio is much greater than that of the controlled object, $L_m L$ will be large at the latter frequency and the feedback system is vulnerable to saturation. Saturation in response to any of the three inputs with content in this frequency range. The last two conclusions dictate that the return ratio must be shaped close to the controlled object after the crossover frequency of the controlled object.

This means that:

The closed-loop system cannot respond much faster than the controlled object! Major improvements in the speed of response can be obtained only with a redesign of the controlled object, as improvements in the cross-over frequency of G_p . This usually involves a boost in power or some other property of the actuators.

Think of it this way: feedback is an information process which insures the automatic system will adjust for disturbance or model uncertainty: Feedback can never be used to circumvent, say, the conservation of mass or energy. Another way to think of this is that a premium is placed on high-bandwidth controlled objects because a faster-responding

feedback-loop can be designed for such objects. Also, the value of a highly accurate and stable sensor is apparent because, in that case, the designer can allow greater deviations of $\text{Lm}(L)$ above $\text{Lm}(G_p)$, greater values of $\text{Lm}(Y)$ at the higher frequencies.

3.9 Problems

3.1 Consider a closed-loop transfer function

$$T = \frac{100K(2.5s - 1)}{(10s + 1)(5s + 1)(2.5s + 1)}$$

that corresponds to a plant

$$G_p = \frac{100(2.5s + 1)}{(10s + 1)(5s + 1)(1.25s + 1)} \quad (3.130)$$

(a) What K will meet the interpolation condition of $T(0) = 1$ for command tracking?

(b) Verify the interpolation conditions for $S = 1 - T$.

3.2 For each controller/plant pair, indicate whether the closed loop system is internally stable and whether it is well-posed.

(a) $G_p = \frac{10(1 - s/2)}{(1 - s/2)}, G_c = \frac{1}{(1 - s/2)}$

(b) $G_p = \frac{1}{(0.25s + 1)^2}, G_c = \frac{\frac{3}{4}(s/4 + 1)^2}{s(s/6 + 1)}$

(c) $G_p = \frac{10(1 - s)}{(1 - s/2)}, G_c = (1 - s/2)$

3.3 Consider the unstable and non-minimum phase plant,

$$G_p = \frac{10(1 + s)}{(1 - s/2)}$$

Consider a Youla

$$Y = \frac{K(\tau_z s + 1)}{(\tau_1 s + 1)(\tau_2 s + 1)}$$

Check the interpolation conditions for this plant and Youla. Is it possible to achieve internal stability with the given Youla parameter? What ranges of τ_z, τ_1 , and τ_2 are required?

3.4 Consider the transfer function

$$G_p = \frac{(10 + s)}{s^3}$$

with proposed Youla

$$Y = \frac{s^3(\tau_z s + 1)}{(\tau + ps + 1)^4} \quad (3.131)$$

Will this Youla be able to meet the interpolation conditions on T ?

CHAPTER 4

FEEDBACK DESIGN FOR SISO: SHAPING, PARAMETRIZATION AND QUANTITATIVE FEEDBACK THEORY

Chapter 4 prologue goes here.

4.1 Closed-Loop Stability Under Uncertain Conditions

To this point, the concept of gain seemed the only important idea for feedback control because it is this parameter which determines the extent to which an input signal is attenuated at the nodes of the feedback loop. Also, it is the gains of the sensitivity functions that characterize the robustness. We demonstrate in this section that it is the phase curve *in combination* with the gain curve that determines the stability of the closed loop.

The use of high gain in the return ratio must be abandoned after some frequency. For one reason, the gain must be lowered in order to attenuate high frequency sensor noise; that is, in order to effect the basic trade-off. For another reason the gain must be lowered in order to establish closed-loop stability. It is the latter fact that is developed in this subsection. In particular, we study the extent to which the gain must be diminished at high frequency in order to obtain a robust stability.

In the final part of this section, we develop the implications of all this for $T(j\omega)$ and $S(j\omega)$ and in that way, the significance of the H_∞ norm is introduced.

The first task is obtain a stability limit condition, a condition written in terms of frequency-response characteristics. The simplest way to do that is to recognize that, at the stability limit, $s = j\omega$, is a closed-loop pole. It then follows from (3.35) (the closed-loop characteristic equation), that

$$1 + L(j\omega) = 0 \quad (4.1)$$

This stability limit condition is the basis of our study of stability in the frequency formalism.

4.1.1 Harmonic Consistency

There is another way to obtain the condition in Equation (4.1), a way that has an intuitive appeal.

Imagine the situation illustrated in Figure 4.1: a closed-loop system is at a stability limit so that, in steady-state, the output has the oscillatory form

$$y(t) = A \cos(\omega t) \quad (4.2)$$

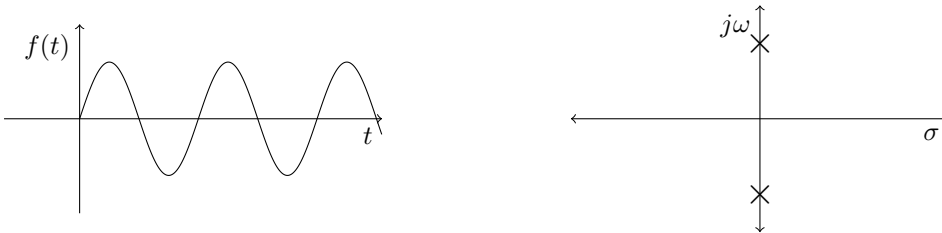


Figure 4.1: Marginally stable system.

for some amplitude “A” and some frequency “ ω ”. Each component in the feedback loop modifies its input with amplitude and phase changes dictated by the frequency response of that component’s transfer function. This sort of modification is shown after each block in Figure 4.1. There is one other important shift in phase: the negative feedback means that a minus sign, a $\pm 180^\circ$ phase shift occurs at the summing junction. The *principal of harmonic consistency* is that the amplitude and phase changes around the loop must be such that the signal leaving the controlled object must match up with the form (4.2) assumed for the output, that is,

$$M_p M_c M_H A = A$$

and

$$\phi_H + \phi_c + \phi_p \pm 180^\circ = 0 \quad (4.3)$$

These conditions can easily be rewritten in terms of the frequency response of the return ratio.

Theorem 4.1 *The stability-limit conditions for the closed loop are*

$$|L(j\omega)| = 1, \quad (4.4)$$

or

$$Lm L(j\omega) = 0 \text{ dB} \quad (4.5)$$

and the phase condition

$$\phi_L = \pm k180^\circ \quad (4.6)$$

where “ k ” is any integer. ■

The conditions of equations (4.4, 4.6) are equivalent to (4.1). Normally, but not always, the system is stabilized by reducing the gain from this condition. Mathematically, $\pm k180^\circ$ are the same angle and so, depending on the software package, one must be alert where any one of these phases occur near a $Lm L = 0 \text{ dB}$ condition. Normally the system is stabilized by keeping the phase less than $+180^\circ$ or greater (less negative) than -180° . Finally, one can be fooled by an uncritical use of this simple criterion: The wrong indication is given if a right half-plane cancellation occurs in the return ratio (more on this further on).

A more precise and a more complete statement of closed-loop stability, the *Nyquist Stability Criterion*, is discussed in the next subsection.

4.1.2 Nyquist Stability Criterion: Heuristic Justification:

There is a more satisfying way to study closed-loop stability, the Nyquist stability criterion. The rigorous proof of this criterion is not provided in this Chapter. However, the utility of the criterion is required at this point so we provide a more intuitive justification here, at a point where the required intricacies of complex analysis are not given.

Rewrite the rational form of the characteristic equation

$$1 + L(s) = 0 \quad (4.7)$$

by placing both terms over a common denominator and factoring both the numerator and denominator polynomials:

$$1 + \frac{\eta(s)}{\delta(s)} = \frac{k(s - p_1)(s - p_2)\dots(s - p_n)}{(s - \delta_1)(s - \delta_2)\dots(s - \delta_n)} = 0 \quad (4.8)$$

The “ δ_k ” are the open-loop poles and the “ p_k ” are closed-loop poles (the zeros of the characteristic equation).

Consider the situation illustrated in Figure 4.2 and imagine carrying the “test point” through one complete clockwise tour around the D-shaped contour which encloses the right half-plane. Clearly, the vector $s - p_k$ will not contribute a net rotation to $1 + L$ if p_k lies in the left half-plane — starting at the origin, for instance, θ_L increases from 0° to $+90^\circ$, then decreases to -90° , and returns to 0° . So, the net change in θ_L is zero. On the other hand there is a net clockwise rotation if p_k lies in the right half-plane if the closed-loop pole, p_k , is in the unstable region — the net change in θ_R will be -360° .

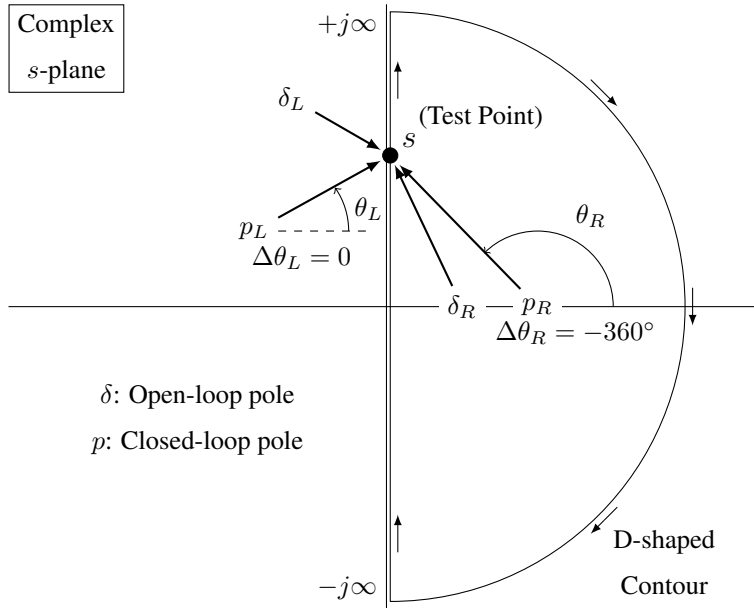


Figure 4.2: An intuitive justification of the Nyquist Stability Criterion. Imagine that the test point moves about the D-shaped contour in the clockwise direction.

The same thing can be said about the vector $s - \delta_k$ except that, since this is a denominator term, $s - \delta_k$ contributes a counter-clockwise rotation to $1 + L$ should the open-loop pole, δ_k , lie in the right half-plane.

Our summary is that the number of clockwise rotations of $1 + L$, as the test point makes one complete clockwise tour (that is, in the direction of increasing frequency) is $N_c - N_o$, where N_c is the number of unstable closed-loop poles and N_o is the number of unstable open-loop poles:

$$N = N_c - N_o \quad (4.9)$$

Here, a negative-integer value of N corresponds to a counter-clockwise rotation of $1 + L$.

The natural thing is to imagine these rotations as the motion of a vector drawn from the origin of the $1 + L$ plane. However, we prefer to imagine the motion of a vector in the L -plane (the polar or Nyquist plot). A rotation of $1 + L$ about its origin (the -1 point) is equivalent to the encirclement by L of the -1 point in the L -plane so we have the following stability criterion:

Theorem 4.2 (Nyquist stability criterion) *The number of unstable close-loop poles is*

$$N_c = N_o + N \quad (4.10)$$

where

$$N_o \triangleq \text{the number of unstable open-loop poles} \quad (4.11)$$

and

$$N \triangleq \text{the number of times the Nyquist locus encircles the } -1 \text{ point in a clockwise manner} \quad (4.12)$$

“ N ” is determined by moving along $L(j\omega)$ in the direction of increasing ω . ■

We assume here that the curved portion of the D-shaped contour maps to the origin of the L -plane; that is, we assume that L is strictly proper. Notice that if there are any unstable closed-loop poles, then we have closed-loop stability if and only if $N = -N_o$; that is, if and only if the Nyquist locus encircles the -1 point in a counter-clockwise manner a number of times exactly equal to the number of unstable poles in the open loop.

We must modify the D-shaped contour where a pole or zero is on the imaginary axis (in either case, there is a discontinuity in phase at that frequency). The usual procedure is to make a small semi-circular indentation in the contour at the complex pole or zero. The indentation is made into the left half-plane. Let “ ϵ ” denote the radius of the semicircle and imagine constructing the Nyquist locus of L on this closed contour. The parameter, ϵ , will have a geometric interpretation on this mapping. Finally, allow $\epsilon \rightarrow 0$ and draw the usual conclusions about stability from the asymptotic Nyquist locus.

It is now clear that the closed loop is stabilized in the manner indicated in Theorem 4.1 only if there are no *open-loop poles in the right half plane*. If the open loop is unstable (as is the case with high-performance aircraft) the closed loop is stabilized by modifying the frequency response of the return ratio, $L(j\omega)$, in quite another way.

How great should we make the adjustment for stability indicated in Theorem 4.1? What are the recommendations that have evolved in this regard? These questions are among those addressed in this subsection.

In the Bode diagrams of minimum-phase systems, an asymptotic phase of -90° is associated with each multiple of slope of -20 dB/decade in the gain curve and, asymptotically, $+90^\circ$ is associated with each multiple of slope of $+20$ dB/decade. This follows from the superposition of the frequency-response curves, as discussed in Chapter 2.

Combine equations (4.4, 4.5) with this principle to draw some conclusions about the closed-loop instability. The goal is to be able to predict instability by examining only the gain curve on the Bode diagram. This can only be done for minimum-phase return ratios because it is for these ratios that this principle can be used to draw the required inferences about phase.

A minimum-phase feedback system is close to the stability limit condition if the gain crosses the 0 dB line with a slope of -40 dB/decade. Normally, a slope more negative than this indicated instability while a slope of more positive than -40 dB/decade indicates stability. If the asymptotic phase has been realized at this cross-over frequency, then the phase margin is 90° .

4.1.3 Stability Margins and Stability Robustness

: An important practical observation must be injected here. It is impossible to have perfect confidence in the predictions of stability made in this way because mathematical models are always an approximation of the truth. Our next explorations of feedback design are based upon that observation. In particular, we wish to provide a closed loop response that is stable even in face of the inevitable uncertainties.

Mathematical models are always an approximation of the truth. This is due to necessity, convenience, or ignorance. It will always be thus. Also, inevitable shortcuts are made in the development of a model; for example, one such shortcut is to neglect bilateral coupling effects. This means that even the true order of the controlled object (the number of poles) is underestimated and uncertain. Usually, the effect can be characterized as an uncertainty in the high-frequency portion of the Bode (or Nyquist) diagram. Since we know next to nothing about this uncertainty, except for its inevitable existence, we call it *unstructured uncertainty*. Another name is parasitic dynamics because it is “parasitic” to the stability of high gain feedback.

Even where we assume the model is of the correct order, we are unsure of the true values of parameters. This is due to uncertainty in measurement or due to the actual change of parameter values due to wear, to warm-up, or due to changes in operating conditions. This kind of uncertainty is different than unstructured uncertainty because, in this instance, we can bound the uncertainties in the magnitude and phase of $G_p(j\omega)$ given estimates of the parameter variations. Also, the bound on phase uncertainty are functions of the bounds on gain uncertainty (with frequency as the intermediate variable). For this reason, we refer to this as *structured uncertainty*.

A special concern is the unstructured uncertainty, for which we must install some kind of factor-of-safety for stability. We begin that development with the following:

Definition 4.1. The phase margin, ϕ_m , is related to a specific value of the phase, ϕ , of $L(j\omega)$; namely, the phase of the return ratio at the cross-over frequency, ω_c :

$$-180^\circ + \phi_m = \phi(\omega_c) \quad (4.13)$$

The gain margin is denoted K_m and is defined by

$$K_m = 1/|L(j\tilde{\omega})| \quad (4.14)$$

where $\tilde{\omega}$ is that frequency where the phase of L is exactly $\pm 180^\circ$. The two margins are referred to, collectively, as *stability margins*.

In some cases, there may be several cross-over frequencies and several values for the stability margins. Also, a particular software package may return $+180^\circ$ where the convention is to use -180° . In that case, the phase margin is

$$\phi_m = 180^\circ - \phi(\omega_c) \quad (4.15)$$

Gain and phase margins are easily visualized and read from the common frequency-response diagrams. For example, the definitions are illustrated in Figure 4.3 for the polar plot. The margins, first of all, indicate how close our design is to satisfying one of the stability limit conditions (4.4, 4.5) and, as such, can be looked upon as factors of safety against instability. Much more is indicated by these stability margins and that is drawn out later.

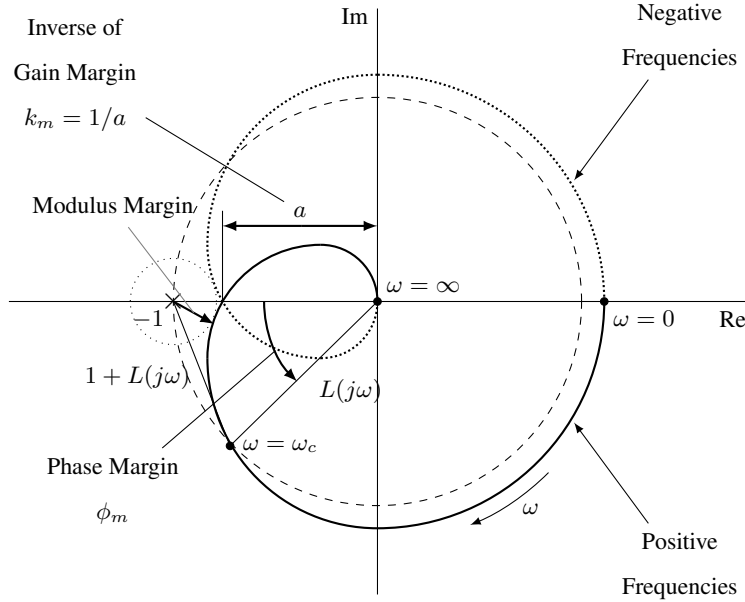


Figure 4.3: Nyquist plot with stability margins.

Another illustration in Figure 4.3 is the intersection of the Bode disc with the unit circle about the origin. It is a simple problem in trigonometry to show that this point of intersection makes an angle of 60° with the real axis of the complex plane L-plane. For various reasons, this value of ϕ_m is the lower bound usually recommended for phase margin: It is recommended that

$$\phi_m \geq 60^\circ \quad (4.16)$$

An important part of any feedback design is to assure minimum values for the gain and phase margins.

We have demonstrated that high gain is desirable at any frequency in order to provide input attenuation and robustness. This means that the steepest possible slopes in the gain curve near cross-over, the -40 dB/decade slopes, are desirable and that should be kept in mind when selecting the compensator.

4.1.4 Margins, $T(j\omega)$ and $S(j\omega)$, and H_∞ Norms (Relationships Between Classical and Neoclassical Approaches)

There is an interesting relationships between the stability margins and the magnitude of the closed-loop transfer functions, $T(j\omega)$ and $S(j\omega)$. Also, there are interesting relationships at the cross-over frequency of the return ratio, $L(j\omega)$.

The cross-over frequency — the frequency at which $|L(j\omega)| = 1$ — is denoted ω_c and the phase margin, ϕ_m , is defined such that $-180^\circ + \phi_m$ is the argument of the return ratio at this frequency. That is,

$$\begin{aligned} L(j\omega_c) &= 1 \cdot e^{j(-(\pi - \phi_m))} \\ &= \cos(\pi - \phi_m) - j \sin(\pi - \phi_m) \\ &= -\cos(\phi_m) - j \sin(\phi_m) \end{aligned} \quad (4.17)$$

2987 Next we observe that, since $T = LS$ and $|L(j\omega_c)| \triangleq 1$,

$$|T(j\omega_c)| = |S(j\omega_c)| \quad (4.18)$$

2988 That is, the intersection of the sensitivity and complimentary sensitivity magnitude curves
2989 occurs at the cross-over frequency of the return ratio.

2990 Equation (4.17) means that the magnitude of the sensitivity is

$$\begin{aligned} |S(j\omega_c)| &= \frac{1}{|1 + L(j\omega_c)|} \\ &= \frac{1}{|1 - \cos(\phi_m) - j \sin(\phi_m)|} \\ &= \frac{1}{\sqrt{[1 - \cos(\phi_m)]^2 + \sin^2(\phi_m)}} \\ &= \frac{1}{\sqrt{2 - 2\cos(\phi_m)}} \end{aligned}$$

2991 At this point, we turn to a table of trigonometric identities, ($\cos(2\theta) = 1 - 2\sin^2(\theta)$), to
2992 establish that

$$\left| S(j\omega_c) \right| = \frac{1}{2 \sin(\phi_m/2)} = \left| T(j\omega_c) \right| \quad (4.19)$$

2993 That a relationship between ϕ_m and $|T|$ should exist is probably obvious from an examina-
2994 tion of Figure 4.3, but the simplicity of the relationship is a surprise. This relationship is
2995 plotted in Figure 4.4.

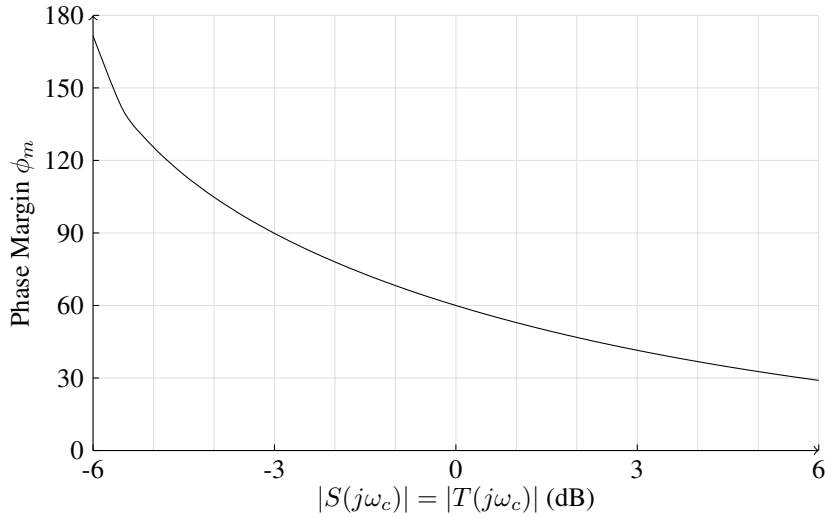


Figure 4.4: The phase margin from the intersection of $|S(j\omega_c)|$ and $|T(j\omega_c)|$ (valid only if G_p is minimum phase).

2996 Consider Figure 4.5 Clearly, the cross-over frequency is where $|L| = 1$; that is, where
 2997 $L(j\omega)$ crosses the unit circle. The value of $|S(j\omega_c)| = |T(j\omega_c)|$ is the ratio of “1” to the
 2998 length $|1 + L|$.

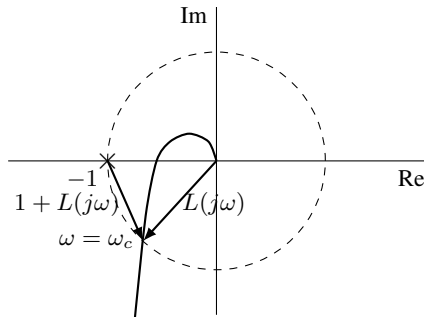


Figure 4.5: Crossover frequency on Nyquist plot.

2999 This observation allows us to conclude with the following:

3000 **Proposition 4.1.** *The phase margin is $> 60^\circ$ if $T(j\omega)$ intersects $S(j\omega)$ at a magnitude*
 3001 *below 1 (0 dB) and is $< 60^\circ$ if $T(j\omega)$ intersects $S(j\omega)$ at a magnitude above 1 (0*
 3002 *dB).* ■

Next, let $\tilde{\omega}$ denote that frequency at which the argument of the return ratio is $\pm 180^\circ$ (where the Nyquist locus crosses the negative real axis). By definition, the Bode-magnitude at this frequency is $1/K_m$ (Figure 4.3), the reciprocal gain-margin. That is to say,

$$L(j\tilde{\omega}) \triangleq -1/K_m \quad (4.20)$$

This leads to the conclusion that

$$T(j\tilde{\omega}) = \frac{1}{1 - K_m} \quad (4.21)$$

so

$$|T(j\tilde{\omega})| = \frac{1}{|K_m - 1|} \quad (4.22)$$

Similarly,

$$|S(j\tilde{\omega})| = \frac{K_m}{|K_m - 1|} \quad (4.23)$$

We usually assume that the closed loop is stable, that $K_m > 1$. The significance of $|T(j\tilde{\omega})|$ for stability robustness is among the features illustrated in Figure 4.5. Clearly,

$$\text{Lm } S(j\tilde{\omega}) = \text{Lm } T(j\tilde{\omega}) + \text{Lm } K_m \quad (4.24)$$

The question arises, how do we find $\tilde{\omega}$ on say, a $T - S$ diagram? (Clearly, $\tilde{\omega}$ is the frequency where $\arg(L(j\omega)) = -180^\circ$ on an L-diagram.) The answer is supplied by an examination of (4.21): If we assume that $K_m > 1$, then $\arg(T(j\tilde{\omega})) = -180^\circ$. That is we seek the frequency where this argument obtains and the gain margin, in dB, is the distance of the sensitivity above the complimentary sensitivity (in log-magnitude).

Incidentally, there is an alternative to the use of these classical stability margins, an alternative in which only a single number is needed to account for unstructured uncertainty.

Neoclassical Approach: Although gain margins and phase margins are useful measures of stability, they do not paint a complete picture regarding stability. These margins measure the amount of gain *or* phase change a system can handle before going unstable, but they do not say anything about how *simultaneous* gain and phase changes can destabilize a system. For instance, consider the Nyquist locus shown in Figure 4.6. This system has an infinite gain margin and has a phase margin near 90° , yet a small amount and gain and phase change at the same time could easily cause the locus to encircle -1 and make the system unstable.

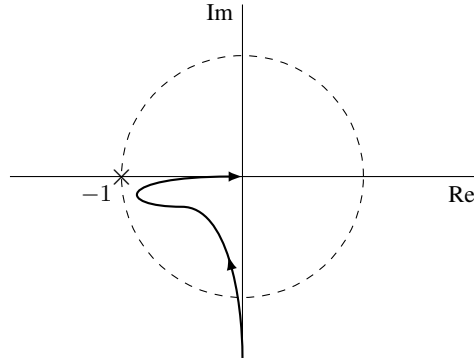


Figure 4.6: Nyquist plot.

For this reason, neoclassical modeling of uncertainties uses a single number to account for unstructured uncertainty.

Definition 4.2. The stability margin, M_2 , is the minimum distance from the plot to the -1 point in the $L(j\omega)$ -plane, as illustrated in Figure 4.3.

It is easy to prove the following:

Proposition 4.2. The M_2 -margin is related to the return difference by

$$M_2 = \min_{\omega} |1 + L(j\omega)| \quad (4.25)$$

and to the sensitivity by

$$1/M_2 = \max_{\omega} |S(j\omega)| \quad (4.26)$$

■

Currently, this measure is gaining favor over the classical phase and gain margins. For one thing, this number is easy to pick off the $S(j\omega)$ curve and, for another, it is a reliable measure of just how close the $L(j\omega)$ locus comes to the -1 point. As discussed earlier, in contrast, it is conceivable that the classical measures could be adequate and the locus still comes dangerously close to the critical point, as shown in Figure 4.6. And besides, by forcing the locus to lie outside some circle about -1, de facto values of phase and gain margin are imposed on the design.

EXAMPLE 4.1

Now, consider a feedback loop for which that maximum magnitude of the sensitivity function, $\max |S(j\omega)|$ is less than 2 (where a magnitude of 2 is roughly 6 dB). Then, show the classical gain margin will be at least 2.0 (6 dB) and the phase margin will be at least 29° .

$$\max_{\omega} |S(j\omega)| < 2$$

Then, from Equation (4.26),

$$M_2 > \frac{1}{2}$$

Then, we can write

$$\begin{aligned} |S(j\tilde{\omega})| &\leq \max_{\omega} |S(j\omega)| \\ \frac{k_m}{k_m - 1} &\leq \frac{1}{M_2} < 2 \\ k_m &< 2(k_m - 1) \\ k_m &> 2 \approx 6\text{dB} \end{aligned}$$

Therefor, if $M_2 > 1/2$, then the gain margin has a lower bound of 2 (about 6 dB). The derivation of the phase margin is left as an exercise.

We can follow this same procedure to find other bounds on the gain margin. For instance, if $\max |S(j\omega)| < 4$ dB, then we can find that $k_m > 2.71$ or approximately 8.7 dB and the phase margin will be at least 36.8° . And where $\max |S(j\omega)| < 3$ dB, the gain margin will be greater than 3.42 (10.7 dB) and the phase margin will be at least 41.5° .

The denominator of every closed-loop transfer function is the return difference so we have

Proposition 4.3. *The frequency that corresponds to M_2 is a closed-loop, resonant frequency.* ■

A new norm is introduced in Chapter 5, the H_∞ -norm for rational functions of a complex variable (or in short ∞ -norm):

$$\|G(s)\|_\infty \triangleq \max_{\omega} |G(j\omega)| \quad (4.27)$$

The significance of this norm begins to become apparent by the following statement:

The H_∞ -norm of the sensitivity, $S(s)$, is the M_2 stability-margin and the frequency of this maximum is a closed-loop resonant frequency.

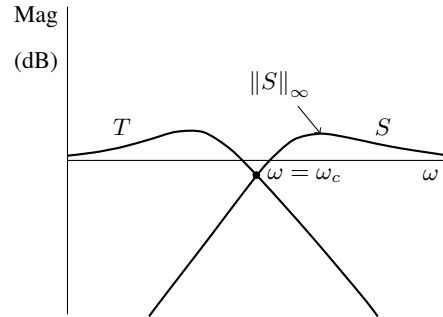


Figure 4.7: Crossover frequency on magnitude plot of S and T .

It should be noted that while this “norm” plays a pivotal role in the neo-classical theory, some of the significance of this number was already appreciated in the classical theory where the H_∞ -norm of $T(s)$ is called the “peak magnitude”. This number plays various purposes in feedback design: Clearly, the H_∞ -norm of $T(s)$ occurs at a closed-loop resonant frequency and can be related, in an approximate way, to the closed-loop damping ratio. The H_∞ -norms of T and S are illustrated in Figure 4.7.

As a result of these observations, it becomes clear that

we always seek to reduce the H_∞ -norms of both $T(s)$ and $S(s)$ in order to obtain sufficient damping, stability margin, and to reduce closed-loop resonances.

We will finish this chapter by discussing and bringing to the attention all the mathematical design constraints. These constraints heavily influence neo-classical feedback design covered in Chapter 4.

4.2 Mathematical Design Constraints

There are certain constraints with more of a mathematical nature than a technical. This is not to say that they are any less real; in fact, these mathematical constraints are equally important considerations in design. Some can be developed in this section but others can only be derived in Chapter 5 because some mathematical background is required for that task. However, it makes sense to at least catalog them here.

4.2.1 Sensitivity/Complimentary-Sensitivity Point-wise Constraints:

Classical feedback-design procedures focus on the return ratio, the open-loop transfer function, in order to achieve closed-loop objectives. Though seemingly indirect, these proce-

dures are quite effective and receive some attention in Chapter 4. In addition, there are the neoclassical procedures which focus directly on the closed-loop transfer functions, $T(s)$, $S(s)$, and $Y(s)$ themselves. These procedures are drawn-out in the next Chapters in this sequel, but this is a good point to introduce some definitions and propositions fundamental to those design methods. Knowing the result (3.104)

$$S(j\omega) = 1 - T(j\omega)$$

and combine this equation with $|S(j\omega)|^2 + |T(j\omega)|^2 = 1$ to establish the Bode-magnitude relations of the

Proposition 4.4.

$$|1 - |T(j\omega)|| \leq |S(j\omega)| \leq 1 + |T(j\omega)| \quad (4.28)$$

Similarly,

$$|1 - |S(j\omega)|| \leq |T(j\omega)| \leq 1 + |S(j\omega)| \quad (4.29)$$

■

For example, where we choose $S(0) = 0$, for good command-following and robustness to parameter changes in the controlled object, inequalities (4.28, 4.29) mean that $T(0) = 1$. This means that we will be susceptible to sensor parameter variation in this low-frequency range.

It follows from this proposition that the gain of either $S(s)$ or $T(s)$ will constrain the value of the other. Roughly speaking, one gain must be near unity where the other is small; for example, those frequencies where error is suppressed are those frequencies of noise which are not suppressed. Often, one can choose to suppress noise at the lower frequencies and attenuate the noise signal at higher frequencies since one normally expects sensor noise to be at relatively higher frequencies. Obviously, some designs can run into difficulty over this point. This elementary consideration is a good example of the tradeoffs that constitute feedback control design.

Other important constraints are the “interpolation conditions” expressed in (3.74, 3.75, 3.76, 3.77). Clearly, the $S(s)$ and $T(s)$ transfer functions are constrained to take on certain values in the complex plane and these constraints influence the values on the imaginary axis. It has been demonstrated that these constraints cannot be removed by a pole-zero cancellation where “z” or “p” lie in the right half-plane, where a factor of the return ratio is non-minimum phase. These positions place constraints on the values that the sensitivity

and the complimentary sensitivity can take on at specific frequencies and, as such, they impose limits to feedback design.¹

One final point on this matter: Where the gain of $L \approx 1$ (this occurs at the cross-over frequency) the gains of both $S(s) = 1/(1 + L)$ and $T(s) = L/(1 + L)$ are greater than 1/2 (-6 dB). Neither gain is very small in this range; that is to say

Feedback is not very effective near the cross-over frequency.

4.2.2 Sensitivity, Complimentary-Sensitivity, and Analytic Constraints

Non-Minimum Phase Constraints on Design: One can often improve a feedback design with the use of a pole-zero cancellation, a cancellation of a pole on the controlled object with a zero in a compensator. It is clear from the principle of internal stability that this is not an option where a pole or a zero lies in the right half-plane, where the controlled-object is characterized by a non-minimum phase zero or an unstable pole. Of course, even if the pole-zero cancellation is effected in the left half-plane, the bandwidth constraints, described in Chapter 3, place a practical constraint on what can be done. As a practical matter, the cancellation should not be effected in the case of stable poles where the canceled pole lies very near the imaginary axis since it may be excited by some disturbance signal. Where cancellation is impossible, improvement must be sought in some other, severely-constrained way.

In this subsection, that constraint is described in more detail. This discussion is aided where we employ factorization for the return ratio:

$$L(s) = L_{mp}(s)L_{ap}(s) \quad (4.30)$$

$L_{mp}(s)$ is a minimum-phase factor and $L_{ap}(s)$ is all-pass and this factorization will provide us with some tools to shape $L(s)$ in case of right half-plane pole or zero or pure delay element. The effect of the all-pass factor on $L(j\omega)$ is merely to contribute frequency-dependent phase-lag while not modifying the modulus in any way. Clearly, this effect is to reduce the stability margins so that a lower cross-over frequency must be selected in order to meet design constraints on stability robustness.

¹In contrast, we have constraints that relate the sensitivity in one frequency range to the sensitivity in another range. Freudenberg and Looze have named these constraints analytical constraints and they are developed in Chapter 6.

The case that is often described is that where the Lm L curve is monotonically decreasing. That argument is repeated here with admonition that things are a bit different where the curve is allowed a more general shape.

4.3 The Neoclassical Approach to Internal Stability

Classical and neoclassical approaches to feedback design are described in this chapter. The emphasis is on the neoclassical. It is likely that, in the 20th century, a major portion of actual designs were performed using classical techniques. The neoclassical principles, much like described here, hold out much promise.

The choice of the “shape” of a frequency-response curve in the G and or Lm G plane ins what we call a shaping design method. The curve being “shaped” is usually Bode magnitude of the Youla parameter, $|Y(j\omega)|$, or of the return ratio, $|L(j\omega)|$. The frequency, ω , is the independent variable in this process. In other words, “shaping” procedures are methods in which a frequency response is manipulated in order to satisfy the design criteria described in Chapter 3. The criteria are the usual: robustness, attenuated effects of unwanted inputs, stability margins, etc... Let’s also not forget improved speed of response and damping to the limited extent that is possible.

An important constraint is that of *internal stability* described in great detail in Chapter 3. That has to do with pole-zero cancellation; in particular, the cancellation of a pole in the right half-plane or on the imaginary axis with a zero in the product

$$L(s) = G_c(s)G_p(s) \quad (4.31)$$

Incidentally, if such a pole-zero cancellation occurs in G_p itself, then nothing can be done; that is:

A necessary condition for successful design of a feedback control is that all pole-zero cancellation in G_p occur strictly in the left half-plane.

This is a restatement of an old (“modern”) control concept: The requirement that an object be *detectable and stabilizable*.

Where “L” is being shaped (the classical method), internal instability is easy to avoid: Simply select G_c so that there is no cancellation of an unstable pole. In this classical approach, one draw inferences about the all-important closed-loop transfer functions, the *sensitivity*

$$S(s) = 1/(1 + L) \quad (4.32)$$

the *complimentary sensitivity*

$$T(s) = L/(1 + L) \quad (4.33)$$

and the *Youla parameter*

$$Y(s) = G_c/(1 + L) \quad (4.34)$$

$$= G_c S \quad (4.35)$$

$$= T/G_p \quad (4.36)$$

In that sense, the classical method is an *indirect method*.

The simple truth is that the superior methods are those associated with the Youla parameter. For one thing, there is much less cutting and fitting with the neoclassical methods. This procedure is a direct method in which the focus is on

$$T(s) = Y(s)G_p(s) \quad (4.37)$$

the neoclassical counter part of (4.31). We call this a “direct” method because both “Y” and “T” are closed loop transfer functions. Notice that *one can easily place closed-loop poles and zeros with the proper choice of Y(s) for a given G_p*. This sort of thing can also be accomplished with the “modern” methods but in an unnecessarily complicated way: The neoclassical procedures are highly transparent and require nothing more than an understanding of some basic mathematics.

Alternatively, one can choose “T” and infer “Y” with this equation. That is fine but remember that in order that Y(s) not be improper, *you must choose the relative degree (number of poles less the number of zeros) of T(s) to be no less than that of G_p(s)*. Also, an absolute requirement is that any nonminimum phase zero of G_p must be included in “T”.

Another appeal of the neoclassical approach is that we directly reduce the vulnerability of the controlled variable to saturation by selecting the Bode magnitude of the parameter, |Y(jω)|, to have low values at all frequencies if possible. In this way, we eliminate Horowitz’ criticism of most feedback-design procedures — the failure of most feedback design procedures to deal directly with bandwidth constraints. Another appealing aspect of this approach is that many other considerations can be automatically fixed during this selection of Y(s). These principles were highlighted in Chapter 3 and are illustrated with examples in this chapter.

In either the classical or neoclassical procedures, the goal is to provide low values of |S(jω)| at “low frequencies” and low values of |T(jω)| at “high frequencies”. “High”

and “low” with respect to the cross-over frequency, ω_c , the frequency where $|S(j\omega_c)| = |T(j\omega_c)|$. That frequency, ω_c , is fixed as the reciprocal of the characteristic response time of the closed-loop. “High” and “low” values of Bode magnitude mean larger or smaller than 1 (0 dB). Of course, all this must be accomplished while keeping $|Y(j\omega)|$ low at all frequencies. Only in this way will the actuators be protected from unreasonable demands in response to the signals input at various points in the feedback loop. Another convenient way to think of ω_c is that frequency where $|Y(j\omega)|$ begins a significant deviation from $1/|G_p(j\omega)|$.

Other details of $|S(j\omega)|$ and $|T(j\omega)|$ are important; in particular we are interested in obtaining low values of the maxima of these functions (the so-called H_∞ norm). The H_∞ norm of $|T(j\omega)|$ is related, in an approximate way, to the damping ratio of the closed loop and for that purpose we use Equation (2.154). The H_∞ norm of $|S(j\omega)|$ is related to the M_2 stability margin (4.26). The use of this margin is gaining favor over the classical *phase and gain* margins. However, the classical margins can be obtained from the Bode diagrams of $|S(j\omega)|$ and $|T(j\omega)|$ as well as M_2 : The phase margin is obtained from the common value of $|S(j\omega_c)|$ and $|T(j\omega_c)|$ at cross-over. If G_p is minimum phase, the phase margin is obtained from Equation (4.19) which is plotted for reader’s convenience in Figure below. (Incidentally, since $S+T = 1$, it is impossible for $|T|$ and $|S|$ to be *simultaneously* less than 1/2, that is less than -6 dB). If G_p is minimum phase, the gain margin is $|S(j\omega_c)| - |T(j\omega_c)|$ at that frequency, ω , where the phase of T is -180° .

Perhaps the most intriguing thing about the neoclassical approach is the idea of parametrization: As demonstrated in Chapter 3, *an internally-stabilizing compensator is parameterized as*

$$G_c(s) = \frac{Y(s)}{1 - Y(s)G_p(s)} \quad (4.38)$$

if and only if $Y(s)$ is any BIBO stable, proper rational-function such that $G_p(1 - YG_p)$ is BIBO stable,; that is, such that Y and $1 - YG_p$ have a zero at each unstable pole of G_p .

This is a very remarkable result! Think of this: One is not assured of a stable feedback loop by placing some restriction on, say the compensator, $G_c(s)$. For example, feedback about a stable compensator in cascade with a stable controlled object can be unstable. This remarkable proposition is one of the motivations for focusing on the Youla parameter, $Y(s)$, during design. There are a number of other motivations.

We demonstrated in Chapter 3 that necessary conditions for $Y(s)$ to be internally stabilizing are

$$Y(p_r) = 0 \quad (4.39)$$

and

$$Y(p_r)G_p(p_r) = 1 \quad (4.40)$$

where “ p_r ” is any pole of G_p in the right half-plane or on the imaginary axis. These conditions are sufficient only if “ p_r ” is not repeated. (4.39, 4.40) may not be sufficient because there is no guarantee that the multiplicity of the zero in $1 - YG_p$ will match the multiplicity of the unstable pole in G_p . Notice that $G_c(s)$ may be stable or unstable. That is to say, a stable $T(s)$ is achieved by placing unstable poles in the compensator (such a thing is always possible). Use of a unstable compensator has double negative effects. First, given a bounded input, the Youla parameter in the feedback loop will be unbounded, this in turn results in actuator saturation. Next, if the feedback loop fails (a sensor or an actuator fails) then the opened-loop system is driven out of bounds by the controller!

If the controlled object is stable, the closed loop is internally stable if and only if $Y(s)$ is stable and proper. This means that, in this special case, the Youla parametrization of all stabilizing compensators is simply the set of all stable, proper rational functions! In other words any G_c given by (4.38) will produce an internally stable closed loop provided that the parameter is stable.

The *interpolation conditions* were derived in Chapter 3: If “ z ” denotes an uncanceled multiplicity a_z of the return ratio, L , and “ p ” denotes an uncanceled pole of multiplicity a_p of the return ratio, then

$$T(p) = 1 \quad S(p) = 0 \quad (4.41)$$

and

$$\frac{d^k}{ds^k} T(p) = 0 \quad \frac{d^k}{ds^k} S(p) = 0 \quad 1 \leq k \leq a_p - 1 \quad (4.42)$$

while

$$T(z) = 0 \quad S(z) = 1 \quad (4.43)$$

and

$$\frac{d^k}{ds^k} T(z) = 0 \quad \frac{d^k}{ds^k} S(z) = 0 \quad 1 \leq k \leq a_z - 1 \quad (4.44)$$

Since unstable poles of L cannot be canceled, these conditions must always be obtained for poles or zeros on the imaginary axis or in the right half-plane. Fixing these conditions with selection of $Y(s)$ is one way to obtain internal stability.

Also, these conditions mean that the closed-loop transfer functions, $S(s)$ and $T(s)$, are constrained to take on certain values in the complex plane. Clearly, these constraints influence the values on the imaginary axis, $S(j\omega)$ and $T(j\omega)$. These constraints cannot be removed by a pole-zero cancellation where “ z ” or “ p ” lies in the right half-plane, where a factor of the return ratio is nonminimum phase. These interpolation conditions impose irrevocable limits to feedback design.

4.4 Feedback Design And Parametrization: Stable Objects

The description of the neoclassical “shaping” of closed-loop transfer functions begins in this section. The treatment is complete and the procedures are quite satisfying in that the vulnerability of the actuators is one of very first considerations in the shaping process. A transfer-function-based procedure for closed-loop, pole/zero assignment is also featured. The main emphasis is on obtaining a satisfactory $S(s)$, $T(s)$, and $Y(s)$.

The shaping of closed-loop transfer functions seems the most direct procedure for finding an internally-stabilizing proper compensator which meets all design criteria. The most obvious form of the closed-loop shaping is described as follows: It is easy to demonstrate that

$$G_c = \frac{1}{G_p} \frac{T}{1 - T} \quad (4.45)$$

The idea is that we can choose the closed-loop transfer function, $T(s)$, we want and then use G_p to calculate the required compensator transfer function. The problem is that we do not assure the feasibility of the procedure because there many important aspects, such as the vulnerability of the actuators, that are concealed and obscured in this formulation. Also, it is likely that G_c would not even be proper (realizable) if calculated in this way. Finally, if G_p is nonminimum phase or unstable, then this form of compensation would produce an internally-unstable closed loop. These criticisms are removed in the alternative procedure we describe in this section.

Incidentally, notice the complete cancellation of the plant in this procedure (4.45); that is, the poles and zeros of G_p are absent in the return ratio, $G_c G_p$. A cancellation of a portion of G_p will also appear in our procedure but that cancellation does not appear in G_c . (Actually most feedback design methods involve some cancellation, even in the state-variable methods where the cancellations are obscured.)

The focus in the method proposed in this section is the Youla parameter

$$Y(s) = u(s)/r(s) = -u(s)/d(s) = -u(s)/n(s) \quad (4.46)$$

This is the closed-loop transfer function that quantifies the vulnerability of the actuators to saturation. We expect $r(t)$ and $d(t)$ to contain low and intermediate frequencies while $n(t)$ will have a relatively high frequency content. That is, the actuators are vulnerable at nearly all frequencies so that $|Y(j\omega)|$ should be kept as low as possible at nearly all frequencies.

It follows from (4.37) that *if G_p is stable, then the closed-loop is internally stable if and only if the parameter, $Y(s)$, is stable.* There is no parallel result in the classical approach; that is, there is no way to ensure even the stability of the closed loop - let alone *the internal stability* — by restricting the choice of $G_c(s)$ in some way! Recall that everywhere that we say “stable” we mean bounded input — bound output (BIBO) stability. This means that a transfer function might be stable only if it is not improper.

We write the parameter as

$$Y(s) = \frac{1}{1 + L(s)} \frac{L(s)}{G_p(s)} \quad (4.47)$$

and conclude that

$$Y(j\omega) \approx 1/G_p(j\omega) \quad \text{where} \quad |L(j\omega)| \gg 1 \quad (4.48)$$

and

$$Y(j\omega) \approx L(j\omega)/G_p(j\omega) = G_c(j\omega) \quad \text{where} \quad |L(j\omega)| \gg 1 \quad (4.49)$$

Recall that we design $|L|$ to have very large values at low frequency in order to attenuate disturbances, to improve command following, and to achieve robustness to variations in the controlled object. We design $|L|$ to have very small values at high frequency in order to attenuate sensor noise and achieve stability robustness. It is clear that:

Our goals are also attained by shaping $|Y(j\omega)|$ to $1/|G_p(j\omega)|$ at low frequencies and by keeping $|Y(j\omega)|$ low at high frequencies.

This is the essence of the shaping procedure. We will examine this procedure by providing several examples in the following sections.

Notice that where we accomplish this shaping, $1/|G_p|$ is the low-frequency asymptote for $|Y(j\omega)|$ and $|L(j\omega)|/|G_p(j\omega)| = |G_c(j\omega)|$ is the high-frequency asymptote. (We set the latter before we even calculate $L(s)$.) This means that the high and low frequency asymptotes for $|Y(j\omega)|$ meet at

$$\omega_{Yb} = \omega_c \quad (4.50)$$

the cross-over frequency of the return ratio! *This means by setting the break frequency of the parameter, we essentially fix the closed-loop response time.*

We begin with an important observation

4.4.1 Renormalization of Gains:

We try to use a high static gain in G_c because that will reduce the vulnerability of the actuators and will help to attenuate the response at the output to the disturbance.

In a sense, any disturbance rejection criterion can be satisfied by magically increasing the gain in any controlled object by redefining the units of the controlled input. For example, suppose you take the units of the controlled input to be millimeters and the gain of G_p is 1.0 but you need that gain to be 10^3 . You can merely redefine the input to be measured in meters and, voilà, the gain is a thousand.

The reader senses that there is something very wrong in this conclusion and in that you are correct. The explanation is that, if before you waved your wand, $u(t)$ saturated at a magnitude of 2, it now saturates at a magnitude of 2×10^{-3} . That is, you now have a more severe restriction on $|Y|$. Think of it this way, you compensated for an apparent attenuation of a disturbance response by a corresponding tightening of the restrictions $Y(s)$. From this point-of-view, it is the nonlinearity (the restriction on u) that gives an absolute meaning to “gain”.

4.4.2 Shaping of The Closed-Loop: Stable SISO:

The basic result is (4.37) and we have a simplification of that in a special case.

Corollary 4.1. *If the controlled object is stable, the closed loop is internally stable if and only if $Y(s)$ is stable (hence, not improper).* ■

This means, as stated earlier, that the Youla parametrization

$$G_c(s) = \frac{Y(s)}{1 - Y(s)G_p(s)}$$

of all internally stabilizing compensators is simply the set of all stable (hence, not improper) rational functions, $Y(s)$! In other words any G_c given by (4.38) will produce an internally stable closed loop provided that the parameter is stable. Equally important, any internally stabilizing compensator can be expressed as in (4.38) for some $Y(s)$. Think of it this way, if you restrict yourself to form (4.38), then no other designer can accomplish a better design using some other form. Of course, this is assuming you will search over all possible $Y(s)$, which is not likely.

Where G_p is stable (possibly non-minimum phase) the formulation is as follows for single-degree-of-freedom design.

a) **Placement of closed-loop poles, zeros, and gain:** Begin with the relation

$$T(s) = Y(s)G_p(s)$$

and “shape” $Y(s)$. You begin that by canceling unwanted poles and zeros of G_p and then including desired location for the closed-loop poles and zeros (we will discuss the impact of the cancellation of poles and zero in the following section). Next make an adjustment in the gain of $Y(s)$ in order to fix a desired value of the closed-loop gain $|T|$. For example, take $Y = 1/G_p$ at low frequency so that $T = 1$ in that range. This gives good low-frequency command following and, since $S = 1 - T$ is thereby forced near zero, there will be good disturbance rejection there as well. (Of course, the mirror image of a non-minimum phase zero of G_p must be used as a pole of $Y(s)$ in that range because $Y(s)$ must be stable). Always remember that $Y(s)$ is a closed-loop transfer function so the necessary and sufficient condition for internal stability (BIBO) is that “ Y ” is stable, hence proper.

b) **Adjust the frequency response of $Y(s)$:** Observe the Bode magnitude of $|Y(j\omega)|$ and recognize that this diagram indicates the vulnerability of your actuators. For example, you pay a price where you cancel a LHP pole of G_p because the corresponding zero in $Y(s)$ will make you more vulnerable to saturation (the slope of the Bode-magnitude $|Y(j\omega)|$ will increase in that way, more on this in the following sections). It may be that you were too ambitious in your selection of the closed-loop poles and so the diagram indicates values for $Lm|Y(j\omega)|$ which are too large. You may have to re-evaluate the choices made in step (a). Generally, it is far better if you can design by canceling only a limited number of LHP poles of G_p near the cross-over frequency.

c) **Adjust the stability margins:** The phase margin is obtained where $|T(j\omega_c)| = |S(j\omega_c)|$ and is related to this common value (see previous sections). Normally, you want this intersection to occur below 0 dB line since that assure a phase margin greater than 60° . Also, this cross-over frequency, ω_c , provides an estimate of the closed loop response time $t_r \approx 1/(4\omega_c)$. Contrary to expectations, this estimate is often more accurate than an estimate based upon the closed-loop poles themselves. The gain margin is the difference

$$Lm|S(j\omega)| - |Lm|T(j\omega)| \quad (4.51)$$

at that frequency, ω , where the phase of $|T(j\omega)|$ is -180° . Another, perhaps better, measure is the M_2 stability margin which is equal to $\max |S(j\omega)|$. Alternatively, recall that unstructured uncertainty can be modeled such that stability margins (stability

robustness) are characterized as constraints on the weighted version of $|T(j\omega)|$. That means these uncertainties will help fix the shape of $|Y(j\omega)|$ at high frequencies.

d) **Shape $Y(s)$ at high frequencies to offset sensor-noise problems: Add high-frequency closed-loop poles.** The constraint $Y(s)$ must be proper or strictly proper will help in this regard, but you may have to take additional steps by adding some high-frequency poles to $Y(s)$ (hence, to $T(s)$).

e) **Evaluate the design in the linear domain:** Examine the Bode magnitudes $|T(j\omega_c)|$ and $|S(j\omega_c)|$ and evaluate the capability of your design for robustness, command following, and unwanted signal attenuation utilizing the same linear plant model G_p used for control design.

f) **Obtain the transfer function of the compensator:** From

$$G_c(s) = \frac{Y(s)}{S(s)} \quad (4.52)$$

This must be synthesized in some way in order to realize the feedback design.

g) **Evaluate the design with the more complete plant model, in the non-linear domain:** Integrate G_c in the complete simulation model and again evaluate the capability of your design for robustness, command following, and unwanted signal attenuation. Are your actuators capable of satisfying the requirements you have placed upon them?

With regard to pole-zero placement in step (a): it is good to cancel a zero in G_p with a pole in Y since that causes a decrease in the slope of $Lm|Y(j\omega)|$, a lessening of the demand on the actuators! Of course, since $Y(s)$ must be stable, that can occur only in the left half-plane. This point illustrates the simple fact that minimum phase zeros in the controlled object can be put to good use in feedback design. In a similar vein, one pays a price for canceling a pole in G_p with a zero in Y because that causes an *increase* in the slope of $Lm|Y(j\omega)|$; an increase of the demand on the actuators. That makes sense because where we cancel an object pole, we are attempting to produce a closed loop that responds faster than the open loop and the cost is immediately made apparent by an increase of the demand on the actuators. Incidentally, since $T = YG_p$ must be stable, any unstable pole of G_p must be canceled in this way. That fact makes very clear the limitation that an unstable pole in the object places on feedback design. *It is inherent that actuators are called upon to do more in the case of unstable objects.* More on the in the following sections.

With regard to the inclusion of **high-frequency poles** in step (d). “High frequency” means poles with break frequencies above cross-over, and that means that these poles will not influence closed-loop transient response. It is probably for that reason that this concept is neglected in some developments of the feedback theory. Notice the way that this procedure keeps

$$|T(j\omega_c)| \approx |G_p(j\omega_c)| \quad (4.53)$$

near crossover since that is the way you prevent excessive growth of $|Y|$. There is another, very good reason to add high-frequency poles. Recall that $u(s) = Y(s)r(s)$ and suppose that $r(s)$ is a step change in command. It follows from the initial-value theorem that if $Y(s)$ has “ k ” more poles than zeros, then $u(0^-)$ and the initial values of $k - 1$ derivatives of $u(t)$ will be zero. Thus, the larger the value of “ k ”, the less usage demanded in the initial range of the actuator output. That is to say, we increase “ k ” to diminish the kind of excessive actuator usage.

We will summarize the procedure to select $Y(j\omega)$ and the impact of pole-zero cancellation in feedback design including several examples in the following section. Before that, however, we lay out the competing objectives and constraints in shaping the response of a system.

4.4.3 Neoclassical Design Principles

Recall the various feedback transfer functions for the system shown in Figure 4.8.

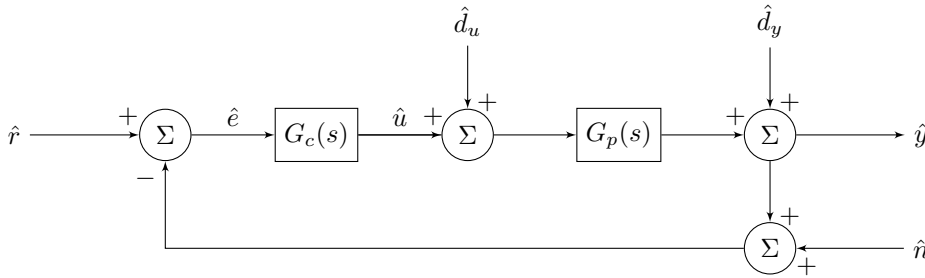


Figure 4.8: Block diagram of a controlled system, $H(s), P(s) \equiv 1$.

$$S = \frac{\hat{e}}{\hat{r}} = -\frac{\hat{e}}{\hat{n}} = \frac{\hat{y}}{\hat{d}_y} \quad (4.54)$$

$$T = \frac{\hat{y}}{\hat{r}} = -\frac{\hat{y}}{\hat{n}} = -\frac{\hat{u}}{\hat{d}_u} \quad (4.55)$$

$$Y = \frac{\hat{u}}{\hat{r}} = -\frac{\hat{u}}{\hat{n}} = -\frac{\hat{u}}{\hat{d}_y} \quad (4.56)$$

The shape of these feedback transfer functions impacts the performance and robustness of the system, however there are generally competing objectives:

1. For disturbance rejection, we must make S small.
2. For noise attenuation, we must make T small.
3. For reference tracking, we must make $T \approx 1$.
4. For reducing control effort, we must make Y small.

Let us consider these objectives in terms of the return ratio L .

1. $S = 1/(1 + L)$, so for small L we need $L \gg 1$.
2. $T = L/(1 + L)^{-1}$, so for small T we need $L \ll 1$.
3. $T = L/(1 + L)^{-1}$, so for $T \approx 1$ we need $L \gg 1$.
4. $Y = G_c/(1 + L)$, so for small Y we need $G_c \ll 1$ at high frequencies.

In meeting these objectives we are subjected to constraints:

▪ Algebraic:

– $S + T = 1$.

– Internal Stability: Interpolation Conditions.

▪ Analytic:

– $\max_{\omega} |T(j\omega)| \geq \left| \frac{p+z}{p-z} \right|$, $\max_{\omega} |S(j\omega)| \geq \left| \frac{z+p}{z-p} \right|$. This is an lower bound on both $\|T\|_{\infty}$ and $\|S\|_{\infty}$ when there are unstable poles and zeros. For instance, a system with an unstable zero at $s = 2$ and an unstable pole at $s = 1$ will have

$$\max_{\omega} |S(j\omega)| \geq \frac{2+1}{2-1} = 3 \approx 10\text{dB} \quad (4.57)$$

To design a controller with the Youla parameterization method, we select a Y considering the above objectives and constraints, calculate $T = YG_p$, then calculate $G_c = Y/S = Y/(1 - T)$. The selection of Y will determine the performance and robustness of the system. A robust system will have S and $T < 0\text{dB}$.

If we are aiming for $Y = 1/G_p$, how do we select $Y(j\omega)$? Do we cancel poles and zeros in G_p ? What do we do with unstable poles and zeros? This will be discussed in the following section.

Another important analytic constraint is related to the “**Bode Integral Theorem**”. These analytic constraints will be further discussed in Chapter 6. In addition, a detail discussion of these constraints are provided in Appendix B.

In the next two sections, the some of the important aspects of the neoclassical feedback control concepts are discussed. In the last section of this chapter, we provide several control design examples.

4.5 Shaping the Youla Parameter

We discussed some of the impact of pole-zero cancellation and shaping the Youla parameter including the consequences of this shaping in feedback control design. This section will detail the basic steps of shaping the Youla parameter to build the desired closed-loop response.

When we speak of pole-zero cancellation to shape the Youla parameter, there are four principal cases that need to be considered. These are cancellation of ...

- a) **Stable Zeros (LHP Zeros) of G_p ,**
- b) **Stable Poles (LHP Poles) of G_p ,**
- c) **Non-minimum Phase Zeros (RHP Zeros) of G_p ,**
- d) **Unstable Poles (RHP Poles) of G_p ,**

4.5.1 LHP Zeros of G_p

Minimum phase zeros of G_p can be canceled by placing a stable (LHP) pole in the Youla parameter, knowing that

$$T(s) = Y(s) \cdot G_p(s) \quad (4.58)$$

But what is the impact of this cancellation and when should we decide to cancel?

If a LHP zero of a plant is located at a lower frequency than its slowest pole (let’s take a simple case of one LHP zero with two poles) then this zero will result in an overshoot due to a step input, and hence the zero dynamics decreases the M_2 stability margin. In addition, as we will show in the example below, not canceling this type of zero results in a

an increase of actuator demand at high frequencies, and sensor noise amplification in this frequency range. If this is the case, then this zero must be canceled by adding a pole to the Youla parameter. Addition of a pole to the Youla parameter has a secondary benefit of lowering the magnitude frequency response and, hence, lowering the actuator demand.

The impact of this cancellation is demonstrated below on the closed loop complementary sensitivity, sensitivity, and Youla transfer functions for a simple case of one pole and one zero G_p . It should be noted that if the T and S vectors are greater than one then the M_2 stability margin decreases. It should also be clear that the vector addition of T and S is equal to one.

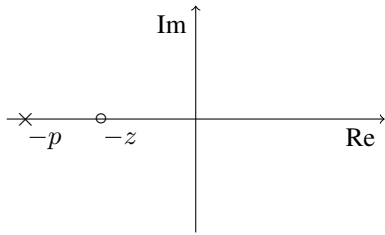
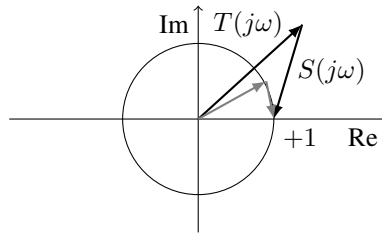
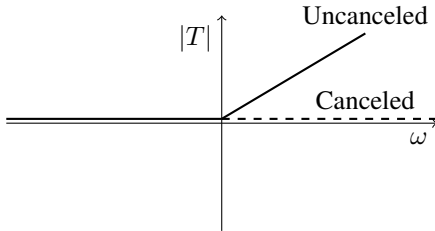
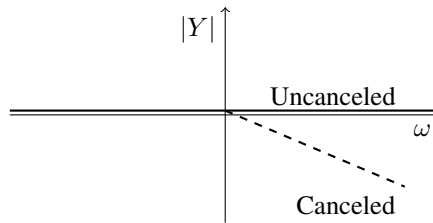


Figure 4.9: s-plane location of LHP zero.

Figure 4.10: S and T for a LHP without cancellation in black and with cancellation in gray.Figure 4.11: $T = YG_p$ when canceling a zero.Figure 4.12: Y when canceling a zero.

EXAMPLE 4.2

Suppose we have a plant, G_p , such that

$$G_p = \frac{s+1}{s+2}. \quad (4.59)$$

We will first consider leaving the zero at $s = 1$ in place. We set $Y_1 = 2$ to ensure accurate command tracking at steady-state ($T = 1$ at low frequencies). This gives a

closed-loop response of

$$T_1 = \frac{s+1}{\frac{1}{2}s+1}. \quad (4.60)$$

Y_1 and T_1 have the magnitude frequency responses shown in Figures 4.13 and 4.14

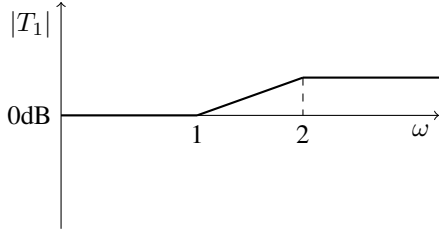


Figure 4.13: Magnitude plot of T_1 .

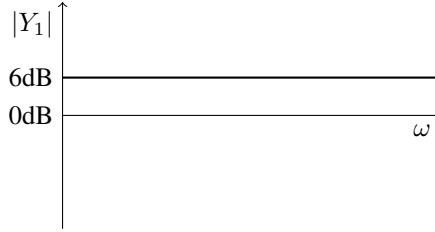


Figure 4.14: Magnitude plot of Y_1 .

If we don't cancel the zero, the closed-loop response will amplify noise. In addition, the M_2 stability margin reduces and actuator demand is constant throughout the entire frequency range. *This is not a good design.* Instead, we shape Y to cancel the zero of the plant, G_p ,

$$Y_2 = \frac{2}{s+1} \quad (4.61)$$

Then,

$$T_2 = \frac{1}{\frac{1}{2}s+1} \quad (4.62)$$

As shown in Figures 4.15 and 4.16, the actuator and closed-loop response both now roll off at high frequencies, limiting the bandwidth of the system so that it will reject the high-frequency sensor noise and reduces actuator demand at high frequency.

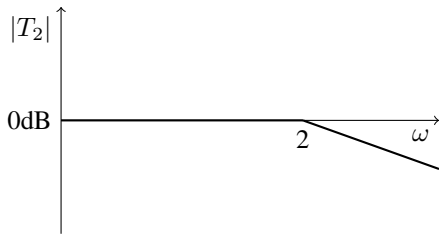


Figure 4.15: Magnitude plot of T_2 .

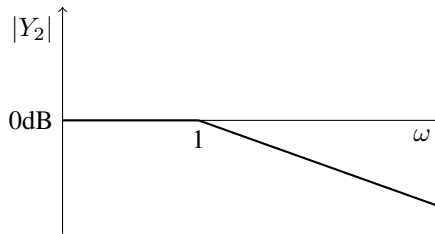


Figure 4.16: Magnitude plot of Y_2 .

4.5.2 Non-minimum Phase Zeros

Canceling a zero in the right-hand plane (that is, an unstable or non-minimum phase zero) would require introducing a pole in the right-hand plane (an unstable pole) to the Youla

transfer function. This would violate internal stability, so cancellations of non-minimum phase zeros is forbidden. Therefore, the existence of a non-minimum phase zero in the plant is bad news and we have to live with this.

In the earlier chapters, we illustrated that a non-minimum phase zero results in a reverse action. For a high gain controller this reverse action could have a destabilizing effect, as at higher gains, poles of a transfer function migrates to its zero. So, if there exists one non-minimum phase zero, at high gain, one pole will cross the $j\omega$ -axis and migrates towards this non-minimum phase zero. This will results in destabilizing the system. Therefore, one can observe that the non-minimum phase zero will have a band-limiting effect on the closed loop feedback control. Is there anything could be done to alleviate this problem?

The answer to this question is yes. Instead of canceling the non-minimum phase zero, we minimize its effects by placing a stable pole, at the same or lower than the non-minimum phase zero frequency, in the Youla transfer function. As mentioned, the non-minimum phase zero in the plant restricts or constrains the the speed of the response or the system bandwidth. The effective lower bound of the time constant τ_p of poles in a non-minimum phase system is given by

$$\tau_p \geq \frac{1}{|z_k|} \quad (4.63)$$

where z_k is the closest non-minimum phase zero to the $j\omega$ -axis. An alternative interpretation of this limit is that a stable closed-loop pole must be placed closer to the $j\omega$ -axis than a non-minimum phase zero. The reason for this is briefly illustrated in Figures 4.17 and 4.18. At first glance, Figure 4.18 could be confusing as the Sensitivity transfer function, S is equal to 2, and we clearly stated that the algebraic constraint, $T + S = 1$. How could this be?

We should remind ourselves that T and S are vectors with magnitudes and phases. So, the algebraic constraint is a vector addition. With the presence of a non-minimum phase zero, the phase of T could drop to -180° while its magnitude at very low frequency, when the magnitude of S is very small, can be equal to 1. Therefore, $-1 + S = 1$ where, S is now equal to 2 or 6 dB. This large magnitude in S drastically reduces the M_2 stability margin. We will discuss this effect by presenting several examples later in this chapter. This is precisely why we need to include a low frequency pole in Y to cancel the negative effect of the plant non-minimum phase zero.

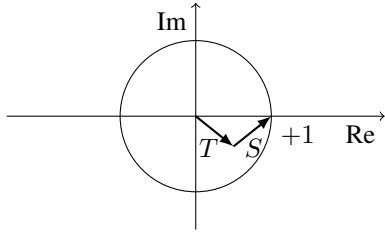


Figure 4.17: S and T with pole placement.

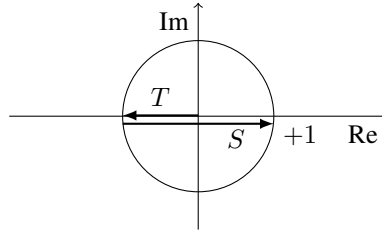


Figure 4.18: S and T without pole placement.

EXAMPLE 4.3

Suppose we have a plant

$$G_p = \frac{1-s}{s+2} \quad (4.64)$$

We want to minimize the effects of the non-minimum phase zero. However, canceling it with a $1-s$ in the denominator would introduce an RHP pole, destabilizing the system. We first set $Y_1 = 2$ and observe the response.

$$T_1 = \frac{s+1}{\frac{1}{2}s+1} \quad (4.65)$$

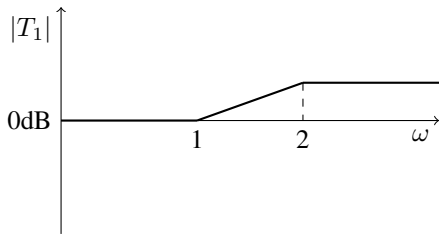


Figure 4.19: Magnitude plot of T_1 .

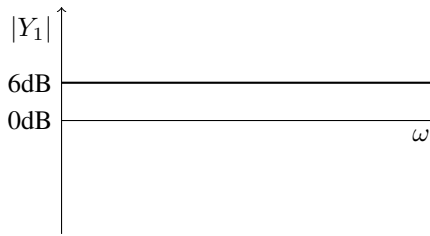
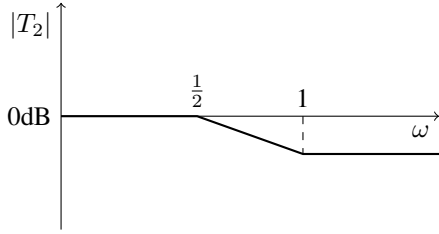
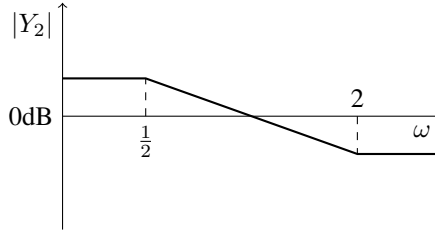


Figure 4.20: Magnitude plot of Y_1 .

We can't cancel the zero, but we need to eliminate the amplification of high-frequency noise. Observe that the time-constant of the close-loop pole is $\tau = \frac{1}{2}$. This is too fast – faster than the time constant of the non-minimum phase zero. We can replace the pole with a slower one, for instance $\tau = 2$.

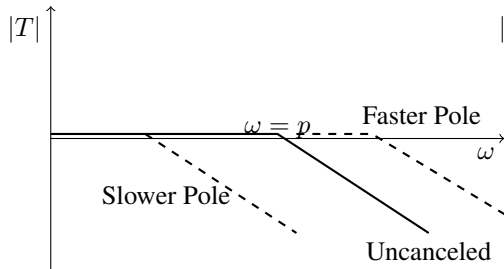
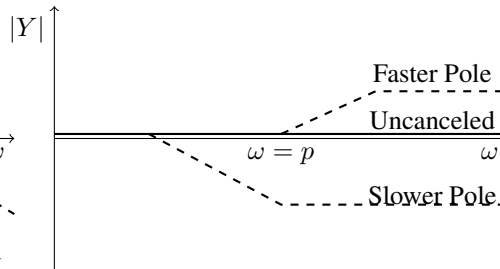
$$Y_2 = \frac{\frac{1}{2}(s+2)}{s+\frac{1}{2}} \quad (4.66)$$

$$T_2 = \frac{1-s}{2s+1} \quad (4.67)$$

Figure 4.21: Magnitude plot of T_2 .Figure 4.22: Magnitude plot of Y_2 .

4.5.3 LHP Poles of G_p

Stable poles of G_p are canceled with minimum-phase zeros in Y . In this case, We have the option of canceling or keeping these stable poles. Therefore, our design requirement including the closed loop desired response, in this case, will determine this cancellation. For instance, we may choose to cancel poles to speed up the response or to improve actuator demand and stability margins as illustrated in Figures 4.23 and 4.24. Faster poles increase the actuator demand and speed up the response, while slower poles have an opposite effect. Y must be proper or strictly proper, so canceled poles of G_p must always be replaced with the same number or more new poles in Y .

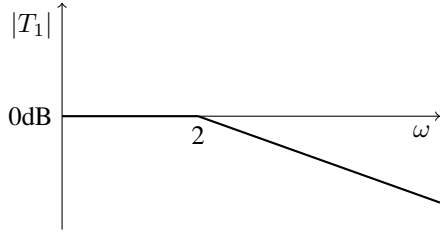
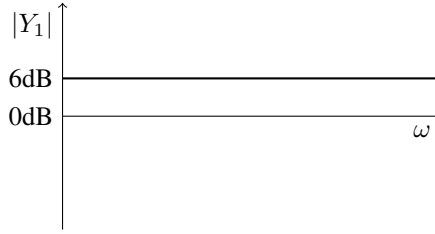
Figure 4.23: $T = YG_p$ when canceling a pole.Figure 4.24: Y un-canceled when canceling a pole or unca.

EXAMPLE 4.4

Suppose $G_p = \frac{1}{s+2}$.

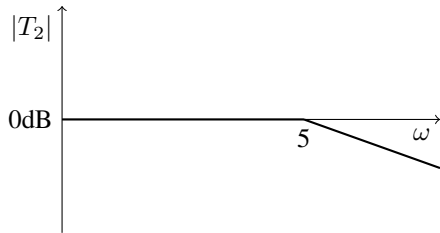
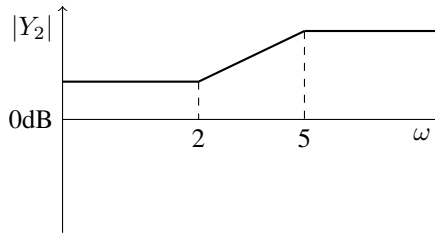
First, set $Y = 2$, yielding no cancellations and zero steady-state error.

$$T_1 = \frac{2}{s+2} \quad (4.68)$$

Figure 4.25: Magnitude plot of T_1 .Figure 4.26: Magnitude plot of Y_1 .

Perhaps we want to increase the bandwidth of this plant. We will cancel the pole and replace it with a pole that has a faster time constant. As can be seen in Figure 4.28, actuator demand has increased.

$$Y_2 = \frac{5(s+2)}{s+5} \Rightarrow T_2 = \frac{5}{s+5} \quad (4.69)$$

Figure 4.27: Magnitude plot of T_2 .Figure 4.28: Magnitude plot of Y_2 .

4.5.4 Unstable Poles

In order to stabilize a plant, we have to cancel its unstable poles. This can be done by introducing zeros in Y , and as these poles are in the RHP, we have no other option but to introduce a non-minimum phase zero in Y for each unstable pole of G_p . Furthermore, we must make sure to respect the interpolation conditions, discussed in the previous chapters, so that the final controller results in an internally stable feedback system. We discuss the case of an unstable plant in the next example.

EXAMPLE 4.5

Suppose $G_p = \frac{1}{s-1}$.

We first select a Y that will cancel the unstable pole and that has tuning parameters that we can use to meet the interpolation conditions. In this example we will use a gain

K and a time constant τ of the new pole as our tuning parameters. Then, the Youla parameter can be written as,

$$Y_1 = \frac{K(s-1)}{\tau s + 1} \quad (4.70)$$

Therefore,

$$T_1 = \frac{K}{\tau s + 1} \quad (4.71)$$

Next, we attempt to verify the interpolation conditions and expose any constraints on K and τ for meeting the interpolation conditions. We have an unstable pole at $s = 1$, so we know that we have to meet the following interpolation condition to guarantee internal stability,

$$T(1) = 1 = \frac{K}{\tau + 1} \quad (4.72)$$

Therefore, we have the following constraint,

$$K = \tau + 1 \quad (4.73)$$

We can select a K, τ pairs and examine the result. For instance, let's consider $\tau = 2$ and, hence, $K = 3$.

$$T_1 = \frac{3}{2s + 1} \quad (4.74)$$

This closed loop transfer function results in a steady-state error due to a step input. To eliminate this steady-state error, we can introduce the constraint $T(0) = 1$ to ensure that L has an integrator.

$$T(0) = 1 = \frac{K}{1} \quad (4.75)$$

This means $K = 1$, which in turn means that $\tau = 0$. This results in ill-posed feedback and, is unacceptable. This means that the original Y has to be modified to meet all the given constraints. It should not be too difficult to observe that we need to add an additional zero to Y to meet the given constraints, but we will discuss this later. By adding an additional zero to Y , we also need to add an extra pole to enforce properness. We propose the following Y .

$$Y_2 = \frac{K(s-1)(s+z)}{(\tau s + 1)^2} \quad (4.76)$$

Therefore,

$$T_2 = \frac{K(s+z)}{(\tau s + 1)^2} \quad (4.77)$$

From the first interpolation condition

$$T(0) = 1 = \frac{K(z+0)}{(\tau \cdot 0 + 1)^2} \quad (4.78)$$

we have the requirement that $Kz = 1$. The next interpolation condition gives

$$T(p) = 1 = \frac{K(z+1)}{(\tau+1)^2} \quad (4.79)$$

From this we can obtain a relationship between K and z in terms of the closed-loop time constant τ .

$$K = (\tau+1)^2 - 1 \quad (4.80)$$

$$z = \frac{1}{(\tau+1)^2 - 1} \quad (4.81)$$

The selected Youla can meet the interpolation conditions and eliminate steady-state error, while affording us some leeway in the design. The interpolation conditions in fact expose an interesting characteristic of the design: If we slow the response time by increasing τ , we move the non-minimum phase zero closer to the origin, which in turn increases the overshoot and can induce oscillations. On the other hand, speeding up the response time by lowering τ , the non-minimum phase zero moves away from the origin, which results in less overshoot or oscillations.

For example, selecting $\tau = 2$ gives us $z = 1/8$ and $K = 8$.

$$Y_2 = \frac{(s-1)(8s+1)}{(2s+1)^2} \quad (4.82)$$

$$T_2 = \frac{8s+1}{(2s+1)^2} \quad (4.83)$$

Having obtained a desirable shape for T , we can then compute the corresponding controller, G_c .

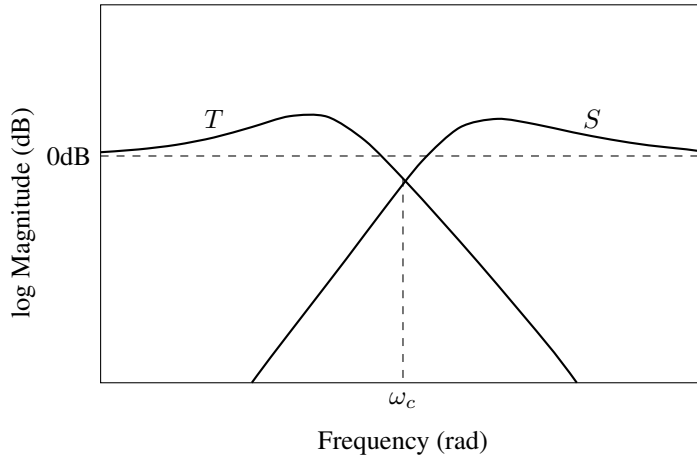
$$G_c = \frac{Y}{1-T} = 2 + \frac{1}{4}s^{-1} \quad (4.84)$$

Interestingly enough, the result is PI control.

4.6 Design Guidelines

In this section, we discuss the impact of our design on the shapes of $T(s)$, $S(s)$, and $Y(s)$. This means that by observing these shapes we learn to decide if our feedback control design is acceptable.

Recall the typical shape of S and T , as shown in Figure 4.29.

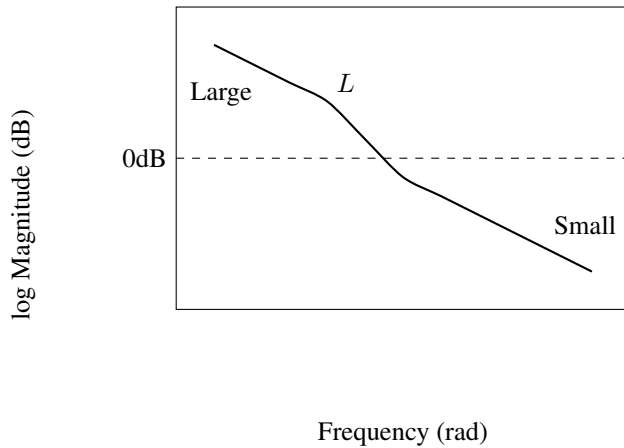
Figure 4.29: S and T for a typical system.

3595 Good design keeps the magnitude of S and T at the crossover frequency less than 0dB.

3596 This ensures phase margins $\geq 60^\circ$.

3597 Recall from Section 3.9 that generally the return ratio is large at low frequencies and

3598 small at high frequencies, as shown in Figure 4.30.

Figure 4.30: Return ratio L for a typical system.

3599 Does it matter, then, where high gain comes from? At low frequencies,

$$T(j\omega) \approx 1 \Rightarrow Y = \frac{1}{G_p} \quad (4.85)$$

3600 Whereas at high frequencies,

$$L \rightarrow 0 \Rightarrow Y = \frac{G_c}{1 + L} \approx G_c \quad (4.86)$$

Suppose that the magnitude of Y must be bounded at low frequencies. If so, then the plant must have high gain. Then, to control the gain of Y at high frequency, we must make the crossover frequency smaller (move it to the left) to slow the response time and we must include a pole (or poles) at high-frequency to ensure roll-off, as shown in Figure 4.31.

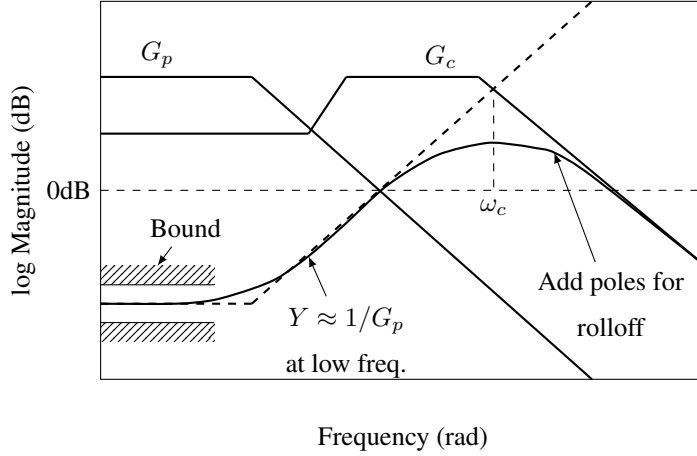


Figure 4.31: Youla design with high-gain plant.

In general, at low frequencies $L(j\omega) \gg 1$, so $Y \approx 1/G_p$. Therefore, at low frequencies, performance is heavily dependent on the plant dynamics (or perhaps, the engineers who designed the plant), rather than your controller. On the other hand, at high frequencies $L(j\omega) \ll 1$, so $Y \approx G_c$. Performance at high frequencies is most heavily dependent on the controller you design.

4.7 Design Examples

This section consists of several example problems of neo-classical design.

EXAMPLE 4.6

Consider a basic plant model.

$$G_p = \frac{10}{s+1}, \quad T = Y \frac{10}{s+1} \quad (4.87)$$

The simplest choice for Y will leave the plant pole in place while ensuring that there is no steady-state error.

$$Y = \frac{1}{10} \quad (4.88)$$

Then,

$$T = \frac{1}{s+1} \quad (4.89)$$

$$S = 1 - T = \frac{s}{s+1} \quad (4.90)$$

The resulting compensator is simply PI control.

$$G_c = \frac{Y}{S} = \frac{\frac{1}{10}}{\frac{s}{s+1}} = \frac{s+1}{10s} = \frac{1}{10} + \frac{1}{10s} \quad (4.91)$$

We then check for internal stability.

$$L = G_c G_p = \frac{1}{10} \frac{s+1}{s} \cdot \frac{10}{s+1} = \frac{1}{s} \quad (4.92)$$

$$1 + L = 1 + \frac{1}{s} = \frac{s+1}{s} \quad (4.93)$$

So, Y is proper and internal stability is met.

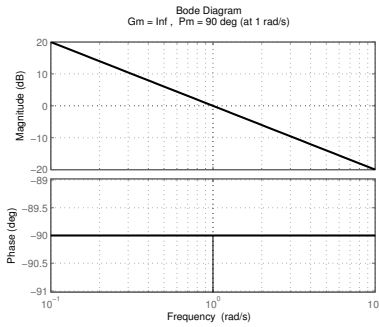


Figure 4.32: Stability margins of the controlled system.

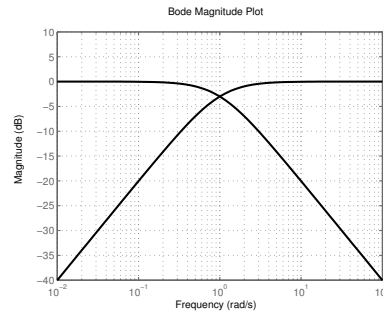


Figure 4.33: Magnitude plot of S and T .

EXAMPLE 4.7

We will consider the same plant, but now choose to speed up the system. We will place the closed-loop pole at $s = -2$ ($\tau = \frac{1}{2}$).

$$Y = \frac{1}{10} \frac{s+1}{\frac{s}{2}+1} \quad (4.94)$$

$$T = \frac{1}{\frac{s}{2}+1}, \quad S = 1 - T = \frac{s}{s+2} \quad (4.95)$$

$$G_c = \frac{Y}{S} = \frac{\frac{1}{5} \frac{s+1}{s+2}}{\frac{s}{s+2}} = \frac{s+1}{5s} = \frac{1}{5} + \frac{1}{5s} \quad (4.96)$$

Again, we check for internal stability:

$$L = G_c G_p = \frac{1}{5} \frac{s+1}{s} \cdot \frac{10}{s+1} = \frac{2}{s} \quad (4.97)$$

$$1 + L = 1 + \frac{1}{s} = \frac{s+2}{s} \quad (4.98)$$

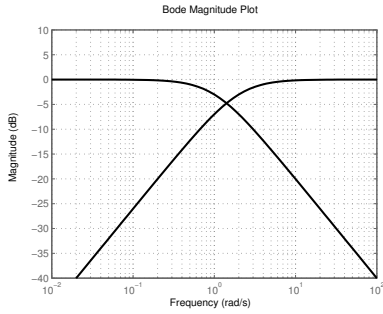


Figure 4.34: Magnitude plot of S and T .

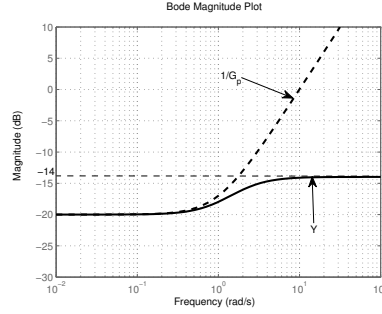


Figure 4.35: Magnitude plot of Y .

ω_c is now at 2rad/s.

An additional pole could be placed at $s = -5$ to attenuate the effect of high-frequency disturbances.

EXAMPLE 4.8

What if our plant does not have high gain, and our system requires small Y at low frequencies?

$$G_p = \frac{1}{s+1} \quad (4.99)$$

For instance, let's say that we require $Y(0) = -20\text{dB}$. Then,

$$Y = \frac{1}{10} \Rightarrow T = \frac{1}{10(s+1)} \quad (4.100)$$

$$T(0) = \frac{1}{10} \quad (4.101)$$

This result is not good: the closed-loop response will have significant steady-state error. The problem arises from low gain in G_p .

$$T(0) = Y(0)G_p(0) \quad (4.102)$$

■ **EXAMPLE 4.9**

Consider a plant with two poles, one at the origin.

$$G_p = \frac{1}{s(s+1)} \quad (4.103)$$

We want to place two closed-loop poles at $s = -2$.

$$Y = \frac{s(s+1)}{(\frac{s}{2}+1)(\frac{s}{2}+1)} \quad (4.104)$$

Then,

$$T(s) = \frac{1}{(\frac{s}{2}+1)(\frac{s}{2}+1)} \quad (4.105)$$

$$S(s) = \frac{s(\frac{s}{4}+1)}{(\frac{s}{2}+2)(\frac{s}{2}+1)} \quad (4.106)$$

$$G_c = \frac{Y}{S} = \frac{4(s+1)}{s+4} \quad (4.107)$$

Verify the interpolation conditions:

$$T(0) = \frac{1}{(\frac{0}{2}+1)(\frac{0}{2}+1)} = 1 \quad (4.108)$$

Verify internal stability:

$$L = G_c G_p = \frac{4(s+1)}{s+4} \cdot \frac{1}{s(s+1)} = \frac{4}{s(s+4)} \quad (4.109)$$

$$1 + L = 1 + \frac{4}{s(s+4)} = \frac{s^2 + 4s + 4}{s^2 + 4s} \quad (4.110)$$

Y is proper, internal stability is met.

■ **EXAMPLE 4.10**

Consider the dynamics of a satellite positioning system.

$$G_{p,1} = G_{p,0} \cdot G_{p,Ap} \quad (4.111)$$

$$G_{p,0} = 0.1 \cdot \frac{1}{s(100s+1)(0.1s+1)} \quad (4.112)$$

$$G_{p,Ap} = \frac{1-100s}{1+100s} \quad (4.113)$$

Classical controller design would require hand-tuning poles, zeros, and integrators for phase correction, high frequency roll-off, boosting low frequency gain, and so on.

With neoclassical control, the process is much simpler.

The actuator delay (the non-minimum phase zero) requires that we slow the system down ($\tau > 100$). So, we can cancel out each of the poles and replace them with slower ones. While it is not necessary to slow down *every* pole, we can obtain better stability margins by doing so.

$$Y = \frac{10(1 + 100s)^2(0.1s + 1)s}{(200s + 1)^4} \quad (4.114)$$

$$T = \frac{1 - 100s}{(200s + 1)^4} \quad (4.115)$$

EXAMPLE 4.11

Consider the following non-minimum phase plant.

$$G_p(s) = \frac{20(1 - \frac{s}{2})}{(s + 1)(\frac{s}{10} + 1)} \quad (4.116)$$

The current closed loop time constants are $\tau = 1$ and $\tau = 1/10$ seconds. We want to design a closed-loop response with time constants $\tau = 1/2$ and $\tau = 1/100$ seconds.

$$T = YG_p = Y \frac{20(1 - \frac{s}{2})}{(s + 1)(\frac{s}{10} + 1)} \quad (4.117)$$

$$Y = \frac{(s + 1)(\frac{s}{10} + 1)}{20(\frac{s}{2} + 1)(\frac{s}{100} + 1)} \quad (4.118)$$

Therefore,

$$T = \frac{1 - \frac{s}{2}}{(\frac{s}{2} + 1)(\frac{s}{100} + 1)} \quad (4.119)$$

$$S = \frac{1.01s(\frac{s}{202} + 1)}{(\frac{s}{2} + 1)(\frac{s}{100} + 1)} \quad (4.120)$$

$$G_c = \frac{Y}{S} = \frac{1}{20.2} \cdot \frac{(s + 1)(\frac{s}{10} + 1)}{s(\frac{s}{202} + 1)} \quad (4.121)$$

Verify the well-posedness and the interpolation conditions:

$$L = G_c G_p = \frac{1}{20.2} \frac{(s + 1)(\frac{s}{10} + 1)}{s(\frac{s}{202} + 1)} \cdot \frac{20(1 - \frac{s}{2})}{(s + 1)(\frac{s}{10} + 1)} = \frac{\frac{20}{20.2}(1 - \frac{s}{2})}{s(\frac{s}{202} + 1)} \quad (4.122)$$

$$1 + L = 1 + \frac{\frac{20}{20.2}(1 - \frac{s}{2})}{s(\frac{s}{202} + 1)} = \frac{\frac{20}{20.2}(\frac{s}{2} + 1)(\frac{s}{100} + 1)}{s(\frac{s}{202} + 1)} \quad (4.123)$$

$$S(2) = \frac{1.01s(\frac{s}{202} + 1)}{(\frac{s}{2} + 1)(\frac{s}{100} + 1)} = 1 \quad (4.124)$$

Y is proper, internal stability and the interpolation conditions are met. However, there are still problems with this design. Consider the frequency and step responses shown in Figures 4.36 and 4.37.

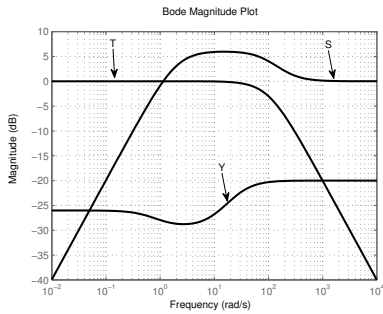


Figure 4.36: Magnitude plot of S , T , and Y .

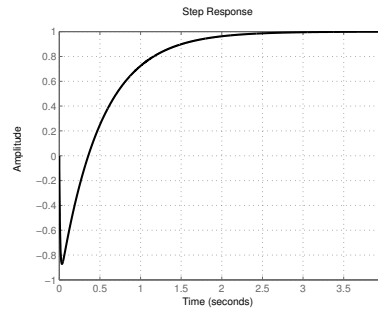


Figure 4.37: Closed-loop step response. The controller produces unacceptably large undershoot.

$\|S\|_{\infty} \approx 2$, which means the M_2 margin is too low. Output disturbances will be amplified and the system will have poor command following (T alone cannot tell you about good command following — one must look at S as well). Overall, it is a poor design.

The pole at $s = -2$ and zero at $s = +2$ both occur at the frequency $\omega = 2$. The response could be corrected (for instance) by adding a pole at $\omega < 2$ to correct the non-minimum phase zero and reduce the M_2 margin.

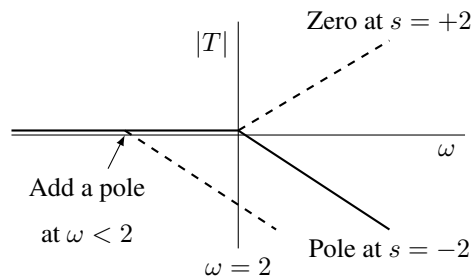


Figure 4.38: $T = YG_p$ when adding a pole.

EXAMPLE 4.12

Consider the plant

$$G_p = \frac{s^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (4.125)$$

Typically we want new poles to have $\tau > 1/|z|$ for non-minimum phase zeros, but here the non-minimum phase zeros are at the origin — an appropriate time constant would be infinitely large. That is not an option, so instead we must settle for a time constant of new poles that is very large.

$$Y = \frac{\omega_n^2 \tau^2}{(1 + \tau s)^2} \quad (4.126)$$

Then,

$$T = \frac{\omega_n^2 \tau^2 s^2}{(1 + \tau s)^2 (s^2 + 2\zeta\omega_n s + \omega_n^2)} \quad (4.127)$$

We can check our interpolation conditions and clearly see that they are met:

$$T(0) = \frac{\omega_n^2 \tau^2 0^2}{(1 + \tau 0)^2 (0^2 + 2\zeta\omega_n 0 + \omega_n^2)} = 0 \quad (4.128)$$

$$\left. \frac{dT}{ds} \right|_{s=0} = 2s(\dots) + s^2(\dots) \Big|_{s=0} = 0 \quad (4.129)$$

(The complete definition of the derivative of T is not given, but it should be clear that all terms will have an s in their numerator.)

$$T(0) \neq 1 \quad (4.130)$$

So we know immediately that this control system will have steady-state error.

Then,

$$S = 1 - T = \frac{(1 + \tau s)^2 (s^2 + 2\zeta\omega_n s + \omega_n^2) - \omega_n^2 \tau^2 s^2}{(1 + \tau s)^2 (s^2 + 2\zeta\omega_n s + \omega_n^2)} \quad (4.131)$$

$$G_c = \frac{Y}{S} = \frac{(s^2 + 2\zeta\omega_n s + \omega_n^2)(\omega_n^2 \tau^2)}{(1 + \tau s)^2 (s^2 + 2\zeta\omega_n s + \omega_n^2) - \omega_n^2 \tau^2 s^2} \quad (4.132)$$

$$= \frac{s^2 + 2\zeta\omega_n s + \omega_n^2}{(\tau^2 s^2 + 2\tau s + 1)(s^2 + 2\zeta\omega_n s + \omega_n^2) - \omega_n^2 \tau^2 s^2} \quad (4.133)$$

$$\Rightarrow G_c = \frac{(s^2 + 2\zeta\omega_n s + \omega_n^2)(\omega_n^2 \tau^2)}{\tau^2 s^4 + (2\zeta\omega_n \tau^2 + 2\tau)s^3 + (1 + 4\tau\zeta\omega_n)s^2 + 2\tau\omega_n^2 s + \omega_n^2} \quad (4.134)$$

EXAMPLE 4.13

Consider the plant

$$G_p = \frac{4(0.4 - s)}{(s + 0.1)(s + 0.2)(s + 0.8)} \quad (4.135)$$

that is subject to actuator saturation design constraints $|u(t)| \leq 0.10$ for noise $|n(t)| \leq 0.01$ at $\omega > 0.1$ rad/sec and input $|r(t)| \leq 1$ at $\omega < 0.1$ rad/sec. First consider the constraint on saturation due to noise:

$$\frac{|\hat{u}|}{|\hat{n}|} = \frac{0.10}{0.01} = 10 = 20\text{dB} \quad (4.136)$$

So, $|Y(\omega)|$ must be less than 20dB at $\omega > 0.1$ rad/sec. For the constraint on saturation due to commanded inputs,

$$\frac{|\hat{u}|}{|\hat{r}|} = \frac{0.10}{1} = 0.1 = -20\text{dB} \quad (4.137)$$

So, $|Y(\omega)|$ must be less than -20dB at $\omega < 0.1$ rad/sec. We start by choosing a shape for Y . We cannot cancel the non-minimum phase zero, but will cancel all three poles and replace them with one first-order pole and one second-order pair.

$$Y = \frac{K(s + 0.1)(s + 0.2)(s + 0.8)}{4(\tau s + 1)(s^2 + 2\zeta\omega_n + \omega_n^2)} \quad (4.138)$$

Then,

$$T = \frac{K(0.4 - s)}{(\tau s + 1)(s^2 + 2\zeta\omega_n + \omega_n^2)} \quad (4.139)$$

Our interpolation conditions are $T(0) = 1$ (for command tracking) and $T(0.4) = 0$. $T(0.4) = 0$ is clearly met, while $T(0) = 1$ requires constraints on the tuning parameters K and ω_n :

$$T(0) = 1 = \frac{0.4K}{\omega_n^2} \quad (4.140)$$

Therefore, $K = \omega_n^2/0.4$.

We can assume that the peak actuator effort will occur at the start of a step response. Then, use the initial value theorem to obtain an additional constraint.

$$\begin{aligned} \mathcal{L}^{-1}[Y(s)]|_{t=0} = 0.1 &= \lim_{s \rightarrow \infty} sY(s) \frac{1}{s} \\ &= \lim_{s \rightarrow \infty} \frac{K(s + 0.1)(s + 0.2)(s + 0.8)}{4(\tau s + 1)(s^2 + 2\zeta\omega_n + \omega_n^2)} = \frac{K}{4\tau} \end{aligned} \quad (4.141)$$

Therefore, $\tau = K/0.4 = (\omega_n/0.4)^2$. At this point we can tune the system with ζ and one of K , τ , or ω until the design requirements are met.

Let's select $K = 1$ and $\zeta = .707$. The resulting Youla magnitude plot is shown in Figure 4.39.

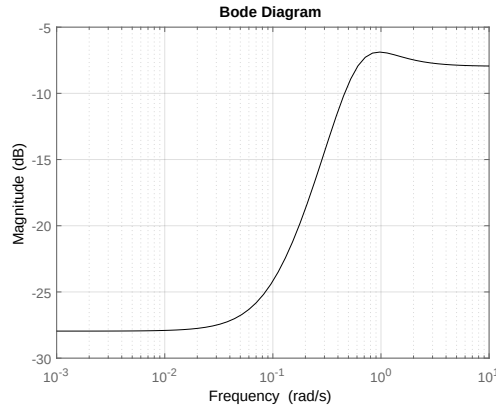


Figure 4.39: Magnitude response of Y .

4.8 Problems

4.1 Consider the unstable and non-minimum phase plant,

$$G_p = \frac{10(1-s)}{(1-s/2)}$$

(a) Design a compensator, G_c , that stabilizes the plant by selecting an appropriate $Y(s)$ to ensure closed loop internal stability using interpolation conditions. Compute T , S , and G_c . Provide Bode magnitude plots of T and Y . Is your design acceptable? Briefly comment on your design.

(b) Propose a new $Y(s)$ that not just ensures internal stability but also provides better performance and command tracking. For this case, compute only T and briefly comment on your design.

4.2 Consider the dynamics of a satellite positioning system,

$$G_p = G_{p,0} \cdot G_{p,Ap}$$

$$G_{p,0} = \frac{0.1}{s(100s+1)(0.1s+1)}$$

$$G_{p,Ap} = \frac{1-100s}{1+100s}$$

Classical controller design would require hand-tuning poles, zeros, and integrators for phase correction, high frequency roll-off, boosted low frequency gain, and so on. With neo-classical control, the process is much simpler.

Use Youla parameterization to design a feedback controller that results in a relatively fast response time, by selecting an appropriate closed loop time constant, with zero percent overshoot, and meeting the internal stability constraint. You need to only provide your choice of $Y(s)$ transfer function and compute $T(s)$. There is no need to provide any Bode plots or compute the controller transfer function.

4.3 This problem considers a system with M_2 stability measure of $M_2 = 1$. Your task is to compute $\max |S(j\omega)|$ in dB and to compute the classical gain margin associated with $M_2 = 1$.

Hint: For computing the classical gain margin, you can do this geometrically by plotting an arbitrary $L(j\omega)$ on a polar plot and then superimposing the $M_2 = 1$ circle on this plot to illustrate the classical gain margin. Or, use the equation that relates $|S(j\omega)|$ to the classical gain margin.

4.4 The transfer function $G_p(s)$ describes the simplified dynamics of a transmission clutch,

$$G_p = \frac{-(150)(s^2 - 0.12s + 144)}{(s + 20)(s^2 + 30s + 10000)}$$

Use Youla parameterization to design a feedback controller that meets the specifications in Table 4.1.

Table 4.1: Requirements

Metric	Specification
Settling time	2.0s
Overshoot	< 10%
Steady-state error	0
Gain margin	> 10dB
Phase margin	> 60°
Crossover frequency	> 5rad/s

Justify your design by providing all the relevant Bode magnitude plots and step responses.

4.5 Consider the block diagram shown in Figure 4.40. Design $G_c(s)$ so that the closed-loop system behaves approximately like a second order system, which, for a unit step input, has

- an overshoot that does not exceed 16% ($\zeta \geq 0.5$),

- a settling time less than or equal to 5.3 seconds ($\omega_n \geq 1.5 \text{ rad/s}$), and
- a steady state error below 5%.

Design the controller using the Youla method. You need to fully justify your choice of compensator and illustrate that the given requirements have been met.

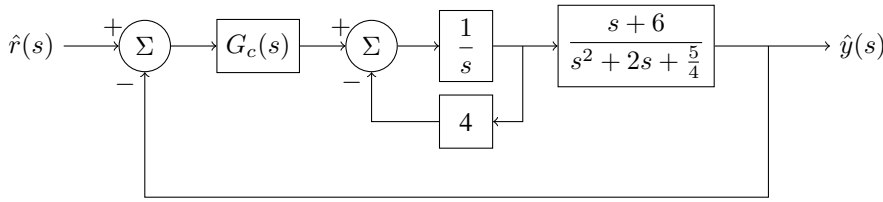


Figure 4.40

4.6 The open-loop block diagram for the active suspension problem in Figure 4.41.

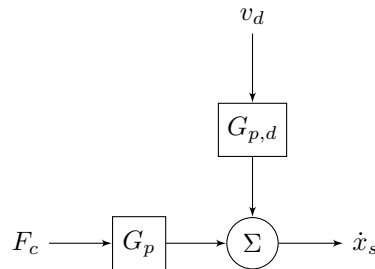


Figure 4.41: Block diagram of the active suspension with sprung mass velocity as an output.

In this problem, your task is to design a Youla controller by hand to minimize the effect of v_d on $\dot{x}_s$. Provide the following Bode plots.

- (a) One plot of T , S , Y , G_p , and G_c .
- (b) One plot of T and $G_{p,d}S$.
- (c) One magnitude plot of $\frac{\dot{x}_s}{v_d}$ (both controlled and uncontrolled).
- (d) One magnitude plot of $\frac{\ddot{x}_u}{v_d}$ (both controlled and uncontrolled).
- (e) One magnitude plot of $\frac{v_{rel}}{v_d}$ (both controlled and uncontrolled).
- (f) One magnitude plot of $\frac{q_s}{v_d}$ (both controlled and uncontrolled).

3762 **4.7** Consider the unstable and non-minimum phase plant,

$$G_p = \frac{e^{-\tau s}}{(s-1)}$$

3763 Design a compensator, G_c , by selecting an appropriate $Y(s)$, where $0 < \tau \leq 0.1$, to
 3764 ensure closed loop internal stability using interpolation conditions and good target follow-
 3765 ing (small steady state errors are acceptable, such as ≤ 0.2), and compute T , S , and G_c .
 3766 Provide Bode plots of T , S , and Y , including your design phase and gain margin, and the
 3767 step response of the closed loop transfer function T . Is your design acceptable? Briefly
 3768 comment on your design.

3769 **4.8** Apply the method of Youla parameterization to the object characterized by

$$G_p = \frac{(10+s)}{s^3}$$

3770 Evaluate the effectiveness of your design by examining the Bode magnitude of $Y(j\omega)$,
 3771 $S(j\omega)$, $T(j\omega)$, and by plotting the closed loop step response.

3772 **Hint:** Instead of interpolation conditions, try to make the four closed loop transfer func-
 3773 tions, Y , YG_p , $1 - YG_p$, $G_p(1 - YG_p)$, BIBO stable to meet internally stability. This can
 3774 be accomplished by enforcing that S transfer function includes s^3 in its numerator.

3775 **4.9** Consider the plant

$$G_p = \frac{s^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

3776 Do your best using Youla parameterization to obtain a controller that can give command
 3777 tracking.

3778 **Note:** You cannot cancel the zeros at $s = 0$, so you will not be able to enforce an interpo-
 3779 lation condition on $T(0) = 1$.

CHAPTER 5

NORMS OF FEEDBACK SYSTEMS

Much of this material is prerequisite for the next chapter, the application of *optimization* in control theory. This application is not new and can be regarded as a branch of mathematics whose goal is to improve a variable in order to maximize a benefit or minimize a cost.

We tried to avoid the necessary mathematical background, especially in complex analysis, for some of developments in this chapter and the chapters to follow. However, there is sufficient background to allow us to retrace certain of our steps and develop rigorous relationships alluded to in only an heuristic manner in previous chapters. In addition, the development of rigorous relationships shed more light into some of the concepts of this chapter and the following chapters. Therefore, we included the background material, “Complex Analysis” and “Real and Complex Analysis in Signal, Dynamic system and Feedback Theory”, in the first two appendices of this book.

In the rest of this chapter, the concentration is on the signal and system norms for SISO system. But, before discussing this topic, we will cover some required concepts in Laplace and Fourier Transform, state-space representations, and controllability and observability.

5.1 The Laplace and Fourier Transform

This section will establish some basic properties of the Laplace and Fourier transforms. These properties will be used throughout the chapter to describe various norms and their uses.

5.1.1 The Inverse Laplace Transform

The Cauchy integral formula, as discussed in Appendix A, is

$$f(\lambda_0) = \frac{1}{2\pi j} \int_{\partial\Omega} \frac{f(\lambda)}{\lambda - \lambda_0} d\lambda \quad (5.1)$$

if $f(\lambda)$ is analytic everywhere on and within the contour “ $\partial\Omega$ ” and if λ_0 is interior to the contour. We say that the value of a function that is analytic in some region is given entirely by the values it attains on the boundary of that region. The counter-clockwise direction is the positive direction of integration in this formula. We wish to apply this formula to the Laplace transform

$$\hat{y}(s) = \int_{0-}^{\infty} e^{-st} y(t) dt \quad (5.2)$$

This integral definition converges for “ s ” in the “region of convergence” which is the half-plane to the right of all poles of $\hat{y}(s)$. That is to say, $\hat{y}(s)$ is analytic everywhere in the region of convergence. This means that where the Cauchy integral formula is applied to (5.1), we have

$$\hat{y}(s) = \frac{1}{2\pi j} \int_D \frac{\hat{y}(\lambda)}{\lambda - s} d\lambda \quad (5.3)$$

Here, “ D ” is the D -shaped contour (Nyquist contour) illustrated in Figure 4.2 and as given in more detail in Appendix A. Imagine taking the inverse transform of both sides of Equation (5.3). The only function of “ s ” on the right side is $\frac{1}{(\lambda - s)}$ and the inverse transform of this term is $-e^{(\lambda t)}$.

$$y(t) = -\frac{1}{2\pi j} \oint_D e^{\lambda t} \hat{y}(\lambda) d\lambda \quad (5.4)$$

We can now replace λ with s and reverse the direction of the path integral.

$$y(t) = \frac{1}{2\pi j} \oint_D e^{st} \hat{y}(s) ds \quad (5.5)$$

Notice that the direction of integration is indicated on the integration symbol itself to serve as a reminder.

The summary of this is given in the following theorem.

Theorem 5.1 (The Laplace transformation)

$$\hat{y}(s) = \int_{0^-}^{\infty} e^{-st} y(t) dt \quad (5.6)$$

3817 *is associated with the inverse Laplace transformation*

$$y(t) = \frac{1}{2\pi j} \oint_D e^{st} \hat{y}(s) ds \quad (5.7)$$

3818 *where the contour “D” encloses the entire half-plane lying to the right of all poles of $\hat{y}(s)$*
 3819 *and the integration is taken in the clockwise direction.* ■

3820 The integral

$$\oint_D e^{st} \hat{y}(s) ds \quad (5.8)$$

3821 is evaluated using Cauchy’s residue theorem. This theorem states that

$$2\pi j \sum_K (\text{Residues of } \hat{y}(s) \text{ at } s = s_K) = \oint_D e^{st} \hat{y}(s) ds = \lim_{\omega \rightarrow \infty} \int_{\sigma - j\omega}^{\sigma + j\omega} \hat{y}(s) e^{st} ds \quad (5.9)$$

3822 σ is a real number so that the contour path of integration is in the region of convergence of
 3823 $\hat{y}(s)$.

3824 ■ **EXAMPLE 5.1**

3825 Use the Nyquist contour to find the value of

$$\mathcal{L}^{-1} \left[\frac{1}{(s+1)(s-2)^2} \right]$$

3826 It is easy to find this particular inverse Laplace Transform using Heaviside expansion
 3827 as covered in Chapter 2; however, this example serves as an illustration of the use of
 3828 the contour integral method. It should be noted that there are examples that there are
 3829 no alternative except using the contour integral method.

3830 Now,

$$\oint_D \frac{e^{st}}{(s+1)(s-2)^2} ds = 2\pi j \sum_K (\text{Residues of } \hat{y}(s) \text{ at } s = s_K)$$

3831 The residue at $s = -1$ is given by

$$\lim_{s \rightarrow -1} (s+1) \frac{e^{st}}{(s+1)(s-2)^2} = \frac{1}{9} e^{-t}$$

3832 The residue at $s = 2$ is given by

$$\lim_{s \rightarrow 2} \frac{d}{ds} \frac{e^{st}}{(s+1)} = \left[\frac{(s+1)te^{st} - e^{st}}{(s+1)^2} \right]_{s=2} = \frac{1}{9} (3te^{2t} - e^{2t})$$

Thus

$$\oint_D \frac{e^{st}}{(s+1)(s-2)^2} ds = 2\pi j \left[\frac{1}{9}(e^{-t} + 3te^{2t} - e^{2t}) \right]$$

Evaluate the integral as the radius of the Nyquist contour goes to ∞ . Thus,

$$\begin{aligned} \mathcal{L}^{-1} \left[\frac{1}{(s+1)(s-2)^2} \right] &= \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} \frac{e^{st}}{(s+1)(s-2)^2} ds \\ &= \frac{1}{9}(e^{-t} + 3te^{2t} - e^{2t}) \end{aligned}$$

A result easily verified by the use of Heaviside expansion.

5.1.2 Parseval's Theorem

There are several extremely important implications of the formulas developed in the previous subsection. These implications are generally called the Parseval theorems.

Begin with the following lemma.

Lemma 5.1. *If $\hat{y}(s)$ is the Laplace transform of a real function, then it is symmetric about the real axis so that*

$$\overline{\hat{y}(j\omega)} = \hat{y}(-j\omega) \quad (5.10)$$

Proof: It follows from (5.6) that

$$\hat{y}(j\omega) = \int_{0-}^{\infty} e^{-j\omega t} y(t) dt = \int_{0-}^{\infty} \cos(\omega t) y(t) dt - j \int_{0-}^{\infty} \sin(\omega t) y(t) dt \quad (5.11)$$

We conclude the lemma. ■

In particular, this result must hold for a transfer function because a transfer function is the Laplace transform of the impulse response.

Often we are interested in an integral called the “energy” of a signal which takes the form

$$E = \int_{0-}^{\infty} y^2(t) dt \quad (5.12)$$

If $\hat{y}(s)$ is not strictly proper, $y(t)$ contains an impulse. Then, the Equation (5.12) for $y(t)$ would be unbounded or undefined:

$$\int_0^{\infty} \delta^2(t) dt = \delta(0) \quad (5.13)$$

3850 The integral for E is finite only if $\hat{y}(s)$ (1) is strictly proper, (2) is characterized by poles
 3851 which lie strictly in the left half-plane. In this case, use (5.7) to write

$$E = \frac{1}{2\pi j} \int_{0^-}^{\infty} y(t) \oint_D e^{st} \hat{y}(s) ds dt \quad (5.14)$$

3852 Reverse the order of integration to find

$$E = \frac{1}{2\pi j} \oint_D \hat{y}(s) \int_{0^-}^{\infty} e^{st} y(t) dt ds \quad (5.15)$$

3853 We have established that the energy of a signal

$$E = \int_{0^-}^{\infty} y^2(t) dt = \frac{1}{2\pi j} \oint_D \hat{y}(s) \hat{y}(-s) ds \quad (5.16)$$

3854 This integral exists where the poles of $\hat{y}(s)$ lie strictly in the left half-plane. The D-
 3855 contour encircles the entire right half-plane and the integration is taken clockwise. The
 3856 other restriction is that $\hat{y}(s)$ be strictly proper.

3857 However, if $\hat{y}(s)$ is strictly proper, the contribution of the curved portion of the con-
 3858 tour to the integral is exactly zero; that is, contour integral (5.16) reduces to an simple
 3859 integration along imaginary axis. On this axis,

$$ds = j d\omega \quad (5.17)$$

3860 and (5.10) means

$$\hat{y}(j\omega) \hat{y}(-j\omega) = |\hat{y}(j\omega)|^2 \quad (5.18)$$

3861 Apply all this to (5.16) to prove:

3862 **Theorem 5.2 (Parseval's Theorem)** *Where it exists, the energy of a signal is*

$$E = \int_{0^-}^{\infty} y^2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\hat{y}(j\omega)|^2 d\omega \quad (5.19)$$

3863 where $\hat{y}(s) \triangleq \mathcal{L}y(t)$. ■

3864 5.1.3 The Fourier Transform

3865 Much of the following requires an understanding of another integral transform, a trans-
 3866 form that can be applied to signals not already encountered in our developments but which
 3867 characterize those which commonly impinge on our control systems.

Definition 5.1. The Fourier transform of a function $v(t)$ is denoted $\mathcal{F}[v(t)]$ or $\hat{v}(j\omega)$ and is given by the improper integral¹

$$\mathcal{F}[v(t)] = \int_{-\infty}^{\infty} e^{-j\omega t} v(t) dt \quad (5.20)$$

We don't restrict our signals to be zero for negative times so the lower limit is taken to be $-\infty$. That is, we expand our horizons beyond that of command signals and begin to study the properties of noise and disturbance signals which have no clear beginning.

Of course, a physical signal at $t = -\infty$ is an idealization, much as it is to say, the idealization of a signal with a discontinuity: The philosophy is the same, it is easier to deal with the idealization that it is to attempt an analysis of the real thing. Of course, the price paid is the manipulation of an improper integral.

An immediate implication is obtained by comparing the definitions of Laplace and Fourier transforms.

Proposition 5.1. Suppose that $v(t) \equiv 0$, $t \leq 0$, and if $\sigma = 0$ is in the domain of convergence of the Laplace transform, $\mathcal{L}[v(t)]$, then

$$\mathcal{F}[v(t)] = \hat{v}(s) \Big|_{s=j\omega} \quad (5.21)$$

■

The point is that we wish to include in our study those signals which do not satisfy the hypothesis of this proposition. This means, for example, that any function with a Laplace transform with all poles strictly in the left half-plane has a Fourier transform, such as in the example below.

EXAMPLE 5.2

Consider the impulse response, $g(t)$, of an asymptotically stable, linear system has a Fourier transform.

$$\mathcal{F}[g(t)] = G(s) \Big|_{s=j\omega} \quad (5.22)$$

¹One encounters variations in the definition of the Fourier Transform. The differences are in the constant-multiplier placed in front of the integral. Obviously, we have chosen 1.0, but, in another text, the choice might be

$\frac{1}{2\pi}$.

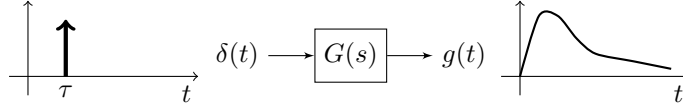


Figure 5.1: Impulse response.

We will demonstrate that we can associate with the Fourier transform, the inverse Fourier transformation

$$v(t) \triangleq \mathcal{F}^{-1}[\hat{v}(j\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{j\omega t} \hat{v}(j\omega) d\omega \quad (5.23)$$

It is usually more convenient to rewrite this as

$$v(t) \triangleq \mathcal{F}^{-1}[\hat{v}(j\omega)] = \frac{1}{2\pi j} \int_{-\infty}^{\infty} e^{j\omega t} \hat{v}(j\omega) d(j\omega) \quad (5.24)$$

So that we can imagine the inverse-transform integral as an integral along the imaginary axis in the s-plane.

This trick will allow us to take advantage of the Cauchy residue theorem. More importantly, we are now able to imagine the Fourier transform as a mapping of sets of time functions onto the imaginary axis of the s-plane. The “domain” (the set of functions to be transformed) is not coincident with the domain of the Laplace transform and neither set is a subset of the other other.

Properties of the Fourier transform: As in the case of the Laplace and Fourier transforms, there are a number of useful facts that are immediate consequences of the definition.

▪ Homogeneity:

$$\mathcal{F}[v(t/\alpha)] = \alpha \hat{v}(\alpha j\omega) \quad \text{if } \alpha > 0 \quad (5.25)$$

▪ Time-shifting:

$$\mathcal{F}[v(t - \tau)] = e^{-j\omega\tau} \hat{v}(j\omega) \quad (5.26)$$

▪ Frequency-shifting:

$$\mathcal{F}[e^{-j\omega_0\tau} v(t)] = \hat{v}(j\omega + \omega_0) \quad (5.27)$$

▪ Symmetry:

$$\hat{v}(-j\omega) = \bar{\hat{v}}(j\omega) \quad (5.28)$$

3907 ■ Differentiation:

$$\mathcal{F}[d^k v(t)/dt^k] = (j\omega)^k \hat{v}(j\omega) \quad (5.29)$$

3908 ■ Convolution:

$$\mathcal{F}\left[\int_{-\infty}^{\infty} v_1(t-\tau)v_2(\tau)d\tau\right] = \hat{v}_1(j\omega)\hat{v}_2(j\omega) \quad (5.30)$$

3909 Where we idealize the signal as extending back to $-\infty$, we have the following generaliza-
3910 tion.

3911 **Theorem 5.3 (Parseval's Theorem)** *Where it exists, the energy of a signal*

$$E = \int_{-\infty}^{\infty} y^2(t)dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\hat{y}(j\omega)|^2 d\omega \quad (5.31)$$

3912 where $\hat{y}(j\omega) \triangleq \mathcal{F}[y(t)]$. ■

3913 **Inverse Fourier-Transformation By Heaviside Expansion:** It is clear that the imaginary
3914 axis must be in the region of convergence of the Fourier transformation. A point that
3915 requires some explanation is how the poles of the transform in the two half-planes are
3916 related to the time-domain signal.

3917 First notice that we can rewrite any signal as

$$v(t) = v_n(t) + v_p(t) \quad (5.32)$$

3918 where

$$v_n(t) = 0, \quad t \geq 0 \quad (5.33)$$

3919 and

$$v_p(t) = 0, \quad t < 0 \quad (5.34)$$

3920 That is, $v_n(t)$ is the signal at the negative time-interval and $v_p(t)$ is the signal at the positive
3921 time-interval, indicated by our choice of subscripts.

3922 Think of this decomposition as converting the Fourier transform into the sum of two
3923 Laplace-like transforms:

$$\mathcal{F}[v(t)] = \int_{-\infty}^0 e^{-j\omega t} v_n(t)dt + \int_0^{\infty} e^{-j\omega t} v_p(t)dt \quad (5.35)$$

3924 The interesting turn-of-events is that the first integral is over negative values of time, t . If
3925 this integral is to exist, it must be true, at the very least, that $v_n(t) \rightarrow 0$ as $t \rightarrow -\infty$.

3926 Replace “ t ” with “ $-t$ ” in the first integral to conclude that

$$\mathcal{F}[v(t)] = \int_0^{\infty} e^{-j\omega t} v_n(-t)dt + \int_0^{\infty} e^{-j\omega t} v_p(t)dt \quad (5.36)$$

This decomposition has the appearance of two Laplace transforms and so we can bring our understanding of the latter transform to develop further our Fourier transform.

Notice that we must have the imaginary axis in the region of convergence of both components of (5.36) and “ t ” is replaced by “ $-t$ ” in the first integral.

This leads to the conclusion that:

The poles of the Fourier transform that lie in the left half-plane are associated with only that portion of $v(t)$ for $t \geq 0$, while those poles in the right half-plane are associated with only that portion of $v(t)$ where $t < 0$.

This observation leads immediately to a Heaviside-expansion scheme for the inverse Fourier transformation.

5.2 Norms of Signals and Systems

Control system analysis and design using optimization techniques requires translation of the system properties and its associated signals into the language of mathematics. One fundamental property of a system and its signal in the analysis and design of control systems is their size. The way we compare the size of different objects mathematically is through real numbers. For example, it is interesting to note that if two real numbers are given, such as 4 and 3, we immediately know 4 is greater than 3. However, if we are given two complex numbers such as $5i$ and $3 + 4i$, an association (a “mapping”) to real numbers, such as the absolute value of a complex number, is required in order to differentiate their sizes. The idea of “minimum” and “maximum” is a reference to a property of a set which can be well-ordered such as a set of real numbers. There are many ways of interpreting the size of a signal or a system. Norms are the geometrical measures of the size of signals and systems that are found to be useful in analysis and design of control systems. These measures of size are the mapping of scalar functions, vector functions, or matrices of rational functions onto the set of real positive numbers.

In this chapter we discuss the properties of scalar signals and SISO systems. We will use the information on Laplace and Fourier transforms presented in the previous section to provide some insight into these norms. We define the related norms that are used for the development of performance/robustness measures for optimal robust controllers.

We will extend and generalize these results to vector signals and MIMO systems in the second volume of this book including several concepts related to systems such as linear-

ity, controllability and observability, and the state-space approach for computation of the related norms.

5.2.1 Signal Norms

Consider the two disturbances shown below. Which is “worse”?

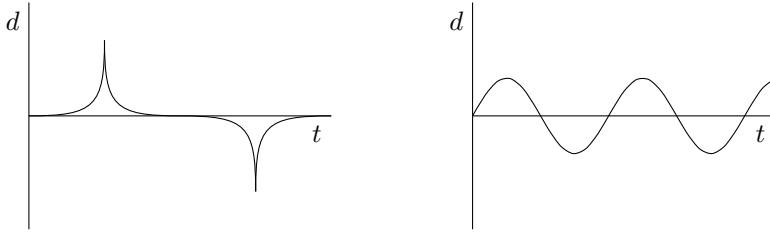


Figure 5.2: Two disturbances.

The answer depends on context. A norm is the measure of the “size” of the signal. To determine how big (or “bad”) a signal is, we need to define what we mean by size. Let $V = \{V_1(\cdot)\}$ be a signal, and let $\|V_1(\cdot)\|$ be a mapping to real number.

Definition 5.2. $\|V\|$ is a semi-norm if it is nonnegative, homogeneous, and does not violate the triangle inequality.

$$\text{Non-Negativity: } \|V\| \geq 0 \quad (5.37)$$

$$\text{Homogeneity: } \|\alpha V\| = |\alpha| \cdot \|V\| \text{ for all } \|V\| < \infty \quad (5.38)$$

$$\text{Triangle Inequality: } \|V + U\| \leq \|V\| + \|U\| \quad (5.39)$$

Definition 5.3. $\|V\|$ is a norm if it is a semi-norm and

$$\|V\| = 0 \text{ only if } V \equiv 0 \quad (5.40)$$

Particular Norms

Definition 5.4. The 1-norm (known as L_1 norm 17.2) measures the integrated absolute error (IAE) and is also a measure of resource consumption.

$$\|y\|_1 = \int_{-\infty}^{\infty} |y(t)| dt \quad (5.41)$$

As explained in the previous section of this chapter, the lower limit is taken here to be $-\infty$.

Definition 5.5. The 2-norm (L_2 norm) measures the integrated square error (ISE) and is also a measure of signal energy.

$$\|y\|_2 = \sqrt{\int_{-\infty}^{\infty} y^2(t) dt} \quad (5.42)$$

Definition 5.6. The ∞ -norm (L_∞ norm) measures peak magnitude.

$$\|y\|_\infty = \sup_t |y(t)| \quad (5.43)$$

where “sup” stands for supremum. It should be noted that if the maximum of a set exists then it is the supremum. However, if the maximum does not exist then the supremum is used.

Definition 5.7. The RMS-norm is a semi-norm (it can be zero for non-zero signals) that is a measure of power in a repeated signal (where τ is the period). It is sometimes written as $\text{pow}(y(t)) = \|y\|_{rms}^2$.

$$\|y\|_{rms} = \sqrt{\lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^\tau y^2(t) dt} \quad (5.44)$$

Using Parseval’s Theorem, we can relate this to a norm in the frequency domain.

$$P_y = \|y\|_{rms}^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_y(\omega) d\omega \quad (5.45)$$

where $S_y(\omega)$ is the power spectral density (PSD) of $y(t)$ such that

$$S_y(\omega) = \lim_{\tau \rightarrow \infty} \frac{|y_\tau(\omega)|^2}{\tau} \quad (5.46)$$

and $\int_{-\infty}^{\infty} y_\tau^2(t) dt = \int_0^\tau y^2(t) dt$, this is the truncated version of the everlasting signal $y(t)$.

EXAMPLE 5.3

Consider the function

$$y(t) = \begin{cases} 0, & t \leq 0 \\ t, & 0 < t \leq 1 \\ 0, & t > 1 \end{cases} \quad (5.47)$$

Then,

$$\|y\|_1 = \int_0^1 t dt = \frac{t^2}{2} \Big|_0^1 = \frac{1}{2} \quad (5.48)$$

$$\|y\|_2 = \sqrt{\int_0^1 t^2 dt} = \sqrt{\frac{t^3}{3} \Big|_0^1} = \frac{1}{\sqrt{3}} \quad (5.49)$$

$$\|y\|_\infty = \sup_t t = 1 \quad (5.50)$$

(This is not a repeated signal, so we will not consider the RMS-norm.)

3989 **EXAMPLE 5.4**

3990 Consider the function

$$y(t) = A \sin t \quad (5.51)$$

This is a repeated signal with period 2π . Therefore,

$$\|y\|_{rms} = \sqrt{\lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^\tau A^2 \sin^2 t dt} \quad (5.52)$$

$$\sin^2 t = \frac{1}{2} - \frac{1}{2} \cos 2t \quad (5.53)$$

$$\frac{1}{\tau} \int_0^\tau A^2 \sin^2 t dt = \frac{1}{\tau} \int_0^\tau \frac{A^2}{2} dt - \underbrace{\frac{1}{\tau} \int_0^\tau A^2 \cos 2t dt}_{=0} \quad (5.54)$$

$$\|y\|_{rms} = \sqrt{\frac{A^2}{2}} = \frac{A}{\sqrt{2}} \quad (5.55)$$

3991 Signals can typically be classified as *energy* or *power* signals. For an energy signal, the
 3992 1-norm, 2-norm, and ∞ -norm are finite and nonzero, while the RMS-norm is zero. On the
 3993 other hand, a power signal has a infinite 1-norm and 2-norm and a finite, nonzero ∞ -norm
 3994 and RMS-norm.

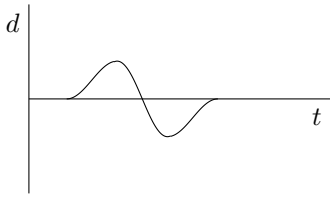


Figure 5.3: An energy signal.

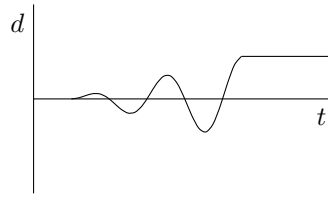


Figure 5.4: A power signal.

3995 **Properties of Norms**

3996 **Proposition 5.2.** The 2-norm is bounded by the product of 1 and ∞ -norms according to

$$\|y\|_2^2 \leq \|y\|_\infty \cdot \|y\|_1 \quad (5.56)$$

Proof:

$$\begin{aligned}
 \|y\|_2^2 &= \int_{-\infty}^{\infty} y^2 dt \\
 &= \int_{-\infty}^{\infty} |y| \cdot |y| dt \\
 \int_{-\infty}^{\infty} |y| \cdot |y| dt &\leq \|y\|_{\infty} \int_{-\infty}^{\infty} |y| dt \\
 \|y\|_{\infty} \int_{-\infty}^{\infty} |y| dt &= \|y\|_{\infty} \|y\|_1 \\
 \|y\|_2^2 &\leq \|y\|_{\infty} \|y\|_1
 \end{aligned}$$

3997

■

3998 **Proposition 5.3.** *If a signal has a finite 2-norm, then the RMS-norm is equal to zero.*

Proof:

$$\begin{aligned}
 \|y\|_{rms}^2 &= \lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^{\tau} y^2(t) dt \\
 &= \left(\lim_{\tau \rightarrow \infty} \frac{1}{\tau} \right) \cdot \left(\lim_{\tau \rightarrow \infty} \int_0^{\tau} y^2(t) dt \right) \\
 &= \|y\|_2^2 \cdot \lim_{\tau \rightarrow \infty} \frac{1}{\tau}
 \end{aligned}$$

3999

■

4000 **Proposition 5.4.** *The power is bounded by $\|y\|_{\infty}$.*

$$\|y\|_{rms}^2 \leq \|y\|_{\infty}^2 \quad (5.57)$$

Proof:

$$\begin{aligned}
 \lim_{t \rightarrow \infty} \frac{1}{\tau} \int_0^{\tau} y^2(t) dt &\leq \|y\|_{\infty}^2 \lim_{t \rightarrow \infty} \frac{1}{\tau} \int_0^{\tau} dt \\
 \|y\|_{rms}^2 &\leq \|y\|_{\infty}^2
 \end{aligned}$$

4001

■

4002 5.2.2 Norms of Dynamic Systems

4003 We will now consider the norms of a dynamic system, $G(s)$. Specifically, we are concerned
 4004 with the 2-norm (H_2) and the ∞ -norm (H_{∞}), as these are the norms which will be used
 4005 when optimization is discussed in the next chapter. Note that “ H_p ” denotes Hardy spaces

— the set of all well behaved functions (analytic functions)². The H_2 and H_∞ norms are defined as follows:

$$H_2 : \|G\|_2 = \sqrt{\frac{1}{2\pi} \int_{-\infty}^{\infty} |G(j\omega)|^2 d\omega} \quad (5.58)$$

$$H_\infty : \|G\|_\infty = \sup_{\omega} |G(j\omega)| \quad (5.59)$$

The 2-norm of a dynamic system is calculated as follows.

$$\|G\|_2^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(j\omega)|^2 d\omega \quad (5.60)$$

Recalling that $G(-j\omega) = \overline{G(j\omega)}$, and $|G(j\omega)|^2 = G(j\omega)\overline{G(j\omega)}$ then

$$\|G\|_2^2 = \frac{1}{2\pi j} \int_{-\infty}^{\infty} G(j\omega)G(-j\omega) d(j\omega) \quad (5.61)$$

Then by replacing the limits of integration with the Nyquist D-contour (this can be done as $G(j\omega)$ is strictly proper, as per the previous section), we can write

$$\|G\|_2^2 = \frac{1}{2\pi j} \int_D G(s)G(-s) ds \quad (5.62)$$

By the residue theorem,

$$\|G\|_2^2 = 2\pi j \left(\frac{1}{2\pi j} \right) \sum_k R_k \quad (5.63)$$

where

$$\int_D G(s)G(-s) ds = 2\pi j \sum_k R_k \quad (5.64)$$

at its poles in the LHP. This leads to the following theorem:

Theorem 5.4 *The 2-norm of a system is*

$$\|G\|_2^2 = \sum_k R_k \quad (5.65)$$

where R_k are the residues of $G(s)G(-s)$ in the LHP.

²we have used L_p and H_p for signals and systems, however, one should note that these norm spaces are equivalent for well behaved functions

EXAMPLE 5.5

Consider the dynamic system $G(s)$.

$$G(s) = \frac{K}{\tau s + 1} \quad (5.66)$$

We can find the 2-norm as follows,

$$G(s)G(-s) = \frac{K^2}{(\tau s + 1)(-\tau s + 1)} \quad (5.67)$$

$$G(s)G(-s) = \frac{K^2}{\tau^2} \cdot \frac{1}{\left(s + \frac{1}{\tau}\right)\left(-s + \frac{1}{\tau}\right)} \quad (5.68)$$

$$\|G\|_2^2 = R_1 = \left(s + \frac{1}{\tau}\right) G(s)G(-s) \Big|_{s=-\frac{1}{\tau}} \quad (5.69)$$

$$R_1 = \frac{K^2}{2\tau} \quad (5.70)$$

$$\|G\|_2 = \frac{K}{\sqrt{2\tau}} \quad (5.71)$$

5.2.3 Input-Output Norms

Transient Inputs (Energy Bounded): What system property (“gain”) relates an output to an input norm?

Suppose that some dynamic system is given an input $u(t) = \delta(t)$ so that $y(t) = g(t)$ where, $g(t)$ is the system impulse response, as shown in Figure 5.5.

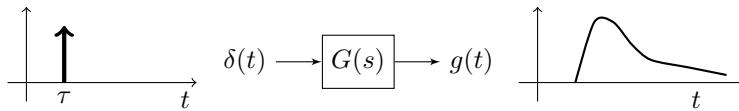


Figure 5.5: Impulse response.

Then,

$$\int_0^\infty y^2(t)dt = \int_0^\infty g^2(t)dt \quad (5.72)$$

So,

$$\|y\|_2^2 = \|g\|_2^2 \quad (5.73)$$

Per Parseval’s Theorem,

$$\|y\|_2^2 = \|G\|_2^2 \quad (5.74)$$

In the case of in impulse function input $u(t) = \delta(t)$, the 2-norm of the output norm is equal to the 2-norm of the dynamic system.

Now let us consider the more general case, where $u(t)$ is any arbitrary input. Then,

$$\|y\|_2^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(j\omega)|^2 |u(j\omega)|^2 d\omega \quad (5.75)$$

$$\leq \frac{\|G\|_{\infty}^2}{2\pi} \int_{-\infty}^{\infty} |u(j\omega)|^2 d\omega \quad (5.76)$$

$$\|y\|_2^2 \leq \|G\|_{\infty}^2 \|u\|_2^2 \quad (5.77)$$

$$(5.78)$$

4030 So, for any input $u(t)$ the “gain” $\|y\|_2 / \|u\|_2$ is bounded by the infinity norm of the dy-
4031 namic system $G(s)$.

$$\frac{\|y\|_2}{\|u\|_2} \leq \|G\|_{\infty} \quad (5.79)$$

4032 Then there exists a some input, u , such that we can define the norm of the dynamic system
4033 in terms of the input and output norms, per Equation (5.79),

$$\|G\|_{\infty} = \sup_{\|u\|_2 \neq 0} \frac{\|y\|_2}{\|u\|_2} \quad (5.80)$$

4034 We can place a bound on the ∞ -norm of $y(t)$ in a similar manner. We begin with

$$y(t) = \int_0^{\infty} g(t - \tau) u(\tau) d\tau \quad (5.81)$$

4035 using convolution to express the behavior of the dynamic system. Then,

$$|y(t)| = \left| \int_0^{\infty} g(t - \tau) u(\tau) d\tau \right| \quad (5.82)$$

We can then use the Cauchy–Schwarz (17.2) Inequality to obtain

$$|y(t)| \leq \sqrt{\int_0^{\infty} g^2(t - \tau) d\tau} \cdot \sqrt{\int_0^{\infty} u^2(\tau) d\tau} \quad (5.83)$$

$$\leq \|g\|_2 \|u\|_2 \quad (5.84)$$

$$\leq \|G\|_2 \|u\|_2 \quad (5.85)$$

4036 Therefore, the maximum value of $y(t)$ is bounded the product of the input $u(t)$ 2-norm and
4037 the dynamic system $G(s)$ 2-norm.

$$\frac{\|y\|_{\infty}}{\|u\|_2} \leq \|G\|_2 \quad (5.86)$$

4038 **Persistent Inputs (Energy Unbounded):** Let $S_u(\omega)$ and $S_y(\omega)$ indicate the Power Spec-
4039 tral Density (PSD) of the input and output signals of a dynamic system with a transfer
4040 function $G(j\omega)$, respectively. Then,

$$S_y(\omega) = |G(j\omega)|^2 S_u(\omega) \quad (5.87)$$

Let's assume white noise input with $S_u(\omega) = 1$. Then,

$$S_y(\omega) = |G(j\omega)|^2 \Leftrightarrow \frac{1}{2\pi} \int_{-\infty}^{\infty} S_y(\omega) d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(j\omega)|^2 d\omega \quad (5.88)$$

Then, the power of the output, $y(t)$, is given by

$$P_y = \|G\|_2^2 \quad (5.89)$$

Therefore, the H_2 norm of a transfer function measures the RMS response of its output when it is driven by a white noise excitation input.

For these persistent signals, we can also write,

$$\|y\|_{rms}^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_y(\omega) d\omega \quad (5.90)$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(j\omega)|^2 S_u(\omega) d\omega \quad (5.91)$$

$$\leq \frac{\|G\|_{\infty}^2}{2\pi} \int_{-\infty}^{\infty} S_u(\omega) d\omega \quad (5.92)$$

$$\|y\|_{rms}^2 \leq \|G\|_{\infty}^2 \|u\|_{rms}^2 \quad (5.93)$$

Therefore,

$$\frac{\|y\|_{rms}}{\|u\|_{rms}} \leq \|G\|_{\infty} \quad (5.94)$$

By concentrating the PSD of the input u near frequencies at which $|G(j\omega)| \approx \|G\|_{\infty}$, then the following is obtained,

$$\|G\|_{\infty} = \sup_{\|u\|_{rms} \neq 0} \frac{\|y\|_{rms}}{\|u\|_{rms}} \quad (5.95)$$

The norms described in this section will be utilized in the next chapter for the controller design using optimization techniques.

5.3 Quantifying Uncertainty

Consider some arbitrary complex system model that could be nonlinear or infinite-dimensional or both, and which is some representation of a real world system. Let w be an input to the system, and let z be an output of the system, as shown in Figure 5.6.

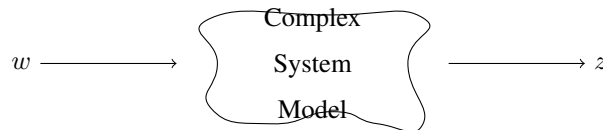


Figure 5.6: An arbitrary complex dynamic system.

Next, consider G to be a simplified model of this complex system, so that we have the dynamics shown in Figure 5.7. By “simplified model” we mean that the original complex model is reduced so that it could be represented by a transfer function for the purpose of control design using the techniques that are laid out in this book.

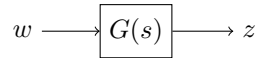


Figure 5.7: A simplified dynamic system.

There are two fundamental questions that we should ask before attempting to design a control system:

1. Is the complex model an exact picture of reality?
2. Is the simplified model a good representation of this complex model?

The short answer to both of these questions is **no**. The following quote by George Box³ is particularly fitting:

“All models are wrong, but some are useful.”

In other words, it is impossible to develop an exact model of reality, no matter how sophisticated and complex our models are. Furthermore, we have to proceed with care when simplifying a model so that we do not discard the very thing that we are after or trying to control in this case. In any act of simplification (or developing a simple model), there is a trade-off between complexity and uncertainty⁴. As we simplify a model of a system, the complexity is reduced but inherently, the uncertainty is increased. However, this is not necessarily a bad thing. If we could quantify and model the uncertainty, we can design or analyze a control system for its performance and robustness to these uncertainties. For instance, we could use the system shown in Figure 5.8, which uses a simplified model G and some model of uncertainty Δ , where v and η are internal signals of G . By utilizing these signals, the Δ model acts as a perturbation to the nominal model G .

³18 October 1919-28 March 2013, was a British statistician, who worked in the areas of quality control, time-series analysis, design of experiments, and Bayesian inference.

⁴Here, what we mean by uncertainty is the level of confidence in our model

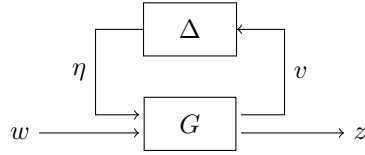


Figure 5.8: A dynamic system with modeled uncertainty Δ .

How do these uncertainties relate to model simplification and how do we characterize and quantify these uncertainties? These are the questions we try to address in this section.

5.3.1 The Characterization of Uncertainty in Models

In Chapter 4 and the last section we discussed M_2 -margin and the H_∞ -norm. We stated that “the H_∞ -norm of the sensitivity, $S(s)$, is the M_2 stability-margin and the frequency of this maximum is a closed-loop resonant frequency.” In addition, we pointed out that “we always seek to reduce the H_∞ -norms of $T(s)$ and $S(s)$ in order to obtain sufficient damping and stability margins and to reduce closed-loop resonances.

It has been pointed out that the specification of the M_2 margin is conservative because it is independent of frequency. That is, it may be that higher values of $\max |S(j\omega)|$ can be tolerated in some frequency ranges. In this section, we begin a development that will end with a more satisfactory treatment of these matters.

First of all, we attempt to depict our uncertainty in a more precise way. We do that by finding some statement that characterizes all possible values of $G(s)$. That is to say, we cannot hope to design without laying down something (say, some bounds) on what might be the range of possibilities for the input-output relation. We are “uncertain” about all aspects of the model of a dynamic system and we choose to characterize that as an uncertainty about the frequency response, the magnitude and phase of $G(j\omega)$.

In characterizing uncertainty about the frequency response, the uncertainties are categorized as follows,

- **Unstructured Uncertainty:** This means a disc is specified in the G -plane (Nyquist plot of G) for each frequency ω . The disc, which is specified in terms of the location of its center and its radius, is such that all variations of $G(j\omega)$ are expected to lie within the disc.
- **Structure Uncertainty:** This means that we specify the specific set of points that represent all possible values of $G(j\omega)$.

Think of it this way: the unstructured-uncertainty disc is a conservative covering of the structured uncertainty. There are many ways to obtain an uncertainty statement. A pragmatic way of defining uncertainty is to roughly divide the frequency response region into:

- **Low-frequency Uncertainty:** An example of this is parameter uncertainty which is a statement of the bounds on the parameters of the state variables or transfer-function model.

“Parameter uncertainty is an example of a structured uncertainty statement”, because one obtains the unique values of the magnitude and phase give the model. In fact, this is the most highly structured form of an uncertainty statement and is considered in the low frequency region.

- **High-frequency Uncertainty:** An example of this is unmodeled dynamics, unmodeled either because high frequency dynamics would have no significant consequences or because the dynamics were too difficult to identify and/or model.

“The greatest kind of uncertainty occurs at high frequency and that is considered an unstructured uncertainty.” We are confident about a structured form of uncertainty only at lower frequencies. Implicit in a statement of parameter uncertainty is the assumption that the model is quite good; so good, in fact, that it is only the precise values of parameters are in the doubt. The unstructured uncertainty at high frequencies, by its very nature, must be based upon the experience of successful designers and may have very little purely scientific foundations.

Here is one point-of-view, the attitude of which adopted for this text:

- Any design method that does not account for high frequency uncertainty (a form of unstructured uncertainty) is not viable.
- Any design method that is based on unstructured uncertainty at all frequencies is overly conservative.
- A design method based on low frequency uncertainty (a form of structured uncertainty) is viable only if there is confidence in the model of the controlled object.

Anyone that accepts this view will recognize that the classical method is viable because the use of stability margins is, in fact, accounting for the high frequency uncertainty. One can argue that the use of margins may not be an adequate accounting, but there is no denial of the intent of the use of these numbers in design.

5.3.2 Weighting Factors and Stability Robustness

The classical stability margins only bound the model uncertainty at two distinct frequencies. Thus, it is possible to imagine situations wherein the margins are accurate measures of the unstructured uncertainty and the margin bounds are met by the designer, but the variations at other frequencies are sufficient to cause instability.

One of the features of the neoclassical design is that a more satisfactory representation is made of the “high frequency/unstructured” uncertainty. We shall explore the case where the statement of uncertainty is unstructured at all frequencies in spite of our observation that this is a very conservative approach. The results are too interesting to ignore.

There are number of ways to represent unstructured uncertainty and it is rewarding to examine all of them. The first formulation begins as follows: Denote the nominal value of the transfer function by $G_p(s)$ and any other possible input-output relation by $\tilde{G}_p(s)$, then

$$\frac{\tilde{G}_p}{G_p} - 1$$

is a measure of the variation of the model. Imagine factoring this rational function as

$$\frac{\tilde{G}_p}{G_p} - 1 = \Delta(s)W_s(s) \quad (5.96)$$

where the factor $\Delta(s)$ is our model of unstructured uncertainty at all frequencies and the factor $W_s(s)$ is a weighting factor that emphasizes the importance of this uncertainty in different frequency ranges. The factors are selected such that the poles of Δ lie strictly in the left half-plane (strictly stable unstructured uncertainty)⁵ and

$$0 \leq |\Delta(j\omega)| \leq 1 \quad \text{or} \quad \|\Delta\|_\infty \leq 1 \quad (5.97)$$

Note that the subscript “s” is used to denote the fact that all this is done to obtain information about the stability robustness of feedback control. Rewrite the defining equation for Δ and W_s :

$$\tilde{G}_p = (1 + \Delta W_s)G_p \quad (5.98)$$

It is in this way that this formulation earns its name, *multiplicative* variation. The impact of the variation on feedback control system is investigated in the next section.

⁵constraining the Δ to be stable simplifies the mathematics, however, equivalently, one can relax this constraint

5.3.3 Robust Stability (Complementary Sensitivity) and Uncertainty

We considered some unstructured uncertainty Δ , constrained to $\|\Delta\|_\infty \leq 1$, paired with frequency weighting W_s . If we have a plant model G_p and the uncertainty as shown in Figure 5.9, then we can say that the “true” plant is

$$\tilde{G}_p = (1 + W_s \Delta) G_p \quad (5.99)$$

derived as *multiplicative* uncertainty in the previous section. Similarly, if the modeled return ratio is

$$L_0 = G_c G_p \quad (5.100)$$

then the “true” return ratio is

$$L = G_c \tilde{G}_p = L_0 (1 + W_s \Delta) \quad (5.101)$$

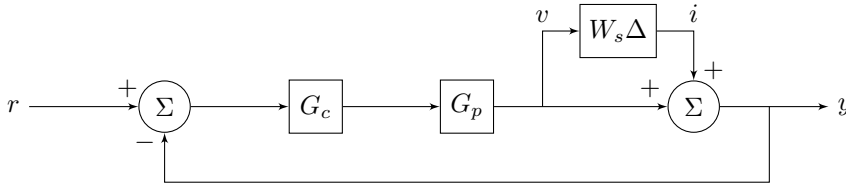


Figure 5.9: True feedback.

Now, let v be the input to the uncertain dynamics $W_s \Delta$ and let i be the output. We then consider the closed-loop transfer function from i to v .

$$v = (-G_c G_p)(i + v) \quad (5.102)$$

So,

$$\frac{v}{i} = -\frac{G_c G_p}{1 + G_c G_p} = -T_0 \quad (5.103)$$

Here, T_0 indicates the closed-loop transfer function when G_p is not subject to any uncertainties. We can then condense the system to the block diagram shown in Figure 5.10.

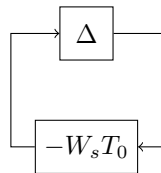


Figure 5.10: Impact of uncertainty on closed-loop.

So, for some bounded perturbation to the closed-loop system, we will capture its magnitude behavior through W_s .

This brings us to the concept of robust feedback: A compensator is robust with respect to a feedback property (e.g. stability) if that property obtains for all G_p in some set of plants called P — that is, all plants within a defined range of uncertainty.

The maximum loop gain in the Figure 5.10 equals $\|-\Delta W_s T_0\|_\infty$ which is < 1 for all allowable Δ 's (e.g. $\|\Delta\|_\infty \leq 1$) if and only if the small-gain condition $\|W_s T_0\|_\infty < 1$ holds. The foregoing discussion is related to the *small-gain theorem* such as $\|W_s T_0\|_\infty < 1$ assures robust stability. A more rigorous proof for robust stability is provided below.

Consider for some unknown perturbation Δ , where

$$\|\Delta\|_\infty \leq 1 \quad (5.104)$$

From Equation (5.99),

$$\frac{\tilde{G}_p(j\omega)}{G_p(j\omega)} - 1 = W_s(j\omega)\Delta \quad (5.105)$$

Accounting for Equation (5.104), we can then say

$$\left| \frac{\tilde{G}_p(j\omega)}{G_p(j\omega)} - 1 \right| \leq |W_s(j\omega)| \quad (5.106)$$

Using the property that norms are homogeneous and multiplying the ratio $\frac{\tilde{G}_p}{G_p}$ by $\frac{G_c}{G_c}$, we can then write

$$\frac{|G_c \tilde{G}_p - G_c G_p|}{|G_c G_p|} = \frac{|L - L_0|}{|L_0|} \leq |W_s(j\omega)| \quad (5.107)$$

Theorem 5.5 (Multiplicative Uncertainty) *The controller G_c provides robust stability for stable $W_s \Delta$ if and only if*

$$\|W_s \Delta T_0\|_\infty < 1$$

and where $\|\Delta\|_\infty \leq 1$, then

$$\|W_s T_0\|_\infty < 1$$

Proof: Let L be the true return ratio and let L_0 be the nominal return ratio. Then,

$$|L(j\omega) - L_0(j\omega)| < |1 + L_0(j\omega)| \text{ for all } \omega \in \mathbb{R}$$

Since L and L_0 have the same right half-plane poles and zeros. This is equivalent to

$$\frac{|L(j\omega) - L_0(j\omega)|}{|L_0(j\omega)|} \cdot \frac{|L_0(j\omega)|}{|L_0(j\omega) + 1|} < 1$$

4188 *As we defined earlier,*

$$\frac{|L - L_0|}{|L_0|} \leq |W_s|$$

4189 *Additionally,*

$$|T_0| = \frac{|L_0|}{|L_0 + 1|}$$

4190 *So,*

$$|W_s \Delta| \cdot |T_0| < 1.$$

4191 *Therefore $\|W_s T_0\|_\infty < 1$ and knowing $\|\Delta\|_\infty \leq 1$, the closed-loop system is robustly*
 4192 *stable. To prove the converse, assume*

$$\|W_s T_0\|_\infty \geq 1$$

4193 *and we prove the result if we can find an allowable “ Δ ” such that the perturbed loop is*
 4194 *unstable. Since “ T_0 ” is strictly proper, there must exist some frequency, ω_1 , such that*

$$\|W_s(j\omega_1)T_0(j\omega_1)\|_\infty = 1$$

4195 *suppose $\omega_1 = 0$ (the more general case is left for the exercises). In that case, $W_s(0)T_0(0)$*
 4196 *is a real number, either +1 or -1. If we take $\Delta = -W_s(0)T_0(0)$, then Δ is allowable and*

$$1 + W_s(0)T_0(0) = 0$$

4197 *the Nyquist locus passes through the critical point so the perturbed loop is not stable;*
 4198 *hence, the closed loop is not internally stable. ■*

4199 The above characterizes robust stability for a multiplicative uncertainty, that is

$$\tilde{G}_p = (1 + W_s \Delta) G_p \quad (5.108)$$

4200 What about other types of uncertainty? Consider the different ways uncertainty can act on
 4201 the plot shown in Figure 5.11.

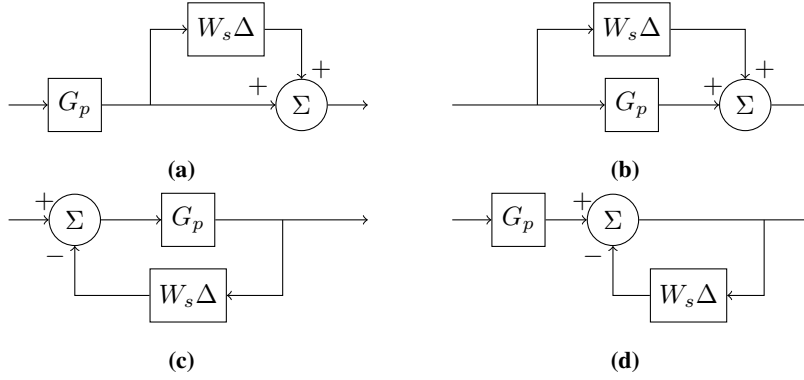


Figure 5.11: Types of uncertainty.

The conditions for robust stability for each of the cases in Figure 5.11 can be derived in a similar manner to the condition for multiplicative uncertainty. These conditions are summarized in Table 5.1.

Table 5.1: Robust stability conditions.

	Characterization	Graphical Interpretation	Robust Stability Condition
a)	$\tilde{G}_p = (1 + W_s\Delta)G_p$	Figure 5.11(a)	$\ W_sT\ _\infty < 1$
b)	$\tilde{G}_p = G_p + W_s\Delta$	Figure 5.11(b)	$\ W_sY\ _\infty < 1$
c)	$\tilde{G}_p = \frac{G_p}{1 + W_s\Delta G_p}$	Figure 5.11(c)	$\ W_sS G_p\ _\infty < 1$
d)	$\tilde{G}_p = \frac{G_p}{1 + W_s\Delta}$	Figure 5.11(d)	$\ W_sS\ _\infty < 1$

To briefly recap, we modeled the unknown high frequency or unstructured uncertainty as Δ . The only assumption that we made regarding this unknown that it is upper bounded (stable) and normalized such that $\|\Delta\|_\infty \leq 1$. This unknown acts as a perturbation to a nominal plant, G_p . The perturbation can generate a family of plants, $\tilde{G}_p$ but because the perturbation is stable, the right half plane poles and zeros of this family of plant are exactly the same as our nominal plant, G_p . For the multiplicative uncertainty model as given in Equation (5.106) and using the *small gain theorem*, we derived a condition for robust stability of the closed loop system, $\|W_sT_0\|_\infty < 1$, where T_0 is the closed loop transfer function and W_s is our filter or weighting function. This weighting function is selected based on our unstructured uncertainty and gives an upper bound to $\frac{\tilde{G}_p(j\omega)}{G_p(j\omega)} - 1$. More on the design of this filter is given in the example below and in the following chapters.

We found if the designed feedback controller G_c satisfies the condition above then not only the feedback loop for the nominal plant would be stable but the stability of the feedback loop for the family of plants due to the perturbation by the unknown uncertainty would

be guaranteed. This is a remarkable outcome. Of course if the norm of the unknown is not normalized such that $\|\Delta\|_\infty \leq \beta$ then the robust stability condition will be equivalently scaled, $\|W_s T_0\|_\infty < \frac{1}{\beta}$.

Assume we design a controller that mathematically meets the robust stability condition. A good question to ask at this stage is, what physical changes occur in the feedback loop so that this controller guarantees robust stability? We will investigate this in Chapter 8 of this text, when we discuss the application-oriented problems and solutions.

We will now consider the following example for designing the robust stability filter, W_s .

EXAMPLE 5.6

Consider a plant subject to uncertain delay.

$$G_p = \frac{1}{s^2} \quad (5.109)$$

$$\tilde{G}_p = e^{-\tau s} G_p, \quad 0 \leq \tau \leq 0.1 \quad (5.110)$$

With $\|\Delta\|_\infty \leq 1$, we want to select W_s so that

$$\left| \frac{\tilde{G}_p}{G_p} - 1 \right| = |e^{-\tau s} - 1| \leq |W_s| \quad \forall \omega, \tau \quad (5.111)$$

Although the work is not shown here, it can be found that

$$|e^{-\tau s} - 1| = \sqrt{2(1 - \cos(\tau\omega))} \quad (5.112)$$

The corresponding magnitude plot of $e^{-\tau s} - 1$ shown in Figure 5.12. Clearly, a high pass filter should be chosen for W_s , such as

$$W_s = \frac{\alpha s}{\tau_f s + 1} \quad (5.113)$$

where α and τ_f are tuning parameters. The magnitude peaks at a value of 2 (or about 6dB), so we need $\alpha > 2\tau_f$ to ensure $|W_s|$ goes to 2 as ω approaches infinity. Additionally, $|W_s|$ must bound $|e^{-\tau s} - 1|$ for the range of possible τ . We can see from Equation (5.112) that as τ increases, the peak magnitude of $|e^{-\tau s} - 1|$ occurs at lower frequencies. Therefore, we shape $|W_s|$ for the worst case delay, where $\tau = 0.1$, the upper limit of τ .

As an example, Figure 5.12 shows $|W_s|$ for $\alpha = 0.25$ and $\tau_f = \tau = 0.1$.

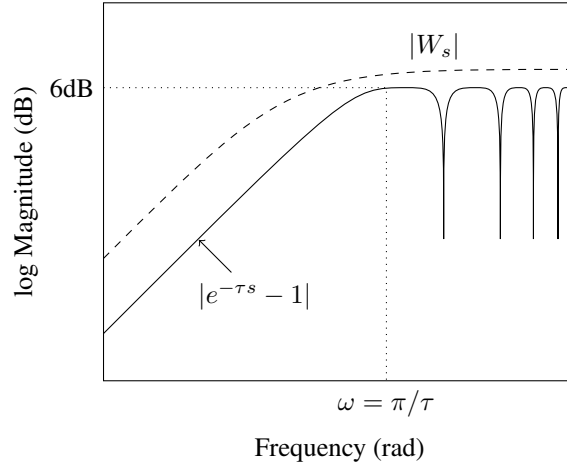


Figure 5.12: Bode plot of $e^{-\tau s} - 1$ and appropriate filter.

5.3.4 Sensitivity and Uncertainty

Consider an error signal

$$e = w - Pu \quad (5.114)$$

Recall that P is a set of plants within a defined range of uncertainty of the nominal plant G_p . As this error signal is a controlled signal that we want to minimize, we will define an output z to be

$$z = W_p e = W_p w - W_p P u \quad (5.115)$$

where W_p is some weighting filter applied to the error signal. We use the subscript p to indicate that this is a *performance* weighting factor. We can write these equations in matrix form.

$$\begin{bmatrix} z \\ e \end{bmatrix} = \begin{bmatrix} W_p & -W_p P \\ 1 & -P \end{bmatrix} \begin{bmatrix} w \\ u \end{bmatrix} \quad (5.116)$$

In this case, the transfer function from w to z is

$$T_{zw} = W_p - W_p P K (1 + P K)^{-1} = W_p (1 - P K (1 + P K)^{-1}) \quad (5.117)$$

letting $e = K u$ define the controller dynamics. That means we have

$$T_{zw} = W_p S \quad (5.118)$$

To minimize the filtered error z , we must minimize $T_{zw} = W_p S$. In particular, we will choose to minimize the infinity norm.

$$\min_K \|T_{zw}\|_\infty = \min_K \|W_p S\|_\infty \quad (5.119)$$

In other words, we want to find the controller K that minimizes the peak magnitude of $W_p S$. The weighting filter W_p can be shaped so that the resulting controller K and sensitivity function S meet the competing design objectives outlined in Section 4.4.3. Conventionally,

$$|W_p S| < 1 \quad (5.120)$$

Therefore,

$$|S| < |W_p^{-1}| \quad (5.121)$$

In other words, the weighting filter W_p should be chosen so that its inverse bounds the desired shape of S .

5.3.5 Performance and Stability

Generally, robust feedback refers to either robust stability or/and robust performance. The H_∞ control problem seeks to find a controller that achieves robust stability while guaranteeing some level of performance. The simplest H_∞ control problem was posed in Equation (5.119) and is restated below.

$$\underset{K}{\text{minimize}} \quad \max_{\omega} |W_p(j\omega)S(j\omega)| \quad (5.122)$$

In other words, find the controller K that minimizes the maximum value of $|W_p S|$ across all frequencies. S was chosen as the transfer function to minimize in order to minimize the error signal.

However, designing around S itself is not enough for good design. We have previously shown that for robust stability, we want

$$\|W_s T\|_\infty < 1 \quad (5.123)$$

Equivalently for S , we have

$$\|W_p S\|_\infty < 1 \quad (5.124)$$

for nominal performance.

Before continuing on, note that we have weighting filter W_p paired with S to affect performance, and weighting filter W_s paired with T to affect stability. From this point onward, we will define W_s as W_Δ to avoid confusion.

To simultaneously address both robust performance and robust stability, we must shape the weighting filters W_p and W_Δ (also called W_1 and W_3 , respectively). Furthermore,

$$\tilde{S} = \frac{1}{1 + \tilde{L}} = \frac{1}{1 + L_0(1 + W_\Delta \Delta)} \quad (5.125)$$

so then,

$$\tilde{S} = \frac{\frac{1}{1+L_0}}{\frac{1}{1+L_0} + (1+W_\Delta\Delta)\frac{L_0}{1+L_0}} = \frac{S_0}{1+W_\Delta\Delta T_0} \quad (5.126)$$

where T_0 was defined in Equation (5.103) and S_0 is similarly defined as

$$S_0 = \frac{1}{1+L_0} \quad (5.127)$$

Given that $\|\Delta\|_\infty \leq 1$, for robust performance we must therefore have

$$\|W_p\tilde{S}\|_\infty = \left\| \frac{W_p S_0}{1+W_\Delta T_0} \right\|_\infty \quad (5.128)$$

Theorem 5.6 *The controller G_c provides robust performance if and only if*

$$\|W_p S_0\|_\infty + \|W_\Delta T_0\|_\infty < 1 \quad (5.129)$$

Theorem 5.6 is often simplified to

$$\|W_p S_0\|_\infty < \frac{1}{2} \quad (5.130)$$

and

$$\|W_\Delta T_0\|_\infty < \frac{1}{2} \quad (5.131)$$

for nominal performance and robust stability in order to enforce robust performance.

In addition, we must penalize the control input u through a third weighting filter W_u or W_2 .

$$\|W_u G_c S_0\|_\infty = \|W_u Y_0\|_\infty < 1 \quad (5.132)$$

where S_0 and Y_0 denote the nominal sensitivity and Youla transfer functions, respectively.

These requirements may be combined into the stacked H_∞ problem:

$$T_{zw} = \begin{bmatrix} W_p S \\ W_u Y \\ W_\Delta T \end{bmatrix}, \quad (5.133)$$

5.4 Problems

5.1 On one hand, we know that signal power can be written as the square of the RMS of a signal such that

$$\|y\|_{RMS} = \sqrt{\lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} y^2 dt}$$

so

$$P_y = \|y\|_{RMS}^2$$

4292 On the other, for a random signal (where signal energy is not finite), signal power can be
4293 written as

$$P_y = 2 \int_0^\infty S_y(2\pi f) df$$

4294 This equation, in discrete case, can be written as,

$$P_{y_i} = (2\Delta f) S_y(2\pi f_i)$$

4295 Given the information above and knowing that a random profile of a road can be approxi-
4296 mated by a superposition of $(N \rightarrow \infty)$ sine waves

$$z_R(t) = \sum_{i=1}^N A_i \sin(\omega_i t - \theta_i)$$

4297 you are asked to find a relation between the discrete road amplitude (A_i) and the road PSD
4298 (Power Spectral Density), $S_{z_R}(2\pi f_i)$ over the narrowband, Δf .

4299 **5.2** A system with the transfer function $\frac{K}{\tau s + 1}$ has a disturbance input, $d(t)$, known to have
4300 the energy bound $\|d\|_2 \leq \alpha$. Suppose that we want to find the best estimate of the ∞ -
4301 norm of the output $y(t)$. Knowing that $\frac{\infty\text{-norm}}{2\text{-norm}} = 2$ -norm of the transfer function. Find
4302 the bound for $\|y\|_\infty$.

4303 **5.3** Apply the method of Youla parameterization to the object characterized by

$$G_p = \frac{4(0.4 - s)}{(s + 0.1)(s + 0.2)(s + 0.8)}$$

4304 Your closed-loop system should behave like a Butterworth filter. Assume that you must
4305 keep $|u(t)| \leq 0.10$ while the sensor noise peaks at $|n(t)| = 0.01$ above 0.25 rad/s. Eval-
4306 uate the effectiveness of your design by examining the Bode magnitude of $Y(j\omega)$, $S(j\omega)$,
4307 $T(j\omega)$. What are the stability margins (phase, gain, and M_2 margins)? Approximate the
4308 closed-loop rise time T_R using the equation

$$\omega_c \times T_R > \frac{1}{4}$$

4309 where ω_c is the cross-over frequency.

CHAPTER 6

FEEDBACK DESIGN BY THE OPTIMIZATION OF CLOSED-LOOP NORMS

Chapter 6 prologue goes here.

6.1 Introduction

The foundation of neoclassical feedback design are developed in the previous chapters.

These basic notions are referred to throughout the remainder of this text.

Feedback design objectives can be stated in terms of three primary rational functions; the sensitivity function, $S = 1/(1 + L)$, the complimentary sensitivity, $T = L/(1 + L)$, and the Youla parameter, $Y = G_c S = T/G_p$. These rational functions are interpreted as closed-loop transfer functions. In particular, disturbance attenuation can be characterized by the sensitivity and the complimentary sensitivity.

To achieve the objective of good feedback design (robustness and attenuation of the effects of unwanted inputs) we arrange things so that the Bode magnitudes of S and Y are as small as possible at all frequencies; however, there are basic trade-offs. We also attempt to arrange things so that the Bode magnitude, $\text{Lm } T$ is small ($\ll 0$ dB) for some purposes and near 0 dB for others. Complicating all this is the fact that Bode magnitudes of these

functions cannot be selected independently since we have a series of relationships such as

$$T(s) + S(s) = 1 \quad (6.1)$$

$$T(s) = Y(s)G_p(s) \quad (6.2)$$

$$S(s) = Y(s)/G_c(s) \quad (6.3)$$

The point is that $S(j\omega)$ and $T(j\omega)$, for example, cannot be varied completely independently so that many of the trade-offs of feedback design are clearly delineated by the Bode diagrams of these rational functions.

It is important to observe that the frequency content is different for the different disturbance and noise signals. This observation indicates an important opportunity, the opportunity to attenuate the effects of some input signals in one frequency range and the effects of other signals in other frequency ranges: A compromise is not required at all frequencies. In the usual case, we expect reference and disturbance signals to be dominated by low frequencies and the sensor noise by relatively high frequency.

There are other aspects of $S(j\omega)$ and $T(j\omega)$ that must be considered in design. First there is the *cross-over frequency*, the frequency ω_c where $|S(j\omega_c)| = |T(j\omega_c)|$. To any extent that we are capable of selecting this frequency, we wish it to separate the sensor-noise frequencies from those frequencies typical of command signals. Also, the reciprocal of the cross-over frequency, $1/\omega_c$, is a measure of the speed of response of the closed loop and that makes things more interesting. Incidentally, there are two other ways to think of the cross-over frequencies:

- ω_c is that frequency where the return ratio $|L| = 1$ (0 dB).
- An approximation of ω_c is that frequency where there is a separation of $|Y(j\omega)|$ from $|1/G_p(j\omega)|$.

The common value of sensitivity and complimentary sensitivity $|S(j\omega_c)| = |T(j\omega_c)|$ (where these two curves intersect) is directly related to the phase margin (see Figure 4.4). Another stability margin is the reciprocal of the peak magnitude (or “ H_∞ norm”) of $S(s)$.

The H_∞ norm of $T(s)$ is a measure of the approximate damping ratio of closed-loop response and for that reason we wish to reduce this number where possible as well.

6.1.1 Frequency Domain Control Design Approaches

The two basic design methods are characterized by a fundamental equation: First, there is the scheme characterized by the equation for the return ratio,

$$L(s) = G_c(s)G_p(s) \quad (6.4)$$

The compensator $G_c(s)$ is selected to “shape” the open-loop (“return ratio”) transfer function. This is an indirect method because one is continually drawing inference about T , S , and Y for each modification made to L with a change in G_c . The classical procedure is described in many control text books, and is based upon fixing or shaping the return ratio, $L(j\omega)$. This is the only alternative where the specific form of the compensator is fixed, say fixed as a PID controller or as a lead-lag compensator.

The designer must also provide for the stability of the feedback loop. If the return ratio is minimum phase, that means the slope of the gain curve, $|L(j\omega)|$, should be -20 dB/decade at the cross-over frequency and -40 dB/decade on either side of the cross-over frequency. The phase margin will likely be between 45° and 90° in this case. The larger value of the phase margin will occur where the cross-over lies near the middle of the -20 dB asymptote.

A larger concept is that of internal stability: It is possible for some closed-loop transfer functions to be stable while the internal stability condition is violated. We say the feedback is internally stable where *all* closed-loop transfer functions are stable. In the classical procedure, internal stability is assured by selecting $G_c(s)$ so that there are no pole-zero cancellations in $L(s)$ in the right half-plane or on the imaginary axis.

It is best not to be too critical of the highly successful classical method; however, it is possible to notice some things that are a bit undesirable. The classical procedure is indirect and that always poses some problems for the beginning student. Also, it is not exactly clear how to select $G_c(s)$ so that the closed loop is stable. Of course, one can use trial and error to eventually stumble upon something that works and there are guides, rules, and tests to aid the designer in this way. The more experienced designer will accomplish this task in an admirably rapid way. However, the procedure is hardly straightforward and this means that after one is finished, there is no guarantee that there is not some easier and better way to achieve the compensation. Also, after stability is assured, the other design criteria must be checked and the procedure started all over where some other kind of problem is indicated. Think of it this way: there is no immediate indication of the trade-offs between stability,

and say, demand on the actuator where the classical approach is used. *The neoclassical procedure suffers from none of these defects.*

The focus of the neoclassical procedure is the equation

$$T(s) = Y(s)G_p(s) \quad (6.5)$$

All feedback design methods involve some form of pole-zero cancellation (this is certainly true in the “modern” control theory — especially where asymptotic observers are employed — but that simple fact is completely obscured in that formulation of things). Some times cancellation is an excellent idea, sometimes a bad idea, and sometimes it is absolutely forbidden. The constraints are internal stability and the demand placed on the actuators (as characterized by $|Y(j\omega)|$). It is *forbidden* to have a pole-zero cancellation in the right half plane or on the axis in the right side of (6.4). On the other hand, it is *absolutely* required that all closed-loop poles of $G_p(s)$ not strictly in the left half-plane be canceled by the zeros selected for $Y(s)$. However, it is *forbidden* to cancel a zero of $G_p(s)$ with a pole selected for $Y(s)$ if that zero is on the axis or in the right half-plane.

So far, we have tried to address all the feedback control design objectives and constraints (some of which are discussed above) through hand-computing an appropriate feedback compensator. In this chapter, we will accomplish feedback control design through various *norm optimization* techniques.

Control design using an optimization approach is discussed in the fourth and final section of this chapter. Before that, we briefly discuss and summarize feedback control design objectives and constraints. In the third section, we introduce Linear Fractional Transformation (LFT), a platform utilized for the design and norm optimization of optimal feedback controllers.

6.2 Optimization Design Objectives and Constraints

In this section, we will discuss control design objectives and constraints. In Section 2 of Chapter 4, we briefly discussed control objectives and constraints for hand computation of the controller using Youla parameter. Similar objectives and constraints are discussed below for the case of control design by norm optimization. The fundamental question is, how does these objectives and constraints translate to the language of mathematics in order to be used by optimization for proper design of a feedback controller? This is addressed below.

As discussed in Chapter 4, the design objectives of neoclassical control are fundamentally accomplished by shaping three closed loop transfer functions, S , T , and Y . However, these design objectives are generally conflicting,

1. Acceptable disturbance rejection and robustness to structured uncertainty (plant parameter variations), $\rightarrow S$ small.
2. Acceptable sensor noise attenuation and robustness to unstructured uncertainty (high frequency unmodeled dynamics) $\rightarrow T$ small.
3. Acceptable reference tracking (good performance), $\rightarrow T \approx 1$.
4. Acceptable control effort, $\rightarrow Y$ small.

It is important to note that Objectives (2) and (3) are in conflict and that these two objectives are both in conflict with (1). In addition, as $Y \approx \frac{1}{G_p}$ in the low frequency region, there is nothing that can be done in this region to reduce control effort!

Let's investigate why such a conflict exists at the first place. Both complementary sensitivity T and sensitivity S transfer functions cannot be kept small simultaneously. This is due to one of the algebraic constraints which will be discussed below. In addition, T cannot simultaneously be kept both small and ≈ 1 , as suggested by objectives 2 and 3. However, because sensor noise attenuation and reference tracking are accomplished in two distinct frequency regions (sensor noise attenuation in high frequency while reference tracking in low frequency regions), it is therefore, possible to avoid this conflict between the objective 2) and 3). We will also see that it is precisely due to this frequency division that the algebraic constraint of S and T could be addressed.

We are in a sense lucky that feedback control design is possible because we can find a suitable compromise between these conflicting design objectives due to this frequency domain division!

In neoclassical feedback control design, there exist two types of constraints, algebraic constraints and analytical constraints. We will address both in this section.

6.2.1 Algebraic Constraints

The following constraints are said to be algebraic constraints.

- The sensitivity and complementary sensitivity transfer functions are two complex vectors in the s-plane when computed at each complex value of "s" with the following

relationship,

$$T(s) + S(s) = 1 \quad (6.6)$$

- The interpolation conditions discussed in Chapter 4 are considered algebraic constraints,

$$T(p) = 1 \quad S(p) = 0 \quad (6.7)$$

and

$$\frac{d^k}{ds^k} T(p) = 0 \quad \frac{d^k}{ds^k} S(p) = 0 \quad 1 \leq k \leq a_p - 1 \quad (6.8)$$

while

$$T(z) = 0 \quad S(z) = 1 \quad (6.9)$$

and

$$\frac{d^k}{ds^k} T(z) = 0 \quad \frac{d^k}{ds^k} S(z) = 0 \quad 1 \leq k \leq a_z - 1 \quad (6.10)$$

where “ z ” denotes an uncanceled zero of the return ratio, L , of multiplicity a_z , and “ p ” denotes an uncanceled pole of the return ratio of multiplicity a_p .

6.2.2 Analytic Constraints

The greatest limitations on stability and disturbance rejection are related to the presence of unstable poles and nonminimum-phase zeros (poles and zeros in the Right Half Plane) and time delays. It should be pointed out, of course, that the rational function approximation of time delay using the Padé approximation results in a transfer functions with zeros in RHP.

The necessary formulas derived in Appendix B can be used to develop the constraints on what can be achieved with feedback.

Non-minimum Phase Effect: The task that is begun here is to explore the limitations and constraints on the design of feedback systems associated with a non-minimum phase controlled object.

The first observation is that it is always possible to factor a non-minimum phase controlled object into the product of a minimum phase and an “all-pass” transfer functions. By “all-pass”, we mean a transfer function for which the Bode magnitude is identically 0 dB

4458 for all frequencies. For example,

$$4459 \quad \text{frac}(1-s)(s+2)(s+3)(s+4) = \overbrace{\frac{(s+1)(s+2)}{(s+3)(s+4)}}^{\text{minimum phase}} \cdot \overbrace{\frac{(1-s)}{(s+1)}}^{\text{all-pass}} \quad (6.11)$$

In general, this means that for any nonminimum phase $T(s)$ and $S(s)$,

$$T(s) = T_{ap}(s)T_{mp}(s) \quad S(s) = S_{ap}(s)S_{mp}(s) \quad (6.12)$$

4459 where the subscript “*ap*” denotes an “all-pass” or *Blaschke factor* and “*mp*” denotes a
4460 *minimum-phase factor*.

4461 Next we call upon the maximum modulus theorem (Appendix A) and apply it to the
4462 D-shaped contour that effectively borders the entire right half plane. We assume that the
4463 design for internal stability is successful so that there are no poles of either $S(s)$ and $T(s)$
4464 in the right half plane. The maximum modulus theorem means that $|S(s)|$ and $|T(s)|$ will
4465 achieve their respective maxima on the contour itself. Specifically speaking, the maxima
4466 of $S(s)$ and $T(s)$ occurs on the imaginary axis.

However, the modulus of an all-pass factor is unity on the imaginary axis, so

$$\max_{\omega} |T(j\omega)| = \max_{\omega} |T_{mp}(j\omega)|$$

4467 Similarly,

$$\max_{\omega} |S(j\omega)| = \max_{\omega} |S_{mp}(j\omega)| \quad (6.13)$$

Now suppose that $L(s)$ (the return ratio) has a right half-plane zero at $s = z$ and a right half-plane pole at $s = p$. Then the all-pass factors have the form

$$T_{ap}(s) = \frac{s-z}{s+z} \quad S_{ap}(s) = \frac{s-p}{s+p} \quad (6.14)$$

4468 On the other hand, interpolation conditions are that $S(z) = 1$ and $T(p) = 1$. So, we have

$$T(p) = 1 = T_{mp}(1)T_{ap}(1) \quad (6.15)$$

4469

$$S(z) = 1 = S_{mp}(1)S_{ap}(1) \quad (6.16)$$

Combine (6.12) and (6.14) to conclude that

$$T_{mp}(p) = T_{ap}^{-1}(p) = \frac{p+z}{p-z} \quad (6.17)$$

$$S_{mp}(z) = S_{ap}^{-1}(z) = \frac{z+p}{z-p} \quad (6.18)$$

4470 It then follows from (6.17) and (6.18) that the peak magnitudes (H_∞ norms)

$$\max_{\omega} |T(j\omega)| \geq \left| \frac{p+z}{p-z} \right| \quad (6.19)$$

4471 and

$$\max_{\omega} |S(j\omega)| \geq \left| \frac{z+p}{z-p} \right| \quad (6.20)$$

4472 If we use frequency weighting functions as we discussed in Chapter 5 to shape both $T(s)$
 4473 and $S(s)$ for performance and robustness objective then we can write the first **Analytic**
 4474 **Constraints** as (6.21) and (6.22) for more information on bounds on the weights W_p and
 4475 W_Δ ,

$$\max_{\omega} |W_\Delta(j\omega)T(j\omega)| \geq \left| W_\Delta(p) \frac{p+z}{p-z} \right| \quad (6.21)$$

4476 and

$$\max_{\omega} |W_P(j\omega)S(j\omega)| \geq \left| W_P(z) \frac{z+p}{z-p} \right| \quad (6.22)$$

4477 Robustness weight (W_Δ) and performance weight (W_p), which are evaluated at the unsta-
 4478 ble pole and zero, enter these constraints. As we saw in chapter 4, this means if an unstable
 4479 pole of a plant is close to an unstable zero then a lower bound is established for the H_∞
 4480 norms. So if a robustness or performance criteria is set such as $\max_{\omega} |W_P(j\omega)S(j\omega)| < 1$,
 4481 and the plant includes an unstable zero and pole, then to achieve this criteria, the perfor-
 4482 mance weight must satisfy $W_p(z) \frac{z+p}{z-p} < 1$. If z and p are close together then this makes
 4483 $W_p z$ very small, which could result in reducing the bandwidth for avoiding a large over-
 4484 shoot in $S(s)$. This is a horrendous constraint, but it must be respected!

4485 **Bode Sensitivity Integral Theorem:** The Bode Sensitivity Integral Theorem is a conser-
 4486 vation theorem formula that quantifies some of the limitations of feedback control systems.
 4487 The consequences of this theorem are the “Area Formula” or “Waterbed Effect” which
 4488 apply to non-minimum phase controlled objects (Appendix B). These consequences are
 4489 elegantly summarized by Bode’s Integral Theorem provided below.

4490 Recall the definition of sensitivity,

$$S \triangleq (1 + L)^{-1} \quad (6.23)$$

4491 Among other things, this closed-loop transfer function characterizes the tracking ability of
 4492 the closed loop. Of course, robustness (disturbance rejection and low frequency parameter
 4493 variations) is another important feedback criterion quantified by this function.

Theorem 6.1 Suppose that the denominator of the return ratio is of degree at least two larger than the numerator and let p_k denote the set of poles of the return ratio which lie strictly in the right half-plane, then the sensitivity is constrained by

$$\int_0^\infty \text{Lm} |S(j\omega)| d\omega = 20\pi e \sum_k p_k \quad (6.24)$$

For the proof, see Appendix B. ■

If the return ratio is stable, however but posses a zero in the RHP (non-minimum phase zero) then,

$$\int_0^\infty \text{Lm} |S(j\omega)| d\omega = 0 \quad (6.25)$$

Figure 6.1: Frequency Response Constraint.

If $L(s)$ is unstable, the area b exceed the area a by $\pi \sum_k p_k$! Any reduction in $|S|$ at low frequencies is purchased at the price of an increase in $|S|$ at high frequencies; overshoot in $|S(j\omega)|$ (and $|T(j\omega)|$) is inevitable.

These algebraic and analytic constraints, at times, put very stringent limitations on feedback control design and they have to be dealt with when designing a feedback controller using either norm optimization or hand computation using Youla parameterization. It is worth mentioning that if the controlled object is stable and minimum phase, the only constraint that we have to deal with is selecting a stable Youla parameter and of course, $T(s) + S(s) = 1$.

6.3 The Linear Fractional Transformation

Linear Fractional Transformation (LFT), as is given by its name, is simply a transformation given by

$$x \rightarrow \frac{ax + b}{cx + d} \quad (6.26)$$

We will discuss this transformation in more detail in the MIMO part of this book. However, it suffices here to say that this transformation interestingly results in a platform which have been utilized for robust optimal control development such as $\|H_2\|$ and $\|H_\infty\|$ norm control optimization. In the next section, we will introduce the LFT and its implication for SISO robust control development.

The Linear Fractional Transformation (LFT) framework for modeling and designing controllers with uncertainty is illustrated in Figure 6.2 and uses the following notation:

- P : Plant or system model.
- K : Controller.
- Δ : The set of all possible uncertainties.
- w : The vector signal of all exogenous inputs (i.e. noises, disturbances, and commands).
- z : The vector signal including all controlled signals and tracking errors.
- u : The controller output (actuator input) signals.
- y : The measurement signals.

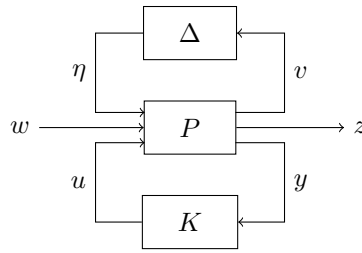


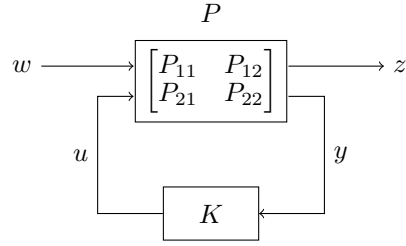
Figure 6.2: An active dynamic system with modeled uncertainty Δ represented by LFT.

This platform is generalized for both MIMO and SISO control development. However, in this section, we will only consider LFT for SISO (where SISO is referred to one actuator input, u , and a single sensor measurement, y) control analysis and design.

Another way of structuring the LFT is to use a matrix form for P , such that

$$\begin{bmatrix} z \\ y \end{bmatrix} = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} \begin{bmatrix} w \\ u \end{bmatrix} \quad (6.27)$$

as shown in Figure 6.3.

Figure 6.3: LFT with matrix representation of P .

4532 Then, the response of z to w and u is

$$z = P_{11}w + P_{12}u \quad (6.28)$$

4533 while the response of y to w and u is

$$y = P_{21}w + P_{22}u \quad (6.29)$$

With $u = Ky$, we can then form the closed-loop definition for u .

$$y = P_{21}w + P_{22}Ky \quad (6.30)$$

$$(I - P_{22}K)y = P_{21}w \quad (6.31)$$

$$y = (I - P_{22}K)^{-1} P_{21}w \quad (6.32)$$

$$u = K(I - P_{22}K)^{-1} P_{21}w \quad (6.33)$$

4534 Using this, we can form the closed-loop response of z to the input w , denoted by the
4535 transfer function matrix T_{zw} .

$$T_{zw} = P_{21} + P_{22}K(I - P_{22}K)^{-1} P_{21} \quad (6.34)$$

4536 This is known as the lower Linear Fractional Transformation (LFT) and is denoted as

$$z = F_l(P, K)w \quad (6.35)$$

4537 We can relate this framework to a conventional block diagram representation as follows.

4538 First, consider the system shown in Figure 6.4.

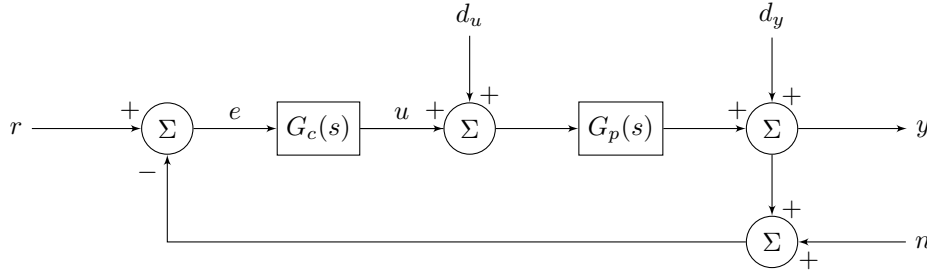


Figure 6.4: Block diagram of a controlled system.

Where without loss of generality, we have replaced the controller K in Figure 6.2 with G_c in the Figure 6.4. In Figure 6.4, we have four inputs to this system: the reference signal input, r , the input disturbance, d_u , the output disturbance, d_y , and sensor noise n . We can then assign $n = w_1$, $d_y = w_2$, $d_u = w_3$, and $r = w_4$ to form the input vector w . The vector of controlled signals z is simply any set of tracked variables, often the same as the measured output y but in this case, it includes the error signal “ e ” and the control output “ u ”. We can then redraw Figure 6.4 as Figure 6.5 and Figure 6.6.

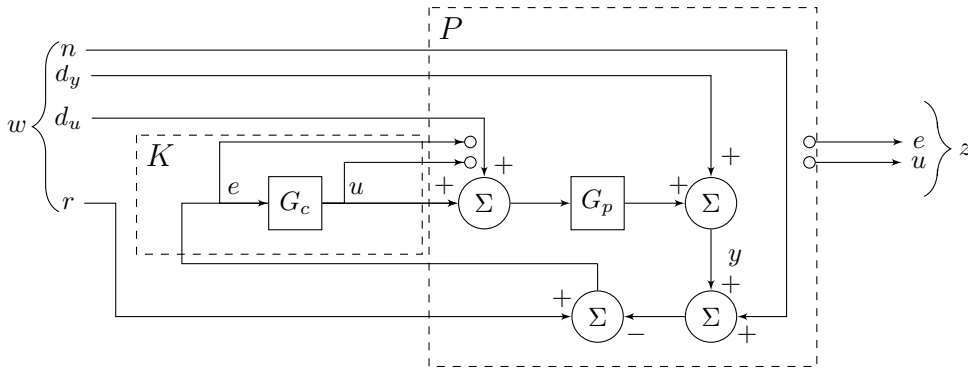


Figure 6.5: Block diagram organized in the LFT framework.

The plant P in Figure 6.5 is the augmented version of G_p and the augment controller K in this case is equal to the original controller G_c . The input-output dynamic equations for the augmented plant P can be written as,

$$z \left\{ \begin{bmatrix} e \\ u \\ y \end{bmatrix} = \left[\begin{array}{ccc|c|c} -1 & -1 & -G_p & 1 & -G_p \\ 0 & 0 & 0 & 0 & 1 \\ -1 & -1 & -G_p & 1 & -G_p \end{array} \right] \begin{bmatrix} n \\ d_y \\ d_u \\ r \\ u \end{bmatrix} \right\} w \quad (6.36)$$

4549 This dynamic system can then be represented by the following LFT, as illustrated in
 4550 Figure 6.6,

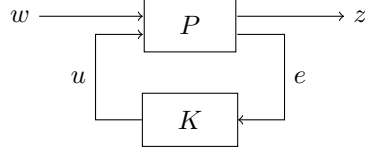


Figure 6.6: The dynamic system manipulated to match the LFT form.

4551 where in Figure 6.6,

$$P = \begin{bmatrix} -1 & -1 & -G_p & 1 & -G_p \\ 0 & 0 & 0 & 0 & 1 \\ -1 & -1 & -G_p & 1 & -G_p \end{bmatrix}, \quad K = G_c, \quad z = \begin{bmatrix} e \\ u \\ e \end{bmatrix}, \quad w = \begin{bmatrix} n \\ d_y \\ d_u \\ r \\ u \end{bmatrix}, \quad u = G_c e$$

4552 Then it is left as an exercise to show that the closed-loop map T_{zw} can be written as

$$z = \begin{bmatrix} -S & -S & -G_p S & S \\ -Y & -Y & -T & Y \end{bmatrix} w \quad (6.37)$$

4553 The transfer function matrix given above includes repetitive transfer functions. Therefore,
 4554 we define a unique and non-repetitive map from the reference and disturbance inputs r and
 4555 d_u to the outputs $z = \begin{bmatrix} e \\ u \end{bmatrix}$ as,

$$\begin{bmatrix} e \\ u \end{bmatrix} = \begin{bmatrix} S & -G_p S \\ Y & -T \end{bmatrix} \begin{bmatrix} r \\ d_u \end{bmatrix} \quad (6.38)$$

4556 Where, as defined in the previous chapters, T is the complimentary transfer function, S
 4557 is the sensitivity transfer function, and Y is the Youla transfer function. These four transfer
 4558 functions can be written as a function of the Youla transfer function,

$$\frac{u}{r} = Y, \quad \frac{e}{r} = 1 - G_p Y, \quad \frac{u}{d_u} = -G_p Y, \quad \frac{e}{d_u} = -G_p (1 - G_p Y)$$

4559 It is important to note that we accomplished both internal stability and SISO control de-
 4560 sign performance and robustness objectives through hand computation of the Youla transfer
 4561 function in the previous chapters. Now, we are ready to accomplish the same objectives
 4562 by shaping these four transfer functions through weighting filters using optimization of the
 4563 system norms as discussed next.

4564 6.4 Setup for Loop-Shaping Optimization

4565 There are several norm optimization methods developed by robust control researchers in
 4566 the past several decades, such as those in references (17.2, 17.2, and 17.2). One of these

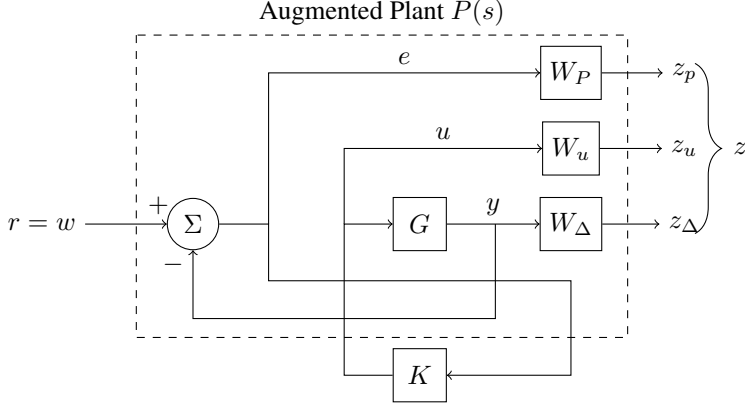
methods is called the “Loop Shaping Technique”, and in this subsection we discuss how LFT is utilized to setup this optimization approach.

In the loop shaping technique, we are interested in minimizing a norm of the closed-loop map T_{zw} , (as given in the previous section) for designing a controller to accomplish all the required feedback control objectives. Before we pose this minimization problem, we need to show how to augment the general feedback loop, as discussed in the previous section, to obtain the closed-loop map T_{zw} .

In the previous section, in Equation (6.38), we showed that there are only four closed-loop transfer functions which need to be shaped for accomplishing the feedback control loop objectives. In the loop shaping technique, three of the four transfer functions (S , T , and Y) are shaped through appropriate selection of shaping or tuning filters (W_P , W_Δ , and W_u) as illustrated in Figure 6.7.

A couple of points are worth mentioning here. First, as discussed in the second section of this chapter, the performance and robustness objectives are divided in such a way in the frequency domain that makes using these weighing filters feasible. The second point is about meeting the internal stability, the most important objective of the feedback control design. In the previous chapters, we discussed that for meeting internal stability when hand computing the Youla transfer function, it is not difficult to guarantee the stability of three out of four transfer functions presented in Equation (6.38). However, to guarantee the stability of the fourth transfer function, $G_p S$ is not an easy task. This was the main reason that we introduced the “interpolation conditions” for assuring internal stability. However, we stated in the paragraph above that this transfer function, $G_p S$, is not even being directly considered in the loop shaping approach. So, how does the loop shaping method guarantees internal stability? We will provide the answer to this question in the last section of this chapter after discussing the solution to the loop shaping approach.

The synthesis of an H_∞ or H_2 controller requires the system to be in a standard form with augmented plant $P(s)$ with the use of LFT, as shown in Figure 6.7. Here, the only input is the reference input r . We will show that by having this input and augmenting the plant with three weighting filters (W_P , W_Δ , and W_u), the norm optimization problem is correctly set up for obtaining the desired shapes of the three transfer functions (S , T , and Y).

Figure 6.7: Controlled system in form for H_∞ synthesis.

This augmented plant can be written in matrix form as,

$$\begin{bmatrix} z_p \\ z_u \\ z_\Delta \\ e \end{bmatrix} = \underbrace{\begin{bmatrix} W_P & -W_P G \\ 0 & W_u \\ 0 & W_\Delta G \\ 1 & -G \end{bmatrix}}_{P(s)} \begin{bmatrix} w \\ u \end{bmatrix} \quad (6.39)$$

It will be shown (through tedious algebra) in the next section that this plant has the following state-space realization,

$$\begin{bmatrix} \dot{x} \\ z \\ e \end{bmatrix} = \begin{bmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{bmatrix} \begin{bmatrix} x \\ w \\ u \end{bmatrix} \quad (6.40)$$

Additionally, because u is dependent on the error e , these representations can be further condensed, by knowing that

$$u = Ke$$

and

$$e = w - Gu$$

therefore

$$\begin{aligned} e &= w - GKe \\ &= (1 + GK)^{-1}w \end{aligned} \quad (6.41)$$

and

$$u = K(1 + GK)^{-1}w \quad (6.42)$$

So then the augmented plant becomes

$$\begin{bmatrix} z_p \\ z_u \\ z_\Delta \\ e \end{bmatrix} = \begin{bmatrix} W_P & -W_P G \\ 0 & W_u \\ 0 & W_\Delta G \end{bmatrix} \begin{bmatrix} w \\ K(1 + GK)^{-1}w \end{bmatrix} \quad (6.43)$$

therefore,

$$z_p = (W_P - W_P GK(1 + GK)^{-1})w = W_P \left(1 - \frac{GK}{1 + GK}\right)w = W_P S w \quad (6.44)$$

$$z_u = (W_u K(1 + GK)^{-1})w = W_u \left(\frac{K}{1 + GK}\right)w = W_u Y w \quad (6.45)$$

$$z_\Delta = (W_\Delta GK(1 + GK)^{-1})w = W_\Delta \left(\frac{GK}{1 + GK}\right)w = W_\Delta T w \quad (6.46)$$

finally,

$$\begin{bmatrix} z_p \\ z_u \\ z_\Delta \end{bmatrix} = \begin{bmatrix} W_P S \\ W_u Y \\ W_\Delta T \end{bmatrix} w. \quad (6.47)$$

which then the closed-loop map (T_{zw}) is the stacked form, $\begin{bmatrix} W_P S \\ W_u Y \\ W_\Delta T \end{bmatrix}$, as is presented in Equation (6.47). This form is the frequency weighting of the three closed-loop transfer functions of interest (S , Y , and T).

With this setup, we are able to simultaneously consider bounds on S (for performance), T (for robustness), and Y (to penalize large actuator usage), by posing the following optimization objective (the H_∞ control problem),

$$\underset{K}{\text{minimize}} \quad \|T_{zw}\|_\infty \quad (6.48)$$

In other words, find the controller K that minimizes the infinity-norm of T_{zw} . Alternatively, we could choose to minimize the 2-norm. Then, the control problem would instead be

$$\underset{K}{\text{minimize}} \quad \|T_{zw}\|_2 \quad (6.49)$$

We will discuss the difficulties, benefits, and the methods for obtaining solutions to these norm optimization problems in the second volume of this book.

6.4.1 Setup for Youla Parameter Loop-Shaping

Our approach in designing controllers proposed in earlier chapters included computing the Youla parameter and deriving a feedback controller utilizing the Youla parameter. However, the proposed loop shaping technique directly searches for an optimal controller that

minimizes the 2- or ∞ -norms given in Equations (6.48 and 6.49), without utilizing the Youla parameter.

In this subsection, we propose an optimization approach where instead of searching directly for an optimal controller, we search for an optimal Youla parameter by minimizing the 2 or ∞ -norms as given below,

$$\underset{Y}{\text{minimize}} \quad \|T_{zw}\|_{2,\infty} \quad (6.50)$$

Youla parameterization is discussed in many articles and textbooks (see 17.2). The solutions proposed in these references include techniques such as co-prime factorization for computing the Youla parameter. However, here we propose a different approach. We are simply searching for an optimization method which would be equivalent to the hand computation technique for Youla parameter presented in the earlier chapters. Then the optimal Youla parameter could be used in computing a feedback controller as proposed earlier. In this context, Internal Model Control (IMC) is an appropriate platform, as illustrated in Figure 6.8, for this task.

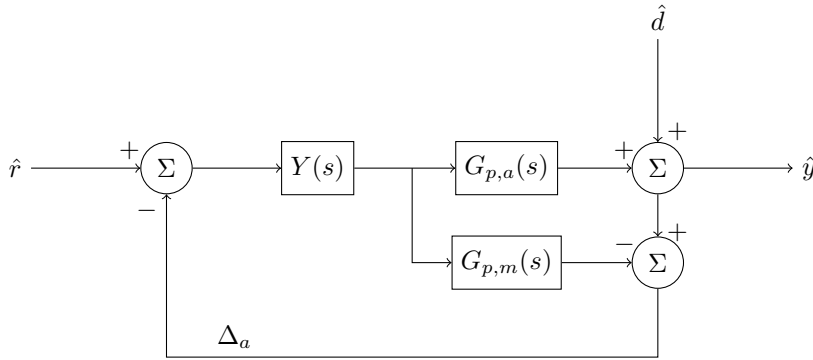


Figure 6.8: Simplified Block Diagram for Internal Model Control.

IMC presents an elegant approach to the principal reason for utilizing a feedback control in the first place. In Figure 6.8, G_{P_a} represents the actual plant, while, the transfer function, G_{p_m} is the nominal plant model. The output disturbance is represented by d , the measurement by y , the reference input by r and the actual plant and the nominal plant model input by u . It can be shown that

$$y = \frac{G_{p_a} Y}{1 + Y(G_{p_a} - G_{p_m})} r + \frac{1 - G_{p_m} Y}{1 + Y(G_{p_a} - G_{p_m})} d \quad (6.51)$$

where the complementary sensitivity, sensitivity, and Youla parameter transfer functions are

$$T_{IMC} = \frac{G_{p_a} Y}{1 + Y(G_{p_a} - G_{p_m})} \quad (6.52)$$

$$S_{IMC} = \frac{1 - G_{p_m} Y}{1 + Y(G_{p_a} - G_{p_m})} \quad (6.53)$$

$$Y_{IMC} = \frac{Y}{1 + Y(G_{p_a} - G_{p_m})} \quad (6.54)$$

respectively.

The difference between the actual plant and nominal plant model outputs includes some important uncertainties. These uncertainties are both at the plant level due to lack of having a perfect nominal plant model and environmental uncertainties which are modeled by the output disturbance, d . Feedback signal Δ_a represents these uncertainties. Here lies the fundamental benefit of the feedback: the feedback signal Δ_a is exactly equal to zero if there are no uncertainties (no output disturbances and a perfect nominal plant model). This states that the only reason to use a feedback loop in control design is simply the existence of uncertainties. Without uncertainties, the IMC is transformed to a feedforward control loop design with Youla parameter serving as a feedforward controller.

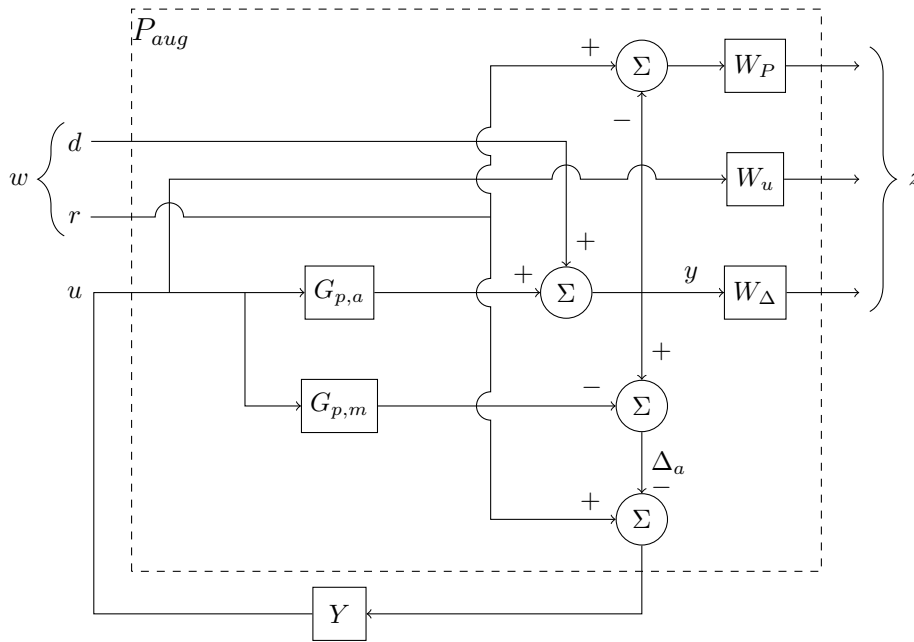


Figure 6.9: Direct Loop Shaping with Youla Parameter.

In this section, we propose two different approaches to optimize the 2- and ∞ -norms of the closed loop as given in Equation (6.50) through directly computing the optimal Youla parameter,

- Loop shaping using IMC structure as illustrated in Figure 6.9.
- Model matching using IMC structure as shown in Figure 6.10.

In the model matching problem given in Figure 6.10 below, defining filters W_P , W_u and W_Δ is optional and not necessary. The computation of the Youla parameter, Y , is performed by minimizing a norm of the closed loop transfer function T_{zw} , which forces our actual plant $G_{p,a}$ to follow our virtual model $G_{p,t}$ in the presence or absence of the uncertainties Δ_a .

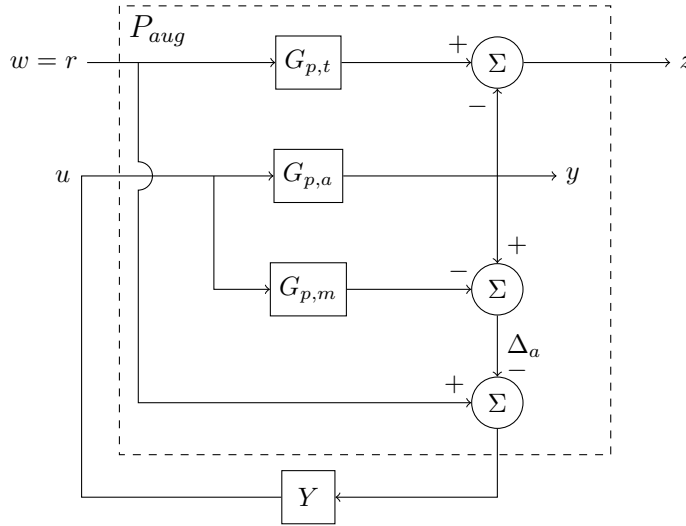


Figure 6.10: Model Matching with Youla Parameter.

In the next section, we briefly discuss the solutions to SISO norm optimization. A more complete solution to the norm optimization will be discussed in the second volume of this book when discussing MIMO control design.

6.5 H_∞ norm optimization problem

The solutions of H_∞ and H_2 norm optimization problems are presented in many textbooks and scholarly articles (see 17.2). We will discuss the solutions to both H_∞ and H_2 solutions in detail in the next volume of this book. However, in this section, our intent is to

provide some insight into the frequency domain solution of H_∞ problem before presenting its equivalent and a more detailed state-space solution in the next volume.

We start this section by first discussing a simplified version of H_∞ problem in the frequency domain:

$$\text{minimize } \left[\max_{\omega} |W(j\omega)S(j\omega)| \right] \quad (6.55)$$

where $S(s)$ is the sensitivity and $W(s)$ is a weighting function. We choose the weighting function $W(s)$ to be proper, stable, and with zeros strictly in LHP. The weighting function is chosen so that that weighting is large at low frequency and small at high frequency so that the solution to Equation (6.55) results in an all pass filter as will be discussed below.

EXAMPLE 6.1

Let

$$W(s) = \frac{\alpha\tau s + 1}{\tau s + 1}$$

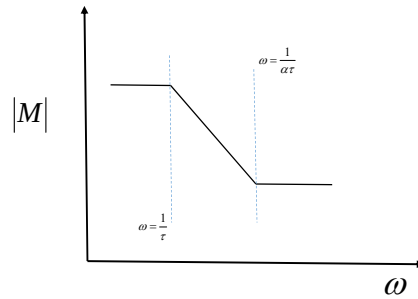


Figure 6.11: Sensitivity Transfer function weighting function.

In this simple problem, we need to consider the following three constraints,

- Well-posedness
- Internal Stability
- Feedback controller must be proper or strictly proper so it is realizable

In the first instance, it is not clear how to insure #3. In fact, we ignore #3 and use an ad-hoc approximation to fix things after the fact.

6.5.1 Solution to a simple optimization problem

Minimizing Equation (6.55) is similar to equating,

$$W(s)S(s) = KS_{ap}(s)$$

where S_{ap} (an all pass filter,) and K are selected to satisfy the interpolation conditions on $S(s)$. Here we must assume that the plant transfer function, $G_p(s)$ has no poles on $j\omega$ -axis, however, we will relax this constraint soon. Therefore,

$$S(s) = \frac{KS_{ap}(s)}{W(s)} \quad (6.56)$$

Interpolation conditions state that,

$$\begin{aligned} S(z_i) &= 1, \quad i = 1, 2, \dots, n \\ S(p_k) &= 0, \quad k = 1, 2, \dots, m \end{aligned}$$

where z_i and p_k are zeros and poles in the RHP, respectively, and where n is the number of zeros in the RHP and m is the number of poles in the RHP. Assuming no repeated poles and zeros in RHP, we therefore have

$$S_{ap} = \frac{(-s + p_1)(-s + p_2) \dots (-s + p_m)}{(s + p_1)(s + p_2) \dots (s + p_m)} B(s) \quad (6.57)$$

If $n = 1$ (where n is the number of zero's in RHP of G_p), then we select $B(s)$ to be also equal to 1, and we select K so that $S(z_i) = 1$. If $n > 1$, then we select,

$$B(s) = \frac{(-s + a_1)(-s + a_2) \dots (-s + a_{n-1})}{(s + a_1)(s + a_2) \dots (s + a_{n-1})} \quad (6.58)$$

Then we select a_k so that

$$S(z_i) = 1$$

EXAMPLE 6.2

Let

$$G_p = \frac{(-s + 1)}{(-s + 2)(s + 3)}$$

and we select $W(s) = \frac{\alpha\tau s + 1}{\tau s + 1}$ then,

$$S(s) = K \frac{S_{ap}(s)}{W(s)}$$

Here, $B(s) = 1$ as $n = 1$. Therefore,

$$S_{ap} = \frac{(-s + 2)}{(s + 2)} B(s) = \frac{(-s + 2)}{(s + 2)}$$

We can write the sensitivity transfer function, $S(s)$ as,

$$S(s) = K \frac{(-s+2)}{(s+2)} \frac{(\tau s+1)}{(\alpha \tau s+1)}$$

and due to interpolation condition, $S(1) = 1$ then $K = 3 \frac{\alpha \tau + 1}{\tau + 1}$. But, we know $T(s) = 1 - S(s)$, so

$$T(s) = 1 - K \frac{(-s+2)}{(s+2)} \frac{(\tau s+1)}{(\alpha \tau s+1)} = \frac{(s+2)(\alpha \tau s+1) - K(-s+2)(\tau s+1)}{(s+2)(\alpha \tau s+1)}$$

The relative degree of T is 0. However, it can be shown that for controller G_c to be realizable (satisfying constraint #3), the relative degree of T must be greater than or equal to 1. Modifying T by adding a fast pole and scaling by K_T to satisfy interpolation condition gives us

$$T(s) = K_T \frac{(s+2)(\alpha \tau s+1) - K(-s+2)(\tau s+1)}{(s+2)(\alpha \tau s+1)(\tau_0 s+1)}$$

We select τ_0 to be small such that,

$$\frac{1}{\tau_0} \gg \frac{1}{\alpha \tau}$$

Next, compute K_T to satisfy interpolation condition, $T(2) = 1$,

$$K_T \frac{1}{2\tau_0 + 1} = 1$$

Therefore, $K_T = 2\tau_0 + 1$.

With this T , it can be shown that we have satisfied all three constraints as discussed above.

What if $G_p(s)$ has poles on $j\omega$ -axis? Then we can write,

$$G_p(s) = \frac{\widetilde{G}_p(s)}{P(s)}$$

where $P(s)$ is the polynomial of poles on $j\omega$ -axis. Thus, $\widetilde{G}_p(s)$ has no poles on $j\omega$ -axis.

We can write,

$$S(s) = P(s)\widetilde{S}(s)$$

Then we select $W(s) = \frac{\widetilde{W}(s)}{P(s)}$. In this case,

$$W(s)S(s) = \frac{\widetilde{W}(s)}{P(s)} P(s)\widetilde{S}(s)$$

Therefore, the solution for $\widetilde{W}(s)\widetilde{S}(s)$ is the same as the solution for $W(s)S(s)$.

■ **EXAMPLE 6.3**

Let

$$G_p = \frac{(-s+1)}{s(s+1)}$$

We select

$$W(s) = \frac{\alpha\tau s + 1}{s(\tau s + 1)}$$

Then,

$$\widetilde{W}(s) = \frac{\alpha\tau s + 1}{\tau s + 1}$$

Knowing that $G_p(s) = \widetilde{G}_p(s)/P(s)$, we have $P(s) = s$. We can then write,

$$\widetilde{W}(s)\widetilde{S}(s) = K S_{ap}(s) \quad (6.59)$$

The interpolation conditions are

$$S(1) = 1 \rightarrow P(1)\widetilde{S}(1) = 1 \cdot \widetilde{S}(1)$$

and

$$S(0) = 0 \rightarrow P(0)\widetilde{S}(0) = 0 \cdot \widetilde{S}(0)$$

Taking Equation (6.59) and knowing $S_{ap}(s) = 1$, we have

$$\widetilde{W}(s)\widetilde{S}(s) = K$$

Therefore,

$$\widetilde{S}(s) = \frac{K(\tau s + 1)}{(\alpha\tau s + 1)}$$

But interpolation condition enforces that $\widetilde{S}(1) = 1$. So, we can compute K to be

$$K = \frac{\alpha\tau + 1}{\tau + 1}$$

We can now compute $S(s)$ and $T(s)$.

$$S(s) = \frac{Ks(\tau s + 1)}{(\alpha\tau s + 1)}$$

It is evident that this transfer function is improper. However, we continue by computing $T(s)$,

$$T(s) = 1 - S(s) = \frac{(\alpha\tau s + 1) - Ks(\tau s + 1)}{(\alpha\tau s + 1)}$$

Given an improper $S(s)$, it is expected that $T(s)$ would also be improper as the relative degree of $T(s) = -1$.

Similar to our previous approach, we modify $T(s)$ by adding two fast poles,

$$T(s) = \frac{K_T}{(\tau_0 s + 1)^2} \left\{ \frac{(\alpha \tau s + 1) - K s(\tau s + 1)}{(\alpha \tau s + 1)} \right\}$$

The interpolation conditions for $T(s)$ are,

$$T(0) = 1 = K_T \{1\} \rightarrow K_T = 1$$

and

$$T(1) = 0 = \frac{K_T}{(\tau_0 + 1)^2} \{0\} \rightarrow T(1) = 0$$

Therefore,

$$T(s) = \frac{(\alpha \tau s + 1) - K s(\tau s + 1)}{(\alpha \tau s + 1)(\tau_0 s + 1)^2} \quad \frac{1}{\tau_0} \gg \frac{1}{\alpha \tau}$$

Then we can recalculate $S(s) = 1 - T(s)$. Knowing that $Y(s) = T(s)/G_p(s)$ and

$G_c(s) = Y(s)/S(s)$, the final controller can be computed.

$$G_c(s) = \frac{Y(s)}{S(s)}$$

6.6 H_∞ Design

Imagine that we devise upper bounds for $|T(j\omega)|$ and $|S(j\omega)|$ as illustrated in the Figure 6.12.

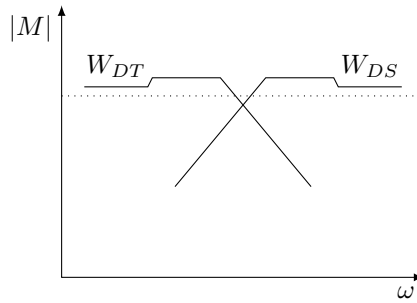


Figure 6.12: Sensitivity and Complementary Sensitivity Transfer functions upper bounds.

The idea is to obtain,

$$|T(j\omega)| \leq |W_{DT}(j\omega)| \quad (6.60)$$

$$|S(j\omega)| \leq |W_{DS}(j\omega)| \quad (6.61)$$

We prefer to write the above equations as,

$$|W_{PT}T(j\omega)| \leq 1, \quad W_{PT} = \frac{1}{W_{DT}} \quad (6.62)$$

$$|W_{PS}S(j\omega)| \leq 1, \quad W_{PS} = \frac{1}{W_{DS}} \quad (6.63)$$

4742 Satisfaction of both these constraints is equivalent to satisfaction of

$$\max_{\omega} \{|W_{PS}S(j\omega)|, |W_{PT}T(j\omega)|\} \leq 1 \quad (6.64)$$

4743 However, the theory cannot easily deal with this type of constraint. Instead, we find
4744 a new constraint which approximates the above, but which the theory (i.e. loop shaping
4745 technique) can handle. This approximate constraint is

$$\left| \frac{W_{PS}S(j\omega)}{W_{PT}T(j\omega)} \right| \leq 1 \quad (6.65)$$

4746 It can be shown that

$$\frac{1}{\sqrt{2}} \left| \frac{W_{PS}S(j\omega)}{W_{PT}T(j\omega)} \right| (j\omega) \leq \max_{\omega} \{|W_{PS}S(j\omega)|, |W_{PT}T(j\omega)|\} \leq \left| \frac{W_{PS}S(j\omega)}{W_{PT}T(j\omega)} \right| (j\omega) \quad (6.66)$$

4747 Meaning, the approximation is reasonable.

4748 To address the actuator constraint, one utilizes a similar weighting function for the Youla
4749 transfer function, for instance $W_{PY} = \frac{1}{W_{DY}}$. However, we know that $Y = \frac{T}{G_p}$, so

$$|W_{PY}Y| = \left| \frac{W_{PT}T}{G_p} \right| \quad (6.67)$$

4750 It is usual to incorporate the weighting transfer functions in the generalized plant (as
4751 discussed in the loop shaping section of this chapter), so that the controlled variable z (the
4752 output of the generalized plant) now satisfies:

$$z = \begin{bmatrix} W_{PS}S \\ W_{PT}T \\ W_{DY}Y \end{bmatrix} w \quad (6.68)$$

4753 So that the nominal performance constraint becomes,

$$\|T_{zw}\|_{\infty} < 1 \quad (6.69)$$

4754 It should be evident that there is a difference between this objective (nominal perfor-
4755 mance constraint) and the one given in the loop shaping in previous section,

$$\underset{K}{\text{minimize}} \quad \|T_{zw}\|_{\infty} \quad (6.70)$$

4756 We will discuss this in the second volume of this book when presenting the MIMO H_∞
4757 solution. Furthermore, we will discuss how the nominal performance constraint has to
4758 change for addressing robust performance and robust stability and performance objectives.

6.7 Problems

6.1 Consider the system shown in Figure 6.13 below.

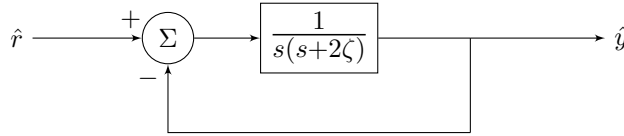


Figure 6.13

Determine the optimal value of the damping ratio ζ so that when the system is subjected to a unit step input $r(t) = 1(t)$, the following performance index is minimized:

$$J = \int_0^{\infty} \underline{x}^T(t) Q \underline{x}(t) dt$$

where

$$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x \\ \dot{x} \end{bmatrix}, \quad Q = \begin{bmatrix} 1 & 0 \\ 0 & M \end{bmatrix} \quad (M > 0)$$

and

$$x_1(0^-) = 0, \quad x_2(0^-) = 0$$

Hint: You need to compute $\underline{P}$ from the Lyapunov function and minimize the performance index with respect to the damping ratio. You also need to note that

$$J = \int_0^{\infty} \underline{x}^T Q \underline{x} dt = -\underline{x}^T P \underline{x} \Big|_0^{\infty} = -\underline{x}^T(\infty)^T P \underline{x}(\infty) + \underline{x}^T(0)^T P \underline{x}(0)$$

Since all eigenvalues of A have negative real parts, we have $\underline{x}(\infty) \rightarrow 0$, therefore

$$J = \underline{x}^T(0) P \underline{x}(0)$$

This problem should be carried out by hand.

6.2 Consider the same quarter car model shown in Figure 6.14. Let $m = 250\text{kg}$, $k = 16000\text{N/m}$, and $b = 1175\text{Ns/m}$.

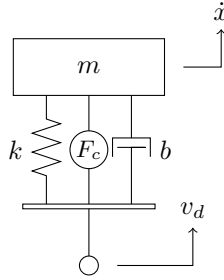


Figure 6.14

4771 Your task is to design an optimal controller with control law $F_c = -k\underline{x}$ using the machin-
 4772 ery of the LQR by minimizing

$$J = \int_0^\infty \underline{x}^T Q \underline{x} + F_c^T R F_c dt$$

4773 Q must be selected so that $\underline{x}^T Q \underline{x}$ represents system energy. You are free to select R .

4774 If you use the bond graph to derive your state model, your states would be $\underline{x} = [p \ q]^T$,
 4775 where p is the momentum of the mass and q is the spring deflection. If you are unfamiliar
 4776 with the bond graph method, you can obtain the equations of motion from a free body
 4777 diagram. Use state variables $q = \int_0^t v_d dt - x$ (so that $\dot{q} = v_d - \dot{x}$) and $p = m\dot{x}$ (so that
 4778 $\dot{p} = m\ddot{x}$).

4779 Use the following parameters:

$$m = 250kg, \ k = 16000N/m, \ b = 1175Ns/m$$

4780 The intent of an active control is to minimize the mass velocity response $\left| \frac{\dot{x}}{v_d} \right|$. You need
 4781 to present your results the magnitude response of $\left| \frac{\dot{x}}{v_d} \right|$ as a function of frequency for the
 4782 controlled and uncontrolled version. In addition, you need to plot $\left| \frac{F_c}{v_d} \right|$. Use 0.01 to 100
 4783 rad/sec for you x -axis (frequency axis).

4784 Hint: When you obtain your state-space model, your B matrix will have two columns – one
 4785 for F_c and one for v_d (denoted B_F and B_v). You can ignore v_d and B_v when designing the
 4786 LQR controller. To determine the frequency responses, let your closed-loop input matrix
 4787 B_{cl} equal B_v and set C_{cl} so that $\dot{x}$ and F_c can be obtained as outputs (D_{cl} will be zero).

4788 **6.3** In this problem, you will explore the application of the LQR optimal controller for the
 4789 linearized automobile suspension system shown in the figure below.

The state vector is $x = [p_s \ q_s \ p_u \ q_t]^T$, where p_s is the momentum of the sprung mass m_s , q_s is the suspension deflection (positive in compression), p_u is the momentum of the unsprung mass m_u , and q_t is the tire deflection (positive in compression).

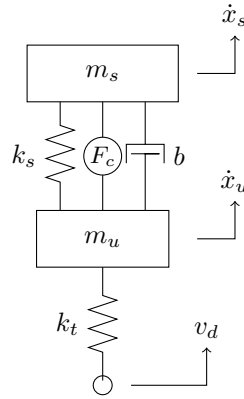


Figure 6.15

The following parameter values are to be used for this problem:

$$m_s = 240\text{kg}, \quad m_u = 36\text{kg}, \quad b = 1,175\text{Nsec/m}, \quad k_s = 16,000\text{N/m}, \quad k_t = 160,000\text{N/m}$$

If you are unfamiliar with the bond-graph method, the state equations resulting from the bond graph are provided below.

$$\begin{bmatrix} \dot{p}_s \\ \dot{q}_s \\ \dot{p}_u \\ \dot{q}_t \end{bmatrix} = \begin{bmatrix} -\frac{b}{m_s} & k_s & \frac{b}{m_u} & 0 \\ -\frac{1}{m_s} & 0 & \frac{1}{m_u} & 0 \\ \frac{b}{m_s} & -k_s & -\frac{b}{m_u} & k_t \\ 0 & 0 & -\frac{1}{m_u} & 0 \end{bmatrix} \begin{bmatrix} p_s \\ q_s \\ p_u \\ q_t \end{bmatrix} + \begin{bmatrix} -1 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} F_c \\ v_d \end{bmatrix}$$

The intent of the active suspension is to keep the motion of the sprung mass small (this is called comfort mode) while keeping the motions of the sprung mass not negatively impacted (in other words, preventing the loss of contact between the tire and ground, the security mode). These two optimization objectives need to be satisfied in the presence of the space constraint (rattle space) between the sprung mass and unsprung mass. The suspension cannot be allowed to bottom-out or over-extend, so the suspension deflection should be kept small.

You must select a weighting matrix Q and scalar R to obtain the best results. A good way to select Q is perhaps by including all the states you wish to minimize and your constraint state (rattle space). The scalar R will require some tuning as the active forces must be

reasonable. You need to present your results by showing magnitude response plots for the following transfer functions:

1. $\frac{\dot{x}_s}{v_d}(s)$

2. $\frac{v_{rel}}{v_d}(s)$ (where $v_{rel} = \dot{x}_u - \dot{x}_s$)

3. $\frac{\dot{x}_u}{v_d}(s)$

4. $\frac{F_c}{v_d}(s)$

5. $\frac{q_s}{v_d}(s)$

Show the magnitude responses for both the controlled and uncontrolled system (put the responses of the controlled and uncontrolled systems on the same figure). It is recommended you use the `bodemag` command to generate the magnitude plots:

```
bodemag(your_tf_uncontrolled,'k-.',your_tf_controlled,'r-',logspace(0,2.3,140))
```

Comment on your findings.

6.4 Repeat Problem 6.3, but this time utilize the mixed sensitivity H_∞ method, where you augment your plant with W_1 , W_2 , and W_3 filters to derive a controller that meets the objectives and constraints stated above. The command in Matlab is `hinfsyn`.

The following filters are suggested:

- $W_1 = W_{1a} \cdot W_{1b}$, with $W_{1a} = G_{p,d}$ where $G_{p,d}$ is the open loop transfer function from v_d (disturbance input) to sprung mass velocity $\dot{x}_s$. You need to find a reasonable W_{1b} . Start simple, e.g. a first order transfer function.
- For W_2 , try a constant such as $\begin{bmatrix} 0 & 0.0005 \\ 0 & 1 \end{bmatrix}$.
- You also need to find a reasonable W_3 . Like with W_1 , start simple.

Keep in mind that this is not a tracking problem, so T (the closed-loop transfer function) does not need to be equal to 1 (0dB) at very low frequencies. T needs to be closed to one at mid-frequencies (0.5Hz to 15Hz). Make S smaller than one in this frequency range for better disturbance rejection capabilities. Note that the sprung mass natural frequency is around 1Hz, while the wheel hop (unsprung mass) natural frequency is around 10Hz.

You need to demonstrate your results by submitting the same plots as in Problem 6.3.

Hint: This is an output (not full-state) feedback H_∞ problem in which you have access to only the sprung mass velocity $\dot{x}_s$.

4836 **6.5** Repeat Problem 6.4, using the same filters, but minimize the H_2 norm instead. Present
 4837 the same result plots as part (a). The command in Matlab is `h2lqg`.

4838 **6.6** Consider the second-order system described by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

4839 Clearly, the equilibrium state is the origin. Assuming a Lyapunov function $V(x) = x^T P x$,
 4840 determine the stability of the equilibrium state.

4841 Hint: Assume $Q = I$ (identity matrix). Check for positive definiteness of P . You also
 4842 need to compute $V(x)$ and its derivative.

4843 **6.7** Repeat Problem 6.6, but this time use the machinery of H_∞ optimization to compute
 4844 the compensator, G_c . You need need to provide $W_P(or W_1)$, $W_\delta(or W_3)$, in addition you
 4845 may need to provide a weighting for the actuator, $W_u(or W_2)$. It is suggested to use a
 4846 simple constant for your actuator weighting. Utilizing the H_∞ controller, compute T, Y,
 4847 and S. Provide Bode plots of T,S, and Y, including your design phase and gain margin, and
 4848 the step response of the closed loop transfer function T. Compare your results to the results
 4849 of the first problem and comment.

4850 Hint: The structure W_3 filter, which is used for compensating delays, is discussed and
 4851 provided to you in the class. You should use this filter for W_3 and then use a first order
 4852 filter to design W_1 .

4853 **6.8** Consider the system show in Figure 6.16.

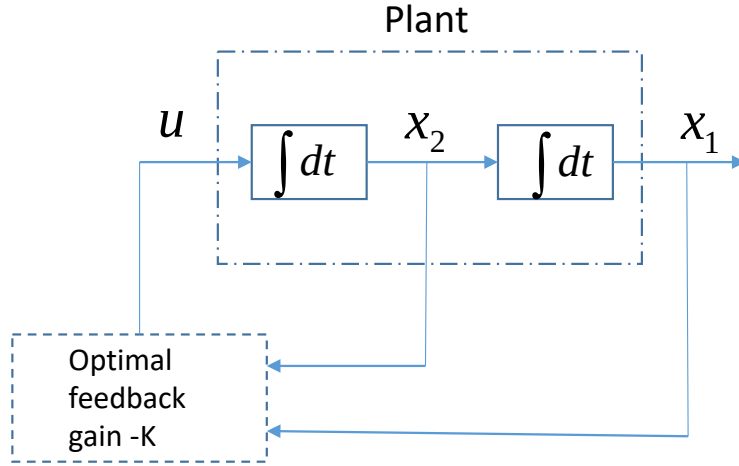


Figure 6.16

4854 Assuming the control signal to be

$$u(t) = -Kx(t)$$

4855 design the optimal feedback gain matrix K such that the following performance index is
4856 minimized:

$$J = \int_0^\infty (x^T Q x + r u^2) dt$$

4857 where

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & \mu \end{bmatrix} \quad (\mu \geq 0)$$

4858 **6.9** A feedback control system can be rearranged as an LFT:

4859 w : All external inputs

4860 u : Control inputs

4861 z : Regulated outputs

4862 y : Measured outputs

4863

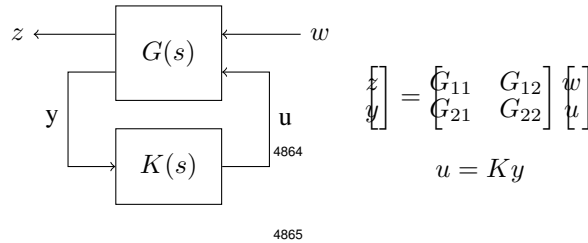


Figure 6.17: LFT.

$$T_{zw} = F_l(G, K) = G_{11} + G_{12}K(I - G_{22}K)^{-1}G_{21}$$

▪ LFT is a useful way to standardize block diagrams for robust control analysis and design

▪ $F_l(G, K)$ is the transfer function between the regulated signals and external inputs. In H_∞ control problems the objective is to minimize $\|F_l(G, K)\|_\infty$

Knowing what you have learned in the class, show the robust performance problem, as shown in Figure ??, has an equivalent infinity norm to the LFT shown above. That is, show that

$$\|F_l(G, K)\|_\infty = \left\| \frac{W_1 S}{W_2 T} \right\|_\infty.$$

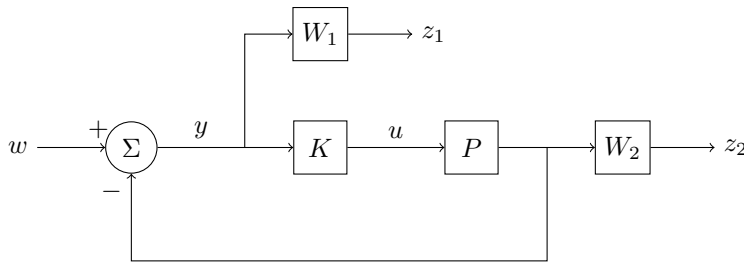


Figure 6.18: Robust Performance.

6.10 Consider the following transfer function.

$$G(s) = \frac{1 - s}{(2 - s)(1 + s)}$$

This system is both unstable and nonminimum phase, so we will have an **analytical constraint** on the maximum value of $\|S(j\omega)\|_\infty$.

$$\max_{\omega} |S(j\omega)| \geq \left| \frac{z + p}{z - p} \right| = \left| \frac{2 + 1}{2 - 1} \right| = 3 \approx 9.5 \text{ dB}$$

$$\max_{\omega} |T(j\omega)| \geq \left| \frac{p + z}{p - z} \right| = \left| \frac{2 + 1}{2 - 1} \right| = 3 \approx 9.5 \text{ dB}$$

Use Youla parameterization to design a controller that will stabilize the system and provide command tracking. Provide the appropriate plots for the frequency response and the closed-loop step response. Do the analytical constraints on $\|S\|_\infty$ and $\|T\|_\infty$ hold true?

Hints:

1. In addition to canceling the $(2 - s)$ pole, you will need to add a zero and three poles.
2. Remember that your only goals here are (1) stability and (2) command tracking.
3. Although it is undesirable to have a controller with unstable poles, it may be unavoidable in this case. It is more important to ensure that internal stability is met.

6.11 Repeat Problem 3, but this time use the H_∞ method. Keep in mind the analytical constraints; you will need to make $W_1 \ll 0$ dB at high frequencies to keep the peak value of $|S|$ small. You could try shaping W_1 by approximating the inverse of the sensitivity transfer function from your Youla method controller. A single pole and zero should suffice.

To make things easier for the H_∞ design, use the `hinfsyn` command in place of `hinf`.

You can also choose the W_2 and W_3 filters to be empty vectors, such as $(W_i = [\])$.

Use the uploaded m-file. Provide the appropriate plots for the frequency response and the closed-loop step response. Do the analytical constraints on $\|S\|_\infty$ and $\|T\|_\infty$ hold true?

Is there a difference in the form of your two controllers (i.e. number of poles and zeros)?

CHAPTER 7

ESTIMATION DESIGN FOR SISO USING PARAMETRIZATION APPROACH

As as been discussed so far, Youla parameterization is a well established technique in deriving SISO (Single-Input Single-Output) and, to a lesser extent, MIMO (Multiple-Input Multiple-Output) controllers [?]. However, the utility of this methodology in estimation design, specifically in the framework of Controller Output Observer [?] (COO), is not established. The fundamental question to be answered is as follows: is it possible to design a deterministic estimation technique using Youla parameterization with the same robust performance, or better, than well established stochastic estimation techniques such as Kalman filtering? To prove this point, at this stage, a comparative analysis between Youla parametrization in estimation and Kalman filtering is performed through simulations only.

In this chapter, we provide an overview of Youla parametrization for estimation design. We develop a deterministic SISO and MIMO Youla estimation technique in the framework of COO and we investigate the utility of this method for two applications in the automotive domain.

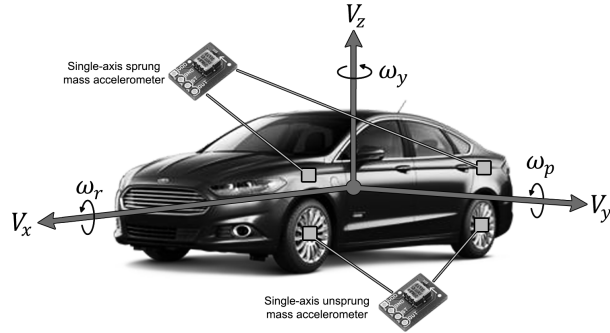


Figure 7.1: Automotive Application for Controller Output Observer

7.1 Introduction

The Controller Output Observer framework is used for estimating system inputs and states that characterizes a physical system, as seen in Figure 7.1, through the use of other measurable outputs [?]. The estimation of unmeasurable quantities was elegantly introduced through the use of classical observers by Luenberger [?], one of the classical deterministic estimation approaches, as illustrated in Figure 6.9(a). In this Figure, A_e , B_e , C_e matrices include the dynamics of the observer/estimator for a strictly proper system with feedthrough matrix D_e assumed to be zero without loss of generality.

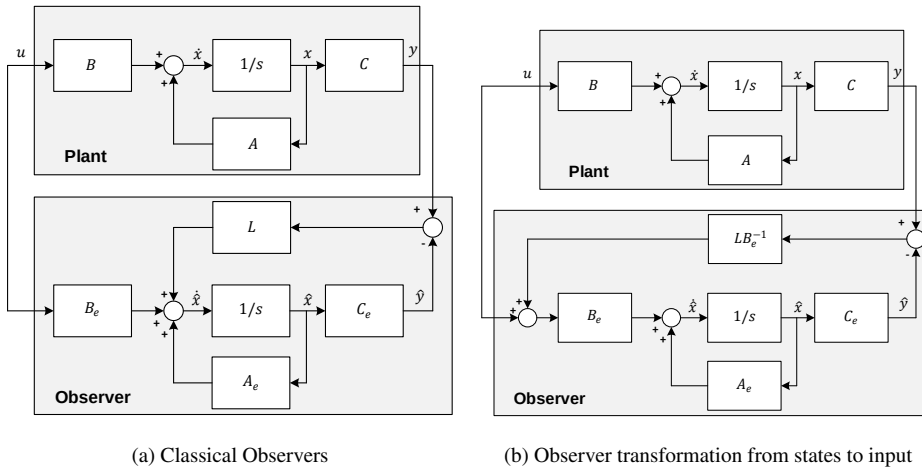


Figure 7.2: Classical observer design

This classical observer assumes that all system inputs are known, and hence, there are no distinctions between disturbance and control inputs. The observer gains, L , are computed

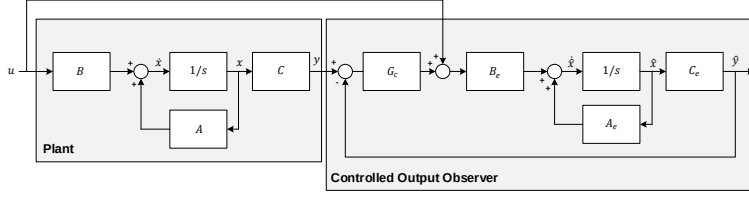


Figure 7.3: General Controller Output Observer

by pole placement techniques. Information about the sensor dynamic response and its noise content are not considered in the design. These limitations drastically reduce the domain of applicability of these types of observers.

We will illustrate that by utilizing a COO with Youla parametrization, these limitations are directly addressed and that the pragmatic implementation of this method is feasible. In the Youla parameterization method, the closed-loop response is shaped directly and the Youla parameter is used to work backwards to form a controller or in this case, an estimator (The Youla parameter is the control effort transfer function, relating a system input to the controller output, given that there is output feedback). Because Youla parameterization allows direct shaping of the closed-loop response, the COO can be designed to be robust to plant uncertainties.

In Figure 7.2(b), a simple transformation from the states to the input illustrates the invertability requirement of the input matrix, B_e . We will remove this required invertability by replacing it with a tunable transfer function, $G_c(s)$.

$$LB_e^{-1} \rightarrow G_c(s) \quad (7.1)$$

Now, Figure 7.2(b) is precisely equivalent to Figure 7.3, where LB_e^{-1} is replaced with $G_c(s)$.

With this type of observer structure, the inputs, as well as states, can be estimated without the need for matrix inversion. In fact, the estimation problem has been transformed to a control design problem and the dynamic controller, G_c , will be designed utilizing the Youla parametrization approach. As stated previously, all plant inputs have to be readily available to the classical observer for proper state estimation. However, there is no such requirement with the new observer structure in Figure 7.3. In fact, the input (from the plant to the observer) acts merely as a disturbance to the COO. It is a requirement that this distur-

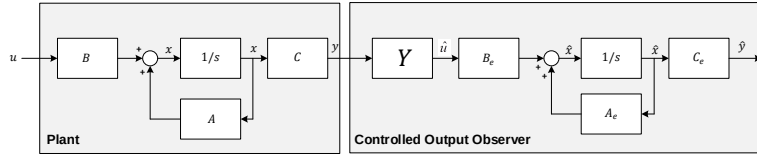


Figure 7.4: Youla Controller Output Observer

bance be properly rejected by the controller such that the observer output follows its input (plant measured output, y) for correct input (controller output, $\hat{u}$) and state estimation.

A final transformation of the COO, from Figure 7.3 to Figure 7.4, illustrates the equivalency of a feedback to a feed-forward controller using the Youla parametrization technique. We name this observer structure YCOO, and the design of the feed forward controller both for SISO and MIMO will be discussed using the YCOO structure.

In comparison, a Kalman filter is formulated using the state-space approach and its main function is to provide an estimate of the system states in presence of both model and measurement uncertainties assuming zero-mean Gaussian white noise disturbances [?]. The YCOO makes no assumption on the statistics of the disturbances except that they must be bounded in terms of their energy or power, similar to the H_∞ control and filter design. Here lies the fundamental difference between YCOO and a Kalman filter as the stochastic processes are now transformed to deterministic ones. The procedure to design an estimator for the YCOO is similar and dual to its traditional feedback control loop counterpart.

In this chapter, we discuss the Controller Output Observer (COO) and its relation with the Luenberger Observer. We provide a simple SISO example of the COO using the Youla approach and we compare the results with a discrete Kalman filter. Step-by-step discussion of the proposed estimation method for the MIMO case is also provided. The final section presents a MIMO Youla COO example for an automotive suspension with both input and state estimation by utilizing the measured outputs of a validated six chassis degree-of-freedom nonlinear vehicle model. The estimation results are then compared to the nonlinear model inputs and states. Additionally, the estimation is shown to work in conjunction with a fully active Skyhook controller and as a road roughness coefficient estimator.

The main contribution of this paper is to introduce a new deterministic Youla parametrization estimation concept utilizing the framework of the COO.

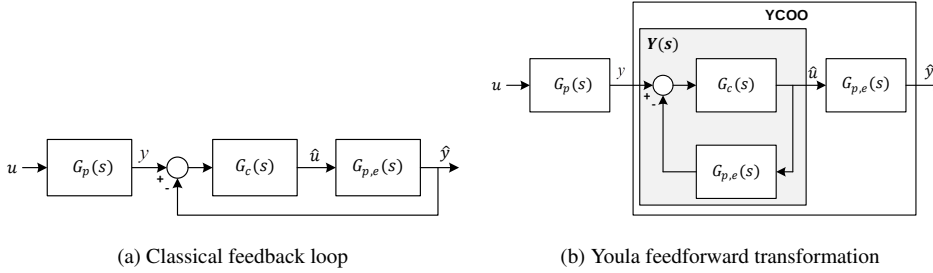


Figure 7.5: Comparison of classical feedback loop and Youla feedforward transformation

7.2 Youla Controller Output Observer Concept

It has been said that the only reason for using feedback in control design is to reduce the effects of uncertainty [?]. If there were no uncertainties, there would be no need for feedback control. In this sense, Youla parameterization is a powerful tool in compensating for the uncertainty and designing a balanced controller by shaping the closed loop transfer functions.

When it comes to SISO control design, hand computation of the controller using Youla parametrization provides an insight into the design of a controller. Performance, internal stability, and robustness are the fundamental measures in control design which are enforced using this approach by directly shaping the closed loop transfer functions [?, ?]. Classical controls address these measures by indirectly shaping the open loop transfer function [?].

In a SISO estimator design, the classical feedback loop (as illustrated in Figure 7.5(a)) is transformed to an equivalent feedforward loop, Figure 7.5(b), to directly shape complementary sensitivity, or closed loop transfer function, by selecting a stable and proper Youla parameter, where $G_{p,e}(s) = C_e(sI - A_e)^{-1}B_e$, assuming the transfer function for the observer is strictly proper without loss of generality. This model is assumed to be always linear. Equivalently, $G_p(s) = C(sI - A)^{-1}B$, represents the actual linear plant model, however, this model could also be nonlinear. Therefore, we have transformed an estimator design to a neoclassical feedback control design utilizing Youla parameterization. It is important to note that the fundamental difference between designing an estimator using this technique and its dual control design is that in the case of an estimator design, the observer model (estimated model of the plant), $G_{p,e}(s)$, is an integral part of the feedback loop. While, in the case of the control design, this model is discarded and the controller is implemented on the actual plant.

From Figure 7.5(b), the relation between Youla, $Y(s)$, and the controller, $G_c(s)$, is derived to be,

$$Y(s) = \frac{G_c(s)}{1 + G_c(s)G_{p,e}(s)} \quad (7.2)$$

Or

$$G_c(s) = \frac{Y(s)}{1 - Y(s)G_{p,e}(s)} \quad (7.3)$$

For a SISO system, it is straightforward to illustrate the procedure for designing a feedback controller to meet the control loop objectives as stated previously. In Figure 7.5(a), the closed loop transfer function (complementary sensitivity transfer function), $T(s)$, is given by,

$$T(s) = \frac{G_c(s)G_{p,e}(s)}{1 + G_c(s)G_{p,e}(s)} \quad (7.4)$$

From the transformation of the feedback to feedforward, by using Youla parameter as shown in Figure 7.5(b), we can compute the closed loop, or the complementary transfer function, as,

$$T(s) = Y(s)G_{p,e}(s) \quad (7.5)$$

Given a particular observer transfer function, $G_{p,e}(s)$, one typically fixes a preferred closed loop transfer function, $T(s)$, as described later in this paper, and then computes a proper and stable Youla, $Y(s)$ through Equation 7.5. If the plant transfer function is stable and, in addition, does not include any non-minimum phase zeros [?], the internal stability is met by only selecting a proper and stable Youla. In the case of an unstable plant or a plant which includes non-minimum phase zeros [?], the following interpolation conditions must be satisfied to guarantee internal stability [?].

If “ z ” denotes a zero of the return ratio (open loop transfer function), L , of multiplicity a_z , and “ p ” denotes a pole of L of multiplicity a_p , then,

$$T(z) = 0 \quad S(z) = 1 \quad (7.6)$$

and

$$\frac{d^k T}{ds^k}(z) = 0 = \frac{d^k S}{ds^k}(z) \quad 1 \leq k \leq a_z \quad (7.7)$$

while

$$S(p) = 0 \quad T(p) = 1 \quad (7.8)$$

5012 and

$$\frac{d^k S}{ds^k}(p) = 0 = \frac{d^k T}{ds^k}(p) \quad 1 \leq k \leq a_p \quad (7.9)$$

5013 Where $S(s)$ is the sensitivity transfer function. This transfer function specifies the ratio
 5014 of plant parameter changes to its nominal parameter values in addition to the ratio of the
 5015 control loop output to the output disturbance. The following algebraic constraint exists
 5016 between the sensitivity and the complementary sensitivity transfer functions.

$$T(s) + S(s) = 1 \quad (7.10)$$

5017 The relations, which are provided in Equations 7.6-7.9, are known as interpolation con-
 5018 ditions. Due to these conditions, the closed-loop transfer functions, $S(s)$ and $T(s)$, are
 5019 constrained to take on certain values in the complex plane and these conditions influence
 5020 these values on the imaginary axis, $S(j\omega)$ and $T(j\omega)$. These conditions cannot be removed
 5021 by a pole-zero cancellation where “z” lies in the right half-plane or on the imaginary axis,
 5022 where a factor of the return ratio is non-minimum phase. Constraining these transfer func-
 5023 tions imposes irrevocable limits to feedback design.

5024 From Equation 7.3, the resulting controller can be written as,

$$G_c(s) = \frac{Y(s)}{S(s)} \quad (7.11)$$

5025 where, $S(s) = 1 - Y(s)G_{p,e}(s)$.

5026 The steps for designing a SISO estimator are given as follows:

- 5027 1. Given an observer transfer function, $G_{p,e}(s)$, select a desired complementary transfer
 5028 function, $T(s)$.
- 5029 2. From Equation 7.5, compute a Youla transfer function, $Y(s)$, which meets the afore-
 5030 mentioned interpolation conditions.
- 5031 3. From the constraint provided in Equation 7.10, compute the sensitivity transfer func-
 5032 tion, $S(s)$.
- 5033 4. Finally, from the expression given in Equation 7.11, compute the controller, $G_c(s)$,
 5034 which satisfies all the performance, robustness, and stability criterion.

5035 To summarize, an estimation problem has been converted to a robust controller output
 5036 observer problem and the guarantee for robust stability and performance, as described in

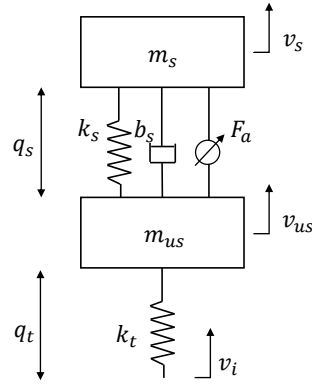


Figure 7.6: Schematic.

many robust control publications, are provided by the selection of the complementary sensitivity transfer function, $T(s)$, and sensitivity transfer function, $S(s)$ [?]. In addition, the well-posedness and the internal stability of the feedback loop are guaranteed by satisfying the interpolation conditions as discussed above.

It is important to note that the Youla parametrization approach discussed above can not be casted in the Internal Model Control (IMC) Paradigm [?], as in the case of the IMC, the controller output has to be simultaneously applied to the actual plant and the observer. In our case, as described above, the actual input to the plant merely acts as a disturbance input, as illustrated in Figure 6.10.

7.3 The SISO Case

This section provides a simple example of a linear time invariant (LTI) quarter car model of a vehicle suspension [?]. A comparison between the Kalman filter design and YCOO is made. The SISO Youla Controller Output Observer (YCOO) concept is used to estimate the vertical road input velocity and all four states of a quarter car model while only requiring access to the sprung mass acceleration measurement. These results are then compared to the results from a discrete Kalman filter.

A simple quarter car model, as shown in Figure 7.6, is used to design the YCOO. The suspension stiffness, k_s , tire stiffness, k_t , and damper, b_s , are assumed linear and the masses, m_s and m_{us} , assumed known. A force actuator, F_a , represents the controllable input for semi-active damping or fully active suspension. However, in this example, this force is assumed zero.

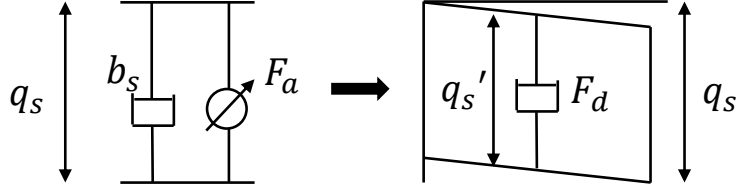


Figure 7.7: Damper.

The suspension stiffness can be obtained from the relationship between vertical wheel stroke, q_s , and spring force, and is therefore represented by a vertical spring. The passive damping force, F_d , is modified by a lever ratio. That is, the deflection is approximately proportional to (by $\frac{\partial q_s'}{\partial q_s}$), but not equal to, the wheel stroke. This is shown in Figure 7.7.

The plant is represented in the state space form as follows,

$$\dot{x} = Ax + Bu \quad (7.12)$$

$$y = Cx + Du$$

The state and input matrices (Equations 7.13 and 7.14) are shown below.

$$A = \begin{bmatrix} b_s/m_s & b_s/m_s & k_s/m_s & 0 \\ b_s/m_{us} & -b_s/m_{us} & -k_s/m_{us} & k_t/m_{us} \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix} \quad (7.13)$$

The states, as depicted in Figure 7.7, are sprung mass velocity, v_s , unsprung mass velocity, v_{us} , suspension deflection, q_s , and tire vertical deflection, q_t .

$$B = \begin{bmatrix} 1/m_s & 0 \\ 1/m_{us} & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \quad (7.14)$$

The two inputs are road velocity disturbance, v_i , and force input, F_a . Note that a physics based modeling technique based on power is used - as a result, inputs to the system include velocities but not displacements.

7.3.1 Output and Feedthrough Matrices

The C and D matrices are shown in Equations 7.15 and 7.16 as the combination of individual output and feedthrough matrices. These individual output matrices are shown in

Equations 7.17 - 7.24. The outputs, as provided by these matrices, are the sprung mass acceleration, the unsprung mass acceleration, the damper force, and the relative velocity between the sprung and unsprung mass.

$$C = [C_{\dot{v}_s} \quad C_{\dot{v}_{us}} \quad C_{F_{de}} \quad C_{v_{rel}}]^T \quad (7.15)$$

$$D = [D_{\dot{v}_s} \quad D_{\dot{v}_{us}} \quad D_{F_{de}} \quad D_{v_{rel}}]^T \quad (7.16)$$

We can obtain the following outputs by properly modifying the general C and D matrices as follows. **It is important to note that the actual plant and the observer models in this example are assumed to be equal, such that $G_p = G_{p,e}$, $A = A_e$, $B = B_e$, $C = C_e$, and $D = D_e$.**

Sprung Mass Acceleration

$$C_{\dot{v}_s} = A_{1,*} \quad (7.17)$$

$$D_{\dot{v}_s} = B_{1,*} \quad (7.18)$$

Unsprung Mass Acceleration

$$C_{\dot{v}_{us}} = A_{2,*} \quad (7.19)$$

$$D_{\dot{v}_{us}} = B_{2,*} \quad (7.20)$$

Damping Force

$$C_{F_{de}} = \left(\frac{\partial q'_s}{\partial q_s} \right)^{-1} [b_s \quad b_s \quad 0 \quad 0] \quad (7.21)$$

$$D_{F_{de}} = \left(\frac{\partial q'_s}{\partial q_s} \right)^{-1} [1 \quad 0] \quad (7.22)$$

Relative Velocity

$$C_{v_{rel}} = [1 \quad 1 \quad 0 \quad 0] \quad (7.23)$$

$$D_{v_{rel}} = [0 \quad 0] \quad (7.24)$$

7.3.2 SISO Estimator Design

The model used for the SISO YCOO includes the road velocity disturbance as an input and the sprung mass acceleration as an output. With this configuration, the states, inputs, and outputs are given by $\hat{x}$, $\hat{u}$, and $\hat{y}$, respectively, and are defined below. This method does

not require access to the force input, F_a , to properly estimate the states and other unknown disturbance inputs. Referring to Figure 7.4, the input to the YCOO is the measured sprung mass acceleration including sensor noise, y_m (y in Figure 7.4), and the output of the YCOO is the road velocity disturbance, $\hat{v}_i$ ($\hat{u}$ in Figure 7.4). In the SISO configuration, the YCOO simultaneously estimates the road velocity disturbance input and the suspension states by only having access to the measured sprung mass acceleration. In the MIMO configuration presented later, road velocity disturbance and the force inputs, and the system states, are estimated while only having access to the measured sprung mass and unsprung mass accelerations. On the other hand, in the SISO configuration, the discrete Kalman filter has to have access to both measured noisy sprung mass acceleration and the force input, F_a , for proper estimation of the suspension states. Classical configuration of the Kalman filter does not lend itself to input estimation.

The plant input, output, and the states for this example are provided below.

$$\hat{\mathbf{x}} = [\hat{q}_s \quad \hat{v}_{us} \quad \hat{q}_s \quad \hat{q}_t]^T \quad (7.25)$$

$$\hat{u} = \hat{v}_i \quad (7.26)$$

$$\hat{y} = \dot{\hat{v}}_s \quad (7.27)$$

Below, we briefly outline the design of the SISO YCOO based on the design steps as provided above. It should be noted that the design of the discrete Kalman filter has been a topic of many papers [?] and is not discussed in this work. In designing the Kalman filter, the road disturbance is treated as process noise and sensor noise is modeled using specifications of a known accelerometer. The selected values for the process noise and the sensor noise are not tuned or changed for different simulations. It should be noted that in practical applications, it is the discrete version of the Kalman filter which is always utilized. In the case of the YCOO, the discretization of the controller has no effect on its performance given a reasonable sampling rate. This is why the continuous version of the YCOO is included in the presented results.

The plant transfer function from the road velocity disturbance input to the sprung mass acceleration is derived to be,

$$G_p(s) = \frac{(\beta_2 s^2 + \beta_1 s)}{(s^2 + \alpha_{22}s + \alpha_{21})(s^2 + \alpha_{21}s + \alpha_{11})} \quad (7.28)$$

$$= sG_p(s)^* \quad (7.29)$$

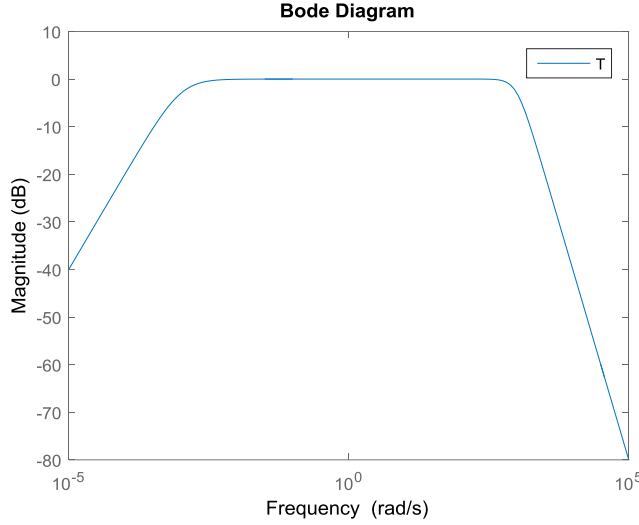


Figure 7.8: Closed loop frequency response.

Asterisks (*) represent the portion of the plant with no zeros on the $j\omega$ -axis or right-half plane, while α 's and β 's correspond to system natural frequencies and damping.

We select the Youla transfer function to be stable and proper based on the procedures given before, and furthermore, the zero of the plant is carried over to complementary/closed loop transfer function for meeting the interpolation conditions (internal stability of the feedback loop), where this condition is given by,

$$T(z) = 0 \text{ where } z \text{ is the zero of the plant on the } j\omega\text{-axis } (z = 0).$$

We have the option to cancel all the stable poles of the plant and shape the closed loop transfer function to closely match the second order Butterworth filter. A Butterworth filter is chosen as it is an optimal, well damped filter with desirable response characteristics. This filter contains ζ and ω_n below and is tuned for good noise rejection. Additionally, a pole, p_1 , is required for reasons that will be explained in the following paragraphs. Note that Youla must be proper. The resulting Youla transfer function is selected as follows,

$$Y(s) = \frac{\omega_n^2}{(s + p_1)(s^2 + 2\zeta\omega_n s + \omega_n^2)G_p(s)^*} \quad (7.30)$$

With this choice of Youla transfer function, the complementary transfer function is computed to be,

$$T(s) = Y(s)G_p(s) \quad (7.31)$$

Which results in the following,

$$T(s) = \frac{s\omega_n^2}{(s + p_1)(s^2 + 2\zeta\omega_n s + \omega_n^2)} \quad (7.32)$$

The $j\omega$ -axis zero of the plant forces the complementary transfer function to be zero at low frequency to meet the interpolation conditions and therefore, to have an internally stable feedback loop. This results in deterioration of the target following/estimation at low frequency as the closed loop transfer function is required to be unity at low frequencies. The pole $(s + p_1)$ in the denominator of the Youla transfer function is added to force the closed loop transfer function, $T(s)$, to act as a bandpass filter as shown in Figure 7.8. Due to selecting this pole to be at a very low frequency, the impact on the target following capability of the closed loop transfer function is minimized.

The Youla transfer function is directly implemented using the configuration of Figure 7.4 to estimate the system states. Given that the models used for estimation are similar to the original plant model both Youla and Kalman estimation methods performed well. This is not shown in this chapter.

In order to reveal the robustness characteristics of this approach, we increased the observer model sprung mass by 50% relative to the actual plant model. The results for two states, wheel stroke and unsprung mass velocity, are depicted in Figure 7.9 for a random road input with a specific power spectral density of a typical D-road (farm road).

Table 7.1: RMS of Unsprung Mass Velocity results for the actual, the Youla and the Kalman Filter estimators

Test	D-road
Actual [RMS, m/s]	0.07
Youla [RMS, m/s]	0.1
Kalman Filter[RMS, m/s]	0.6

In addition, the RMS values of the unsprung mass velocities, for the actual plant, the Youla and the Kalman filter estimators, are provided in Table 7.1.

From Figure 7.9 and the results for the RMS values of unsprung mass velocities for the actual plant, and the Youla and the Kalman filter estimators provided in Table 7.1, we can conclude that the Youla estimation method is more robust than the Kalman filter.

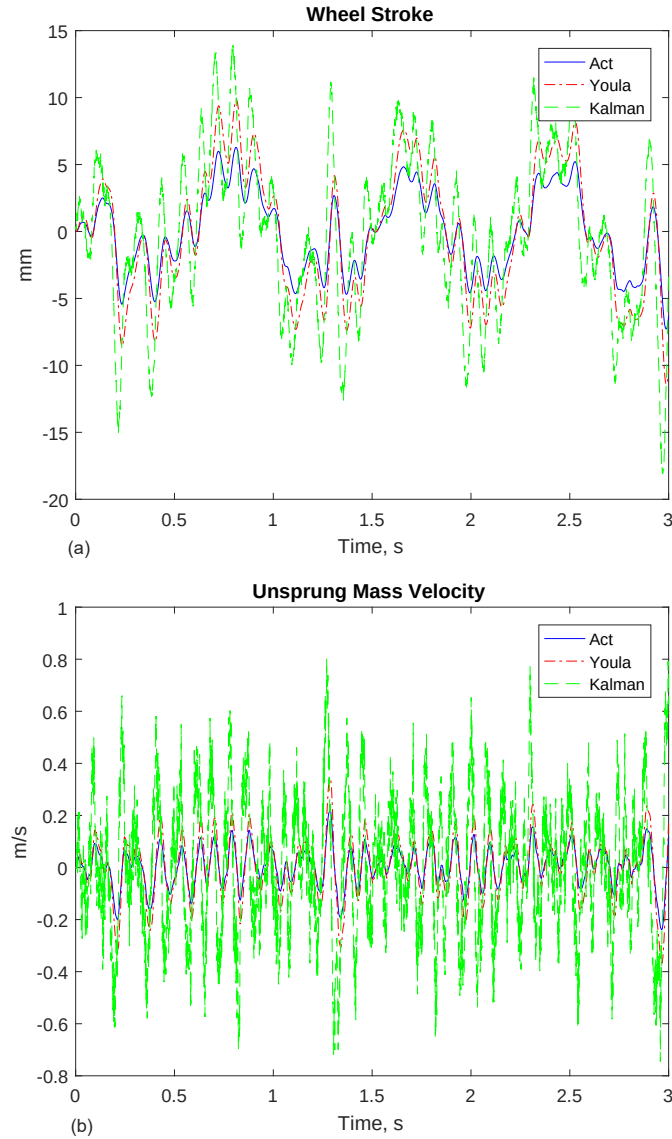


Figure 7.9: SISO state estimation with D-road PSD input (a) Wheel stroke (b) Unsprung mass velocity.

5145 It should also be noted that the Kalman filter has access to the input force and the noisy
 5146 sprung mass acceleration signal while the Youla approach only has access to the noisy
 5147 sprung mass acceleration signal. In addition, although not shown, the road disturbance
 5148 input is also estimated by the Youla approach without any modifications to the base ap-
 5149 proach. In the multivariable case, as discussed in the next sections, the two inputs, road

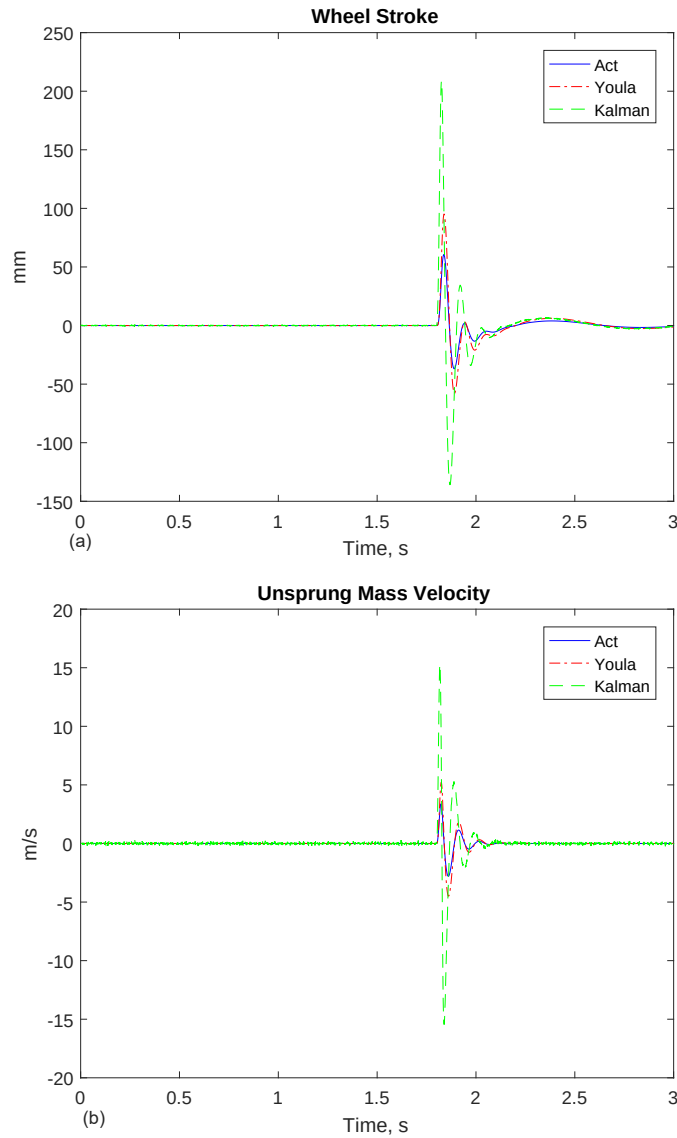


Figure 7.10: SISO state estimation with a speed bump input (a) Wheel stroke (b) Unsprung mass velocity.

disturbance and input force, in addition to all the states, will be estimated by the MIMO YCOO approach.

As the final comparative analysis for the SISO case, using the same quarter car model, Figure 7.10 presents the results for similar state estimates but this time with a prescribed speed bump as a road input. A similar conclusion can be drawn from these results.

7.4 Conclusions and Future Work

Youla parametrization has a long history in control development. Using both linear and nonlinear vehicle suspension models, it was shown that this approach could be equivalently utilized for state and input estimation. The comparative results between YCOO and Kalman filter were presented in the linear domain including robustness characteristics of YCOO versus Kalman filtering method.

There are many open questions that remain to be answered. The performance and robustness characteristics of YCOO in the nonlinear domain and its comparative analysis against established nonlinear estimation techniques, such as Extended Kalman and Unscented Kalman filters, Sliding Mode Observers, and Particle filters, require further investigation. The implication of using established analytical and numerical optimization approaches to replace hand computation will be addressed in future work.

CHAPTER 8

PRACTICAL APPLICATIONS

In this chapter, we present three practical applications for which we investigate SISO Youla control development using hand computation, SISO H_∞ control design and finally, Youla loop shaping control technique using Internal Model Control (IMC). These applications include Yaw Stability Control (YSC) development using Active Limited Slip Differential (ALSD), Anti-lock Braking System (ABS) control development using electric motors, and vehicle side-slip estimation using Youla Controlled Output Observer and we compare the results with both linear and non-linear Kalman filtering methods.

8.1 Yaw Stability Control with Active Limited Slip Differential

Yaw stability control systems have been and are addressed using various actuators in the automotive industry. The majority of current yaw stability systems, such as DSC (Dynamic Stability Controller), utilize brake interventions at individual wheels to develop yaw moments to ensure vehicle stability at times when it would otherwise be compromised. However, the focus of the premium vehicle manufacturers is not just to improve stability but also to increase driver enjoyment while driving both below or at the limits of adhesion. Brake

based stability systems tend to be somewhat intrusive, causing decelerations and hence, loss of vehicle speed. Alternative actuation systems are therefore being considered that may provide increased stability, without the intrusiveness of brake-based systems.

There are different versions of an active differential, here, an electric motor driven active limited slip differential (ALSD) is considered, with a wet multi-disc friction clutch effectively providing a connection between the two driveshafts, as illustrated in Figure (8.1). By controlling the clamping force on the clutch torque transfer can be modulated. At maximum capacity, it is capable of providing 2500 Nm locking torque, which is equivalent to 1250 Nm of torque transfer from one driven wheel to the other.

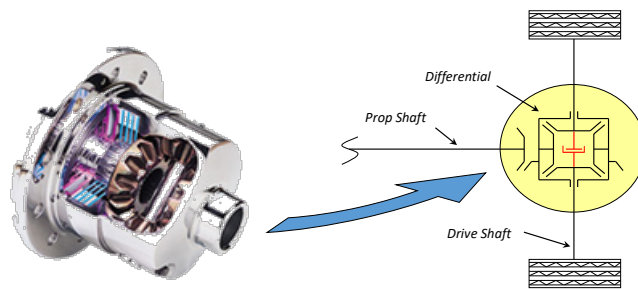


Figure 8.1: E-Locker.

One of the key limitations of an ALSD is that only control over the magnitude of torque transfer is possible. The direction of the torque transfer is determined by the wheel speed difference as torque is always transferred from faster to the slower wheel. As a result, the actuator is limited, in most cases, to generating more relative understeer and therefore, it is only capable of stabilizing an oversteering vehicle, as shown in Figure (8.2).

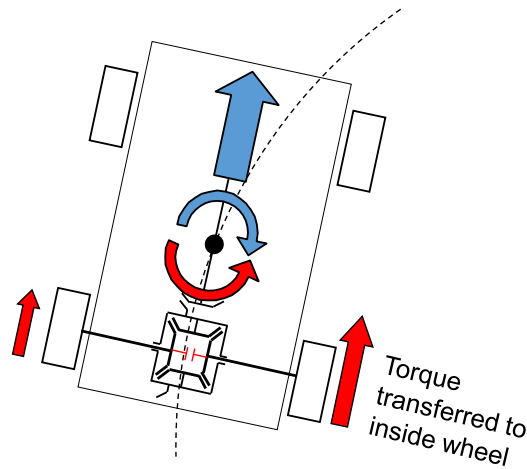


Figure 8.2: A schematic of the torque transfer capability and limitation of of the E-Locker.

8.1.1 Model and Control Design

In this section, we describe the vehicle model (which is utilized for the control development), our simplified assumption for the actuator model, and our control development strategies including their comparison.

To reveal the basics of the vehicle dynamics with steady-state cornering and where the longitudinal velocity is assumed to be constant, one can use the so-called “linear bicycle model.” This model is composed of one front and one rear tire linked by a rigid body, as illustrated in Figure 8.3. In this representation, the front and rear axles are squashed together. This model represents the yaw dynamics of a vehicle and its simplicity lends it to use for control design purposes.

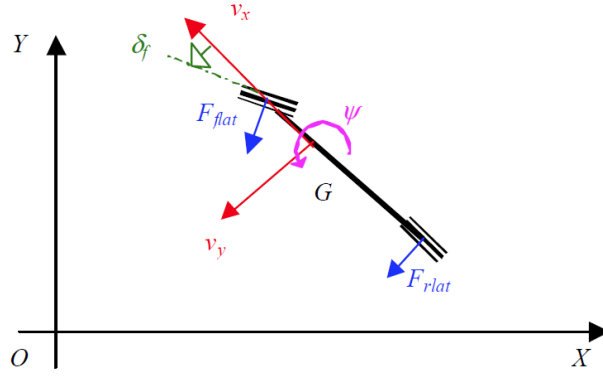


Figure 8.3: Bicycle model.

5205 The linear bicycle model includes two degrees-of-freedom. The degrees of freedom are
 5206 yaw and sideslip velocity. In state-space form, this model may be written as

$$\begin{bmatrix} \dot{V} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} Y_V/M & Y_r/M - U \\ N_V/I_{zz} & N_r/I_{zz} \end{bmatrix} \begin{bmatrix} V \\ r \end{bmatrix} + \begin{bmatrix} 0 \\ 1/I_{zz} \end{bmatrix} [T_{yaw}] + \begin{bmatrix} Y_\delta/M \\ N_\delta/I_{zz} \end{bmatrix} [\delta] \quad (8.1)$$

5207 or

$$\dot{X} = AX + Bu + Gw, \quad (8.2)$$

where

$$\begin{aligned} Y_V &= \frac{-C_{\alpha f} - C_{\alpha r}}{U}, & Y_r &= \frac{-aC_{\alpha f} + bC_{\alpha r}}{U}, & Y_\delta &= C_{\alpha f} \\ N_V &= \frac{-aC_{\alpha f} + bC_{\alpha r}}{U}, & N_r &= \frac{-a^2C_{\alpha f} - b^2C_{\alpha r}}{U}, & N_\delta &= aC_{\alpha f} \end{aligned}$$

5208 where M is the vehicle mass, I_{zz} is the mass moment of inertia around z-axis, $C_{\alpha f}$ is the
 5209 front tire cornering stiffness, $C_{\alpha r}$ is the rear tire cornering stiffness, U is the longitudinal
 5210 velocity, a is the distance from CG to the front axle, and b is the distance from CG to the
 5211 rear axle.

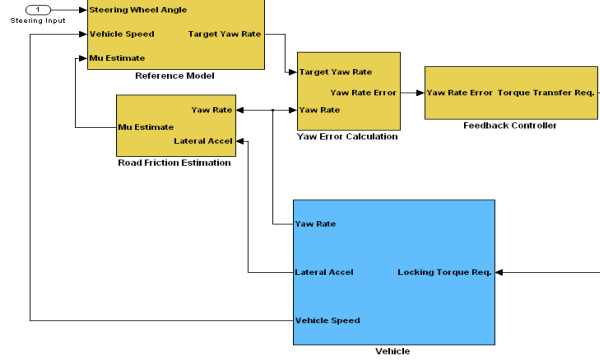


Figure 8.4: Controller structure.

The general structure of the controller implemented on-board vehicle is shown in Figure 8.4. To design the feedback controller using Youla parameterization and the techniques discussed in the previous chapters, we will utilize the following feedback control loop.

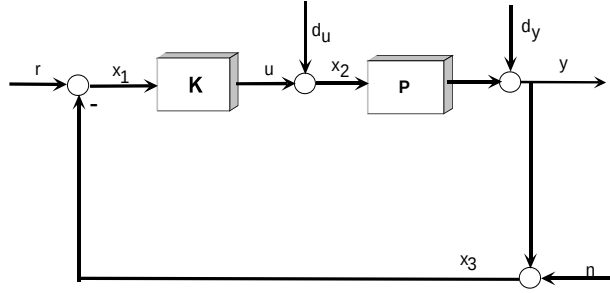


Figure 8.5: Feedback control loop.

The feedback control design will control the yaw rate r by affecting the yaw torque T_{yaw} . From the above state-space representation, we can write the transfer function.

$$\frac{r}{T_{yaw}} = \frac{\frac{1}{I_{zz}} \left(s + \frac{C_{\alpha f} + C_{\alpha r}}{MU} \right)}{s^2 + \left(\frac{C_{\alpha f} + C_{\alpha r}}{MU} + \frac{a^2 C_{\alpha f} + b^2 C_{\alpha r}}{I_{zz} U} \right) s + \left(\frac{-a C_{\alpha f} + b C_{\alpha r}}{I_{zz}} + \frac{C_{\alpha f} C_{\alpha r} (a + b)^2}{I_{zz} MU^2} \right)} \quad (8.3)$$

Additionally, we will consider actuator dynamics. Then, the feedback control design is performed after having obtained the plant transfer function from the actuator input (the locking torque request) to the vehicle yaw rate as illustrated in the Figure below.

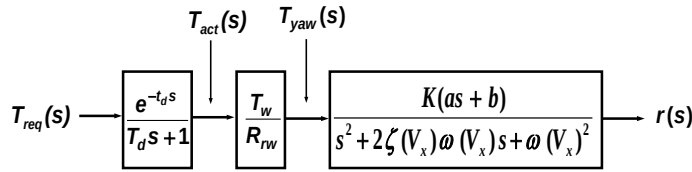


Figure 8.6: A 3rd order plant model.

Where t_d is the actuator model pure time delay, T_d is the actuator model first order time constant, T_w is track width, and R_{rw} is the wheel radius. This means we have actuator dynamics

$$\frac{T_{yaw}}{T_{req}} = \frac{T_w}{R_{rw}I_{zz}} \cdot \frac{e^{-t_d s}}{T_d s + 1} \quad (8.4)$$

The vehicle model from above can be rewritten as a second order denominator where parameters ζ and ω are functions of vehicle forward velocity V_x . For the control design V_x is assumed to be constant. Then, we have the complete transfer function from input torque request to output yaw rate.

$$\frac{r}{T_{req}} = \frac{T_w}{R_{rw}I_{zz}} \times \frac{e^{-t_d s}}{T_d s + 1} \times \frac{\left(s + \frac{C_{\alpha f} + C_{\alpha r}}{MU}\right)}{s^2 + \left(\frac{C_{\alpha f} + C_{\alpha r}}{MU} + \frac{a^2 C_{\alpha f} + b^2 C_{\alpha r}}{I_{zz}U}\right)s + \left(\frac{-aC_{\alpha f} + bC_{\alpha r}}{I_{zz}} + \frac{C_{\alpha f}C_{\alpha r}(a+b)^2}{I_{zz}MU^2}\right)} \quad (8.5)$$

It should be pointed out that in order to model the time delay associated with this actuator (electric motor and clutch plates), a pure time delay, t_d , and a first order low pass filter with a time constant T_d is utilized as shown in Figure 8.6. The pure time delay and the time constant associated with the actuator model is obtained from vehicle measurement as depicted in Figure below.

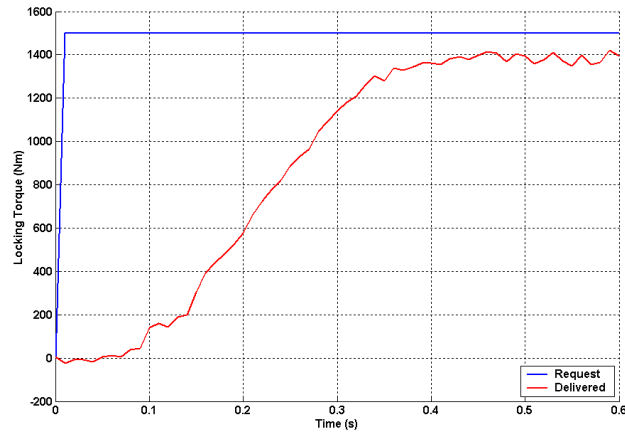


Figure 8.7: Actuator model.

```

5228
5229      %% Vehicle Data:
5230      mf = 994.5260; % Total mass at front axle (kg)
5231      mr = 882.1203; % Total mass at rear axle (kg)
5232      i_zz = 3552; % Yaw moment of inertia ($kg m^2$)
5233      l = 2.909; % Wheelbase (m)
5234      rr = 0.329; % Rolling radius (m)
5235      trr = 1.53; % Front and Rear track (m)
5236
5237      % Understeer Gradient (rad/g):
5238      K\_under = (1.8*pi/180); % measured UG
5239
5240      % Cornering Stiffnesses:
5241      Car = 1.5679e+005; % Rear (N/rad)
5242      Caf = (mf*9.81) / (Kunder + (mr*9.81/Car)); % Front
5243              (N/rad)
5244
5245      %% VEHICLE SPEED (m/s):
5246      V_x = 25;
5247
5248      %% PRELIMINARY CALCULATIONS:
5249      g = 9.81; % Acceleration due to gravity (m/s^2)
5250      m = mf + mr; % Total mass (kg)
5251
5252      % Weight distribution
5253
5254      a = mr*l/(mf+mr); % Distance from CG to front axle
5255              (m)
5256      b = mf*l/(mf+mr); % Distance from CG to rear axle
5257              (m)

```

5258 The pure delay, t_d is approximately 100ms while the a time constant, T_d is assumed
5259 to be 167ms. Given the parameters above the transfer function from the locking torque

5260 request $T_{req}(s)$ to the yaw rate $r(s)$ is given by,

$$\frac{r(s)}{T_{req}(s)} = \frac{-s + 20}{s + 20} \frac{1}{0.167s + 1} \frac{4.65(0.00028s + 0.0016)}{s^2 + 12.31s + 60.6} \quad (8.6)$$

5261 where the pure delay handled with a 1st-Order Padè approximation.

5262 Given the plant transfer function above, we derive three different SISO controllers using
5263 the techniques discussed in the previous chapters,

- 5264 ▪ Youla control design using hand computation
- 5265 ▪ H_∞ control design using loop shaping technique

5266

5267 **Youla control design using hand computation:**

5268 The Youla transfer function is selected so that the closed loop complementary transfer
 5269 function would approach,

$$T = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

5270 where $\omega_n^2 = 60.6$ and $2\zeta\omega_n = 12.31$

5271 However, knowing that there is a pure delay, therefore, a nonminimum phase zero, and
 5272 the Youla transfer function cannot be selected to cancel these nonminimum phase zero, the
 5273 Youla transfer function is modified to be,

$$Y = K \frac{\alpha s + \beta}{T_d}$$

5274 Where $K = 4.65$, $\alpha = 0.00028$, $\beta = 0.0016$, and $T_d = 1/0.167s + 1$.

5275 This selection of Youla transfer function gives the following closed loop complementary
 5276 transfer function,

$$T = \frac{-s + \gamma_1}{s + \gamma_2} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

5277 Where $\gamma_1 = \gamma_2 = 20$ (it should be clear that $\gamma_2 \leq \gamma_1$ to limit the bandwidth due to the
 5278 nonminimum phase zero), and ω_n and $\zeta\omega_n$ are the same as given above.

5279 The following m-file is used to design this controller,

5280

```
5281 s=tf('s');
```

5282

```
5283 % Youla hand computation method
```

5284

```
5285 % Declaring plant model
```

```
5286 td=(-s+20)/(s+20);
```

```
5287 Td=1/(0.167*s+1);
```

```
5288 G= (4.65*(0.00028*s+0.0016))/(s^2+12.31*s+60.6);
```

5289

```
5290 Gp= td*Td*G;
```

5291

```
5292 % Selecting Youla parameter
```

```
5293 Y=(1/Td)*(60.6/(4.65*(0.00028*s+0.0016)))*(1/(1/50*s+1)^2);
5294
5295 % Computing closed loop transfer functions
5296 T=minreal(Y*Gp);
5297 S=minreal(1-T);
5298 Gc=(Y/S);
5299 Gc=minreal(Gc);
5300
5301 % Plotting results
5302 figure(1)
5303 bode(T);
5304 hold on
5305 bode(S);
5306 bode(Y);
5307 legend('T','S','Y')
5308 grid on
5309
5310 figure(2)
5311 step(Y)
5312 legend('Actuator')
5313 grid on
5314
5315
5316 figure(3)
5317 step(T)
5318 legend('Yaw rate response')
5319 grid on
```

5320 Two high frequency poles are added to the Youla transfer function to better shape its
5321 frequency response at high frequency as can be seen in the m-file above.

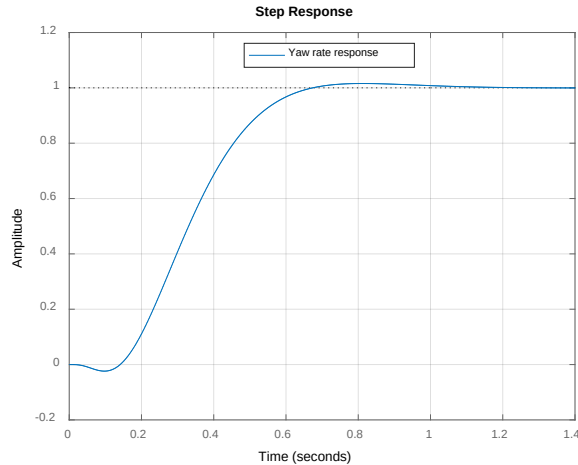


Figure 8.8: Yaw rate response to the locking torque step input request

The Figure 16.3 illustrates the yaw rate response to a locking torque step request. The non-minimum phase zero behavior is evident from this response. It should be clear that to control the non-minimum phase behavior, the constant γ_2 could have been taken to be much lower than γ_1 . In this case, the impact of the non-minimum phase zero would be minimized while it decreases the closed loop transfer function bandwidth, hence, it increases the rise time and settling time.

The locking torque capacity for this actuator is highly nonlinear and it depends on variables such as clutch plate and road coefficient of frictions, wheel speed difference, locking and clutch plate temperature to name a few. This actuator is capable of providing a peak locking torque of 2500Nm. Therefore, it is interesting to investigate the impact of the Youla linear control design on the actuator effort.

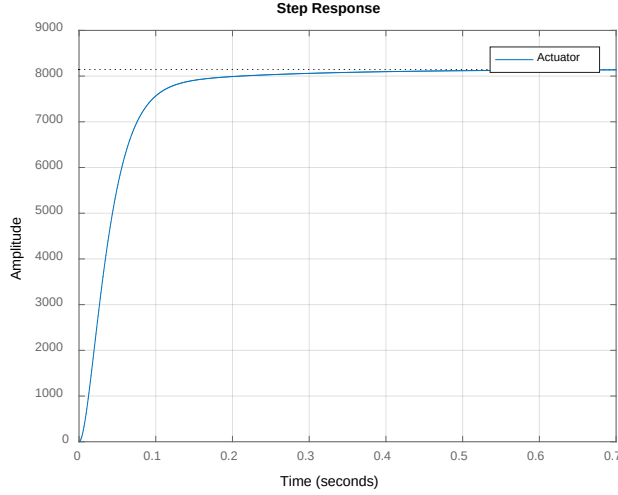


Figure 8.9: Actuator effort to the locking torque step input request

It should be noted that the longitudinal velocity for this linear model is assumed to be 25m/s. The Figure 16.4 shows that, given this longitudinal velocity, for good target following, the actuator requires almost more than 3 times its peak capacity at steady state. This means at high speed (25m/s or greater) stability maneuvers, the clutch pack of this actuator fully locks the two half shafts.

H_∞ control design using loop shaping technique:

The loop shaping approach as discussed in the previous chapters requires augmentation of the nominal plant by shaping filters, as illustrated in Figure 6.7. The filters that are used for the nominal plant augmentation are given as follows,

$$W_P = \frac{K_{W_P} \omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$

Where $K_{W_P} = 15$, $\omega_n = 0.1$, and $\zeta = 3.0$.

As pointed out in the Chapter 6, this filter weights the sensitivity transfer function. The shaping filter, W_Δ , for the closed loop complementary transfer function is given as follows,

$$W_\Delta = \frac{K_{W_\Delta} s}{s + \frac{K_{W_\Delta}}{T_{max}}}$$

This filter is used as weighting on the complementary transfer function so that the closed loop stability is guaranteed in the presence of the pure time delay of T_{max} , due to the actuator dynamics. In order to design the weighting filter W_Δ with parameter K_{W_Δ} and $T_{max} = 0.1s$ that will ensure robustness in the presence of a pure time delay $\leq 0.1s$, we utilize multiplicative perturbation¹. Suppose that nominal plant transfer function is P and consider perturbed plant transfer function of the form $\hat{P} = (1 + \Delta W_\Delta)P$. Then knowing that $\|\Delta\|_\infty \leq 1$, then

$$\left| \frac{\hat{P}(j\omega)}{P(j\omega)} - 1 \right| \leq |W_\Delta(j\omega)| \quad \forall \omega,$$

So $|W_\Delta(j\omega)|$ provides the uncertainty profile. This inequality describes a disk in the complex plane, where at each frequency the point $\hat{P}/P$ lies in the disk with center 1, and radius of $|W_\Delta|$. Typically, $|W_\Delta(j\omega)|$ is a increasing function of ω as uncertainty increases with increasing frequency. For the case of pure delay,

$$\frac{\hat{P}(j\omega)}{P(j\omega)} = e^{-jT_{max}\omega}$$

Therefore, the filter should be designed so that $|e^{-jT_{max}\omega} - 1| \leq |W_\Delta(j\omega)|$. A Bode plot of $|e^{-jT_{max}\omega} - 1|$ and $|W_\Delta(j\omega)|$ illustrate that the designed weighting filter satisfies the condition in Figure below when $K_{W_\Delta} = 2.5$,

¹We have shown in the previous chapters that multiplicative perturbation is equivalent to weighting the nominal complementary transfer function - small-gain theorem

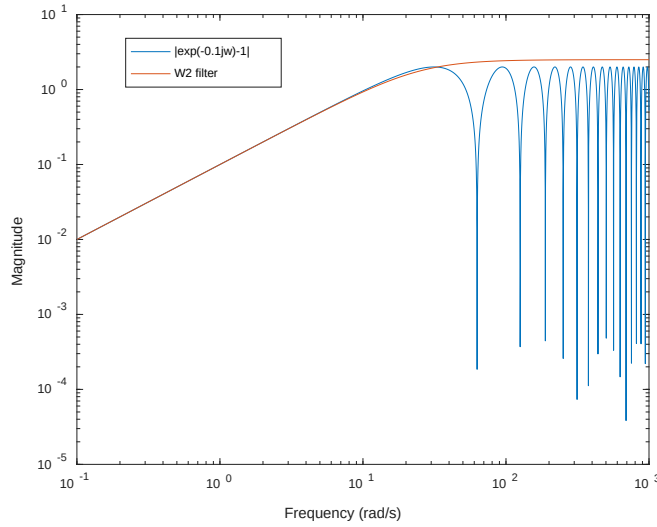


Figure 8.10: Bode plot of $|e^{-j0.1\omega} - 1|$ and W_{Δ} (W_2) filter

5360 The final weighting filter is used to limit the actuator effort and is given by

$$W_u = \frac{1}{\rho}$$

5361 Where ρ is a constant and is equivalent to the maximum locking torque generated by
 5362 the actuator (in this case, based on our previous analysis, this constant is taken to be equal
 5363 to 8000).

5364 The following m-file is used to design this controller.

5365

5366 `s=zpk('s');`

5367

5368 `% Hinf method`

5369

5370 `% Declaring plant model`

5371 `td=(-s+20)/(s+20);`

5372 `Td=1/(0.167*s+1);`

5373 `G= Td*(4.65*(0.00028*s+0.0016))/(s^2+12.31*s+60.6);`

5374

5375 `Gp= td*Td*G;`

```

5376
5377 % Shaping Filters
5378 wn=0.10;
5379 zeta=3.0;
5380 Kw1=15.0;
5381 Kw3=2.5;
5382 Tmax=0.1;
5383 W1=(Kw1*wn^2)/(s^2+2*zeta*wn*s+wn^2);
5384 W2=1/8000; W3=(Kw3*s)/(s+Kw3/Tmax);
5385
5386 % Plant augmentation with Filters
5387 P=augw(G,W1,W2,W3);
5388 % Controller computation
5389 [K,CL,GAM]=hinfsyn(P);
5390 %
5391 K=minreal(K);
5392 Gc=minreal(tf(K));
5393
5394 % Computing closed loop transfer functions
5395 Yhinf=minreal(Gc/(1+Gc*Gp));
5396 Thinf=minreal(Gp*Yhinf);
5397 Shinf=minreal(1-Thinf);
5398
5399 % Plotting results
5400 figure(1)
5401 bode(Thinf);
5402 hold on
5403 bode(Shinf);
5404 hold off
5405 legend('sensitivity','complementary sensitivity')
5406 grid on
5407
5408 figure(2)
5409 bode(Gp)

```

```
5410 hold on
5411 bode(Yhinf)
5412 bode(Gc)
5413 hold off
5414 legend('Plant','Youla','Gc')
5415 grid on
5416
5417 figure(3)
5418 step(Thinf)
5419 legend('Yaw rate response ,Hinf')
5420 grid on
5421
5422 figure(4)
5423 bode(Shinf)
5424 hold on
5425 bode(W1)
5426 bode(1/W1)
5427 hold off
5428 legend('sensitivity','Wp','1/Wp')
5429 grid on
5430
5431 figure(5)
5432 step(Yhinf)
5433 legend('Actuator effort, Hinf' )
5434 grid on
```

5435 The yaw rate response to a locking torque step request is shown in Figure 16.6 below.

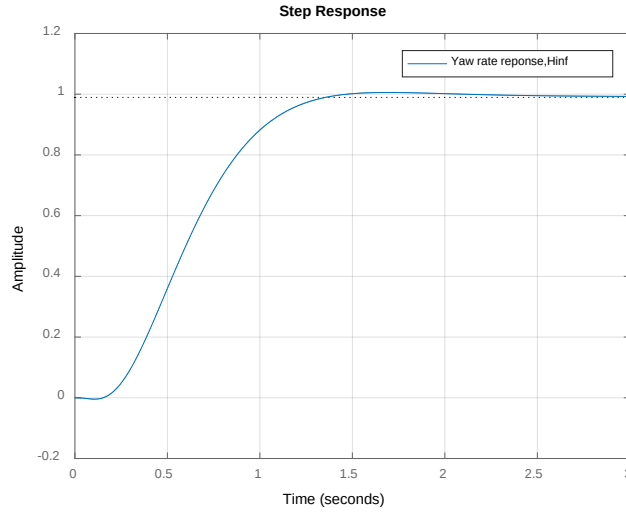


Figure 8.11: Yaw rate response to the locking torque step input request with Hinf Controller

Comparing the yaw rate responses of the H_∞ controller with the Youla hand computation method, it is obvious the results of the H_∞ controller is more conservative with respect to dealing with the pure time delay and response time. In addition, this response is more oscillatory than the yaw response from Youla hand computation results. Furthermore, Youla hand computation technique did not require any shaping filters and tuning these filters is not necessarily a straightforward task.

It would be informative to compare the actuator efforts of the H_∞ method with the Youla hand computation method.

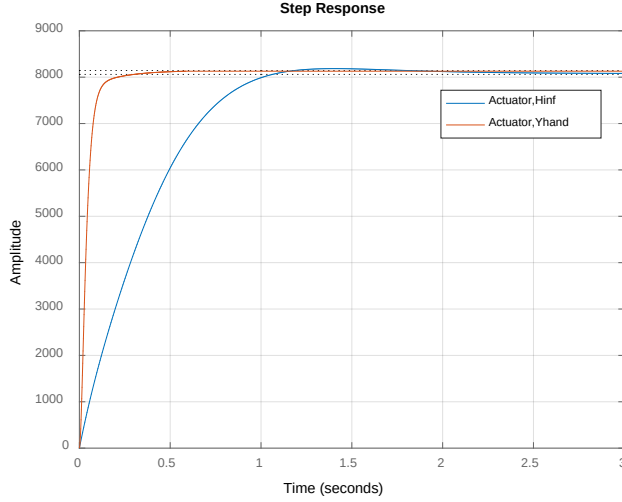


Figure 8.12: Actuator effort to the locking torque step input request comparison

Comparing the actuator effort results of the H_∞ controller with the Youla hand computation, it can be observed that the steady-state values are similar but the transient behaviors (rise time) are different. The response time for H_∞ is slower with slightly higher overshoot. This demonstrates that H_∞ method is conservative as it sacrifices performance (response time) for robust stability. It would be interesting to tune the Youla hand computation actuator effort transient behavior so that it is similar to H_∞ method then compare yaw rate responses. However, before addressing this, we present the results of one final control method as discussed in the previous chapter, this method utilizes the structure of IMC for designing a Youla controller using H_∞ optimization. This will be presented next.

In Chapter 6, we discussed the method of designing a Youla controller with model matching using IMC structure. This method has significant benefits over loop shaping technique as it does not require shaping filters. Furthermore, a Youla controller is computed directly using an optimization method rather than hand computation and only a target model is needed for this optimization. According to our previous discussion, this target model is selected to be,

$$T_t = \frac{-s + \gamma_1}{s + \gamma_2} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

We have selected these constants to be the same as above.

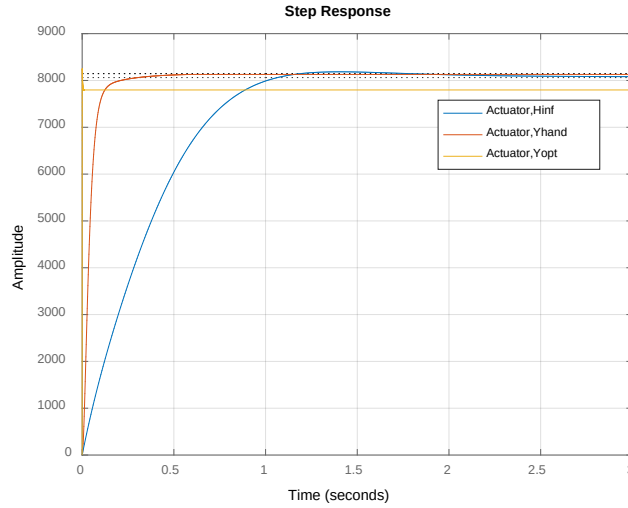


Figure 8.13: Actuator effort to the locking torque step input request comparison of all the proposed methods

In Figure above, we have added the result of Youla optimization actuator effort to Figure 16.7. It is interesting to observe that the actuator effort response of the Youla optimization method differs from Youla hand computation technique. It should be noted that we included two high frequency poles when selecting the Youla transfer function to better control the high frequency response of the Youla parameter at high frequency when we hand computed the Youla parameter. These two high frequency poles had an impact on the closed loop transfer function of the Youla controller by hand computation. However, when selecting the target model for the Youla optimization method, we did ignore these high frequency poles. We will add these high frequency poles to the target model and we will also limit the bandwidth of both Youla methodologies to the H_∞ method so that the actuator effort transient responses for all three strategies would be similar. Then we compare the results of these methods.

Let's assume a target model for the Youla Optimization method as follows,

$$T_t = \left(\frac{\gamma_2}{\gamma_1} \right) \frac{-s + \gamma_1}{s + \gamma_2} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \frac{1}{(\alpha s + 1)^2}$$

The constant γ_1 is the same as before but this time to limit the bandwidth $\gamma_2 = 3$. In addition, the constant $\alpha = 1/50$. The rest of the constant is the same as before. The ratio $\frac{\gamma_2}{\gamma_1}$ forces the closed loop transfer function target to be one at low frequency or dc. The Youla

transfer function for hand computation is selected so that the closed loop complementary transfer function to be precisely the same as this target model.

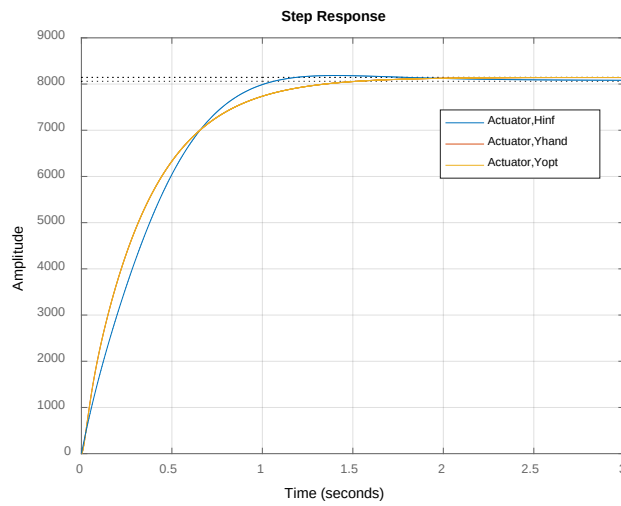


Figure 8.14: Actuator effort response to the locking torque request comparison of all the proposed methods with high frequency poles and limited bandwidth

The actuator effort results are illustrated in Figure above. The results for Youla hand computation and Youla optimization method are the same. This should be no surprise as we tried to force the closed loop complementary transfer functions for both methods to be the same. We also limited the bandwidth so that the actuator effort for the H_∞ optimization would be similar to the Youla methods.

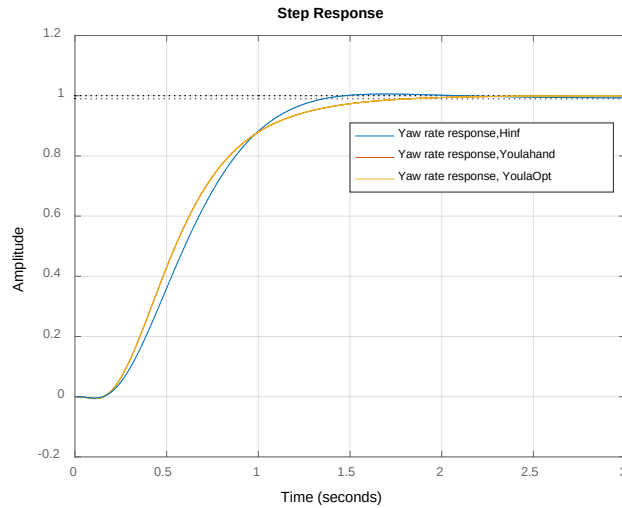


Figure 8.15: Yaw rate response to the locking torque request comparison of all the proposed methods with high frequency poles and limited bandwidth

By observing the Figure above, two things are evident,

- Youla control design using hand computation perform exactly the same as Youla control design using IMC structure with H_∞ optimization.
- By limiting the bandwidth of Youla control design techniques, we can obtain very similar results to H_∞ loop shaping control method.

These control methodologies could be implemented on-board vehicle for yaw stability control with active limited slip differential.

8.2 Vehicle Yaw Rate and Side-Slip Estimation

In this section, we will utilize the development of Chapter 7 in a vehicle dynamics application for yaw rate and side-slip estimation.

In the previous years, there are numerous articles for which many experts have attempted to estimate yaw rate and side-slip, e.g. see ([?]). Particularly, vehicle side-slip estimation has important benefits in Yaw Stability Control (YSC) applications. The standard estimation techniques which have been pursued in the past to estimate these variables were non-linear and unscented Kalman filtering techniques. Although, the authors have shown successful estimation of these variables using these approaches but there is no discussion about robustness characteristics or calibration overhead, two most important implementation requirements, using these standard techniques.

Here, we will utilize Youla Controller Output Observer (YCOO) as discussed in Chapter 7, to estimate vehicle yaw rate and side-slip. The YCOO turns out to have lower calibration overhead with better robustness characteristics in terms of many important vehicle dynamics parameters as will be investigated in this example.

Figure below illustrates a block diagram of COO with the acceleration measurement. The measured lateral acceleration including sensor noise, n , are used as a feedback loop reference target and the output consists of lateral acceleration from state-space representation of the plant model.

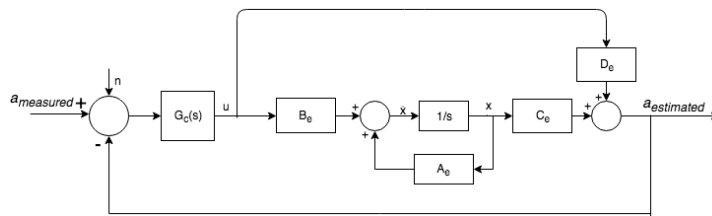


Figure 8.16: Block Diagram of Controller Output Observer (COO) using acceleration measurement

Kalman Filters:

(Linear) Kalman Filtering (LKF) is an estimation technique which utilizes a statistical approach to estimate the mean and co-variance of the states. A plant model and sensor measurements are used to find the best estimation of the states. However, LKF uses linear

state space representation as a model, therefore there are some limitations in the applications where plants are highly nonlinear or sensor measurements are of poor quality. For these applications, other types of Kalman Filters are suitable, such as Extended Kalman Filter (EKF) and Unscented Kalman Filter (UKF). A linearization of the system equations at each time step is required for EKF and the linearized Jacobian matrices are utilized to estimate the variance matrix. This technique is widely used in state, as well as certain parameter estimation. However, since the Jacobian matrices are required to be computed at each time step, the computational overhead can be high especially for systems with large number of states. UKF uses unscented transformation to sample the statistics associated with the plant model and sensor uncertainties to better represent the mean and the covariance at each time step without the need for linearization. Therefore, UKF requires less computational overhead than EKF and it also has wide applications.

In this example, we will make a comparison of SISO YCOO, LKF, EKF, and UKF, mainly in terms of their robustness characteristics, in estimation of vehicle yaw rate and side-slip. However, before designing the estimators, we require a dynamic model of the vehicle and that is discussed next.

Vehicle Model - Nonlinear Bicycle Model with Pacejka Tire Model:

One way to estimate the yaw rate is to use the lateral acceleration measurement at the vehicle center of gravity (c.g.) and using the dynamic relationship between the yaw rate and lateral acceleration. For a front steering vehicle, the system diagram of a simple vehicle dynamic model, using a two degrees of freedom bicycle model, is shown in Figure 16.12. Where U is the forward velocity, which is assumed to be constant. Parameters m and J are vehicle mass and moment of inertia. The distances from the vehicle c.g. to the front and rear axles are respectively shown as a and b . The front and rear tire lateral forces are indicated by F_f and F_r respectively. The front wheel steering angle input is illustrated by δ . The model includes two states, the lateral velocity V , and the yaw rate ω .

With body fixed coordinates, the equations of motion of the bicycle model are written as,

$$\dot{V} = \frac{F_f \cos \delta + F_r}{m} - U\omega$$

$$\dot{\omega} = \frac{F_f \cos \delta \cdot a + F_r \cdot b}{J}$$

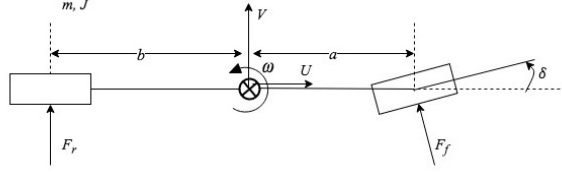


Figure 8.17: Front Steering Bicycle Model System Diagram

The lateral forces exerted on front and rear tires, F_f and F_r , are functions of the slip angles of front and rear tires. The slip angles of front and rear are related to the system input and the states,

$$\alpha_f = \delta - \tan^{-1}\left(\frac{V + a\omega}{U}\right)$$

$$\alpha_r = \tan^{-1}\left(\frac{b\omega - V}{U}\right)$$

The tire slip angles result in generating lateral tire forces and their relationship has been studied, for many decades, for developing an appropriate mathematical tire model. There are many tire models which relate the tire slip angles to the lateral tire forces. One commonly used and popular model is the Pacejka tire model, also known as the “Magic Formula” tire model. From [?], the basic formulas for lateral forces are,

$$F_f = D_f \cdot \sin\left(C \cdot \arctan\left(B_f \cdot \left((1 - E) \cdot \alpha_f + \frac{E}{B_f} \cdot \arctan(B_f \alpha_f)\right)\right)\right)$$

$$F_r = D_r \cdot \sin\left(C \cdot \arctan\left(B_r \cdot \left((1 - E) \cdot \alpha_r + \frac{E}{B_r} \cdot \arctan(B_r \alpha_r)\right)\right)\right)$$

where B , C , D and E are various tire parameters, such as B is the stiffness factor, C is the shape factor, D is the peak factor which is related to the tire normal force (F_z) and tire-road friction coefficient (μ), and E is the curvature factor. These coefficients are extracted from various tire testing and measurement data. We have utilized a typical tire data in this paper.

Given a steering angle input and system state initial conditions, tire slip angles are computed from equations above, where the relationship between slip angles and the system input and the states is defined. Knowing the tire slip angles, lateral tire forces are computed from the Pacejka tire model. Having the lateral tire forces, the equations of motion for lateral velocity and yaw rate are solved to obtain solutions of the new states. This cycle is then repeated at each time step.

The lateral acceleration at c.g. is related to the front and rear tire forces and is given by,

$$a_y = \frac{F_f \cos \delta + F_r}{m}$$

This SISO model then consists of one input, the front road wheel angle, δ , and one output, the vehicle lateral acceleration at the c.g., a_y .

This non-linear model is used to represent the actual vehicle for testing the robustness and performance of the proposed estimators while the linearized version of this model is used for developing YCOO and LKF estimators.

Linearizing the Bicycle Model:

Both YCOO and LKF require a linearized model of the plant. Therefore, we need to linearize the vehicle and tire models to obtain a state space representation of a the plant to be used for YCOO and LKF estimators. For linearizing a bicycle model, all angles were assumed to be small, and the state equations are re-written as

$$\begin{aligned}\dot{V} &= \frac{F_f + F_r}{m} - U\omega \\ \dot{\omega} &= \frac{F_f \cdot a + F_r \cdot b}{J}\end{aligned}$$

For linearizing a tire model, it is assumed that the lateral tire forces are proportional to slip angles with front and rear cornering stiffness coefficients C_f and C_r ,

$$F_f = C_f \alpha_f$$

$$F_r = C_r \alpha_r$$

Then the state-space representation of the equations of motion is written as,

$$\begin{pmatrix} \dot{V} \\ \dot{\omega} \end{pmatrix} = \begin{pmatrix} -\frac{C_f + C_r}{mU} & -(\frac{aC_f - bC_r}{mU} + U) \\ -\frac{aC_f - bC_r}{JU} & -\frac{a^2C_f + b^2C_r}{JU} \end{pmatrix} \begin{pmatrix} V \\ \omega \end{pmatrix} + \begin{pmatrix} C_f \\ \frac{aC_f}{J} \end{pmatrix} \delta$$

The lateral acceleration at c.g a_y can be expressed as,

$$a_y = \begin{pmatrix} \frac{C_f + C_r}{mU} & -\frac{aC_f - bC_r}{mU} \end{pmatrix} \begin{pmatrix} V \\ \omega \end{pmatrix} + \left(\frac{C_f}{m}\right)\delta$$

Uncertainties:

The acceleration sensor measurements in the simulation environment are made from a non-linear model, a representation of an actual vehicle, as stated previously. While, the YCOO and LKF utilize a linear model to estimate the states of this nonlinear plant. Therefore, for these estimators, model uncertainties include the differences between linear and nonlinear plants used for state estimation and acceleration measurements respectively. In addition,

to this inherent uncertainty, we will examine parameter and environmental uncertainties, mass and tire-road coefficient of friction variations, for all estimator approaches.

For simulationg sensor noise, data sheet of a single axis accelerometer SCA830-D05 [?] is utilized. From the data sheet, we can assume that the sensor noise has a zero mean and a variance of $R = 2.5 \cdot 10^{-3}$.

State Estimation

In the state estimation for the SISO case, a front wheel steering vehicle with the front road wheel angle and one accelerometer at center of gravity of the vehicle for acceleration measurement are utilized to estimate the vehicle yaw rate as shown in Figure 16.12. The input, in this case, is the front wheel steering angle (δ), and the output is the lateral acceleration at c.g (a_y).

Youla Parameterization Estimator Design:

The transfer function of the plant is the ratio between lateral acceleration and the road wheel angle,

$$G_p(s) = \frac{a_y}{\delta}(s) \quad (8.7)$$

There is no unstable poles and zeros, based on the Youla parameterization technique in [?], the Youla parameter can be computed by ²,

$$Y = \frac{1}{G_p} \cdot T$$

Where T is the desired closed-loop transfer function, and we select this desired transfer function to be a second order well damped filter with two additional poles at $s = 1/\tau$ for better shaping the Youla parameter at high frequency,

$$T = \frac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)(\tau s + 1)^2}$$

Then the Youla parameter is written as

$$Y = \frac{1}{G_p} \cdot \frac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)(\tau s + 1)^2}$$

The sensitivity transfer-function S of the feedback loop can be derived using the algebraic constraint between this transfer function and the closed-loop transfer function, T ,

$$S = 1 - T$$

²It should be noted if the plant included unstable poles or non-minimum phase zeros, the interpolation conditions could be used to handle these types of plants.

and finally the controller G_c , to be used for state estimation using *YCOO*, can be calculated by,

$$G_c = Y/S$$

With proper choice of ζ and ω_n , the controller, G_c , is designed so that the closed loop transfer function, T , is at 0 dB at low frequency which indicates good target following capability, and the sensitivity transfer function, S , is small at low frequency which indicates good disturbance rejection capability. In selecting a bandwidth for the closed loop transfer function, T , a trade off between the transient response and the overshoot should be made.

Figure 16.13 shows the bode plot of the linear plant used in *YCOO*. The plant has high gain of 40 dB at lower frequency and the gain drops after frequency of 1 rad/s and reach the minimum at 20 dB around 8 rad/s. If the bandwidth of the controller/estimator is greater than 1 rad/sec, the system will have an overshoot due to the change in plant gain.

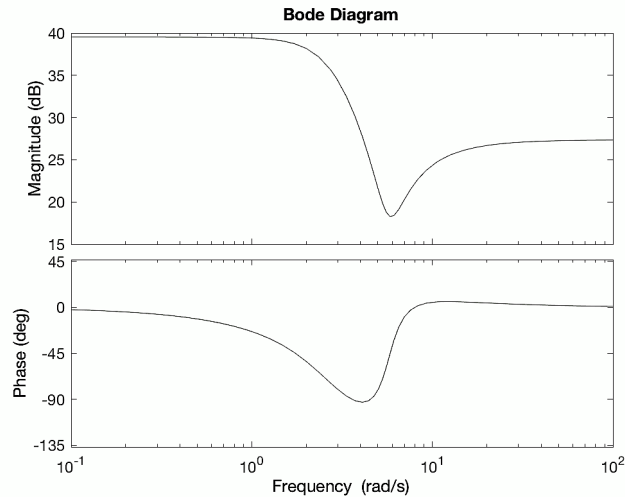


Figure 8.18: Bode plot of linear plant.

As shown in Figure 16.14, T is at 0 dB for a wide range of frequency. For the case of yaw rate estimation, due to designing a closed loop with large bandwidth, the yaw rate transient response will include a large overshoot. This overshoot can only be reduced or eliminated by reducing the closed loop bandwidth. Regarding the bandwidth requirement/constraint, a set of objective requirements for the estimation results should be decided to reduce the subjectivity of our findings. For our comparative analysis in this paper, we will use the results of the *YCOO* with the highest bandwidth.

5625 *It is important to point out that the bandwidth constraint in both control and estimation*
 5626 *design has a significant importance. This constraint is directly addressed in the YCOO*
 5627 *technique while in all Kalman filtering strategies, the bandwidth is dealt with as an af-*
 5628 *terthought.*

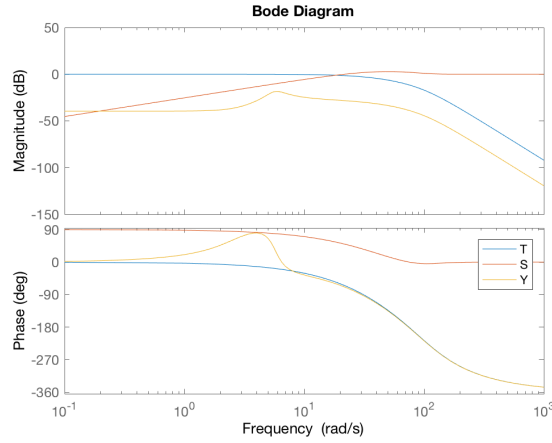


Figure 8.19: Bode plot for SISO case with large bandwidth.

5629 Another way to look at the design is to look at the bode plot of the closed loop transfer
 5630 function of estimating the yaw rate T_ω . The transfer function of the yaw rate ω to the wheel
 5631 angle input δ is

$$G_{p_\omega} = \frac{\omega}{\delta}(s) = \frac{\frac{aC_f}{J}s + \frac{(a+b)C_fC_r}{mJU}}{s^2 + \left(\frac{C_f+C_r}{mU} + \frac{a^2C_f+b^2C_r}{JU}\right)s + \frac{(a+b)^2C_fC_r}{mJU^2} + \frac{bC_r-aC_f}{J}}$$

5632 With the controller transfer function designed G_c , we can obtain the complementary
 5633 closed loop transfer function as,

$$T_\omega = \frac{G_c G_{p_\omega}}{1 + G_c G_{p_\omega}}$$

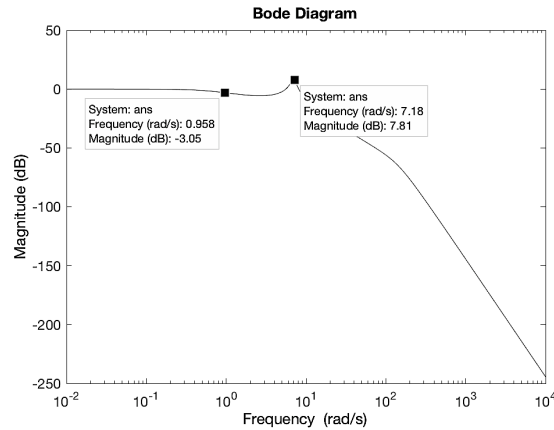


Figure 8.20: Bode plot of closed loop transfer function of yaw rate estimation.

Looking at the bode plot of T_ω (Figure 16.15), the gain of T_ω is at 0 dB at low frequency, which means a time response to a step input with zero steady-state error. However, around 7 rad/s the closed loop transfer function has a peak magnitude of around 8 dB, which means a transient response with a large overshoot. If we decrease the bandwidth for T to 1 rad/s or even smaller, the yaw rate estimation will follow the target very well with no overshoot, but it will drastically slow down the estimation response time.

Simulation Results:

Figure 16.16 and 16.17 show a comparison of the lateral acceleration at the c.g. and the yaw rate estimations using YCOO, LKF, EKF, and UKF with 2° road wheel angle step input. For both the lateral acceleration and the yaw rate estimates, all these strategies perform well in both sensor noise rejection and target following. Figures 16.18 and 16.19 illustrates the results of the yaw rate estimations with 6° and 10° road wheel angle step input. As we increase the wheel angle input, we can observe that LKF will have a steady state error as the vehicle model (tire forces) starts operating in the nonlinear region. It should be noted that the lateral tire forces are linear at small tire slip angles and as the slip angle increases, the tire forces begin to operate in the nonlinear region and eventually at the very high tire slip angles, the lateral tire forces will saturate. As the tire slip angles increase, purely linear tire models will provide larger lateral forces than their nonlinear counterpart, and as a result, LKF overestimates the yaw rate. It is interesting to observe that in the case of YCOO, although this strategy utilizes a linear tire model, its robustness characteristics

5654 results in excellent steady state target following. The robustness characteristics of YCOO
 5655 will be further investigated in the next section. However, as discussed in the last section,
 5656 the closed loop bandwidth of YCOO is selected in such a way that its transient response
 5657 due to the overshoot is not as good as the other proposed methods. The EKF and UKF
 5658 perform well in both these cases as they are utilizing the same nonlinear vehicle model
 5659 used for the reference generator (generating the lateral acceleration reference) as for the
 5660 estimation. We will investigate the performance of the EKF and UKF in the robustness test
 5661 of the next section when the model used for estimation differs from its nominal values.

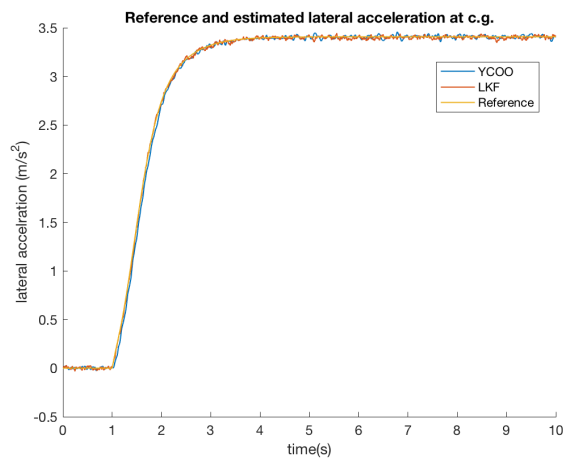


Figure 8.21: Lateral acceleration estimation using different strategies with 2° road wheel angle step input

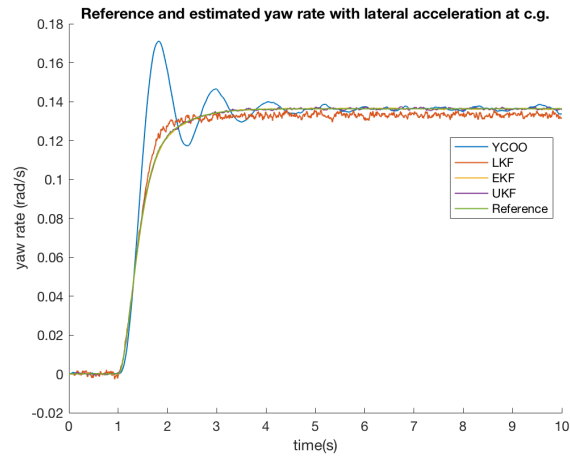


Figure 8.22: Yaw rate estimation using different strategies with 2° road wheel angle step input

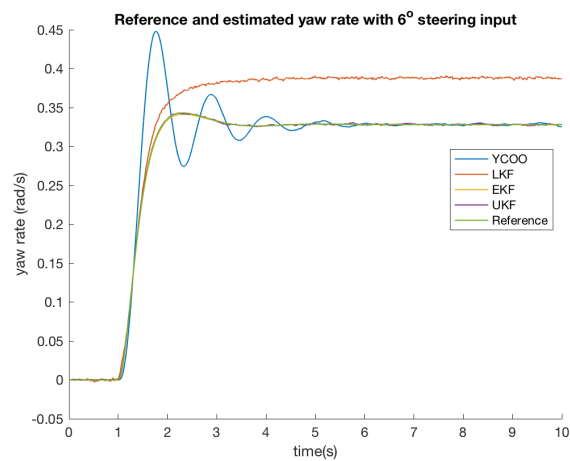


Figure 8.23: Yaw rate estimation using different strategies with 6° road wheel angle step input

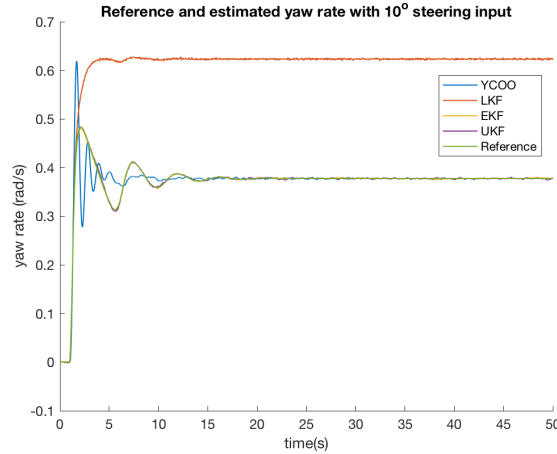


Figure 8.24: Yaw rate estimation using different strategies with 10° road wheel angle step input

Robustness Test

Vehicle Mass variation: Figure 16.20 shows the yaw rate estimation with different strategies with an addition of 30% change in the vehicle mass and a 2° road wheel angle step input. It illustrates that all Kalman filter approaches are sensitive to the vehicle mass variation and do not have good robustness characteristics with respect to this parameter. These results suggest that if the Kalman Filters are tuned to have good sensor noise rejection capabilities and hence, more confidence is given to the model and if the sensitivity of the filters are high with respect to the model parameter changes, then this will negatively impact the estimation performance. But, it is interesting to observe that the YCOO performance has not been impacted with respect to this parameter variation. It could be argued that the plant parameter variations occur in the low frequency region where the YCOO is designed to have a closed loop sensitivity transfer function with a small magnitude, hence, resulting in a good robustness characteristic.

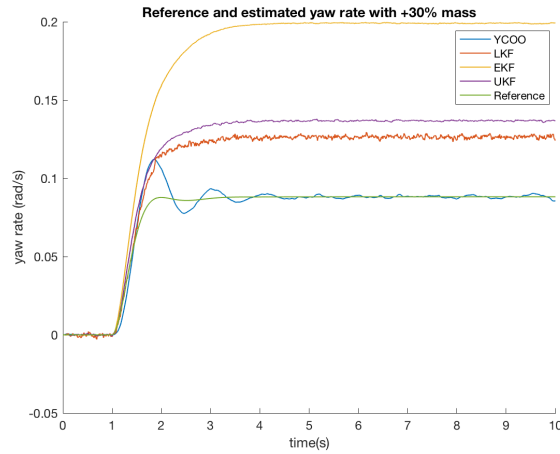


Figure 8.25: Yaw rate estimation using different strategies with +30% mass

5675 **Tire-road coefficient of friction:** The yaw rate estimations with a reduced tire-road co-
 5676 efficient of friction $\mu = 0.5$ (the coefficient for dry asphalt is equal to $0.9 - 1$) and a 2°
 5677 road wheel angle step input are shown in Figure 16.21. The results indicate that all strate-
 5678 gies are not very sensitive to the changes in μ as they are to the vehicle mass variations.
 5679 Comparatively, EKF is more sensitive to μ than the other strategies.

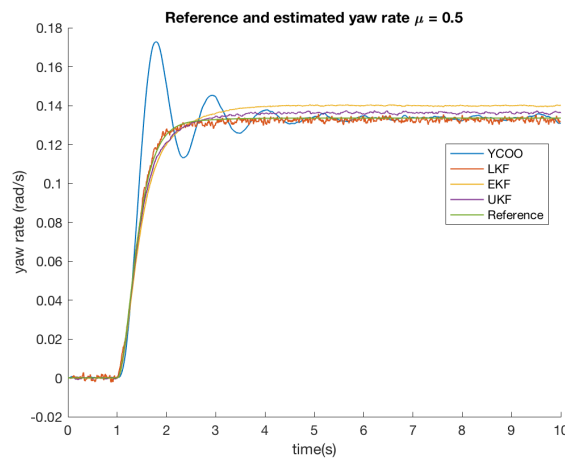


Figure 8.26: Yaw rate estimation using different strategies with $\mu = 0.5$

5680 The estimation results presented above suggest that the YCOO is an estimation method
5681 with both good performance and robustness characteristics. But, most importantly, it re-
5682 quires very little to no tuning when compared to the standard estimation techniques such
5683 as Kalman filtering.

5684 It is worth mentioning that all the estimation approaches did show sensitivity to one
5685 important variable, the vehicle longitudinal velocity. Our future plan is to utilize Rein-
5686 forcement Learning techniques with the YCOO to address this deficiency.

CONTROL DESIGN USING YOU LA PARAMETRIZATION MULTI INPUT MULTI OUTPUT (MIMO)

CHAPTER 9

INTRODUCTION TO MULTIVARIABLE FEEDBACK CONTROL

9.1 An Introduction to Multivariable Feedback Control

The multivariable control design becomes sensible when there are more than one input and output exist in a dynamic system and there is relatively a strong coupling between these inputs and outputs.

Control system design, in the late 1800s, was presented in the form of differential equations [1]. Then, in 1940s, through the work of Bode and Nyquist, the frequency domain approach for feedback control design was introduced. In 1960s, with the advancement of the space program and through the work of Kalman and others, the benefits of differential equations through state-space representation of dynamic system was introduced [2]. Interestingly enough, once again in the late 1970s and early 80s through the work of Zames, Horowitz, Doyle and others, the importance of uncertainty quantification in control design was demonstrated through the frequency domain approach and this approach once again found its way into the main stream control design community [3].

The multivariable or MIMO feedback control design include a vast area of many different techniques. The original work for multivariable control design using frequency domain was pioneered by Ronsenbrock with the introduction of Nyquist-array method [[]]. Multivariable control design methodologies in frequency domain include methods such as sequential loop closures, the characteristic-locus, reversed-frame normalization, Nyquist-array, and quantitative feedback theory [[]]. However, these techniques rely more on hand computation and engineering insights than deriving MIMO controllers directly using numerical optimization approaches. Furthermore, when it comes to MIMO control design, these aforementioned techniques could deviate from optimal design and the multivariable design part is an extension of these techniques.

Optimal control techniques such as LQR (Linear Quadratic Regulators) and LQG (Linear Quadratic Gaussian) under the modern control domain which were introduced in the late 50's and early 60's are inherently multivariable strategies [[]]. These optimal control techniques are derived in the time domain. The equivalency between time and frequency domain is well understood. But, it begs the question that if optimal controllers are designed to not only stabilize the system but have a certain bandwidth, or satisfy any one of the desirable constraints associated with classical control (e.g. disturbance rejection, target following, robustness), and result in the best possible system of a particular type - hence, the word optimal, then why is it necessary to discuss and design robust control techniques?

In the rest of this section, we discuss non-optimal and optimal control strategies, such as LQR and LQG, and robust control and attempt to answer the above question. In the next and final section of this chapter, we introduce Schur complement and provide an interpretation of poles and zeros. We end this chapter by introducing transfer function matrices.

9.2 Non-Optimal, Optimal and Robust Control

In this section, we briefly review our choice of models for the model-based control development. We then review MIMO non-optimal methods and in the last part of this section, we discuss optimal controllers such as LQR, LQG and robust control including the fundamental reasons for designing controllers using robust control approach.

Given a MIMO complex system (G) as shown in Figure 9.1, and let's assume this system is represented by nonlinear partial differential equations, we ask the following question, what are the choices to develop model based controllers for this complex system ?

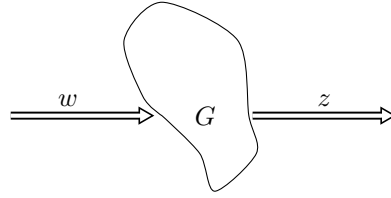


Figure 9.1: General Input-Output System

We can globally divide these choices into two methods:

- Direct: control of the complex system using nonlinear PDEs. This is doable but suffers from robustness issues and complexity.
- Indirect: if the linearization of this complex system would be possible and PDE's could be transformed to ordinary differential equations, then it is possible to write this model as Linear Time Invariant (LTI) ODEs, and a controller could be designed using this simplified model as shown in Figure 9.2.

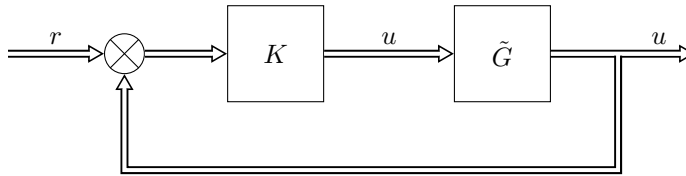


Figure 9.2: LTI Input-Output System

It is shown [] that the trade-off for this simplification is the addition of uncertainty. Therefore, there is always a trade-off between simplicity and uncertainty. We will discuss uncertainties in Robust Control part of this section.

With respect to MIMO control design methods, these can be divided to two distinct approaches, Non-Optimal and Optimal methods:

9.2.1 Non-Optimal Control Methods

Non-Optimal MIMO control strategies have been thoroughly discussed in the following references [] and []. Below, we merely list these methods.

- Sequential Loop Closures
- Nyquist Array
- Characteristic Locus Method
- Quantitative Feedback Theory (QFT)

9.2.2 Optimal Control Methods

Classical control system design is generally a trial-and-error process in which various methods of analysis are used iteratively to determine the design parameters to satisfy the control requirements.

The objective of optimal control theory is to minimize/maximize a performance criterion while simultaneously satisfying the constraints (e.g. state and control signals physical limits) provided in the performance criterion. In a sense, optimal control replaces engineering judgment and a trial-and-error process with an organized mathematical process for solving a control system design problem.

Fundamentally, there are two methods of accomplishing the minimization of a given performance criterion, “the minimum Pontryagin principle” and “the Bellman dynamic programming”. All optimal control laws start with either of these two methods. Interestingly, the minimum Pontryagin principle could be derived from calculus of variations or the dynamic programming []. The performance objective of this optimal control law could be posed as a quadratic cost function for a linear dynamic system and this is known as “Linear Quadratic Regulator” or “LQR”. The infinite horizon solution (as time approaches infinity) of this deterministic problem results in a full state feedback controller with constant gains. In addition, LQR turns out to have very good classical phase (60°) and gain ($\geq 6db$) margins. As intuitive the development of the LQR is, it turns out that this method at times is neither practical nor accounts for model uncertainties and sensor noise (two very practical neoclassical disturbances for addressing robustness). In addition, the LQR requires having access to the entire system states and at times this is not possible.

One of the most popular and practical version of these optimal control methods is “Linear Quadratic Gaussian” [1]. In a real feedback system, there are process and sensor noises, this was addressed by Kalman in the 1960s [2]. Two additional inputs were added to the system resulting in Figure 9.2.

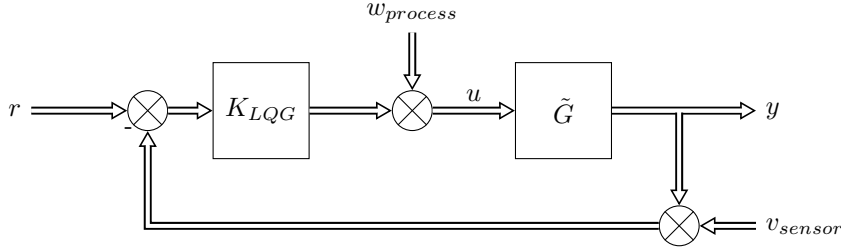


Figure 9.3: Full LTI Input-Output System

The LQG assumes that $w_{process}$ and v_{sensor} are statistically independent white Gaussian noises. The idea of LQG optimal control is illustrated in Figure 9.4.

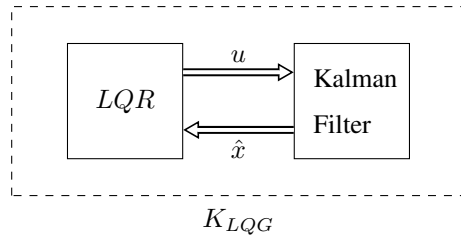


Figure 9.4: LTI Input-Output System

We stated that the LQR is robust in a classical sense (60° phase margin and at least 6 db-gain margin) and Kalman filter is also robust in a similar classical fashion as the LQR. The LQG optimal controller utilizes the LQR and Kalman filtering concepts as illustrated in both Figures 9.3 and 9.4 and the solution to the LQG problem is determined by what is known as **separation principle**, where unlike the LQR, the dynamic system states are not directly measured, they are estimated using Kalman filtering. Then the estimated states are utilized as the true states to synthesize a state feedback controller similar to the LQR technique. The LQG is a dynamic controller which could be posed as an output feedback

as illustrated in Figure 9.3.

We have stated that both the LQR and Kalman filter have excellent classical robustness margins. However, it is surprising that as proposed by Doyle [1] through a simple example, that LQG design can display poor stability margins. Fortunately, the stability margin could be recovered through an ad hoc method called **Loop Transfer Recovery** where this approach is discussed in many papers and books, such as [2]. Simply put, this methodology indirectly re-shapes sensitivity transfer function to recover some of the lost stability margins, but one important issue which either ignored or not mentioned in this development is the compromise between performance and stability. Interestingly enough, this compromise was directly addressed when utilizing frequency domain loop-shaping either through Youla parameterization technique or H_2 and H_∞ optimal robust control techniques as discussed in the first volume of this book.

Deriving an equivalency between the LQG (time domain) and H_2 (Freq. Domain)[3], and use of H_∞ norm and its role in shaping sensitivity, introduced by G. Zames in [4], paved the way for understanding how these norms play a significant role in control system design. It became evident that the lack of stability margin in the LQG could be directly addressed by minimizing H_2 cost function and hence, directly shaping sensitivity and complementary sensitivity transfer functions. In addition, it was shown H_∞ norm, which again was introduced by G. Zames in [5], could be effectively utilized for plant uncertainty modeling. An era in robust control was started.

9.2.3 Optimal Robust Control

In this volume, we concentrate on designing MIMO Optimal Robust controllers. These controllers will be dynamic and constant gains.

Similar to the first volume, we start our development by introducing necessary mathematical tools so we would be able to uniquely design

- Robust non-optimal MIMO controllers using Youla parameterization (hand computation)

We will then introduce H_2 and H_∞ norms for MIMO systems so that we would be able to design

- Optimal robust MIMO controllers by minimizing cost functions related to the aforementioned norms (using control optimization toolboxes)

One of the most important aspects of the feedback control design is how to manage uncertainties. We have discussed uncertainties in the first volume of this book but in this volume, we will extend these ideas for MIMO control synthesis.

Before ending this section, it would be helpful to say a few words about the benefits of norm optimization especially, H_∞ norm.

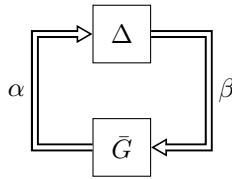


Figure 9.5: General Plant Configuration with Structured Uncertainty

As illustrated in Figure 9.5, the power of H_∞ norm optimization, comes from the following:

- Modeling uncertainties and revealing the trade off between dealing with complexity (e.g. highly complex plant models) and uncertainty
- Dealing with uncertainties (e.g. disturbance signals) in deterministic manner by assuming that disturbance signals are energy bounded

Our concentration in this book is to introduce a unique method to design MIMO controllers using Youla parameterization. Synthesizing controllers using norm optimization is included only for comparative basis.

Let's start by introducing the necessary mathematical tools first.

In this section, we start with reviewing deriving SISO transfer function from its state-space matrices. We will introduce Schur complement and Rosenbrock system matrix and their application in deriving transfer function and transfer function matrices from state-

space matrices. Then we provide an interpretation of poles, zeros, and transmission blocking zeros of transfer function. We will end this section by introducing transfer function matrices including some of its basic aspects.

9.3 Review of the SISO Transfer Function

We derived the SISO transfer function from its state-space matrices in the first volume of this book. To refresh our memory, this derivation is provided below. Given A , B , C and D matrices as defined previously, the first order LTI (Linear Time Invariant) ODEs (ordinary differential equations) describing the system dynamics are written as,

$$\dot{x} = Ax + Bu \quad (9.1)$$

$$y = C^T x + Du \quad (9.2)$$

where x are the system states. Transfer function from the input, u to the output, y is derived assuming zero state input (all I.C.s (Initial Conditions) are set to zero) using Laplace transform,

$$sX \equiv sIX = AX + DU \quad (9.3)$$

$$(sI - A)X = BU \quad (9.4)$$

$$X = (sI - A)^{-1}BU \quad (9.5)$$

$$(9.6)$$

$$(9.7)$$

where s is the Laplace operator, and X and U are functions of s in Laplace domain. Then the output is written as a function of the input,

$$Y = C^T X + DU \quad (9.8)$$

$$Y = (C^T(sI - A)^{-1}B + D)U \quad (9.9)$$

where

$$(sI - A)^{-1} = \frac{\text{adj}(sI - A)}{\det(sI - A)} \quad (9.10)$$

and finally, the transfer function describing the input/output relationship is written as,

$$G(s) = \frac{Y}{U} = \frac{1}{\det(sI - A)} (C^T \text{adj}(sI - A)B + \det(sI - A)D) \quad (9.11)$$

where "zeros" are the zeros of the numerator polynomial and the "poles" are the zeros of the denominator.

9.3.1 Schur Complement

In this section, we introduce Schur Complement and Rosenbrock's system matrix in computing system transfer function. Let's start by introducing the matrix Z with the given block dimensions,

$$Z = \left[\begin{array}{c|c} \underbrace{M}_{m \times m} & \underbrace{E}_{m \times n} \\ \hline \underbrace{H}_{n \times m} & \underbrace{N}_{n \times n} \end{array} \right] \quad (9.12)$$

Now, let's define,

$$U = M - EN^{-1}H \quad (9.13)$$

$$W = N - HM^{-1}E \quad (9.14)$$

The matrix U is defined to be the Schur Complement of N in Z and the matrix W is defined to be the Schur Complement of M in Z . The Schur Complement has many applications and properties in matrix manipulations and optimization [[]].

One of the properties of the Schur Complement is as follows,

$$\det(N) \det(U) = \det \left[\begin{array}{c|c} \underbrace{M}_{m \times m} & \underbrace{E}_{m \times n} \\ \hline \underbrace{H}_{n \times m} & \underbrace{N}_{n \times n} \end{array} \right] = \det(M) \det(W) \quad (9.15)$$

Proof: Let's assume M is invertible, then

$$\begin{bmatrix} M & E \\ H & N \end{bmatrix} = \begin{bmatrix} I_m & 0 \\ HM^{-1} & I_n \end{bmatrix} \begin{bmatrix} M & 0 \\ 0 & N - HM^{-1}E \end{bmatrix} \begin{bmatrix} I_m & M^{-1}E \\ 0 & I_n \end{bmatrix} \quad (9.16)$$

Therefore

$$\det \begin{bmatrix} M & E \\ H & N \end{bmatrix} = \det M \det(N - HM^{-1}E) = \det M \det W \quad (9.17)$$

■

Rosenbrock's System matrix is given by,

$$P(s) = \left[\begin{array}{c|c} \underbrace{sI - A}_{n \times n} & \underbrace{-B}_{n \times 1} \\ \hline \underbrace{C^T}_{1 \times n} & \underbrace{D}_{1 \times 1} \end{array} \right] \quad (9.18)$$

If we define the block dimension in the Rosenbrock's matrix as shown above (such as for SISO system) then we can write,

$$W = D - C^T(sI - A)^{-1}(-B) = G(s) \quad (9.19)$$

Using the property of the Schur Complement given above, the transfer function $G(s)$ can be written as,

$$G(s) = \frac{\det P(s)}{\det(sI - A)} \quad (9.20)$$

Most students and control practitioners are familiar with the concept of poles and zeros of a transfer function. But, what might be an unfamiliar concept is the difference between a zero (what is also known as transmission zero) and transmission blocking zero. Therefore, it would be helpful to review these concepts.

9.3.2 Interpretation of poles and zeros of a transfer function

Poles An interpretation of a pole of a transfer function is acquired by the zero input or what is known as unforced response. The state-space equation is then written as,

$$\dot{x} = Ax, \quad x(0) = x_0 \quad (9.21)$$

where the solution to this differential equation is

$$x(t) = x_0 e^{\lambda t} \quad (9.22)$$

and λ could be a real or complex number and it is called the eigenvalue. Inserting $x(t)$ into Equation 9.21 yields

$$\lambda x_0 e^{\lambda t} = A x_0 e^{\lambda t} \quad (9.23)$$

Dividing out similar terms and moving everything to the left side yields

$$[\lambda I - A]x_0 = 0 \quad (9.24)$$

Knowing that the initial condition, x_0 , is not equal to 0, then,

$$\det[\lambda I - A] = 0 \quad (9.25)$$

This equation allows you to solve for the eigenvalue, λ , which is equal to a pole of the transfer function or equivalently the roots of the characteristic equation as discussed in the example below.

EXAMPLE 9.1

Take a simple spring damper system in two different configurations as shown in Figures below.

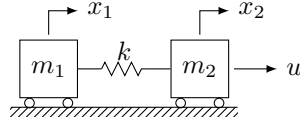


Figure 9.6: Example 1 - System Configuration 1

Controlling the position x_2 with the control force u in the first configuration as shown in Figure above results in the following transfer function,

$$\frac{X_2}{U} = \frac{\frac{1}{m_2}(s^2 + \frac{k}{m_1})}{s^2(s^2 + (\frac{k}{m_1} + \frac{k}{m_2}))} \quad (9.26)$$

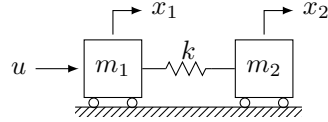


Figure 9.7: Example 1 - System Configuration 2

Controlling the position x_2 by moving the control force u in the second configuration as shown in Figure above results in the following transfer function,

$$\frac{X_2}{U} = \frac{\frac{k}{m_1 m_2}}{s^2(s^2 + (\frac{k}{m_1} + \frac{k}{m_2}))} \quad (9.27)$$

This example illustrates that the location of the actuator input, and in the same manner, sensor location, does not impact the location of the eigenvalues or the roots of the characteristic equation. In both of these configurations, there are four eigenvalues, two are located at zero and two are complex conjugates and are located on the $j\omega$ -axis. However, as briefly discussed in the first volume, the location of the actuator input, or sensor location, impacts the location of the zeros of the transfer function as can be seen in these two configurations.

9.3.3 Zeros (Transmission Zeros, TZ) vs. Transmission Blocking Zeros (TBZ)

An interpretation of a zero of a transfer function is acquired by the zero state or what is known as the forced response. Let's select an input to the a dynamic system to be,

$$u(t) = u_0 e^{\mu t} \quad (9.28)$$

nonzero input. Then, we ask the following question, is it possible to have the output of this system precisely cancels this input?

The answer is yes, if the system transfer function has a zero at location equal to μ . This is simply called a zero of the system transfer function.

However, a zero of a transfer function is different than a transmission blocking zero. We say a zero have a transmission blocking property when assuming the same input as before, there is a specific initial condition associated with the zero input response,

$$x(t) = x_0 e^{\mu t} \quad \text{where} \quad x(0) = x_0 \quad (9.29)$$

so that the addition of the zero state response (forced response) and the zero input response (unforced or natural response) has precisely the following output response,

$$y(t) \equiv 0 \quad (9.30)$$

Given that the LTI ODE is,

$$\dot{x} = Ax + Bu \quad (9.31)$$

The total response becomes

$$\mu x_0 e^{\mu t} = Ax_0 e^{\mu t} + Bu_0 e^{\mu t} \quad (9.32)$$

Knowing that

$$y = C^T x + Du \quad (9.33)$$

Therefore,

$$0 \equiv C^T x_0 e^{\mu t} + Du_0 e^{\mu t} \quad (9.34)$$

From Equation 9.32, we derive

$$(\mu I - A)x_0 = Bu_0 \quad (9.35)$$

Dividing Equation 9.34 by $e^{\mu t}$, we can write

$$0 \equiv C^T x_0 + Du_0 \quad (9.36)$$

5932 The last two equations can be written in the matrix form as

$$\begin{bmatrix} \mu I - A & -B \\ C^T & D \end{bmatrix} \begin{bmatrix} x_0 \\ u_0 \end{bmatrix} = 0 \quad (9.37)$$

5933 which when x_0 and u_0 are not equal to zero then,

$$\det \begin{bmatrix} \mu I - A & -B \\ C^T & D \end{bmatrix} = 0 \quad (9.38)$$

5934 from the Schur complement, the determinant of the matrix above can be written as,

$$\det \begin{bmatrix} \mu I - A & -B \\ C^T & D \end{bmatrix} = \det(\mu I - A) \det(C^T (\mu I - A)^{-1} B + D) = \det(\mu I - A) \det(G(\mu)) \quad (9.39)$$

5935 If this determinant is precisely equal to zero, then a zero has transmission blocking prop-
5936 erty.

5937 An example below will help in shedding some light into this finding.

5938 ■ EXAMPLE 9.2

5939 Let

$$\hat{G} = \frac{s + 2}{s^2 + 7s + 12} \quad (9.40)$$

In controllable canonical form, the system takes the form

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 12 & -7 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \quad (9.41)$$

$$y = \begin{bmatrix} 2 & 1 \end{bmatrix} x \quad (9.42)$$

5940 *Note:* that $D = 0$ in this example and therefore the system is strictly proper. If the
5941 system were just proper, i.e. $O(Num) = O(Den)$, use long division to get the re-
5942 mainder which is used to obtain D for the canonical form.

5943

5944 The Rosenbrock system matrix (R) is used to define the response at a transmission
5945 zero (μ).

$$\begin{bmatrix} \mu I - A & -B \\ C^T & D \end{bmatrix} \begin{bmatrix} x_0 \\ u_0 \end{bmatrix} = 0 \quad (9.43)$$

5946 Because we are assuming that x_0 and u_0 are non-zero, this leaves the condition

$$\det \begin{bmatrix} \mu I - A & -B \\ C^T & D \end{bmatrix} = 0 \quad (9.44)$$

This implies that

$$\det(\mu I - A) \det(C^T(\mu I - A)^{-1}B) = 0 \quad (9.45)$$

$$\det(\mu I - A) \det(\hat{G}) = 0 \quad (9.46)$$

$$\det(\mu I - A) \frac{\mu + 2}{\mu^2 + 7\mu + 12} = 0 \quad (9.47)$$

$$\mu + 2 = 0 \quad (9.48)$$

Which implies that $\mu = -2$.

Obviously this is not very interesting because we could tell this from $\hat{G}$. However, further information can be gained from this. Using Equation (9.43) the following relationship exists

$$\begin{bmatrix} \mu & -1 & 0 \\ 12 & \mu + 7 & -1 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_{01} \\ x_{02} \\ u_0 \end{bmatrix} = 0 \quad (9.49)$$

Letting $\mu = -2$

$$\begin{bmatrix} -2 & -1 & 0 \\ 12 & 5 & -1 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_{01} \\ x_{02} \\ u_0 \end{bmatrix} = 0 \quad (9.50)$$

Which results in the following

$$2x_{01} + x_{02} = 0 \quad (9.51)$$

$$12x_{01} + 5x_{02} - u_0 = 0 \quad (9.52)$$

This results in the null space of the matrix in Equation (9.50) which is

$$\mathcal{N}(R) = k \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \quad (9.53)$$

where $k \in \mathbb{R}$.

Recall that for a linear system, the total response is a sum of two different parts

$$\hat{Y} = \hat{Y}_1 + \hat{Y}_2 \quad (9.54)$$

where $\hat{Y}_1$ is the forced or zero-state response of the system and $\hat{Y}_2$ is the initial condition or zero-input response of the system. $\hat{Y}_1$ and $\hat{Y}_2$ can be found with the following equations

$$\hat{Y}_1 = C^T(sI - A)^{-1}B\hat{U} \quad (9.55)$$

$$\hat{Y}_2 = C^T(sI - A)^{-1}x_0 \quad (9.56)$$

$$(9.57)$$

5957 Taking the state-space system from equations (9.41)-(9.42) and $u(t) = u_0 e^{\mu t} = 2e^{-2t}$,
 5958 $\hat{U} = \frac{2}{s+2}$ and $\hat{Y}_1$ becomes

$$\hat{Y}_1 = \frac{s+2}{s^2+7s+12} \cdot \frac{2}{s+2} = \frac{2}{s^2+7s+12} \quad (9.58)$$

5959 now taking $x_0 = \begin{bmatrix} 1 & -2 \end{bmatrix}^T$ results in $\hat{Y}_2$ being

$$\hat{Y}_2 = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} s & -1 \\ 2 & s+7 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \frac{-2}{s^2+7s+12} \quad (9.59)$$

5960 Therefore $\hat{Y}_1 + \hat{Y}_2 = 0$ for the input $u(t) = u_0 e^{-2t}$ with x_{01} , x_{02} , and u_0 defined using
 5961 Equation (9.53). This is a transmission blocking zero that will result in no output for the
 5962 system. Note that if only the transmission zero is used, the only thing that happens is the
 5963 zeros are canceled in the transfer function.

5964 9.4 Basic Aspects of Transfer Matrices

5965 Define “ r ” as the sensor dimension, “ m ” as the actuator dimension, and “ n ” as the state
 5966 dimension. This means that $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, $C \in \mathbb{R}^{r \times n}$, and $D \in \mathbb{R}^{r \times m}$.

5967

5968 Recall, to obtain the transfer functions of the system, set all of the initial conditions to
 5969 zero and use the following equation

$$\hat{Y} = [C^T(sI - A)^{-1}B + D]\hat{U} \quad (9.60)$$

5970 Letting $C^T(sI - A)^{-1}B + D = \hat{G}$, $\hat{G}$ is now a rational transfer function matrix and with
 5971 dimension $r \times m$. Further, $\hat{G}$ is either strictly proper and

$$\lim_{s \rightarrow \infty} \hat{G} = 0_{r \times m} \quad (9.61)$$

5972 or $\hat{G}$ is proper and

$$\lim_{s \rightarrow \infty} \hat{G} = K_{r \times m} \quad (9.62)$$

5973 where $k_{ij} \in \mathbb{R}$ and $k_{ij} < \infty$.

CHAPTER 10

MATRIX FRACTIONAL DESCRIPTION

10.1 Transfer Function Matrices

In this section, we discuss the matrix fraction description (MFD). Furthermore, we introduce the polynomial matrix properties and their relation to the MFDs.

Recall a MIMO dynamic system with n states, m actuators, and r sensors.

$$\dot{\underline{x}} = \underset{(n \times n)}{A} \underline{x} + \underset{(n \times m)}{B} \underline{u}$$

$$\underline{y} = \underset{(r \times n)}{C} \underline{x} + \underset{(r \times m)}{D} \underline{u}$$

As stated previously, we can follow the same process as for a SISO system to obtain the transfer functions.

$$s\underline{X} = sI\underline{X} = A\underline{X} + B\underline{U}$$

$$\underline{X} = (sI - A)^{-1} B\underline{U}$$

$$\underline{Y} = C^T \underline{X} + D\underline{U}$$

$$\underline{Y} = \left(C^T (sI - A)^{-1} B + D \right) \underline{U}$$

$$G(s) = C^T (sI - A)^{-1} B + D$$

$$G(s) = \frac{C^T \text{adj} (sI - A) B + D \det(sI - A)}{\det(sI - A)}$$

However, in this formulation, $G(s)$ is now a rational matrix (i.e., a matrix of rational functions).

$$\begin{matrix} \underline{Y} \\ (r \times 1) \end{matrix} = \begin{matrix} G(s) \\ (r \times m) \end{matrix} \begin{matrix} \underline{U} \\ (m \times 1) \end{matrix}$$

$G(s)$ is either strictly proper

$$\lim_{s \rightarrow \infty} G(s) = 0$$

or proper

$$\lim_{s \rightarrow \infty} G(s) < \infty$$

10.2 Matrix Fraction Description

Every rational matrix can be written as a ratio of two polynomial matrices, N_R and D_R or N_L and D_L . So, we prefer to write

$$G(s) = N_R(s) D_R^{-1}(s)$$

This is known as a matrix fraction description (MFD). $N_R(s)$ is a right numerator polynomial matrix such that

$$N_R(s) = (C^T \text{adj} (sI - A) B + D \det(sI - A))$$

and D_R is a right denominator polynomial matrix such that

$$D_R(s) = \det(sI - A) I_m$$

We could just as well write

$$G(s) = D_L^{-1}(s) N_L(s)$$

such that N_L is a left numerator polynomial matrix and D_L is a left denominator polynomial matrix. In this case,

$$N_L = N_R, \quad D_R(s) = \det(sI - A) I_r$$

6001 Although the left and right numerator matrices are equal, the left and right denominator
 6002 matrices are a different size and are therefore *not* equal.

$$\begin{matrix} D_L^{-1} & N_L \\ (r \times r) & (r \times m) \end{matrix} = \begin{matrix} N_R & D_R^{-1} \\ (r \times m) & (m \times m) \end{matrix}$$

MFD's are never unique. Suppose F is any $m \times m$ polynomial matrix. Then,

$$G(s) = N_R D_R^{-1} \quad (10.1)$$

$$= N_R F F^{-1} D_R^{-1} \quad (10.2)$$

$$= (N_R F) (D_R F)^{-1} \quad \text{A new MFD!} \quad (10.3)$$

6003

6004

6005 ■ EXAMPLE 10.1

6006 Consider a left MFD as given below,

$$G(s) = D_L^{-1}(s) N_L(s),$$

6007 where now the denominator polynomial matrix enters $G(s)$ from the left.

6008 Consider the RLC circuit shown below.

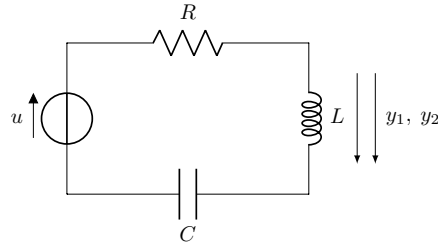


Figure 10.1: RLC circuit.

6009 The state equations are given by

$$\begin{bmatrix} \dot{y}_1 \\ \dot{y}_2 \end{bmatrix} = \begin{bmatrix} -\frac{R}{L} & -\frac{1}{C} \\ \frac{1}{L} & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u,$$

6010 where $\dot{y}_1$ is the inductor voltage and $\dot{y}_2$ is the current. y_1 and y_2 are both state variables

6011 and the outputs.

$$\begin{bmatrix} s + \frac{R}{L} & \frac{1}{C} \\ -\frac{1}{L} & s \end{bmatrix} \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} U$$

6012 Assuming $\dot{y}_1$ and $\dot{y}_2$ are the outputs,

$$\begin{bmatrix} s + \frac{R}{L} & \frac{1}{C} \\ -\frac{1}{L} & s \end{bmatrix} \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = \begin{bmatrix} s \\ 0 \end{bmatrix} U$$

6013 We can then write the above relationship in matrix fraction description as,

$$\begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = \begin{bmatrix} s + \frac{R}{L} & \frac{1}{C} \\ -\frac{1}{L} & s \end{bmatrix}^{-1} \begin{bmatrix} s \\ 0 \end{bmatrix}$$

$$D_L = \det(sI - A)I_r = (s^2 + \frac{R}{L}s + \frac{1}{CL}) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

6014

$$N_L = \text{adj}(sI - A) \begin{bmatrix} s \\ 0 \end{bmatrix} = \begin{bmatrix} s^2 \\ s \end{bmatrix}$$

10.3 Polynomial Matrix Properties

Consider a polynomial matrix $P(s)$ that is $r \times m$ in size.

Definition 10.1. The degree of $P(s)$ (denoted $\deg P(s)$) is the highest degree of all nonzero minors of maximal order (usually, $\min(r, m)$). For a square matrix, this means $\deg \det P(s)$.

Definition 10.2. A polynomial matrix is unimodular if it is square and $\det P(s)$ is constant.

Definition 10.3. (Rank). The rank of a polynomial matrix, P , is the number of linearly independent columns or rows.

Definition 10.4. (Regular). A matrix, P , is regular iff $\text{rank}(P) = \min\{r, m\}$

Definition 10.5. $P(s)$ is a unit if its inverse is a polynomial.

Proposition 10.1. A polynomial matrix is a unit if and only if it is unimodular.

Proof: By example:

$$P_1 = \begin{bmatrix} s & s \\ 0 & 1 \end{bmatrix} \Rightarrow P_1^{-1} = \begin{bmatrix} \frac{1}{s} & -1 \\ 0 & 1 \end{bmatrix} \quad \text{Is this a polynomial?} \quad (10.4)$$

P_1^{-1} is not a polynomial and therefore, P_1 is not a unit as it is not unimodular. ■

Definition 10.6. (Zero). $s = z$ is a zero of a polynomial matrix $P(s)$ if $\text{rank}(P(z)) < \text{rank}(P(s))$.

EXAMPLE 10.2

Consider the following polynomial matrix,

$$P(s) = \begin{bmatrix} 1 & s \\ 0 & 1 \end{bmatrix}$$

$\det P(s) = 1$, so $P(s)$ is unimodular and $\deg P(s) = 0$. In fact all unimodular polynomials have degree = 0.

6034 **EXAMPLE 10.3**

6035 In this example, we examine the meaning of the ‘highest degree of all nonzero minors
6036 of maximal order’, consider the following polynomial matrix,

$$P(s) = \begin{bmatrix} 1 & s & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

6037 Identify all minors of maximal order for this matrix,

$$\text{minors of maximal order: } \begin{bmatrix} 1 & s \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} s & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

6038 Compute the degree of this polynomial matrix,

$$\deg P(s) = \max \left(\deg \det \begin{bmatrix} 1 & s \\ 0 & 1 \end{bmatrix}, \deg \det \begin{bmatrix} s & 1 \\ 1 & 1 \end{bmatrix}, \deg \det \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \right)$$

$$\deg P(s) = \max(\deg(1), \deg(s-1), \deg(1))$$

6040 So, $\deg P(s) = 1$. $P(s)$ is not unimodular.

6041 **EXAMPLE 10.4**

6042 If $P(s)$ is scalar, then unimodular polynomials are the scalars.

6043 **EXAMPLE 10.5**

6044 Let

$$P_1 = \begin{bmatrix} 1 & 0 \\ 0 & s \end{bmatrix} \text{ and } P_2 = \begin{bmatrix} s & s \\ 1 & 1 \end{bmatrix} \quad (10.5)$$

6045 $\text{rank}(P_1) = 2$ and $\text{rank}(P_2) = 1$ therefore, P_1 is regular and P_2 is not regular.

6046 **EXAMPLE 10.6**

6047 using P_1 from above with $\text{rank} = 2$,

$$P_1(0) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad (10.6)$$

6048 and $\text{rank}(P_1(0)) = 1$. Therefore $s = 0$ is a zero. Defining

$$P_3(s) = \begin{bmatrix} 1 & s \\ 0 & 1 \end{bmatrix} \quad (10.7)$$

6049 $\text{rank}(P_3(s)) = \text{rank}(P_3(0)) = 2$ so therefore $s = 0$ is not a zero of P_3 .

10.3.1 Minimum-Degree Factorization

Every rational transfer function matrix can be written as a ratio of a polynomial matrix numerator and an identity matrix times the transfer function matrix's characteristic polynomial. For example, consider the following matrix $G(s)$,

$$G(s) = \begin{bmatrix} \frac{1}{s-1} & 1 \\ 0 & \frac{1}{s-2} \end{bmatrix} = \frac{\begin{bmatrix} (s-2) & (s-1)(s-2) \\ 0 & (s-1) \end{bmatrix}}{(s-1)(s-2)} \quad (10.8)$$

$$= \underbrace{\begin{bmatrix} (s-2) & (s-1)(s-2) \\ 0 & (s-1) \end{bmatrix}}_{N_R} \underbrace{\begin{bmatrix} (s-1)(s-2) & 0 \\ 0 & (s-1)(s-2) \end{bmatrix}^{-1}}_{D_R} \quad (10.9)$$

For this example, $\deg N_R = 2$ and $\deg D_R = 4$. But can we obtain a lower degree factorization (analogous to checking for pole/zero cancellations in a SISO system)?

A different factorization for $G(s)$ is

$$G(s) = \begin{bmatrix} \frac{1}{s-1} & 1 \\ 0 & \frac{1}{s-2} \end{bmatrix} = \begin{bmatrix} 1 & s-2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{s-1} & 0 \\ 0 & \frac{1}{s-2} \end{bmatrix} \quad (10.10)$$

$$= \underbrace{\begin{bmatrix} 1 & s-2 \\ 0 & 1 \end{bmatrix}}_{N_R, \deg=0} \underbrace{\begin{bmatrix} s-1 & 0 \\ 0 & s-2 \end{bmatrix}^{-1}}_{D_R, \deg=2} \quad (10.11)$$

This factorization results in the $\deg N_R = 0$ and $\deg D_R = 2$, so we can firmly state that the original factorization was not minimum-degree.

In the following sections, we will discuss the conditions for the minimum-degree factorization. But, first, in the next section, we introduce the equivalency condition between polynomial matrices.

Equivalency of Polynomial Matrices

Definition 10.7. (Equivalent). Two matrices are equivalent if one can be converted to the other using a combination of

1. The interchange of columns or rows
2. The multiplication of one column or row by a scalar
3. The addition of a polynomial multiple of a row or column to another row or column

EXAMPLE 10.7 Interchange

$$P = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \sim \begin{bmatrix} 2 & 1 & 3 \\ 4 & 4 & 6 \\ 8 & 7 & 9 \end{bmatrix} = Q \quad (10.12)$$

or

$$P \xrightarrow{\sim} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \sim \begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{bmatrix} = R \quad (10.13)$$

Let

$$D = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (10.14)$$

Then

$$PD = Q \quad (10.15)$$

$$DP = R \quad (10.16)$$

EXAMPLE 10.8 Scalar multiplication

$$P = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \sim \begin{bmatrix} 4 & 4 & 3 \\ 4 & 10 & 6 \\ 7 & 16 & 9 \end{bmatrix} = Q \quad (10.17)$$

or

$$P = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 8 & 10 & 12 \\ 7 & 8 & 9 \end{bmatrix} = R \quad (10.18)$$

Let

$$D = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (10.19)$$

Then

$$PD = Q \quad (10.20)$$

$$DP = R \quad (10.21)$$

EXAMPLE 10.9 Addition of Polynomial Product

Using P from above,

(1st col.) $-s \times$ (2nd col.)

$$P = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \sim \begin{bmatrix} 1-2s & 2 & 3 \\ 4-5s & 5 & 6 \\ 7-8s & 8 & 9 \end{bmatrix} = Q \quad (10.22)$$

or (2nd row) $-s \times$ (1st row)

$$P = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ -s & 5-2s & 6-3s \\ 7 & 8 & 9 \end{bmatrix} = R \quad (10.23)$$

Let

$$D = \begin{bmatrix} 1 & 0 & 0 \\ -s & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (10.24)$$

Then

$$PD = Q \quad (10.25)$$

$$DP = R \quad (10.26)$$

In this section, we introduce the concept of Smith form.

10.4 Smith Canonical Form

In the analysis and design of SISO controllers, the locations of zeros and poles of a transfer function model are critical. Therefore, when we are dealing with MIMO dynamic systems and, hence, transfer function matrices one can ask how we could obtain the location of zeros and poles. Before, introducing generalized transfer function matrices and their associated poles and zeros, we start with introducing Smith canonical form. One benefit of Smith canonical form is easy identification of the zeros of polynomial matrices.

We generalize the polynomial matrices equivalency by the following statement.

Matrices $P(s)$ and $Q(s)$ (both $r \times m$) are in an equivalence relationship if and only if there exist unimodular U_R and U_L such that

$$\underset{(r \times m)}{P} = \underset{(r \times r)}{U_L} \underset{(r \times m)}{Q} \underset{(m \times m)}{U_R}$$

Let's suppose that $r = m$ and P is regular. Then,

$$\det P = \underbrace{\det U_L}_{\text{constant}} \det Q \underbrace{\det U_R}_{\text{constant}}$$

So, P and Q have the same degree and same zeros (this is also true for $r \neq m$).

Theorem 10.1 (Smith Canonical Form) Any $r \times m$ polynomial matrix P is in an equivalence relationship to a diagonal matrix S whose diagonal elements are

$$S_{kk} = \begin{cases} \epsilon_k(s), & k \leq \text{rank } P \\ 0, & k > \text{rank } P \end{cases}$$

6093 where each $\epsilon_k(s)$ is monic and divides into $\epsilon_{k+1}(s)$ without remainder.

6094 *Proof:* [] ■

6095 **Definition 10.8.** A polynomial is monic if the coefficient of the highest degree is equal to
6096 one.

6097 For square, full rank $P(s)$ with rank $P(s) = p$,

$$P \sim S = \begin{bmatrix} \epsilon_1(s) & 0 & \cdots & 0 \\ 0 & \epsilon_2(s) & & \\ \vdots & & \ddots & \\ 0 & & & \epsilon_p(s) \end{bmatrix}$$

6098 Add zeros as needed for $P(s)$ not full rank or $P(s)$ not square.

6099 ■ **EXAMPLE 10.10**

6100 Consider the matrix P .

$$P = \begin{bmatrix} s & 1 \\ 0 & s \end{bmatrix}$$

6101 We will put P into Smith Canonical Form by performing four transformations:

- 6102 1. Exchanging columns 1 and 2
- 6103 2. Subtract s times the first row from the second row into second row
- 6104 3. Subtract s times the first column from the second column into second column
- 6105 4. Multiply the second row by -1

$$P(s) = \begin{bmatrix} s & 1 \\ 0 & s \end{bmatrix} \xrightarrow{\text{Swap Columns}} \begin{bmatrix} 1 & s \\ s & 0 \end{bmatrix} \xrightarrow{\text{2nd Row} - s \times \text{1st Row}} \begin{bmatrix} 1 & s \\ 0 & -s^2 \end{bmatrix} \xrightarrow{\text{2nd Col} - s \times \text{1st Col}} \begin{bmatrix} 1 & 0 \\ 0 & -s^2 \end{bmatrix} \xrightarrow{-1 \times \text{2nd Col}} \begin{bmatrix} 1 & 0 \\ 0 & s^2 \end{bmatrix} = N(s)$$

6106 For every action or matrix manipulation, there is an associated unimodular matrix,
6107 such that

- 6108 1. Exchanging columns 1 and 2,

$$U_R(s) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- 6109 2. Subtract s times the first row from the second row,

$$U_L(s) = \begin{bmatrix} 1 & 0 \\ -s & 1 \end{bmatrix}$$

3. Subtract s times the first column from the second column,

$$U_R(s) = \begin{bmatrix} 1 & -s \\ 0 & 1 \end{bmatrix}$$

4. Multiply the second row by -1 ,

$$U_L(s) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Therefore, we can multiply these unimodular matrices by $P(s)$ in one operation as follows,

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \left(\left(\begin{bmatrix} 1 & 0 \\ -s & 1 \end{bmatrix} \left(P(s) \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right) \right) \begin{bmatrix} 1 & -s \\ 0 & 1 \end{bmatrix} \right) = \begin{bmatrix} 1 & 0 \\ 0 & s^2 \end{bmatrix}$$

$P(s)$ includes two zeros at $s = 0$ as given by $N(s)$, this was not obvious from $P(s)$, unless, we compute the determinant of this matrix, but, this would not be possible if $P(s)$ is not a square matrix.

As shown in the example above, $P(s) \sim N(s)$ means that there exist right and left unimodular matrices U_R and U_L such that

$$U_L P U_R = N$$

But, what is the procedure to find U_R and U_L ? To answer this question, we will setup the $P(s)$ and $N(s)$ matrices as shown below

$$\left[\begin{array}{c|c} I_r & P \\ \hline & I_m \end{array} \right] \sim \left[\begin{array}{c|c} U_L & N \\ \hline & U_R \end{array} \right]$$

(the bottom left quadrant can be neglected).

Let's now repeat the same example as above by assuming P to be

$$P = \begin{bmatrix} s & 1 \\ 0 & s \end{bmatrix}$$

We know how to compute $N(s)$ from $P(s)$, by performing the same row and column operations. But this time, every time we perform a row or column operation, we will keep track of them in I_r and I_m .

So we must simply carry out the same steps on the larger matrix.

$$\left[\begin{array}{cc|cc} 1 & 0 & s & 1 \\ 0 & 1 & 0 & s \\ \hline & & 1 & 0 \\ & & 0 & 1 \end{array} \right] \sim \left[\begin{array}{cc|cc} 1 & 0 & 1 & s \\ 0 & 1 & s & 0 \\ \hline & & 0 & 1 \\ & & 1 & 0 \end{array} \right] \sim \left[\begin{array}{cc|cc} 1 & 0 & 1 & s \\ -s & 1 & 0 & -s^2 \\ \hline & & 0 & 1 \\ & & 1 & 0 \end{array} \right]$$

6128

$$\sim \left[\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ -s & 1 & 0 & -s^2 \\ \hline & & 0 & 1 \\ & & 1 & -s \end{array} \right] \sim \left[\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ s & -1 & 0 & s^2 \\ \hline & & 0 & 1 \\ & & 1 & -s \end{array} \right]$$

6129 Verify the result:

$$U_L = \begin{bmatrix} 1 & 0 \\ s & -1 \end{bmatrix}, \quad U_R = \begin{bmatrix} 0 & 1 \\ 1 & -s \end{bmatrix}$$

Both U_R and U_L are unimodular.

$$\begin{aligned} \begin{bmatrix} 1 & 0 \\ s & -1 \end{bmatrix} \begin{bmatrix} s & 1 \\ 0 & s \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & -s \end{bmatrix} &= \begin{bmatrix} 1 & 0 \\ s & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ s & -s^2 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & s^2 \end{bmatrix} \\ &= N(s) \quad \checkmark \end{aligned}$$

6130 **EXAMPLE 10.11**

6131 Consider the matrix

$$P = \begin{bmatrix} s & s^2 + s \\ 1 & s \end{bmatrix}$$

6132 We will find the equivalent Smith canonical form matrix.

$$P(s) = \begin{bmatrix} s & s^2 + s \\ 1 & s \end{bmatrix} \sim \begin{bmatrix} 1 & s \\ s & s^2 + s \end{bmatrix} \xrightarrow{\text{Swap Rows}} \begin{bmatrix} 1 & s \\ 0 & s \end{bmatrix} \xrightarrow{\substack{\text{2nd Row} - \\ s \times \text{1st Row}}} \begin{bmatrix} 1 & s \\ 0 & s \end{bmatrix} \xrightarrow{\substack{\text{1st Row} - \\ \text{2nd Row}}} \begin{bmatrix} 1 & 0 \\ 0 & s \end{bmatrix} = N(s)$$

- 6133 1. Swap the first and second rows
- 6134 2. Subtract $s \times$ the first row from the second row
- 6135 3. Subtract the second row from the first row

6136 There is one zero at $s = 0$. Computation of the unimodular matrices are left as an
6137 exercise.

6138 **EXAMPLE 10.12**

6139 Consider the matrix

$$P = \begin{bmatrix} s+3 & -1 \\ -4 & 1 \end{bmatrix}$$

6140 We will find the equivalent Smith canonical form matrix.

$$P(s) = \begin{bmatrix} s+3 & -1 \\ -4 & 1 \end{bmatrix} \xrightarrow{\text{Swap Columns}} \begin{bmatrix} -1 & s+3 \\ 1 & -4 \end{bmatrix} \xrightarrow{\substack{\text{Add 1st Row} \\ \text{to 2nd Row}}} \begin{bmatrix} -1 & s+3 \\ 0 & s-1 \end{bmatrix} \xrightarrow{\substack{\text{Add } (s+3) \times \\ \text{1st Column to} \\ \text{2nd Column}}} \begin{bmatrix} 1 & 0 \\ 0 & s-1 \end{bmatrix} = N(s)$$

- 6141 1. Swap the first and second columns
- 6142 2. Add the first row to the second row
- 6143 3. Add $(s+3) \times$ the first column to the second column

6144 There is one zero at $s = 1$. Computation of the unimodular matrices are left as an
6145 exercise.

6146 In the last section, we introduced Smith form of matrices and discussed that one of the
6147 advantages of this form is determining the zeros of polynomial matrices.

6148 In this section, we will introduce Smith-McMillan form, this form of transfer function
6149 matrices is beneficial in determining the location of both poles and zeros these matrices,
6150 and hence it will be utilized in later section for the MIMO control design. Furthermore, we
6151 discuss the relation between the MFD's and Smith-McMillan form.

10.4.1 Smith-McMillan Form

Theorem 10.2 Every transfer function matrix $G(s)$ of a general size of $r \times m$ can be transformed to a pseudo-diagonal matrix through row and column operations,

$$G(s) \underset{(r \times m)}{\sim} \begin{bmatrix} \frac{\eta_1}{\delta_1} & 0 & \cdots & 0 & 0 \\ 0 & \frac{\eta_2}{\delta_2} & & & \\ \vdots & & \ddots & & \\ 0 & & & \frac{\eta_p}{\delta_p} & \\ 0 & \cdots & \cdots & \cdots & 0 \end{bmatrix} = M$$

M is an $r \times m$ rational matrix whose first p (where $p = \min(r, m)$) diagonal elements ratios of polynomials and all other elements are 0. η_k divides into η_{k+1} without remainder, while δ_k divides into δ_{k-1} without remainder.

Proof: Assuming $r = m$ without loss of generality, we can always write $G(s) = \frac{p(s)}{\delta(s)}$, where $p(s)$ is a polynomial matrix and $\delta(s)$ is the greatest common multiple of the denominators $G_{ij}(s)$. Using row and column operations, $p(s)$ can be reduced to Smith form

$$G(s) \sim \frac{1}{\delta} \begin{bmatrix} \varepsilon_1 & 0 & \cdots & 0 \\ 0 & \varepsilon_2 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & \varepsilon_p \end{bmatrix} \quad (10.27)$$

After canceling any factors between δ and ε_i and rearranging to meet the division requirements

$$\frac{1}{\delta} \begin{bmatrix} \varepsilon_1 & 0 & \cdots & 0 \\ 0 & \varepsilon_2 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & \varepsilon_p \end{bmatrix} = \begin{bmatrix} \frac{\varepsilon_1}{\delta_1} & 0 & \cdots & 0 \\ 0 & \frac{\varepsilon_2}{\delta_2} & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{\varepsilon_p}{\delta_p} \end{bmatrix} = M \quad (10.28)$$

By transitivity, $G(s) \sim M$. ■

There are different approaches to compute Smith-McMillan form. Two of the most popular approaches are listed below,

- Factor $G(s) = \frac{p(s)}{\delta(s)}$ then transform $p(s)$ into Smith form
- Define a determinantal divisor $D_k =$ greatest common divisor (gcd) of all $k \times k$ minors and $D_0 = 1$ with the following steps,

a Factor $G(s) = \frac{p(s)}{\delta(s)}$

- b Find all minors of $p(s)$
- c Compute D_k for each $k \times k$ minor.
- d Calculate $\varepsilon_k = \frac{D_k}{D_{k-1}}$ for each $i \in \{1, \dots, p\}$
- e Find M using $M = \frac{1}{\delta(s)} \text{diag}(\varepsilon_k)$

We demonstrate the first approach by the following example.

EXAMPLE 10.13

Consider the dynamic system defined in state-space form by

$$A = \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$C^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{aligned} G(s) &= \frac{1}{\det(sI - A)} \begin{bmatrix} (s+3)(s+1) & (s+3) \\ (s+3) & (s+3) \end{bmatrix} \\ &= \frac{1}{s(s+3)^2} \underbrace{\begin{bmatrix} (s+3)(s+1) & (s+3) \\ (s+3) & (s+3) \end{bmatrix}}_{p(s)} \end{aligned}$$

Transform $p(s)$ into Smith form by the following row and column manipulations:

1. Swap the first and second rows.
2. Subtract $(s+1)$ times the first row from the second row.
3. Subtract the first column from the second column.

$$\begin{aligned} p(s) &= \begin{bmatrix} (s+3)(s+1) & (s+3) \\ (s+3) & (s+3) \end{bmatrix} \sim \begin{bmatrix} (s+3) & (s+3) \\ (s+3)(s+1) & (s+3) \end{bmatrix} \\ &\sim \begin{bmatrix} (s+3) & (s+3) \\ 0 & s(s+3) \end{bmatrix} \sim \begin{bmatrix} (s+3) & 0 \\ 0 & s(s+3) \end{bmatrix} \end{aligned}$$

Then, the Smith-McMillan form of $G(s)$ is

$$G(s) \sim \frac{1}{s(s+3)^2} \begin{bmatrix} (s+3) & 0 \\ 0 & s(s+3) \end{bmatrix} = \begin{bmatrix} \frac{1}{s(s+3)} & 0 \\ 0 & \frac{1}{s+3} \end{bmatrix}$$

6187 We will now look at the second or a more procedural approach to compute the Smith-
 6188 McMillan form. Let determinantal divisors $D_0 = 1$, D_k = the greatest common divisor of
 6189 all $k \times k$ minors. It can be shown that

$$\epsilon_k = \frac{D_k}{D_{k-1}}$$

6190 We repeat the example above but this time, we use the second approach to compute the
 6191 Smith-McMillan form.

6192 **EXAMPLE 10.14**

$$p(s) = \begin{bmatrix} (s+3)(s+1) & (s+3) \\ (s+3) & (s+3) \end{bmatrix}$$

$$1 \times 1 \text{ Minors: } (s+3)(s+1), (s+3), (s+3), (s+3)$$

$$2 \times 2 \text{ Minors: } \det \begin{bmatrix} (s+3)(s+1) & (s+3) \\ (s+3) & (s+3) \end{bmatrix}$$

$$D_0 = 1$$

$$D_1 = \gcd((s+3)(s+1), (s+3))$$

$$= s+3$$

$$D_2 = s(s+3)^2$$

6193

$$\Rightarrow \epsilon_1 = \frac{D_1}{D_0} = s+3, \quad \epsilon_2 = \frac{D_2}{D_1} = s(s+3)$$

6194 **EXAMPLE 10.15**

6195 Let's repeat the Example 10.14 but this time with a different B matrix. Select B to be

$$B = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix} \Rightarrow G(s) = \frac{1}{s(s+3)^2} \underbrace{\begin{bmatrix} (s+1)(s+3) & (s+3) \\ (s+3) & \frac{1}{2}(s+2)(s+3) \end{bmatrix}}_{p(s)}$$

$$1 \times 1 \text{ Minors: } \det(s+3)(s+1), \det(s+3), \det(s+3), \frac{1}{2} \det(s+3)(s+2)$$

$$2 \times 2 \text{ Minors: } \det \begin{bmatrix} (s+1)(s+3) & (s+3) \\ (s+3) & \frac{1}{2}(s+2)(s+3) \end{bmatrix}$$

$$D_0 = 1$$

$$D_1 = \gcd((s+3)(s+1), (s+3), \frac{1}{2}(s+3)(s+2))$$

$$= s+3$$

$$D_2 = \frac{1}{2}s(s+3)^3$$

$$\Rightarrow \epsilon_1 = \frac{D_1}{D_0} = s+3, \quad \epsilon_2 = \frac{D_2}{D_1} = \frac{1}{2}s(s+3)^2$$

$$G(s) \sim \frac{1}{s(s+3)^2} \begin{bmatrix} (s+3) & 0 \\ 0 & \frac{1}{2}s(s+3)^2 \end{bmatrix} = \begin{bmatrix} \frac{1}{s(s+3)} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$$

It is not as obvious as the previous example, but in this example δ_k divides δ_{k-1} , this means, $s(s+3) = 2 \times \frac{1}{2} \times s(s+3)$ is a factor.

10.4.2 MFD's and their relations to Smith-McMillan Form

Recall left and right Matrix Fraction Descriptions for MIMO systems,

$$G(s) = D_L^{-1} N_L = N_R D_R^{-1} \quad (10.29)$$

We discussed the idea of coprimeness for SISO transfer functions. Here, We extend this idea to MIMO systems,

Definition 10.9. (Right Common Factor). W is a right common factor of N_R and D_R if

$$N_R = N'_R W \text{ and } D_R = D'_R W \text{ where } N'_R, D'_R, \text{ and } W \text{ are polynomial matrices.}$$

Left common factors are equivalently defined.

Definition 10.10. (Coprimeness). N_R and D_R are right coprime if all right common factors, W , are unimodular. Left common factors are equivalently defined.

Definition 10.11. (Irreducible MFD). $N_R D_R^{-1}$ is an irreducible matrix fraction description (MFD) of $G(s)$ if N_R and D_R are right coprime. Left coprime is equivalently defined.

We will now consider finding a MFD from the Smith-McMillan form of matrix G .

$$G \sim M \Rightarrow U_L G U_R = M = \text{diag} \left(\frac{\eta_1}{\delta_1}, \frac{\eta_2}{\delta_2}, \dots, \frac{\eta_p}{\delta_p}, 0, \dots, 0 \right)$$

So,

$$G = U_L^{-1} \begin{bmatrix} \eta_1 & 0 & \cdots & 0 & | & 0 \\ 0 & \eta_2 & & & & \\ \vdots & & \ddots & & & \\ 0 & & & \eta_p & & \\ \hline 0 & \cdots & \cdots & \cdots & 0 \end{bmatrix} \begin{bmatrix} \delta_1 & 0 & \cdots & 0 & | & 0 \\ 0 & \delta_2 & & & & \\ \vdots & & \ddots & & & \\ 0 & & & \delta_p & & \\ \hline 0 & \cdots & \cdots & \cdots & 1 \end{bmatrix}^{-1} U_R^{-1}$$

Therefore,

$$G = N_R D_R^{-1} \quad \text{where} \quad \begin{matrix} N_R = U_L^{-1} \text{diag}(\eta_1, \dots, \eta_p, 0, \dots, 0) \\ D_R = U_R \text{diag}(\delta_1, \dots, \delta_p, 1, \dots, 1) \end{matrix}$$

However,

$$\begin{bmatrix} \eta_1 & 0 & \cdots & 0 & | & 0 \\ 0 & \eta_2 & & & | & \\ \vdots & & \ddots & & | & \\ 0 & & & \eta_p & | & 0 \\ \hline 0 & \cdots & \cdots & \cdots & | & 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} \delta_1 & 0 & \cdots & 0 & | & 0 \\ 0 & \delta_2 & & & | & \\ \vdots & & \ddots & & | & \\ 0 & & & \delta_p & | & \\ \hline 0 & \cdots & \cdots & \cdots & | & 1 \end{bmatrix}$$

are coprime because each η_k/δ_k is irreducible. Hence, this MFD, N_R and D_R , will be coprime.

Poles of $G(s)$ are the zeros of D_R , so the pole polynomial is

$$\delta_1(s)\delta_2(s)\cdots\delta_p(s)$$

The degree of the pole polynomial is called the Smith-McMillan (SM) degree. **It is important to note that the SM form is the minimum realization of G** All poles are the eigenvalues of A matrix in the state-space representation.

Zeros of $G(s)$ are zeros of N_R . The zero polynomial is therefore

$$\eta_1(s)\eta_2(s)\cdots\eta_p(s)$$

It is also interesting to note if one could compute a coprime MFD and Smith-MacMillan form, then from relationship above, the left and right unimodular matrices could be computed. This will be presented in an example later in this section. Computation of the unimodular matrices become important during MIMO control design computation as will be presented in the later chapters.

10.4.3 Computing an Irreducible (coprime) Matrix Fraction Description

Given N_R and D_R (right MFD in this case, left MFD is equivalent) which are not coprime, we ask the following question, how can we find the common divisor of greatest degree for computing a coprime MFD? Imagine row operations on

$$\begin{bmatrix} (m \times m) \\ D_R \\ \vdots \\ N_R \\ (r \times m) \end{bmatrix} \sim \begin{bmatrix} (m \times m) \\ W \\ \vdots \\ 0 \\ (r \times m) \end{bmatrix}$$

Now, we can consider a matrix V such that

$$\begin{bmatrix} D_R \\ N_R \end{bmatrix} = V \begin{bmatrix} W \\ \bar{0} \end{bmatrix} = \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix} \begin{bmatrix} W \\ \bar{0} \end{bmatrix}$$

So,

$$D_R = V_{11}W, \quad N_R = V_{21}W$$

or conversely,

$$V_{11} = D_R W^{-1}, \quad V_{21} = N_R W^{-1}$$

Then,

$$G = V_{21}V_{11}^{-1}$$

is an irreducible factorization of G , where V_{11} and V_{21} are coprime.

In the next two examples we will illustrate the procedure above for computing a coprime MFD. The first example is for the SISO case and the second example is for the MIMO case.

EXAMPLE 10.16

Consider the transfer function

$$G(s) = \frac{s^2 + s}{s^2 + 3s + 2} = \frac{s(s+1)}{(s+1)(s+2)}$$

Which we know is not coprime due to the common factor of $(s+1)$ but we will verify this below. We select the numerator and the denominator of this transfer function to be $N_R = s(s+1)$ and $D_R = (s+1)(s+2)$. Then given the procedure above, we can write,

$$\begin{bmatrix} D_R \\ N_R \end{bmatrix} = \underbrace{\begin{bmatrix} s^2 + 3s + 2 \\ s^2 + s \end{bmatrix}}_{(a)} \sim \underbrace{\begin{bmatrix} 2s + 2 \\ s^2 + s \end{bmatrix}}_{(b)} \sim \underbrace{\begin{bmatrix} 2s + 2 \\ 0 \end{bmatrix}}_{(c)} \sim \underbrace{\begin{bmatrix} s + 1 \\ 0 \end{bmatrix}}_{(d)} = \begin{bmatrix} W \\ \bar{0} \end{bmatrix}$$

$a \rightarrow b$ Subtract the second row from the first.

$b \rightarrow c$ Subtract $\frac{1}{2}s$ times the first row from the second row.

$c \rightarrow d$ Divide the first row by 2.

We find that $W = s + 1$. W is not unimodular (scalar), so N_R and D_R are not coprime. ■

Now, we convert N_R and D_R to a coprime MFD,

$$N_R = V_{21}W \Rightarrow N'_R = V_{21} = N_R W^{-1} = s$$

6253

6254

$$G = \underbrace{N'_R D'^{-1}_R}_{\text{Coprime}} = \frac{s}{s+2}$$

6255

EXAMPLE 10.17

6256

Consider the following non-coprime MFD,

$$G(s) = N_R D_B^{-1} \quad (10.30)$$

6257

With the following N_B and D_B ,

$$N_R = \begin{bmatrix} s-2 & (s-1)(s-2) \\ 0 & s(s-1) \end{bmatrix}, \quad D_R = \begin{bmatrix} (s-1)(s-2) & 0 \\ 0 & (s-1)(s-2) \end{bmatrix}$$

6258

Now, we compute the common divisor of greatest degree, W matrix, using the proce-

6259

dure above as follows,

$$\begin{bmatrix} D_R \\ N_R \end{bmatrix} = \begin{bmatrix} (s-1)(s-2) & 0 \\ 0 & (s-1)(s-2) \\ \hline s-2 & (s-1)(s-2) \\ 0 & s(s-1) \end{bmatrix} \sim \begin{bmatrix} (s-1)(s-2) & 0 \\ 0 & -2(s-1) \\ \hline s-2 & -2(s-1) \\ 0 & s(s-1) \end{bmatrix} \sim$$

6260

$$\left[\begin{array}{cc} (s-1)(s-2) & 0 \\ 0 & (s-1) \\ \hline s-2 & -2(s-1) \\ 0 & 0 \end{array} \right] \sim \left[\begin{array}{cc} (s-1)(s-2) & 0 \\ 0 & (s-1) \\ \hline s-2 & 0 \\ 0 & 0 \end{array} \right] \sim$$

6261

$$\begin{bmatrix} \begin{array}{cc} -(s-2) & 0 \\ 0 & (s-1) \end{array} \\ \hline \begin{array}{cc} s-2 & 0 \end{array} \\ \begin{array}{cc} 0 & 0 \end{array} \end{bmatrix} \sim \begin{bmatrix} \begin{array}{cc} s-2 & 0 \\ 0 & s-1 \end{array} \\ \hline \begin{array}{cc} 0 & 0 \end{array} \\ \begin{array}{cc} 0 & 0 \end{array} \end{bmatrix}$$

6262

$a \rightarrow b$ Subtract the last row from the 2nd and 3rd rows.

6263

$b \rightarrow c$ Divide the 2nd row by -2 , subtract s times the second row from the last row.

6264

$c \rightarrow d$ Add 2 times the 2nd row to the 3rd row.

6265

$d \rightarrow e$ Subtract s times the 3rd row from the 1st row.

6266

$e \rightarrow f$ Multiply the first row by -1 and then subtract the 1st row from the 3rd row.

$$W = \begin{bmatrix} s-2 & 0 \\ 0 & s-1 \end{bmatrix} \Rightarrow W^{-1} = \begin{bmatrix} \frac{1}{s-2} & 0 \\ 0 & \frac{1}{s-1} \end{bmatrix}$$

$$\begin{aligned}
N'_R = V_{21} = N_R W^{-1} &= \begin{bmatrix} s-2 & (s-1)(s-2) \\ 0 & s(s-1) \end{bmatrix} \begin{bmatrix} \frac{1}{s-2} & 0 \\ 0 & \frac{1}{s-1} \end{bmatrix} = \begin{bmatrix} 1 & s-2 \\ 0 & s \end{bmatrix} \\
D'_R = V_{11} = D_R W^{-1} &= \begin{bmatrix} (s-1)(s-2) & 0 \\ 0 & (s-1)(s-2) \end{bmatrix} \begin{bmatrix} \frac{1}{s-2} & 0 \\ 0 & \frac{1}{s-1} \end{bmatrix} \\
&= \begin{bmatrix} s-1 & 0 \\ 0 & s-2 \end{bmatrix} \\
\Rightarrow G(s) &= \underbrace{\begin{bmatrix} 1 & s-2 \\ 0 & s \end{bmatrix} \begin{bmatrix} s-1 & 0 \\ 0 & s-2 \end{bmatrix}^{-1}}_{\text{Coprime}}
\end{aligned}$$

In the next example, we compute the Smith-McMillan form of a transfer function matrix.

In the last example, we demonstrate how to compute the unimodular matrices which map this transfer function matrix to its Smith-McMillan form. We accomplish this by first computing a coprime MFD, and then use this MFD with its Smith-McMillan form to compute the unimodular matrices.

EXAMPLE 10.18

Consider the following transfer function matrix

$$\begin{aligned}
G &= \begin{bmatrix} \frac{1}{(s+1)^2} & \frac{1}{(s+1)(s+2)} \\ \frac{1}{(s+1)(s+2)} & \frac{s+3}{(s+2)^2} \end{bmatrix} \\
&= \begin{bmatrix} (s+2)^2 & (s+1)(s+2) \\ (s+1)(s+2) & (s+1)^2(s+3) \end{bmatrix} \frac{1}{(s+1)^2(s+2)^2}
\end{aligned}$$

then using the second approach for computing the Smith-McMillan form, the determinantal divisors are written as,

$$\begin{aligned}
D_0 &= 1, \quad D_1 = 1, \quad D_2 = (s+1)^2(s+2)^3 \\
\Rightarrow \epsilon_1 &= \frac{D_1}{D_0} = 1, \quad \epsilon_2 = \frac{D_2}{D_1} = (s+1)^2(s+2)^3
\end{aligned}$$

Therefore, the SM form can be written as,

$$G \sim \begin{bmatrix} \frac{1}{(s+1)^2(s+2)^2} & 0 \\ 0 & s+2 \end{bmatrix}$$

The zero polynomial of $G(s)$ is therefore $s+2=0$, and the pole polynomial is $(s+1)^2(s+2)^2=0$. $G(s)$ has a Smith-McMillan degree of 4.

Therefore, all other MFD's are irreducible if and only if their $\deg = 4$.

Note:

1) Not all the entries of Smith-McMillan form are proper transfer functions.

It is important to know that there is no restriction on the properness of the S-M forms, they simply reveal the poles and zeros of MIMO systems.

2) The poles and zeros of $\det(G)$ are not similar to the poles and zeros of the Smith-McMillan form of G .

A MIMO system is directional based, so zeros and poles of the same magnitude but different direction do not cancel each other. But this is not the same as when one computes the determinant of a matrix as the act of computing the determinant eliminates directional sensitivity, and hence all the poles and zeros with the same magnitude are then canceled.

EXAMPLE 10.19

Consider the same transfer function matrix as the above example, G can be written as multiplication of a numerator, N_R and denominator, D_R matrices as follows,

$$\begin{aligned}
 G &= \begin{bmatrix} \frac{1}{(s+1)^2} & \frac{1}{(s+1)(s+2)} \\ \frac{1}{(s+1)(s+2)} & \frac{s+3}{(s+2)^2} \end{bmatrix} \\
 &= \begin{bmatrix} (s+2) & (s+2) \\ (s+1) & (s+1)(s+3) \end{bmatrix} \begin{bmatrix} \frac{1}{(s+1)^2(s+2)} & 0 \\ 0 & \frac{1}{(s+1)(s+2)^2} \end{bmatrix} \\
 &= \underbrace{\begin{bmatrix} (s+2) & (s+2) \\ (s+1) & (s+1)(s+3) \end{bmatrix}}_{N_R} \underbrace{\begin{bmatrix} (s+1)^2(s+2) & 0 \\ 0 & (s+1)(s+2)^2 \end{bmatrix}^{-1}}_{D_R: \deg=6}
 \end{aligned}$$

It is obvious that this MFD is not coprime as the degree of D_R is greater than 4. Hence, we can compute a coprime factorization by calculating the W matrix,

$$\begin{aligned}
 \begin{bmatrix} D_R \\ N_R \end{bmatrix} &= \begin{bmatrix} (s+1)^2(s+2) & 0 \\ 0 & (s+1)(s+2)^2 \\ \hline s+2 & s+2 \\ s+1 & (s+1)(s+3) \end{bmatrix} \sim \begin{bmatrix} 0 & -(s+1)^2(s+2) \\ 0 & (s+1)(s+2)^2 \\ \hline s+2 & s+2 \\ s+1 & (s+1)(s+3) \end{bmatrix} \sim \\
 &\quad (a) \qquad (b) \\
 &\quad \begin{bmatrix} 0 & -(s+1)^2(s+2) \\ 0 & (s+1)(s+2) \\ \hline s+2 & s+2 \\ s+1 & (s+1)(s+3) \end{bmatrix} \sim \begin{bmatrix} 0 & 0 \\ 0 & (s+1)(s+2) \\ \hline s+2 & s+2 \\ s+1 & (s+1)(s+3) \end{bmatrix} \sim \\
 &\quad (c) \qquad (d)
 \end{aligned}$$

$$\begin{aligned}
& \begin{bmatrix} 0 & 0 \\ 0 & (s+1)(s+2) \\ s+2 & s+2 \\ s+1 & s+1 \end{bmatrix} \underset{(e)}{\sim} \begin{bmatrix} 0 & 0 \\ 0 & (s+1)(s+2) \\ 1 & 1 \\ s+1 & s+1 \end{bmatrix} \underset{(f)}{\sim} \\
& \begin{bmatrix} 0 & 0 \\ 0 & (s+1)(s+2) \\ 1 & 1 \\ 0 & 0 \end{bmatrix} \underset{(g)}{\sim} \begin{bmatrix} 1 & 1 \\ 0 & (s+1)(s+2) \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \underset{(h)}{\sim}
\end{aligned}$$

6296 $a \rightarrow b$ Subtract $(s+1)^2$ times the 3rd row from the 1st row.

6297 $b \rightarrow c$ Add the 1st row to the 2nd row.

6298 $c \rightarrow d$ Add $(s+1)$ times the 2nd row to the 1st row.

6299 $d \rightarrow e$ Subtract the 2nd row from the 4th row.

6300 $e \rightarrow f$ Subtract the 4th row from the 3rd row.

6301 $f \rightarrow g$ Subtract $(s+1)$ times the 3rd row from the 4th row.

6302 $g \rightarrow h$ Swap the 1st and 3rd row.

6303 So, the right common factor, W , is

$$W = \begin{bmatrix} 1 & 1 \\ 0 & (s+1)(s+2) \end{bmatrix} \Rightarrow W^{-1} = \begin{bmatrix} 1 & -\frac{1}{(s+1)(s+2)} \\ 0 & \frac{1}{(s+1)(s+2)} \end{bmatrix}$$

$$\begin{aligned}
N'_R &= N_R W^{-1} = \begin{bmatrix} (s+2) & (s+2) \\ (s+1) & (s+1)(s+3) \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{(s+1)(s+2)} \\ 0 & \frac{1}{(s+1)(s+2)} \end{bmatrix} \\
&= \begin{bmatrix} s+2 & 0 \\ s+1 & 1 \end{bmatrix}
\end{aligned}$$

$$\begin{aligned}
D'_R &= D_R W^{-1} = \begin{bmatrix} (s+1)^2(s+2) & 0 \\ 0 & (s+1)(s+2)^2 \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{(s+1)(s+2)} \\ 0 & \frac{1}{(s+1)(s+2)} \end{bmatrix} \\
&= \begin{bmatrix} (s+1)^2(s+2) & -(s+1) \\ 0 & s+2 \end{bmatrix}
\end{aligned}$$

6304 So, an irreducible realization of $G(s)$ is

$$G(s) = N'_R D'^{-1}_R = \underbrace{\begin{bmatrix} s+2 & 0 \\ s+1 & 1 \end{bmatrix}}_{\text{Zeros}=-2} \underbrace{\begin{bmatrix} (s+1)^2(s+2) & -(s+1) \\ 0 & s+2 \end{bmatrix}^{-1}}_{\text{Poles}=-2, -2, -1, -1}$$

Note that the degree of this denominator polynomial matrix, D'_R , is 4, precisely equal to the SM degree.

Next, we must find U_R and U_L , by knowing both the SM form and a coprime MFD,

$$G = \begin{bmatrix} s+2 & 0 \\ s+1 & 1 \end{bmatrix} \begin{bmatrix} (s+1)^2(s+2) & -(s+1) \\ 0 & s+2 \end{bmatrix}^{-1} \sim \underbrace{\begin{bmatrix} \frac{1}{(s+1)^2(s+2)^2} & 0 \\ 0 & s+2 \end{bmatrix}}_{\text{SM form}}$$

we know,

$$N'_R = U_L^{-1} \text{diag}(\eta_1, \eta_2, \dots, \eta_p)$$

$$\begin{bmatrix} s+2 & 0 \\ s+1 & 1 \end{bmatrix} = U_L^{-1} \begin{bmatrix} 1 & 0 \\ 0 & s+2 \end{bmatrix}$$

therefore,

$$U_L = \begin{bmatrix} 1 & 0 \\ 0 & s+2 \end{bmatrix} \begin{bmatrix} s+2 & 0 \\ s+1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{s+2} & 0 \\ -\frac{1}{s+1} & s+2 \end{bmatrix}$$

and

$$D_R = U_R \text{diag}(\delta_1, \delta_2, \dots, \delta_p)$$

$$\begin{bmatrix} (s+1)^2(s+2) & -(s+1) \\ 0 & s+2 \end{bmatrix} = U_R \begin{bmatrix} (s+1)^2(s+2)^2 & 0 \\ 0 & 1 \end{bmatrix}$$

so,

$$U_R = \begin{bmatrix} (s+1)^2(s+2) & -(s+1) \\ 0 & s+2 \end{bmatrix} \begin{bmatrix} (s+1)^2(s+2)^2 & 0 \\ 0 & 1 \end{bmatrix}^{-1}$$

$$= \begin{bmatrix} \frac{1}{s+2} & -(s+1) \\ 0 & s+2 \end{bmatrix}$$

Knowing the unimodular matrices, U_L and U_R , we now verify if we could obtain the same Smith-McMillan form computed above, directly from the transfer function matrix, G , as follows,

$$U_L G U_R = M$$

$$\begin{bmatrix} \frac{1}{s+2} & 0 \\ -s-1 & s+2 \end{bmatrix} \begin{bmatrix} \frac{1}{(s+1)^2} & \frac{1}{(s+1)(s+2)} \\ \frac{1}{(s+1)(s+2)} & \frac{s+3}{(s+2)^2} \end{bmatrix} \begin{bmatrix} \frac{1}{s+2} & -s-1 \\ 0 & s+2 \end{bmatrix} = \begin{bmatrix} \frac{1}{(s+1)^2(s+2)^2} & 0 \\ 0 & s+2 \end{bmatrix}$$

This is the same SM form computed above. ■

Note:

1) U_L and U_R in the example above are not polynomial matrices, but their determinants are constants and are equal to 1, as some of the entries are rational

functions.

2) Unimodular matrices U_L and U_R can be computed using hermite form (as is done in Matlab and not included here) to obtain the following matrices,

$$U_L = \begin{bmatrix} 1 & -(s^2 + 4s + 5) \\ s + 1 & -(s^3 + 5s^2 + 9s + 6) \end{bmatrix}$$

$$U_R = \begin{bmatrix} s + 2 & -(s^5 + 9s^4 + 32s^3 + 55s^2 + 44s + 13) \\ -1 & s^4 + 7s^3 + 18s^2 + 19s + 6 \end{bmatrix}$$

Multiplication of these unimodular matrices with G results in the same M matrix! (try it).

In this section, we discuss controllability and observability as they relate to co-prime factorization. We then briefly present the SISO state-space realization before discussing the steps required for the computation of MIMO state-space realization in controllable-form (observable-form).

10.5 MIMO Controllability and Observability

Consider the state equations for a MIMO system:

$$\left. \begin{array}{l} \dot{\underline{x}} = A\underline{x} + B\underline{u} \\ \underline{y} = C^T \underline{x} + D\underline{u} \end{array} \right\} \begin{array}{l} G(s) = C^T (sI - A)^{-1} B + D = \begin{matrix} N_R & D_R^{-1} \\ (r \times m) & (m \times m) \end{matrix} = \begin{matrix} D_L^{-1} & N_L \\ (r \times r) & (r \times m) \end{matrix} \end{array}$$

These two realizations are strictly comparable if and only if $\deg D_L$ or $\deg D_R = n$. We present the controllability and observability of MIMO dynamic systems as follows, The realization is controllable (inputs can effect all states) if

$$\text{rank} \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix} = n$$

($n \times nm$)

The realization is observable (outputs are affected by all states) if

$$\text{rank} \begin{bmatrix} C^T & C^T A & \cdots & C^T A^{n-1} \end{bmatrix}^T = n$$

($rn \times n$)

We have previously discussed the idea of coprimeness of an MFD or factorization. We can show observability and controllability if and only if N_R and D_R (or N_L and D_L) are coprime. We know that eigenvalues of A are poles of $G(s)$ (and zeros of D_R or D_L) and the transmission zeros of (A, B, C^T, D) are zeros of $G(s)$ (zeros of N_R or N_L).

If the realization is uncontrollable or unobservable, then there is at least one transmission zero on an eigenvalue. The common factor in D and N will also have zeros at the same location.

The minimum/irreducible realization is controllable and observable if $\deg D_R$ or $\deg D_L = n$ and where N_R and D_R (or N_L and D_L) are coprime.

10.5.1 State-space Realization

We have shown how to convert a state-space system representation to a transfer function (or transfer function matrix), but how do we convert a transfer function (or transfer function matrix) to state-space form? Before, we discuss state-space representation for MIMO systems, we present this realization for SISO systems which has been covered in various texts but we present this material for the sake of completeness.

SISO System We can easily express a transfer function in state-space form by first expressing the transfer function as a differential equation, and then rewriting it as a set of first-order differential equations. We will consider three cases for the transfer function $G(s)$:

1. $G(s)$ is strictly proper and has a numerator equal to unity.
2. $G(s)$ is strictly proper.
3. $G(s)$ is proper.

A third-order system will be used for the sake of example, but the method can be generalized to systems of any order.

Case 1:

$$G(s) = \frac{1}{s^3 + a_1 s^2 + a_2 s + a_3} = \frac{Y}{U}(s)$$

First, write as a differential equation,

$$\ddot{y} + a_1 \dot{y} + a_2 y + a_3 y = u$$

Then, express in matrix form as a system of first-order equations,

$$\begin{bmatrix} \dot{y} \\ \ddot{y} \\ \ddot{y} \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_1 & -a_2 & -a_3 \end{bmatrix}}_A \cdot \begin{bmatrix} y \\ \dot{y} \\ \ddot{y} \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_B \cdot u$$

$$y = \underbrace{[1 \ 0 \ 0]}_{C^T} \cdot \begin{bmatrix} y \\ \dot{y} \\ \ddot{y} \end{bmatrix} + \underbrace{[0]}_D \cdot u$$

Case 2:

$$G(s) = \frac{b_1 s^2 + b_2 s + b_3}{s^3 + a_1 s^2 + a_2 s + a_3} = \frac{Y}{U}(s)$$

First, we define a new transfer function Y/V , where V is a pseudo state, such that

$$\frac{Y}{U} = \frac{Y}{V} \cdot \frac{V}{U}$$

We define Y/V to be the numerator of the transfer function and V/U to be the denominator.

Then,

$$\frac{Y}{V} = b_1 s^2 + b_2 s + b_3, \quad \frac{V}{U} = \frac{1}{s^3 + a_1 s^2 + a_2 s + a_3}$$

Clearly, V/U is identical to the problem shown in Case 1. However, we now have a new differential equation to define the system output, which will be realized as a new C^T matrix.

$$\ddot{v} + a_1 \dot{v} + a_2 v + a_3 v = u$$

$$y = b_1 \ddot{v} + b_2 \dot{v} + b_3 v$$

Expressed in matrix form:

$$\begin{bmatrix} \dot{v} \\ \ddot{v} \\ \ddot{v} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_1 & -a_2 & -a_3 \end{bmatrix} \cdot \begin{bmatrix} v \\ \dot{v} \\ \ddot{v} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \cdot u$$

$$y = [b_3 \quad b_2 \quad b_1] \cdot \begin{bmatrix} v \\ \dot{v} \\ \ddot{v} \end{bmatrix} + [0] \cdot u$$

Case 3:

$$G(s) = \frac{b_0 s^3 + b_1 s^2 + b_2 s + b_3}{s^3 + a_1 s^2 + a_2 s + a_3} = \frac{Y}{U}(s)$$

We can rewrite $G(s)$ as

$$G(s) = \frac{c_1 s^2 + c_2 s + c_3}{s^3 + a_1 s^2 + a_2 s + a_3} + d$$

where $d = G(\infty)$.

The problem is then analogous to Case 2, but with an additional feed-through element, d , expressed in the output equation.

Expressed in matrix form:

$$\begin{bmatrix} \dot{v} \\ \ddot{v} \\ \ddot{v} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_1 & -a_2 & -a_3 \end{bmatrix} \cdot \begin{bmatrix} v \\ \dot{v} \\ \ddot{v} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \cdot u$$

$$y = [c_3 \quad c_2 \quad c_1] \cdot \begin{bmatrix} v \\ \dot{v} \\ \ddot{v} \end{bmatrix} + [d] \cdot u$$

The state-space representations for the above cases have a special form known as controllable-form or controllable canonical form, where controllability in this representation is guaranteed.

Now, we are in the position to discuss this realization for MIMO systems.

MIMO System It would be informative to first illustrate how we can quickly convert a transfer function matrix to its equivalent state-space representation using two approaches before discussing its controllable form.

Consider the following general transfer function matrix, $G(s)$,

$$G(s) = \begin{bmatrix} g_{11}(s) & \cdots & g_{1m}(s) \\ \vdots & \ddots & \vdots \\ g_{r1}(s) & \cdots & g_{rm}(s) \end{bmatrix}$$

The simplest approach to transform this transfer function matrix to its equivalent state-space representation would be to develop a state-space model for each element of $G(s)$, similar to the SISO case, in the form of $(A_{ij}, B_{ij}, C_{ij}, D_{ij})$ and then assemble the complete state-space representation. If $G(s)$ is $r \times m$, then

$$A = \begin{bmatrix} A_{11} & & 0 & & & \\ & \ddots & & & & \\ 0 & & A_{1m} & & & 0 \\ \hline & & & \ddots & & \\ & & & & A_{r1} & 0 \\ \hline & 0 & & & 0 & \ddots \\ & & & & 0 & A_{rm} \end{bmatrix}, \quad B = \begin{bmatrix} B_{11} & & 0 \\ & \ddots & \\ 0 & & B_{1m} \\ \hline & & \vdots \\ B_{r1} & & 0 \\ \hline & & \vdots \\ 0 & & B_{rm} \end{bmatrix}$$

$$C = \begin{bmatrix} C_{11} & \cdots & C_{1m} & \vdots & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ 0 & & 0 & \vdots & C_{r1} & \cdots & C_{rm} \end{bmatrix}, \quad D = \begin{bmatrix} D_{11} & \cdots & D_{1m} \\ \vdots & \ddots & \vdots \\ D_{r1} & \cdots & D_{rm} \end{bmatrix}$$

However, this method is inefficient and most definitely not a minimal realization.

Another approach, discussed in [?], called Gilbert's realization, requires rewriting the transfer function matrix as,

$$G(s) = \frac{p(s)}{\delta(s)}$$

Then, finding the Smith-McMillan form and the McMillan degree, and finally, applying partial fraction expansion to each of the elements of $G(s)$ and collect residues for each

6399 distinct pole,

$$G(s) = \sum_i^m \frac{R_i}{s - p_i}, \quad \text{where } R_i = \lim_{s \rightarrow p_i} (s - p_i)G(s)$$

6400 The residues, R_i , are matrices in the MIMO case, and the sum of the ranks of matrices R_i
 6401 gives the McMillan degree.

6402

6403 The steps for forming the state-space representations are detailed below,

6404

- 6405 1. Set $R_i = C_i B_i$ and find appropriate B_i and C_i .
- 6406 2. Form A_i by placing the poles on the diagonal as many times as needed (determined
 6407 by rank R_i).

6408 The minimal state-space realization is then written as,

$$\begin{aligned} \dot{\underline{x}} &= \begin{bmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_m \end{bmatrix} \underline{x} + \begin{bmatrix} B_1 \\ \vdots \\ B_m \end{bmatrix} \underline{u} \\ \underline{y} &= [C_1 \quad \cdots \quad C_m] \underline{x} + D \underline{u} \end{aligned}$$

6409

6410 **EXAMPLE 10.20**

$$G(s) = \begin{bmatrix} \frac{1}{s+2} & \frac{1}{s+2} \\ \frac{1}{s-2} & \frac{1}{(s+1)(s+2)} \end{bmatrix}$$

$$= \frac{1}{(s+1)(s+2)(s-2)} \begin{bmatrix} (s+1)(s-2) & (s+1)(s-2) \\ (s+1)(s+2) & (s-2)^2 \end{bmatrix}$$

We then find the SM form (left here as an exercise), and realize the characteristic equation is a fourth-order system with poles at $s = -2, -2, -1, 2$.

The transfer function matrix, $G(s)$ is written as a ratio of $\frac{p(s)}{\delta(s)}$,

$$G(s) = \frac{\begin{bmatrix} (s+1)(s-2) & (s+1)(s-2) \\ (s+1)(s+2) & (s-2)^2 \end{bmatrix}}{(s+1)(s+2)(s-2)}$$

and using partial fraction expansion, $G(s)$ is written as follows,

$$G(s) = \frac{R_1}{s+1} + \frac{R_2}{s-2} + \frac{R_3}{s+2}$$

6411 then the residue matrices are computed,

$$R_1 = \lim_{s \rightarrow -1} (s+1)G(s) = \begin{bmatrix} 0 & 0 \\ 0 & -3 \end{bmatrix}$$

$$R_2 = \lim_{s \rightarrow 2} (s-2)G(s) = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$R_3 = \lim_{s \rightarrow -2} (s+2)G(s) = \begin{bmatrix} 1 & 1 \\ 0 & 4 \end{bmatrix}$$

6414 now, $G(s)$ can be written as,

$$G(s) = \frac{1}{s+1} \begin{bmatrix} 0 & 0 \\ 0 & -3 \end{bmatrix} + \frac{1}{s-2} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + \frac{1}{s+2} \begin{bmatrix} 1 & 1 \\ 0 & 4 \end{bmatrix}$$

6415 Note that the last residue matrix is rank 2, so, there will be two poles at $s = -2$ to
6416 make the system order 4, equal to the SM degree.

6417 It is evident from $G(s)$ that the number of inputs and outputs are equal 2. In ad-
6418 dition, knowing that the system order is 4, we can then state that the dimensions for
6419 matrices, A is 4×4 , B is 4×2 , C is 2×4 , and D is 2×2 . Next, we find the elements
6420 of the B and C matrices from the previously computed residue matrices. And easy
6421 way to form B and C is to set $C_i = 1$ or 0 as follows,

$$\begin{bmatrix} 0 & 0 \\ 0 & -3 \end{bmatrix} = \begin{bmatrix} C_{11} \\ C_{21} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \end{bmatrix} \Rightarrow \begin{matrix} B_{11}C_{11} = 0 & B_{12}C_{11} = 0 \\ B_{11}C_{21} = 0 & B_{12}C_{21} = -3 \end{matrix}$$

$$\Rightarrow \begin{matrix} B_{11} = 0 & C_{11} = 0 \\ B_{12} = -3 & C_{21} = 1 \end{matrix} \Rightarrow B_1 = \begin{bmatrix} 0 & -3 \end{bmatrix}, \quad C_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} C_{12} \\ C_{22} \end{bmatrix} \begin{bmatrix} B_{21} & B_{22} \end{bmatrix} \Rightarrow \begin{array}{ll} B_{21}C_{12} = 0 & B_{22}C_{12} = 0 \\ B_{21}C_{22} = 1 & B_{22}C_{22} = 0 \end{array}$$

$$\Rightarrow \begin{array}{ll} B_{21} = 1 & C_{12} = 0 \\ B_{22} = 0 & C_{22} = 1 \end{array} \Rightarrow B_2 = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad C_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} C_{13} & C_{14} \\ C_{23} & C_{24} \end{bmatrix} \begin{bmatrix} B_{31} & B_{32} \\ B_{41} & B_{42} \end{bmatrix}$$

$$\Rightarrow \text{Set } C_3 = eye(2) \Rightarrow B_3 = \begin{bmatrix} 1 & 1 \\ 0 & 4 \end{bmatrix}$$

The A matrices can then be formed by placing poles on the diagonal.

$$A_1 = -1, \quad A_2 = 2, \quad A_3 = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}$$

The D matrix is equal to $G(\infty)$. So, the complete state-space system is

$$A = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & -3 \\ 1 & 0 \\ 1 & 1 \\ 0 & 4 \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

10.5.2 Controllable form state-space realization of MIMO System

We start by introducing the concept of column reduced matrix as applied to D_R matrix in the right fraction.

Let $K_{c,i}$ be the maximum degree in column i . We say D_R is column-reduced if

$$\deg(D_R) = \deg(\det D_R) = \sum_i K_{c,i}$$

We will show, in the mathematical details section below, that the column reduced D_R matrix results in the controllable form state-space realization. Equivalently if D_L in the left fraction is row reduced, not discussed here, this left fraction results in observable form realization, see [?] for proofs and other details.

A proper transfer function matrix when expressed as a right coprime column reduced fraction, then its state-space realization will automatically be both controllable and observable.

6440 **EXAMPLE 10.21**

$$D_R = \begin{bmatrix} s^2 + s & s + 1 \\ s + 2 & 1 \end{bmatrix} : \text{ where } K_{c,1} = 2, \quad K_{c,2} = 1, \quad K_{c,1} + K_{c,2} = 3$$

$$\det D_R = s^2 + s - s^2 - 2s - 2 = -2s - 2 \Rightarrow \deg(\det D_R) = 1$$

6442 Therefore, D_R is not column-reduced, as $\deg(D_R) > \deg(\det D_R)$.

6443 We can make it column-reduced by using column operations. Subtract s times the 2nd
6444 column from the 1st column:

$$\begin{bmatrix} s^2 + s & s + 1 \\ s + 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -s & 1 \end{bmatrix} = \begin{bmatrix} 0 & s + 1 \\ 2 & 1 \end{bmatrix}$$

$$K_{c,1} = 0, \quad K_{c,2} = 1, \quad K_{c,1} + K_{c,2} = 1 : \text{ column reduced}$$

6446 Let D' be the reduced form and U be the appropriate transformation matrix. Then, we
6447 say

$$D_R U = D' \Rightarrow D_R = D' U^{-1}$$

6448 Similarly,

$$N_R U = N' \Rightarrow N_R = N' U^{-1}$$

6449 Then,

$$G = N_R D_R^{-1} = N' U^{-1} (D' U^{-1})^{-1} = N' D'^{-1}$$

6450 **Mathematical Details** To obtain the state-space realization in controllable form, we, first,
6451 introduce a pseudo state ζ with length $m \times 1$, where m is the number of column of D_R .
6452 These developments are compared to the general SISO state-space realization for compar-
6453 ative and clarification purposes.

6454 **S/SO** First, we consider the single-input, single-output state space realization.

$$G = \frac{NUM}{DEN} \quad (10.31)$$

$$\frac{Y}{U} = \frac{NUM}{DEN} \quad (10.32)$$

$$\frac{Y V}{V U} = \frac{NUM}{DEN} \quad (10.33)$$

6457 where V is the Laplace transform of v , the pseudo state,

$$DEN V = U \quad (10.34)$$

$$Y = NUM V \quad (10.35)$$

6459 The general nth order characteristic equation is written as,

$$DEN = s^n + a_1 s^{n-1} + a_2 s^{n-2} \dots + a_{n-1} s + a_n \quad (10.36)$$

6460 Then, (10.34) is written in differential equation form as follows,

$$\frac{d^n V}{dt^n} = -a_1 \frac{d^{n-1} V}{dt^{n-1}} - a_2 \frac{d^{n-2} V}{dt^{n-2}} \dots - a_{n-1} \frac{dV}{dt} - a_n V + u \quad (10.37)$$

setting,

$$x_1 = \frac{d^{n-1} V}{dt^{n-1}} = \dot{x}_2 \quad (10.38)$$

$$x_2 = \frac{d^{n-2} V}{dt^{n-2}} = \dot{x}_3 \quad (10.39)$$

$$\vdots \quad (10.40)$$

$$x_{n-1} = \frac{dV}{dt} = \dot{x}_n \quad (10.41)$$

$$x_n = V \quad (10.42)$$

6461 and, therefore, $\dot{x}_1 = \frac{d^n V}{dt^n}$, we can write

$$\dot{x}_1 = -a_1 x_1 - a_2 x_2 \dots - a_{n-1} x_{n-1} - a_n x_n + u \quad (10.43)$$

6462 Finally,

$$\Psi(s)_{SISO} V = X \quad (10.44)$$

6463 where $\Psi(s)_{SISO} = [s^{n-1}; s^{n-2}; \dots; s^2; s; 1]$.

6464 *MIMO* The multi-input, multi-output state-space realization will logically follow from
6465 the SISO realization.

$$G = N_R D_R^{-1} \quad (10.45)$$

6466

$$Y = N_R D_R^{-1} U \quad (10.46)$$

6467

$$Y = N_R Z (D_R Z)^{-1} U \quad (10.47)$$

6468 where Z is the Laplace transform of ζ and

$$D_R Z = U \quad (10.48)$$

6469

$$Y = N_R Z \quad (10.49)$$

6470 Next we define

$$\Gamma = \text{diag} (s^{K_{c1}}, s^{K_{c2}}, \dots, s^{K_{cm}}) \quad (10.50)$$

where K_{ck} is the column degrees of D_R . Then,

$$\Psi(s) = \begin{bmatrix} s^{K_{c1}-1} & 0 & \\ s^{K_{c1}-2} & 0 & \\ \vdots & \vdots & \\ s & 0 & \\ 1 & 0 & \\ \hline 0 & s^{K_{c2}-1} & \\ 0 & s^{K_{c2}-2} & \text{etc...} \\ \vdots & \vdots & \\ 0 & s & \\ 0 & 1 & \\ \hline \vdots & \vdots & \\ 0 & 0 & \end{bmatrix}$$

$$= \text{block diag} \left\{ \begin{bmatrix} s^{K_{ci}-1} \\ s^{K_{ci}-2} \\ \vdots \\ s \\ 1 \end{bmatrix}, i = 1, \dots, m \right\}$$

where

$$\Psi(s)Z = X \quad (10.51)$$

and X are the system states in Laplace domain. We can always write

$$D_R = \Delta_h \Gamma + \Delta_p \Psi, \quad (10.52)$$

where Δ_h is non-singular if and only if D_R is column reduced.

The above developments should manifest the equivalency between the MIMO and SISO realizations.

Next, we illustrate this mathematical development through a simple example.

Example

$$D = \begin{bmatrix} s^2 + 3s & 0 \\ 0 & s^2 + 3s \end{bmatrix}$$

$$K_{c1} = 2, \quad K_{c2} = 2$$

$$D = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{\Delta_h} \underbrace{\begin{bmatrix} s^2 & 0 \\ 0 & s^2 \end{bmatrix}}_{\Gamma} + \underbrace{\begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 \end{bmatrix}}_{\Delta_p} \underbrace{\begin{bmatrix} s & 0 \\ 1 & 0 \\ 0 & s \\ 0 & 1 \end{bmatrix}}_{\Psi}$$

To complete this mathematical development, next, we define the state variables ($X = \Psi(s)Z$) as,

$$\begin{aligned} x_1 &= \frac{d^{K_{c1}-1}}{dt^{K_{c1}-1}} \zeta_1 \\ x_2 &= \frac{d^{K_{c1}-2}}{dt^{K_{c1}-2}} \zeta_1 \\ &\vdots \\ x_{K_{c1}} &= \zeta_1 \\ x_{K_{c1}+1} &= \frac{d^{K_{c2}-1}}{dt^{K_{c2}-1}} \zeta_2 \\ &\vdots \\ x_{K_{c1}+K_{c2}} &= \zeta_2 \end{aligned}$$

6480

6481 How many states are there?

$$K_{c1} + K_{c2} + \dots + K_{cm} = \deg D_R \quad (10.53)$$

6482 so, the total number of states are equal to the $\deg D_R$.

6483 Recall that we want

$$\dot{\underline{x}} = A\underline{x} + B\underline{u} \quad (10.54)$$

6484

$$\underline{y} = C^T \underline{x}, \quad (10.55)$$

6485 considering $G(s)$ to be strictly proper without loss of generality,

$$D_R Z = (\Delta_h \Gamma + \Delta_p \Psi) Z = U \quad (10.56)$$

6486 So,

$$\Gamma Z = -\Delta_h^{-1} \Delta_p \Psi Z + \Delta_h^{-1} U \quad (10.57)$$

6487 We know $\Psi Z = X$, so we make the substitution into the above equation,

$$\Gamma Z = -\Delta_h^{-1} \Delta_p X + \Delta_h^{-1} U \quad (10.58)$$

6488 Let's use the previous example for revealing the utility of this equation for deriving a state-
6489 space realization for a MIMO system.

6490 **EXAMPLE 10.22**

$$\begin{bmatrix} s^2 & 0 \\ 0 & s^2 \end{bmatrix}_{\Gamma} Z = -\Delta_h^{-1} \Delta_p \Psi Z + \Delta_h^{-1} U$$

6491 where,

$$X = \begin{bmatrix} s & 0 \\ 1 & 0 \\ 0 & s \\ 0 & 1 \end{bmatrix}_{\Psi} \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix}$$

6492 and therefore,

$$\begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix} = \begin{bmatrix} sZ_1 \\ Z_1 \\ sZ_2 \\ Z_2 \end{bmatrix}$$

6493 furthermore,

$$\begin{bmatrix} s^2 & 0 \\ 0 & s^2 \end{bmatrix}_{\Gamma} \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_3 \end{bmatrix}$$

6494 we then can write,

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -3 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 \\ & -\Delta_h^{-1} \Delta_p & & \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{\Delta_h^{-1}} u$$

6495 It is evident for the example above that not all the states could be included in this derivation
 6496 as the state derivatives include only the highest order derivative of each column and not all
 6497 state derivatives. Knowing this, we can then write,

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_{K_{c1}+1} \\ \dot{x}_{K_{c1}+K_{c2}+1} \\ \vdots \end{bmatrix} = \underbrace{-\Delta_h^{-1} \Delta_p}_{m \text{ rows of } A} x + \underbrace{\Delta_h^{-1}}_{m \text{ rows of } B} u \quad (10.59)$$

6498 while, in order to complete this realization, we need to add other rows which are,

$$\frac{dx_k}{dt} = x_{k-1} \quad (10.60)$$

Next we can write the outputs (Y) as,

$$\begin{aligned} Y &= N_R Z \\ &= N_l \Psi X \end{aligned} \quad (10.61)$$

Therefore,

$$y = N_l x \quad (10.62)$$

and we can then write the output matrix as,

$$C^T = N_l \quad (10.63)$$

EXAMPLE 10.23

$$G = \begin{bmatrix} \frac{4s-10}{2s+1} & \frac{3}{s+2} \\ \frac{1}{(2s+1)(s+2)} & \frac{s+2}{(s+2)^2} \end{bmatrix} = G(\infty) + G_{sp} =$$

$$\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} \frac{-12}{2s+1} & \frac{3}{s+2} \\ \frac{1}{(2s+1)(s+2)} & \frac{s+2}{(s+2)^2} \end{bmatrix}$$

where 'sp' stands for strictly proper. The strictly proper part of G is written as a MFD as follows,

$$G_{sp} = \begin{bmatrix} -6s-12 & -9 \\ 0.5 & 1 \end{bmatrix} \begin{bmatrix} s^2 + 2.5s + 1 & 2s+1 \\ 0 & s+2 \end{bmatrix}^{-1}$$

$N_R \qquad D_R$

$$K_{c1} = 2, \quad K_{c2} = 1$$

so D_R is column reduced.

$$D_R = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} s^2 & 0 \\ 0 & s \end{bmatrix} + \begin{bmatrix} 2.5 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} s & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$\Delta_h \qquad \Gamma \qquad \Delta_p \qquad \Psi$

$$-\Delta_h^{-1} \Delta_p = \begin{bmatrix} -2.5 & -1 & 3 \\ 0 & 0 & -2 \end{bmatrix}$$

We know that there are total of three states and from Equation (10.59),

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -2.5 & -1 & 3 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} u$$

From Equation (10.60),

$$\dot{x}_2 = x_1$$

Therefore,

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -2.5 & -1 & 3 \\ 1 & 0 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 & -2 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} u$$

$A \qquad B$

From Equation (10.61),

$$N_R = \begin{bmatrix} -6 & -12 & -9 \\ 0 & 0.5 & 1 \end{bmatrix} \begin{bmatrix} s & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

N_l Ψ

So we can write the output equation as,

$$y = \begin{bmatrix} -6 & -12 & -9 \\ 0 & 0.5 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} u$$

C^T D

10.6 Straightforward Computational Procedures

Previous mathematical approach for the computation of the state-space matrices can be made easier by defining the following matrices,

$$A_0 = \mathbf{block\ diag} \left\{ \begin{bmatrix} 0 & & 0 \\ 1 & \ddots & \\ 0 & 1 & 0 \end{bmatrix}, K_{ci} \times K_{ci}, i = 1, \dots, m \right\}$$

such when $K_{ci} = 2$, for two blocks, we have,

$$A_0 = \begin{bmatrix} 0 & 0 & & & \\ 1 & 0 & & & \\ \hdashline & & & & \\ & & 0 & 0 & \\ & & 1 & 0 & \end{bmatrix}$$

and when $K_{ci} = 3$ for two blocks, we can write,

$$A_0 = \begin{bmatrix} 0 & 0 & 0 & & & \\ 1 & 0 & 0 & & & \\ 0 & 1 & 0 & & & \\ \hdashline & & & & & \\ & & & 0 & 0 & 0 \\ & & & 1 & 0 & 0 \\ & & & 0 & 1 & 0 \end{bmatrix}$$

and

$$B_0 = \mathbf{block\ diag} \left\{ \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, K_{ci} \times 1, i = 1, \dots, m \right\}$$

Then it can be shown that the state-space matrices can be written as,

$$A = A_0 - B_0 \Delta_h^{-1} \Delta_p \quad (10.64)$$

$$B = B_0 \Delta_h^{-1} \quad (10.65)$$

$$C^T = N_l \quad (10.66)$$

$$D = G(\infty) \quad (10.67)$$

EXAMPLE 10.24

$$G = \begin{bmatrix} s+1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} s^2+3s & 0 \\ 0 & s^2+3s \end{bmatrix}^{-1}$$

$$K_{c1} = 2, \quad K_{c2} = 2$$

$$D_R = \underset{\Delta_h}{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}} \underset{\Gamma}{\begin{bmatrix} s^2 & 0 \\ 0 & s^2 \end{bmatrix}} + \underset{\Delta_p}{\begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 \end{bmatrix}} \underset{\Psi}{\begin{bmatrix} s & 0 \\ 1 & 0 \\ 0 & s \\ 0 & 1 \end{bmatrix}}$$

$$-\Delta_h \Delta_p = \begin{bmatrix} -3 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 \end{bmatrix}$$

$$N_R = \begin{bmatrix} s+1 & 1 \\ 1 & 1 \end{bmatrix} = \underset{N_l}{\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}} \underset{\Psi}{\begin{bmatrix} s & 0 \\ 1 & 0 \\ 0 & s \\ 0 & 1 \end{bmatrix}}$$

Using the equations above

$$A_0 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad B_0 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Therefore,

$$A = \begin{bmatrix} -3 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad C^T = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}, \quad D = 0$$

EXAMPLE 10.25

$$G = \begin{bmatrix} s+2 & 0 \\ s+1 & 1 \end{bmatrix} \begin{bmatrix} (s+1)^2(s+2) & -(s+1) \\ 0 & s+2 \end{bmatrix}^{-1}$$

$$K_{c1} = 3, \quad K_{c2} = 1$$

$$D_R = \underset{\Delta_h}{\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}} \underset{\Gamma}{\begin{bmatrix} s^3 & 0 \\ 0 & s \end{bmatrix}} + \underset{\Delta_p}{\begin{bmatrix} 4 & 5 & 2 & -1 \\ 0 & 0 & 0 & 2 \end{bmatrix}} \underset{\Psi}{\begin{bmatrix} s^2 & 0 \\ s & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}}$$

$$-\Delta_h \Delta_p = \begin{bmatrix} -4 & -5 & -2 & -1 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$

$$N_R = \underset{N_l}{\begin{bmatrix} 0 & 1 & 2 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}} \underset{\Psi}{\begin{bmatrix} s^2 & 0 \\ s & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}} = \begin{bmatrix} s+2 & 0 \\ s+1 & 1 \end{bmatrix}$$

Then,

$$A_0 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad B_0 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Therefore,

$$A = \begin{bmatrix} -4 & -5 & -2 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad C^T = \begin{bmatrix} 0 & 1 & 2 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}, \quad D = 0$$

CHAPTER 11

EIGENVALUES AND SINGULAR VALUES

In this chapter we discuss some of the properties (eigenvectors and eigenvalues) of matrices and our concentration will be on square matrices. We will then discuss singular value decomposition (SVD) and generalize the results for both square and non-square matrices. There are many many texts and technical publications [?] which discuss these topics in details. Our hope here is to provide some insight into these topics which are very relevant in MIMO control design. We start this chapter by discussing eigenvalues and eigenvectors next.

11.1 Eigenvalues & Eigenvectors

Suppose we have a square matrix A . Matrices act on vectors like a function: in goes vector $\underline{x}$ and out comes vector $A\underline{x}$ (like $T(\underline{x})$, but multidimensional). In particular, we are interested in vectors that come out parallel to the vector that went in.

$$A\underline{x} = \lambda\underline{x} \quad \text{such that} \quad A\underline{x} \parallel \underline{x}$$

Such vectors are known as eigenvectors, while the values λ are known as eigenvalues. So, if the multiplication of a vector by a general matrix results in simultaneously stretching/shrinking and rotating the vector, then if this vector is an eigenvector, the outcome would be equal to only stretching/shrinking this vector by a scalar called an eigenvalue.

We can state the above in a different way by first assuming the following transformation, $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ then,

$$T(\underline{x}) = \lambda \underline{x} \quad (11.1)$$

if and only if $\underline{x}$ is an eigenvector. if $T(\underline{x}) = A\underline{x}$ where A is any matrix, then,

$$A\underline{x} = \lambda \underline{x} \quad (11.2)$$

However, we have two unknowns (the scalar λ and the eigenvector $\underline{x}$) but only one equation. So how can we find both the eigenvalues and eigenvectors?

First, let's consider the case where $\lambda = 0$: $A\underline{x} = 0$, so A is singular (meaning it takes some vector and maps it to zero) and the eigenvectors form the basis of the null space of A , denoted $\eta(A)$.

Let's now consider another example, this time a permutation matrix. Let

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

We are looking for vectors $\underline{x}$ and scalars λ such that $A\underline{x} = \lambda \underline{x}$. Intuitively,

$$\underline{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \lambda_1 = 1, \quad \Rightarrow \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\underline{x}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \lambda_2 = -1, \quad \Rightarrow \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = -1 \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

It is important to note that for two-dimensional matrices, we have shortcuts for finding the eigenvalues and eigenvectors,

$$\lambda_1 + \lambda_2 = \text{trace}(A) = \sum \text{diag}(A), \quad \det(A) = \lambda_1 \lambda_2$$

How do we use this to solve for the eigenvalues and eigenvectors? Let's rearrange Equation 11.2 by writing,

$$A\underline{x} - \lambda I\underline{x} = 0 \quad \Rightarrow \quad \underbrace{(A - \lambda I)}_{\text{Singular}} \underline{x} = 0 : \quad \underline{x} \in \eta(A - \lambda I)$$

We will demonstrate the above computations for eigenvalues and eigenvectors through the following three examples,

EXAMPLE 11.1

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \Rightarrow \text{trace}(A) = \lambda_1 + \lambda_2 = 6, \quad \det(A) = \lambda_1 \lambda_2 = 8 \Rightarrow \lambda_1 = 2, \quad \lambda_2 = 4$$

$$\lambda_1 = 2 : \left(\overbrace{\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}}^{A - \lambda_1 I} \right) \underline{x}_1 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \underline{x}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \underline{x}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\lambda_2 = 4 : \left(\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \right) \underline{x}_2 = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \underline{x}_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \underline{x}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

EXAMPLE 11.2

Rotation matrix:

$$Q = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Let $\theta = 90^\circ$ (90° rotation).

$$Q = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \Rightarrow \lambda_1 + \lambda_2 = 0, \quad \lambda_1 \lambda_2 = 1 \Rightarrow \lambda_1 = i, \quad \lambda_2 = -i$$

It is interesting to note that for rotation matrices the eigenvalues are imaginary and there are no real eigenvectors.

EXAMPLE 11.3

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix} \Rightarrow \lambda_1 + \lambda_2 = 6, \quad \lambda_1 \lambda_2 = 9 \Rightarrow \lambda_1 = 3, \quad \lambda_2 = 3$$

$$\lambda_1 = 3 : \left(\begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \right) \underline{x}_1 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \underline{x}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \underline{x}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

What happens for λ_2 ? There is no other linearly independent eigenvector, so $\underline{x}_2$ does not exist. This is a degenerate case. This should be evident by looking at the matrix A as the vector (first column) $\begin{bmatrix} 3 \\ 0 \end{bmatrix}$ is one dimensional and as it stretches or shrinks, the vector remains on its own span.

In the previous section, we discussed the concepts of eigenvalues and eigenvectors. In this section, we briefly introduce the application of eigenvalues and eigenvectors in matrix diagonalization including some important properties of matrices.

11.2 Matrix Diagonalization

We are interested in diagonalizing a matrix (or alternately, eigenvalue decoupling).

Suppose a matrix A has n independent eigenvectors/eigenvalues. Then, $A\underline{x}_i = \lambda_i\underline{x}_i$ for all $i = 1, \dots, n$. We can express this equation for all eigenvalues and eigenvectors by using a matrix representation.

$$\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n), \quad S = [\underline{x}_1, \dots, \underline{x}_n]$$

where, Λ is a diagonal matrix of all eigenvalues and S is a matrix consists of all eigenvectors. Then,

$$AS = A \begin{bmatrix} | & | & | \\ \underline{x}_1 & \cdots & \underline{x}_n \\ | & | & | \end{bmatrix} = [\lambda_1\underline{x}_1, \dots, \lambda_n\underline{x}_n] = [\underline{x}_1, \dots, \underline{x}_n] \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} = S\Lambda$$

To diagonalize matrix A , S must be invertible. If so, then we can pre-multiply both sides by S^{-1} ,

$$S^{-1}AS = \Lambda$$

Diagonalization of matrix A with dimension of $n \times n$ could be possible if matrix S consists of n independent eigenvectors, in addition to A must be square. There are some matrices that do not contain n independent eigenvectors (consider the degenerate case example). Therefore, we can ask the following question,

Which matrices are diagonalizable?

Matrix A must have n linearly independent eigenvectors. If there are no repeated eigenvalues, A is diagonalizable. If A has repeated eigenvalues, then A may or may not have sufficient linearly independent eigenvectors.

If matrix A is diagonalizable its exponential can be computed as follows,

$$\begin{aligned} A\underline{x} &= \lambda\underline{x} \\ A^2\underline{x} &= \lambda A\underline{x} = \lambda^2\underline{x} \\ A^2 &= (S\Lambda S^{-1})(S\Lambda S^{-1}) = S\Lambda^2 S^{-1} \\ A^k &= S\Lambda^k S^{-1} \end{aligned}$$

It can be shown that for symmetric real matrices, where $A = A^T$, their eigenvectors are orthogonal ($Q^{-1} = Q^T$ where, Q is the normalized eigenvector matrix, in addition,

the eigenvalues of symmetric matrices are all real. We could always then diagonalize a symmetric matrix $A = Q\Lambda Q^T$, as shown in the following example,

■ **EXAMPLE 11.4**

$$\begin{aligned}
 A &= \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \Rightarrow \lambda_1 + \lambda_2 = 6, \quad \lambda_1 \lambda_2 = 8 \Rightarrow \lambda_1 = 4, \quad \lambda_2 = 2 \\
 \lambda_1 = 4 : \quad &\left(\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \right) \underline{x}_1 = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \underline{x}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \underline{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\
 \lambda_2 = 2 : \quad &\left(\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \right) \underline{x}_2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \underline{x}_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \underline{x}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\
 \Lambda &= \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}, \quad S = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \Rightarrow Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \\
 &\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}
 \end{aligned}$$

We can generalize the diagonalizability criterion for a matrix A but before we do that, we need to introduce the algebraic and geometric multiplicities.

Definition 11.1. Algebraic Multiplicity

Let $(A - \lambda I)\underline{x} = 0$, where, $\underline{x} \neq 0 \Rightarrow \det(A - \lambda I) = 0 = \underbrace{\delta(\lambda)}_{\text{Polyn of deg. } n}$,

where

$\delta(\lambda) = (\lambda - \lambda_1)^{a_1} (\lambda - \lambda_2)^{a_2} \cdots (\lambda - \lambda_d)^{a_d}$, $\lambda_1, \lambda_2, \dots, \lambda_d$ are distinct eigenvalues.

Then

“ a_k ”, $k \in 1, \dots, d$ is the **Algebraic Multiplicity**.

Definition 11.2. Geometric Multiplicity

Find the eigenvector $\underline{x}_k$ from $(A - \lambda I)\underline{x} = 0$, $\underline{x} \neq 0$, then, compute $g_k = n - \text{rank}(A - \lambda_k I)$,

where

“ g_k ” is the **Geometric Multiplicity**.

Geometric Multiplicity is the null space of $(A - \lambda_k I)$ or $g_k \in \eta(A - \lambda_k I)$. Independent solutions can show $1 \leq g_k \leq a_k$. There are classes of matrices that $g_k \equiv a_k$, these classes are called “non-defective” and it can always be diagonalized, i.e. symmetric matrices.

6623 **EXAMPLE 11.5**

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \Rightarrow \lambda_1 + \lambda_2 = 4, \quad \lambda_1 \lambda_2 = 4 \Rightarrow \lambda_1 = 2, \quad \lambda_2 = 2$$

$$\lambda_1 = 2 : \left(\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \right) \underline{x}_1 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \underline{x}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \underline{x}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

6625 But, there is no second linearly independent eigenvector. So, the algebraic multiplicity
6626 is 2 ($a_k = 2$, number of repeated eigenvalues) but the geometric multiplicity is 1
6627 ($g_k = 1$, number of independent eigenvectors). Note that

$$1 \leq g_k \leq a_k$$

6628 When $g_k = a_k$, the matrix is diagonalizable.

6629 **11.2.1 Classes of Diagonalizable Matrices**

6630 If a matrix A is non-defective, then

$$T = [\underline{x}_1 | \underline{x}_2 | \cdots | \underline{x}_n] \quad (11.3)$$

6631 is non-singular.

6632 An important class is the set of ‘normal’ matrices which is non-defective.

6633 Let’s first introduce some notations.

6634 We say $\underline{W}$ and $\underline{V}$ are orthogonal if

$$\langle \underline{W}^*, \underline{V} \rangle = \sum_K \bar{W}_K V_K = 0 \quad (11.4)$$

6635 And pseudo-orthogonal if

$$\langle \underline{W}^T, \underline{V} \rangle = \sum_K W_K V_K = 0 \quad (11.5)$$

6636 Where the bracket represents the inner product, the star represents complex conjugate
6637 transpose, the bar represents complex conjugate, and T represents transpose.

6638 **EXAMPLE 11.6**

6639 Let $\underline{W} = \begin{bmatrix} 1 \\ j \end{bmatrix}$ then $\underline{W}^T \underline{W} = [1 \quad j] \begin{bmatrix} 1 \\ j \end{bmatrix} = 0$ where this is pseudo-orthogonal.

6640 **Definition 11.3.** Matrix $\underline{N}$ is Normal if it commutes with its own adjoint (complex con-
6641 jugate transpose),

$$\underline{N} \underline{N}^* = \underline{N}^* \underline{N} \quad (11.6)$$

This class has the following properties,

1) non-defective

$$g_k \equiv a_k$$

2) can select $\underline{x}_k$ such that

$$\underline{x}_k^* \underline{x}_i = 0, i \neq k$$

EXAMPLE 11.7

Assume $\underline{Q}$ is real and symmetric then $\underline{Q} = \underline{Q}^T$ and

$$\underline{Q}\underline{Q}^* = \underline{Q}\underline{Q}^T = \underline{Q}^2 = \underline{Q}^*\underline{Q} = \underline{Q}^T\underline{Q}$$

Therefore, $\underline{Q}$ is Normal. Additionally, eigenvalues are real. What if $\underline{Q}$ is complex and symmetric? Then it could be shown that $\underline{Q}$ is not Normal. This is left as an exercise.

Definition 11.4. A complex matrix $\underline{H}$ is Hermitian if

$$\underline{H}\underline{H}^* = \underline{H}^2 = \underline{H}^*\underline{H} \text{ then}$$

$\underline{H}$ is Normal. Additionally the eigenvalues of $\underline{H}$ is real as is illustrated in the example below.

EXAMPLE 11.8

A Hermitian matrix is a matrix for which $H = H^*$. For instance,

$$A = \begin{bmatrix} 3 & 2-i & -3i \\ 2+i & 0 & 1-i \\ 3i & 1+i & 0 \end{bmatrix}, \quad \bar{A} = \begin{bmatrix} 3 & 2+i & 3i \\ 2-i & 0 & 1+i \\ -3i & 1-i & 0 \end{bmatrix}$$

$$\Rightarrow A^* = \bar{A}^T = \begin{bmatrix} 3 & 2-i & -3i \\ 2+i & 0 & 1-i \\ 3i & 1+i & 0 \end{bmatrix}$$

So, A is Hermitian.

EXAMPLE 11.9

In this example, we will show that the eigenvalues of a Hermitian matrix are real. We write,

$$\underline{H}\underline{x} = \lambda\underline{x} \tag{11.7}$$

We obtain the complex conjugate of the Equation (11.7),

$$\underline{x}^* \underline{H}^* = \bar{\lambda} \underline{x}^* \quad (11.8)$$

then as H is Hermitian, we can write,

$$\underline{x}^* \underline{H} = \bar{\lambda} \underline{x}^* \quad (11.9)$$

From Equation (11.7), it follows,

$$\underline{x}^* \underline{H} \underline{x} = \lambda \underline{x}^* \underline{x} \quad (11.10)$$

From Equation (11.9), it follows,

$$\underline{x}^* \underline{H} \underline{x} = \bar{\lambda} \underline{x}^* \underline{x} \quad (11.11)$$

Subtract Equation (11.11) from (11.10),

$$0 = (\lambda - \bar{\lambda}) \underline{x}^* \underline{x} \quad (11.12)$$

Knowing that $\underline{x}^* \underline{x} = \sum_K \bar{x}_k x_K = \sum (x_K^2) > 0$, then,

$(\lambda = \bar{\lambda})$ which means eigenvalues are real

EXAMPLE 11.10

Let's assume matrix A is given as,

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

then

$$\delta(\lambda) = (1 - \lambda)^2$$

where $\lambda = 1$ and the algebraic multiplicity $a = 2$.

This matrix is diagonal with algebraic multiplicity of 2. Therefore, the geometric multiplicity of this matrix has to be equal to 2 and this is both evident from the matrix A and can also be computed from the null space of $([A - \lambda_k I])$.

Let's check to find out if A is normal,

6684 $A^* = A^T$ and $A^T A = A A^T \rightarrow$ Normal In addition, the eigenvectors are

$$\underline{x}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

6685 and

$$\underline{x}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

6686 which are orthogonal.

6687 **EXAMPLE 11.11**

6688 Let's assume matrix A is given as,

$$6689 \quad A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

6690
6691 then

$$6692 \quad \delta(\lambda) = (\lambda)^2$$

6693 where $\lambda = 0$ and the algebraic multiplicity $a = 2$.

6694 Let's check to find out if A is normal,

$$6695 \quad A A^T = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$6696 \quad 6697 \quad A^T A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \neq A A^T$$

6698 therefore, A is not normal. In addition, the eigenvector is

$$\underline{x}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

6699 and there is no second linearly independent eigenvector, so the geometric multiplicity
6700 is 1.

Definition 11.5. A complex matrix $\underline{U}$ is Unitary if

$$\underline{U}^* = \underline{U}^{-1}$$

For this matrix, we can write,

$$\underline{U}^* \underline{U} = \underline{U}^{-1} \underline{U} = \underline{I} = \underline{U} \underline{U}^*$$

In addition, unitary matrices are Normal, non-defective, eigenvectors are orthogonal,

and

$$|\lambda| = 1$$

■ **EXAMPLE 11.12**

For instance,

$$A = \begin{bmatrix} \frac{1+i}{2} & \frac{1-i}{2} \\ \frac{1-i}{2} & \frac{1+i}{2} \end{bmatrix}$$

where $A^* = A^{-1}$.

■ **EXAMPLE 11.13**

A real matrix $\underline{T}$ is Unitary if

$$\underline{T}^T = \underline{T}^{-1}$$

For this matrix, we can write,

$$\underline{T}^* \underline{T} = \underline{T}^T \underline{T} = \underline{T}^{-1} \underline{T} = \underline{I} = \underline{T} \underline{T}^*$$

In addition, $\underline{T}$ is Normal, non-defective, eigenvectors are orthogonal, and

$$|\lambda_k| = 1$$

There are many other important matrix sub-classes. Suppose that $\underline{H}$ is Hermitian so that,

$$\underline{x}_i^* \underline{x}_k = 0 \quad \text{for all } i \neq k$$

suppose we normalize $\underline{x}_k$ such that,

$$\underline{x}_k^* \underline{x}_k = \sum_k |\underline{x}_k|^2 = 1$$

and

$$\underline{U} = [\underline{x}_1 | \underline{x}_2 | \underline{x}_3 \cdots | \underline{x}_n]$$

where $\underline{U}$ is known as the Modal matrix. Then,

$$\underline{U}^* \underline{U} = \begin{bmatrix} \underline{x}_1^* \\ \underline{x}_2^* \\ \vdots \\ \underline{x}_n^* \end{bmatrix} [\underline{x}_1 | \underline{x}_2 \cdots | \underline{x}_n] = \underline{I}$$

then $U^* = U^{-1}$, hence, U is unitary. It should be noted that the modal matrix of any normal matrix is unitary.

One may question that why is it necessary to decouple/diagonalize a matrix? Diagonalization of a matrix generally reduces computational overhead, e.g. numerical optimization. But, most importantly, as will be pointed out in the rest of this book, it plays a significant role in MIMO control system design. But, not all matrices can be diagonalized using the approaches presented in the last section.

In this section, we present an approach which is general enough to be applied for diagonalization of all matrices.

11.3 Singular Value Decomposition

MIMO systems often do not result in square matrices, nor have independent eigenvectors, and so on. So, we turn our attention to singular value decomposition (SVD) to analyze these systems.

The origin of SVD comes from eigenvector analysis of symmetric matrices $A = A^T$, which have orthogonal eigenvectors. If we have a non-symmetric matrix, is it possible to transform this matrix to a symmetric one? For instance,

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \Rightarrow AA^T = \begin{bmatrix} 5 & 11 \\ 11 & 25 \end{bmatrix} \text{ Symmetric!}$$

So, any real non-symmetric matrices could be transformed to a symmetric one by multiplying it to its transpose. Furthermore, AA^T or $A^T A$ is always a square matrix regardless of the A matrix. Therefore, we can always find an eigenvalue decomposition for AA^T or $A^T A$.

11.3.1 What is a Singular Value Decomposition?

All square or non-square matrices can be decomposed into a diagonal Σ and an orthonormal basis (normalized orthogonal eigenvectors.)¹ U, V .

$$A_{(r \times m)} = U_{(r \times r)} \Sigma_{(r \times m)} V^T_{(m \times m)}$$

¹Normalized orthogonal eigenvectors always form an orthonormal basis, but basis vectors do not always form eigenvectors

6746 where the left singular vectors are the column vectors of U , the right singular vectors
 6747 are the row vectors of V^T , and Σ is a diagonal matrices of singular values. The singular
 6748 values are always real if the matrix A is real.

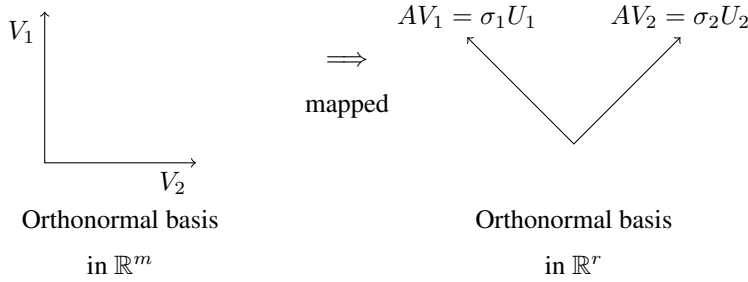


Figure 11.1: SVD orthonormal basis.

6749 The SVD and the eigenvalue decomposition have similarities which we investigate next.
 6750 Because U and V are orthonormal, we have

$$AV = U \underbrace{\Sigma V^T V}_{=1} = U \Sigma, \quad U^T A = \underbrace{U^T U}_{=1} \Sigma V^T = \Sigma V^T$$

6751 That is to say, $V^T = V^{-1}$ and $U^T = U^{-1}$, therefore, these matrices are unitary. For a
 6752 singular value σ_i , V_i is defined as the right singular vector and U_i is defined as the left
 6753 singular vector.

$$AV_i = \sigma_i U_i$$

6754

(Analogous to $A\underline{x} = \lambda\underline{x}$)

6755 This mapping is illustrated in Figure (??). The size of the matrix Σ is the same as the
 6756 matrix A , while U and V are square. The null space of A presents itself as zeros on the
 6757 diagonal of the matrix Σ , as shown below,

$$A[V_1 \ V_2 \ \cdots \ V_m] = [U_1 \ U_2 \ \cdots \ U_r] \Sigma$$

6758

$$\text{if } r = m \text{ then } \Sigma = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}$$

6759

$$\text{if } r < m \text{ then } \Sigma = \begin{bmatrix} \sigma_1 & & & & \\ & \ddots & & & \\ & & \sigma_r & 0 & \cdots & 0 \end{bmatrix}$$

6760

$$\text{if } r > m \text{ then } \Sigma = \begin{bmatrix} \sigma_1 & & & \\ & \ddots & & \\ & & \sigma_m & \\ & & 0 & \\ & & \vdots & \\ & & 0 & \end{bmatrix}$$

6761 Typically, singular values are ordered from the largest to the smallest. That is, $\sigma_1 \geq \cdots \geq$

6762 $\sigma_{\min(r,m)}$

How do singular values relate to eigenvalues and singular vectors relate to eigenvectors?

We can write the following relationships,

$$AV = U\Sigma$$

$$A = U\Sigma V^{-1} = U\Sigma V^T$$

$$A^T A = V\Sigma^T U^T U\Sigma V^T$$

$$\underbrace{A^T A}_{\text{square}} = V\Sigma^T \Sigma V^T$$

From the above development, we can state that the right singular vectors V are exactly the eigenvectors of $A^T A$ and the square of the singular values $\Sigma^T \Sigma$, where the non-zero value of this matrix is equal to σ^2 , are exactly the eigenvalues of $A^T A$. Similarly, we can write relate the left singular vectors U to the eigenvectors by writing the following expressions,

$$A^T U = V\Sigma^T$$

$$A^T = V\Sigma^T U^{-1} = V\Sigma^T U^T$$

$$AA^T = U\Sigma V^T V\Sigma^T U^T$$

$$AA^T = U\Sigma\Sigma^T U^T$$

6763 Therefore, the left singular vectors, U are the eigenvectors of AA^T and the square of the
6764 singular values $\Sigma\Sigma^T$ are the eigenvalues of AA^T .

6765  **EXAMPLE 11.14**

$$A = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix}$$

6766 V_1, V_2 are in row space $\mathbb{R}^2$. U_1, U_2 are in column space $\mathbb{R}^2$. $\sigma_1 > 0, \sigma_2 > 0$.

$$A^T A = \begin{bmatrix} 4 & -3 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} = \begin{bmatrix} 25 & 7 \\ 7 & 25 \end{bmatrix}$$

6767

$$\lambda_1 = 32 \Rightarrow V_{1n} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow V_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

6768

$$\lambda_2 = 18 \Rightarrow V_{2n} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \Rightarrow V_2 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

6769

$$AA^T = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 32 & 0 \\ 0 & 18 \end{bmatrix}$$

$$\lambda_1 = 32 \Rightarrow U_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \lambda_2 = 18 \Rightarrow U_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\sigma_1 = \sqrt{\lambda_1} = \sqrt{32}, \quad \sigma_2 = \sqrt{\lambda_2} = \sqrt{18}$$

$$\begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{32} & 0 \\ 0 & \sqrt{18} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \dots & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \dots & \frac{1}{\sqrt{2}} \end{bmatrix}$$

6770 11.3.2 Orthonormal Vectors

6771 A set of vectors are normal if

$$q_i^T q_j = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

6772 for

$$Q = [q_1 \quad \dots \quad q_n]$$

6773 so that

$$Q^T Q = \begin{bmatrix} q_1^T \\ \vdots \\ q_n^T \end{bmatrix} [q_1 \quad \dots \quad q_n] = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix} = I_n$$

6774 If Q is square, then $Q^T Q = I$, which tells us that $Q^T = Q^{-1}$.

6775 For any independent vectors a and b , Gram-Schmidt process (17.2) could be to find

6776 orthonormal basis vectors.

6777 **EXAMPLE 11.15**

$$A = \begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix}$$

6778 The rows and columns of A are not linearly independent, so, in this example, there
 6779 will be a singular value (square root of an eigenvalue) at 0, and only one linearly
 6780 independent right and one linearly independent left singular vectors. The second left
 6781 and right singular vectors, as will be shown in this example, will be computed from
 6782 the null space of $A(\eta(A))$.

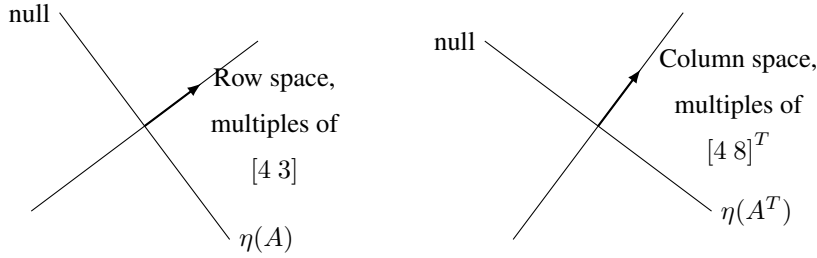


Figure 11.2: Row and column spaces of A .

6783 What should V be? We can define V_1 as basis of the row space of A .

$$V_1 = \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}$$

6784 Then, the only orthogonal vector V_1 is the basis of the null space of A ($\eta(A)$).

$$V_2 = \begin{bmatrix} 3 \\ 5 \\ -4 \end{bmatrix}$$

6785 Similarly, we choose U to contain the basis of the column space of A and the basis of
 6786 the null space of A^T ($\eta(A^T)$).

$$U_1 = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}, \quad U_2 = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} \end{bmatrix}$$

6787 We can find the singular values from either $A^T A$ or AA^T .

$$AA^T = \begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix} \begin{bmatrix} 4 & 8 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 25 & 50 \\ 50 & 100 \end{bmatrix} \Rightarrow \sigma_1^2 = \lambda_1 = 125, \quad \sigma_2^2 = \lambda_2 = 0$$

6788 So,

$$\begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \sqrt{125} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{4}{5} & \frac{3}{5} \\ \frac{3}{5} & -\frac{4}{5} \end{bmatrix}$$

6789 **EXAMPLE 11.16**

6790 In this example, we illustrate a different approach in the calculation of the left singular
6791 vectors U after having obtained the right singular vectors V .

6792 Consider a matrix C .

$$C = \begin{bmatrix} 5 & 5 \\ -1 & 7 \end{bmatrix}$$

We want to find U and V such that $C = U\Sigma V^T$, we can then write,

$$C^T C = V\Sigma^T \Sigma V^T \quad (1)$$

and then use this equation to compute right singular vectors V , furthermore,

$$CV = U\Sigma \quad (2)$$

So, U can be computed from (2), by multiplying both sides of (2) with Σ^{-1} ,

$$CV\Sigma^{-1} = U \quad (3)$$

6793 This computation is provided below,

$$C^T C = \begin{bmatrix} 26 & 18 \\ 18 & 74 \end{bmatrix} \Rightarrow \det(C^T C - \lambda I) = \lambda^2 - 100\lambda + 1600 = (\lambda - 20)(\lambda - 80)$$

$$\lambda_1 = 80, \quad \lambda_2 = 20$$

$$C^T C - 80I = \begin{bmatrix} -54 & 18 \\ 18 & -6 \end{bmatrix} \Rightarrow V_1 = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$C^T C - 20I = \begin{bmatrix} 6 & 18 \\ 18 & 54 \end{bmatrix} \Rightarrow V_2 = \frac{1}{\sqrt{10}} \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

$$V = \begin{bmatrix} \frac{1}{\sqrt{10}} & -\frac{3}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 4\sqrt{5} & 0 \\ 0 & 2\sqrt{5} \end{bmatrix}$$

6798 So, with C , Σ , and V known, we can work backwards to find U with the correct signs.

$$CV = U\Sigma \Rightarrow CV\Sigma^{-1} = U$$

$$U = \begin{bmatrix} 5 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{10}} & -\frac{3}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix} \begin{bmatrix} 4\sqrt{5} & 0 \\ 0 & 2\sqrt{5} \end{bmatrix}^{-1}$$

$$\Rightarrow U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

6801 We will end this section by summarizing our development and discussion about the SVDs.

6802 We will generalize the SVDs for all square and non-square, real and imaginary matrices.

11.4 Singular Value Properties

Let M be a $r \times m$ matrix that may have complex elements. k will denote the k th singular value, left singular vector, or right singular vector, as appropriate. M^* denotes the complex conjugate transpose of M . Without loss of generality, to be more specific, we will change our notation from V_k for the right singular vector to r_k and similarly, for the left singular vector from U_k to l_k .

Definition 11.6. σ_k is a singular value, r_k is a right singular vector, and l_k is a left singular vector of M if

$$1. \sigma_k \text{ is real and } \geq 0.$$

$$2. Mr_k = l_k \sigma_k, \quad k \leq \text{rank } M$$

$$Mr_k = 0, \quad k > \text{rank } M$$

$$3. M^*l_k = r_k \sigma_k, \quad k \leq \text{rank } M$$

$$M^*l_k = 0, \quad k > \text{rank } M$$

where M^* is the complex conjugate transpose of M .

Note: Consider M^* times Mr_k .

$$M^*Mr_k = M^*l_k \sigma_k$$

$$M^*Mr_k = \sigma_k^2 r_k$$

similar to the eigenvalue problem.

Consider a matrix M and vectors $\underline{x}$ and $\underline{y}$ such that

$$\underline{y} = M\underline{x}$$

We can define a scalar q such that

$$q = \underline{x}^* M^* M \underline{x}$$

$$q = \underline{y}^* \underline{y} = \sum_{k=1}^m |y_k|^2 \geq 0$$

So, q is positive definite or positive semi-definite (non-negative definite) and

$$\lambda_k(M^*M) \geq 0$$

Therefore, we can always find real singular values.

$$\sigma_k(M^*M) = +\sqrt{\lambda_k(M^*M)}$$

We can now define the procedure to find singular values and singular vectors:

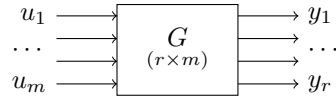
1. From M , form M^*M and find the eigenvalues λ_k and normalized eigenvectors $\underline{e}_k$.
2. Set $\sigma_k = \sqrt{\lambda_k}$, $\underline{r}_k = \underline{e}_k$ and generate $\underline{l}_k = M\underline{r}_k\sigma_k^{-1}$ and, if necessary, $\underline{l}_k$ from $M^*\underline{l}_k = 0$.

So far, we have discussed both eigenvalue and singular value decompositions. In this section, we will further discuss and compare these two decompositions to develop a better understanding and insight in how to use these methodologies in practical applications.

11.5 Comparison of Eigenvalue and Singular Value Decompositions

The development below is provided in [?], we reiterate this here for completeness.

Consider a static system with m inputs and r outputs.



Let $g_{ij}(\omega)$ represent the harmonic (sinusoidal) response from input j to output i .

$$u_j(t) = u_{j0} \sin(\omega t + \alpha_j) \Rightarrow y_i(t) = y_{i0} \sin(\omega t + \beta_i)$$

where the amplification (gain) and the phase shift may be obtained from the complex number $g_{ij}(j\omega)$ as follows.

$$|g_{ij}(j\omega)| = \frac{y_{i0}}{u_{j0}}, \quad \angle g_{ij}(j\omega) = \beta_i - \alpha_j$$

In phasor notation,

$$y_i(\omega) = g_{ij}(j\omega)u_j(\omega)$$

where

$$u_j(\omega) = u_{j0}e^{j\alpha_j}, \quad y_i(\omega) = y_{i0}e^{j\beta_i}$$

The overall response to simultaneous input signals, of the same frequency, in several channels is

$$y_i(\omega) = g_{i1}(j\omega)u_1(\omega) + g_{i2}(j\omega)u_2(\omega) + \dots = \sum_{j=1}^m g_{ij}(j\omega)u_j(\omega)$$

In matrix form,

$$y(\omega) = G(j\omega)u(\omega)$$

$$u(\omega) = \begin{bmatrix} u_1(\omega) \\ u_2(\omega) \\ \vdots \\ u_m(\omega) \end{bmatrix}, \quad y(\omega) = \begin{bmatrix} y_1(\omega) \\ y_2(\omega) \\ \vdots \\ y_r(\omega) \end{bmatrix}$$

It can be stated that eigenvalues $|\lambda(G(j\omega))|$ are a poor measure of gain as they do not provide a useful means of generalizing the SISO gain. As mentioned, eigenvalues can only be computed for square systems, and even then they can be very misleading. For instance, take $y = Gu$ with

$$G = \begin{bmatrix} 0 & 10 \\ 0 & 0 \end{bmatrix}$$

which has both eigenvalues equal to zero. However, to conclude from the eigenvalues that the system gain is zero is clearly misleading.

$$\text{Input: } u = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow \text{Output: } y = \begin{bmatrix} 10 \\ 0 \end{bmatrix}$$

We can ask, what seems to be the problem? The eigenvalues measure the gain for the special case when the inputs and the outputs are in the same direction, namely in the direction of the eigenvectors.

Let t_i be the eigenvector of G and consider $u = t_i$. Then,

$$y = Gt_i = \lambda_i t_i$$

where λ_i is the corresponding eigenvalue. We can write,

$$\frac{\|y\|}{\|u\|} = \frac{\|\lambda_i t_i\|}{\|t_i\|} = |\lambda_i|$$

The eigenvalues are a good measure for system stability but not for system performance.

11.5.1 System Gain

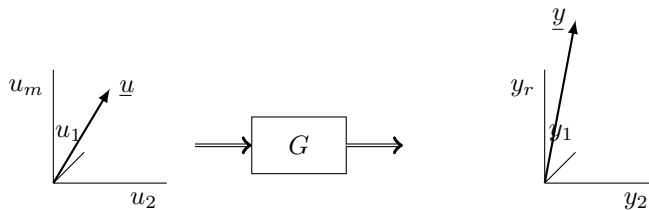


Figure 11.3: Multidimensional input/output map.

The magnitude of the input/output, as illustrated in Figure ??, is the Euclidean distance.

$$\|\underline{u}\|_E = \sqrt{\sum_k u_k^2}, \quad \|\underline{y}\|_E = \sqrt{\sum_k y_k^2}$$

“Gain”, in linear systems, is independent of input magnitude (Euclidean distance) and depends on the input frequency (similar to SISO systems),

$$Gain = \frac{\|\underline{y}\|_E}{\|\underline{u}\|_E} = \frac{\|G(j\omega)\underline{u}(\omega)\|_E}{\|\underline{u}(\omega)\|_E}$$

but now for a MIMO system, there are additional degrees of freedom and the gain depends on the direction of each input u_k or the relative magnitude of each input, e.g. u_1/u_2 , as well as the input frequency.

Let’s get a better understanding of what we mean by input direction through analyzing a static system in the next example.

EXAMPLE 11.17

For a system with two inputs

$$\underline{u} = \begin{bmatrix} u_{10} \\ u_{20} \end{bmatrix},$$

the gain is in general different for the following five inputs.

$$\begin{aligned} \underline{u}_1 &= \begin{bmatrix} 1 \\ 0 \end{bmatrix}, & \underline{u}_2 &= \begin{bmatrix} 0 \\ 1 \end{bmatrix}, & \underline{u}_3 &= \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}, \\ \underline{u}_4 &= \begin{bmatrix} 0.707 \\ -0.707 \end{bmatrix}, & \underline{u}_5 &= \begin{bmatrix} 0.6 \\ -0.8 \end{bmatrix} \end{aligned}$$

In each of these cases, the input magnitude $\|\underline{u}\|_E = 1$, but with a different relative magnitude or direction.

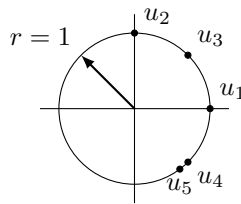


Figure 11.4: Projection of a vector onto a plane.

We can determine the outputs for the following static system G .

$$G = \begin{bmatrix} 5 & 4 \\ 3 & 2 \end{bmatrix},$$

Then,

$$y_1 = \begin{bmatrix} 5 \\ 3 \end{bmatrix}, \quad y_2 = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \quad y_3 = \begin{bmatrix} 6.36 \\ 3.53 \end{bmatrix},$$

$$y_4 = \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}, \quad y_5 = \begin{bmatrix} -0.2 \\ 0.2 \end{bmatrix}$$

$$\|y_1\|_E = 5.83, \quad \|y_2\|_E = 4.47, \quad \|y_3\|_E = 7.30,$$

$$\|y_4\|_E = 1, \quad \|y_5\|_E = 0.28$$

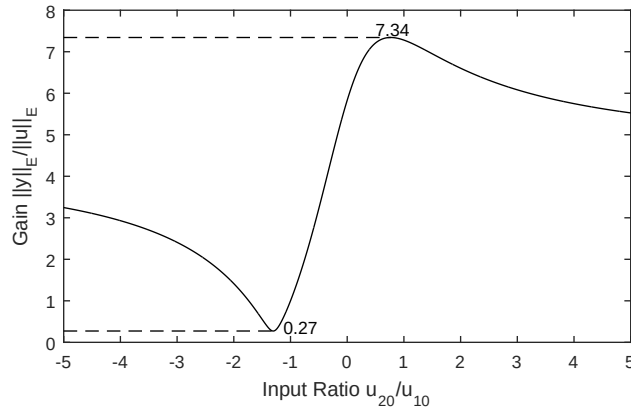


Figure 11.5: Gain as a function of input direction.

So, the singular values (σ) are precisely the system gains given an input direction, and they can be written as,

$$\max_{\underline{u} \neq 0} \frac{\|G\underline{u}\|_E}{\|\underline{u}\|_E} = \max_{\|\underline{u}\|_E=1} \|G\underline{u}\|_E = \bar{\sigma}(G)$$

$$\min_{\underline{u} \neq 0} \frac{\|G\underline{u}\|_E}{\|\underline{u}\|_E} = \min_{\|\underline{u}\|_E=1} \|G\underline{u}\|_E = \underline{\sigma}(G)$$

and hence,

$$\underline{\sigma}(G) \leq \frac{\|G\underline{u}\|_E}{\|\underline{u}\|_E} \leq \bar{\sigma}(G)$$

In MIMO systems, input directions are determined by right singular vectors and the gains by singular values. We stated that singular vectors are unitary therefore they have unit gains, so they only affect the direction while the singular values affect the magnitude.

Let $\underline{u} = u_0 \underline{r}_k$, where u_0 is the input magnitude and $\underline{r}_k$ is the direction, a right singular vector of G . We know the singular vectors are unitary,

$$\underline{r}_k^* \underline{r}_k = 1 = \underline{l}_k^* \underline{l}_k$$

6887 What is $\|\underline{u}\|_E$?

$$\|\underline{u}\|_E = \sqrt{\underline{u}^* \underline{u}} = \sqrt{u_0^2 \underline{r}_k^* \underline{r}_k} = u_0 = |\underline{u}|$$

6888 What is $\|\underline{y}\|_E$?

$$\underline{y} = G\underline{u} = u_0 G \underline{r}_k$$

6889 But, $G \underline{r}_k = \underline{r}_k \sigma_k$, from definition of SVD in the previous section. So,

$$\underline{y} = u_0 \underline{r}_k \sigma_k, \quad \|\underline{y}\|_E = \sqrt{\sigma_k^2 u_0^2 \underline{r}_k^* \underline{r}_k} = \sigma_k |\underline{u}|$$

6890 So, for $\underline{u} = u_0 \underline{r}_k$,

$$\frac{\|\underline{y}\|_E}{\|\underline{u}\|_E} = \sigma_k$$

6891 This again illustrates that singular values determine the system gains.

6892 In general, any real matrix can always be written. in this case, 2×2 , in the form of,

$$G = \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 \\ \sin \theta_1 & \cos \theta_1 \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix} \begin{bmatrix} \cos \theta_2 & \sin \theta_2 \\ -\sin \theta_2 & \cos \theta_2 \end{bmatrix}$$

6893 where the matrix in the right is the right singular vectors, r , and the matrix in the left is
 6894 the left is the left singular vectors and the middle matrix are the singular values. So, the left
 6895 and right singular vectors purely rotate the input vector and the singular values stretch the
 6896 input vector. We will generalize our findings for non-square matrices in the next section.

6897 11.6 Generalized Singular Value Decomposition

6898 Below, we will generalize the singular value decomposition for any non-square matrix M
 6899 with dimension $r \times m$. We will start by writing the following expression,

$$\begin{bmatrix} \underline{l}_1^* \\ \vdots \\ \underline{l}_r^* \end{bmatrix}_{(r \times r)} M_{(r \times m)} \begin{bmatrix} \underline{r}_1 & \cdots & \underline{r}_m \end{bmatrix}_{(m \times m)}$$

6900 where,

$$\begin{bmatrix} \underline{l}_1^* \\ \vdots \\ \underline{l}_r^* \end{bmatrix}_{(r \times r)} = U^*$$

6901 the left singular column vectors, and

$$\begin{bmatrix} \underline{r}_1 & \cdots & \underline{r}_m \end{bmatrix}_{(m \times m)} = V$$

6902 the right singular row vectors. Then we can write,

$$U^*MV = U^* [M_{r_1} \mid M_{r_2} \mid \cdots \mid M_{r_m}]$$

where we learned that $M_{r_k} = \sigma_k l_k$, hence, we can write the following relationship,

$$\begin{aligned} U^* [M_{r_1} \mid M_{r_2} \mid \cdots \mid M_{r_m}] &= U^* [\sigma_1 l_1 \mid \sigma_2 l_2 \mid \cdots \mid \sigma_m l_m] \\ &= \begin{bmatrix} l_1^* \\ l_2^* \\ \vdots \\ l_r^* \end{bmatrix} [\sigma_1 l_1 \mid \sigma_2 l_2 \mid \cdots \mid \sigma_m l_m] \end{aligned}$$

we know: $U^* = U^{-1}$ and $V^* = V^{-1}$, therefore,

$$\begin{bmatrix} l_1^* \\ l_2^* \\ \vdots \\ l_r^* \end{bmatrix} [\sigma_1 l_1 \mid \sigma_2 l_2 \mid \cdots \mid \sigma_m l_m] = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_m, 0, \dots) = \Sigma$$

So,

$$U^*MV = \Sigma \Leftrightarrow M = U\Sigma V^*$$

A reminder note on dimensions:

- If $m > r$, columns of Σ (let's call it j) $j > r$ and Σ is going to be a wide matrix with $j - r$ column zeroed.
- If $m < r$, rows of Σ (let's call it i) $i > m$ and Σ is going to be a tall matrix with $i - m$ row zeroed.

EXAMPLE 11.18

In the field of integrated vehicle dynamics and controls (IVDC), it is well known that control integration of active-chassis systems has many benefits, such as an increase in agility and stability (to name a few). The following transfer function matrix is computed at low frequency when active differential and active roll control are integrated.

$$G = \begin{bmatrix} -0.0342 & -0.1195 \\ 0.0058 & 0.0214 \end{bmatrix}$$

The following singular vectors and singular values are calculated for this transfer function matrix (TFM).

$$G = \underbrace{\begin{bmatrix} -0.9845 & 0.1756 \\ 0.1756 & 0.9845 \end{bmatrix}}_U \underbrace{\begin{bmatrix} l_1 & l_2 \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} 0.2747 & 0.9615 \\ -0.9615 & 0.2747 \end{bmatrix}}_{\substack{r_1 \\ r_2 \\ V^*}}$$

This system has two inputs $\underline{u} = [u_1, u_2]^T$, and two outputs, $\underline{y} = [y_1, y_2]^T$. In particular, u_1 is the roll torque distribution γ , and u_2 is the yaw torque T_ψ , y_1 is the vehicle lateral velocity V_y , and y_2 is the vehicle yaw rate V_ψ .

This turns out to be not an easy plant to control, let's first investigate the directional properties of this plant. From the first right singular vector r_1 (the first vector in V^*), we conclude that if both inputs are increased then the gain is 0.1263. However, from the second right singular vector r_2 (the second vector in V^*), we conclude that the gain is only 0.0003 when one input is increased and the other is decreased. We will define the condition number ($\kappa(G)$) as follows,

$$\text{Condition Number, } \kappa(G) = \frac{\bar{\sigma}(G)}{\underline{\sigma}(G)} = \frac{0.1263}{0.0003} = 469$$

A large condition number means that the matrix is ill-conditioned. Control of ill-conditioned plants is difficult, as some combinations of the inputs have a strong effect on the outputs, whereas other combinations have a weak effect on the outputs. Hence, based on the aforementioned information, with our current actuators, under certain conditions (low adherence conditions), it becomes extremely difficult to satisfy both lateral velocity and yaw rate desired targets.

Consider the following possible inputs:

$$\underline{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow \underline{y} = \begin{bmatrix} -0.0342 \\ 0.0058 \end{bmatrix}, \quad \underline{u} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow \underline{y} = \begin{bmatrix} -0.1195 \\ 0.0214 \end{bmatrix}$$

It is obvious from that the first input, roll torque distribution input, has much less authority over the outputs relative to the second input, the yaw torque input. Now let's consider simultaneous change of both inputs in the same and opposite directions as follows,

$$\underline{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow \underline{y} = \begin{bmatrix} -0.1537 \\ 0.0272 \end{bmatrix}, \quad \underline{u} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \Rightarrow \underline{y} = \begin{bmatrix} -0.0853 \\ 0.0156 \end{bmatrix}$$

So, if we apply both inputs in the same direction then the authority on the outputs are much greater than when we apply the inputs in different direction. This also has

a physical significance, if both actuators are working together to generate understeer or oversteer, their impacts are greater, versus when one actuator generates understeer and the other oversteer. These results can also be concluded by looking at the first and second right singular vectors. The first right singular vector, r_1 is associated with the largest singular value, σ_1 , this means if the inputs are in the direction of the first singular vector, which means the inputs are in the same direction, then they will have much stronger effect on the outputs. However, if they are selected to be in the direction of the second singular vector, r_2 , which is associated with the smallest singular value, σ_2 , and this means the inputs are in the opposite direction and hence, they have much weaker effect on the outputs.

Finally, the direction of the left singular vectors impacts the output direction such that the direction of the first left singular vector, l_1 , pulls the outputs in different direction, while, the direction of the second left singular vector, l_2 , results in forcing the outputs to be in the same direction. This says that the actuators could have a bigger impact on the outputs with the decrease of lateral velocity and the increase of yaw rate or vice versa.

11.6.1 The Scalar Case

Consider the scalar case such that M is a complex number, then the singular value of M is,

$$\sigma(M) = \sqrt{\lambda(M^*M)} = \sqrt{\lambda(\bar{M}M)} = \sqrt{\bar{M}M} = |M|$$

In other words, in the scalar case, the singular value of M is the modulus (used in the Bode magnitude plot).

11.6.2 Input and Output Spaces

We say that $\{r_1, \dots, r_m\}$, the right singular vectors, is a “basis” for the input space. Similarly, $\{l_1, \dots, l_m\}$, the left singular vectors, is a basis for the output space. In other words, the input $\underline{u}$ can be written as some linear combination of the right singular vectors and the output $\underline{y}$ can equivalently be written as some linear combination of the left singular vectors.

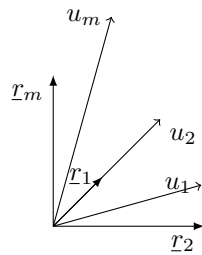


Figure 11.6: Altered view of the input space.

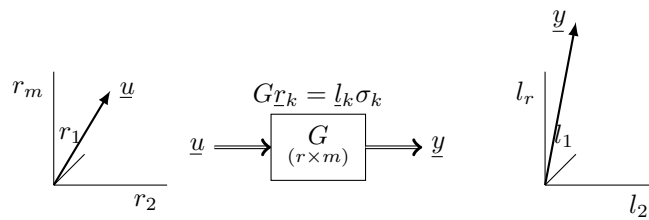


Figure 11.7: Input and output spaces of a dynamic system.

■ **EXAMPLE 11.19**

Suppose matrix G is $r \times m$, where $r = 2$, $m = 3$. Then, the right singular vectors consist of r_1 , r_2 and r_3 but r_3 is in the null space of G .

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} g_{11} & g_{12} & g_{13} \\ g_{21} & g_{22} & g_{23} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

The right and left singular vectors can be written as,

$$V = [r_1 \mid r_2 \mid r_3], \quad U = [l_1 \mid l_2]$$

While the singular value matrix, Σ is a wide matrix with the last column equal to zero,

$$GV = \begin{bmatrix} \vdots & \vdots & 0 \\ \vdots & \vdots & 0 \end{bmatrix} \Rightarrow U^*GV = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \end{bmatrix}$$

Any component along r_3 is annihilated, and hence the output space is two dimensional,

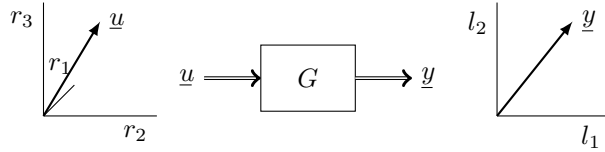


Figure 11.8: A system with 3 inputs and 2 outputs.

■ **EXAMPLE 11.20**

Consider matrix G with $r \times m$ size, but this time, suppose $r = 3$, $m = 2$, then,

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \\ g_{31} & g_{32} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

The right and left singular vectors can be written as,

$$V = [r_1 \mid r_2], \quad U = [l_1 \mid l_2 \mid l_3]$$

While, this time, the singular value matrix, Σ is a tall matrix with the last row equal to zero,

$$GV = \begin{bmatrix} \vdots & \vdots \\ 0 & 0 \end{bmatrix} \Rightarrow U^*GV = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \\ 0 & 0 \end{bmatrix}$$

The input space is two dimensional and even-though, the output space is three dimensional, all outputs rest along l_1 and l_2 .

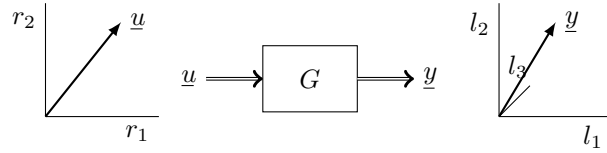


Figure 11.9: A system with 2 inputs and 3 outputs.

In the last part of this section, we present some of the SVD properties.

11.7 SVD properties

Given matrix, M with the right singular vector, $\underline{r}_k$, the left singular vector, $\underline{l}_k$, and the singular value, σ_k , then we can write, $M\underline{r}_k = \underline{l}_k\sigma_k$, then knowing this relationship, we can write the following equivalencies,

- $M^*\underline{l}_k = \sigma_k\underline{r}_k$
- $\sigma(M) = \sigma(M^*)$
- $\underline{r}_k(M^*) = \underline{l}_k(M)$ (and $\underline{l}_k$ of M is an $\underline{r}_k$ of M^*)
- $\underline{l}_k(M^*) = \underline{r}_k(M)$ (and $\underline{r}_k$ of M is an $\underline{l}_k$ of M^*)

Suppose that M is square with $M\underline{r}_k = \underline{l}_k\sigma_k$, then,

$$M^{-1}M\underline{r}_k = \sigma_k M^{-1}\underline{l}_k \Rightarrow M^{-1}\underline{l}_k = \frac{1}{\sigma_k}\underline{r}_k$$

This implies that

- $\sigma(M^{-1}) = (\sigma(M))^{-1}$
- $\underline{r}_k(M^{-1}) = \underline{l}_k(M)$
- $\underline{l}_k(M^{-1}) = \underline{r}_k(M)$

11.8 Norms

How *big* is a vector or matrix? We can use norms to find out. For example, consider the vector (or euclidean) norm:

$$\|\underline{x}\| = \sqrt{\underline{x}^*\underline{x}} = \sqrt{\sum_k |x_k|^2}$$

In more general terms, norms obey the following rules. For vector norms,

- $\|\underline{x}\| \geq 0$ (with equality only where $\underline{x} = \underline{0}$)
- $\|\alpha \underline{x}\| = |\alpha| \cdot \|\underline{x}\|$ (where α is any complex number)
- $\|\underline{x} + \underline{y}\| \leq \|\underline{x}\| + \|\underline{y}\|$ (triangle inequality)

While for matrix norms,

- $\|M\| \geq 0$ (with equality only where $M = 0$)
- $\|\alpha M\| = |\alpha| \cdot \|M\|$ (where α is any complex number)
- $\|M + N\| \leq \|M\| + \|N\|$ (triangle inequality)

11.8.1 The Spectral Norm

Consider a dynamic system with input $\underline{x}$, output $\underline{y}$.

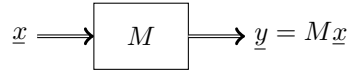


Figure 11.10: A multivariable dynamic system.

The size of the input is $\|\underline{x}\|_E$, while the size of the output is given by $\|\underline{y}\|_E = \|M\underline{x}\|_E$.

We define the spectral norm of a matrix M to be

$$\|M\|_S = \max_{\text{all } \underline{x}} \frac{\|M\underline{x}\|_E}{\|\underline{x}\|_E}$$

It can be shown that

$$\|M\|_S = \max_{\text{all } \underline{x}} \frac{\|M\underline{x}\|_E}{\|\underline{x}\|_E} = \sqrt{\lambda_{\max}(M^*M)} = \sigma_{\max}$$

Also,

$$\sigma_{\min} \leq \frac{\|M\underline{x}\|_E}{\|\underline{x}\|_E} \leq \sigma_{\max}$$

If M is square, then

$$\sigma_{\min} \leq |\lambda(M)| \leq \sigma_{\max}$$

CHAPTER 12

MIMO FEEDBACK PRINCIPALS

In this chapter, we discuss MIMO feedback principals before introducing how to design MIMO controllers using Youla parameterization.

In this section, we derive relationships for MIMO return ratio and return difference, complementary sensitivity, sensitivity, and Youla transfer functions matrices. We finish this section by introducing some state-space realization short cuts.

12.1 Multivariable Closed-Loop Transfer Functions

Consider the multi-variable feedback control loop, with their associated vector signal and matrices dimensions, shown below. In a SISO system, the controller and plant were both scalar, but now they are represented by transfer function matrices. This means we will have to re-think how we derive the closed-loop transfer function matrices, as $G_c G_p$ is no longer equal to $G_p G_c$.

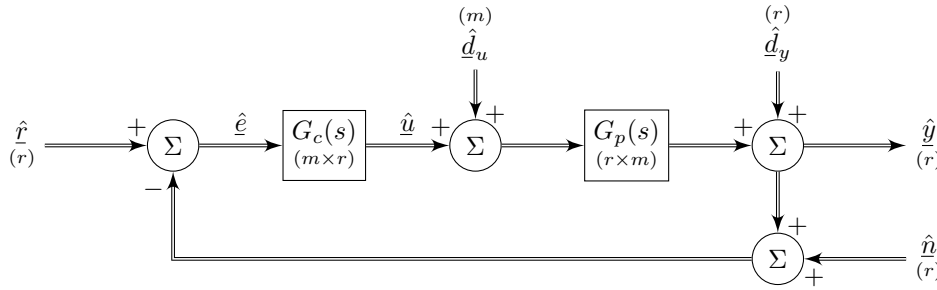


Figure 12.1: Block diagram of a multi-variable control system.

7028 Consider the state feedback system shown below. Let $G_c(s)$ denote an $m \times r$ matrix
 7029 of controller transfer functions and let $G_p(s)$ denote an $r \times m$ matrix of plant transfer
 7030 functions.

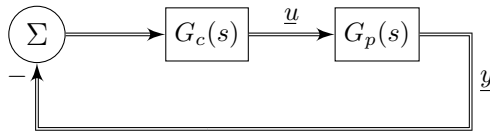


Figure 12.2: State feedback control system.

7031 We want to derive the return ratio (or loop gain) transfer function matrix, $L(s)$. One
 7032 way to measure the loop gain is to break the loop at the output or the input of the plant. A
 7033 good question to ask is if this return ratio matrix stays the same no matter where we break
 7034 the loop. Let's investigate this by first breaking the loop at the output of the plant as shown
 7035 below.

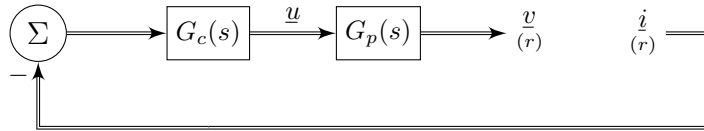


Figure 12.3: Open loop gain from the plant output.

7036 We can then derive the return ratio matrix taken from the plant input, $\underline{i}$, to the plant
 7037 output, $\underline{v}$ denoted by $L_y(s)$ using the following relationships,

$$\underline{v} = G_p(s)\underline{u} = -G_p(s)G_c(s)\underline{i}$$

$$L_y(s) = -\frac{\underline{v}}{\underline{i}}(s) = G_p(s)G_c(s)$$

$(r \times r)$
 $(r \times m)$
 $(m \times r)$

Similarly, we can break the loop at the plant input.

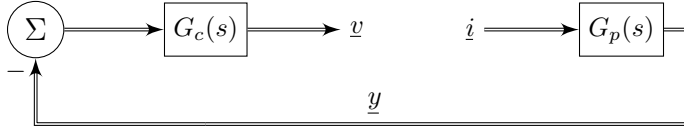


Figure 12.4: Open loop gain from the plant input.

We can then derive the return ratio taken from the plant input to output, similar to our previous derivation, however, this time, we denote the return ratio by $L_u(s)$ and using the following relationships,

$$\begin{aligned} \underline{v} &= -G_c(s)\underline{y} = -G_c(s)G_p(s)\underline{i} \\ L_u(s) &= -\frac{\underline{v}}{\underline{i}}(s) = G_c(s)G_p(s) \end{aligned}$$

$(m \times m)$ $(r \times m)$

We stated that for MIMO transfer function matrices, $G_c G_p$ is no longer equal to $G_p G_c$, hence the return ratio, $L_y \neq L_u$, so in MIMO feedback control loop, unlike the SISO counterpart, we have two distinct return ratios.

We can now form the closed-loop transfer functions for these two distinct return ratios.

Transfer Function Matrix, from $\underline{r}$ to $\underline{y}$ as shown in Figure (12.1):

$$\begin{aligned} \underline{y} &= \underbrace{G_p G_c}_{L_y} \underline{r} - \underbrace{G_p G_c}_{L_y} \underline{y} \\ (I + L_y)\underline{y} &= L_y \underline{r} \\ \underline{y} &= (I + L_y)^{-1} L_y \underline{r} = T_y \underline{r} \\ \Rightarrow T_y &= (I + L_y)^{-1} L_y \end{aligned}$$

$(r \times r)$

The transfer function matrix T_y is known as the output complementary sensitivity or closed loop transfer function matrix.

Transfer Function Matrix from $\underline{d}_y$ to $\underline{y}$ as shown in Figure (12.1)::

$$\begin{aligned} \underline{y} &= \underline{d}_y - L_y \underline{y} \\ (I + L_y)\underline{y} &= \underline{d}_y \\ \underline{y} &= (I + L_y)^{-1} \underline{d}_y = S_y \underline{d}_y \\ \Rightarrow S_y &= (I + L_y)^{-1} \end{aligned}$$

$(r \times r)$

Where the transfer function matrix S_y is known as the output sensitivity transfer function matrix.

Note:

$$\begin{aligned} S_y + T_y &= (I + L_y)^{-1} + (I + L_y)^{-1} L_y \\ &= (I + L_y)^{-1} (I + L_y) \\ \Rightarrow S_y + T_y &= I \end{aligned}$$

This is similar to the SISO case where $S + T = 1$.

Transfer Function Matrix from $\underline{r}$ to $\underline{e}$:

$$\begin{aligned} \underline{e} &= \underline{r} - \underbrace{G_p G_c}_{L_y} \underline{e} \\ \underline{e} &= (I + L_y)^{-1} \underline{r} = S_y \underline{r} \end{aligned}$$

Transfer Function from $\underline{r}$ to $\underline{u}$:

$$\begin{aligned} \underline{u} &= G_c \underline{r} - \underbrace{G_c G_p}_{L_u} \underline{u} \\ (I + L_u) \underline{u} &= G_c \underline{r} \\ \underline{u} &= (I + L_u)^{-1} G_c \underline{r} = Y \underline{r} \\ \Rightarrow Y_{(m \times r)} &= (I + L_u)^{-1} G_c \end{aligned}$$

This is known as the Youla transfer function matrix. We can use this transfer function matrix to derive a relationship for the input closed loop transfer function matrix, T_u ,

$$\begin{aligned} T_u &= (I + L_u)^{-1} L_u \\ &= (I + L_u)^{-1} G_c G_p \\ \Rightarrow T_{u(m \times m)} &= Y_{(m \times r)} G_{p(r \times m)} \end{aligned}$$

A similar definition for T_y can be obtained by using the following identity.

$$\begin{aligned} G_p(I + G_c G_p) &= G_p + G_p G_c G_p \\ &= (I + G_p G_c) G_p \end{aligned}$$

Now, multiplying both sides first by $(I + G_p G_c)^{-1}$ on the left, then by $(I + G_c G_p)^{-1}$ on the right,

$$\Rightarrow (I + L_y)^{-1} G_p = G_p (I + L_u)^{-1}$$

Now we use this property in deriving T_y as a function of Youla transfer function matrix as follows,

$$\begin{aligned} T_y &= (I + L_y)^{-1} L_y \\ &= (I + L_y)^{-1} G_p G_c \\ &= G_p (I + L_u)^{-1} G_c \\ \Rightarrow T_y &= \underset{(r \times r)}{G_p} \underset{(r \times m)}{Y} \underset{(m \times r)}{} \end{aligned}$$

Additionally,

$$\begin{aligned} Y &= (I + G_c G_p)^{-1} G_c \\ \Rightarrow Y &= S_u G_c \end{aligned}$$

Where the transfer function matrix S_u is known as the input sensitivity transfer function matrix. we can also write,

$$\begin{aligned} Y &= (I + G_c G_p)^{-1} G_c \\ &= G_c (I + G_p G_c)^{-1} \\ \Rightarrow Y &= G_c S_y \end{aligned}$$

7049 12.1.1 Realization Tricks

Since we know that the general MIMO sensitivity transfer function matrix $S = I - T$ where T is the complimentary sensitivity transfer function matrix and has the following state-space realization,

$$\dot{x} = A_T x + B_T u \quad (12.1)$$

$$y = C_T^T x + D_T u \quad (12.2)$$

7050 which the transfer function matrix for this realization can be written as,

$$\hat{G}_T = C_T^T (sI - A_T)^{-1} B_T + D_T \quad (12.3)$$

7051 Inserting this into the equation $S = I - T$ (yields the following

$$\hat{G}_S = I - [C_T^T (sI - A_T)^{-1} B_T + D_T] \quad (12.4)$$

which yields an equivalent state-space realization for the sensitivity transfer function matrix as given below,

$$A_S = A_T \quad (12.5)$$

$$B_S = B_T \quad (12.6)$$

$$C_S = -C_T \quad (12.7)$$

$$D_S = I - D_T \quad (12.8)$$

Therefore, knowing the state-space realization for either complementary sensitivity or sensitivity transfer function is sufficient to obtain the state-space realization for the other by using the equivalency provided above.

In this section, we start by introducing the concept of well-posedness for MIMO systems.

12.2 Well-Posedness of MIMO Systems

Let's refresh our memory about the definition and meaning of well-posedness for SISO systems. When a SISO feedback system is well-posed, all the closed-loop transfer functions are either proper or strictly proper. In this case, the mathematical condition for well-posedness was stated to be $1 + L(\infty) \neq 0$. Let's pause for a moment and discuss why such a condition as $1 + L(\infty) \neq 0$. Consider the Heaviside expansion of the return ratio:

$$L(s) = L(\infty) + L'(s)$$

where $L'(s)$ is strictly proper and has the same denominator as $L(s)$. It follows that all closed-loop transfer functions, such as,

$$\frac{e(s)}{r(s)} = \frac{1}{1 + L(s)} = S(s)$$

include $1 + L(s)$ as their denominators. It is clear that if the denominator, $1 + L(s)$, is strictly proper (if $1 + L(\infty) = 0$) then $\frac{1}{1 + L(s)}$ is improper. Hence, for the properness of all closed loop transfer functions, we have to enforce the mathematical condition $1 + L(\infty) \neq 0$.

Now we are in a position to derive a mathematical condition for well-posedness of MIMO systems. Consider the following system.

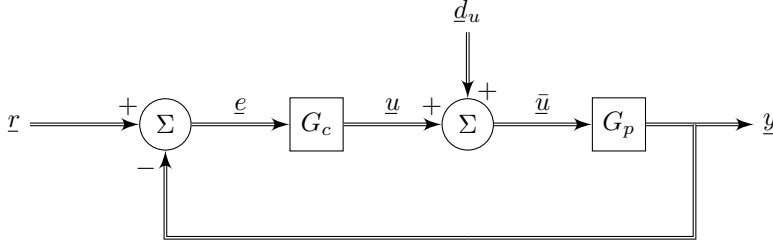


Figure 12.5: Block diagram of a multi-variable control system.

The state-space representation of the compensator G_c is written as,

$$\dot{\underline{x}}_c = A_c \underline{x}_c + B_c \underline{e}$$

$$\underline{u} = C_c \underline{x}_c + D_c \underline{e}$$

and the controlled Object state-space representation is given by,

$$\dot{\underline{x}}_p = A_p \underline{x}_p + B_p \underline{\bar{u}}$$

$$\underline{y} = C_p \underline{x}_p + D_p \underline{\bar{u}}$$

The relationship between the inputs, $\underline{u}$ and $\underline{d}_u$, and the independent nodes, $\underline{\bar{u}}$ and $\underline{e}$ can be written as:

$$\underline{e} = \underline{r} - \underline{y},$$

$$\underline{\bar{u}} = \underline{u} + \underline{d}_u$$

Substituting the node equations into the output equations, $\underline{u}$ and $\underline{y}$ above, gives us,

$$\underline{e} = \underline{r} - C_p \underline{x}_p - D_p \underline{\bar{u}}$$

$$\underline{\bar{u}} = C_c \underline{x}_c + D_c \underline{e} + \underline{d}_u$$

We can reorganize and rewrite these equations as,

$$\underline{e} + D_p \underline{\bar{u}} = \underline{r} - C_p \underline{x}_p$$

$$\underline{\bar{u}} - D_c \underline{e} = \underline{d}_u - C_c \underline{x}_c$$

These equations can be re-written in matrix form,

$$\begin{bmatrix} I & D_p \\ -D_c & I \end{bmatrix} \begin{bmatrix} \underline{e} \\ \underline{\bar{u}} \end{bmatrix} = \begin{bmatrix} -C_p & 0 \\ 0 & C_c \end{bmatrix} \begin{bmatrix} \underline{x}_p \\ \underline{x}_c \end{bmatrix} + \begin{bmatrix} \underline{r} \\ \underline{d}_u \end{bmatrix}$$

(nodes) (states) (inputs)

Definition 12.1. We say that a MIMO feedback is well-posed if the nodes are uniquely determined by the inputs and the states. The feedback is ill-posed otherwise. That is to say, the feedback is well-posed if and only if,

$$\det \begin{bmatrix} I & D_p \\ -D_c & I \end{bmatrix} \neq 0$$

$$\det(I_m + D_c D_p) = \det(I_r + D_p D_c) \neq 0$$

If either D_p or D_c is zero, then the return ratio is strictly proper, and the feedback is well-posed.

This condition is very similar to the well-posedness condition for a SISO feedback given above where $1 + L(\infty) \neq 0$.

In this section, we introduce state-space representation for the return ratio, L , for MIMO systems. This representation play an important role when presenting MIMO well-posedness and stability concepts.

12.3 State Variable Compositions

Given the matrices, A , B , C , and D of a MIMO dynamic system, we know that the easiest way to obtain the plant transfer function matrix $G_p(s)$ is,

$$G_p(s) = C(sI - A)^{-1}B + D$$

But the question is how do we obtain a state-space realization for the return ratio L , where we computed the return ratio in the last section? We start with $L = G_p G_c$ and state the realizations for G_c and G_p as follows,

$$\left. \begin{aligned} \dot{\underline{x}}_c &= A_c \underline{x}_c + B_c \underline{e} \\ \underline{u} &= C_c \underline{x}_c + D_c \underline{e} \end{aligned} \right\} \quad G_c \text{ Realization}$$

where, $\underline{e}$ is the error signal in the feedback loop, and $\underline{x}_c$ is the controller, G_c state variables and $\underline{u}$ is its output or the plant G_p input.

$$\left. \begin{aligned} \dot{\underline{x}}_p &= A_p \underline{x}_p + B_p \underline{u} \\ \underline{y} &= C_p \underline{x}_p + D_p \underline{u} \end{aligned} \right\} \quad G_p \text{ Realization}$$

where, $\underline{x}_p$ is the plant, G_p , state variables and $\underline{y}$ is the plant output. Then, by substituting for $\underline{u}$, $\underline{u} = C_c \underline{x}_c + D_c \underline{e}$, into G_p state-space realization we obtain the followings,

$$\dot{\underline{x}}_p = A_p \underline{x}_p + B_p C_c \underline{x}_c + B_p D_c \underline{e}$$

7104

$$\underline{y} = C_p \underline{x}_p + D_p C_c \underline{x}_c + D_p D_c \underline{e}$$

7105 So, this can be written in the matrix form as,

$$\frac{d}{dt} \begin{bmatrix} \underline{x}_c \\ \underline{x}_p \end{bmatrix} = \begin{bmatrix} A_c & \vdots & 0 \\ B_p C_c & \vdots & A_p \end{bmatrix} \begin{bmatrix} \underline{x}_c \\ \underline{x}_p \end{bmatrix} + \begin{bmatrix} B_c \\ B_p D_c \end{bmatrix} \underline{e}$$

7106

$$\underline{y} = [D_p C_c \quad C_p] \begin{bmatrix} \underline{x}_c \\ \underline{x}_p \end{bmatrix} + [D_p D_c] \underline{e}$$

7107 We can write this as

$$\dot{\underline{x}}_L = A_L \underline{x}_L + B_L \underline{i}$$

7108

$$-\underline{v} = C_L \underline{x}_L + D_L \underline{i}$$

7109 where $\underline{i}$ and $\underline{v}$ are given in the last section. The loop is closed for $\underline{v} = \underline{i}$. Then,

$$-\underline{i} = C_L \underline{x}_L + D_L \underline{i}$$

7110 we can write,

$$(-I - D_L) \underline{i} = C_L \underline{x}_L$$

7111 and

$$\underline{i} = -(I + D_L)^{-1} C_L \underline{x}_L$$

7112 We will show in the previous section of this chapter that $(I + D_L)^{-1}$, where $D_L = D_p D_c$,7113 always exist if and only if the feedback loop is well-posed. Substituting for $\underline{i}$ into the7114 differential equation for $\underline{x}_L$, we obtain the closed loop form,

$$\dot{\underline{x}}_L = (A_L - B_L(I + D_L)^{-1} C_L) \underline{x}_L$$

7115 So, the closed-loop characteristic polynomial (C.P.), $\Delta(s)$, can be written as,

$$\Delta(s) = \det(sI - A_L + B_L(I + D_L)^{-1} C_L)$$

7116 where the closed-loop characteristic polynomial is

$$\Delta(s) = \underbrace{\det(sI - A_L)}_{\text{Open-loop C.P.}} \det\left(I + (sI - A_L)^{-1} B_L(I + D_L)^{-1} C_L\right)$$

7117 where the open loop C.P. is $\delta(s)$, or,

$$\Delta(s) = \delta(s) \det(I + C_L(sI - A_L)^{-1} B_L(I + D_L)^{-1})$$

7118 The ratio of closed loop C.P. to the open loop C.P. is given by,

$$\frac{\Delta(s)}{\delta(s)} = \frac{(\det(I + D_L + C_L(sI - A_L)^{-1}B_L))}{\det(I + D_L)}$$

7119 where $\det(I + D_L)^{-1} = \frac{1}{\det(I + D_L)}$. Similar to the state-space representation of $G_p(s)$,
7120 the state-space representation of the return ratio, $L(s)$ is given by,

$$L(s) = D_L + C_L(sI - A_L)^{-1}B_L$$

7121 Therefore,

$$\frac{\Delta(s)}{\delta(s)} = \frac{\det(I + L(s))}{\det(I + D_L)}$$

7122 If the system is well-posed then $\det(I + D_L)$ exist and it is a constant which has no effect
7123 on the system stability and stability is only determined by,

$$\det(I + L(s))$$

7124 which will be discussed in the next section.

7125 This derivation for MIMO stability is again analogous to SISO systems, where

$$\frac{\Delta(s)}{\delta(s)} = 1 + L(s)$$

7126 where, the zeros of $1 + L(s)$ are the poles of the closed-loop, and the poles of $1 + L(s)$
7127 are the poles of the open-loop.

7128 Similar to SISO systems if the feedback system is well-posed and stable then this does not
7129 necessarily guarantee *system internally stability*, such that if there is a pole-zero cancella-
7130 tion on the $j\omega$ -axis or in the right-half plane by the controller and the plant (the description
7131 of internal stability for SISO systems is provided in Chapter 3). Our interest in internal
7132 stability here lies in the input-output maps rather than system state-space representation as
7133 will be discussed in the last section of this Chapter.

7134 12.4 Nyquist Criterion for MIMO Systems

7135 We start with the general relationship for Nyquist Criterion, avoiding the details of deriva-
7136 tions and noting that $\det(I + L(j\omega))$ is a transfer function with zeros and poles, for system
7137 stability,

$$N_c = N_o + N$$

Where:

- N_c is the closed-loop poles in the right-hand plane (positive zeros of $\det(I + L(j\omega))$).

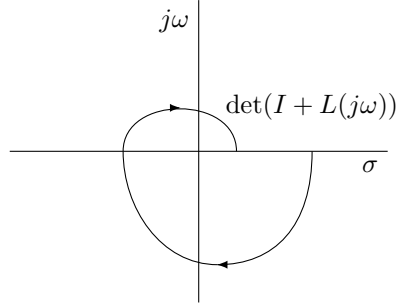


Figure 12.6: Nyquist plot for a MIMO system.

- N_o is the open-loop poles in the right-hand plane (positive poles of $\det(I + L(j\omega))$).
- N is the number of clockwise encirclements of the origin by the Nyquist plot of $\det(I + L(j\omega))$.

Then, the Nyquist criterion states: For a closed-loop system to be stable, $N_c = 0$, N_o , number of unstable open loop poles, must be precisely equal to $-N$, number of counter-clockwise encirclements of the origin by the Nyquist plot of $\det(I + L(j\omega))$. Another intuitive way to compute $\det(I + L(j\omega))$ is with the use of *Characteristic Gains* as described next.

12.4.1 Characteristic Gains

The characteristic gains or characteristic gain functions, $\lambda_k(j\omega)$, of a square matrix M , are the eigenvalues of M as a function of frequency.

Note: The characteristic gains can be found in MATLAB with the function `cglloci(M)` where M is a transfer function matrix.

The characteristic gains have the following property:

$$\sigma_{\min}(M) \leq |\lambda_k| \leq \sigma_{\max}(M) \quad (12.9)$$

Next we will discuss two identities which could be used to illustrate the use of characteristic gains in computing $\det(I + L(j\omega))$.

First, for any general square matrix M ,

$$\det(M) = \prod_{k=1}^n \lambda_k$$

where λ_k is the eigenvalue of matrix M .

Secondly, knowing that $\lambda(I + M) = 1 + \lambda(M)$ (see Appendix A).

Now applying these two eigenvalue identities, we can write,

$$\det[I + L(s)] = \prod_k \lambda[I + L(s)] = \prod_k (1 + \lambda[L(s)]) \quad (12.10)$$

This yields a Nyquist array and by applying the Nyquist criterion across all of the resulting characteristic gains,

$$N_c = N_o + \sum_k n_k \quad (12.11)$$

where n_i is the number of clockwise encirclements of -1 by λ_i . Now, this could be utilized to compute system stability using Nyquist criterion, such as for stable open loop system where $N_o = 0$, for the closed loop to be stable, $\sum_k n_k = 0$.

In this section, we discuss the important topic of *internal stability* for MIMO systems. Before, addressing this topic, we will discuss poles and zeros of MIMO systems using matrix factorization to provide some insight into the interpretation of poles and zeros of MIMO systems.

12.4.2 Poles and Zeros

Recall that the complementary sensitivity and sensitivity transfer function matrices can be written as,

$$T = (I + L)^{-1}L = L(I + L)^{-1} \quad (12.12)$$

$$S = (I + L)^{-1} \quad (12.13)$$

Using the right factorization of L ,

$$L = ND^{-1} \quad (12.14)$$

Then, T and S can be written as,

$$\begin{aligned} T &= ND^{-1}(I + ND^{-1})^{-1} \\ &= ND^{-1}((D + N)D^{-1})^{-1} \\ &= N(D + N)^{-1} \end{aligned} \quad (12.15)$$

Equivalently, it can be shown that

$$S = D(D + N)^{-1} \quad (12.16)$$

7172 Similar development for the left factorization which is not going to be repeated here. Re-
 7173 member if this is the coprime factorization, it is the minimal realization of the system, i.e.
 7174 there is no pole/zero cancellation.

7175

Definition 12.2. (Coprime). Suppose W is a right factor of maximal degree such that

$$N = N'W \quad (12.17)$$

$$D = D'W \quad (12.18)$$

We know if W is unimodular, then (N, D) is coprime. In general we can write,

$$L = N'W(D'W)^{-1} \quad (12.19)$$

$$= N'(D')^{-1} \quad (12.20)$$

and then,

$$T = N'(D' + N')^{-1} \quad (12.21)$$

$$S = D'(D' + N')^{-1} \quad (12.22)$$

Therefore, uncanceled zeros of L are the zeros of T and uncanceled poles of L are the zeros of S .

Recall the SISO interpolation conditions:

$$S + T = 1 \quad (12.23)$$

$$T(z) = 0 \rightarrow S(z) = 1 \quad (12.24)$$

$$S(p) = 0 \rightarrow T(p) = 1 \quad (12.25)$$

For a MIMO system, these interpolation conditions become

$$S + T = I \quad (12.26)$$

$$\det T(z) = 0 \text{ if } z \text{ is an uncanceled zero of } L \quad (12.27)$$

$$\det S(p) = 0 \text{ if } p \text{ is an uncanceled pole of } L \quad (12.28)$$

However, note that $S + T = I$ does not yield a similar condition for MIMO systems, i.e.

$$S + T = I \quad (12.29)$$

$$\det(T(z)) = 0 \not\Rightarrow \det(S(z)) = 1 \quad (12.30)$$

$$\det(S(p)) = 0 \not\Rightarrow \det(T(p)) = 1 \quad (12.31)$$

These MIMO interpolation conditions are for single unstable poles and zeros and repeated poles and zeros are not discussed above. For repeated poles or zeros use the same derivative constraints as a SISO system. This works because derivatives are a linear operator, $\det[\frac{d^k}{ds^k} T(z)] = 0$.

However, we will present a more straightforward and computationally less intensive method than the MIMO interpolation conditions given above. This method takes advantage of matrix diagonalization and S-M form to enforce internal stability as will be discussed in the following Chapters but first let's provide an overview of the internal stability of MIMO systems.

12.4.3 Internal Stability

It is important to remember that internal stability refers to the initial condition response in the absence of generalized inputs; in this regard it coincides with the standard notion of *asymptotic stability* of dynamic systems. However, it has immediate implications on the input-output properties of the system. For the SISO feedback system, we pointed out that all closed-loop transfer functions can be reduced to only four to address internal stability. We will extend this concept to the MIMO system, by finding the closed loop maps from r and d_u to y and $\bar{u}$ as illustrated in Figure below ¹.

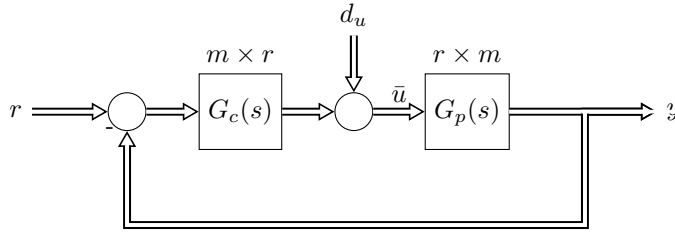


Figure 12.7: MIMO Feedback System without Noise and Output Disturbance Injection

recall the following transfer function matrices

- Output complementary sensitivity

$$y = (I + G_p G_c)^{-1} G_p G_c r \quad (12.32)$$

$$= T_y r \quad (12.33)$$

- Youlla Parameter

$$\bar{u} = (I + G_p G_c)^{-1} G_c r \quad (12.34)$$

$$= Y r \quad (12.35)$$

- Output Sensitivity

$$y = (I + G_p G_c)^{-1} G_p d_u \quad (12.36)$$

$$= S_y G_p d_u \quad (12.37)$$

¹In place of the input r , we could equivalently use the disturbance input d_y or the noise input n (not shown), furthermore, in place of the output y , we could use the error signal e , the conclusion would have been the same

▪ Input Sensitivity

$$\bar{u} = (I + G_c G_p)^{-1} d_u \quad (12.38)$$

$$= S_u d_u \quad (12.39)$$

7195 If these 4 transfer function matrices are BIBO stable then the system is internal stability.

7196 Note, that well-posedness is a necessary condition for BIBO stability.

$$M = \begin{bmatrix} Y & S_u \\ T_y & S_y G_p \end{bmatrix} \quad (12.40)$$

These four transfer function matrices can be written in terms of Youla parameter, Y . Recall that

$$\bar{u} = Yr \quad (12.41)$$

$$\bar{u} = S_u d_u = (I - T_u) d_u = (I - Y G_p) d_u \quad (12.42)$$

$$y = T_y r = G_p Y r \quad (12.43)$$

$$y = G_p S_u d_u = G_p (I - Y G_p) d_u \quad (12.44)$$

7197 Then

$$\begin{bmatrix} \bar{u} \\ y \end{bmatrix} = \underbrace{\begin{bmatrix} Y & I - Y G_p \\ G_p Y & G_p (I - Y G_p) \end{bmatrix}}_M \begin{bmatrix} r \\ d_u \end{bmatrix} \quad (12.45)$$

7198 The closed-loop TFM is parameterized by Y and G_p . Each entry of matrix M has to be

7199 BIBO stable to guarantee the *internal stability* of the closed loop system.

7200 Further, if G_p is stable, then the closed loop is internally stable if and only if Y is stable.

7201 All internally stabilizing compensators are parameterized by

$$G_c = (I - Y G_p)^{-1} Y \quad (12.46)$$

7202 We will further discuss the topic of internal stability and the criteria for enforcing internal stability in the the following Chapter.

7204 It is important to discuss and develop objective measures to aid in both MIMO control design and analysis using frequency shaping techniques, similar to the SISO case, before fully present the MIMO Youla parametrization control design method.

7207 12.5 MIMO Performance and Robustness Criteria

For the full MIMO system in Figure 14.10, we are unable to utilize Bode gain and phase plots, as were done in the SISO frequency domain control method. Instead we use singular

values of the relevant closed loop transfer function matrices, such as T , S , and Y as will be discussed in this section.

Note: The MATLAB command `sigma(A, B, C, D, w)`, where w is a frequency vector, gives the singular values as a function of ω .

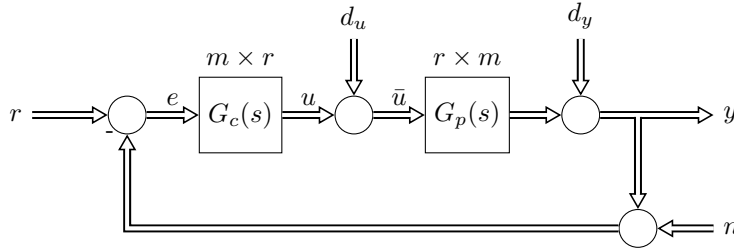


Figure 12.8: MIMO Feedback System

We start with the transfer function matrices which map the input signals, r , n , and d_y , to the actuator inputs, u ,

$$u = Y d_y \quad (12.47)$$

$$u = Y r \quad (12.48)$$

$$u = -Y n \quad (12.49)$$

Typically, we ideally design a controller for which $\sigma_{max}(Y) \ll 1$ at all frequencies to protect the actuator and keep it from saturating. The frequency shape of Y along with all the existing constraints will be discussed later in this section.

For the output sensitivity transfer function matrix, S_y which maps the input signals r and d_y to the error signals e and the output signals y as follows,

$$e = S_y r \quad (12.50)$$

$$y = S_y d_y \quad (12.51)$$

We design a controller so that $\sigma_{max}(S_y) \ll 1$. This will result in good disturbance rejections and good target following. These are the control design performance measures.

Finally to ensure robust stability in presence of high frequency disturbances such as sensor noise and unstructured model uncertainties, as discussed in SISO control design, the

following measures are imposed at high frequencies,

$$\lim_{\omega \rightarrow \infty} T_y(j\omega) \rightarrow 0_{r \times r} \Rightarrow \sigma_{max}(T_y) \ll 1 \quad (12.52)$$

but, as $T_y + S_y = I$ (algebraic constraint), this means,

$$\lim_{\omega \rightarrow \infty} S_y(j\omega) \rightarrow I_{r \times r} \Rightarrow \sigma_{max}(S_y) \rightarrow 1 \quad (12.53)$$

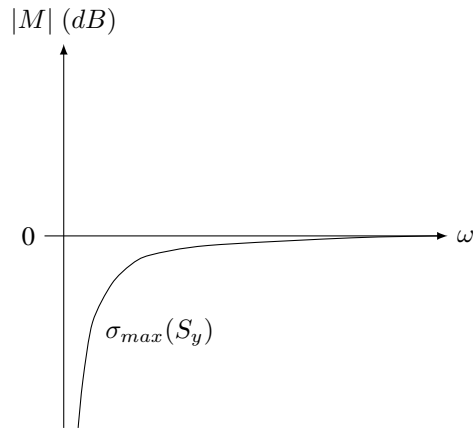


Figure 12.9: Frequency plot of $\sigma_{max}(S_y)$

as illustrated in Figure (12.9).

Therefore, here a clear trade-off between performance and robustness (robust stability) is given. As we try to make the loop robustly stable to uncertainties at high frequency, such that $\sigma_{max}(T_y) \ll 1$, we are forced to compromise disturbance rejection and target following capabilities or the loop performance in this frequency range, such that $\sigma_{max}(S_y) \rightarrow 1$.

Let's look at the same measures (performance and robust stability) in terms of the transfer function matrices which map the feedback loop reference and disturbance inputs to the output, we can then write,

$$y = T_y r \quad (12.54)$$

$$y = -T_y n \quad (12.55)$$

$$y = S_y d_y \quad (12.56)$$

$$y = S_y G_p d_u \quad (12.57)$$

MIMO control design using frequency shaping technique is similar to the SISO case. However, as mentioned above, in the MIMO case, we shape the loop singular values rather than Bode gains.

The fundamental question is how do we want to shape the transfer function matrices T_y and S_y given the equations above. The secondary question, which will be discussed later in this section, is the frequency shape of Y and the role of T_u and S_u in this design.

First, how do we want to frequency shape the singular values of T_y and S_y ? If $y = T_y r$ then $T_y = I$ yields perfect tracking but by looking at Equation (12.55), we are unable to keep $T_y = I$ at all frequencies, as if this was the case the sensor noise at higher frequencies will not be rejected at the output. Here again lies the fundamental trade-off, as discussed above, between performance and robustness. Therefore, we shape $T_y(0) = I$ (means T_y at zero frequency and without abuse of notation, we also imply T_y at very low frequency) for good target following. By enforcing $T_y(0) = I$, this results $\sigma_{max}(T_y(0)) = \sigma_{min}(T_y(0)) = 1$.

²

Knowing the algebraic constraint, $S + T = I$, Figure (12.10) provides our target frequency shapes of T_y and S_y singular values.

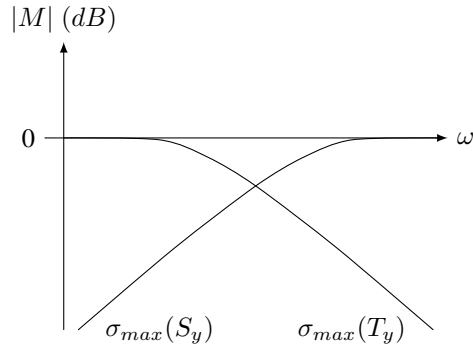


Figure 12.10: Frequency plot of $\sigma_{max}(S_y)$ and $\sigma_{max}(T_y)$

²We will discuss the benefits of having minimum and maximum singular values of T_y to be equal in the next Chapters.

Next, we map the closed loop performance and robust stability measures to the equivalent open loop (return ratio) measures. This way, by plotting the singular values of the return ratio, we could decide if our design measures are acceptable.

12.6 Open Loop Singular Values

Recall that

$$S = (I + L)^{-1} \quad (12.58)$$

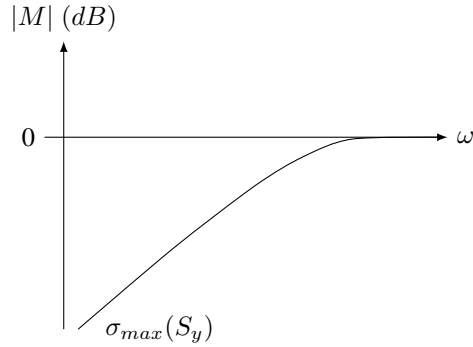


Figure 12.11: Frequency plot of $\sigma_{\max}(S_y)$

Figure above illustrates an acceptable frequency shape of the sensitivity transfer function matrix singular value. From previous sections, we learned that,

$$\sigma_{\max}(M^{-1}) = \frac{1}{\sigma_{\min}(M)} \quad (12.59)$$

Therefore,

$$\sigma_{\max}(S) = \sigma_{\max}((I + L)^{-1}) = \frac{1}{\sigma_{\min}(I + L)} \quad (12.60)$$

Another property of SVD (discussed in the Appendix A) is,

$$|\sigma_{\min}(M) - \sigma_{\min}(N)| \leq \sigma_{\min}(M + N) \leq \sigma_{\min}(M) + \sigma_{\min}(N) \quad (12.61)$$

Substituting I and L in place of M and N yields,

$$|\sigma_{\min}(I) - \sigma_{\min}(L)| \leq \sigma_{\min}(I + L) \leq \sigma_{\min}(I) + \sigma_{\min}(L) \quad (12.62)$$

$$|1 - \sigma_{\min}(L)| \leq \sigma_{\min}(I + L) \leq 1 + \sigma_{\min}(L) \quad (12.63)$$

Therefore,

$$\sigma_{max}(S) = \frac{1}{\sigma_{min}(I + L)} \leq \frac{1}{|1 - \sigma_{min}(L)|} \quad (12.64)$$

The relationship in Equation (12.64) to translate the closed loop measures to the open loop measures.

For example, at low frequencies, the performance measure enforces $\sigma_{max}(S) \ll 1$ which would imply that $|1 - \sigma_{min}(L)| \gg 1$ or $\sigma_{min}(L) \gg 1$. Therefore, at low frequency, we can write,

$$\sigma_{max}(S) \leq \frac{1}{\sigma_{min}(L)} \quad (12.65)$$

Similarly, at high frequency, where $\sigma_{max}(S) \rightarrow 1$, this implies $\sigma_{max}(L) \ll 1$. The proof of this is straightforward and left as an exercise.

So, in terms of the return ratio transfer function matrix, our performance and robust stability measures dictate at low frequency, $\sigma_{min}(L) \gg 1$, while at high frequency, $\sigma_{max}(L) \ll 1$. Next, we discuss how the maximum and minimum singular values of L should behave at the crossover frequency.

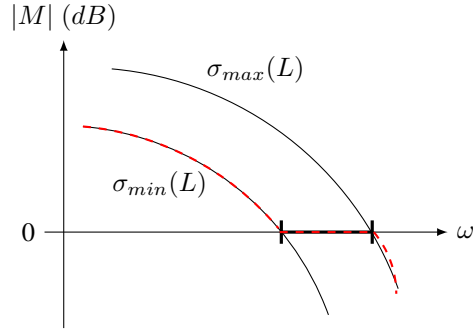
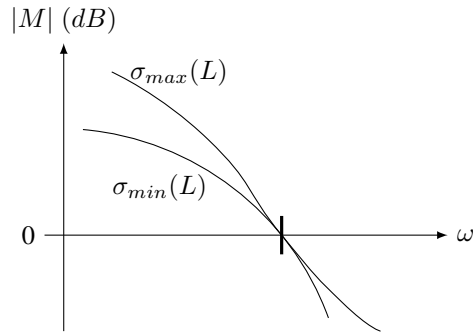
12.6.1 Crossover Frequency

It is important to note that at the crossover frequency, we start to lose all of the benefits of feedback. Therefore, at crossover frequency, the gap between $\sigma_{min}(L)$ and $\sigma_{max}(L)$ should be avoided, as the larger the gap, the more vulnerable the feedback system to noise (It can be shown that the closed loop complimentary transfer function matrix singular values do not decrease fast enough below 0 db when this gap exists). We can equivalently look at the SISO design as shown in dashed red line, where this design would not be acceptable as the magnitude of the complimentary closed loop transfer function, T , below 0 db does not decrease fast enough and this will result in sensor noise amplification and negatively impact minimizing or suppressing the unstructured uncertainty. There are other benefits for having this gap reduced and this will be discussed when presenting the condition numbers and their role in MIMO control design approach. An example problem is provided in the problem set to provide a better understanding of the impact of minimizing this gap.

Therefor, for a MIMO system we want $\sigma_{min}(L) \cong \sigma_{max}(L)$ around the crossover frequency, to reduce the impact of noise in the feedback loop.

Recall the condition number of a matrix, where for L we can write,

$$\frac{\sigma_{max}(L)}{\sigma_{min}(L)} = \mathcal{K}(L) \quad (12.66)$$

Figure 12.12: Comparison of plot of $\sigma_{max}(L)$ and $\sigma_{min}(L)$ Figure 12.13: Comparison of plot of $\sigma_{max}(L)$ and $\sigma_{min}(L)$

7270 where $\mathcal{K}(L) \approx 1$ around the crossover frequency. Next, we discuss the important topic of
 7271 bandwidth and its constraints similar to the SISO systems.

7272 12.6.2 Bandwidth Constraints

7273 As discussed previously, ideally, we want $\sigma_{max}(Y)$ to be low at all frequencies for min-
 7274 imizing actuator effort and avoiding actuator saturation. But, as we will find next, this is
 7275 usually not possible for all frequencies.

7276

We know the following relationships,

$$T_y = G_p Y \quad (12.67)$$

$$T_u = Y G_p \quad (12.68)$$

7277 We will use the following singular value property,

$$\sigma_{\min}(M)\sigma_{\max}(N) \leq \sigma_{\max}(MN) \leq \sigma_{\max}(M)\sigma_{\max}(N) \quad (12.69)$$

using this property, we can write the following inequality,

$$\sigma_{\min}(G_p)\sigma_{\max}(Y) \leq \sigma_{\max}(T_y) \quad (12.70)$$

$\Rightarrow$

$$\sigma_{\max}(Y) \leq \frac{\sigma_{\max}(T_y)}{\sigma_{\min}(G_p)} \quad (12.71)$$

7278 From this relationship, we can state that it is undesirable to have $\sigma_{\max}(T_y)$ to be much
 7279 greater than $\sigma_{\min}(G_p)$, as this results in large maximum singular values of Y . Let's in-
 7280 vestigate this maximum singular value at different frequency ranges, such as low and high
 7281 frequencies.

7282 **At low frequencies** a good target following is obtained by $\sigma_{\max}(T_y) \cong 1$ and combining
 7283 this with Equation (12.71) yields

$$\sigma_{\max}(Y) \leq \frac{1}{\sigma_{\min}(G_p)} \quad (12.72)$$

7284 This is an interesting result, similar to the SISO case, we are unable to shape the control
 7285 effort Y at low frequencies! If the plant is low gain it will then a large actuator effort would
 7286 be required and then the actuator will be vulnerable to saturation and there is nothing we
 7287 could do to avoid this without compromising a good target following requirement!

7288

7289 *Recall that for SISO systems this changes to $T \cong 1$ and $Y \cong \frac{1}{G_p}$ at low frequencies.*

7290

7291 **At high frequencies**, $\sigma_{\max}(S_y) \cong 1$ due to the algebraic constraint $T + S = I$, where we
 7292 want $\sigma_{\max}(T_y) < 1$ and we have the following relationship,

$$Y = G_c S_y \quad (12.73)$$

Combining this with the singular value property above,

$$\sigma_{\max}(Y) = \sigma_{\max}(G_c S_y) \quad (12.74)$$

$\Rightarrow$

$$\sigma_{\max}(S_y)\sigma_{\min}(G_c) \leq \sigma_{\max}(Y) \leq \sigma_{\max}(S_y)\sigma_{\max}(G_c) \quad (12.75)$$

7293 so at high frequencies with $\sigma_{\max}(S_y) \cong 1$, the following expression is obtained,

$$\sigma_{\min}(G_c) \leq \sigma_{\max}(Y) \leq \sigma_{\max}(G_c) \quad (12.76)$$

Hence, at high frequencies, the control effort Y could be shaped, unlike at the low frequencies, by assuring the singular values of G_c are not large. A typical plot of $\sigma_{max}(Y)$ will follow something similar to the following Figure, if $\sigma_{min}(G_p)$ is below 0 db, Next,

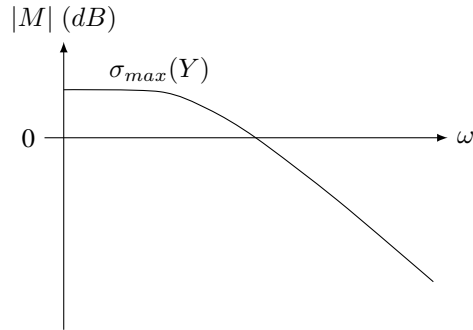


Figure 12.14: Frequency plot of $\sigma_{max}(Y)$

we discuss the condition number and its role in MIMO control design.

12.7 Condition Number and Its Role in MIMO Control Design

The condition number of a general matrix M is given by,

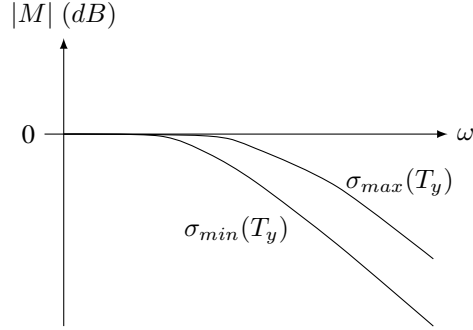
$$\mathcal{K}(M) = \frac{\sigma_{max}(M)}{\sigma_{min}(M)} \quad (12.77)$$

where it could be stated that M is well conditioned if $\mathcal{K} \sim 1$ and ill conditioned if $\mathcal{K} \gg 1$.

12.7.1 Condition Numbers and Decoupling

We stated that for good target following at low frequencies, we are required to design a MIMO controller so that $T(0) = I$ (this means $\sigma_{max}(T(0)) = \sigma_{min}(T(0))$, which in turn, this results in closed loop input-output decoupling). Having $T(0) = I$ provides perfect reference tracking and zero steady state error and input-output decoupling. In MIMO control design decoupling could be very desirable as every reference input is only affecting one output, this could make controller tuning and calibration an easier task.

The question becomes, how do we recognize this on a MIMO singular value plot? $\sigma_{max}(T_y) = \sigma_{min}(T_y)$ at low frequencies, as shown in Figure below.

Figure 12.15: Frequency plot of $\sigma(T_y)$

It is important to note that it could be shown large $\sigma_{min}(L)$ at low frequencies with $\mathcal{K}(L) \cong 1$ also results in input-output decoupling.

Before we summarize our performance and robust stability requirements both for the return ratio L and the closed loop transfer functions matrices, it would be informative to discuss the role of the closed loop transfer function matrices T_u and S_u in MIMO control design.

12.7.2 Role of T_u and S_u Input Closed Loop TFMs in MIMO feedback design

The important question to be asked here is, do we need to consider more than one node when designing MIMO controllers? This means in addition, to T_y and S_y do we need to consider T_u and S_u . Referring again to Figure 12.8, we can write,

$$y = S_y G_p d_u \quad (12.78)$$

$$\bar{u} = S_u d_u \quad (12.79)$$

Recognize that

$$G_p S_u = S_y G_p \quad (12.80)$$

$$S_u G_c = G_c S_y \quad (12.81)$$

using the singular value properties stated above, the following expressions are given for Equation (12.80),

$$\sigma_{max}(S_u) \sigma_{min}(G_p) \leq \sigma_{max}(G_p S_u) = \sigma_{max}(S_y G_p) \leq \sigma_{max}(S_y) \sigma_{max}(G_p) \quad (12.82)$$

$$\sigma_{max}(S_y) \sigma_{min}(G_p) \leq \sigma_{max}(S_y G_p) = \sigma_{max}(G_p S_u) \leq \sigma_{max}(S_u) \sigma_{max}(G_p) \quad (12.83)$$

Dividing Equation (12.82) by $\sigma_{max}(G_p)$,

$$\begin{aligned} \frac{\sigma_{max}(S_u)}{\mathcal{K}(G_p)} &\leq \sigma_{max}(S_y) \Rightarrow \\ \sigma_{max}(S_u) &\leq \sigma_{max}(S_y)\mathcal{K}(G_p) \end{aligned} \quad (12.84)$$

Dividing Equation (12.83) by $\sigma_{max}(G_p)$,

$$\frac{\sigma_{max}(S_y)}{\mathcal{K}(G_p)} \leq \sigma_{max}(S_u) \quad (12.85)$$

7321 Combining these two yields the equation,

$$\frac{\sigma_{max}(S_y)}{\mathcal{K}(G_p)} \leq \sigma_{max}(S_u) \leq \sigma_{max}(S_y)\mathcal{K}(G_p) \quad (12.86)$$

7322 Similar analysis for Equation (12.81) yields,

$$\frac{\sigma_{max}(S_y)}{\mathcal{K}(G_c)} \leq \sigma_{max}(S_u) \leq \sigma_{max}(S_y)\mathcal{K}(G_c) \quad (12.87)$$

7323 We can ask, under what conditions will $\sigma_{max}(S_y) \cong \sigma_{max}(S_u)$? When $\mathcal{K}(G_p)$ or
 7324 $\mathcal{K}(G_c) \cong 1$. Therefore, if either $\mathcal{K}(G_p)$ or $\mathcal{K}(G_c) \cong 1$ then feedback design would
 7325 be sufficient using one node (S_y). If neither of these conditions are true then one must
 7326 check S_u when designing MIMO feedback control to make sure the shape of S_u follows
 7327 S_y !

7328 12.8 Summary of Requirements

7329 12.8.1 Closed Loop Requirements

- 7330 ▪ Disturbance rejection: $\sigma_{max}(S_y)$ as small as possible at low frequencies.
- 7331 ▪ Sensor noise rejection: $\sigma_{max}(T_y)$ as small as possible at high frequencies.
- 7332 ▪ Tracking: $\sigma_{min}(T_y) \cong \sigma_{max}(T_y) \cong 1 \rightarrow \mathcal{K}(T_y) \cong 1$ at low frequencies.
- 7333 ▪ Minimize Control Effort: $\sigma_{max}(Y)$ as small as possible at high frequencies.
- 7334 ▪ Robust Stability: $\sigma_{max}(T_y)$ as small as possible at high frequencies (model uncer-
 7335 tainty increases as $\omega \rightarrow \infty$).

7336 12.8.2 Open Loop Requirements

- 7337 ▪ Disturbance rejection: $\sigma_{min}(L)$ as large as possible at low frequencies.

- 7338 ▪ Sensor noise rejection: $\sigma_{max}(L)$ as small as possible at high frequencies and $\mathcal{H}(L) \approx$
7339 1 at crossover frequency.
- 7340 ▪ Tracking: $\sigma_{min}(L)$ as large as possible at low frequencies.
- 7341 ▪ Minimize Control Effort: $\sigma_{max}(G_c)$ as small as possible at high frequencies.
- 7342 ▪ Robust Stability: $\sigma_{max}(L)$ as small as possible at high frequencies.

CHAPTER 13

YOULA PARAMETERIZATION FOR FEEDBACK SYSTEMS

13.1 Neo-Classical Control for MIMO Systems

In this section, similar to the SISO feedback loop, we discuss the role of neo-classical control for MIMO feedback control design. We present equivalent MIMO feedback loops which Youla parametrization technique could be utilized either directly (*Internal Model Control*) or utilized to derive a MIMO feedback controller, G_c .

We begin our discussion with a MIMO feedback control loop as illustrated in Figure below.

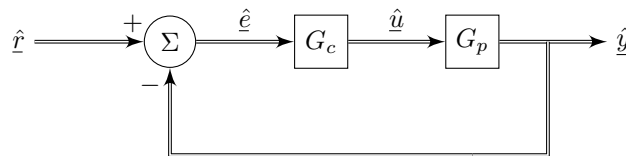


Figure 13.1: Block diagram of a multi-variable control system.

In neo-classical control, as discussed in the previous chapter, the performance and robustness criteria are achieved by directly frequency shaping three different closed-loop transfer function matrices, Y , Youla, S , sensitivity, and T , complementary sensitivity. directly. Then, utilizing the Youla transfer function matrix, we compute the corresponding G_c , MIMO feedback controller, as presented below. From Figure (13.1) the following relationships are written,

$$\underline{u} = Y\underline{r} = G_c\underline{r} - L_u\underline{u}$$

where, $L_u = G_c G_p$,

$$\Rightarrow Y = (I + L_u)^{-1} G_c = S_u G_c$$

we can then write,

$$Y = (1 + G_c G_p)^{-1} G_c = G_c (I + G_p G_c)^{-1} = G_c S_y$$

Given the realization for Y , the realization for G_c is computed as follows,

$$Y = (I + G_c G_p)^{-1} G_c$$

$$G_c = (I + G_c G_p) Y$$

$$G_c - G_c G_p Y = Y$$

$$G_c (I - G_p Y) = Y$$

$$\Rightarrow G_c = Y (I - G_p Y)^{-1} = (I - Y G_p)^{-1} Y$$

In terms of a feedback control loop, Youla parametrization can be interpreted as an equivalent open-loop control by re-drawing Figure (13.1) as,

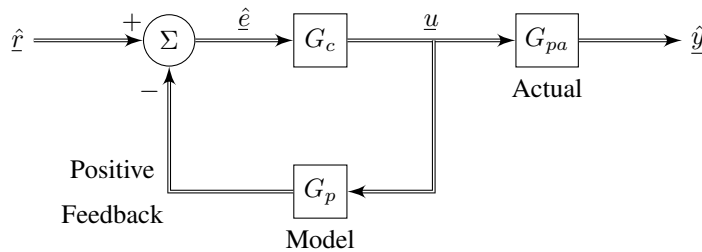


Figure 13.2: Interpretation of a neo-classical control system.

Now knowing that $Y = (1 + G_c G_p)^{-1} G_c$, then the following equivalent control loop can be drawn,

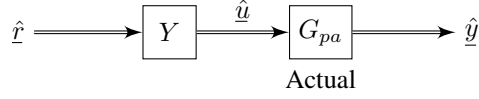


Figure 13.3: Open-loop Youla control.

13.1.1 Internal Model Control

It should be obvious that in terms of feedback control design, a transformation to an open-loop control using Youla parameterization, as illustrated in Figure (13.3) is therootically insightful and novel and practically straightforward. It should also be clear that the open-loop transformation is only for computation and not implementation purposes. A good question to ask at this stage if Youla transfer function matrix could be directly used in a feedback control loop rather than only utilized for computation of the feedback controller? This question is previously answered through Internal Model Control or IMC. Let's again find an equivalent feedback control loop to Figure (13.1) by modifying this loop to include Y as illustrated in the Figure below,

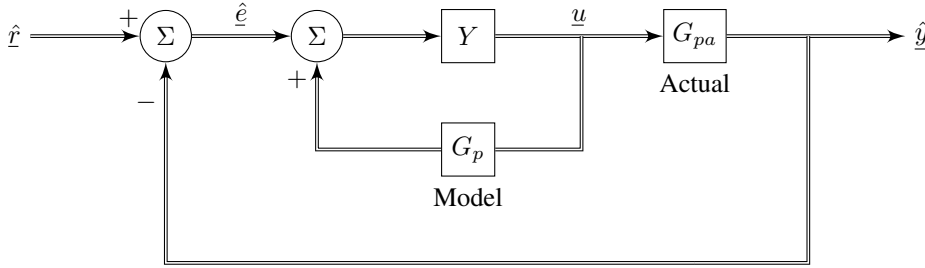


Figure 13.4: Internal model control.

An equivalent feedback control loop now could be obtained, known as IMC [17.2], from Figure above,

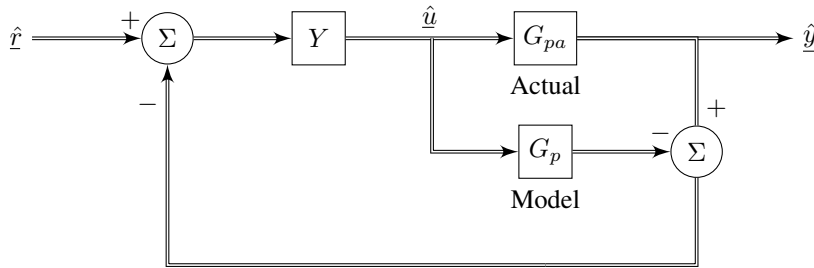


Figure 13.5: Internal model control.

In the above feedback control loop, Y is directly utilized in place of feedback controller G_c . But the benefits of IMC goes beyond just directly utilize Y , some of these benefits are discussed in the previous Chapters when IMC is used in the SISO control design using optimization techniques. Similar optimization techniques could be used for MIMO controller design using this feedback control loop.

13.2 MIMO Feedback Control Design for Stable Plants

We formulate the MIMO control design for two cases,

a) Stable plants, G_p .

b) Unstable plants, G_p .

13.2.1 Procedure to Find the MIMO Controller, G_c

Starting with $G_p(s) \in \mathbb{C}^{r \times m}$,

$$\left[\begin{array}{c|c} I_{r \times r} & G_p \\ \hline * & I_{m \times m} \end{array} \right] \sim \left[\begin{array}{c|c} U_L & M_p \\ \hline * & U_R \end{array} \right] \quad (13.1)$$

where M_p is the Smith-McMillan form of G_p , and U_L and U_R are left and right unimodular matrices,

$$U_L G_p U_R = M_p \Rightarrow G_p = U_L^{-1} M_p U_R^{-1} \quad (13.2)$$

This diagonalizes G_p , and furthermore, aids in decoupling the control design as follows,

We choose,

$$Y = U_R M_Y U_L \quad (13.3)$$

$$(13.4)$$

where M_Y is a diagonal matrix of stable proper transfer functions. Then the closed loop transfer functions matrices at the two nodes can be written as,

$$T_y = G_p Y = U_L^{-1} M_p M_Y U_L \quad (13.5)$$

$$T_u = Y G_p = U_R M_Y M_p U_R^{-1} \quad (13.6)$$

Therefore, we assign the closed-loop poles and zeros simply by our choices for the diagonal elements of M_Y .

13.2.2 Interpolation Conditions

From the SISO feedback control systems, we know if “ z ” is any non-minimum phase zeros, in $G_p(s)$, then “ z ” must be a zero of T_y .

Furthermore, If “ p ” is a pole of $G_p(s)$ on the $j\omega$ -axis or in the right half plane, then “ p ” must be a zero of S_y .

But, we also know if the G_p is stable, then closed-loop BIBO stability is satisfied only by selecting a stable and proper Y transfer function matrix. As stated above, this can be done through proper selection of the diagonal elements of M_Y . Then, we write the following relationship,

$$S_y = I - T_y = I - U_L^{-1} M_P M_Y U_L = U_L^{-1} (I - M_P M_Y) U_L \quad (13.7)$$

Finally, the MIMO controller $G_c = (I - Y G_p)^{-1} Y$ which yields

$$\begin{aligned} G_c &= (I - U_R M_Y U_L U_L^{-1} M_P U_R^{-1})^{-1} U_R M_Y U_L \\ &= (U_R U_R^{-1} - U_R M_Y M_P U_R^{-1})^{-1} U_R M_Y U_L \\ &= U_R (I - M_Y M_P)^{-1} M_Y U_L \end{aligned} \quad (13.8)$$

13.3 MIMO Feedback Control Design Examples

EXAMPLE 13.1

Let G_P to be a stable and proper transfer function matrix,

$$G_P = 20 \begin{bmatrix} \frac{1}{(s+2)^2} & \frac{s+1}{s+2} \\ -\frac{s+1}{s+2} & \frac{s+1}{s+2} \end{bmatrix} \quad (13.9)$$

$$= \frac{20}{(s+2)^2} \begin{bmatrix} 1 & (s+1)(s+2) \\ -(s+1)(s+2) & (s+1)(s+2) \end{bmatrix} \quad (13.10)$$

$$= \frac{20}{(s+2)^2} G_{p_n} \quad (13.11)$$

7403 We start with

$$\left[\begin{array}{c|c} I_{r \times r} & G_{p_n} \\ \hline * & I_{m \times m} \end{array} \right] \quad (13.12)$$

7404 where then we can write,

$$\left[\begin{array}{cc|cc} 1 & 0 & 1 & (s+1)(s+2) \\ 0 & 1 & -(s+1)(s+2) & (s+1)(s+2) \\ \hline * & * & 1 & 0 \\ * & * & 0 & 1 \end{array} \right] \quad (13.13)$$

7405 Using the standard row/column operations on the numerator matrix

- 7406 1. Add C_1 to C_2 into C_2
- 7407 2. Multiply R_1 by $(s+1)(s+2)$ and add to R_2 into R_2
- 7408 3. Multiply C_1 by $(s+1)(s+2)$ and subtract from C_2 into C_2
- 7409 4. Subtract C_1 from C_2 into C_2

7410 These operations result in

$$\sim \left[\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ (s+1)(s+2) & 1 & 0 & (s+1)(s+2)(s^2+3s+3) \\ \hline * & * & 1 & -(s+1)(s+2) \\ * & * & 0 & 1 \end{array} \right] \quad (13.14)$$

Using this,

$$U_L = \begin{bmatrix} 1 & 0 \\ (s+1)(s+2) & 1 \end{bmatrix}, U_L^{-1} = \begin{bmatrix} 1 & 0 \\ -(s+1)(s+2) & 1 \end{bmatrix} \quad (13.15)$$

$$U_R = \begin{bmatrix} 1 & -(s+1)(s+2) \\ 0 & 1 \end{bmatrix}, U_R^{-1} = \begin{bmatrix} 1 & (s+1)(s+2) \\ 0 & 1 \end{bmatrix} \quad (13.16)$$

$$(13.17)$$

7411 Then we write S-M form using $M_p = U_L G_p U_R$,

$$M_p = 20 \begin{bmatrix} \frac{1}{(s+2)^2} & 0 \\ 0 & \frac{(s+1)(s^2+3s+3)}{s+2} \end{bmatrix} = \begin{bmatrix} G_{p1} & 0 \\ 0 & G_{p2} \end{bmatrix} \quad (13.18)$$

7412 From above, this system as

7413 poles at: $-2, -2, -2$

7414 zeros at: $-1, -\frac{3}{2} \pm j\sqrt{\frac{3}{2}}$

7415 It should be noted that the coefficient of the highest order polynomial in M_p is not
 7416 equal to one, as the gain is equal 20. This is simply a scaling problem and does
 7417 not impact the MIMO control design. Now using the S-M form, the Youla transfer
 7418 function for each channel can be selected independently. Let's assume simple form of
 7419 T_1 and recognizing that the desired $T_1(0) = 1$,

$$T_1 = \frac{4}{(s+2)^2} \quad (13.19)$$

7420 Then Y_1 can computed by,

$$T_1 = Y_1 G_{p1} \Rightarrow Y_1 = \frac{1}{5} \quad (13.20)$$

7421 This yields the desired first closed loop control channel. Ideally, we are interested in
 7422 having $T_2 = T_1$ at all frequencies (based on our previous discussions on the benefits
 7423 of decoupling), so we will do the following, Therefore

$$T_1 = Y_2 G_{p2} \Rightarrow Y_2 = \frac{1}{5(s+1)(s+2)(s^2+3s+3)} \quad (13.21)$$

Now $T_y = G_p Y$ and $T_y = U_L^{-1} M_P M_Y U_L$, where $M_Y = \begin{bmatrix} Y_1 & 0 \\ 0 & Y_2 \end{bmatrix}$,

$$T_y = \begin{bmatrix} 1 & 0 \\ -(s+1)(s+2) & 1 \end{bmatrix} \underbrace{\begin{bmatrix} \frac{4}{(s+2)^2} & 0 \\ 0 & \frac{4}{(s+2)^2} \end{bmatrix}}_{M_P M_Y} \begin{bmatrix} 1 & 0 \\ (s+1)(s+2) & 1 \end{bmatrix} \quad (13.22)$$

$$= \begin{bmatrix} \frac{4}{(s+2)^2} & 0 \\ 0 & \frac{4}{(s+2)^2} \end{bmatrix} \quad (13.23)$$

$$S_y = I - T_y = \frac{s(s+4)}{(s+2)^2} I \quad (13.24)$$

Now the MIMO controller $G_c = U_R(I - M_Y M_p)^{-1} M_Y U_L$,

$$G_c = \begin{bmatrix} 1 & -(s+1)(s+2) \\ 0 & 1 \end{bmatrix} \underbrace{\begin{bmatrix} \frac{(s+2)^2}{5s(s+4)} & 0 \\ 0 & \frac{s+2}{5s(s+1)(s+4)(s^2+3s+3)} \end{bmatrix}}_{(I - M_Y M_p)^{-1} M_Y} \begin{bmatrix} 1 & 0 \\ (s+1)(s+2) & 1 \end{bmatrix} \quad (13.25)$$

$$= \frac{1}{5} \begin{bmatrix} \frac{(s+2)^2}{s(s+4)(s^2+3s+3)} & -\frac{(s+2)^2}{s(s+4)(s^2+3s+3)} \\ \frac{(s+2)^2}{5s(s+4)(s^2+3s+3)} & \frac{(s+2)}{s(s+1)(s+4)(s^2+3s+3)} \end{bmatrix} \quad (13.26)$$

In this case, the shape of the final controller is determined based on our selection of M_Y , and, the selection of M_Y is achieved by our choice for the desired complementary sensitivity transfer function matrix.

Now, we are in a position to place this MIMO controller, G_c , in the feedback loop with the plant, G_p as shown below,

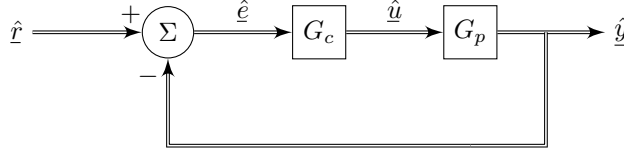


Figure 13.6: Block diagram of a multi-variable control system.

We then investigate the performance and robustness criteria for MIMO feedback controller design given in the previous Chapter and provided below as a reminder,

Summary of Closed Loop Requirements

- Disturbance rejection: $\sigma_{max}(S_y)$ as small as possible at low frequencies.
- Sensor noise rejection: $\sigma_{max}(T_y)$ as small as possible at high frequencies.
- Tracking: $\sigma_{min}(T_y) \cong \sigma_{max}(T_y) \cong 1 \rightarrow \mathcal{K}(T_y) \cong 1$ at low frequencies.
- Minimize Control Effort: $\sigma_{max}(Y)$ as small as possible at high frequencies.
- Robust Stability: $\sigma_{max}(T_y)$ as small as possible at high frequencies (model uncertainty increases as $\omega \rightarrow \infty$).

Summary of Open Loop Requirements

- Disturbance rejection: $\sigma_{min}(L)$ as large as possible at low frequencies.
- Sensor noise rejection: $\sigma_{max}(L)$ as small as possible at high frequencies.
- Tracking: $\sigma_{min}(L)$ as large as possible at low frequencies.
- Minimize Control Effort: $\sigma_{max}(G_c)$ as small as possible at high frequencies.
- Robust Stability: $\sigma_{max}(L)$ as small as possible at high frequencies.

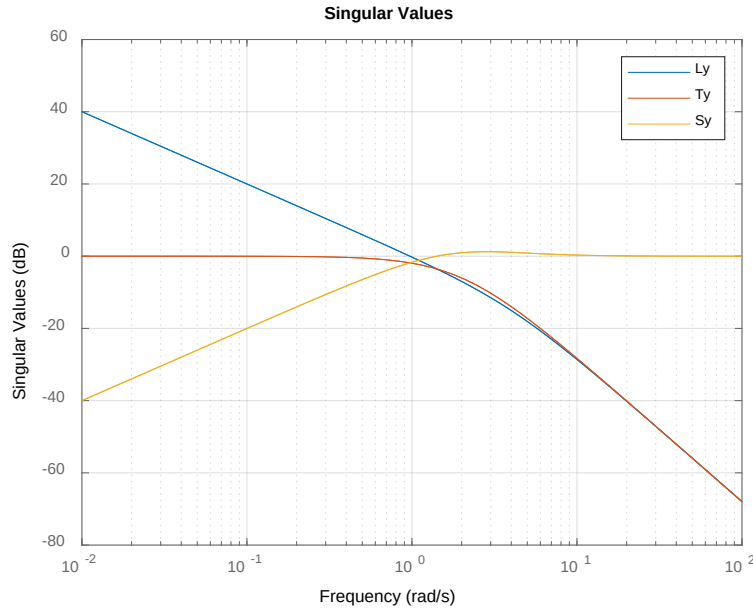


Figure 13.7: Closed Loop and Return Ratio Singular Values.

The closed loop complementary, sensitivity and return ratio singular values are illustrated in Figure above. The maximum and minimum singular values for all these transfer functions matrices are exactly equal. This is the result of our design which enforced a decoupling of the complementary transfer function, T_y . This decoupling is in all frequencies as evident from this Figure. As a result, the return ratio transfer function matrix maximum and minimum singular values have become equal in all frequencies including the cross-over frequency. It is important to point out that this decoupling also results in $S_y = S_u$ and hence, there is no need for considering the second node or S_u in this design. It should be noted that the closed loop speed of response or the closed loop bandwidth has been arbitrarily selected in this example.

All the closed loop requirements are met through this design. The control effort requirement will be discussed next. The important take away here is that with this method, all the tuning knobs for this design are under our control, and unlike many other black box optimization approaches, which we will discuss in the next Chapters, this method is very much a white box approach. In addition, the closed-loop transfer function matrix decoupling, as discussed above, has several benefits, and hence, when

possible, it is recommended to decouple the closed-loop transfer function matrix when designing MIMO controls.

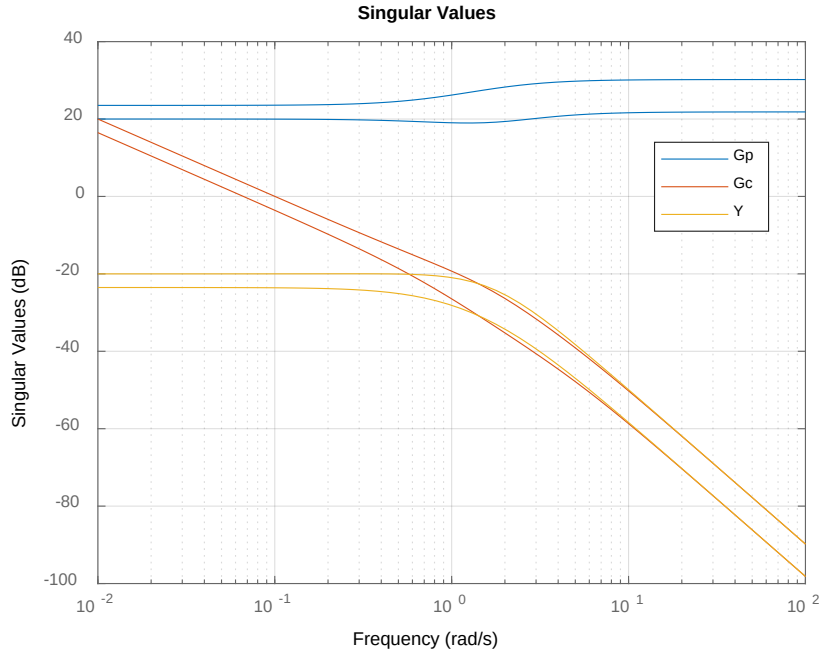


Figure 13.8: MIMO controller, Plant and Youla Transfer Function Matrices Singular Values.

In Figure (13.8), the plant, G_p , the Youla, Y , and the MIMO controller, G_c singular values are illustrated. In this example, the plant is high gain and therefore, the Youla transfer function matrix is low gain at low frequency as

$$\sigma(Y) \rightarrow \frac{1}{\sigma(G_p)} \quad \text{when } \omega \rightarrow 0$$

This is similar to the SISO case as discussed in the previous Chapters. In addition, as it can be seen from Figure (13.8),

$$\sigma(Y) \rightarrow \sigma(G_c) \quad \text{when } \omega \rightarrow \infty$$

Another important point, which is the result of a new finding as presented in the SISO control development, is the frequency at which the feedback loses all its benefits. This frequency is specified to be the frequency when the three closed loop transfer

function matrices, T_y , S_y and Y are within $\pm 2\%$ to $\pm 5\%$ their final values,

$$T_y \rightarrow L_y$$

$$S_y \rightarrow 1$$

$$Y \rightarrow G_c$$

7463 This frequency is around 2 rad/sec.

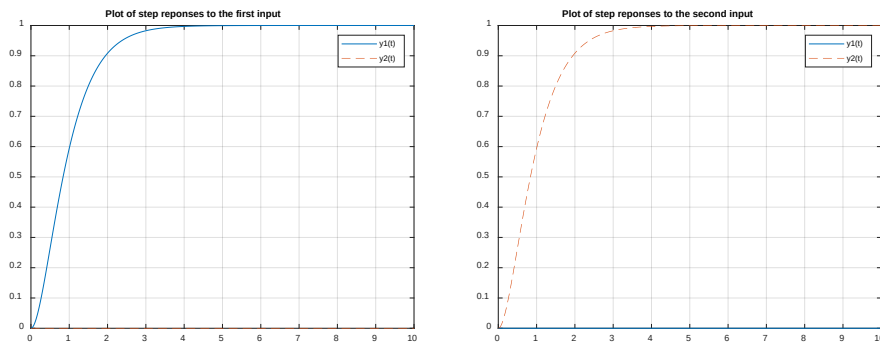


Figure 13.9: Step Input Responses.

7464 Figure (13.9), step input responses of the closed loop transfer function, T_y . Again,
 7465 the decoupling is evident from this Figure. Only the first output responds to the first
 7466 unit step input, and the second output is exactly equal to zero. While, the response of
 7467 the first output to the second unit step input is precisely equal to zero.

7468 Computing the unimodular matrices and the S-M from using row and column operations
 7469 are not practical, especially when matrix sizes become relatively large. In the next exam-
 7470 ple, we will use the same plant transfer function matrix as above but this time, we only
 7471 compute the S-M form and the unimodular matrices based on our development in section
 7472 3 of Chapter 10.

7473 ■ EXAMPLE 13.2

Let G_p to be a stable and proper transfer function matrix,

$$G_p = 20 \begin{bmatrix} \frac{1}{(s+2)^2} & \frac{s+1}{s+2} \\ -\frac{s+1}{s+2} & \frac{s+1}{s+2} \end{bmatrix} \quad (13.27)$$

7474 We can compute a right MFD form for G_p as follows,

$$G_p = N_R D_R^{-1} \quad (13.28)$$

where we select N_R and D_R to be,

$$D_R = \begin{bmatrix} (s+2)^2 & 0 \\ 0 & s+2 \end{bmatrix}$$

$$N_R = 20 \begin{bmatrix} 1 & s+1 \\ -(s+1)(s+2) & s+1 \end{bmatrix}$$

7475 As we made this transformation arbitrary, we don't know if N_R and D_R are co-
 7476 prime. Next, we compute the S-M form using the following steps given in section 3
 7477 of Chapter 10, There are different approaches to compute Smith-McMillan form, the
 7478 most common approach discussed previously is as follows,

7479 ■ Define a determinantal divisor $D_k = \text{greatest common divisor (gcd) of all } k \times k$
 7480 minors and $D_0 = 1$ with the following steps,

7481 a Factor $G_p(s) = \frac{p(s)}{\delta(s)}$

7482 b Find all minors of $p(s)$

7483 c Compute D_k for each $k \times k$ minor.

7484 d Calculate $\varepsilon_k = \frac{D_k}{D_{k-1}}$ for each $i \in \{1, \dots, p\}$

7485 e Find M using $M = \frac{1}{\delta(s)} \text{diag}(\varepsilon_k)$

We choose $\frac{1}{\delta(s)} = \frac{20}{(s+2)^2}$ and then

$$p(s) = \begin{bmatrix} 1 & (s+1)(s+2) \\ -(s+1)(s+2) & (s+1)(s+2) \end{bmatrix}$$

we can then write the determinantal divisors as,

$$D_0 = 1, \quad D_1 = 1, \quad D_2 = (s+1)(s+2)(s^2 + 3s + 3)$$

$$\Rightarrow \epsilon_1 = \frac{D_1}{D_0} = 1, \quad \epsilon_2 = \frac{D_2}{D_1} = (s+1)(s+2)(s^2 + 3s + 3)$$

Therefore, the S-M form can be written as,

$$M_p = 20 \begin{bmatrix} \frac{1}{(s+2)^2} & 0 \\ 0 & \frac{(s+1)(s^2 + 3s + 3)}{s+2} \end{bmatrix} \quad (13.29)$$

This is exactly the same S-M form that we found in the previous example using row-column operations.

The intent here to compute the unimodular matrices, U_L and U_R . But, before we accomplish this, we need to check if N_R and D_R in the previous MFD are coprime or not. This is simply accomplished by comparing the zeros of N_R to the zeros of M_p (S-M form) and zeros of D_R to poles of the M_p . It is obvious that the poles and zeros of the S-M form are exactly the same as zeros of D_R and N_R . So, we conclude that N_R and D_R are coprime.

So, an irreducible realization of $G(s)$ is

$$G_p = N_R D_R^{-1} = 20 \underbrace{\begin{bmatrix} 1 & s+1 \\ -(s+1)(s+2) & s+1 \end{bmatrix}}_{\text{Zeros} = -1, -\frac{3}{2} \pm j\sqrt{\frac{3}{2}}} \underbrace{\begin{bmatrix} (s+2)^2 & 0 \\ 0 & s+2 \end{bmatrix}^{-1}}_{\text{Poles} = -2, -2, -2}$$

Note that the degree of this denominator polynomial matrix, D_R , is 3, precisely equal to the SM degree.

Next, we must find U_R and U_L , by knowing both the SM form and a coprime MFD,

$$G_p = \begin{bmatrix} 1 & s+1 \\ -(s+1)(s+2) & s+1 \end{bmatrix} \begin{bmatrix} (s+2)^2 & 0 \\ 0 & s+2 \end{bmatrix}^{-1} \sim \underbrace{\begin{bmatrix} \frac{1}{(s+2)^2} & 0 \\ 0 & \frac{(s+1)(s^2+3s+3)}{s+2} \end{bmatrix}}_{\text{S-M form}}$$

we know,

$$N_R = U_L^{-1} \text{diag}(\eta_1, \eta_2, \dots, \eta_p)$$

$$\begin{bmatrix} 1 & s+1 \\ -(s+1)(s+2) & s+1 \end{bmatrix} = U_L^{-1} \begin{bmatrix} 1 & 0 \\ 0 & (s+1)(s^2+3s+3) \end{bmatrix}$$

therefore,

$$U_L = \begin{bmatrix} 1 & 0 \\ 0 & (s+1)(s^2+3s+3) \end{bmatrix} \begin{bmatrix} 1 & s+1 \\ -(s+1)(s+2) & s+1 \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{s^2+3s+3} & \frac{-1}{s^2+3s+3} \\ (s+1)(s+2) & 1 \end{bmatrix}$$

and

$$D_R = U_R \text{diag}(\delta_1, \delta_2, \dots, \delta_p)$$

$$\begin{bmatrix} (s+2)^2 & 0 \\ 0 & s+2 \end{bmatrix} = U_R \begin{bmatrix} (s+2)^2 & 0 \\ 0 & s+2 \end{bmatrix}$$

so,

$$U_R = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Now we can verify that,

$$G_p = U_L^{-1} M_p U_R^{-1}$$

The computed U_L is not a polynomial matrix, we could use the "smithForm" command in Matlab (this command only works for square transfer function matrices, we will discuss computation of unimodular matrices for non-square transfer function matrices in the next section) to find the two unimodular matrices U_L and U_R , which are polynomials, as follows,

$$U_L = \begin{bmatrix} 1 & 0 \\ -(s^2 + 3s + 2) & -1 \end{bmatrix}$$

$$U_R = \begin{bmatrix} 1 & s^2 + 3s + 2 \\ 0 & -1 \end{bmatrix}$$

13.4 MIMO Feedback Control Design: Unstable Plants

In this section, we extend our MIMO control design approach to unstable plants. As shown

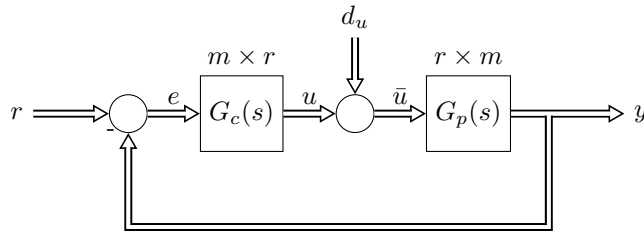


Figure 13.10: MIMO Feedback System without Noise Injection

previously, there are 4 independent transfer function matrices that need to be analyzed for internal stability,

$$\begin{bmatrix} \bar{u} \\ y \end{bmatrix} = \begin{bmatrix} Y & I - YG_p \\ G_p Y & G_p(I - YG_p) \end{bmatrix} \begin{bmatrix} r \\ d_u \end{bmatrix} \quad (13.30)$$

Necessary conditions for these transfer functions include,

▪ Every uncanceled pole of L must be a zero of S

▪ Every uncanceled zero of L must be a zero of T

In MIMO systems, the “location” of zeros of T and S is important. In addition, Y , S , T , and G_c must be proper. This is complicated by the fact that equivalence transformations (utilizing S-M form for our MIMO control design) do not preserve properness. This must be checked separately!!!.

13.4.1 The Proposed Control Design Method

1. Compute the Smith-McMillan form (M_p) and a MFD ($N_R D_R^{-1}$) of the plant.
2. Find U_L and U_R using either column and row operations or using Smith-McMillan form and coprime factorization ($N'_R D'^{-1}_R$), or other methods, which will be discussed in this section.
3. Choose M_Y to shape $M_T = M_p M_Y$, where M_T is decoupled T_y , or select the desired M_T and then compute M_Y . Note that $Y = U_R M_Y U_L$, check and make sure all entries of Y are proper.
4. Check the equivalent SISO interpolation conditions (discussed below) for each diagonal entry of M_T .
5. Find $T_y = U_L^{-1} M_p M_Y U_L$.
6. Find $S_y = I - T_y = U_L^{-1} (I - M_p M_Y) U_L$.
7. Compute $G_c = U_R (I - M_Y M_p)^{-1} M_Y U_L$.
8. Find $S_y G_p = U_L^{-1} (I - M_p M_Y) M_p U_R^{-1}$ and make sure its shape meets the requirement.
9. Check the, $\mathcal{K}$, condition numbers of G_p and G_c to make sure designing for one node does meet the appropriate shapes for T_u , and S_u .

Note that $M_p = \text{diag}\{G_1, G_2, \dots\}$, $M_T = \text{diag}\{T_1, T_2, \dots\}$, and $I - M_T = \text{diag}\{S_1, S_2, \dots\}$.

The SISO interpolation conditions that are applied to each transfer function G_k .

- If G_k has a $j\omega$ -axis or a RHP pole at p_k , then $S_k(p_k) = 0$ and $T_k(p_k) = 1$.
- If G_k has a non-minimum phase zero at z_k , then $S_k(z_k) = 1$ and $T_k(z_k) = 0$.
- if p_k or z_k is of order m then, in addition to the above conditions, the first to the $m^{\text{th}} - 1$ derivatives of S_k and T_k evaluated at p_k or z_k have to be equal to zero, as discussed in the SISO control design part.

The following examples should provide some insight into the proposed MIMO control design method.

EXAMPLE 13.3

$$G_p = 100 \begin{bmatrix} \frac{1}{s+1} & -\frac{1}{s(s+1)} \\ \frac{1}{s+1} & \frac{1}{s+1} \end{bmatrix} \quad (13.31)$$

$$= 100 \underbrace{\begin{bmatrix} 1 & 0 \\ s & 1 \end{bmatrix}}_{U_L^{-1}} \underbrace{\begin{bmatrix} \frac{1}{s(s+1)} & 0 \\ 0 & \frac{s+2}{s+1} \end{bmatrix}}_{M_p} \underbrace{\begin{bmatrix} s & -1 \\ 1 & 0 \end{bmatrix}}_{U_R^{-1}} \quad (13.32)$$

Computation of U_L and U_R are not shown and left as an exercise.

Pole are: $-1, -1, 0$

Zeros at: -2 We select the following Y ,

$$Y = \frac{1}{100} \underbrace{\begin{bmatrix} 0 & 1 \\ 1 & s \end{bmatrix}}_{U_R} \underbrace{\begin{bmatrix} Y_1 & 0 \\ 0 & Y_2 \end{bmatrix}}_{M_Y} \underbrace{\begin{bmatrix} s & 0 \\ 1 & 1 \end{bmatrix}}_{U_L} \quad (13.33)$$

$$M_p M_Y = \begin{bmatrix} \frac{Y_1}{s(s+1)} & 0 \\ 0 & \frac{Y_2(s+2)}{s+1} \end{bmatrix} \quad (13.34)$$

Therefore,

$$T_1 = \frac{Y_1}{s(s+1)} \text{ and } T_2 = \frac{Y_2(s+2)}{s+1} \quad (13.35)$$

Let

$$Y_1 = \frac{s}{\tau_1 s + 1} \text{ and } Y_2 = \frac{1}{(\tau_1 s + 1)(s + 2)} \quad (13.36)$$

This yields the complementary sensitivity functions

$$T_1 = \frac{1}{(\tau_1 s + 1)(s + 1)} \text{ and } T_2 = \frac{1}{(\tau_1 s + 1)(s + 1)} \quad (13.37)$$

$T_1(0) = 1$ so it meets the required interpolation condition, as the first entry in M_p is not BIBO stable.

Letting $\tau_1 = 1$ and inserting Y_1 and Y_2 into Equation 17.11 yields

$$Y = \frac{1}{100} \begin{bmatrix} 0 & 1 \\ -1 & s \end{bmatrix} \begin{bmatrix} \frac{s}{s+1} & 0 \\ 0 & \frac{1}{(s+1)(s+2)} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ s & 1 \end{bmatrix} \quad (13.38)$$

$$= \frac{1}{100} \begin{bmatrix} \frac{s}{(s+1)(s+2)} & \frac{1}{(s+1)(s+2)} \\ \frac{-2s}{(s+1)(s+2)} & \frac{s}{(s+1)(s+2)} \end{bmatrix} \quad (13.39)$$

From here it is straight forward to compute G_c . *Checking the conditions 8 and 9 as given above is left as an exercise.*

EXAMPLE 13.4

$$G_p = \begin{bmatrix} 2 & -\frac{1}{s+1} \\ \frac{1}{s^2} & \frac{1}{s^2(s+1)} \end{bmatrix} \quad (13.40)$$

$$= \underbrace{\begin{bmatrix} -s^2 & 1 \\ 1 & 0 \end{bmatrix}}_{U_L^{-1}} \underbrace{\begin{bmatrix} \frac{1}{s^2(s+1)} & 0 \\ 0 & 1 \end{bmatrix}}_{M_p} \underbrace{\begin{bmatrix} -(s+1) & 1 \\ 1 & 0 \end{bmatrix}}_{U_R^{-1}} \quad (13.41)$$

Again the computation of U_L and U_R are not shown and left as an exercise. For this example, the pole are: $-1, 0, 0$, and there are no zeros.

So, the two of the poles are at origin and the plant is BIBO unstable. We write the following expression for Y .

$$Y = \underbrace{\begin{bmatrix} 0 & 1 \\ 1 & s+1 \end{bmatrix}}_{U_R} \underbrace{\begin{bmatrix} Y_1 & 0 \\ 0 & Y_2 \end{bmatrix}}_{M_Y} \underbrace{\begin{bmatrix} 0 & 1 \\ 1 & s^2 \end{bmatrix}}_{U_L} \quad (13.42)$$

Knowing that,

$$T_1 = \frac{Y_1}{s^2(s+1)} \text{ and } T_2 = Y_2 \quad (13.43)$$

Now, internal stability dictates the shape of Y_2 , let

$$Y_1 = \frac{s^2}{\delta_1(s)} \Rightarrow T_1 = \frac{1}{\delta_1(s)(s+1)} \quad (13.44)$$

Where $\delta_1(s)$ is some polynomial of s . The interpolation conditions for the double pole at 0 are $T_1(0) = 1$ and $T_1'(0) = 0$, where T_1' is the first derivative of T_1 with respect to s , $T_1'(0) = \frac{dT_1(0)}{ds}$. Taking the derivative of T_1 with respect to s ,

$$T_1'(s) = \frac{-(\delta_1'(s)(s+1) + \delta_1(s))}{[\delta_1(s)(s+1)]^2} \quad (13.45)$$

Applying the interpolation conditions yields the following set of conditions that must be met for $\delta_1(s)$. From Equation (13.44),

$$T_1(0) = 1 \Rightarrow \delta_1(0) = 1 \quad (13.46)$$

From Equation (13.45),

$$T_1'(0) = 0 \Rightarrow \delta_1(0) = -\delta_1'(0) \quad (13.47)$$

How can we meet these conditions for assuring internal stability?

We will answer this question by selecting a simple polynomial for $\delta_1(s)$. Letting $\delta_1(s) = as + b$, (from the above conditions, $\delta_1(0) = b = 1$, and $\delta_1'(0) = a = -1$), this means that $\delta_1(s) = 1 - s$ which is not stable, obviously not meeting stability requirements of $T_1(s)$ (it should be noted that the higher order polynomials will have the same stability issue).

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Try a new selection for Y_1 such as,

$$Y_1 = \frac{s^2 \eta_1(s)}{\delta_1(s)} \Rightarrow T_1(s) = \frac{\eta_1(s)}{\delta_1(s)(s+1)} \quad (13.48)$$

Then by using the interpolation conditions above, Equations (??) and (13.45), the following conditions exist

$$T_1(0) = \frac{\eta_1(0)}{\delta_1(0)} = 1 \Rightarrow \eta_1(0) = \delta_1(0) \quad (13.49)$$

$$T_1'(0) = \frac{\eta_1'(0)\delta_1(0) - \eta_1(0)[\delta_1'(0) + \delta_1(0)]}{\delta_1^2(0)} = 0 \Rightarrow \eta_1'(0) = \delta_1'(0) + \delta_1(0) \quad (13.50)$$

It is not difficult to show that by setting

$$\eta_1(s) = \cdots + (q+p)s + p \quad (13.51)$$

$$\delta_1(s) = \cdots + qs + p \quad (13.52)$$

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will meet the interpolation conditions.

Another concern is the properness of Y , using the transformation

$$Y = \begin{bmatrix} 0 & 1 \\ 1 & s+1 \end{bmatrix} \begin{bmatrix} \frac{s^2 \eta_1(s)}{\delta_1(s)} & 0 \\ 0 & \frac{1}{\delta_2(s)} \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & s^2 \end{bmatrix} \quad (13.53)$$

$$= \begin{bmatrix} \frac{1}{\delta_2(s)} & \frac{s^2}{\delta_2(s)} \\ \frac{s+1}{\delta_2(s)} & s^2 \left(\frac{s+1}{\delta_2(s)} + \frac{\eta_1(s)}{\delta_1(s)} \right) \end{bmatrix} \quad (13.54)$$

By ensuring that Y is selected to be proper and stable, then all of the other closed transfer function matrices will also be proper.

For the properness of Y , the two requirements that must be met, from Equation 13.54, are

$$\deg \delta_2(s) \geq 2 \quad (13.55)$$

$$\deg \delta_1(s) + \deg \delta_2(s) \geq \max\{2 + \deg \eta_1(s) + \deg \delta_2(s), 3 + \deg \delta_1(s)\} \quad (13.56)$$

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The degree constraints in Equations (13.55) and (13.56) come from $\frac{s^2}{\delta_2(s)}$ and $\frac{s^2(s+1)\delta_1(s) + s^2\eta_1(s)\delta_2(s)}{\delta_1(s)\delta_2(s)}$, the entries of the TFM in Equation 13.54.

If we select $\delta_1(s) = \delta_2(s) = \delta(s)$, then the constraints reduce to

$$\deg \delta(s) \geq \max\{\deg \eta_1(s) + 2, 3\} \quad (13.57)$$

Hence, by selecting $\deg \eta_1(s) = 1$ makes the minimal degree of $\delta(s)$ to be 3. Setting $\delta(s) = (s + 1)^3 = s^3 + 3s^2 + 3s + 1$ with the interpolation conditions of Equations (13.51) and (13.52) makes $\eta_1(s) = 4s + 1$.

Therefore,

$$Y_1 = \frac{s^2(4s + 1)}{(s + 1)^3}, T_1 = \frac{4s + 1}{(s + 1)^4} \quad (13.58)$$

$$Y_2 = \frac{1}{(s + 1)^3}, T_2 = \frac{1}{(s + 1)^3} \quad (13.59)$$

Now using $S = I - T$

$$S_1 = \frac{s^2(s^2 + 4s + 6)}{(s + 1)^4} \quad (13.60)$$

$$S_2 = \frac{s(s^2 + 3s + 3)}{(s + 1)^3} \quad (13.61)$$

Finding T_y using the transformation $T_y = U_L^{-1} M_y U_L$

$$T_y = \begin{bmatrix} -s^2 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{4s + 1}{(s + 1)^4} & 0 \\ 0 & \frac{1}{(s + 1)^3} \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & s^2 \end{bmatrix} \quad (13.62)$$

$$= \begin{bmatrix} \frac{1}{(s + 1)^3} & \frac{-3s^3}{(s + 1)^4} \\ 0 & \frac{4s + 1}{(s + 1)^4} \end{bmatrix} \quad (13.63)$$

Observe that $T_y(0) = I$ which means decoupling at low frequency.

We can then compute the controller G_c . *Checking the conditions 8 and 9 as given above is left as an exercise.*

So far, we used our proposed method to design MIMO controllers for square plant matrices. In the next example, we generalize our method to a non-square plant matrix. In the last chapter of this volume, we apply our MIMO technique to both square and non-square plants which are derived from practical applications.

7591 **EXAMPLE 13.5**

$$G_p = 100 \begin{bmatrix} \frac{s}{s+1} & \frac{s}{s+1} & \frac{1}{s+1} \\ \frac{1}{s+1} & \frac{1}{s+1} & \frac{1}{s+1} \end{bmatrix} \quad (13.64)$$

$$= 100 \underbrace{\begin{bmatrix} -1 \\ 0 \end{bmatrix}}_{U_L^{-1}} \underbrace{\begin{bmatrix} \frac{1}{s+1} & 0 & 0 \\ 0 & \frac{1-s}{s+1} & 0 \end{bmatrix}}_{M_p} \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}}_{U_R^{-1}} \quad (13.65)$$

7592 Computation of U_L and U_R are done using column-row operations, this is left as an
 7593 exercise. Below, we will compute U_L and U_R by finding first the S-M form using gcd
 7594 of all $k \times k$ minors, as described in Chapter 10, and coprime MFDs.

7595 First, we find the S-M form by writing $G_p = \frac{p(s)}{\delta(s)}$ as,

$$G_p = 100 \frac{1}{s+1} \begin{bmatrix} s & 1 \\ 1 & 1 \end{bmatrix} \quad (13.66)$$

Then the determinantal divisors are written,

$$D_0 = 1, \quad (13.67)$$

The determinantal divisors D_1 is obtained by computing all 1×1 minors of $p(s)$ as,

$$D_1 = \begin{vmatrix} s & 1 \\ 1 & 1 \end{vmatrix} = 1 \quad (13.68)$$

The determinantal divisor, D_2 is obtained by computing all 2×2 minors of $p(s)$ as,

$$D_2 = \begin{vmatrix} s & 1 \\ 1 & 1 \end{vmatrix} = [0, s-1, s-1] = s-1 \quad (13.69)$$

$$\Rightarrow \epsilon_1 = \frac{D_1}{D_0} = 1, \quad \epsilon_2 = \frac{D_2}{D_1} = s-1$$

$$M_p = 100 \frac{1}{s+1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & s-1 & 0 \end{bmatrix} = 100 \begin{bmatrix} \frac{1}{s+1} & 0 & 0 \\ 0 & \frac{s-1}{s+1} & 0 \end{bmatrix}$$

7597 We should know that M_p is the same size of G_p but with the last column of zeros.

7598 From S-M form, we know, Pole are: $-1, -1$

7599 Zeros at: 1

7600 Now, we write G_p as multiplication of a numerator, N_R and denominator, D_R
 7601 matrices as follows,

$$G_p = 100 \begin{bmatrix} \frac{s}{s+1} & \frac{s}{s+1} & \frac{1}{s+1} \\ \frac{1}{s+1} & \frac{1}{s+1} & \frac{1}{s+1} \end{bmatrix} \quad (13.70)$$

$$= 100 \underbrace{\begin{bmatrix} s & s & 1 \\ 1 & 1 & 1 \end{bmatrix}}_{N_R} \underbrace{\begin{bmatrix} s+1 & 0 & 0 \\ 0 & s+1 & 0 \\ 0 & 0 & s+1 \end{bmatrix}}_{D_R: \deg=3}^{-1} \quad (13.71)$$

We know this MFD is not coprime as the degree of D_R is higher than the number of poles in S-M form, M_p .

We use our previously presented procedure in chapter 10 to compute the right common factor, W (this computation is not shown here and is left as an exercise). The right common factor is given below,

$$W = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ s+1 & 0 & 0 \end{bmatrix} \Rightarrow W^{-1} = \begin{bmatrix} 0 & 0 & \frac{1}{s+1} \\ 1 & 0 & -\frac{1}{s+1} \\ 0 & 1 & 0 \end{bmatrix}$$

$$N'_R = N_R W^{-1} = \begin{bmatrix} s & s & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & \frac{1}{s+1} \\ 1 & 0 & -\frac{1}{s+1} \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} s & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$D'_R = D_R W^{-1} = \begin{bmatrix} s+1 & 0 & 0 \\ 0 & s+1 & 0 \\ 0 & 0 & s+1 \end{bmatrix} \begin{bmatrix} 0 & 0 & \frac{1}{s+1} \\ 1 & 0 & -\frac{1}{s+1} \\ 0 & 1 & 0 \end{bmatrix} \quad (13.72)$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ s+1 & 0 & -1 \\ 0 & s+1 & 0 \end{bmatrix} \quad (13.73)$$

$$\Rightarrow G_p = 100 \underbrace{\begin{bmatrix} s & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ s+1 & 0 & -1 \\ 0 & s+1 & 0 \end{bmatrix}^{-1}}_{\text{Coprime}} \quad (13.74)$$

Now that we have the coprime factorization and the S-M form of G_p , we can compute U_L and U_R matrices as follows,

$$\begin{aligned} N'_R &= U_L^{-1} \text{diag}(\eta_1, \dots, \eta_p, 0, \dots, 0) \\ D'_R &= U_R \text{diag}(\delta_1, \dots, \delta_p, 1, \dots, 1) \end{aligned} \quad (13.75)$$

where,

$$\text{diag}(\eta_1, \dots, \eta_p, 0, \dots, 0) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & s-1 & 0 \end{bmatrix}$$

and

$$\text{diag}(\delta_1, \dots, \delta_p, 1, \dots, 1) = \begin{bmatrix} s+1 & 0 & 0 \\ 0 & s+1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

From Equation (13.75), we can compute U_L and U_R ,

$$U_L = \begin{bmatrix} \frac{1}{s-1} & \frac{-1}{s-1} \\ -1 & s \end{bmatrix} \quad (13.76)$$

$$U_R = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \quad (13.77)$$

these are not the same U_L and U_R as given at the beginning of this problem as the methods of the two computation differ but one can check that,

$$M_p = U_L G_p U_R$$

Now the S-M form can be computed from the unimodular matrices, U_L and U_R , and G_p , as shown above, to check if we obtain the same S-M form as given above.

Unfortunately, the left unimodular matrix U_L can not be utilized to compute a MIMO controller based on our proposed method, as this matrix is not stable (BIBO stability).

We introduce the following robust method to compute the unimodular matrices for both tall and wide matrices. It can be shown that for the tall matrices, where the number of sensors are greater than the actuators (underactuated plants), using the following method, one can automatically compute the unimodular matrices. This method utilizes the hermite form to compute the left unimodular matrix U_L from $p(s)$, where $G_p = \frac{p(s)}{\delta(s)}$. This could be done using the Matlab following command,

$$[U_L, H] = \text{hermiteForm}(p(s))$$

Where U_L is the left unimodular matrix and H is the upper triangular matrix. Then knowing U_L , U_R could be computed from,

$$U_R = G \backslash M_p$$

Where $G = U_L G_p$ and $\backslash$ stands for the solution of the linear equations $M_p = G U_R$.

When the plant matrix is wide, where the number of sensors are less than the actuators (overactuated plants), such as this case. It can be shown that directly using the above approach results in rank deficiency of the right unimodular matrix, U_R . So, we use a simple trick to get around this problem as follows,

Matrix $p(s)$ is computed as previously but we transpose $p(s)$ before computing the left unimodular matrix, U_L , when using the hermite form given above. Let's call this U_L, U_{L_w} , where w stands for wide matrix.

$$p(s) = p(s)'$$

Then, we compute the hermite form as,

$$[U_{L_w}, H] = \mathbf{hermiteForm}(p(s))$$

To compute the U_{R_w} , we also need to transpose both the plant matrix G_p and the S-M form M_p , given above,

$$G_p = G_p'$$

$$M_p = M_p'$$

We can then compute the U_{R_w} as given above,

$$U_{R_w} = G \backslash M_p$$

Where $G = U_{L_w} G_P$ and the final unimodular matrices U_L and U_R to be used in control design are computed as follows,

$$U_R = U_{L_w}'$$

and

$$U_L = U_{R_w}'$$

Using the above approach, we compute U_L and U_R to be

$$U_L = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \quad (13.78)$$

$$U_R = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & -1 \\ 1 & s & 0 \end{bmatrix} \quad (13.79)$$

Now these unimodular matrices could be utilized to compute a MIMO controller based on our proposed method. This is left as an exercise.

13.5 Another approach for MIMO Controller Design

If the plant TFM G_p is minimum phase, stable, and square:

We can design the MIMO controller very quickly using the following steps,

- Select $Y = G_p^{-1}T_{yd}$ where T_{yd} is a desired proper stable closed-loop transfer function matrix.
- Then $T_y = G_p Y$.
- Find $S_y = I - T_y$.
- Finally $G_c = Y S_y^{-1}$.

EXAMPLE 13.6

As above, we us

$$G_p = 20 \begin{bmatrix} 1 & s+1 \\ \frac{(s+2)^2}{s+1} & \frac{s+2}{s+1} \\ \frac{s+1}{s+2} & \frac{s+1}{s+2} \end{bmatrix} \quad (13.80)$$

We want

$$Y = G_p^{-1}T_{yd} \quad (13.81)$$

$$= G_p^{-1} \left(\frac{4}{(s+2)^2} \right) I \quad (13.82)$$

$$= \frac{1}{5} \begin{bmatrix} \frac{1}{s^2+3s+3} & -\frac{1}{s^2+3s+3} \\ \frac{1}{s^2+3s+3} & \frac{1}{s^2+3s+3} \\ \frac{1}{s^2+3s+3} & \frac{1}{(s+1)(s+2)(s^2+3s+3)} \end{bmatrix} \quad (13.83)$$

Now

$$T_y = G_p Y = \frac{4}{(s+2)^2} I = T_{yd} \quad (13.84)$$

and

$$S_y = I - T_y = \frac{s(s+4)}{(s+2)^2} I \quad (13.85)$$

Finally

$$G_c = Y S_y^{-1} = \frac{1}{5s(s+4)(s^2+3s+3)} \begin{bmatrix} (s+2)^2 & -(s+2)^2 \\ (s+2)^2 & \frac{s+2}{s+1} \end{bmatrix} \quad (13.86)$$

CHAPTER 14

NORMS OF FEEDBACK SYSTEMS

14.1 Norms

In this chapter, we introduce the norms, which discussed in the SISO section of this book, for vector signals with the applications to MIMO controller design. Similar to SISO systems, we concentrate on two fundamental norms, $\mathcal{H}_2$ and $\mathcal{H}_\infty$. In addition, we revisit LFT (Linear Fractional Transformation) with application to MIMO controller design optimization. We finish this chapter by introducing the concept of uncertainties, as this discussed in the SISO part of of this book, for MIMO systems.

These topics have been discussed in many textbooks and we introduce them again so that we could provide comparative results, in the last chapter of this book, between control concepts such as MIMO $\mathcal{H}_\infty$ with previously introduced MIMO Youla control method.

A norm measures the “size” of signals or systems but what do we mean by size? let $V_s = \{V_1(\cdot), \dots, V_k(\cdot)\}$ where $V_i \in \mathbb{R}^n$ and $\|V_i(\cdot)\|$ is a mapping $\mathbb{R}^n \rightarrow \mathbb{R}$.

Definition 14.1. (Norm). $\|V_i(\cdot)\|$ is a norm if it meets the following requirements

1. Nonnegative: $\|V_i(\cdot)\| \geq 0$ for all k
2. Homogeneity: $\|\alpha V_i(\cdot)\| = |\alpha| \|V_i(\cdot)\|$
3. Triangle Inequality: $\|V_i(\cdot) + V_j(\cdot)\| \leq \|V_i(\cdot)\| + \|V_j(\cdot)\|$

14.1.1 Signal Norms, the Discrete Case

The mean of a signal V is given by

$$\mu_V \triangleq \frac{1}{N} \sum_{n=0}^{N-1} V_n \quad (14.1)$$

If signal V was a stationary random signal then the calculation of its mean, as given in Equation (14.1) still applies but it would be an estimate of its mean[17.2]. The total energy of V is defined as

$$E_V \triangleq \sum_{n=0}^{N-1} |V_n|^2 \quad (14.2)$$

The average power of the signal is defined as

$$P_V \triangleq \frac{1}{N} \sum_{n=0}^{N-1} |V_n|^2 \quad (14.3)$$

The variance of V is defined as the power of the signal with its mean removed

$$\sigma_V^2 \triangleq \frac{1}{N} \sum_{n=0}^{N-1} |V_n - \mu_V|^2 \quad (14.4)$$

For real signals this reduces to $\sigma_V^2 = P_V - \mu_V^2$

the 2-norm of a signal is defined as

$$\|V\|_2 \triangleq \sqrt{E_V} = \sqrt{\sum_{n=0}^{N-1} |V_n|^2} \quad (14.5)$$

We could choose a normalized 2-norm (rms)

$$\|V\|_{rms} \triangleq \sqrt{P_V} = \sqrt{\frac{1}{N} \sum_{n=0}^{N-1} |V_n|^2} \quad (14.6)$$

more generally we can define a p -norm as

$$\|V\|_p \triangleq \left[\sum_{n=0}^{N-1} |V_n|^p \right]^{1/p} \quad (14.7)$$

The most popular norms are $p = 1, 2$, and ∞ with ∞ -norm is defined as

$$\|V\|_\infty \triangleq \max_{0 \leq n < N} \{|V_n|\} \quad (14.8)$$

14.1.2 System Norms

For a dynamic system of the form

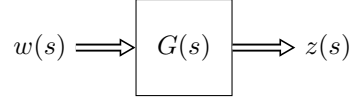


Figure 14.1: Standard I/O Relationship

An induced system norm is defined as

$$\max_{w \neq 0} \frac{\overbrace{\|Gw\|_p}^{\|z\|_p}}{\|w\|_p} = \|G\|_{p,i} \quad (14.9)$$

$\|G\|_{p,i}$ is known as an induced p norm.

EXAMPLE 14.1

In this example, we illustrate computation of an induced 2-norm for a static matrix A .

Consider a static matrix A , then by definition,

$$\max_{w \neq 0} \frac{\|Aw\|_2}{\|w\|_2} = \|A\|_{2,i} \quad (14.10)$$

From SVD, $A = U\Sigma V^T$, then, we know that the largest possible gain of A is in the direction of the first right singular vector. Let's assume $w = r_1$, where r_1 is first right singular vector, we can then write,

$$Ar_1 = \sigma_1 l_1$$

plug this Equation into (14.10),

$$\frac{\|Ar_1\|_2}{\|r_1\|_2} = \frac{\|\sigma_1 l_1\|_2}{\|r_1\|_2} = \sigma_1$$

We know that $\sigma_1 = \sigma_{max}$.

Consider a transfer function matrix $G(s)$ with each element being strictly proper and stable.

Then the $\mathcal{H}_2$ norm of the TFM is given by

$$\|G\|_2^2 = \frac{1}{2\pi} \text{tr} \int_{-\infty}^{\infty} G(j\omega)G^*(j\omega)d\omega \quad (14.11)$$

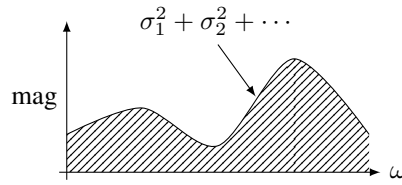
For a 2-norm,

$$GG^* = \sum_k l_k \sigma_k^2 l_k^*$$

and therefore

$$\|G\|_2^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \sum_k \sigma_k^2(\omega) d\omega \quad (14.12)$$

So $\|G\|_2^2$ is the equivalent to the area under the curve of the sum of the squares of the



7681

7682 singular values of G .

7683

7684 Again referring to the system of Figure 15.8 where $G(s)$ is the closed loop transfer function

7685 $T_{zw}(s)$, then

$$z(t) = \int_0^T G(t - \tau)w(\tau)d\tau \quad (14.13)$$

7686 Let each input be a zero-mean white noise with unit variance, such that, $E[ww^T] =$

7687 $I_m \delta(t - \tau)$ where $\delta(t - \tau)$ is a Dirac delta function and it has a value when $t = \tau$ and

7688 equal to zero when $t \neq \tau$, and the expected value of a discrete Random variable X is,

7689 $E[X] = \sum_{i=1}^k x_i p_i$. Then it can be show that,

$$E[z^T z] = \text{tr} \int_0^\infty G^T(t)G(t)dt \quad (14.14)$$

7690 where $E[z^T z] = \lim_{T \rightarrow \infty} E[\frac{1}{T} \int_0^T z^T z dt]$.

Theorem 14.1 (Parseval's Theorem)

$$\text{tr} \int_0^\infty G(t)^T G(t)dt = \frac{1}{2\pi} \text{tr} \int_{-\infty}^\infty G^*(j\omega)G(j\omega)d\omega \quad (14.15)$$

7691 where $G(t)$ is the system impulse response and

$$G(t) = Ce^{At}B \quad (14.16)$$

7692 $e^{At} = \Phi(t)$ is defined as the state transition matrix and $\Phi(t)$ is the inverse Laplace trans-

7693 form $e^{At} = \mathcal{L}^{-1}(sI - A)^{-1}$

Using Parseval's theorem,

$$E[z^T z] = \frac{1}{2\pi} \text{tr} \int_{-\infty}^{\infty} G^*(j\omega) G(j\omega) d\omega = \|G\|_2^2 \quad (14.17)$$

Aside: Minimizing the $\mathcal{H}_2$ norm of a system TFM is equivalent to minimizing the power of the outputs when the inputs are driven by zero-mean white noise.

14.1.3 The $\mathcal{H}_2$ -Norm

This norm can be defined as

- Power of the system output when the input is driven by zero-mean white noise (LQG or $\mathcal{H}_2$) (persistent or everlasting input).
- Energy of the system output (LQR or $\mathcal{H}_2$) when the input is transient and not persistent.

Computation of the $\mathcal{H}_2$ -norm can be done in both frequency, as presented in the SISO section, and state-space form using the system impulse response. Next, we present this computation in the state-space form. Let $G(t)$ be the system impulse response or

$$G(t) = Ce^{At}B \quad (14.18)$$

Matrices A , B , and C are as defined previously, and

$$\|G\|_2^2 = \text{tr} \int_0^{\infty} G(t)G^T(t)dt \quad (14.19)$$

$$= \text{tr} \int_0^{\infty} Ce^{At}BB^Te^{At}C^Tdt \quad (14.20)$$

$$= \text{tr}C \underbrace{\left[\int_0^{\infty} e^{At}BB^Te^{At}dt \right]}_P C^T \quad (14.21)$$

P is defined as the controllability grammian, and it satisfies the Lyapunov equation

$$AP + PA^T + BB^T = 0 \quad (14.22)$$

Proof: See “Linear Controller Design - Limits of Performance” ■

We could utilize the MATLAB command `P=lyap(A,Q)` with $Q = BB^T$ yields the above solution.

Similar to the controllability grammian, the observability grammian could be used for this computation, where the $\mathcal{H}_2$ -norm is written as,

$$\|G\|_2^2 = \text{tr} \int_0^\infty G(t)^T G(t) dt \quad (14.23)$$

7711 This is left as an exercise.

7712 14.1.4 The $\mathcal{H}_\infty$ -Norm

7713 Recall that $\mathcal{H}_\infty$ -Norm for MIMO systems provides the maximum singular value of a system across all frequencies, or $\|G\|_\infty = \max_\omega \bar{\sigma}[G(j\omega)]$

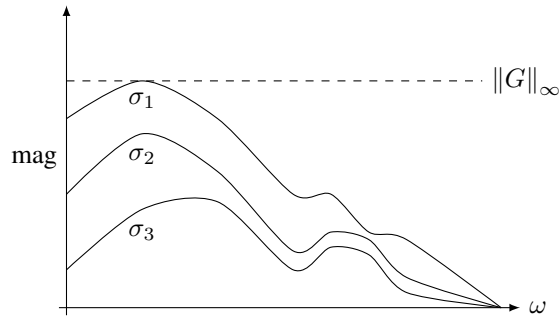


Figure 14.2: $\mathcal{H}_\infty$ -Norm of MIMO systems.

7714

7715 Also recall that

$$\sigma_k = \frac{\|z\|_2}{\|w\|_2} \quad (14.24)$$

7716 **Theorem 14.2** *The ∞ -norm is an induced norm, i.e.*

$$\|G\|_\infty = \max_{w \neq 0} \frac{\|z\|_2}{\|w\|_2} \quad (14.25)$$

Proof:

$$\|z\|_2^2 = \|z(j\omega)\|_2^2 \quad (14.26)$$

$$= \frac{1}{2\pi} \text{tr} \int_{-\infty}^{\infty} |G(j\omega)w(j\omega)|^2 d\omega \quad (14.27)$$

$$= \frac{1}{2\pi} \text{tr} \int_{-\infty}^{\infty} |G(j\omega)|^2 |w(j\omega)|^2 d\omega \quad (14.28)$$

$$\leq \|G\|_\infty^2 \frac{1}{2\pi} \text{tr} \int_{-\infty}^{\infty} |w(j\omega)|^2 d\omega \quad (14.29)$$

$$\leq \|G\|_\infty^2 \|w\|_2^2 \quad (14.30)$$

7717 Now take the maximum of the left hand side to obtain

$$\|G\|_\infty = \max_{w \neq 0} \frac{\|z\|_2}{\|w\|_2} \quad (14.31)$$

7718

■

7719 In other words if

$$\|G\|_\infty \leq \gamma \quad (14.32)$$

7720 where $\gamma \in \mathbb{R}_+$ then

$$\frac{\|z\|_2}{\|w\|_2} \leq \gamma \quad (14.33)$$

7721 This relation can be written as

$$\|z\|_2^2 - \gamma^2 \|w\|_2^2 \leq 0 \quad (14.34)$$

7722 Writing the system ∞ -norm as proposed in Equation (14.34) leads to stat-space solution
7723 for the H_∞ problem as discussed later.

7724 A physical interpretation of the H_∞ -norm is appropriate here. Recall that the square
7725 of the 2-norm is equivalent to signal energy. Therefore, by specifying a bound or limit γ
7726 on the ratio of the output signal energy z to the disturbance energy w , we will therefore
7727 minimize the impact of worst possible disturbance on the system output when minimizing
7728 the H_∞ -norm.

14.2 Linear Fractional Transformations (LFT)

In this section, we generalize the Linear Fractional Transformation concept which is discussed in the SISO part of this book to MIMO optimization and control design.

We generalize linear control design by the following representation,

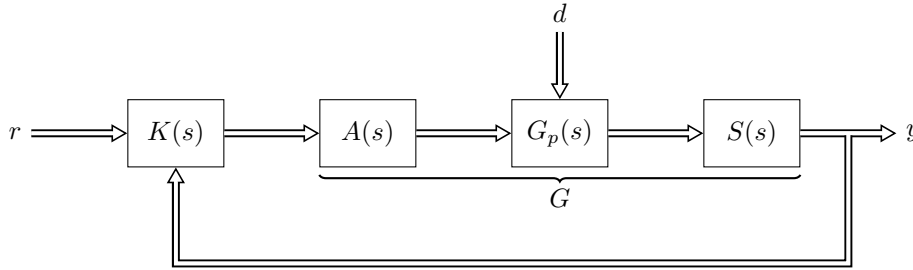


Figure 14.3: General Linear Control Feedback Design

Where $A(s)$ represents the actuator dynamics, $G_p(s)$ is the physical plant, $S(s)$ represents the sensors, $K(s)$ is the controller, with one or more degrees of freedom as discussed below, and d represents the vector of system disturbances. Here $G = SG_pA$.

The vector disturbance signals d can be decomposed as follows,

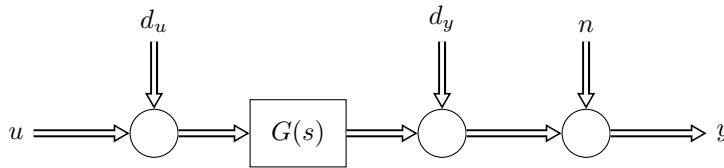
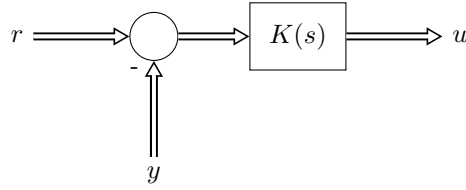


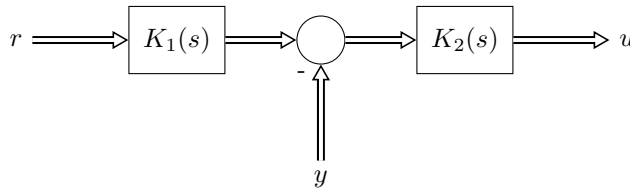
Figure 14.4: Disturbance Decomposition

Where d_u are the vector process disturbances at the input, d_y are the output vector disturbances (low frequency), and n is the vector sensor noise (high frequency).

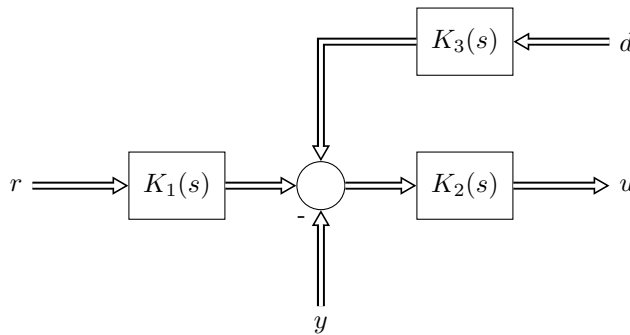
In general, control design will follow one of the following typologies,



(a) 1 Degree of Freedom Controller



(b) 2 Degree of Freedom Controller



(c) 3 Degree of Freedom Controller

Figure 14.5: Degrees of Freedom for Control Design

It is important to note the following points about the higher degrees of freedom controller,

1. The prefilter ($K_1(s)$) in 2 degrees of freedom controller could aid in relaxing the target following constraint ($T(0) = 1$) for the feedback control design. For example, the feedback controller ($K_2(s)$) could be only designed to stabilize the loop and the prefilter ($K_1(s)$) could then be designed to meet the target following constraint.
2. In 3 degrees of freedom controller, $K_3(s)$ controller/estimator, could help in increasing the feedback loop performance and decreasing the actuator efforts.

We can design various degrees of freedom controllers with the use of a framework called the Linear Fractional transformation (LFT) of the system, more specifically this is called the lower LFT, as opposed to the upper LFT, which will be discussed later in this section. All types of controllers with various degrees of freedom, as discussed in the previous section, can be transformed to this framework for the application of norm optimization and control design. This framework is illustrated in the following Figure,

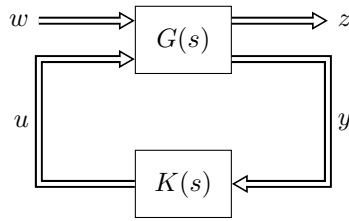


Figure 14.6: Lower LFT on systems

Where,

- The vector w is composed of the exogenous inputs including: references, disturbances, sensor noises, etc.
- The vector z is composed of the signals that we wish to regulate: tracking errors, actuator effort signals that must be kept small and within certain limits, and system states that must be kept within certain limits. These are known as the regulated outputs.
- The vector y is the sensor output measurements that are available to the controller.
- The vector u is the controlled inputs of the system.

14.3 Linear Fractional Transformation Explained

In this section, we will provide some insight into the LFT concept. Assuming a rational function of the form,

$$F(s) = \frac{as + b}{cs + d} \quad (14.35)$$

7771 after manipulation, it can be shown that

$$F(s) = e + fs(1 - gs)^{-1}h \quad (14.36)$$

where e , f , g , and h are constants and

$$e = bd^{-1} \quad (14.37)$$

$$f = a - bd^{-1}c \quad (14.38)$$

$$g = -d^{-1}c \quad (14.39)$$

$$h = d^{-1} \quad (14.40)$$

This results in the following lower LFT,

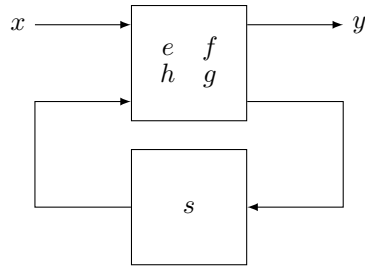


Figure 14.7: Lower LFT

7772

In general form, the lower LFT looks like,

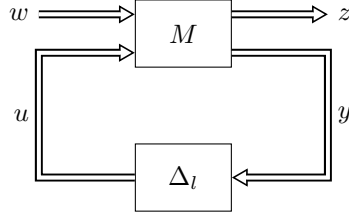


Figure 14.8: Lower MIMO LFT

7773

7774 Where the matrix M can be written as,

$$\begin{bmatrix} z \\ y \end{bmatrix} = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} \begin{bmatrix} w \\ u \end{bmatrix} \quad (14.41)$$

7775 Then, the closed loop map from w to z , using Equation 14.36, can be written as,

$$F_l(M, \Delta_l) = M_{11} + M_{12}\Delta_l(I + M_{22}\Delta_l)^{-1}M_{21} \quad (14.42)$$

7776

Proof: Starting with the relationships,

$$z = M_{11}w + M_{12}u \quad (14.43)$$

$$y = M_{21}w + M_{22}u \quad (14.44)$$

$$u = \Delta_l y \quad (14.45)$$

7777 Combining Equations (14.44) and (14.45) yields,

$$y = (I - M_{22}\Delta_l)^{-1}M_{21}w \quad (14.46)$$

7778 Substituting this into Equation (14.45) yields,

$$u = \Delta_l(I - M_{22}\Delta_l)^{-1}M_{21}w \quad (14.47)$$

7779 Finally substituting this expression for u into Equation (14.43) yields,

$$z = (M_{11} + M_{12}\Delta_l(I - M_{22}\Delta_l)^{-1}M_{21})w \quad (14.48)$$

7780

■

7781 As you might expect, if there is a lower LFT then there is also an upper LFT. This takes

7782 the form,

$$\begin{bmatrix} \tilde{y} \\ z \end{bmatrix} = \begin{bmatrix} M_{u11} & M_{u12} \\ M_{u21} & M_{u22} \end{bmatrix} \begin{bmatrix} \tilde{w} \\ w \end{bmatrix} \quad (14.49)$$

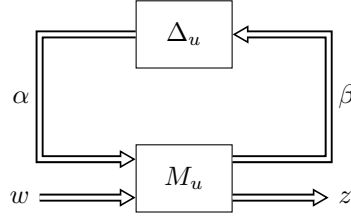


Figure 14.9: Upper system LFT

The closed loop map from w to z is

$$F_u(M_u, \Delta_u) = M_{u22} + M_{u21}\Delta_u(I - M_{u11}\Delta_u)^{-1}M_{u12} \quad (14.50)$$

The upper LFT is used for modeling uncertainties while the lower LFT is used for control design, as will be discussed in this and the following section.

14.3.1 LFTs in Control Design

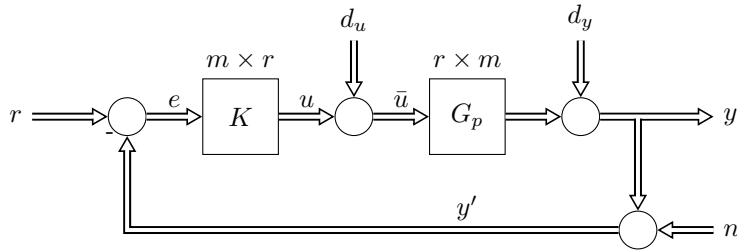


Figure 14.10: MIMO Feedback System

With this system,

- The exogenous inputs are $w = \begin{bmatrix} w_1 & w_2 & w_3 \end{bmatrix}^T = \begin{bmatrix} d & n \end{bmatrix}^T$, where $d = \begin{bmatrix} d_u & d_y \end{bmatrix}$
- The regulated outputs are selected to be $z_1 = e = r - y'$, $z_2 = \bar{u}$,
- The measured output is $y' = y + n$
- Controlled input is u

This is equivalent to the following block diagram

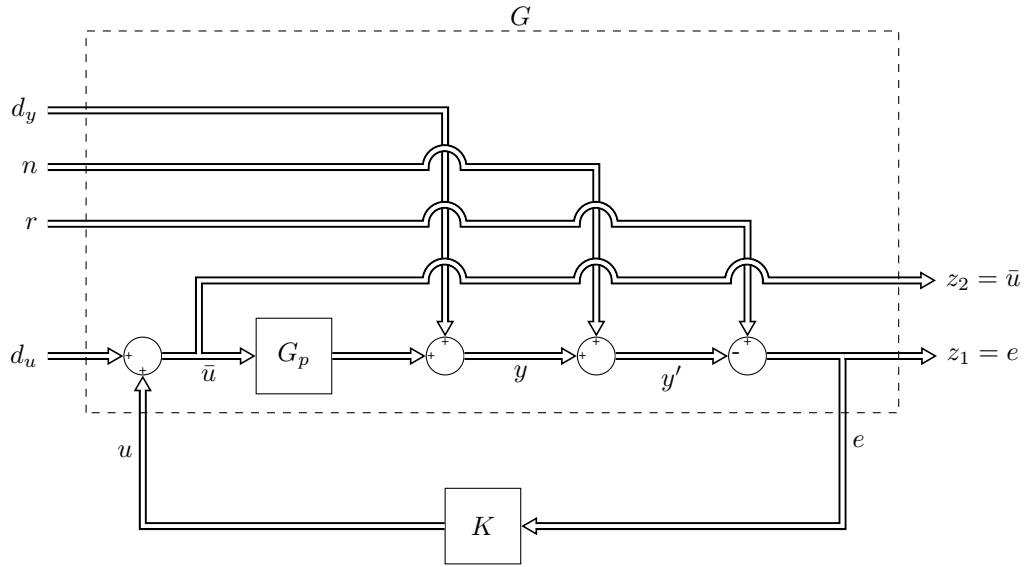


Figure 14.11: LFT Block Diagram

7794 The relationships become

$$\begin{bmatrix} z \\ e \end{bmatrix} = \begin{bmatrix} \bar{f} - n - y' \\ \bar{u} \\ r - n - y' \end{bmatrix} = \begin{bmatrix} \bar{f} - n - d_y - G_p d_u - G_p u \\ u + d_u \\ r - n - d_y - G_p d_u - G_p u \end{bmatrix} \quad (14.51)$$

7795 or

$$\begin{bmatrix} z_1 \\ z_2 \\ e \end{bmatrix} = \begin{bmatrix} -I & -I & I & -G_p & -G_p \\ 0 & 0 & 0 & I & I \\ -I & -I & I & -G_p & -G_p \end{bmatrix} \begin{bmatrix} d_y \\ n \\ r \\ d_u \\ u \end{bmatrix} \quad (14.52)$$

7796 However, as discussed in the SISO section, for meeting performance and robustness
 7797 measures, we must augment this plant with the required filters. This is done so that the
 7798 optimization techniques, which will be discussed later, could be utilized to compute a
 7799 controller so that the closed loop singular values meet the specified requirements. This is
 7800 known as *MIMO Mixed Sensitivity Method*.

7801 The simplified version of the MIMO feedback loop, where $w = n$ and the other exoge-
 7802 nous inputs are ignored, shown in Figure 14.10 is augmented as follows,

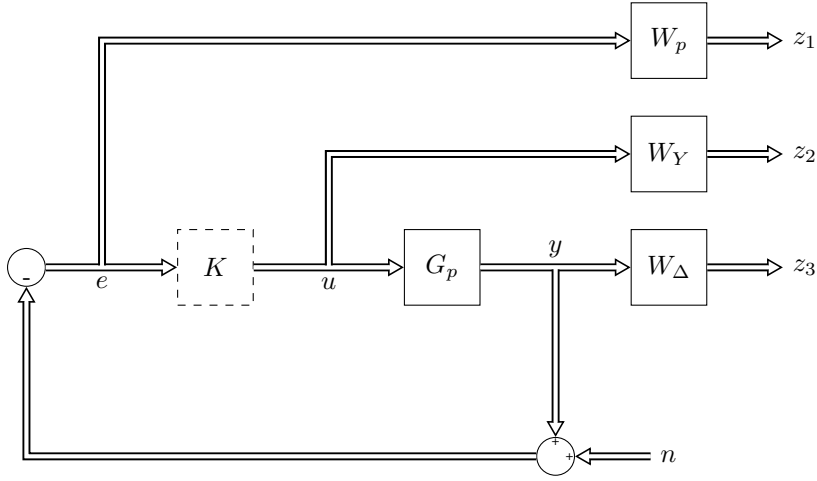


Figure 14.12: Augmented closed-loop feedback system

where

▪ Exogenous Input: $w = n$

▪ Regulated Outputs: $z = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} W_p(y + w) \\ W_Y u \\ W_\Delta y \end{bmatrix} = \begin{bmatrix} W_p(G_p u + n) \\ W_Y u \\ W_\Delta G_p u \end{bmatrix}$

▪ Measured Outputs: $e = -y - w = -G_p u - n$

▪ Controlled Input: u

The input-output relationships for the generalized plant are:

$$\begin{bmatrix} z \\ e \end{bmatrix} = G \begin{bmatrix} w \\ u \end{bmatrix} = \begin{bmatrix} W_p & -W_p G_p \\ 0 & W_Y \\ 0 & W_\Delta G_p \\ -I & -G_p \end{bmatrix} \begin{bmatrix} w \\ u \end{bmatrix} \quad (14.53)$$

We will utilize the LFT framework for norm optimization and control design in the later sections. In addition, we will discuss some physical interpretation of this framework with bond graphs (17.2) when discussing MIMO control design by norm optimization.

7814 14.4 Modeling Uncertainties

7815 14.4.1 Uncertainties

7816 Every dynamic model includes a certain amount of uncertainty. The reason for not being
7817 able to derive a precise dynamic model could be one of the followings or a combination of
7818 the followings,

7819

7820 a) It could be time prohibitive or even not possible to derive a precise model for a highly
7821 complex dynamic system.

7822 b) Precise or high fidelity models are computationally intensive.

7823 c) High fidelity models are not necessarily useful for the control design applications.

7824

7825 These statements are true across all areas of research and design.

7826 Control systems are designed using models which are not exact; they merely approximate
7827 the real system. It is then necessary to account for model error in the design process. Mod-
7828 eling requires quantification of the model error, even if this can not be done precisely. A
7829 reasonable design allows a 'margin' for model error; a design with this feature is called
7830 robust.

7831 One approach to deal with impreciseness of our dynamic models or model error is to in-
7832 troduce uncertainties to these models. In general, uncertainties of dynamic systems in
7833 engineering fall into two categories,

7834

7835 1. Unstructured uncertainty (this could be due to ignoring high frequency dynamics in
7836 our models (complex-valued uncertainty));

7837

- 7838 ▪ Uses a specific single perturbation of the entire system.
- 7839 ▪ The conditions for robust stability can be expressed as bounds on the TFM, i.e.
7840 $\bar{\sigma}(T)$, $\bar{\sigma}(S)$, $\bar{\sigma}(Y)$ or a combination of these principal gains.
- 7841 ▪ results in conservative closed loop robustness, i.e. sacrificing closed loop perfor-
7842 mance.

7843 Introducing this type of uncertainty and compensating for it guarantees robust stability
7844 and nominal performance, as discussed in the SISO section.

2. Structured uncertainty (this could be due to the combination of low frequency parameter variations of our models (real-valued uncertainty) and high frequency uncertainties similar to unstructured uncertainty):

- The individual sources of uncertainty are identified.
- Does not result in conservative closed loop robustness.
- The disadvantage of this procedure results in non-analytical solution, i.e. solved by numerical optimization.

Compensation for structured uncertainty guarantees both robust performance and stability [see ??].

In this section, we will only cover unstructured uncertainty and deal indirectly to obtain robust performance for the MIMO feedback design in place of using numerical optimization.

14.4.2 Descriptions of Unstructured Uncertainty

There are different ways of describing uncertainty in a system,

- Additive uncertainty where the modeled plant $\tilde{G}_p(s)$ is the sum of the nominal plant $G_{p0}(s)$ and function $L_A(s)$ where $\tilde{G}_p(s) = G_{p0}(s) + L_A(s) \Rightarrow L_A(s) = \tilde{G}_p(s) - G_{p0}(s)$ and is represented by the block diagram

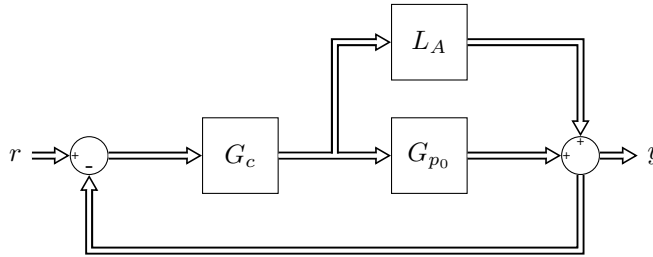


Figure 14.13: Additive Uncertainty Topology

- Multiplicative output uncertainty where the modeled plant $\tilde{G}_p(s)$ is the product of the nominal plant $G_{p0}(s)$ and the sum $I + L_\Delta(s)$ where $\tilde{G}_p(s) = (I + L_\Delta(s))G_{p0}(s) \Rightarrow L_\Delta(s) = (\tilde{G}_p(s) - G_{p0}(s))G_{p0}^{-1}(s)$ and is represented by the block diagram

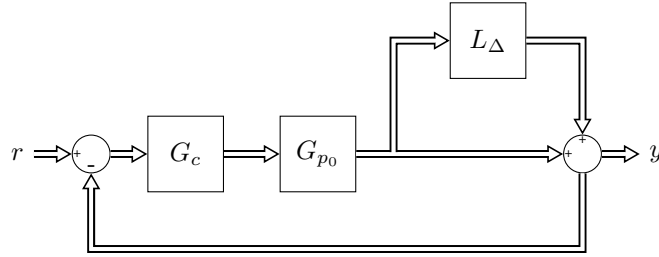


Figure 14.14: Multiplicative Uncertainty Topology

- Inverse multiplicative output uncertainty where the modeled plant $\tilde{G}_p(s)$ is the ratio of the nominal plant $G_{p0}(s)$ and sum $I + L_I(s)$ where $\tilde{G}_p(s) = (I + L_I(s))^{-1} G_{p0}(s) \Rightarrow L_I(s) = (G_{p0}(s) - \tilde{G}_p(s)) \tilde{G}_p^{-1}(s)$ and is represented by the block diagram

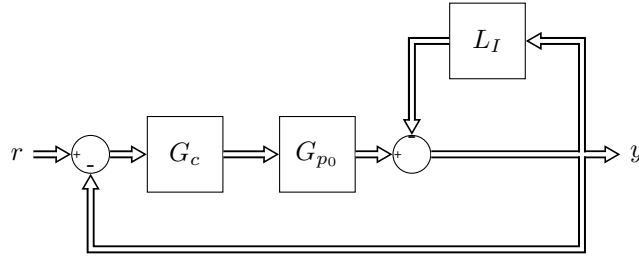


Figure 14.15: Multiplicative Uncertainty Topology

The conditions for robust stability depends on which perturbation is chosen to describe the uncertainty. Only the case of multiplicative uncertainty is discussed in this section.

This type of modeling for uncertainty is abstract and counterintuitive, after all if there is uncertainty then what is the meaning of modeling for uncertainty?! If through all this development, we keep in mind that our perturbed plant, $\tilde{G}_p$ is the actual plant and the nominal plant or the model that we use for control design is G_{p0} , then perhaps the above analysis would make more sense.

14.5 General Robust Stability Theorem

Due to its practical importance, in this section, we investigate the impact of multiplicative uncertainties on the closed loop robust stability. Now if $L_\Delta = W_\Delta \Delta$ with $\bar{\sigma}(\Delta) \leq 1$,

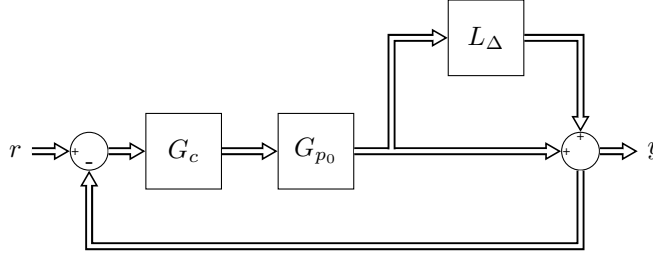


Figure 14.16: Multiplicative Uncertainty Topology

where the bound W_Δ can be interpreted as a weight on a normalized perturbation Δ , the topology becomes

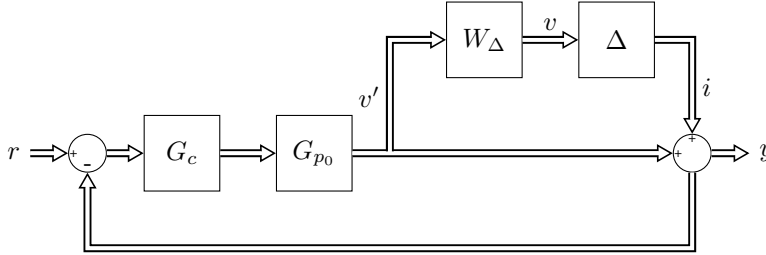


Figure 14.17: Multiplicative Uncertainty Topology with factored L_o

Now we are interested in how the uncertainty Δ affects the closed loop behavior of the system. First, find the closed loop function from i to v'

$$v' = -G_{p0} G_c (i + v') \Rightarrow (I + G_{p0} G_c) v' = -G_{p0} G_c i \quad (14.54)$$

letting $L = G_{p0} G_c$ shows that

$$v' = -(I + L)^{-1} L i \quad (14.55)$$

$$= T_o i \quad (14.56)$$

$$v = W_\Delta T_o i \quad (14.57)$$

Which conveniently fits into the framework, and where T_o is the nominal closed loop transfer function matrix,

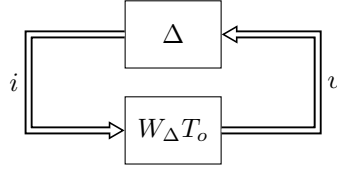


Figure 14.18: Upper LFT

Theorem 14.3 (Small Gain or Small Signal Theorem) Assume $W_\Delta T_o$ is stable, then the actual closed loop system is robustly stable for all Δ with $\bar{\sigma}(\Delta) \leq 1$ if and only if $\bar{\sigma}(W_\Delta T_o \Delta) < 1 \Leftrightarrow \det(I - W_\Delta T_o \Delta) \neq 0$.

Proof: See “lectures in Feedback Design for MIMO Systems” by Alberto Isidori ■

Other descriptions of uncertainties result in bounds on different closed loop transfer matrices singular values.

14.5.1 SVD Properties Applied

To find some of the conditions for robust stability, we use properties of the SVD,

$$\bar{\sigma}(W_\Delta T_o \Delta) \leq \bar{\sigma}(W_\Delta) \bar{\sigma}(T_o) \underbrace{\bar{\sigma}(\Delta)}_{\leq 1} \leq \bar{\sigma}(W_\Delta) \bar{\sigma}(T_o) \quad (14.58)$$

So the condition for robust stability becomes

$$\bar{\sigma}(W_\Delta) \bar{\sigma}(T_o) < 1 \text{ or } \bar{\sigma}(T_o) < \frac{1}{\bar{\sigma}(W_\Delta)} \quad (14.59)$$

Equivalently for nominal performance, the the conditions are

$$\bar{\sigma}(W_p) \bar{\sigma}(S_o) < 1 \text{ or } \bar{\sigma}(S_o) < \frac{1}{\bar{\sigma}(W_p)} \quad (14.60)$$

where W_p is a weighting matrix on the plant.

Then, the control design constraints can be written as,

- Nominal Performance: $\bar{\sigma}(S_o) < \frac{1}{\bar{\sigma}(W_p)}$

- Robust Stability (multiplicative Uncertainty): $\bar{\sigma}(T_o) < \frac{1}{\bar{\sigma}(W_\Delta)}$

The inverse of shaping filters can be interpreted as upper bounds for $\bar{\sigma}(T_o)$ and $\bar{\sigma}(S_o)$.

Interpret these with graphical representations result in the following Figure,

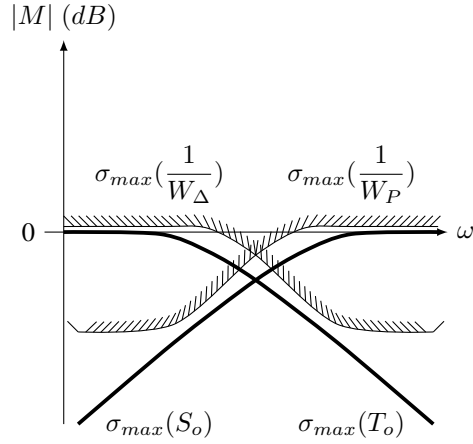


Figure 14.19: Frequency plot of MIMO loop-shaping technique

14.5.2 Robust performance

Robust performance means that the performance does not deteriorate for not only the nominal plant but for a set of plants as defined by the uncertainty bound. The robust performance is dealt with by giving a structure to the unstructured uncertainty (such as the multiplicative uncertainty for robust stability) with an addition of a performance uncertainty, such that the uncertainty block is written as, $\Delta_S = \begin{bmatrix} \bar{\Delta} & 0 \\ 0 & \Delta_P \end{bmatrix}$ then the technique such as μ synthesis [see 17.2] could be utilized to guarantee robust performance and stability. But, as mentioned, techniques such as μ synthesis does not have analytical solutions. Hence, in this section, we will present a conservative solution to the mixed sensitivity technique to guarantee robust performance.

We start with the constraints on both S_o and T_o which can be written as $\bar{\sigma}(W_\Delta T_o) < 1$ and $\bar{\sigma}(W_p S_o) < 1$, the condition for simultaneously achieving nominal performance and robust stability is,

$$\max\{\bar{\sigma}(W_\Delta T_o), \bar{\sigma}(W_p S_o)\} < 1 \quad (14.61)$$

It is shown for the SISO control systems [see 17.2], which this condition is modified here for the MIMO control systems that a necessary and sufficient condition for robust performance is,

$$\bar{\sigma}(W_\Delta T_o) + \bar{\sigma}(W_p S_o) < 1 \quad (14.62)$$

Since,

$$\max\{\bar{\sigma}(W_\Delta T_o), \bar{\sigma}(W_p S_o)\} \leq \bar{\sigma}(W_\Delta T_o) + \bar{\sigma}(W_p S_o) \leq 2\max\{\bar{\sigma}(W_\Delta T_o), \bar{\sigma}(W_p S_o)\} \quad (14.63)$$

and in addition,

$$\frac{2}{\sqrt{2}} \bar{\sigma} \begin{bmatrix} W_\Delta T_o \\ W_p S_o \end{bmatrix} \leq 2\max\{\bar{\sigma}(W_\Delta T_o), \bar{\sigma}(W_p S_o)\} \leq 2\bar{\sigma} \begin{bmatrix} W_\Delta T_o \\ W_p S_o \end{bmatrix} \quad (14.64)$$

So the condition for simultaneously achieving *nominal performance and robust stability* is written as,

$$\bar{\sigma} \begin{bmatrix} W_p S_o \\ W_\Delta T_o \end{bmatrix} < 1 \quad (14.65)$$

While the *robust performance* condition can be written as,

$$\bar{\sigma} \begin{bmatrix} W_p S_o \\ W_\Delta T_o \end{bmatrix} < \frac{1}{2} \quad (14.66)$$

It is usual to incorporate the weighting filters in the generalized plant so that the regulated variable z satisfies

$$z = \underbrace{\begin{bmatrix} W_p S_o \\ W_\Delta T_o \end{bmatrix}}_{T_{zw}} w \quad (14.67)$$

The augmented plant, G , can be obtained by including the weighting filters to the general feedback loop below and finding the closed loop TFM,

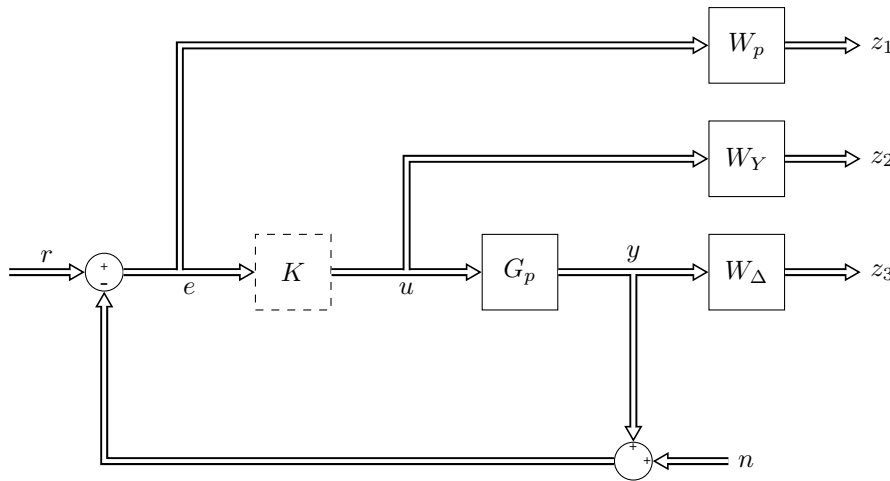


Figure 14.20: Augmented Plant Block Diagram

$$z = \underbrace{\begin{bmatrix} W_p S_o \\ W_Y Y_o \\ W_\Delta T_o \end{bmatrix}}_{T_{zw}} w \quad (14.68)$$

7932 where $w = r$ and $z = \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \tilde{x}_3 \end{bmatrix}$. The optimization objective can be written as,

$$\bar{\sigma} \begin{bmatrix} W_p S_o \\ W_Y Y_o \\ W_\Delta T_o \end{bmatrix} < 1 \quad (14.69)$$

CHAPTER 15

OPTIMAL CONTROL IN MIMO SYSTEMS

In this chapter, we present three practical applications for which we investigate a comparison between MIMO Youla control design and MIMO H_∞ , mixed sensitivity, control design. These applications include active suspension control (ASC) development, fuel injection control for engine start-up for hybrid diesel powertrain with a combination of an electric motor and a diesel engine, and a free-flying space robot.

In this chapter, we utilize the norms, which discussed in the previous chapter, to derive MIMO controllers. As the discussion of $\mathcal{H}_2$ and $\mathcal{H}_\infty$ MIMO control design are topics of Robust Optimal Control domain, it would be beneficial to start with having a brief discussion on LQG (Linear Quadratic Gaussian) control, which is a well established a MIMO technique. Our intent is to discuss what the main drawbacks of LQG method were and how the techniques such as $\mathcal{H}_\infty$ control method addresses these drawbacks. It should be noted that all the Optimal Controllers are derived based on the linearity and infinite horizon assumptions, as $t \rightarrow \infty$ (17.2).

Again, these topics have been discussed in many textbooks and publications and we introduce them for the sake of completeness and so that we would be able to provide

a comprehensiveness comparative results between these control concepts such as MIMO $\mathcal{H}_\infty$ with previously introduced MIMO Youla control method.

15.1 Output Feedback Control

The first linear optimal controller, LQR (Linear Quadratic Regulator), is derived given the following cost function,

$$J_{LQR} = \int_0^\infty x^T Q x + u^T R u \quad (15.1)$$

where u is the control input and x are the system states. It has shown (17.2) that by solving this cost function, utilizing algebraic Riccati equation, results in a control input/inputs as linear function of the states, $u = -K_c x$. In addition, it has shown that this optimal control method has very good robustness characteristics such as 60° phase margin and from 6 to ∞ gain margin. However, this method requires having access to the entire dynamic system states. It should be obvious that this last requirement is not very practical at times.

The general dynamics system in a differential equation form is written ,

$$\begin{bmatrix} \dot{x} \\ \dot{z} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} A & B_1 & B_2 \\ C_1 & 0 & D_{12} \\ I & D_{21} & 0 \end{bmatrix} \begin{bmatrix} x \\ w \\ u \end{bmatrix} \quad (15.2)$$

Where I is the identity matrix, D_{11} and D_{22} are assumed to be equal to zero. As discussed before, the H_2 norm of the closed loop will not be finite if D_{11} is not precisely equal to zero. The second assumption, $D_{22} = 0$, is done without loss of generality and make the derivation of the equations easier. However, we will remove this second assumption, when discussing minimization of H_2 norm of the closed loop system.

Now assume that all of the states are not available, then the dynamic system above is re-written as,

$$\begin{bmatrix} \dot{x} \\ \dot{z} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} A & B_1 & B_2 \\ C_1 & 0 & D_{12} \\ C_2 & D_{21} & 0 \end{bmatrix} \begin{bmatrix} x \\ w \\ u \end{bmatrix} \quad (15.3)$$

Or

$$G = \begin{bmatrix} A & B_1 & B_2 \\ C_1 & 0 & D_{12} \\ C_2 & D_{21} & 0 \end{bmatrix} \quad (15.4)$$

Where the identity matrix I is replaced by matrix C_2 or the read out/output matrix.

In order to get around this constraint, the researchers in 1960s derived two approaches for

solving this problem. The first approach, which will be developed in Appendix B provides a general solution to static output feedback problem using nonlinear optimization method. This approach results in a controller with static gains, similar to the LQR problem.

In the second approach a new optimal control method based on estimation of the states with use of Kalman Filtering technique is derived (assuming the dynamic system is observable/stabilizable) and then the estimated states are used in place of the measured states for devising an optimal feedback controller similar to the LQR method. This technique is known as the LQG (Linear Quadratic Gaussian, as the noise signals introduced in the system assumed to be normally/Gaussian distributed). It is interesting that although the differential equations written to solve LQG are stochastic (see next section), but this problem can be solved deterministically (see Appendix C). It is shown that the optimal solution for the computation of control gains is given by the set of equations

$$u = -K_c \hat{x} \quad (15.5)$$

$$K_c = R^{-1} B_2^T P_c \quad (15.6)$$

$$0 = P_c A + A^T P_c - P_c B_2 R^{-1} B_2^T P_c + Q \quad (15.7)$$

7970 With the only difference here being that $\hat{x}$ is the set of states estimated from the measured
7971 outputs y (this assumes the separation between state estimation and state feedback control
7972 design). The matrix P is a non-negative definite matrix and a solution to the algebraic
7973 Riccati equation above. The general Linear Quadratic Gaussian (LQG) is discussed next.

7974 15.1.1 LQG Control

For a general stochastic systems with process and sensor noise, which are assumed to be Gaussian distributed, the dynamic system differential equation, using the system differential equations 15.2, is written as,

$$\dot{x} = Ax + B_2 u + w_d \quad (15.8)$$

$$y = C_2 x + w_n \quad (15.9)$$

$$z = C_1 x + D_{12} u \quad (15.10)$$

7975 Where w_d and w_n are stochastic process and sensor noises modeled by assuming zero-
7976 mean Gaussian white noise, where $w_d = B_1 w$ and $w_n = D_{21} w$.

7977 The LQG cost is similar to the LQR cost and is given by

$$J_{LQG} = E\{z^T z\} \quad (15.11)$$

where this time, due to the stochastic nature of the problem, as expectation operator has been used and $z^T z = x^T C_1^T C_1 x + u^T D_{12}^T D_{12} u$ with cross terms equal to zero without loss of generality. It is discussed in the Appendix B that for the infinite horizon case and assuming a stabilizing feedback controller, this cost function can be written as $E\{z^T z\} = \lim_{T \rightarrow \infty} E \left[\frac{1}{T} \int_0^T z^T(t) z(t) dt \right]$, and, furthermore,

$$E \left\{ \begin{bmatrix} w_d(t) \\ w_n(t) \end{bmatrix} \begin{bmatrix} w_d^T(t) & w_n^T(t) \end{bmatrix} \right\} = \begin{bmatrix} W & 0 \\ 0 & V \end{bmatrix} \delta(t - \tau) \quad (15.12)$$

where W and V are variances (strength of the impulse) and $\delta(t - \tau) = 0$ when $t \neq \tau$ and, obviously $\delta(t - \tau) = \delta(0)$ when $t = \tau$, hence, these noise signals are assumed to be uncorrelated. In addition, it could be shown that these variances are equal to the PSDs (Power Spectral Densities) of these signals, e.g., $S_{w_d}(\omega) = \sigma_{w_d}^2 = W$

For a signal $w(s)$ (zero-mean white noise) with the following I/O relationship

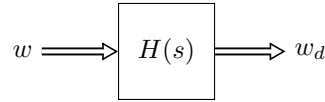


Figure 15.1: PSD I/O Relationship

The PSD of the output signal w_d can be written as

$$S_{w_d}(j\omega) = |H(j\omega)|^2 S_w(j\omega) \quad (15.13)$$

Therefore, with $S_w = I$, this implies that $S_{w_d} = W$, if $|H(j\omega)| = W^{1/2} = B_1$ (same applies to w_n , where $|H(j\omega)| = V^{1/2} = D_{21}$). In addition, if one selects $C_1 = Q^{1/2}$ where Q is a weighting matrix, and $D_{12} = R^{1/2}$ where R is another weighting matrix, then the following relationships hold for an LQG type control system,

$$\begin{bmatrix} w_d \\ w_n \end{bmatrix} = \begin{bmatrix} W^{1/2} & 0 \\ 0 & V^{1/2} \end{bmatrix} w \quad (15.14)$$

$$z = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} Q^{1/2} & 0 \\ 0 & R^{1/2} \end{bmatrix} \begin{bmatrix} x \\ u \end{bmatrix} \quad (15.15)$$

Then the LFT framework is given by the following system diagram,

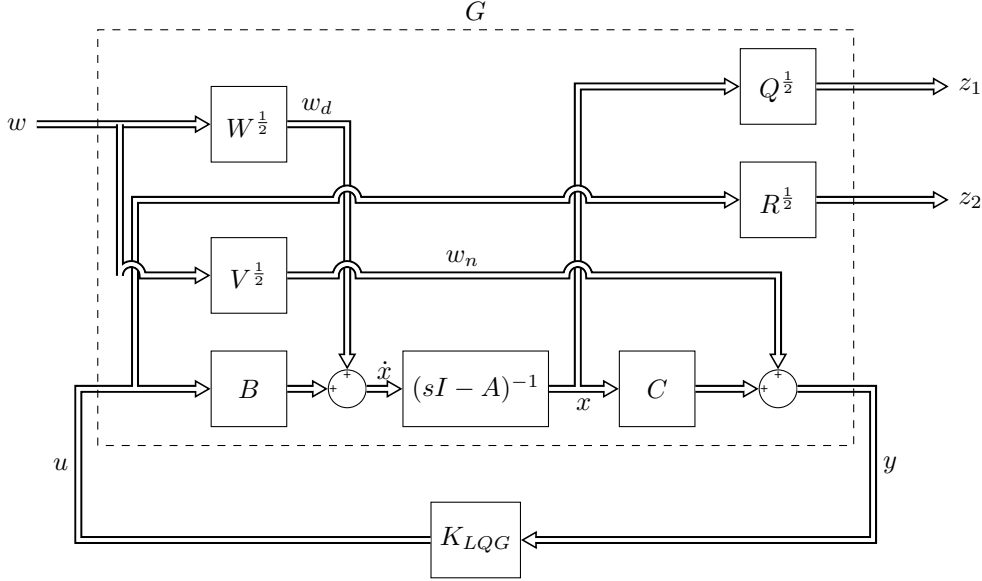


Figure 15.2: LQG using LFT framework

This gives the relationship,

$$\begin{bmatrix} \tilde{x} \\ z_1 \\ z_2 \\ y \end{bmatrix} = \begin{bmatrix} A & W^{\frac{1}{2}} & B \\ Q^{\frac{1}{2}} & 0 & 0 \\ 0 & 0 & R^{\frac{1}{2}} \\ C & V^{\frac{1}{2}} & 0 \end{bmatrix} \begin{bmatrix} \tilde{x} \\ w \\ u \end{bmatrix} \quad (15.16)$$

Now the square of the 2-norm of the closed loop transfer function, $T_{zw} = F_l(G, K)$, where the input, w is white noise with the PSD of $S_w = I$, can be written as,

$$J_{LQG} = E\{z^T z\} = \|T_{zw}\|_2^2 \quad (15.17)$$

Now the LQG cost is written as,

$$J_{LQG} = \lim_{T \rightarrow \infty} E \left[\frac{1}{T} \int_0^T z^T(t) z(t) dt \right] \quad (15.18)$$

$$= \lim_{T \rightarrow \infty} E \left[\frac{1}{T} \int_0^T x^T Q x + u^T R u \right] \quad (15.19)$$

The linear controller K_{LQG} include two separable parts. The first part is the feedback control portion similar to LQR. It can be shown (see Appendix C) that the optimal cost V^o , in this case, can be written as,

$$V^0 = x^T P_c x + C \quad (15.20)$$

where, P_c is a solution to the algebraic Riccati equation and $x = \hat{x} + \tilde{x}$, where, $\hat{x}$ are the estimated states and $\tilde{x}$ are the error states. Note the presence of the additional constant term C which is independent of x ; this is the additional cost caused by the white noise disturbance. But, because it is independent of x , the optimal controller is the same as in the deterministic case, LQR. The second part of the controller K_{LQG} is the Kalman Filter, which is discussed next.

15.1.2 Kalman Filter

To construct the estimator we use the following system differential equation,

$$\dot{\hat{x}} = A\hat{x} + Bu + K_f(y - \hat{y}) \quad (15.21)$$

where, as stated earlier, $x = \hat{x} + \tilde{x}$, and this system differential equation corresponds to the following topology,

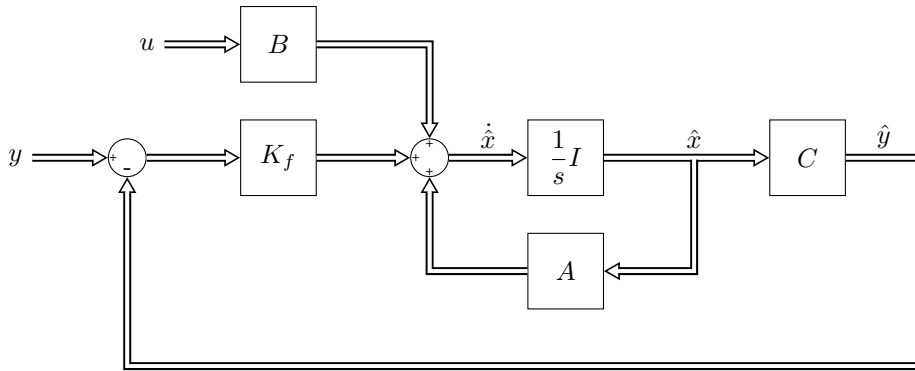


Figure 15.3: Kalman Filter Topology

The filter feedback gains K_f are chosen to minimize the variance of the estimation error, $\tilde{x} = x - \hat{x}$. Now let, P_f denote the variance $P_f = E\{\tilde{x}^T(t)\tilde{x}(t)\}$. The Riccati equation from the LQR design can be used to find P_f with the following substitutions $A \rightarrow A^T$, $B \rightarrow C^T$, $Q \rightarrow W$, and $R \rightarrow V$. This results in the following equations

$$K_f = P_f C^T V^{-1} \quad (15.22)$$

$$0 = P_f A^T + A P_f - P_f C^T V^{-1} C P_f + W \quad (15.23)$$

8008 Adding this to the full-state feedback controller above results in the following feedback
8009 diagram,
8010

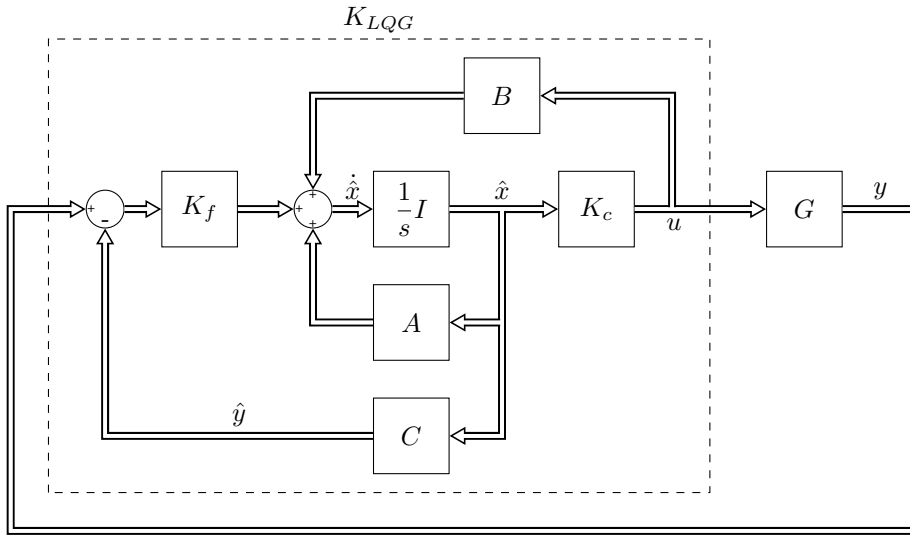


Figure 15.4: Full LQG feedback diagram

The control K_{LQG} can be redrawn as shown in Figure below,

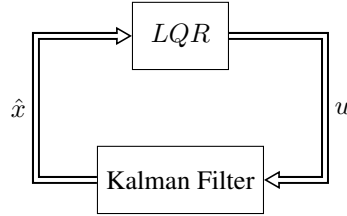


Figure 15.5: K_{LQG}

LQR is an optimal controller while Kalman Filter is an optimal estimator. Both LQR and Kalman Filter have very good robustness characteristics. It is then natural for one to assume that by integrating two optimal control and estimator results for generating the LQG controller K_{LQG} , the outcome is robust as well. But, interestingly, it has been shown (??) that this is not strictly true. The work for addressing the robustness of LQG controller has been referred to as the Loop Transfer Recovery or LTR. In the LTR, one tries to recover the robustness characteristics of the LQR controller when designing the LQG controls. One approach is to modify the Kalman filter gains, K_f so that the return ratio for the LQG control approaches the return ratio of the LQR control. This will be shown in Appendix D. This approach works well for stable minimum phase plants (no right half plane zeros). Newer procedures have been derived for the case when $G(s)$ is not minimum phase. It is important to keep in mind that by changing the Kalman filter gains we will trade-off robustness with optimality or performance. This is unfortunate but it is a price that we have to pay for this recovery.

It should be pointed out that $\mathcal{H}_2$ control method is the deterministic version of the LQG control as will be discussed next. Furthermore, LTR using $\mathcal{H}_2$ control is much more straightforward when comparing to LQG control approach.

15.1.3 $\mathcal{H}_2$ Control

We now have all the necessary ingredients for exploiting LQG optimal control theory to solve multivariable control design problems where the objectives are robust stability and performance.

The basic design objectives have been formulated as constraints on $\|S\|_2$, $\|Y\|_2$ and $\|T\|_2$. As discussed earlier, we may transform this, at least approximately, into a constraint on $\|T_{zw}\|$, the norm of the transfer function from w to z , where w are the exogenous (dis-

turbance/reference) signals and z the regulated (costed/performance) outputs; this transfer function is,

$$T_{zw} = \begin{bmatrix} W_p S_0 \\ W_Y Y_0 \\ W_\Delta T_0 \end{bmatrix}$$

The weights are chosen to reflect the constraints. For example, in the scalar case, W_p is chosen so that $|W_p^{-1}(j\omega)|$ is the upper bound for $|S_0(j\omega)|$ so that the constraint on S_0 is equivalent to the constraint,

$$|W_p(j\omega)S_0(j\omega)| < 1$$

for all (relevant) ω . This means that W_p is chosen so that $W_p(j\omega)$ is large at low frequencies and small at high frequencies. The weighting functions W_Y and W_Δ are similarly chosen to reflect the constraints on Y and T .

We next employ LQG optimal control theory to minimize the H_2 norm of T_{zw} . As discussed in the LQG control part of this section, this problem is equivalent to a suitably formulated LQG problem; we give the detailed formulation below. Now, LQG control does not enforce strict satisfaction of the design constraints at each ω , but it does tend to make $\|T_{zw}\|$ small and constant in the pass band of the optimal system, and this, in turn, enforces approximate satisfaction of the constraints.

The weighting functions are dynamic systems, so that the system G_p must be suitably augmented. How this is done is shown in Figure below. The augmented plant G whose controlled output z looks at $e = Sw$, $u = Yw$ and $y = Tw$. The output $y = e$ is fed back to the controller K ; this minor change does not effect our conclusions.

The LQG/ H_2 problem corresponding to Figure above is,

$$\min \lim_{T \rightarrow \infty} \int_0^T z^T z dt \quad (15.24)$$

subject to

$$\dot{x} = Ax + B_1 w + B_2 u \quad (15.25)$$

$$z = C_1 x + D_{11} w + D_{12} u \quad (15.26)$$

$$y = C_2 x + D_{21} w + D_{22} u \quad (15.27)$$

where x is the state of the augmented plant, $B_1 w$ is the process noise, $D_{11} = 0$ (as H_2 norm from w to z will be infinite when this term is included), and $D_{21} w$ is the sensor noise.

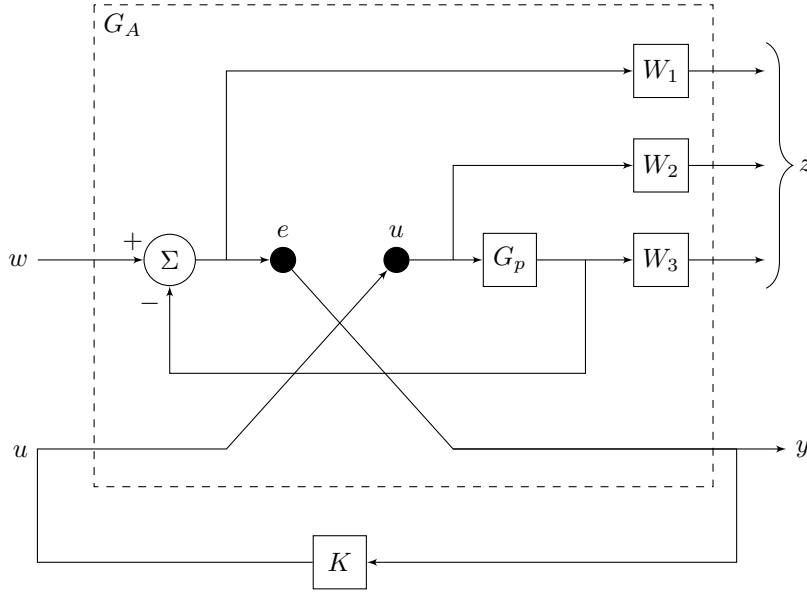


Figure 15.6: Augmented plant

It should be noted that the cost is no longer stochastic, it is deterministic as the disturbance inputs are considered impulsive now.

Making use of the definition of z , the LQG/ H_2 problem may be written in the form, and assuming $w(t)$ is impulsive. Then,

$$J_{LQR} = \|T_{zw}\|_2^2 = \lim_{T \rightarrow \infty} \int_0^T z^T(t)z(t)dt \quad (15.28)$$

where $T_{zw} = F_l(G, K)$, the closed loop map from w to z . Using the equations above,

$$z^T z = x^T C_1^T C_1 x + u^T D_{12}^T D_{12} u + 2x^T C_1^T D_{12} u \quad (15.29)$$

Without loss of generality, assume that $C_1^T D_{12} = 0$, this makes the cross product of u and x to be zero, where

$$z = \begin{bmatrix} C_1 x \\ D_{12} u \end{bmatrix} \quad (15.30)$$

$$z^T = \begin{bmatrix} x^T C_1^T & u^T D_{12}^T \end{bmatrix} \quad (15.31)$$

then,

$$Q = C_1^T C_1 \quad (15.32)$$

$$R = D_{12}^T D_{12} \quad (15.33)$$

$$(15.34)$$

8058 The variance matrix $\Sigma_{w_d w_n}$ of the process noise, $w_d = B_1 w$ and the sensor noise,
8059 $w_n = D_{21} w$, assuming $E(w(t)w^T(\tau)) = I\delta(t - \tau)$, is:

$$\Sigma_{w_d w_n} = \Sigma = \begin{bmatrix} B_1 B_1^T & B_1 D_{21}^T \\ D_{21} B_1^T & D_{21} D_{21}^T \end{bmatrix} \quad (15.35)$$

8060 Where the process noise variance, W , as defined before is equal to $B_1 B_1^T$ and the sensor
8061 noise variance, V is equal to $D_{21} D_{21}^T$. If we assume the cross correlation terms are zero
8062 then $B_1 D_{21}^T = D_{21} B_1^T = 0$.

8063 **(i) Kalman Filter** The dynamic model for the estimation is given as,

$$\dot{\hat{x}} = A\hat{x} + B_2 u + K_f(y - C_2 \hat{x} - D_{22} u) \quad (15.36)$$

8064 The solution to the Kalman filter gains for the general case of nonzero cross correlation
8065 noise terms are written as,

$$K_f = (P_f C_2^T + \Sigma_{12}) \Sigma_{22}^{-1} = (P_f C_2^T + B_1 D_{21}^T) (D_{21} D_{21}^T)^{-1} \quad (15.37)$$

8066 where P_f satisfies ARE,

$$P_f A^T + A P_f + \Sigma_{11} - (P_f C_2^T + \Sigma_{12}) \Sigma_{22}^{-1} (C_2 P_f + \Sigma_{21}) = 0 \quad (15.38)$$

If we assume zero for the cross correlation terms and knowing $\Sigma_{22} = (D_{21} D_{21}^T) = V$
then,

$$K_f = P_f C_2^T V^{-1}$$

8067 This is similar to Equation 15.22, which is derived previously.

8068

8069 **(ii) State Feedback** Then the H_2 control cost is exactly equal to LQR control cost.

8070 Therefore, the optimal control is given by $u = -K_c \hat{x}$ where, for the general case with the
8071 cross product term $N = C_1^T D_{12}$, we can write,

$$K_c = R^{-1} (B_2^T P_c + N^T) = (D_{12}^T D_{12})^{-1} (B_2^T P_c + D_{12}^T C_1) \quad (15.39)$$

Where P_c is computed from the following algebraic Ricatti equation,

$$P_c A + A^T P_c - (P_c B_2 + N)(D_{12}^T D_{12})^{-1}(B_2^T P_c + N^T) + C_1^T C_1 = 0 \quad (15.40)$$

The complete controller is given as,

$$K \sim \begin{bmatrix} A - K_f C_2 - B_2 K_c + K_f D_{22} K_c & K_f \\ K_c & 0 \end{bmatrix} \quad (15.41)$$

which maps $\begin{bmatrix} \hat{x} \\ y \end{bmatrix} \rightarrow \begin{bmatrix} \hat{x} \\ u \end{bmatrix}$.

It can be shown that by letting $D_{21} \rightarrow 0$, which is letting the sensor noise decrease in comparison with the process noise (or the process noise increase in comparison with the sensor noise) full state feedback loop transfer recovery is obtained.

For this design procedure to work the following conditions must be satisfied,

- (A, B_2) is stabilizable.
- (A, C_2) is detectable.
- D_{11} must be zero.
- D_{12} and D_{21} must have full rank.

In this section, we generalize the Linear Fractional Transformation concept which is discussed in the SISO part of this book to MIMO optimization and control design. We only discuss the infinite horizon case (as $T \rightarrow \infty$) as the controller gains in this case are constants and hence, time independent.

We start the section by assuming access to the full states of a dynamic system (this is also known as the full information case), then we introduce H_∞ filter for state estimation where access to full states is not possible (this is also known as the partial information case). Finally, we combine H_∞ control and filter design (similar to the LQG problem), and present the solution for this combined case.

Similar to LQR and LQG optimal control topics, these material are fully covered in other texts and technical papers. We present these topics again for completeness and comparative analysis with the proposed Youla MIMO control design method.

15.2 $\mathcal{H}_\infty$ Control Design

Understanding that the closed loop system of using lower LFT, as shown in Figure below,

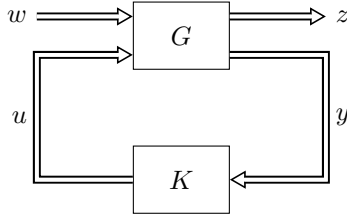


Figure 15.7: Lower LFT

is equivalent to the following Figure,

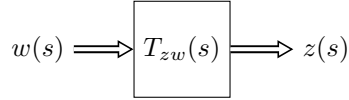


Figure 15.8: Standard I/O Relationship

where $z(s) = T_{zw}(s)w(s)$. For H_∞ optimal robust control design, the objective can be written as,

$$\min_K \|T_{zw}\|_\infty \quad (15.42)$$

where the H_∞ norm of the closed loop transfer function, T_{zw} , is the objective function.

The minimization of this objective function turns out to be nontrivial and hence, a very difficult problem. The best that could be done is to find a lower bound for this norm optimization, and therefore, a sub-optimal solution as discussed next.

Recall that

$$\|T_{zw}\|_\infty = \max_w \frac{\|z\|_2}{\|w\|_2} \quad (15.43)$$

in words this is the largest possible gain over all possible disturbances or the gain for the largest disturbance input. If we assume that $\|T_{zw}\|_\infty \leq \gamma$ for $\gamma \in \mathbb{R}_+$, this implies that for any arbitrary disturbances,

$$\frac{\|z\|_2}{\|w\|_2} \leq \gamma \quad (15.44)$$

or

$$\|z\|_2^2 - \gamma^2 \|w\|_2^2 \leq 0 \quad (15.45)$$

This bound reduces (instead of minimizing) the impact of disturbance energy (this is actually signal energy not a true disturbance energy (17.2)) on the output energy (again this is signal energy).

Now we can write the previous objective function as,

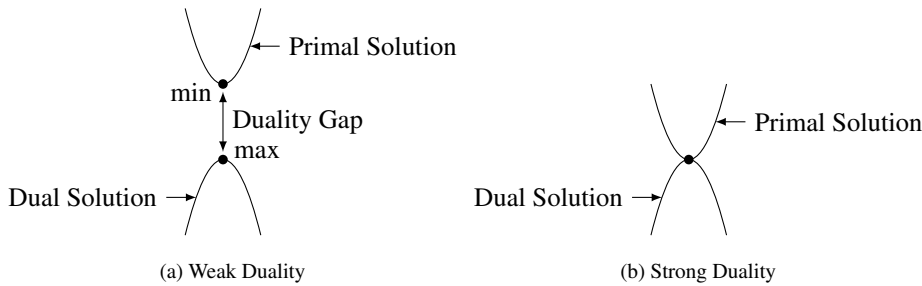
$$J(x_0, w) = \int_0^\infty (z^T z - \gamma^2 w^T w) dt \quad (15.46)$$

where x_0 is the initial condition. If we ignore the disturbance input in the above objective function then this cost or objective function would be precisely equal to the H_2 or LQR cost function. The bound γ acts a weighting to the disturbance input, at very large γ , the effect of disturbance input becomes insignificant and the solution to this optimal control problem approaches LQR (H_2) solution. It is important to note that disturbance input appears with a negative sign in this cost/objective function. Therefore, at small γ where the disturbance input become significant, it tries to maximize this cost function. The H_∞ problem is a zero-sum game. We can write the optimal control problem as,

$$\min_u \max_w J(x, u, w) \quad (15.47)$$

In the simplest setting a zero-sum game with two players u and w involves a cost $J(u, w)$; player u seeks to minimize J and player w to maximize J (or minimize $-J$). One player's gain is the other player's loss, the sum of the two objectives (J and $-J$) being zero; hence its name.

Every optimization problem can be cast either in the primal minimization form or the dual maximization, the solution of which provides a lower bound for the solution of the primal problem. If the two solutions are equal this is called strong duality, where a saddle point exists. If a duality gap exists, this is known as weak duality. Graphically, this is represented by the following Figures, For this application in optimal control problems, the



8134 following is always true,

$$J(u^*, w) \leq J(u^*, w^*) \leq J(u, w^*) \quad (15.48)$$

8135 for all $u \in U$, for all $w \in W$, the game admits a saddle point. It should be noted that not
8136 every game has a saddle point. However,

8137 **Theorem 15.1** *if U and W are convex, J is continuous, convex in u for all $w \in W$, and*
8138 *concave in w for all $u \in U$, then the game admits a saddle point. The saddle point is*
8139 *unique if J is strictly convex-concave.*

8140 In other words, we have transformed the original minimization problem of the H_∞ norm
8141 of the closed loop transfer function matrix to a problem which can be analytically solved,
8142 by assuming a bound on the infinity norm of the closed loop transfer function matrix, and
8143 the state-space solution is discussed next.

8144

8145 15.2.1 State Feedback (Full Information) $\mathcal{H}_\infty$ Control Design

8146 Consider the previous dynamic game defined by,

$$\min_u \max_w J(x_0, 0; u, w) \quad (15.49)$$

8147 where:

$$J(x_0, 0; u, w) = \int_0^\infty (z^T z - \gamma^2 w^T w) dt \quad (15.50)$$

8148 where $x(\cdot)$ is the solution of,

$$\dot{x} = Ax + B_2 u + B_1 w \quad (15.51)$$

$$z = C_1 x + D_{12} u \quad (15.52)$$

$$y = x \quad (15.53)$$

8149 The following assumptions are made about the dynamic system to solve this optimal
8150 control problem,

8151

8152 ▪ (A, B_2) are stabilizable (Necessary)

8153 ▪ (C_1, A) is detectable (Necessary)

- 8154 ■ $C_1^T D_{12} = 0$ (orthogonal) and $D_{12}^T D_{12} = I$ (Neither are necessary)

From the LQR/LQG solutions, we assume that the worst case disturbance w and the best case control u have the following structure,

$$w(t) = K_{d\infty}x(t) \quad (15.54)$$

$$u(t) = K_{c\infty}x(t) \quad (15.55)$$

Now

$$z = C_1x + D_{12}u \quad (15.56)$$

$$= (C_1 + D_{12}K_{c\infty})x \quad (15.57)$$

substituting this into Equation (15.50),

$$J = \int_0^\infty x^T \underbrace{(C_1^T C_1 + K_{c\infty}^T K_{c\infty} - \gamma^2 K_{d\infty}^T K_{d\infty})}_{Q_c} x dt \quad (15.58)$$

$$= \int_0^\infty x^T Q_c x dt \quad (15.59)$$

where $Q_c \geq 0$, a positive semi-definite matrix, and

$$\dot{x} = Ax + B_1 K_{d\infty}x + B_2 K_{c\infty}x \quad (15.60)$$

$$= \underbrace{(A + B_1 K_{d\infty} + B_2 K_{c\infty})}_{A_c} x \quad (15.61)$$

Now, the derivations of the two Equations (15.58 and 15.61) transforms the min-max problem to the general LQR like optimal control problem.

Next, we discuss how to compute the gains $K_{c\infty}$ and $K_{d\infty}$ such that the conditions for minimization of the worst disturbance of the cost function J is met and the solution is stable.

We know from Lyapunov analysis that for an energy function $V(x) = x^T P_\infty x$ and $P_\infty \geq 0$,

$$\dot{V}(x) = \dot{x}^T P_\infty x + x^T P_\infty \dot{x} \quad (15.62)$$

$$= x^T A_c^T P_\infty x + x^T P_\infty A_c x \quad (15.63)$$

$$= x^T (A_c^T P_\infty + P_\infty A_c) x \quad (15.64)$$

Now for a stable solution, we have to enforce that rate of change of the Lyapunov function is negative, therefore, we select,

$$A_c^T P_\infty + P_\infty A_c = -Q_c \quad (15.65)$$

Substituting the values from Equations (15.58) and (15.61) results in,

$$\begin{aligned} (A + B_1 K_{d\infty} + B_2 K_{c\infty})^T P_\infty + P_\infty (A + B_1 K_{d\infty} + B_2 K_{c\infty}) \\ + C_1^T C_1 + K_{c\infty}^T K_{c\infty} - \gamma^2 K_{d\infty}^T K_{d\infty} = 0 \end{aligned} \quad (15.66)$$

because this is an infinite horizon problem, we know that P_∞ is a constant matrix, and the following hold,

$$\nabla P_\infty = 0 \quad (15.67)$$

Taking the gradient of Equation (15.66) with respect to $K_{d\infty}$ yields

$$A_c^T \underbrace{\nabla_{K_{d\infty}} P_\infty}_{=0} + B_1^T P_\infty + \underbrace{\nabla_{K_{d\infty}} P_\infty A_c}_{=0} + P_\infty B_1 - 2\gamma^2 K_{d\infty} = 0 \quad (15.68)$$

Solving for $K_{d\infty}$

$$K_{d\infty} = \frac{1}{\gamma^2} B_1^T P_\infty \quad (15.69)$$

Following similar development for $K_{c\infty}$ results in

$$K_{c\infty} = -B_2^T P_\infty \quad (15.70)$$

These two gains are the min-max and stabilizing solution to J .

For computing P_∞ , we substitute these two gains into Equation (15.66), which results in the following Algebraic Ricatti Equation (ARE):

$$P_\infty A + A^T P_\infty + C_1^T C_1 - P_\infty \left(B_2 B_2^T - \frac{1}{\gamma^2} B_1 B_1^T \right) P_\infty = 0 \quad (15.71)$$

It can be shown that the Algebraic Ricatti Equation will always have a stabilizing solution, P_∞ , when $\gamma \gg 1$ and in this case, the optimal $u = K_c x$ where $K_c = -B_2^T P_\infty$, i.e. this is an equivalent LQR solution!

8167

The minmax optimization routine is solved iteratively, through 2 principal stages, The condition for the 1st stage is: $\|T_{zw}\|_\infty < \gamma$, for any arbitrary disturbances, where γ is selected to be a large value, and this implies,

8171

8172 1. The ARE has a solution such that $P_\infty \geq 0$

8173 2. $A_c = A + B_1 K_{d\infty} + B_2 K_{c\infty}$ is Hurwitz (stable).

8174 The second stage iterates until γ_{min} is found which still implies that $\|T_{zw}\|_\infty < \gamma_{min}$.

8175 This is done by iterating until there is no solution to the ARE and/or A_c is not Hurwitz.

8176

8177

15.2.2 $\mathcal{H}_\infty$ Filtering

In order to compensate for missing states, $\mathcal{H}_\infty$ filter is utilized to estimate the states. We assume that the governing equations for generating signal y is,

$$\dot{x} = Ax + \overbrace{B_1 w}^{\text{Process Noise, } w_d} + B_2 u \quad (15.72)$$

$$y = C_2 x + \underbrace{D_{21} w}_{\text{Sensor Noise, } w_n} \quad (15.73)$$

$$z = C_1 x \quad (15.74)$$

where $D_{21} D_{21}^T = I$ and $d = (w_d, w_n)$ is assumed to be energy bounded (2-norm is finite) with no assumption on the statistics of these noise signals.

The objective is to find an estimate $\hat{z}$ of $z = C_1 x$ which minimizes $\|T_{ed}\|_\infty$ where $e = z - \hat{z}$.

As with the $\mathcal{H}_\infty$ control, this filter is designed in 2 stages. In the first stage, the estimate $\hat{z} = C_1 x$ is chosen to ensure,

$$\|e\|_2^2 = \|z - \hat{z}\|_2^2 < \gamma^2 \|d\|_2^2 \quad (15.75)$$

such that

$$\|T_{ed}\|_\infty < \gamma \quad (15.76)$$

In the second stage, γ^* (the smallest possible γ) is found such that a solution to the first stage exists.

The filter $K_{f\infty}$ corresponds to γ^* is the solution to the H_∞ filtering problem. Assume the following estimation model,

$$\dot{\hat{x}} = A\hat{x} + B_2 u + K_{f\infty}(y - \hat{y}) \quad (15.77)$$

$$\hat{y} = C_2 \hat{x} \quad (15.78)$$

$$\hat{z} = C_1 \hat{x} \quad (15.79)$$

$$(15.80)$$

Then the error dynamics for $\tilde{x} = x - \hat{x}$ are

$$\dot{\tilde{x}} = (A - K_{f\infty} C_2) \tilde{x} + (B_1 - K_{f\infty} D_{21}) w \quad (15.81)$$

$$e = C_1 \tilde{x} \quad (15.82)$$

We can rewrite these equations in term of noise vector input, d ,

$$\dot{\tilde{x}} = (A - K_{f\infty}C_2)\tilde{x} + [I \quad -K_{f\infty}]d \quad (15.83)$$

$$e = C_1\tilde{x} \quad (15.84)$$

Where d is a column vector of w_d and w_n . The problem is to choose $K_{f\infty}$ to minimize the induced 2-norm (∞ norm) of the system mapping d to e .

Similar to LQG problem, it can easily be shown that this is a dual problem to the H_∞ control problem where,

$$\blacksquare A \rightarrow A^T$$

$$\blacksquare B_2 \rightarrow C_2^T$$

$$\blacksquare B_1 \rightarrow C_1^T$$

This results in the following ARE for the estimation filter:

$$P_{f\infty}A^T + AP_{f\infty} + B_1B_1^T - P_{f\infty}\left(C_2^TC_2 - \frac{1}{\gamma^2}C_1^TC_1\right)P_{f\infty} = 0 \quad (15.85)$$

and the filter gains are

$$K_{f\infty} = P_{f\infty}C_2^T \quad (15.86)$$

More precisely, the above solution ensures, for given γ ,

$$\|T_{ed}\|_\infty < \gamma \quad (15.87)$$

provided that a solution to the ARE Equation (15.85) exists; the solution to the H_∞ estimation problem is then obtained by determining the minimum γ for which this solution exists.

An important fact to observe is that the solution of the H_∞ filtering problem, unlike that of the corresponding LQG/ H_2 filtering problem, independent of the input noise.

15.3 $\mathcal{H}_\infty$ - Robust Optimal Control

In this section, we combine the results of H_∞ state feedback and filtering to develop a output feedback (partial state feedback) H_∞ control. Similar to the LQG control, the partial state/output feedback H_∞ utilizes an H_∞ filter for state estimation. However, unlike Kalman Filter, the H_∞ filter is a deterministic filter, which, no assumptions regarding the

statistics of the noise inputs are made, except their energy boundedness. Furthermore, similar to the LQG , two ARE (Algebraic Ricatti Equations) needs to be solved for finding an appropriate solution to infinite horizon case. However, unlike the LQG , the two AREs are coupled as the worst disturbance input, as discussed earlier, plays a role both in control and filtering problems. Therefore, in the most basic form of the H_∞ output feedback control, the separation of control and filtering, as was the case for LQG, is not provided. In addition and most importantly, the LTR (Loop Transfer Recovery), as discussed earlier for LQG controllers, for H_∞ output feedback control design becomes unnecessary.

As stated earlier, finding all possible controllers $K(s)$ which minimizes $\|T_{zw}\|_\infty$ is a difficult problem to solve.

In general, this is solved using a suboptimal solution, as discussed earlier, such that:

- a) Choose a $\gamma > 0$.
- b) Find a controller such that $\|T_{zw}\|_\infty < \gamma$.
- c) If such a controller exists, then decrease γ otherwise, increase γ .

The following general state-space representation of the dynamic system is utilized to solve this problem,

$$\dot{x} = Ax + B_2u + B_1w \quad (15.88)$$

$$z = C_1x + D_{11}w + D_{12}u \quad (15.89)$$

$$y = C_2x + D_{21}w + D_{22}u \quad (15.90)$$

The following assumptions are made,

- A1) Pair $(A, B_2 C_2)$ is stabilizable and detectable. (Necessary)
- A2) $D_{11}=0$ and $D_{22}=0$. (Not necessary for H_∞ but necessary for H_2)
- A3) D_{12} and D_{21} have full rank. (Necessary)
- A4) $D_{12}^T C_1^T \begin{bmatrix} B_1 & D_{12} \end{bmatrix} = \begin{bmatrix} 0 & I \end{bmatrix}$ (Not necessary)
- A5) $\begin{bmatrix} B_1 \\ D_{21} \end{bmatrix} D_{21}^T = \begin{bmatrix} 0 \\ I \end{bmatrix}$ (Not necessary)

A6) Pair $(A, B_1 C_1)$ is stabilizable and detectable. (Necessary)

(A1) is for the existence of stabilizing controllers.

(A2) is the conventionality in H_2 control. $D_{11}=0$ makes G_{11} the open loop transfer function from w to z strictly proper. $D_{22}=0$ makes G_{22} the open loop transfer function from u to y strictly proper and simplifies the formulas in the algorithms.

(A3) is to ensure the controllers are proper and hence realizable.

(A4) means that C_1 and D_{12} are orthogonal so that the penalty on $z = C_1 x + D_{12} u$ includes a nonsingular normalized penalty on the control u .

Conditions (A4) and (A5) drastically simplify the mathematical equations but they are not necessary.

(A6) is to ensure that the optimal controller does not try to cancel poles or zeros on the imaginary axis which would result in closed-loop instability.

Remark: If there is a control input with no weighting filter, then D_{12} does not have full rank. If this happens, then we need to add a small weight on u to make D_{12} full rank. In addition no $j\omega$ poles for H_∞ computation are allowed.

These assumption will result in the following general state-space representation of the dynamic system,

$$\dot{x} = Ax + B_2 u + B_1 w \quad (15.91)$$

$$z = C_1 x + D_{12} u \quad (15.92)$$

$$y = C_2 x + D_{21} w \quad (15.93)$$

The synthesis of the output feedback H_∞ control is based on the following two AREs (Algebraic Riccati Equations), where the first is for the synthesis of the full state feedback and the second is for the synthesis of H_∞ state estimation,

$$P_{c\infty} A + A^T P_{c\infty} + P_{c\infty} (\gamma^{-2} B_1 B_1^T - B_2 B_2^T) P_{c\infty} + C_1^T C_1 = 0 \quad (15.94)$$

$$A P_{f\infty} + P_{f\infty} A^T + P_{f\infty} (\gamma^{-2} C_1^T C_1 - C_2^T C_2) P_{f\infty} + B_1 B_1^T = 0 \quad (15.95)$$

These two AREs are quadratic equations in terms of the two unknowns matrices $P_{c\infty}$ and $P_{f\infty}$. These matrices are symmetric with dimension $n \times n$, where n is dimension of the state space. Both equations depend on the value of γ . We are interested in the stabilizing

solutions of these two quadratic equations. As the worst disturbance plays a role in both AREs, to find at least one stabilizing solution to these equations, it can be shown that the solution to H_∞ state estimation, $P_{f\infty}$, has to be scaled such that,

$$P_{f\infty_S} = P_{f\infty} Z_\infty \quad (15.96)$$

where, $Z_\infty = (I - \gamma^{-2} P_{c\infty} P_{f\infty})^{-1}$. Again, similar to control solution, we see that the state estimation solutions, scaled vs. non-scaled, become equal when γ is very large.

We conclude that a solution to the output feedback H_∞ control exists if and only if there exists a stabilizing solution to the control ARE, $P_{c\infty}$, and the filtering ARE, $P_{f\infty_S}$. From (15.96) this is equivalent to the existence of solutions for $P_{c\infty}$ and $P_{f\infty}$ and the coupling condition that $(I - \gamma^{-2} P_{c\infty} P_{f\infty})^{-1}$ is non singular.

The conditions for the general output feedback H_∞ control is summarized in the following theorem.

Theorem 15.2 *There exists a linear stabilizing feedback controller for the system defined by (15.91) to (15.93) that satisfies*

$$\|T_{zw}\|_\infty < \gamma$$

if and only if

1. The Algebraic Riccati Equation (15.94) has a stabilizing solution $P_{c\infty} = P_{c\infty}^T \geq 0$.

2. The Algebraic Riccati Equation (15.95) has a stabilizing solution $P_{f\infty} = P_{f\infty}^T \geq 0$.

3. $\bar{\sigma}(P_{c\infty} P_{f\infty}) < \gamma^2$.

Items 2 and 3 are equivalent to the existence of a solution $P_{f\infty_S}$. Moreover, in this case, the dynamics of one such controller is written as,

$$\dot{\hat{x}} = (A + B_1 K_{d\infty} + B_2 K_{c\infty} + Z_\infty K_{f\infty} C_2) \hat{x} - Z_\infty K_{f\infty} y \quad (15.97)$$

$$u = K_{c\infty} \hat{x} \quad (15.98)$$

Next, using LFT framework, we compare the structure of H_2 and H_∞ controllers:

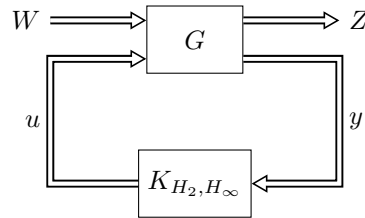


Figure 15.9: H_2 and H_∞ LFT

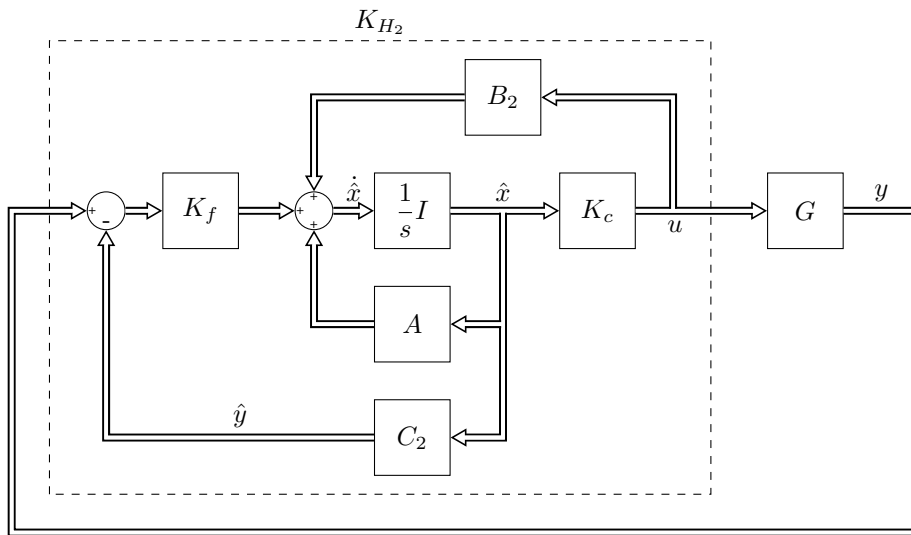
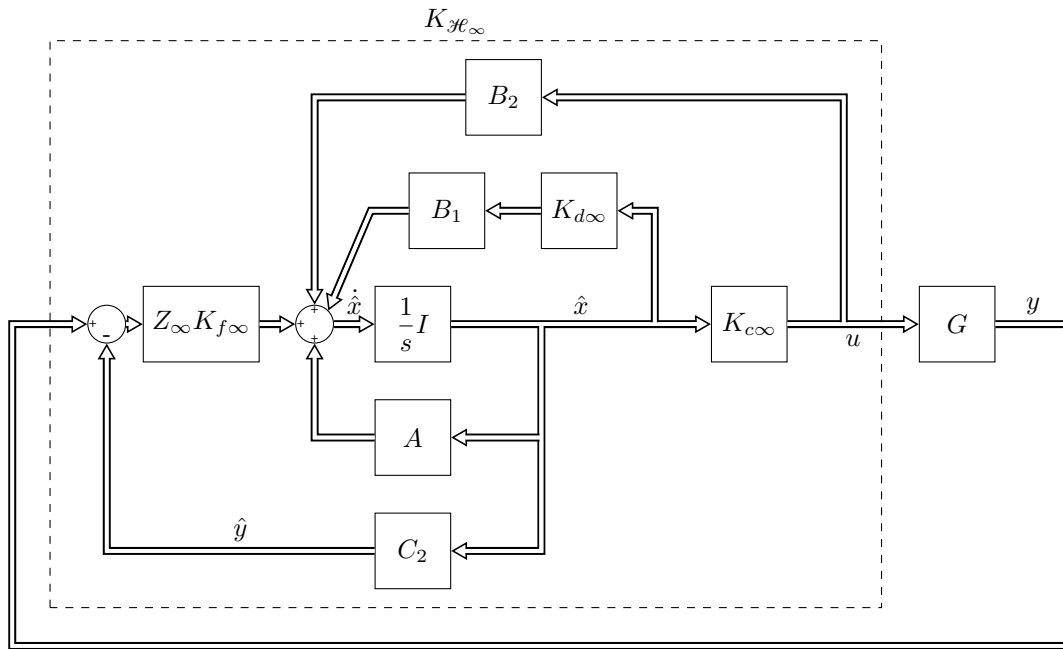


Figure 15.10: LQG/ H_2 Block Diagram

8280

8281 Given the dynamic equations for the H_∞ output feedback in (15.98) and (15.97), the fol-
8282 lowing control block diagram is given,

Figure 15.11: $\mathcal{H}_\infty$ Block Diagram

CHAPTER 16

APPLICATIONS

In this chapter, we present three practical applications for which we investigate a comparison between MIMO Youla control design and MIMO H_∞ , mixed sensitivity, control design. These applications include active suspension control (ASC) development, fuel injection control for engine start-up for hybrid diesel powertrain with a combination of an electric motor and a diesel engine, and a free-flying space robot.

16.1 Active Suspension

Active vibration control has been extensively studied and its applications stretch across a wide range of fields. Precision industrial process including semiconductor development relying on lasers to develop nanoscale devices. Vibration control is essential to these precision devices due to their nanoscales. Precision motion control systems for such applications in the semiconductor manufacturing have been developed utilizing conventional PID controls (17.2). Application of vibration control in the automotive industry utilizes active dampers (known as semi-active suspension control) to improve occupant comfort and vehicle handling.

Simple quarter-car mathematical models for vehicle suspension systems have been developed and utilized for the purpose control design (17.2). Furthermore, quarter-car models of active suspension systems utilizing a multitude of control systems have been studied. Control methodologies such as velocity feedback, LQR and H_∞ are among some of the popular control methods which have been used for active control of suspension system. The minimization of disturbances in automotive suspension systems utilizing robust control methods has proven to be an effective approach in increasing vehicle comfort and handling. Furthermore, active vehicle suspension design is among some of the most commonly studied MIMO systems.

Table 16.1: Nomenclature

symbols	Description	Units
m_s	Quarter of vehicle body Mass	kg
m_u	Wheel Mass	kg
x_s	Vehicle body displacement	m
x_u	Wheel displacement	m
x_in	Road disturbance input displacement	m
k_s	Suspension spring stiffness	N/m
k_t	Tire spring stiffness	N/m
b_s	Suspension damping Coefficient	N/(m/s)
F_{c1}	Suspension actuator active force	kN
F_{c2}	Tire actuator active force	kN

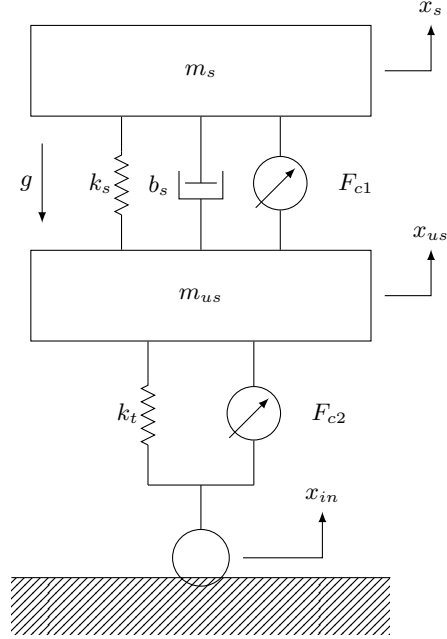


Figure 16.1: Quarter car model with two active control forces.

In this part, we will focus on a two active force control actuators using a simple quarter car model as show in Figure (16.1). The first actuator, which is used in many previous studies, is located between the unsprung and sprung masses (wheel and vehicle body). The second actuator, which is unique in this development and has only been a topic of one previous study (17.2), is positioned between the road disturbance input and the unsprung mass, and it represents active tire pressure control.

The objectives for the control design are to minimize the acceleration of the sprung mass, therefore improving the comfort of passengers, by not violating the suspension deflection constraint (the available space between the sprung and unsprung masses). This constraint not only put a limit on the suspension deflection but it would also indirectly help in road holding and hence, improving vehicle handling.

16.1.1 Model and Control Design

In this section, we provide the state equations for the quarter car model, which is utilized for the control development, our simplified assumption for the actuator model, and our control development strategies including their comparisons.

The equations of motion for the quarter car model are written as follows,

$$m_s \ddot{x}_s = k_s(x_{us} - x_s) + b_s(\dot{x}_{us} - \dot{x}_s) + 10^3 F_{c1} \quad (16.1)$$

$$m_{us} \ddot{x}_{us} = k_s(x_s - x_{us}) + b_s(\dot{x}_s - \dot{x}_{us}) - 10^3 F_{c1} + 10^3 F_{c2} + k_t(x_{in} - x_{us}) \quad (16.2)$$

The state variables are defined as follows,

$$x_1 = x_s, \quad x_2 = \dot{x}_s, \quad x_3 = x_{us}, \quad x_4 = \dot{x}_{us}. \quad (16.3)$$

The measured outputs used for control design are the suspension deflection, $s_d = x_1 - x_3$ and the sprung mass acceleration, $a_s = \ddot{x}_2$.

The state-space representation of this system is as follows,

$$\dot{x} = Ax + B_w w + B_u u, \quad (16.4)$$

$$y = Cx + D_w w + D_u u, \quad (16.5)$$

where w is the road disturbance input, x_{in} and u is the vector of active control forces, $\begin{bmatrix} F_{c1} \\ F_{c2} \end{bmatrix}$.

The state space representation is re-written as,

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{-k_s}{m_s} & \frac{-b_s}{m_s} & \frac{k_s}{m_s} & \frac{b_s}{m_s} \\ 0 & 0 & 0 & 1 \\ \frac{k_s}{m_{us}} & \frac{b_s}{m_{us}} & \frac{-k_s - k_t}{m_{us}} & \frac{-b_s}{m_{us}} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ \frac{10^3}{m_s} & 0 \\ 0 & 0 \\ \frac{-10^3}{m_{us}} & \frac{10^3}{m_{us}} \end{bmatrix} \begin{bmatrix} F_{c1} \\ F_{c2} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{k_t}{m_{us}} \end{bmatrix} [x_{in}]$$

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ \frac{-k_s}{m_s} & \frac{-b_s}{m_s} & \frac{k_s}{m_s} & \frac{b_s}{m_s} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \frac{10^3}{m_s} & 0 \end{bmatrix} \begin{bmatrix} F_{c1} \\ F_{c2} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} [x_{in}]$$

The following suspension parameter values are considered for our proposed model,

Table 16.2: Nomenclature

Parameter	Value
m_s	300 kg
m_u	60 kg
k_s	16000 N/m
k_t	190,000 N/m
b_s	1000 N/(m/s)

The general block diagram for the design of the feedback controller is shown in Figure (16.2),

Figure 16.2: General MIMO Feedback control loop.

Youla parameterization, as discussed in the previous chapters, is a frequency domain control method. Therefore, the plant (quarter car model) as given in Equations (16.4 and 16.5) has to be Laplace transformed to obtain its transfer function matrix. Before making this transformation, we will rewrite the state-space representations as,

$$\dot{x} = Ax + B_2u, \quad (16.6)$$

$$y = C_2x + D_2u, \quad (16.7)$$

where $C_2 = \begin{bmatrix} 1 & 0 & -1 & 0 \\ -\frac{k_s}{m_s} & -\frac{b_s}{m_s} & \frac{k_s}{m_s} & \frac{b_s}{m_s} \end{bmatrix}$ and $D_2 = \begin{bmatrix} 0 & 0 \\ \frac{10^3}{m_s} & 0 \end{bmatrix}$. We can then compute the transfer function matrix, G_p , using the following,

$$G_p = C_2(sI - A)^{-1}B_2 + D_2 \quad (16.8)$$

The transfer function G_p is written as,

$$G_p = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$$

where the inputs are $\begin{bmatrix} F_{c1} \\ F_{c2} \end{bmatrix}$ and the outputs are $\begin{bmatrix} s_d \\ a_{ms} \end{bmatrix}$.

This transfer function matrix is used in the next section for designing a MIMO Youla controller.

MIMO Youla control design:

The MIMO Youla control design approach, introduced in this book, starts by first computing the Smith-McMillan (SM) form of the plant transfer function matrix.

For computing the SM form, we first compute a polynomial matrix numerator, $P(s)$, and a denominator matrix of the plant, $\delta(s)$, such as,

$$G_p(s) = P(s)\delta(s)^{-1} \quad (16.9)$$

The plant transfer function matrix is,

$$G_p(s) = \begin{bmatrix} \frac{20(s^2+527.8)}{(s^2+2.864s+49.85)*(s^2+17.14s+3388)} & \frac{-16.667(s^2)}{(s^2+2.864s+49.85)*(s^2+17.14s+3388)} \\ \frac{3.3333s^2(s^2+3167)}{(s^2+2.864s+49.85)*(s^2+17.14s+3388)} & \frac{55.556s^2(s+16)}{(s^2+2.864s+49.85)*(s^2+17.14s+3388)} \end{bmatrix}$$

where $P(s) = \begin{bmatrix} 20(s^2 + 527.8) & -16.667(s^2) \\ 3.3333s^2(s^2 + 3167) & 55.556s^2(s + 16) \end{bmatrix}$ and
 $\delta(s) = (s^2 + 2.864s + 49.85) * (s^2 + 17.14s + 3388)$.

Knowing polynomial matrix $P(s)$ and the denominator $\delta(s)$, we can compute the SM form, M_p , using the greatest common denominator of minors of $P(s)$ as follows,

GCD of the 1x1 minor of $P(s)$, D_1 : 1
 GCD of the 2x2 minor of $P(s)$, D_2 : $s^2\delta(s)$

SM form for this plant is computed as following,

$$M_p = \frac{1}{\delta(s)} \begin{bmatrix} \epsilon_1 & 0 \\ 0 & \epsilon_2 \end{bmatrix}$$

where $\epsilon_i = \frac{D_i}{D_{i-1}}$ for $i = 1, 2$, and knowing that $D_0 = 1$.

Therefore, we can write M_p as,

$$M_p = \begin{bmatrix} \frac{1}{\delta(s)} & 0 \\ 0 & s^2 \end{bmatrix}$$

The roots of $\delta(s)$ are all in the LHP and hence are stable. The SM form has four stable poles and two zeros at $s = 0$. These zeros are on the $j\omega$ -axis and can not be canceled. The right and left unimodular matrices are found using a method described in the last example in section 3 of Chapter 13.

The decoupled Youla transfer function matrix, M_Y , for MIMO control design is written as,

$$M_Y = \begin{bmatrix} Y_1 & 0 \\ 0 & Y_2 \end{bmatrix}$$

We also know that the decoupled complimentary sensitivity transfer function matrix, M_T is,

$$M_T = M_p M_Y = \begin{bmatrix} \frac{Y_1}{\delta(s)} & 0 \\ 0 & Y_2 s^2 \end{bmatrix}$$

We select Y_1 and Y_2 so that the diagonal entries are exactly the same, this makes designing this MIMO controller easier. The transfer function matrix, M_T is written as,

$$M_T = \begin{bmatrix} \alpha \frac{\tau^2 s^2}{\delta(s)(\tau s + 1)^2} & 0 \\ 0 & \alpha \frac{\tau^2 s^2}{\delta(s)(\tau s + 1)^2} \end{bmatrix}$$

As pointed out earlier, the two zeros of the SM form cannot be canceled with the selection of Y_2 , otherwise, the internal stability will be compromised. The interpolation conditions in this case for meeting internal stability enforces that,

$$T(0) = 0 \quad \text{and} \quad \frac{dT(0)}{ds} = 0$$

which are both met if the zeros of M_p are not canceled.

The constant α is used as a tuning parameter.

$$T = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

where $\omega_n^2 = 60.6$ and $2\zeta\omega_n = 12.31$

However, knowing that there is a pure delay, therefore, a nonminimum phase zero, and the Youla transfer function cannot be selected to cancel these nonminimum phase zero, the Youla transfer function is modified to be,

$$Y = K \frac{\alpha s + \beta}{T_d}$$

Where $K = 4.65$, $\alpha = 0.00028$, $\beta = 0.0016$, and $T_d = 1/0.167s + 1$.

This selection of Youla transfer function gives the following closed loop complementary transfer function,

$$T = \frac{-s + \gamma_1}{s + \gamma_2} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Where $\gamma_1 = \gamma_2 = 20$ (it should be clear that $\gamma_2 \leq \gamma_1$ to limit the bandwidth due to the non-minimum phase zero), and ω_n and $\zeta\omega_n$ are the same as given above.

The following m-file is used to design this controller,

```
s=tf('s');

% Youla hand computation method

% Declaring plant model
td=(-s+20)/(s+20);
Td=1/(0.167*s+1);
G=(4.65*(0.00028*s+0.0016))/(s^2+12.31*s+60.6);
```



```

8393
8394 Gp= td*Td*G;
8395
8396 % Selecting Youla parameter
8397 Y=(1/Td)*(60.6/(4.65*(0.00028*s+0.0016)))*(1/(1/50*s+1)^2);
8398
8399 % Computing closed loop transfer functions
8400 T=minreal(Y*Gp);
8401 S=minreal(1-T);
8402 Gc=(Y/S);
8403 Gc=minreal(Gc);
8404
8405 % Plotting results
8406 figure(1)
8407 bode(T);
8408 hold on
8409 bode(S);
8410 bode(Y);
8411 legend('T','S','Y')
8412 grid on
8413
8414 figure(2)
8415 step(Y)
8416 legend('Actuator')
8417 grid on
8418
8419
8420 figure(3)
8421 step(T)
8422 legend('Yaw rate response')
8423 grid on

```

8424 Two high frequency poles are added to the Youla transfer function to better shape its
8425 frequency response at high frequency as can be seen in the m-file above.

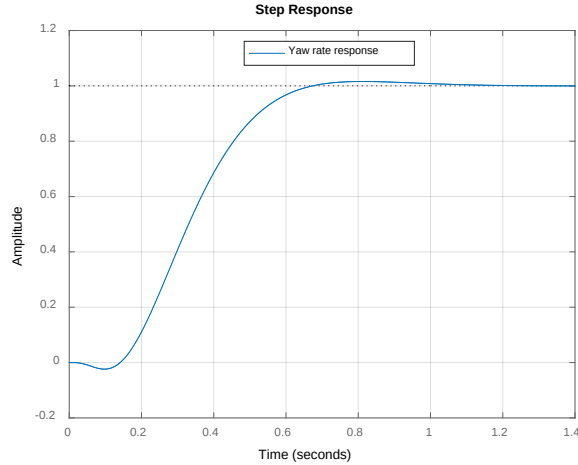


Figure 16.3: Yaw rate response to the locking torque step input request

The Figure (16.3) illustrates the yaw rate response to a locking torque step request. The non-minimum phase zero behavior is evident from this response. It should be clear that to control the non-minimum phase behavior, the constant γ_2 could have been taken to be much lower than γ_1 . In this case, the impact of the non-minimum phase zero would be minimized while it decreases the closed loop transfer function bandwidth, hence, it increases the rise time and settling time.

The locking torque capacity for this actuator is highly nonlinear and it depends on variables such as clutch plate and road coefficient of frictions, wheel speed difference, locking and clutch plate temperature to name a few. This actuator is capable of providing a peak locking torque of 2500Nm. Therefore, it is interesting to investigate the impact of the Youla linear control design on the actuator effort.

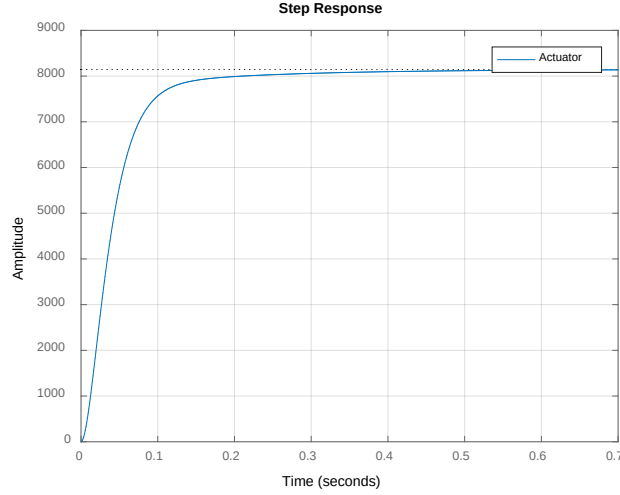


Figure 16.4: Actuator effort to the locking torque step input request

It should be noted that the longitudinal velocity for this linear model is assumed to be 25m/s. The Figure (16.4) shows that, given this longitudinal velocity, for good target following, the actuator requires almost more than 3 times its peak capacity at steady state. This means at high speed (25m/s or greater) stability maneuvers, the clutch pack of this actuator fully locks the two half shafts.

H_∞ control design using loop shaping technique:

The loop shaping approach as discussed in the previous chapters requires augmentation of the nominal plant by shaping filters, as illustrated in Figure (6.7). The filters that are used for the nominal plant augmentation are given as follows,

$$W_P = \frac{K_{W_P} \omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$

Where $K_{W_P} = 15$, $\omega_n = 0.1$, and $\zeta = 3.0$.

As pointed out in the Chapter 6, this filter weights the sensitivity transfer function. The shaping filter, W_Δ , for the closed loop complementary transfer function is given as follows,

$$W_\Delta = \frac{K_{W_\Delta} s}{s + \frac{K_{W_\Delta}}{T_{max}}}$$

8451 This filter is used as weighting on the complementary transfer function so that the closed
 8452 loop stability is guaranteed in the presence of the pure time delay of T_{max} , due to the
 8453 actuator dynamics. In order to design the weighting filter W_Δ with parameter K_{W_Δ} and
 8454 $T_{max} = 0.1s$ that will ensure robustness in the presence of a pure time delay $\leq 0.1s$, we
 8455 utilize multiplicative perturbation¹. Suppose that nominal plant transfer function is P and
 8456 consider perturbed plant transfer function of the form $\hat{P} = (1 + \Delta W_\Delta)P$. Then knowing
 8457 that $\|\Delta\|_\infty \leq 1$, then

$$\left| \frac{\hat{P}(j\omega)}{P(j\omega)} - 1 \right| \leq |W_\Delta(j\omega)| \quad \forall \omega,$$

8458 So $|W_\Delta(j\omega)|$ provides the uncertainty profile. This inequality describes a disk in the
 8459 complex plane, where at each frequency the point $\hat{P}/P$ lies in the disk with center 1, and
 8460 radius of $|W_\Delta|$. Typically, $|W_\Delta(j\omega)|$ is a increasing function of ω as uncertainty increases
 8461 with increasing frequency. For the case of pure delay,

$$\frac{\hat{P}(j\omega)}{P(j\omega)} = e^{-jT_{max}\omega}$$

8462 Therefore, the filter should be designed so that $|e^{-jT_{max}\omega} - 1| \leq |W_\Delta(j\omega)|$. A Bode
 8463 plot of $|e^{-jT_{max}\omega} - 1|$ and $|W_\Delta(j\omega)|$ illustrate that the designed weighting filter satisfies
 8464 the condition in Figure below when $K_{W_\Delta} = 2.5$,

¹We have shown in the previous chapters that multiplicative perturbation is equivalent to weighting the nominal complementary transfer function - small-gain theorem

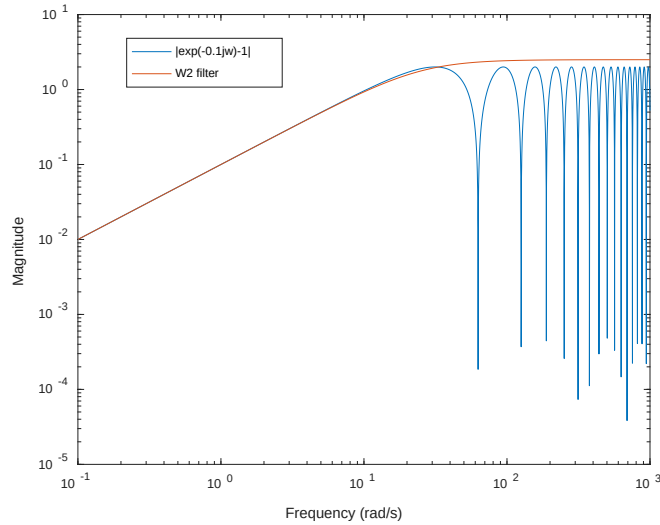


Figure 16.5: Bode plot of $|e^{-j0.1\omega} - 1|$ and W_{Δ} (W_2) filter

The final weighting filter is used to limit the actuator effort and is given by

$$W_u = \frac{1}{\rho}$$

Where ρ is a constant and is equivalent to the maximum locking torque generated by the actuator (in this case, based on our previous analysis, this constant is taken to be equal to 8000).

The following m-file is used to design this controller.

```
s=zpk('s');
% Hinf method
% Declaring plant model
td=(-s+20)/(s+20);
Td=1/(0.167*s+1);
G= Td*(4.65*(0.00028*s+0.0016))/(s^2+12.31*s+60.6);
```

```
8481
8482 Gp= td*Td*G;
8483
8484 % Shaping Filters
8485 wn=0.10;
8486 zeta=3.0;
8487 Kw1=15.0;
8488 Kw3=2.5;
8489 Tmax=0.1;
8490 W1=(Kw1*wn^2)/((s^2+2*zeta*wn*s+wn^2));
8491 W2=1/8000; W3=(Kw3*s)/(s+Kw3/Tmax);
8492
8493 % Plant augmentation with Filters
8494 P=augw(G,W1,W2,W3);
8495 % Controller computation
8496 [K,CL,GAM]=hinfsyn(P);
8497 %
8498 K=minreal(K);
8499 Gc=minreal(tf(K));
8500
8501 % Computing closed loop transfer functions
8502 Yhinf=minreal(Gc/(1+Gc*Gp));
8503 Thinf=minreal(Gp*Yhinf);
8504 Shinf=minreal(1-Thinf);
8505
8506 % Plotting results
8507 figure(1)
8508 bode(Thinf);
8509 hold on
8510 bode(Shinf);
8511 hold off
8512 legend('sensitivity','complementary sensitivity')
8513 grid on
8514
```

```

8515 figure(2)
8516 bode(Gp)
8517 hold on
8518 bode(Yhinf)
8519 bode(Gc)
8520 hold off
8521 legend('Plant','Youla','Gc')
8522 grid on
8523
8524 figure(3)
8525 step(Thinf)
8526 legend('Yaw rate response ,Hinf')
8527 grid on
8528
8529 figure(4)
8530 bode(Shinf)
8531 hold on
8532 bode(Wl)
8533 bode(1/Wl)
8534 hold off
8535 legend('sensitivity','Wp','1/Wp')
8536 grid on
8537
8538 figure(5)
8539 step(Yhinf)
8540 legend('Actuator effort, Hinf' )
8541 grid on

```

8542 The yaw rate response to a locking torque step request is shown in Figure below.

8543

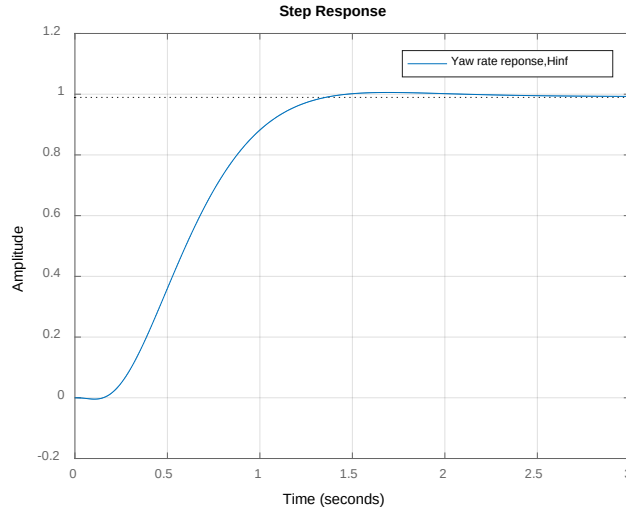


Figure 16.6: Yaw rate response to the locking torque step input request with Hinf Controller

Comparing the yaw rate responses of the H_∞ controller with the Youla hand computation method, it is obvious the results of the H_∞ controller is more conservative with respect to dealing with the pure time delay and response time. In addition, this response is more oscillatory than the yaw response from Youla hand computation results. Furthermore, Youla hand computation technique did not require any shaping filters and tuning these filters is not necessarily a straightforward task.

It would be informative to compare the actuator efforts of the H_∞ method with the Youla hand computation method.

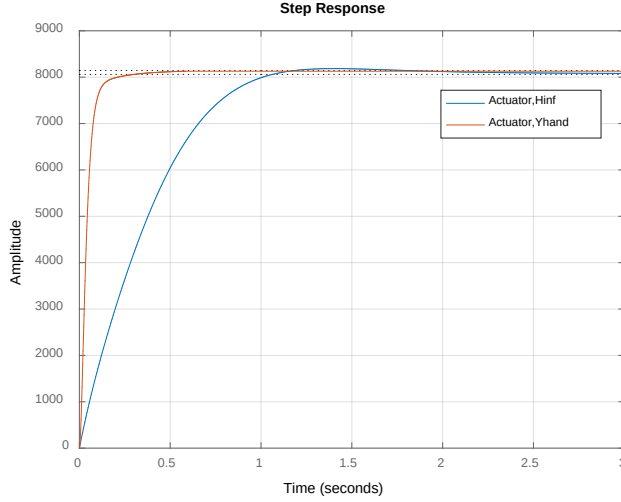


Figure 16.7: Actuator effort to the locking torque step input request comparison

Comparing the actuator effort results of the H_∞ controller with the Youla hand computation, it can be observed that the steady-state values are similar but the transient behaviors (rise time) are different. The response time for H_∞ is slower with slightly higher overshoot. This demonstrates that H_∞ method is conservative as it sacrifices performance (response time) for robust stability. It would be interesting to tune the Youla hand computation actuator effort transient behavior so that it is similar to H_∞ method then compare yaw rate responses. However, before addressing this, we present the results of one final control method as discussed in the previous chapter, this method utilizes the structure of IMC for designing a Youla controller using H_∞ optimization. This will be presented next.

In Chapter 6, we discussed the method of designing a Youla controller with model matching using IMC structure. This method has significant benefits over loop shaping technique as it does not require shaping filters. Furthermore, a Youla controller is computed directly using an optimization method rather than hand computation and only a target model is needed for this optimization. According to our previous discussion, this target model is selected to be,

$$T_t = \frac{-s + \gamma_1}{s + \gamma_2} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

We have selected these constants to be the same as above.

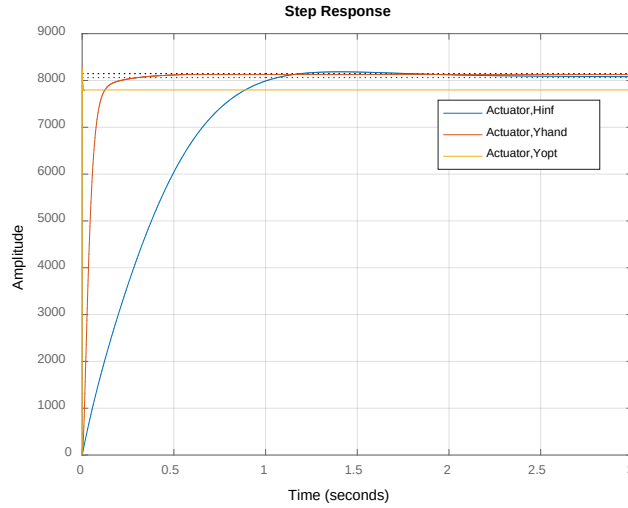


Figure 16.8: Actuator effort to the locking torque step input request comparison of all the proposed methods

In Figure above, we have added the result of Youla optimization actuator effort to Figure (16.7). It is interesting to observe that the actuator effort response of the Youla optimization method differs from Youla hand computation technique. It should be noted that we included two high frequency poles when selecting the Youla transfer function to better control the high frequency response of the Youla parameter at high frequency when we hand computed the Youla parameter. These two high frequency poles had an impact on the closed loop transfer function of the Youla controller by hand computation. However, when selecting the target model for the Youla optimization method, we did ignore these high frequency poles. We will add these high frequency poles to the target model and we will also limit the bandwidth of both Youla methodologies to the H_∞ method so that the actuator effort transient responses for all three strategies would be similar. Then we compare the results of these methods.

Let's assume a target model for the Youla Optimization method as follows,

$$T_t = \left(\frac{\gamma_2}{\gamma_1} \right) \frac{-s + \gamma_1}{s + \gamma_2} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \frac{1}{(\alpha s + 1)^2}$$

The constant γ_1 is the same as before but this time to limit the bandwidth $\gamma_2 = 3$. In addition, the constant $\alpha = 1/50$. The rest of the constant is the same as before. The ratio $\frac{\gamma_2}{\gamma_1}$ forces the closed loop transfer function target to be one at low frequency or dc. The Youla

transfer function for hand computation is selected so that the closed loop complementary
transfer function to be precisely the same as this target model.

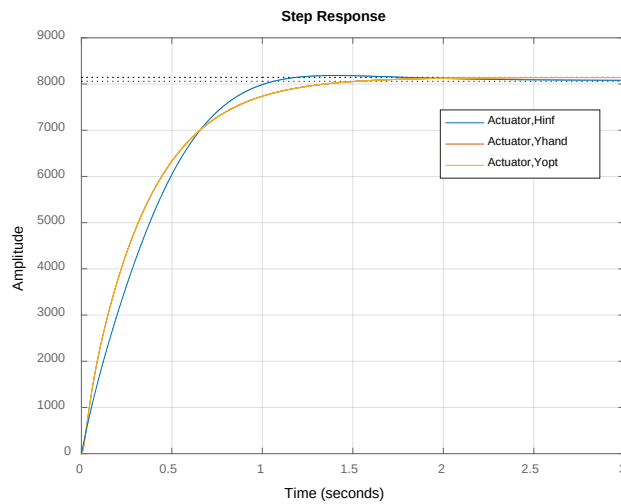


Figure 16.9: Actuator effort response to the locking torque request comparison of all the proposed methods with high frequency poles and limited bandwidth

The actuator effort results are illustrated in Figure above. The results for Youla hand
computation and Youla optimization method are the same. This should be no surprise as
we tried to force the closed loop complementary transfer functions for both methods to be
the same. We also limited the bandwidth so that the actuator effort for the H_∞ optimization
would be similar to the Youla methods.

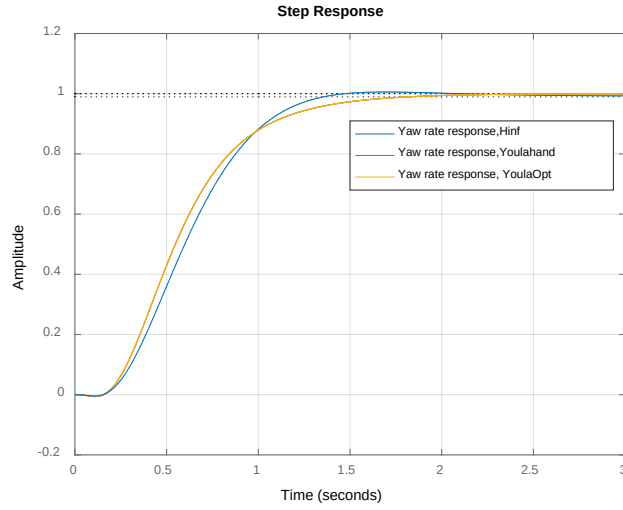


Figure 16.10: Yaw rate response to the locking torque request comparison of all the proposed methods with high frequency poles and limited bandwidth

By observing the Figure above, two things are evident,

- Youla control design using hand computation perform exactly the same as Youla control design using IMC structure with H_∞ optimization.
- By limiting the bandwidth of Youla control design techniques, we can obtain very similar results to H_∞ loop shaping control method.

These control methodologies could be implemented on-board vehicle for yaw stability control with active limited slip differential.

16.2 Vehicle Yaw Rate and Side-Slip Estimation

In this section, we will utilize the development of Chapter 7 in a vehicle dynamics application for yaw rate and side-slip estimation.

In the previous years, there are numerous articles for which many experts have attempted to estimate yaw rate and side-slip, e.g. see ([?]). Particularly, vehicle side-slip estimation has important benefits in Yaw Stability Control (YSC) applications. The standard estimation techniques which have been pursued in the past to estimate these variables were non-linear and unscented Kalman filtering techniques. Although, the authors have shown successful estimation of these variables using these approaches but there is no discussion about robustness characteristics or calibration overhead, two most important implementation requirements, using these standard techniques.

Here, we will utilize Youla Controller Output Observer (YCOO) as discussed in Chapter 7, to estimate vehicle yaw rate and side-slip. The YCOO turns out to have lower calibration overhead with better robustness characteristics in terms of many important vehicle dynamics parameters as will be investigated in this example.

Figure below illustrates a block diagram of COO with the acceleration measurement. The measured lateral acceleration including sensor noise, n , are used as a feedback loop reference target and the output consists of lateral acceleration from state-space representation of the plant model.

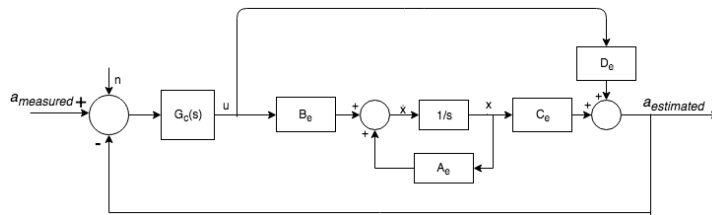


Figure 16.11: Block Diagram of Controller Output Observer (COO) using acceleration measurement

Kalman Filters:

(Linear) Kalman Filtering (LKF) is an estimation technique which utilizes a statistical

approach to estimate the mean and co-variance of the states. A plant model and sensor measurements are used to find the best estimation of the states. However, LKF uses linear state space representation as a model, therefore there are some limitations in the applications where plants are highly nonlinear or sensor measurements are of poor quality. For these applications, other types of Kalman Filters are suitable, such as Extended Kalman Filter (EKF) and Unscented Kalman Filter (UKF). A linearization of the system equations at each time step is required for EKF and the linearized Jacobian matrices are utilized to estimate the variance matrix. This technique is widely used in state, as well as certain parameter estimation. However, since the Jacobian matrices are required to be computed at each time step, the computational overhead can be high especially for systems with large number of states. UKF uses unscented transformation to sample the statistics associated with the plant model and sensor uncertainties to better represent the mean and the co-variance at each time step without the need for linearization. Therefore, UKF requires less computational overhead than EKF and it also has wide applications.

In this example, we will make a comparison of SISO YCOO, LKF, EKF, and UKF, mainly in terms of their robustness characteristics, in estimation of vehicle yaw rate and side-slip. However, before designing the estimators, we require a dynamic model of the vehicle and that is discussed next.

Vehicle Model - Nonlinear Bicycle Model with Pacejka Tire Model

One way to estimate the yaw rate is to use the lateral acceleration measurement at the vehicle center of gravity (c.g.) and using the dynamic relationship between the yaw rate and lateral acceleration. For a front steering vehicle, the system diagram of a simple vehicle dynamic model, using a two degrees of freedom bicycle model, is shown in Figure 16.12. Where U is the forward velocity, which is assumed to be constant. Parameters m and J are vehicle mass and moment of inertia. The distances from the vehicle c.g. to the front and rear axles are respectively shown as a and b . The front and rear tire lateral forces are indicated by F_f and F_r respectively. The front wheel steering angle input is illustrated by δ . The model includes two states, the lateral velocity V , and the yaw rate ω . With body fixed coordinates, the equations of motion of the bicycle model are written as,

$$\dot{V} = \frac{F_f \cos \delta + F_r}{m} - U\omega$$

$$\dot{\omega} = \frac{F_f \cos \delta \cdot a + F_r \cdot b}{J}$$

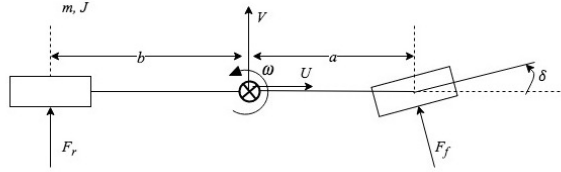


Figure 16.12: Front Steering Bicycle Model System Diagram

The lateral forces exerted on front and rear tires, F_f and F_r , are functions of the slip angles of front and rear tires. The slip angles of front and rear are related to the system input and the states,

$$\alpha_f = \delta - \tan^{-1}\left(\frac{V + a\omega}{U}\right)$$

$$\alpha_r = \tan^{-1}\left(\frac{b\omega - V}{U}\right)$$

The tire slip angles result in generating lateral tire forces and their relationship has been studied, for many decades, for developing an appropriate mathematical tire model. There are many tire models which relate the tire slip angles to the lateral tire forces. One commonly used and popular model is the Pacejka tire model, also known as the “Magic Formula” tire model. From [?], the basic formulas for lateral forces are,

$$F_f = D_f \cdot \sin\left(C \cdot \arctan\left(B_f \cdot \left((1 - E) \cdot \alpha_f + \frac{E}{B_f} \cdot \arctan(B_f \alpha_f)\right)\right)\right)$$

$$F_r = D_r \cdot \sin\left(C \cdot \arctan\left(B_r \cdot \left((1 - E) \cdot \alpha_r + \frac{E}{B_r} \cdot \arctan(B_r \alpha_r)\right)\right)\right)$$

where B , C , D and E are various tire parameters, such as B is the stiffness factor, C is the shape factor, D is the peak factor which is related to the tire normal force (F_z) and tire-road friction coefficient (μ), and E is the curvature factor. These coefficients are extracted from various tire testing and measurement data. We have utilized a typical tire data in this paper. Given a steering angle input and system state initial conditions, tire slip angles are computed from equations above, where the relationship between slip angles and the system input and the states is defined. Knowing the tire slip angles, lateral tire forces are computed from the Pacejka tire model. Having the lateral tire forces, the equations of motion

for lateral velocity and yaw rate are solved to obtain solutions of the new states. This cycle is then repeated at each time step.

The lateral acceleration at c.g. is related to the front and rear tire forces and is given by,

$$a_y = \frac{F_f \cos \delta + F_r}{m}$$

This SISO model then consists of one input, the front road wheel angle, δ , and one output, the vehicle lateral acceleration at the c.g., a_y .

This non-linear model is used to represent the actual vehicle for testing the robustness and performance of the proposed estimators while the linearized version of this model is used for developing YCOO and LKF estimators.

Linearizing the Bicycle Model:

Both YCOO and LKF require a linearized model of the plant. Therefore, we need to linearize the vehicle and tire models to obtain a state space representation of a the plant to be used for YCOO and LKF estimators. For linearizing a bicycle model, all angles were assumed to be small, and the state equations are re-written as

$$\begin{aligned}\dot{V} &= \frac{F_f + F_r}{m} - U\omega \\ \dot{\omega} &= \frac{F_f \cdot a + F_r \cdot b}{J}\end{aligned}$$

For linearizing a tire model, it is assumed that the lateral tire forces are proportional to slip angles with front and rear cornering stiffness coefficients C_f and C_r ,

$$F_f = C_f \alpha_f$$

$$F_r = C_r \alpha_r$$

Then the state-space representation of the equations of motion is written as,

$$\begin{pmatrix} \dot{V} \\ \dot{\omega} \end{pmatrix} = \begin{pmatrix} -\frac{C_f + C_r}{mU} & -(\frac{aC_f - bC_r}{mU} + U) \\ \frac{aC_f - bC_r}{JU} & -\frac{a^2C_f + b^2C_r}{JU} \end{pmatrix} \begin{pmatrix} V \\ \omega \end{pmatrix} + \begin{pmatrix} \frac{C_f}{m} \\ \frac{aC_f}{J} \end{pmatrix} \delta$$

The lateral acceleration at c.g a_y can be expressed as,

$$a_y = \left(\frac{C_f + C_r}{mU} \quad -\frac{aC_f - bC_r}{mU} \right) \begin{pmatrix} V \\ \omega \end{pmatrix} + \left(\frac{C_f}{m} \right) \delta$$

Uncertainties:

The acceleration sensor measurements in the simulation environment are made from a nonlinear model, a representation of an actual vehicle, as stated previously. While, the YCOO and LKF utilize a linear model to estimate the states of this nonlinear plant. Therefore, for these estimators, model uncertainties include the differences between linear and nonlinear plants used for state estimation and acceleration measurements respectively. In addition, to this inherent uncertainty, we will examine parameter and environmental uncertainties, mass and tire-road coefficient of friction variations, for all estimator approaches. For simulationg sensor noise, data sheet of a single axis accelerometer SCA830-D05 [?] is utilized. From the data sheet, we can assume that the sensor noise has a zero mean and a variance of $R = 2.5 \cdot 10^{-3}$.

State Estimation

In the state estimation for the SISO case, a front wheel steering vehicle with the front road wheel angle and one accelerometer at center of gravity of the vehicle for acceleration measurement are utilized to estimate the vehicle yaw rate as shown in Figure 16.12. The input, in this case, is the front wheel steering angle (δ), and the output is the lateral acceleration at c.g (a_y).

Youla Parameterization Estimator Design:

The transfer function of the plant is the ratio between lateral acceleration and the road wheel angle,

$$G_p(s) = \frac{a_y}{\delta}(s) \quad (16.10)$$

There is no unstable poles and zeros, based on the Youla parameterization technique in [?], the Youla parameter can be computed by ²,

$$Y = \frac{1}{G_p} \cdot T$$

²It should be noted if the plant included unstable poles or non-minimum phase zeros, the interpolation conditions could be used to handle these types of plants.

Where T is the desired closed-loop transfer function, and we select this desired transfer function to be a second order well damped filter with two additional poles at $s = 1/\tau$ for better shaping the Youla parameter at high frequency,

$$T = \frac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)(\tau s + 1)^2}$$

Then the Youla parameter is written as

$$Y = \frac{1}{G_p} \cdot \frac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)(\tau s + 1)^2}$$

The sensitivity transfer-function S of the feedback loop can be derived using the algebraic constraint between this transfer function and the closed-loop transfer function, T ,

$$S = 1 - T$$

and finally the controller G_c , to be used for state estimation using YCOO, can be calculated by,

$$G_c = Y/S$$

With proper choice of ζ and ω_n , the controller, G_c , is designed so that the closed loop transfer function, T , is at 0 dB at low frequency which indicates good target following capability, and the sensitivity transfer function, S , is small at low frequency which indicates good disturbance rejection capability. In selecting a bandwidth for the closed loop transfer function, T , a trade off between the transient response and the overshoot should be made. Figure 16.13 shows the bode plot of the linear plant used in YCOO. The plant has high gain of 40 dB at lower frequency and the gain drops after frequency of 1 rad/s and reach the minimum at 20 dB around 8 rad/s. If the bandwidth of the controller/estimator is greater than 1 rad/sec, the system will have an overshoot due to the change in plant gain.

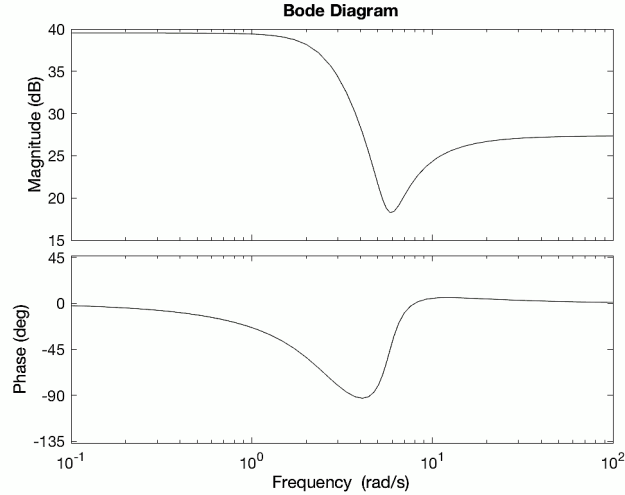


Figure 16.13: Bode plot of linear plant.

As shown in Figure 16.14, T is at 0 dB for a wide range of frequency. For the case of yaw rate estimation, due to designing a closed loop with large bandwidth, the yaw rate transient response will include a large overshoot. This overshoot can only be reduced or eliminated by reducing the closed loop bandwidth. Regarding the bandwidth requirement/constraint, a set of objective requirements for the estimation results should be decided to reduce the subjectivity of our findings. For our comparative analysis in this paper, we will use the results of the YCOO with the highest bandwidth.

It is important to point out that the bandwidth constraint in both control and estimation design has a significant importance. This constraint is directly addressed in the YCOO technique while in all Kalman filtering strategies, the bandwidth is dealt with as an afterthought.

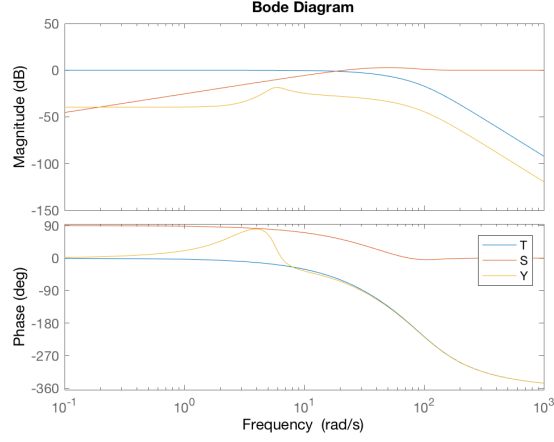


Figure 16.14: Bode plot for SISO case with large bandwidth.

8754 Another way to look at the design is to look at the bode plot of the closed loop transfer
 8755 function of estimating the yaw rate T_ω . The transfer function of the yaw rate ω to the wheel
 8756 angle input δ is

$$G_{p_\omega} = \frac{\omega}{\delta}(s) = \frac{\frac{aC_f}{J}s + \frac{(a+b)C_fC_r}{mJU}}{s^2 + \left(\frac{C_f+C_r}{mU} + \frac{a^2C_f+b^2C_r}{JU}\right)s + \frac{(a+b)^2C_fC_r}{mJU^2} + \frac{bC_r-aC_f}{J}}$$

8757 With the controller transfer function designed G_c , we can obtain the complementary
 8758 closed loop transfer function as,

$$T_\omega = \frac{G_c G_{p_\omega}}{1 + G_c G_{p_\omega}}$$

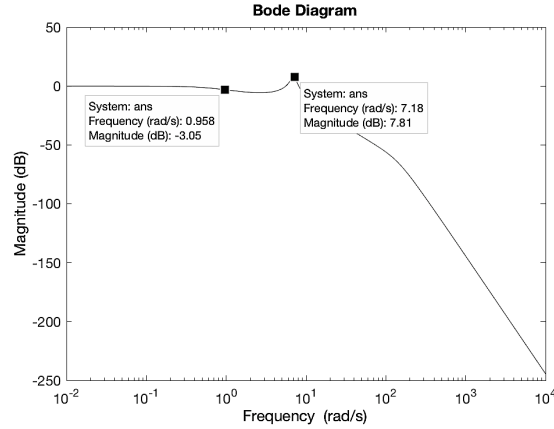


Figure 16.15: Bode plot of closed loop transfer function of yaw rate estimation.

Looking at the bode plot of T_ω (Figure 16.15), the gain of T_ω is at 0 dB at low frequency, which means a time response to a step input with zero steady-state error. However, around 7 rad/s the closed loop transfer function has a peak magnitude of around 8 dB, which means a transient response with a large overshoot. If we decrease the bandwidth for T to 1 rad/s or even smaller, the yaw rate estimation will follow the target very well with no overshoot, but it will drastically slow down the estimation response time.

Simulation Results

Figure 16.16 and 16.17 show a comparison of the lateral acceleration at the c.g. and the yaw rate estimations using YCOO, LKF, EKF, and UKF with 2° road wheel angle step input. For both the lateral acceleration and the yaw rate estimates, all these strategies perform well in both sensor noise rejection and target following. Figures 16.18 and 16.19 illustrates the results of the yaw rate estimations with 6° and 10° road wheel angle step input. As we increase the wheel angle input, we can observe that LKF will have a steady state error as the vehicle model (tire forces) starts operating in the nonlinear region. It should be noted that the lateral tire forces are linear at small tire slip angles and as the slip angle increases, the tire forces begin to operate in the nonlinear region and eventually at the very high tire slip angles, the lateral tire forces will saturate. As the tire slip angles increase, purely linear tire models will provide larger lateral forces than their nonlinear counterpart, and as a result, LKF overestimates the yaw rate. It is interesting to observe that in the case

of YCOO, although this strategy utilizes a linear tire model, its robustness characteristics results in excellent steady state target following. The robustness characteristics of YCOO will be further investigated in the next section. However, as discussed in the last section, the closed loop bandwidth of YCOO is selected in such a way that its transient response due to the overshoot is not as good as the other proposed methods. The EKF and UKF perform well in both these cases as they are utilizing the same nonlinear vehicle model used for the reference generator (generating the lateral acceleration reference) as for the estimation. We will investigate the performance of the EKF and UKF in the robustness test of the next section when the model used for estimation differs from its nominal values.

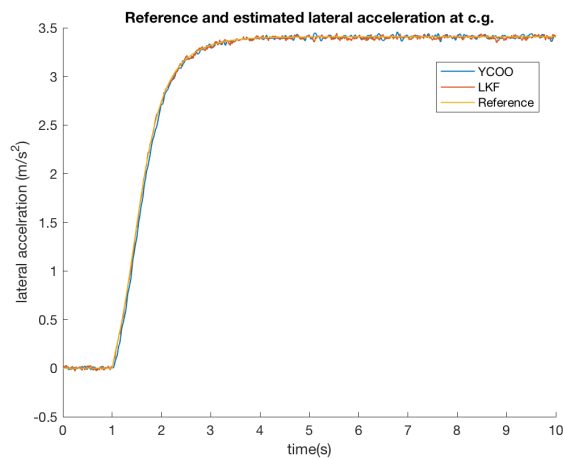


Figure 16.16: Lateral acceleration estimation using different strategies with 2° road wheel angle step input

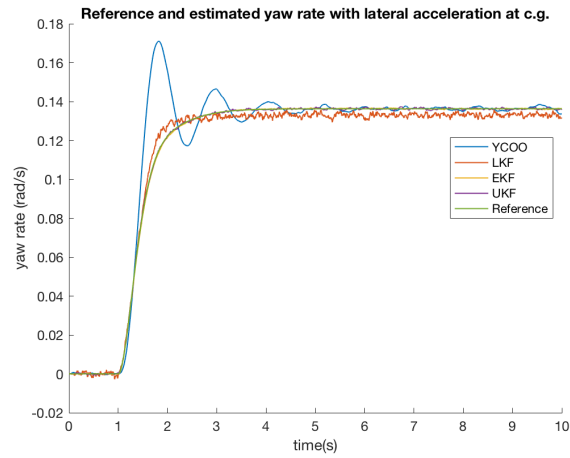


Figure 16.17: Yaw rate estimation using different strategies with 2° road wheel angle step input

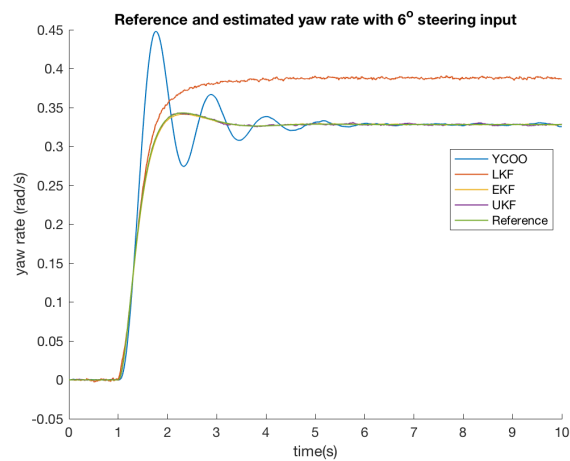


Figure 16.18: Yaw rate estimation using different strategies with 6° road wheel angle step input

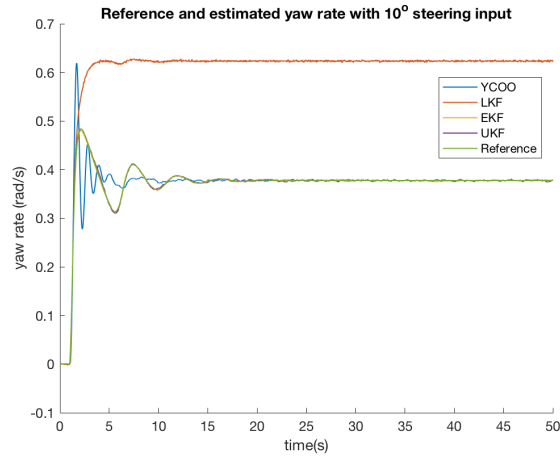


Figure 16.19: Yaw rate estimation using different strategies with 10° road wheel angle step input

Robustness Test:

Vehicle Mass variation,

Figure 16.20 shows the yaw rate estimation with different strategies with an addition of 30% change in the vehicle mass and a 2° road wheel angle step input. It illustrates that all Kalman filter approaches are sensitive to the vehicle mass variation and do not have good robustness characteristics with respect to this parameter. These results suggest that if the Kalman Filters are tuned to have good sensor noise rejection capabilities and hence, more confidence is given to the model and if the sensitivity of the filters are high with respect to the model parameter changes, then this will negatively impact the estimation performance. But, it is interesting to observe that the YCOO performance has not been impacted with respect to this parameter variation. It could be argued that the plant parameter variations occur in the low frequency region where the YCOO is designed to have a closed loop sensitivity transfer function with a small magnitude, hence, resulting in a good robustness characteristic.

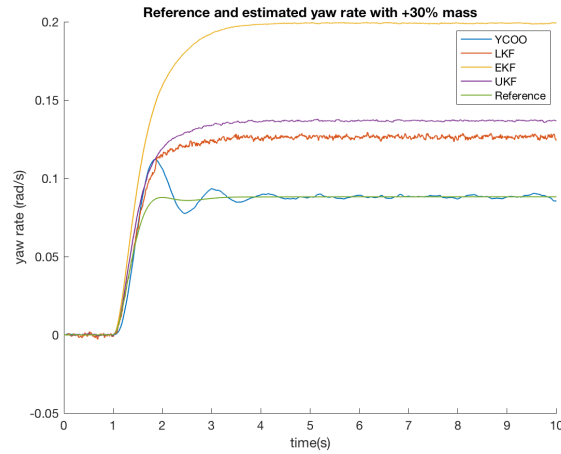


Figure 16.20: Yaw rate estimation using different strategies with +30% mass

8806 Tire-road coefficient of friction,

8807 The yaw rate estimations with a reduced tire-road coefficient of friction $\mu = 0.5$ (the
 8808 coefficient for dry asphalt is equal to $0.9 - 1$) and a 2° road wheel angle step input are
 8809 shown in Figure 16.21. The results indicate that all strategies are not very sensitive to
 8810 the changes in μ as they are to the vehicle mass variations. Comparatively, EKF is more
 8811 sensitive to μ than the other strategies.

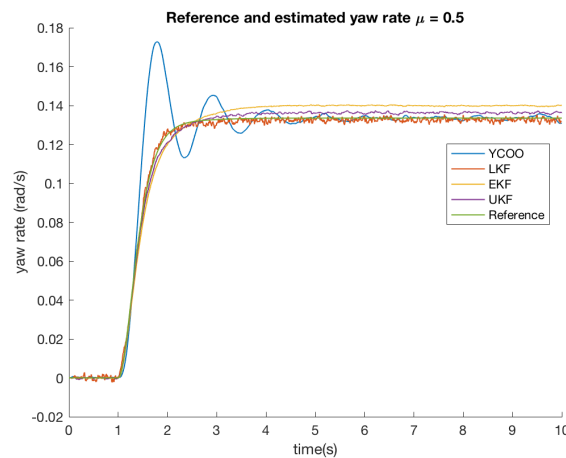


Figure 16.21: Yaw rate estimation using different strategies with $\mu = 0.5$

8812 The estimation results presented above suggest that the YCOO is an estimation method
8813 with both good performance and robustness characteristics. But, most importantly, it re-
8814 quires very little to no tuning when compared to the standard estimation techniques such
8815 as Kalman filtering.

8816 It is worth mentioning that all the estimation approaches did show sensitivity to one impor-
8817 tant variable, the vehicle longitudinal velocity. Our future plan is to utilize Reinforcement
8818 Learning techniques with the YCOO to address this deficiency.

CHAPTER 17

ESTIMATION DESIGN FOR MIMO USING PARAMETERIZATION APPROACH

17.1 Introduction

In Chapter 7 of Volume 1, we demonstrated how the Youla Parameterization approach could be adapted for state feedback for a single-input, single-output system. Here, we will expand this method to include multi-input, multi-output systems.

The general structure for a MIMO YCOO is shown in Figure 17.1. Here, controllable inputs, $\hat{u}$, are used to minimize the error between the estimated signals, $\hat{y}$, and the measured signals, y , with the use of the feedforward controller, $Y(s)$. The estimated states, $\hat{x}$, are also shown in this structure. The matrices A_e , B_e , C_e , and D_e were discussed previously. The left partition represents signals coming from an actual vehicle (containing sensor noise) while the right is contained entirely within the computer for estimation purposes.

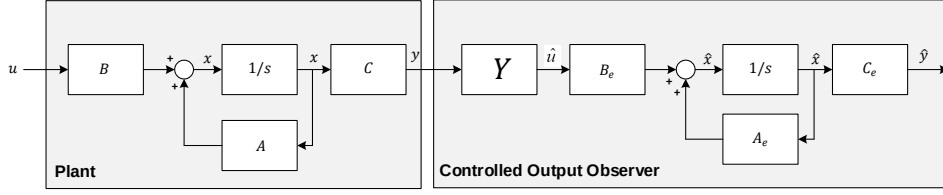


Figure 17.1: Youla Controller Output Observer

17.2 MIMO Estimator Design

We will use the suspension system from the last SISO example to derive a MIMO estimator. In this example, both the road disturbance and the force input are treated as two inputs and the sprung and unsprung mass accelerations are the measured outputs. Therefore, the system includes two inputs and two outputs. The block diagram in Figure 17.1 will be utilized to design this estimator using the previously developed quarter car suspension model. However, this time, the results of the state estimation are compared to a validated nonlinear six chassis degree-of-freedom vehicle model, G_p . Due to the space constraints, the development of this model is not discussed in this paper [?]. It is important to note that in this example the plant model, G_p , and the observer model, $G_{p,e}$ are not the same as it was the case for the SISO design. In fact, G_p is a nonlinear model as mentioned above and the observer model, $G_{p,e}$, as is described in the following paragraphs, is a linear transfer function matrix.

In Chapter 7, it was stated that the road input velocity, v_i , and the force input, F_a , are the controlled inputs while the sprung mass, $\dot{v}_s$, and unsprung mass, $\dot{v}_{us}$, accelerations are the signals we wish to track. This is represented by $G_{p,e}$ in Equation 17.1.

$$\begin{bmatrix} \dot{v}_s \\ \dot{v}_{us} \end{bmatrix} = G_{p,e} \begin{bmatrix} F_a \\ v_i \end{bmatrix} \quad (17.1)$$

where, $G_{p,e} = C_e(sI - A_e)^{-1}B_e + D_e$

SISO Youla transfer functions are found for the diagonals of the Smith McMillan form.

$$G_{p,e}^{SM}(s) = \begin{bmatrix} G_{sm1} & 0 \\ 0 & G_{sm2} \end{bmatrix}; \quad (17.2)$$

These diagonal entries are defined below. Asterisks (*) represent the portion of the plant with no unstable zeros.

$$G_{sm1} = \frac{\beta_1 s}{(s^2 + \alpha_{21}s + \alpha_{22})(s^2 + \alpha_{12}s + \alpha_{11})} \quad (17.3)$$

$$= sG_{sm1}^* \quad (17.4)$$

$$G_{sm2} = \beta_2 s^2 \quad (17.5)$$

$$= s^2 G_{sm2}^* \quad (17.6)$$

17.3 State Estimation

17.3.1 First Decoupled System (G_{sm1})

Based on the procedures outlined for SISO design, the Youla transfer function, Y_1 , is shaped to give a desirable complementary sensitivity transfer function, T_1 . Normally, T_1 is shaped as a low-pass Butterworth filter where the magnitude is one at low frequencies and decreases at high frequencies for good tracking and noise rejection, respectively. However, in this case, it is impossible to achieve a DC magnitude of one because there exists an unstable zero at the origin. Therefore, T_1 is designed similar to a bandpass filter where the magnitude is one except at very low frequencies. This process was applied in the earlier SISO control development. A pole, p_1 , is placed near the origin to approximately cancel the effect of the unstable zero. A first order lag and a second order filter are designed to achieve the required performance and robustness characteristics. To ensure the properness of Y_1 , a high frequency pole, p_2 , is included. A gain was chosen so that the magnitude of T_1 is equal to one in the low frequency region up to the closed loop bandwidth. The results are summarized in Equation 17.7 - 17.8 and Figure 17.2.

$$Y_1 = \frac{k_1}{(s + p_1)(s^2 + 2\zeta\omega_n s + \omega_n^2)(s + p_2)} \frac{1}{G_{sm1}^*} \quad (17.7)$$

$$\begin{aligned} T_1 &= Y_1 G_{sm1} \\ &= \frac{k_1 s}{(s + p_1)(s^2 + 2\zeta\omega_n s + \omega_n^2)(s + p_2)} \end{aligned} \quad (17.8)$$

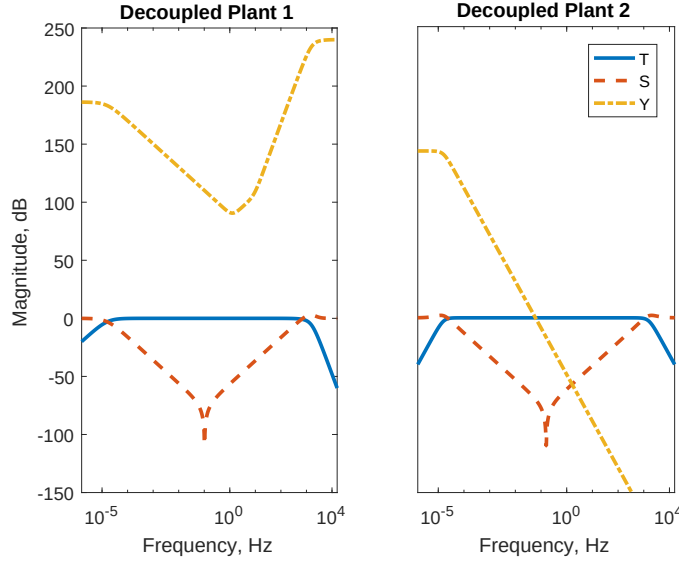


Figure 17.2: Bode for decoupled system.

17.3.2 Second Decoupled System (G_{sm2})

A similar process is performed for the second decoupled system, G_{sm2} . With this new plant, there are now two unstable zeros at the origin. The two unstable zeros are dealt with as before, this time using a complex set of poles (defined by ζ_1 and ω_{n1}) at a very low frequency. Then, a complex set of poles (defined by ζ_2 and ω_{n2}) are used to shape T_2 after the closed loop bandwidth and at a higher frequency. A gain was chosen so that the magnitude of T_2 is equal to one in the low frequency region up to the closed loop bandwidth. The results are summarized in Equation 17.9 - 17.10 and Figure 17.2.

$$Y_2 = \frac{k_2}{(s^2 + 2\zeta_1\omega_{n1}s + \omega_{n1}^2)(s^2 + 2\zeta_2\omega_{n2}s + \omega_{n2}^2)} \frac{1}{G_{sm2}^*} \quad (17.9)$$

$$\begin{aligned} T_2 &= Y_2 G_{sm2} \\ &= \frac{k_2 s^2}{(s^2 + 2\zeta_1\omega_{n1}s + \omega_{n1}^2)(s^2 + 2\zeta_2\omega_{n2}s + \omega_{n2}^2)} \end{aligned} \quad (17.10)$$

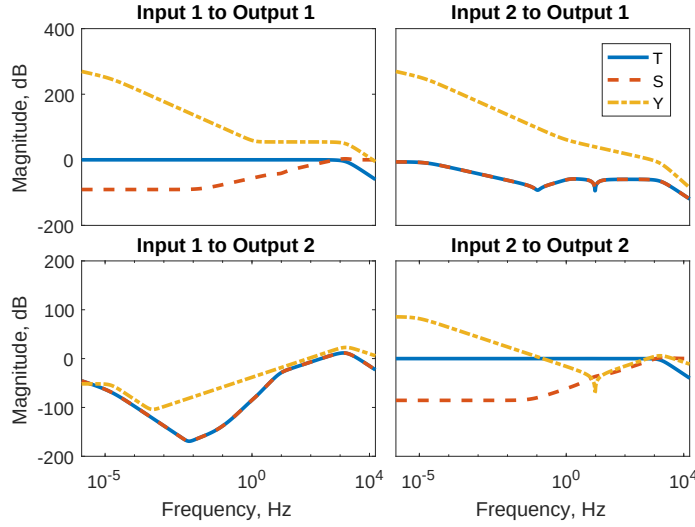


Figure 17.3: Bode for the coupled system.

17.3.3 Coupled System

The decoupled Youla transfer functions are coupled by utilizing the previously computed unimodular matrices. Complementary sensitivity, sensitivity, and the closed loop controller transfer functions for the coupled system are derived in Equations 17.11-17.14.

$$Y = U_R M_y U_L \quad (17.11)$$

$$T = G_p Y \quad (17.12)$$

$$S = I_2 - T \quad (17.13)$$

$$G_c = Y S^{-1} \quad (17.14)$$

Where

$$M_y = \begin{bmatrix} Y_1 & 0 \\ 0 & Y_2 \end{bmatrix} \quad (17.15)$$

The coupled system Bode plots are shown in Figure 17.3.

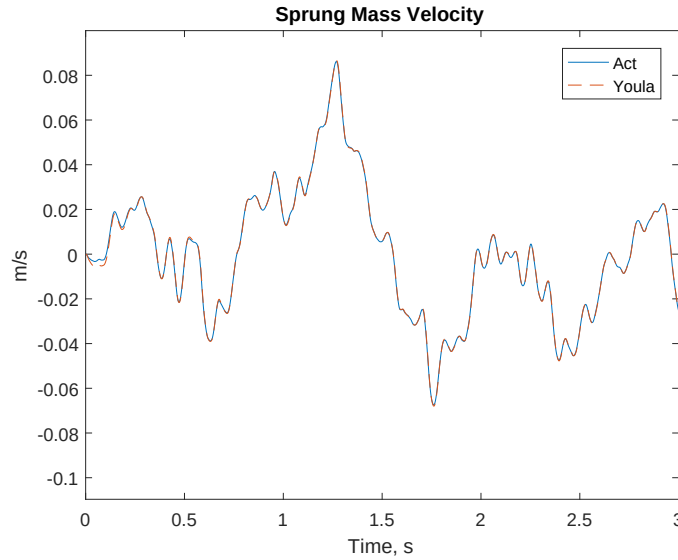


Figure 17.4: MIMO state estimation with D-Road PSD input - Sprung mass velocity.

17.4 Applications

The utility of this estimator is made apparent by two example applications: Skyhook based damping force estimation for closed loop control in active/semi-active suspensions, and road roughness coefficient estimation.

17.4.1 States Estimation: Four States

As with the SISO case, state estimation is performed over a typical D-road (Farm road). The states include the sprung and unsprung mass velocity (Figures 17.4 and 17.5), wheel stroke (Figure 17.6) and tire deflection (Figure 17.7). The estimates are comparable to the actual signals generated from the nonlinear vehicle model.

17.4.2 Input Estimation: Skyhook Based Control

The estimator is applied to a Skyhook controller, a common strategy with variants deployed on many vehicle brands. It was proposed that an optimal suspension is one in which the sprung mass is connected through a linear damper to some inertial reference [?]. Due to the inability to build such a contraption, this ideal damping force is instead applied within the suspension between sprung and unsprung masses through either force actuator or control of the damper. Although sub-optimal, this strategy has proven highly effective. The ideal

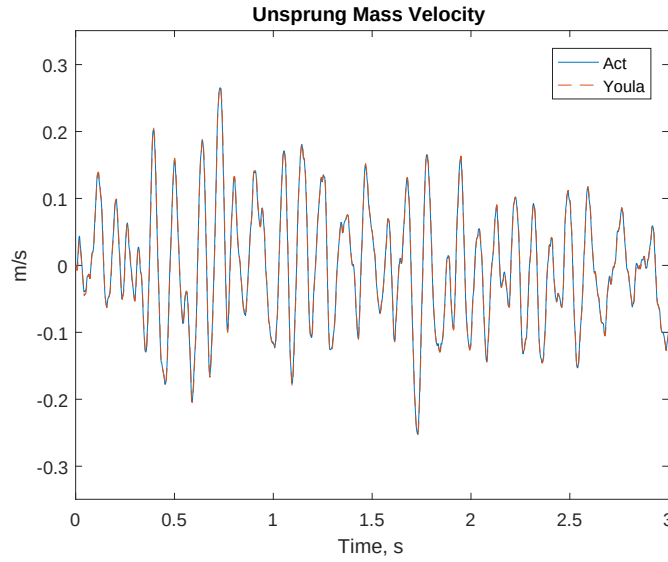


Figure 17.5: MIMO state estimation with D-Road PSD input - Unsprung mass velocity.

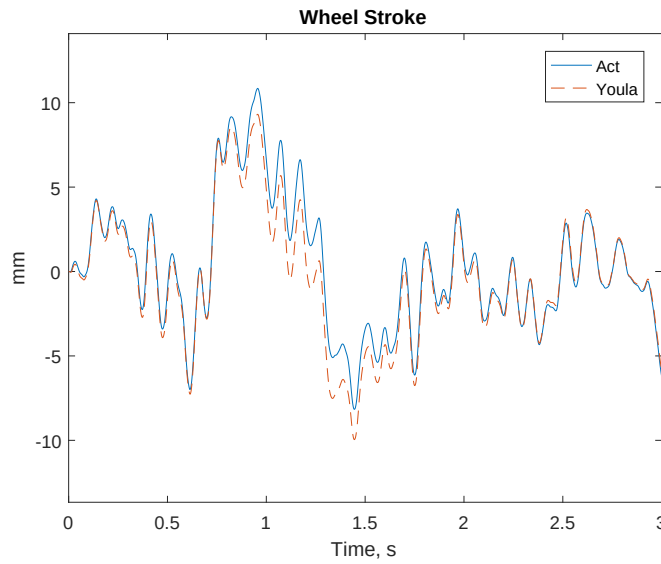


Figure 17.6: MIMO state estimation with D-Road PSD input - Wheel stroke.

8896 damping force is given by Equation 17.16, where b_{SH} is a tuning parameter and v_s is the
8897 sprung mass vertical velocity.

$$F_{d,SH} = b_{SH}v_s \quad (17.16)$$

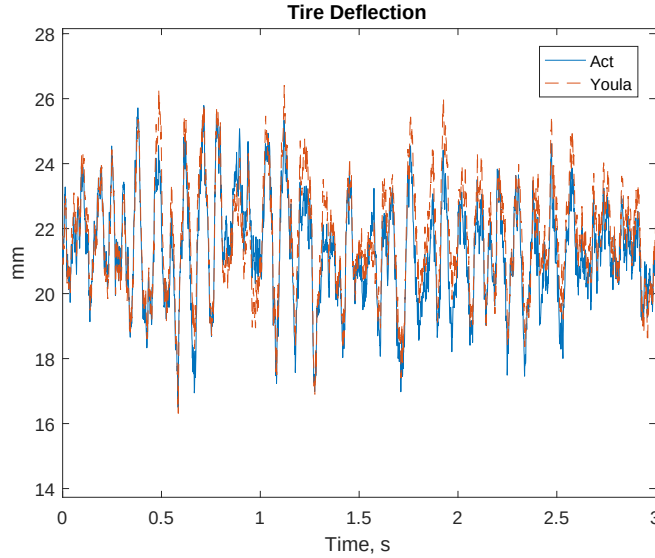


Figure 17.7: MIMO state estimation with D-Road PSD input - Tire deflection.

An idealized fully active suspension using the Skyhook strategy is applied to a high order vehicle model developed in [?, ?]. Each corner suspension is modeled as a nonlinear spring and damper (lookup tables) with an active element capable of applying ideal forces.

For optimizing the active or semi-active suspension performance, it is recommended to utilize the damper force in the feedback loop. Therefore, either a measurement or estimation of this signal is required. In this section, comparative performance results are obtained for scenarios where ideal access to the damping force measurement is available versus when an estimation of this force using the YCOO concept is made. It will be shown that the influence of the estimator does very little to inhibit the performance of the controller when compared to the ideal case of having access to the measured damping force.

The control structure in Figure 17.8 is proposed and described in this paragraph. A simple controller, $G_{c,SH}$, different from the YCOO discussed previously, is developed that drives the effective damping force towards the ideal Skyhook damping force, $F_{d,SH}$, using the controlled input force, F_i . This effective damping force is the combination of both passive damping and the controllable damping force. The estimate of the damping force, $\hat{F}_d$, will inherently contain both these forces. Note that d represents additional system inputs and the YCOO block represents the estimator.

Success of the estimator is measured by the level of degradation in root-mean-square (RMS) heave acceleration as this value is strongly correlated with ride comfort. A com-

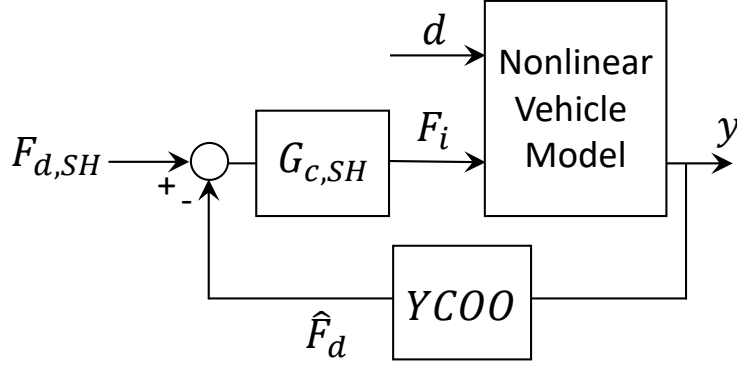


Figure 17.8: Damping force feedback control structure.

Table 17.1: Heave acceleration results for the uncontrolled and controlled case, with and without the estimator

Test	Motorway	Bump
No control [RMS, m/s^2]	0.0279	0.1192
Control with ideal measurement [% change]	-52.3176	-30.2342
Control with estimator [% change]	-46.5230	-22.0213

parison is made between uncontrolled, controlled with no estimator, and controlled with estimator. This is done in two scenarios: (1) driving over a motorway at 100 KPH and (2) driving over a large bump (0.1 m tall) at 40 KPH. The results are summarized in Table 17.1 where RMS heave acceleration is shown for the uncontrolled case and percent change for cases with control. Both control scenarios are shown: with the ideal case of direct damping force feedback, and with the estimated damping force feedback. It is apparent that little degradation in RMS heave acceleration exists due to the inclusion of the estimator versus direct measurement feedback. Additionally, Figure 17.9 shows the estimated damping force versus that generated from the nonlinear model for motorway conditions.

17.4.3 Input Estimation: Road Roughness

Road roughness estimation is important for suspension parameter adaptation. Expensive sensors may be used to measure the road directly, but the method proposed in this paper needs only sprung and unsprung mass accelerations, which are readily available on production vehicles. A road roughness coefficient, G_0 , is a parameter that fits a special curve, as shown in Equation 17.17, to a Power-Spectral-Density (PSD) plot of the road - v is the

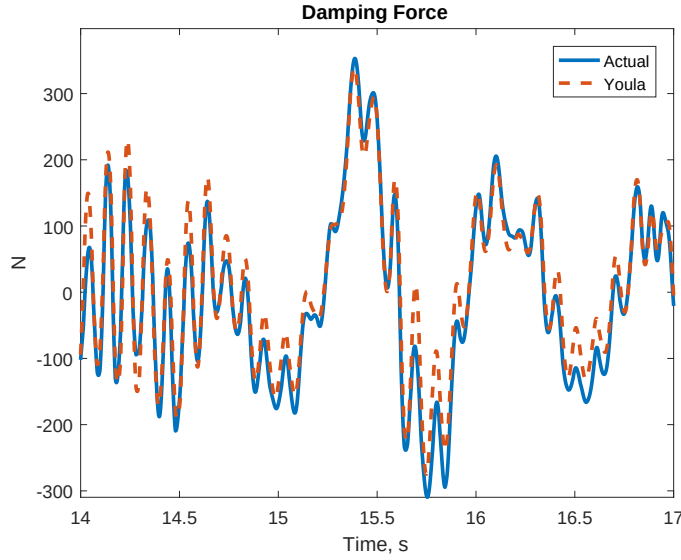


Figure 17.9: Actual versus estimated damping force over a motorway at 100 KPH.

8932 spatial frequency and v_0 is the cutoff spatial frequency, both in rad/m , and G_z is the road
 8933 PSD in $m^2/(rad/m)$ [?].

$$G_z(v) = G_0[1 + (v_0/v)^2]/(2\pi v)^2 \quad (17.17)$$

8934 Such a plot is shown in Figure 17.10 where results are shown for a motorway (bottom)
 8935 and farm road (top). Shown in gray, blue, and red are the PSD from the actual road data,
 8936 the best fit of the actual road PSD, and a best fit of the estimated road PSD, respectively.
 8937 The general shape of this plot is a result of decreasing road variation over smaller distances
 8938 (larger spatial frequency). It is apparent from Figure 17.10 that the best fit of both the actual
 8939 and estimated road data are very similar over a large frequency range.

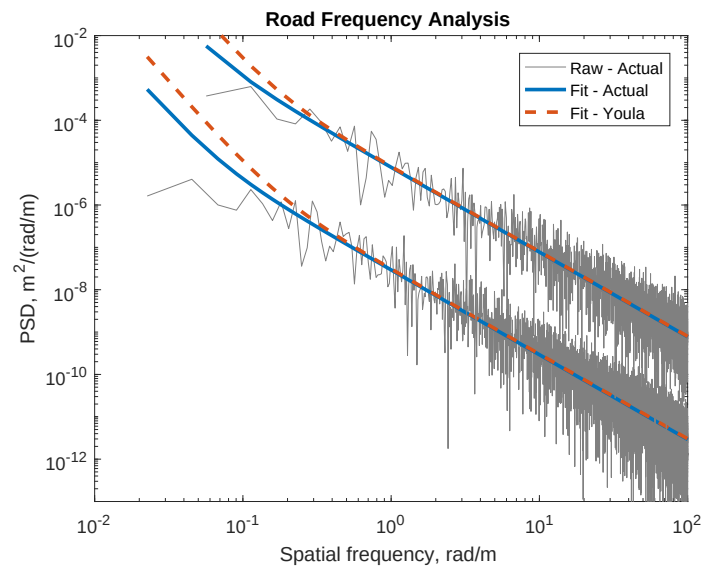


Figure 17.10: Power Spectral Density of road profile for a motorway (bottom) and farm road (top).

CHAPTER 18

PRACTICAL APPLICATIONS

APPENDIX A
